# Proof Transfers and Global Classifications

Complete Integrated Edition
Spectral Design, Statistical Recovery, Recurrences,
Period Geometry, Arithmetic, and Exact Cone Gluing

KEIO

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**Abstract**

This volume brings together the nine research manuscripts developed in the present research sequence. The collection begins with transfers to exact recurrence complexity, random-matrix statistics, spectral network design, Hamiltonian dynamics, stopping boundaries, and hidden-state recovery. It continues through entropy–transport methods for mixture likelihood, complete support and noise regimes, and sharp graph-design comparisons. The differential and arithmetic parts study weighted Franel series, anisotropic lattice and banana periods, differential Galois groups, algebraic quantiles, arithmetic subsequences, and Apéry limits. Part VIII contains the expanded entropy–transport and Laplace-monodromy manuscript, including optimal coefficient exponents, complete testing transitions, Bessel determinant modules, tableau values, and split-beta orthogonality. The final part develops exact cone gluing, all-collision criteria, and sharp Jacobi regularity, and incorporates the subsequent algebraicity obstructions with their proofs. All nine mathematical texts are retained with their proofs, examples, limitations, and supporting appendices. A single notation framework, part-qualified cross-references, a unified bibliography, and an editorial map make the progression of results explicit. Earlier special cases and alternative arguments remain available for comparison; statements of open directions in those earlier texts are interpreted with the later closures indicated in this volume.

**Research status.** This is a consolidated research draft prepared with AI assistance. Its component manuscripts underwent the internal checks described in their texts; compilation and integration checks do not constitute external mathematical peer review. Historical priority and publication-level significance are not certified by this collection.

## Editorial guide and coverage

The scope of the collection is the nine manuscripts produced during this research sequence, including the expanded entropy–transport revision and the subsequent cone-gluing manuscript. The older numbered manuscripts supplied as methodological source material remain bibliographical sources, rather than additional constituent papers. Each part preserves the mathematical body of one manuscript. Part IX also contains the later proved algebraicity corollaries. Repeated supporting proofs have deliberately been retained where they make a part self-contained or document a different argument. Duplicate front matter and repeated reference lists are consolidated.

| Part | Constituent manuscript | Main coverage |
|---|---|---|
| I | Exact Complexity and Arithmetic Transfers | Six transfers, from lattice moments and Jacobi ensembles to clocks, stopping, and hidden-state recovery. |
| II | Sharp Bounds and Connection Formulas Beyond Existing Restrictions | Trees, alternating binomial sums, nonresonant trace connections, and long hidden trajectories. |
| III | Entropy, Likelihood Geometry, and Sharp Proof Transfers | Corrupted reports, unequal depths, posterior transport, all connected graphs, and phase weights. |
| IV | Global Parameter Theorems from Entropy, Transport, and Spectral Structure | All-support Poisson recovery, resonant trace connections, block graphs, and a robustness obstruction. |
| V | Global Algebraicity and Exact Complexity | Weighted Franel algebraicity, hypercubic rank, quantiles, Gaussian likelihood, and graph inflation. |
| VI | Global Galois Groups and Complete Parameter Classifications | Galois groups, colliding thresholds, half-integer quantiles, cubic weights, and all-scale Gaussian recovery. |
| VII | Beyond Generic Parameters: Global Classifications | Remaining orthogonal groups, all-exponent recurrence widths, complementary beta quantiles, and a noisy-observation obstruction. |
| VIII | Entropy–Transport, Critical Values, and Laplace Monodromy | The latest expanded manuscript: coefficient optimality, statistical transitions, subsequences, local Fourier bounds, and arithmetic classifications. |
| IX | Exact Cone Gluing and Collision Tests | General graphical connections, resonant logarithms, sharp regularity, finite collision decisions, and algebraicity obstructions. |

Part-qualified numbering makes every theorem, equation, and appendix reference unambiguous. For example, a reference beginning with VIII belongs to the entropy–transport manuscript; IX belongs to the cone-gluing extension. Appendix letters remain local to their part. Each part starts with its original abstract and an integration note. All cited reference entries are collected at the end. The accompanying TEX source contains the full volume and requires no external manuscript, bibliography database, image, or included source file.

### How to read the successive closures

The chapter sequence records research development, so a statement such as “not proved here” in an earlier part describes that part’s argument. It must be read together with the following developments. The nonresonant Jacobi trace connection in Part II is continued to resonant parameters in Part IV. The tree design result in Part II is followed by connected-graph, block-graph,

and dense-inflation comparisons. The all-support Poisson theorem of Part IV is incorporated with its proof into Part VIII. The support-free growing-support Gaussian result in Part V is complemented by a known-support all-scale theorem in Part VI. These use different optimization domains: Part VI also explains why the unrestricted fit cannot adapt to arbitrarily small known intervals. The heteroscedastic obstruction in Part VII concerns a different observation model.

The phase and cubic-weight recurrence classifications in Parts III and VI are followed by the all-exponent, all-weight result in Part VII and the arithmetic-subsequence theorem in Part VIII. Differential Galois groups left open by the rank argument in Part V are addressed in Parts VI and VII. Part VIII gives the local Fourier formulation used to repair the Laplace–Jordan argument and states its regularity hypotheses explicitly. The precise replacement is Theorem VIII.8.3; the complete subsequence classification is Theorem VIII.7.2. All algebraicity and transcendence statements retain the precise parameter assumptions and exceptions in their individual theorems.

Minimum recurrence width is not the number of nonzero recurrence terms. The sparse-term lower bounds for alternating and unit-circle weights in Parts II and III therefore remain additional assertions, with their own positivity and Descartes arguments. They are not replaced by the later width-only classifications, even when those later results cover more weights or more subsequences.

Part VIII does not supersede the collection's unrelated results in network optimization, Hamiltonian arithmetic, optimal stopping, hidden trajectories, robust likelihood, or Jacobi trace connections. Those arguments are therefore retained in full. No new theorem is claimed merely from placing the manuscripts in a single volume.

## The final extension and the meaning of a global conclusion

Part IX recovers the ordinary Jacobi connection of Part IV from a direct local contour identity and extends it to unequal positive trace weights, heterogeneous endpoint exponents, nonnegative graphical repulsion, and a positive analytic factor. The generalized identity is local at every real knot; it does not assert unrestricted complex continuation for an arbitrary analytic factor. Its polynomial-weight theorem gives a finite decision at every collision and removes the distinct-subset-sum assumption in that regime. These statements have different scopes and are not counted as duplicate independent solutions of the original connection problem.

Section IX.7 records the later discussion: for $N \geq 2$, piecewise algebraicity of the ordinary trace density forces rational $a, b, \beta$, while the family $N = 2$, $a = b = -1/2$ has nonalgebraic physical pieces for every $\beta \geq 0$. These are proved corollaries of the connection and logarithm formulas, not a complete algebraicity classification.

The nine constituent mathematical bodies are retained. Repeated title pages and bibliographies have been consolidated; overlapping results are grouped once in the editorial ranking. A separate proof is preserved when it documents a distinct argument or makes a constituent part self-contained.

## Ten principal contributions: an editorial ranking

This ranking weighs scope, removal of substantive restrictions, precision of classification, and reuse of the proof method. It groups successive extensions of one result together. It is an editorial assessment of the statements and proofs in this research draft, not a certification of historical priority, external acceptance, or journal level.

1. **Physical Jacobi connections and their cone-gluing extension.** The all-parameter formula identifies the physical density, including resonance (Theorem IV.3.1). Part IX extends the local identity to weighted graphical models, decides complete polynomial collisions, and derives sharp regularity and algebraicity obstructions (Theorems IX.2.5, IX.6.1, and IX.7.1). The generalized analytic-factor case is local at every real knot, not unrestricted complex continuation.

2. **Franel recurrences for every complex weight and arithmetic subsequence.** The minimum eventual shift width is classified without a generic-weight restriction (Theorem VIII.7.2). The alternating sparse-term bound is retained separately (Theorem II.3.1); Theorem VIII.7.6 classifies the algebraic-number-valued tail solution limits. Width and number of nonzero terms are distinct invariants.

3. **Global Galois classifications for hypercubic and banana periods.** Mass multiplicities determine exact ranks, and the symplectic or orthogonal groups are identified including colliding thresholds (Theorems V.3.1, VI.3.1, VII.2.1, VI.4.1, and VII.3.1). These are functional classifications; they do not determine arithmetic special values or all multiplicities of a scalar leading coefficient.

4. **The entire Poisson-mixture minimax curve for the ordinary NPMLE.** For a known interval $[0, B]$, the same estimator achieves the sharp Wasserstein rate over all support scales and sample sizes (Theorem VIII.5.1; the earlier version is in Part IV). Known support and the stated common-exposure observation model remain part of the theorem.

5. **Chebyshev–Bernstein coefficient bounds with sharp exponential scales.** The constant-one bound $|C(t, m, j)| \leq \exp(m^2/t)$ is proved, and $K2^{m^2/t}$ has the optimal exponential constant among fixed-prefactor bounds (Theorems VIII.2.1 and VIII.3.1). The latter assertion does not set $K = 1$.

6. **Sharp contamination robustness of the ordinary binomial NPMLE.** The estimator attains the minimax recovery curve without knowing the report-replacement budget (Theorem III.2.1). The adversary acts on reported counts in the specified experiment; kernel-dependent counterexamples prevent an unrestricted robustness claim.

7. **All-scale Gaussian-mixture recovery and its adaptation boundary.** The known-support likelihood estimator achieves the complete minimax curve (Theorem VI.7.1). In the large-support regime an unrestricted NPMLE attains the rate without support information (Theorem V.5.1); these have different optimization domains. The small-support and heterogeneous-noise obstructions are retained.

8. **Sharp separation of Lewis and A-optimal graph designs.** The worst tree ratio has leading term $\sqrt{n}/8$, the general connected graph order is $\Theta(\sqrt{n})$, and block graphs admit exact designs and a sharp leading constant (Theorems II.2.2, III.5.5, and IV.4.4). The unrestricted connected-graph leading constant is not determined.

9. **Arithmetic classifications for beta quantiles.** Half-integer beta laws admit an explicit exception list (Theorem VI.5.1); complementary rational exponents and polynomial modifications admit an algebraic/transcendental dichotomy (Theorem VII.5.1). This is a classification of the stated families, not all incomplete-beta parameters.

10. **Tableau values and derivatives at every nonzero algebraic argument.** For each stated height, the relevant finite jet has transcendence degree two and every pair of distinct jet coordinates is algebraically independent (Theorem VIII.10.4). Exact ordinary-point control and an $E$-function specialization theorem are both used; the conclusion is not inferred from functional Galois information alone.

Other retained contributions include split-beta Angelesco arithmetic, Hamiltonian clocks, optimal-stopping boundary arithmetic, exact central-trinomial product widths, and hidden-population observation thresholds. Their absence from this ten-item ranking is not an exclusion from the integrated manuscript.

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# Part I
# Exact Complexity and Arithmetic Transfers

**Abstract**

We transfer proof mechanisms from recurrence minimality, period arithmetic, weighted Laplacian spectra, and mixture identification to six concrete problems. Products of separated central-trinomial moments have exact recurrence width $\prod_i (m_i + 1)$. Jacobi transforms have exact differential order $N + 1$, increasing to $2^N$ under arbitrarily small separated weights at even nonnegative Dyson index. For every inverse-power spectral loss, we determine the unique optimal conductances when a matching of links is forbidden. Holomorphic time differentials imply simultaneous transcendence of Hamiltonian coordinates at nonzero algebraic-logarithmic clock times and a harmonic exception for action-quantized energies. For a mean-reverting square-root diffusion, we verify power-call stopping for every positive real power, classify algebraic exercise boundaries by an exact discount resonance, and prove transcendence of zero-state values at positive integer discount ratios. Finally, a known emission family with finite-dimensional mixing defect admits hidden-population recovery from three visits despite unknown persistence; an explicit positive analytic family shows both the failure of two visits and the sharp risk $\Theta(\min\{1, (r^2\sqrt{N})^{-1}\})$ on a specified submodel. The proofs identify the assumptions and classical inputs of each transfer.

**Integration note.** This opening manuscript supplies six distinct transfers. Its methodological sources are the older numbered manuscripts cited in its introduction. The nonlinear clock, stopping, spectral-design, and hidden-state results are retained here in full.

## I.1 What is transferred, and what is proved

The starting collection contains several distinct kinds of proof: positive integrals make singularities visible; a finite-dimensional representation gives an upper bound that can be matched by those singularities; an exact spectral region converts inverse existence into optimization; and arithmetic separation of Abelian integrals from logarithms constrains dynamical motion. A separate observation-theoretic principle distinguishes local differential relations from the kernel of a global mixing operator. We use these mechanisms as starting points, and reconstruct the needed arguments for the target problems below.

There are three different senses of an exact conclusion in this paper. A *complexity theorem* supplies matching upper and lower bounds. A *classification theorem* states the exceptional parameter set as well as the generic alternative. An *observation theorem* proves both a sufficient experiment and an obstruction to a smaller or noisier one. These claims must not be conflated with historical priority.

The target problems are deliberately concrete. They ask for a minimal recurrence for products of central-trinomial moments, the change in Jacobi-trace differential complexity when weights are separated, an optimal weighted network when several links are forbidden, arithmetic constraints on polynomial Hamiltonian motion, the optimal exercise boundary of a mean-reverting diffusion, and recovery of a hidden population with unknown temporal persistence. The problem definitions are part of the statements; none of the conclusions is obtained merely by relabelling a constant from the source collection.

Classical inputs include Selberg and Kummer identities, analytic subgroup and $E$-function theorems, finite-dimensional linear algebra, and elementary statistical testing. Whenever a result relies on such an input, it is stated at the point of use. In the mechanical application, action quantization defines semiclassical levels; the exact quantum spectrum is a separate problem.

The source methods were selected from the supplied manuscripts [8, 9, 50, 10, 11, 30, 12, 49]. Their role is methodological provenance; the target results below include their own arguments and identify any external theorem they use.

| Mechanism | Target field | Conclusion proved |
|---|---|---|
| Visible critical values and tensor bounds | Lattice return probabilities | Exact shift width, including repeated parameters |
| Distinct endpoint phases | Random-matrix linear statistics | Exact differential order and a symmetry-breaking jump |
| Spectral feasibility and strict convexity | Network design | Unique optimum for every inverse-power loss |
| Holomorphic time forms | Classical and semiclassical mechanics | Arithmetic clocks and exceptional quantization levels |
| $E$-function specialization | Mean-reverting optimal stopping | Discount-resonance classification of a genuine exercise boundary |
| Mixture-kernel repair and temporal moments | Hidden-state statistics | Minimal observation count and persistence-dependent minimax rate |

## I.2 Exact recurrence widths for anisotropic return probabilities

### I.2.1 A transfer from critical values to Hadamard products

The singularity-counting argument for Franel sequences has an application to products of classical moment sequences and to return probabilities of finite-range random walks. It determines an exact recurrence width in every dimension, including all positive integer multiplicities of the parameters. The result is not presented as the resolution of a separately named published conjecture: the literature checked below does not supply such a conjecture for this family. Its contribution here is a uniform, explicit lower bound matching the standard symmetric-power upper bound.

For real numbers $p > 1$ define

$$L_p(x) = (p-1) + \frac{p+1}{2}(x + x^{-1}), \qquad u_p(n) = \mathrm{CT}_x\, L_p(x)^n \quad (n \geq 0). \tag{I.2.1}$$

The generalized central trinomial coefficient is

$$T_n(b,c) = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j}\binom{2j}{j} b^{n-2j} c^j.$$

Thus $u_p(n) = T_n(p-1, (p+1)^2/4)$. In particular $u_p(n)$ is a positive integer when $p$ is an odd positive integer. The generating function and three-term recurrence are classical; see [1, 2]. For completeness the special case needed here will be derived below.

Fix $s \geq 1$, distinct odd primes $p_1, \ldots, p_s$, and integers $m_1, \ldots, m_s \geq 1$. Put

$$D = \sum_{i=1}^{s} m_i, \qquad N = \prod_{i=1}^{s} (m_i + 1), \qquad A_n = \prod_{i=1}^{s} u_{p_i}(n)^{m_i}. \tag{I.2.2}$$

A recurrence of shift width $r$ means an identity

$$\sum_{j=0}^{r} q_j(n) A_{n+j} = 0, \qquad q_j \in \mathbb{C}(n), \quad q_0 q_r \neq 0,$$

valid for all sufficiently large integers $n$. No degree bound is imposed on its rational coefficients.

**Theorem I.2.1** (Exact product width)**.** *For the sequence in* (I.2.2)*, the least recurrence width over* $\mathbb{C}(n)$ *is exactly*

$$\boxed{\prod_{i=1}^{s} (m_i + 1).}$$

*A recurrence of this width exists over* $\mathbb{Q}(n)$*. The same conclusion holds for multiplicatively independent rational parameters* $p_i > 1$*, without the assertion that every* $A_n$ *is an integer.*

The conclusion measures the width in the discrete variable $n$. It is not a claim about the minimum differential order of the generating function. Repeated parameters contribute $m_i + 1$, whereas distinct independent parameters contribute multiplicatively. In particular the width can be $2^D$ even though the underlying Laurent polynomial has degree at most one in each individual variable.

### I.2.2 Visibility and the recurrence budget

We recall the precise analytic principle, including the condition that prevents cancellation on an interior critical fiber.

**Lemma I.2.2** (Definite-fiber visibility)**.** *Let $M$ be a compact real-analytic manifold of positive dimension $d$ with a positive analytic density $d\mu$, and let $\Phi : M \to \mathbb{R}$ be analytic. Suppose $v \neq 0$ is a critical value whose critical fiber contains finitely many critical points, all nondegenerate and all local maxima, or all nondegenerate and all local minima. Regular points on the same fiber are allowed. If*

$$a_n = \int_M \Phi^n \, d\mu, \qquad F(t) = \sum_{n \geq 0} a_n t^n,$$

*then a branch continued from the germ of $F$ at $0$ is singular at $t = 1/v$. Consequently every rational-coefficient recurrence for $(a_n)$ has width at least the number of distinct nonzero values with this property.*

*Proof.* The analytic Morse lemma and polar coordinates show that the pushforward $\Phi_* \mu$ has, near $v$, density

$$\rho(v + x) = r(x) + \mathbf{1}_{\{\epsilon x > 0\}} |x|^{d/2-1} h(|x|), \qquad h(0) > 0,$$

where $r$ is smooth and $h$ is analytic. Here $\epsilon = 1$ at minima and $\epsilon = -1$ at maxima. To justify this with regular points on the fiber, take disjoint Morse neighborhoods and cutoffs equal to one near each critical point. The complement is a compact submersion region over a sufficiently small interval about $v$, whose pushforward is smooth. Inside a Morse chart, the density contribution at distance $w > 0$ from $v$ is

$$\frac{1}{2} w^{d/2-1} \int_{\mathbb{S}^{d-1}} g(\sqrt{w}\, \omega) \, d\omega.$$

The spherical average is analytic in $w$, with positive constant term. Every critical point contributes on the same side, so the leading coefficients add. The resulting one-sided term is not smooth, for either integer or half-integer exponent, and cannot be canceled by $r$.

The Cauchy transform

$$G(z) = \int_M \frac{d\mu(x)}{z - \Phi(x)}$$

is holomorphic in the upper half-plane and satisfies $G(z) = z^{-1} F(z^{-1})$ near infinity. Its boundary distribution obeys $-\pi^{-1} \operatorname{Im} G(x + i0) = \Phi_* \mu$ by the Poisson kernel. If this branch extended holomorphically through $v$, its imaginary part on the real axis would be analytic, contradicting the nonsmooth density. Inversion gives the asserted singularity of the continued germ of $F$.

For the last assertion clear the denominators of a width-$r$ recurrence, writing its coefficients as $q_j(n) \in \mathbb{C}[n]$. With $\vartheta = t\, d/dt$, summation gives

$$\left( \sum_{j=0}^{r} t^{r-j} q_j(\vartheta - j) \right) F(t) = R(t),$$

where $R$ is a polynomial of initial terms. If $e = \max_j \deg q_j$ and $c_j = [n^e] q_j(n)$, the coefficient of the highest ordinary derivative is $t^e H(t)$, with

$$H(t) = \sum_{j=0}^{r} c_j t^{r-j} \neq 0, \qquad \deg H \leq r.$$

An ordinary point of this inhomogeneous linear differential equation cannot be a singularity of any continued solution branch. Thus every nonzero finite singularity just obtained is a zero of $H$, proving the bound. □

### I.2.3 All nonzero critical fibers are definite

Use one angle $\theta_{i,j}$ for each pair $1 \le i \le s$, $1 \le j \le m_i$, and normalized Haar measure on $\mathbb{T}^D$. Constant-term extraction gives

$$A_n = \int_{\mathbb{T}^D} \Phi(\theta)^n \, d\theta, \qquad \Phi(\theta) = \prod_{i=1}^{s} \prod_{j=1}^{m_i} ((p_i - 1) + (p_i + 1)\cos\theta_{i,j}). \tag{I.2.3}$$

The convention in the integral includes normalization by $(2\pi)^{-D}$.

**Lemma I.2.3** (The complete critical-value list)**.** *All critical points with $\Phi \neq 0$ have every angle equal to $0$ or $\pi$. For $k_i$ angles equal to $0$ in block $i$, their value is*

$$v_{\boldsymbol{k}} = 2^D (-1)^{D - \sum_i k_i} \prod_{i=1}^{s} p_i^{k_i}, \qquad 0 \le k_i \le m_i. \tag{I.2.4}$$

*These $N$ values are distinct. Every critical point over a positive $v_{\boldsymbol{k}}$ is a nondegenerate maximum, and every one over a negative $v_{\boldsymbol{k}}$ is a nondegenerate minimum.*

*Proof.* At a nonzero value no factor in (I.2.3) vanishes. The logarithmic derivative with respect to $\theta_{i,j}$ is

$$-\frac{(p_i + 1)\sin\theta_{i,j}}{(p_i - 1) + (p_i + 1)\cos\theta_{i,j}}.$$

It vanishes exactly when $\theta_{i,j} \in \{0, \pi\}$. The corresponding factor values are $2p_i$ and $-2$, respectively, giving (I.2.4). Unique factorization shows that equality of two absolute values forces equality of every $k_i$. Multiplicative independence gives the same argument for rational $p_i$.

At a corner, every off-diagonal Hessian entry is zero. The diagonal entry for angle $\theta_{i,j}$ is

$$-v_{\boldsymbol{k}} \frac{(p_i + 1)\cos\theta_{i,j}}{(p_i - 1) + (p_i + 1)\cos\theta_{i,j}}.$$

The quotient is $(p_i + 1)/(2p_i) > 0$ at $0$, and $(p_i + 1)/2 > 0$ at $\pi$. Thus all Hessian eigenvalues are nonzero and have sign $-\operatorname{sgn}(v_{\boldsymbol{k}})$. This also proves that all critical points on any one nonzero fiber use the same side; their contributions cannot cancel in Lemma I.2.2. □

The lemma and the visibility principle give a lower bound of $N$ over $\mathbb{C}(n)$. This argument does not require irreducibility of a telescoper, numerical guessing, or noncancellation of complex saddle-point expansions.

### I.2.4 A matching upper bound from symmetric powers

The one-coordinate generating function is

$$\sum_{n \ge 0} u_p(n) t^n = \frac{1}{\sqrt{1 - 2(p-1)t - 4pt^2}}. \tag{I.2.5}$$

For example, integrate the geometric series for $1/(1 - t((p-1) + (p+1)\cos\theta))$; the constant-term binomial expansion gives the same identity formally. Differentiating (I.2.5) and comparing coefficients proves

$$(n+1)u_p(n+1) = (2n+1)(p-1)u_p(n) + 4pn\, u_p(n-1) \quad (n \ge 1), \tag{I.2.6}$$

with $u_p(0) = 1$ and $u_p(1) = p - 1$.

For a fixed exponent $m$, the $m+1$ monomials

$$u_p(n)^{m-j}u_p(n+1)^j, \qquad 0 \le j \le m,$$

form a vector whose shift is a rational matrix times the original vector: replace $u_p(n+2)$ using (I.2.6) and expand binomially. For rational $p$ all matrix entries lie in $\mathbb{Q}(n)$. Taking the tensor product of these symmetric-power systems over the $s$ blocks gives an $N$-dimensional system containing $A_n$ as a coordinate. The $N+1$ coordinate rows expressing $A_n, \dots, A_{n+N}$ in terms of the state at $n$ are linearly dependent over $\mathbb{Q}(n)$. Consequently $A_n$ satisfies a recurrence of width at most $N$, allowing finitely many initial exceptions when rational denominators vanish. Combined with the lower bound already proved, this completes the proof of Theorem I.2.1.

**Corollary I.2.4** (Powers of the usual central trinomial coefficients)**.** *Let $C_n = \mathrm{CT}_x(1+x+x^{-1})^n$. For every integer $m \ge 1$, the least recurrence width of $C_n^m$ over $\mathbb{C}(n)$ is $m+1$.*

*Proof.* For $p=3$, $L_3(x) = 2(1+x+x^{-1})$ and $u_3(n) = 2^n C_n$. Multiplication of a sequence by a nonzero geometric sequence preserves its minimum recurrence width: absorb the geometric factors into the individual coefficients. Apply the main theorem with one block. □

### I.2.5 Rigidity of the leading recurrence symbol

**Theorem I.2.5** (Exact singularity set and leading symbol)**.** *Let $F(t) = \sum_{n\ge 0} A_n t^n$ for (I.2.2). The set of nonzero finite points at which some branch continued from the germ of $F$ is singular is exactly*

$$\{v_{\boldsymbol{k}}^{-1} : 0 \le k_i \le m_i\}.$$

*For any width-$N$ polynomial-coefficient recurrence $\sum_{j=0}^{N} q_j(n) A_{n+j} = 0$, put $e = \max_j \deg q_j$ and $c_j = [n^e]q_j(n)$. There is a nonzero constant $c$ such that*

$$\sum_{j=0}^{N} c_j t^{N-j} = c \prod_{0\le k_i \le m_i} (1 - v_{\boldsymbol{k}} t). \tag{I.2.7}$$

*Proof.* The budget polynomial on the left is nonzero and has degree at most $N$. The $N$ distinct reciprocal critical values are all zeros of it, by the visibility lemma. Its degree must therefore be exactly $N$, and these are all its roots, proving (I.2.7). The associated inhomogeneous differential equation is ordinary away from these points and 0, and therefore has no other nonzero finite singularities for any branch of its solution $F$. Visibility already proves the reverse inclusion. □

This is an exact constraint on every recurrence of minimum width, including ones with larger polynomial degrees than necessary. It can be used to reject an incorrectly guessed recurrence without first factoring its Ore operator. It also records every secondary exponential scale, although a single dominant asymptotic ratio sees only $2^D \prod_i p_i^{m_i}$.

### I.2.6 A completely explicit two-parameter certificate

Take $p_1 = 3$, $p_2 = 5$, $m_1 = m_2 = 1$. Then

$$A_n = T_n(2,4)T_n(4,9) = 1, 8, 408, 15680, 750880, 36334848, 1844830464, \dots.$$

The exact recurrence is

$$\begin{aligned} 0 = {} & 57600(n+1)^2(2n+7)A_n \\ & - 1920(2n+3)^2(2n+7)A_{n+1} \\ & - 32(2n+5)(49n^2+245n+261)A_{n+2} \\ & - 8(2n+3)(2n+7)^2A_{n+3} + (n+4)^2(2n+3)A_{n+4}, \qquad n \ge 0. \end{aligned} \tag{I.2.8}$$

It has minimum width four, since the visible critical values are $4, -12, -20, 60$. The leading-symbol identity is

$$115200t^4 - 15360t^3 - 3136t^2 - 64t + 2 = 2(1-4t)(1+12t)(1+20t)(1-60t).$$

Here is an exact algebraic certificate of the recurrence, independent of checking initial values. Set

$$M_p(n) = \begin{pmatrix} 0 & 1 \\ 4p(n+1)/(n+2) & (p-1)(2n+3)/(n+2) \end{pmatrix}, \qquad K(n) = M_3(n) \otimes M_5(n),$$

and let $e_1 = (1,0,0,0)$. If $q_0, \dots, q_4$ are the coefficients displayed in (I.2.8), direct rational arithmetic gives the row identity

$$q_0(n)e_1 + \sum_{j=1}^{4} q_j(n)e_1K(n+j-1)\cdots K(n) = 0.$$

Multiplication by $(u_3(n), u_3(n+1))^{\mathsf{T}} \otimes (u_5(n), u_5(n+1))^{\mathsf{T}}$ proves the recurrence. The accompanying exact-arithmetic certificate verifies this rational matrix identity; initial-term checks serve only as additional diagnostics.

### I.2.7 A Markov-chain interpretation

Consider a discrete-time walk on $\mathbb{Z}^D$. At each step all coordinate increments are independent. A coordinate in block $i$ has distribution

$$\mathbb{P}(\Delta = 0) = \frac{p_i - 1}{2p_i}, \qquad \mathbb{P}(\Delta = 1) = \mathbb{P}(\Delta = -1) = \frac{p_i + 1}{4p_i}. \tag{I.2.9}$$

The step distribution has finite support, positive holding probability, and positive probabilities for both signs in every coordinate. Let $R_n$ be its probability of returning to the origin after $n$ steps.

**Corollary I.2.6** (Exact width of anisotropic return probabilities)**.** *For the walk* (I.2.9)*,*

$$R_n = \frac{A_n}{(2^D \prod_i p_i^{m_i})^n}, \qquad \operatorname{width}_{\mathbb{C}(n)}(R) = \prod_i (m_i + 1).$$

*Consequently anisotropic walks with one coordinate for each of $D$ distinct odd primes have exact recurrence width $2^D$. When all coordinates have the same parameter, their exact width is $D+1$.*

*Proof.* The one-coordinate probability generating Laurent polynomial is $L_{p_i}(x)/(2p_i)$. Independence, followed by constant-term extraction for the origin, proves the formula for $R_n$. A nonzero geometric normalization preserves recurrence width, as in Corollary I.2.4. □

This return-probability statement is an application of the exact-width theorem, not a second independent solution of an open problem. The sharp change from $D+1$ to $2^D$ quantifies how distinct coordinate parameters remove the symmetric-power reduction.

### I.2.8 Literature boundary and remaining work

Generalized central trinomial coefficients, their Legendre-polynomial representations, moment properties, and three-term recurrence are classical; the studies [1, 2] concern this established family. Closure under Hadamard products and the symmetric-power construction are also

standard. The substantive assertion proved here is the matching lower bound for the explicit independent-parameter family, together with the complete nonzero finite singularity set and forced leading symbol. The proof is self-contained once elementary analytic Morse theory and the ordinary-point theorem for linear differential equations are available.

The wider methodological context is recurrence minimality. Straub and Zudilin [74] explicitly distinguish minimum telescoping order from minimum order among all rational-coefficient recurrences for Franel sums. The present lower bound addresses the latter notion for a different family. We do not claim that their Franel problem is newly settled by this section, nor that any existing central-trinomial congruence or real-rootedness question has been resolved here. A more exhaustive priority search for the exact product-width formula remains appropriate before journal submission. Multiplicatively dependent distinct parameters can cause critical-value collisions; the proof still supplies a lower bound equal to the number of distinct values, but does not assert a complete minimum-width formula on those collision loci.

## I.3 Jacobi ensembles: exact differential complexity under symmetry breaking

The transfer in this section replaces the special Bessel equations of the tableau problem by arbitrary beta-integral factors. The main lower bound comes from the Fourier asymptotics of the *particular integral*; it does not assume that a differential system is irreducible. The existence of the unweighted Jacobi differential system is due to Forrester and Kumar. What is proved here is exact minimality, and an exact increase from $N+1$ to $2^N$ after a small perturbation of the weights in the linear statistic. These are mathematical deductions; historical priority for the resulting formulations is not asserted.

For $N \geq 1$, real $a_i, b_i > -1$, $c_i > 0$, and $\beta \geq 0$, put

$$Z_{\mathbf{a},\mathbf{b},\beta,\mathbf{c}}(t) = \int_{[0,1]^N} e^{t\sum_i c_i u_i} |\Delta(u)|^\beta \prod_{i=1}^N u_i^{a_i}(1-u_i)^{b_i}\, du, \qquad \Delta(u) = \prod_{i<j}(u_j - u_i). \tag{I.3.1}$$

The convention at $\beta = 0$ is $|\Delta|^0 = 1$. The integral is an entire function, since its weight is integrable and the integration domain is compact. Dividing it by $Z(0) > 0$ gives the moment generating function of the corresponding weighted linear statistic. Such a constant normalization does not change its minimal linear differential order over $\mathbb{C}(t)$.

### I.3.1 An endpoint principle for a single integral

For $A \subseteq \{1, \dots, N\}$ and exponents $q_i > -1$, define

$$d_A(\mathbf{q}) = \sum_{i\in A}(q_i+1) + \frac{\beta}{2}|A|(|A|-1), \qquad \mathcal{L}_A(\mathbf{q},\mathbf{c}) = \int_{(0,\infty)^A} e^{-\sum_{i\in A} c_i x_i} |\Delta(x_A)|^\beta \prod_{i\in A} x_i^{q_i}\, dx_A.$$

For the empty set, set $d_\varnothing = 0$ and $\mathcal{L}_\varnothing = 1$. Every nonempty integral on the right is finite and strictly positive: the endpoint powers are integrable, the Vandermonde has no singularity for $\beta \geq 0$, and the exponential controls infinity.

**Lemma I.3.1** (All endpoint frequencies occur)**.** *For a subset $S \subseteq \{1, \dots, N\}$, write*

$$\lambda_S = \sum_{i\in S} c_i, \qquad d_{S,-} = d_{S^c}(\mathbf{a}), \qquad d_{S,+} = d_S(\mathbf{b}), \qquad \gamma_S = d_{S,-} + d_{S,+}.$$

*As $R \to +\infty$, the function in (I.3.1) has a complete oscillatory expansion*

$$Z(iR) \sim \sum_{S\subseteq[N]} e^{i\lambda_S R} R^{-\gamma_S} \sum_{k\geq 0} A_{S,k} R^{-k}, \tag{I.3.2}$$

*where*

$$A_{S,0} = e^{\frac{\pi i}{2}(d_{S,-}-d_{S,+})}\mathcal{L}_{S^c}(\mathbf{a},\mathbf{c})\mathcal{L}_S(\mathbf{b},\mathbf{c}) \neq 0. \tag{I.3.3}$$

*Complete means that truncating each series after $k = K-1$ leaves $\sum_S O(R^{-\gamma_S-K})$; expansions to any larger prescribed order are available. For every fixed $j \geq 0$ the same assertion holds for $Z^{(j)}(iR)$, with the leading coefficient at the $S$-th frequency equal to $\lambda_S^j A_{S,0}$. All the expansions can therefore be used in any fixed polynomial-coefficient differential expression in $Z$.*

*Proof.* The corner indexed by $S$ has $u_i = 1$ for $i \in S$ and $u_i = 0$ otherwise. Near it put $x_i = u_i$ for $i \notin S$ and $y_i = 1-u_i$ for $i \in S$. The phase is

$$\lambda_S + \sum_{i\notin S} c_i x_i - \sum_{i\in S} c_i y_i.$$

The weight is the product of the two homogeneous functions

$$H_-(x) = |\Delta(x_{S^c})|^\beta \prod_{i\notin S} x_i^{a_i}, \qquad H_+(y) = |\Delta(y_S)|^\beta \prod_{i\in S} y_i^{b_i},$$

and a real analytic function $B(x,y)$ with $B(0,0) = 1$. Indeed, the cross-cluster differences are $1-x_i-y_j$, which remain positive, and the remaining endpoint factors are powers of $1-x_i$ or $1-y_j$. The degrees of $H_-$ and $H_+$, including their measures, are $d_{S,-}$ and $d_{S,+}$.

Here is an elementary conical version of Watson's endpoint argument. For a homogeneous, locally integrable function $H$ on a positive orthant, of degree $d-r$ in $r$ variables with $d > 0$, use the radial coordinate $v = \sum c_i x_i$. Integration over the compact simplex $\sum c_i x_i = 1$ gives

$$\int H(x) f\Big(\sum c_i x_i\Big)\, dx = C_H \int_0^\infty v^{d-1} f(v)\, dv, \qquad C_H = \frac{\int e^{-\sum c_i x_i} H(x)\, dx}{\Gamma(d)}.$$

With a smooth cutoff equal to one at zero, the one-dimensional endpoint expansion of the last integral, for $f(v) = e^{\pm iRv}$, starts with $\Gamma(d)e^{\pm\pi id/2}R^{-d}$. This follows, for example, by first inserting $e^{-\epsilon v}$, evaluating $\int_0^\infty e^{-(\epsilon\mp iR)v}v^{d-1}dv$, and integrating by parts outside zero. Taylor expansion of a smooth amplitude gives the expansion to every order with a remainder of the stated order. For two nonempty clusters, choose the corner cutoff as $\chi_-(v)\chi_+(w)$, where $v = \sum_{i\notin S} c_i x_i$ and $w = \sum_{i\in S} c_i y_i$. After integration over the two compact simplices, the remaining amplitude is jointly smooth in $v, w$; all its derivatives are dominated by the integrable angular weights. The same endpoint argument therefore applies to the two radial variables independently. Each monomial of $B(x,y)$ increases the total exponent by its nonnegative integer total degree. This proves the corner expansion and (I.3.3). An empty cluster simply contributes the factor one.

It remains to exclude contributions away from the corners, including points on diagonals where the weight need not be smooth. At any such point at least one cluster of equal coordinates lies strictly between zero and one. In a sufficiently small neighbourhood, translate all coordinates in that cluster together. The associated constant vector field leaves all within-cluster differences unchanged. Differences between distinct clusters and endpoint factors of the moving cluster are smooth and nonzero there. All remaining singular factors are independent of this translation. Its derivative of the phase is $\sum_{i \text{ in the cluster}} c_i > 0$. A smooth partition of unity and repeated integration by parts in this direction consequently give $O(R^{-M})$ for every $M$. There are no boundary terms in the moving direction: the cutoff is compactly supported in that neighbourhood, and the direction is tangent to the collision faces. All differentiated amplitudes are bounded by integrable endpoint and collision weights. Compactness gives a finite cover.

Finally, differentiation in $t$ inserts the polynomial $(\sum c_i u_i)^j$. Its value at the corner is $\lambda_S^j$, and the same Taylor and integration-by-parts arguments apply to it. This establishes the derivative assertion, including the case $\lambda_S = 0$, in which a displayed leading coefficient can vanish. □

**Lemma I.3.2** (Frequency lower bound)**.** *Suppose an entire function and all its fixed-order derivatives have complete expansions of the form above with $M$ distinct real frequencies $\lambda_1, \dots, \lambda_M$, real power exponents, and nonzero leading amplitudes. Suppose the derivative leading amplitudes are multiplied by $\lambda_j^k$. Every nonzero rational-function-coefficient differential operator annihilating the function has order at least $M$. More precisely, if*

$$L = \sum_{k=0}^{r} p_k(t)\partial_t^k, \qquad D = \max_k \deg p_k, \qquad \chi_L(X) = \sum_k [t^D]p_k(t)X^k,$$

*then $\prod_{j=1}^{M}(X - \lambda_j)$ divides $\chi_L(X)$.*

*Proof.* Clear denominators. The top coefficient at frequency $\lambda_j$ in $(Lf)(iR)$ is $i^D A_{j,0}\chi_L(\lambda_j)$ times $R^{D-\gamma_j}e^{i\lambda_j R}$. A complete finite-frequency power expansion has unique coefficients. For clarity, take its largest remaining real power, divide by that power, and average against $e^{-i\lambda_j R}$ on $[T, 2T]$. Terms of smaller power tend to zero, and distinct frequencies average to zero. The coefficient at the chosen frequency must therefore vanish. Repeating in decreasing powers is legitimate because only finitely many exponents occur above any fixed bound. Thus every formal coefficient of an identically zero expression vanishes. In particular $\chi_L(\lambda_j) = 0$ for all $j$. The polynomial $\chi_L$ is nonzero by the definition of $D$ and has degree at most $r$, proving both conclusions. □

### I.3.2 Exact order for the ordinary Jacobi trace

Write $Z_{N,a,b,\beta}$ for (I.3.1) when all $a_i = a$, all $b_i = b$, and all $c_i = 1$.

**Theorem I.3.3** (Exact Jacobi trace order)**.** *For every $N \geq 1$, $a, b > -1$ and $\beta \geq 0$, the minimal linear differential order of $Z_{N,a,b,\beta}$ over $\mathbb{C}(t)$ is exactly $N + 1$. For any primitive polynomial-coefficient minimal operator $L = \sum_{k=0}^{N+1} p_k(t)\partial_t^k$, if $D = \max_k \deg p_k$, then*

$$\sum_{k=0}^{N+1} [t^D]p_k(t)X^k = C\prod_{j=0}^{N}(X - j), \qquad C \neq 0.$$

*In particular* $\deg p_{N+1} = D$.

*Proof.* For the upper bound, we recall the published Jacobi differential system with the parameter conventions made explicit. Set

$$H_p(x) = \binom{N}{p}^{-1} \int_{[0,1]^N} e^{-x\sum u_i} \prod u_i^a (1 - u_i)^b |\Delta(u)|^\beta e_p(1 - u_1, \dots, 1 - u_N)\, du.$$

Forrester–Kumar, Corollary 2.5, equations (2.19)–(2.20) of `arXiv:2011.00787v2`, gives

$$(N - p)xH_{p+1} = ((N - p)x + B_p)H_p + xH_p' - D_pH_{p-1},$$

where

$$B_p = p\left(a + b + 2 + \frac{\beta}{2}(2N - p - 1)\right), \qquad D_p = p\left(b + 1 + \frac{\beta}{2}(N - p)\right).$$

Its parameters there are $a = a$ and the parameter denoted by $b$ in that system is $b + 1$ here. The same identity at $\beta = 0$ follows directly by integration by parts, or by continuity in $\beta$. These identities hold for all real endpoint parameters in the stated integrability range.

Let $\mathscr{D} = \partial_x$, $Q_0 = 1$, $Q_1 = 1 + \mathscr{D}/N$, and recursively, for $1 \leq p < N$, set

$$(N - p)Q_{p+1} = (\mathscr{D} + N - p + B_p/x)\, Q_p - (D_p/x)Q_{p-1}.$$

Then $H_p = Q_p H_0$, and $Q_p$ has order $p$ and leading coefficient $(N-p)!/N!$. The last equation of the system is

$$((\mathscr{D} + B_N/x)Q_N - (D_N/x)Q_{N-1})\, H_0 = 0.$$

Its order is $N+1$, with leading coefficient $1/N!$. Since $Z(t) = H_0(-t)$, this proves the upper bound.

For the lower bound, the corners with $|S| = j$ all have the same frequency $j$, the same exponent

$$\gamma_j = (N-j)(a+1) + j(b+1) + \frac{\beta}{2}((N-j)(N-j-1) + j(j-1)),$$

and the same nonzero leading amplitude in Lemma I.3.1. Their amplitudes add with the positive integer multiplier $\binom{N}{j}$; no cancellation occurs. Thus all $N+1$ distinct frequencies $0, \dots, N$ occur. Apply Lemma I.3.2. For a minimal operator the degree of $\chi_L$ is at most $N+1$, so its $N+1$ prescribed roots give the displayed exact polynomial. □

### I.3.3 Weighted statistics and an exponential increase in order

**Theorem I.3.4** (Exact anisotropic order at even Dyson index)**.** *Let $N \geq 1$, $m \in \mathbb{Z}_{\geq 0}$, $a_i, b_i > -1$ and $c_i > 0$. Assume that the $2^N$ subset sums $\lambda_S = \sum_{i \in S} c_i$ are pairwise distinct. Then $Z_{\mathbf{a},\mathbf{b},2m,\mathbf{c}}$ has minimal linear differential order exactly $2^N$ over $\mathbb{C}(t)$. For every polynomial-coefficient minimal operator, its top-degree coefficient polynomial satisfies*

$$\chi_L(X) = C \prod_{S \subseteq [N]} (X - \lambda_S), \qquad C \neq 0.$$

*The assertion includes $m = 0$, that is, independent beta variables.*

*Proof.* When $\beta = 2m$, the Vandermonde factor is the polynomial $\Delta(u)^{2m}$. Expand it as a finite linear combination of monomials $u_1^{\nu_1} \cdots u_N^{\nu_N}$. Put

$$f_i(t) = \int_0^1 e^{c_i t u} u^{a_i} (1-u)^{b_i}\, du.$$

Integration by parts of $e^{c_i t u} u^{a_i+1}(1-u)^{b_i+1}$, whose endpoint values vanish, gives the Kummer equation

$$t f_i'' + (a_i + b_i + 2 - c_i t) f_i' - c_i(a_i+1) f_i = 0.$$

Consequently every derivative of $f_i$ is in the $\mathbb{C}(t)$-span of $f_i, f_i'$. The vector space spanned by the $2^N$ products

$$\prod_{i=1}^{N} f_i^{(\epsilon_i)}, \qquad \epsilon_i \in \{0,1\},$$

is closed under differentiation. The monomial expansion of $\Delta^{2m}$ expresses the integral as a linear combination of products $\prod_i c_i^{-\nu_i} f_i^{(\nu_i)}$, so the integral lies in this space. Its derivative span therefore has dimension at most $2^N$. This proves an annihilator of order at most $2^N$, without any assumption about the Galois groups of the factors.

By Lemma I.3.1, each of the $2^N$ corners has a nonzero leading amplitude. The subset-sum hypothesis keeps their frequencies distinct, so Lemma I.3.2 gives the matching lower bound. It also proves the top-degree polynomial identity, as in Theorem I.3.3. □

**Corollary I.3.5** (An arbitrarily small perturbation changes exact order)**.** *Fix $N \geq 2$, $a, b > -1$ and $m \geq 0$. At equal weights $c_i = 1$ the order is $N+1$. For every $0 < \epsilon < 1$, set*

$$c_i(\epsilon) = 1 + \epsilon 2^{-i}, \qquad 1 \leq i \leq N.$$

*The order at these perturbed weights is exactly $2^N$. Thus weights arbitrarily close to equal weights change the exact differential order by $2^N - N - 1$.*

*Proof.* For two subsets of different cardinality, the difference of their subset sums has a nonzero integer part and a perturbation of absolute value less than $\epsilon < 1$, so it is nonzero. For subsets of the same cardinality, uniqueness of finite binary sums shows that their perturbations are different. Apply the two theorems. Clearly $\max_i |c_i(\epsilon) - 1| \to 0$ as $\epsilon \to 0^+$. □

**An explicit order-three to order-four example.** For $N = 2$, $a = b = 0$ and $\beta = 2$, direct integration gives

$$Z_{(1,1)}(t) = \frac{2e^{2t} - (2t^2 + 4)e^t + 2}{t^4},$$

whereas weights $(1, 2)$ give

$$Z_{(1,2)}(t) = \frac{3e^{3t} - (2t^2 + 2t + 3)e^{2t} - (2t^2 - 2t + 3)e^t + 3}{4t^4}.$$

Both expressions have removable singularities at zero, with value $1/6$. The exact minimal orders are respectively three and four. These formulas follow by writing $(u - v)^2 = u^2 - 2uv + v^2$ and differentiating $(e^x - 1)/x$; thus they are exact elementary checks of both the tensor upper bound and the endpoint lower bound.

*Remark* I.3.6 (Scope and what has not been inferred). The upper bound for the unweighted trace is a consequence of the Forrester–Kumar system, not a new discovery of that system. The lower bounds above are for the specific transforms, not merely for ambient differential systems. No irreducibility, full differential Galois group, absence of apparent singularities, or arithmetic nature of special values follows from the present argument. In particular integer endpoint parameters can give elementary exponential expressions and reducible minimal operators; the exact-order conclusions remain valid. For general non-even $\beta$ and unequal weights the endpoint argument still gives the same $2^N$ lower bound under the subset-sum hypothesis, but no matching upper bound is claimed here. At colliding subset sums, the unequal-parameter amplitudes can combine, so the anisotropic theorem is not extended silently across that locus.

**Primary source.** P. J. Forrester and S. Kumar, *Differential recurrences for the distribution of the trace of the $\beta$-Jacobi ensemble*, Physica D **434** (2022), 133220; `arXiv:2011.00787v2`, Corollary 2.5. `https://doi.org/10.1016/j.physd.2022.133220`.

## I.4 Exact statistical network design with missing interactions

The inverse spectral results of the earlier manuscript have consequences for experimental design, electrical networks, and stochastic consensus. The general interpretation is classical: total effective resistance is $n \operatorname{tr} L^+$, and its optimization over conductances is a convex problem; see Ghosh–Boyd–Saberi, *SIAM Review* **50** (2008), 37–66, doi:10.1137/050645452. Symmetry reduction is also standard. The results below are explicit solutions for all inverse spectral powers on a missing-edge graph and on an arbitrary matching-defect graph. They are new deductions in this manuscript, without a historical-priority claim for these particular graph families.

Let $L$ be a nonnegatively weighted graph Laplacian on $n$ labeled vertices, with connected positive support. Write its positive eigenvalues as $\lambda_2, \dots, \lambda_n$, fix

$$T = \operatorname{tr} L = 2 \sum_e w_e > 0, \qquad J_p(L) = \sum_{i=2}^{n} \lambda_i^{-p} \quad (p > 0).$$

Disconnected positive supports have $J_p = +\infty$ by convention; this is the extended objective for estimation of all vertex contrasts. All optimizations below permit zero weights on allowed edges, but their optimizers will have strictly positive weights on every allowed edge.

**Lemma I.4.1** (Strict convexity and symmetry reduction)**.** *On the connected Laplacian cone with fixed vertex set, $J_p$ is strictly convex for every $p > 0$. So is $-\log\det'(L)$, where $\det'(L)$ is the product of the positive eigenvalues. A minimizer under a fixed trace and fixed allowed graph is unique and invariant under every automorphism of that graph.*

*Proof.* Restrict $L$ to $\mathbf{1}^\perp$, obtaining a positive definite matrix $A$. For a strictly convex twice differentiable scalar function $f$, the second derivative of $\operatorname{tr} f(A)$ in a nonzero symmetric direction $H$, in an eigenbasis of $A$, is

$$\sum_i f''(\lambda_i)H_{ii}^2 + 2\sum_{i<j}\frac{f'(\lambda_i)-f'(\lambda_j)}{\lambda_i-\lambda_j}H_{ij}^2,$$

with the continuous interpretation at repeated eigenvalues. This is strictly positive for $f(x) = x^{-p}$ and $f(x) = -\log x$. The passage from weights to a Laplacian is injective, since its off-diagonal entries recover all the weights. The fixed-trace weight simplex is compact, and the objectives diverge on disconnected limits; thus a minimizer exists and is unique whenever a connected weighting exists. Averaging over the finite automorphism group preserves feasibility and cannot increase the objective. Uniqueness gives invariance. □

### I.4.1 One missing edge: every inverse power and the exact design penalty

**Theorem I.4.2** (All-power optimality with one forbidden interaction)**.** *Let $n \geq 3$, $p > 0$, and $T > 0$. Put*

$$c = \frac{n}{n-2}, \qquad k = n-3, \qquad \rho_p = \left(\frac{1+c}{1+c^{-p}}\right)^{1/(p+1)}, \qquad x = \frac{T}{1+c+k\rho_p}.$$

*Among all connected noncomplete weighted graphs on $n$ vertices with trace $T$, the minimum of $J_p$ is*

$$\boxed{T^{-p}\left[k + \left((1+c)^p(1+c^{-p})\right)^{1/(p+1)}\right]^{p+1}.} \tag{I.4.1}$$

*Every minimizing graph is $K_n - e$. With the missing edge labeled 12, the unique minimizing weighting is*

$$w_{1j} = w_{2j} = \alpha := \frac{x}{n-2} \quad (j \geq 3), \qquad w_{ij} = \beta := \frac{x}{n-2}\left(\rho_p - \frac{2}{n-2}\right) \quad (3 \leq i < j \leq n). \tag{I.4.2}$$

*The second family is absent when $n = 3$. Its spectrum is*

$$\{0, x, (\rho_p x)^{(n-3)}, cx\}.$$

*For a fixed forbidden edge, the optimizer is unique even when all additional allowed edges may be assigned zero weight.*

*Proof.* We reproduce the spectral obstruction needed for the lower bound. If $L_{12} = 0$, set $x_0 = \lambda_2$, $z_0 = \lambda_n$, $a = (x_0 + z_0)/2$, $d = (z_0 - x_0)/2$, and $H = I - \mathbf{1}\mathbf{1}^\mathsf{T}/n$. On $\mathbf{1}^\perp$, $\|L - aH\| \leq d$. For $u = He_1$ and $v = He_2$, the missing edge gives

$$a/n = u^\mathsf{T}(L - aH)v \leq d\|u\|\|v\| = d(n-1)/n.$$

Consequently $z_0 \geq cx_0$.

Minimize $\sum \lambda_i^{-p}$ over the larger convex spectral set

$$0 < \lambda_2 \leq \cdots \leq \lambda_n, \qquad \sum \lambda_i = T, \qquad c\lambda_2 - \lambda_n \leq 0.$$

The candidate in the statement has $1 < \rho_p < c$, so its intermediate eigenvalues lie strictly between its extreme eigenvalues. For the trace multiplier choose $\nu = p(\rho_p x)^{-p-1}$ and for the last inequality choose $\mu = p((\rho_p x)^{-p-1} - (cx)^{-p-1}) > 0$. The stationarity equations for the two endpoints reduce to

$$(1+c)(\rho_p x)^{-p-1} = x^{-p-1} + c(cx)^{-p-1},$$

which is exactly the defining formula for $\rho_p$. All middle-coordinate stationarity equations hold as well. The order constraints need no multiplier. Convexity proves optimality and strict convexity proves uniqueness of this spectral list. For $n = 3$ the two-coordinate problem gives $x = T/(1+c)$ directly; the same stated value follows.

To realize the list, put $m = n - 2$ and use the weighting (I.4.2). The endpoint difference has eigenvalue $m\alpha = x$, the vector constant on the middle vertices and opposite on the endpoints has eigenvalue $n\alpha = cx$, and middle zero-sum vectors have eigenvalue $2\alpha + m\beta = \rho_p x$. All allowed weights are positive: for $n \geq 4$, $2/(n-2) \leq 1 < \rho_p$. The trace and the desired objective therefore agree with the lower bound. Algebraic simplification gives (I.4.1).

For each fixed missing pair, Lemma I.4.1 gives uniqueness on the entire weight simplex supported on $K_n - e$. Since its unique optimizer makes every other edge positive, any global optimizer with an additional absent edge is impossible. This proves the support classification as well as uniqueness. □

**Corollary I.4.3** (Exact resistance and uncertainty-volume costs)**.** *Under the same trace constraint, every connected noncomplete graph obeys*

$$\operatorname{tr} L^+ \geq \frac{1}{T}\left(n - 3 + \sqrt{\frac{n}{n-2}} + \sqrt{\frac{n-2}{n}}\right)^2. \tag{I.4.3}$$

*Equality has the unique weighting of Theorem I.4.2 with $\rho_1 = \sqrt{c}$. The maximum spanning-tree partition function is*

$$\tau(L) := \frac{\det'(L)}{n} \leq \frac{T^{n-1}(n-2)}{(n-1)^{n+1}}. \tag{I.4.4}$$

*Here equality holds uniquely, up to the missing-edge label, at*

$$\alpha_D = \frac{T}{(n-1)^2}, \qquad \beta_D = \frac{T(n-3)}{(n-2)(n-1)^2},$$

*with the core-edge formula omitted for $n = 3$. Relative to the complete-graph optimum, the loss in the maximum determinant is exactly the factor $1 - (n-1)^{-2}$.*

*Proof.* The first assertion is the case $p = 1$. For the determinant, maximize $\log x + \log z + (n-3)\log t$ under the same spectral constraints. The strictly concave maximization has $z = cx$ and $t = (1+c)x/2$; the multiplier check is the logarithmic version of the preceding proof. Thus

$$x = \frac{T(n-2)}{(n-1)^2}, \quad t = \frac{T}{n-1}, \quad z = \frac{Tn}{(n-1)^2}.$$

The same explicit realization and strict convexity prove uniqueness. Multiply these eigenvalues and use the weighted matrix-tree theorem. The complete-graph maximum is $(T/(n-1))^{n-1}$ for $\det'(L)$, which gives the claimed exact ratio. □

For example, take $n = 4$, $T = 1$, and forbid 12. The resistance-optimal design has

$$x = \frac{1}{3+\sqrt{2}}, \qquad \alpha = \frac{1}{2(3+\sqrt{2})}, \qquad \beta = \frac{\sqrt{2}-1}{2(3+\sqrt{2})},$$

and $\operatorname{tr} L^+ = (1+\sqrt{2}+1/\sqrt{2})^2$. The determinant-optimal design instead has $\alpha_D = 1/9$, $\beta_D = 1/18$, and $\tau = 2/243$. Thus two standard statistical criteria already prescribe different conductance allocations on this smallest nontrivial example.

### I.4.2 Any number of disjoint missing interactions

Write $M_r$ for a matching of size $r$ and let $G = K_n - M_r$. The $2r$ vertices meeting the matching are called paired, and the other $s = n - 2r$ vertices are called unpaired. This graph is connected for the parameters used below.

**Theorem I.4.4** (All-power optimizer for arbitrary matching defects)**.** *Let $r \geq 2$, $s \geq 1$, $n = 2r+s$, $p > 0$, and $q = p + 1$. There is a unique $t \in (1, r/(r-1))$ satisfying*

$$\frac{n\{r-(r-1)t\}}{s} = \left(\frac{n}{r+(n-r)t^{-q}}\right)^{1/q}. \tag{I.4.5}$$

*Define*

$$h = \frac{n\{r-(r-1)t\}}{s}, \qquad v = \left(\frac{2}{1+t^{-q}}\right)^{1/q}, \qquad B = r+(r-1)t+(s-1)v+h, \qquad d = T/B.$$

*The unique minimizer of $J_p$ over all weights on $K_n - M_r$ of trace $T$ is the following three-orbit weighting:*

$$\begin{aligned} a &= \tfrac{d}{2}(t-1) && \text{between distinct paired pairs,} \\ b &= dh/n && \text{between paired and unpaired vertices,} \\ c_0 &= \frac{d}{s}\left(v - \frac{2r}{n}h\right) && \text{between distinct unpaired vertices.} \end{aligned} \tag{I.4.6}$$

*The last family is absent for $s = 1$. All present weights are positive. Its positive spectrum, with multiplicities, is*

$$\{d^{(r)}, (td)^{(r-1)}, (vd)^{(s-1)}, hd\}, \tag{I.4.7}$$

*so the minimum is explicitly*

$$(B/T)^p\{r+(r-1)t^{-p}+(s-1)v^{-p}+h^{-p}\}. \tag{I.4.8}$$

*The case $r = 1$ is Theorem I.4.2. If $s = 0$ and $n = 2r \geq 4$, the unique minimizer has the uniform edge weight $T/[n(n-2)]$ and positive spectrum*

$$\{(T/n)^{(r)}, (T/(n-2))^{(r-1)}\}.$$

*Proof.* The left side of (I.4.5) is strictly decreasing, from $n/s > 1$ to 0, and the right side is strictly increasing and positive, with value 1 at $t = 1$. This proves existence and uniqueness of the root. In particular $a, b > 0$. Put $\theta = 2r/n \in (0, 1)$. The root equation gives

$$\left(\frac{v}{h}\right)^q = \frac{2\{r+(n-r)t^{-q}\}}{n(1+t^{-q})} > \theta.$$

Since $q > 1$, this implies $v/h > \theta^{1/q} > \theta$, and hence $c_0 > 0$ whenever that edge family is present.

By Lemma I.4.1, it suffices to optimize among the three automorphism orbits just described. For orbit weights $(a, b, c_0)$, decompose the vertex space into pair differences, pairwise constant vectors whose paired-coordinate sum vanishes, unpaired zero-sum vectors, the paired/unpaired quotient mode, and the global constant mode. The positive eigenvalues are respectively

$$D = 2(r-1)a+sb\ (r \text{ times}), \quad U = 2ra+sb\ (r-1 \text{ times}), \quad V = 2rb+sc_0\ (s-1 \text{ times}), \quad W = nb.$$

These subspaces are orthogonal and their dimensions sum to $n$ after including the constant mode. The trace constraint is

$$4r(r-1)a + 4rsb + s(s-1)c_0 = T. \tag{I.4.9}$$

Substitution of (I.4.6) gives $(D, U, V, W) = d(1, t, v, h)$ and the trace $dB = T$.

Divide the stationarity multiplier for $J_p$ by $p$ and call it $\mu$. The $a$-equation is

$$\mu = \tfrac{1}{2}(D^{-q} + U^{-q}).$$

When $s \geq 2$, the $c_0$-equation is $\mu = V^{-q}$. After these substitutions, the $b$-equation is equivalent to

$$nW^{-q} = rD^{-q} + (n-r)U^{-q}. \tag{I.4.10}$$

For $s = 1$ there is no $c_0$-equation and its coefficient in the $b$-equation is zero; the same displayed relation follows directly. The definitions of $v$ and $h$ give exactly these stationarity relations. Thus the stated positive point is a constrained stationary point of a strictly convex objective on the orbit-weight simplex. It is its unique minimum, and symmetry reduction makes it the unique minimum on the full weight simplex. The spectral value formula follows immediately.

For $s = 0$, the graph is edge-transitive, so uniqueness and symmetry force all weights to agree. There are $n(n-2)/2$ edges. The eigenvalues follow from $L = w(nH - \sum_{j=1}^{r}(e_{2j-1} - e_{2j})(e_{2j-1} - e_{2j})^{\mathsf{T}})$. □

**Corollary I.4.5** (Quartic resistance equation and quadratic determinant equation)**.** *For $r \geq 2$ and $s = n - 2r \geq 1$, the resistance-optimal parameter is the unique root in $(1, r/(r-1))$ of*

$$s^2t^2 = n(rt^2 + n - r)\{r - (r-1)t\}^2. \tag{I.4.11}$$

*For maximizing* $\det'(L)$*, the same weighting formulas hold with*

$$q = 1, \qquad t = \frac{1 - s + \sqrt{(n-1)^2 - 4s}}{2(r-1)}, \qquad v = \frac{2t}{1+t}, \qquad h = \frac{nt}{rt + n - r}.$$

*The determinant maximum equals $d^{n-1}t^{r-1}v^{s-1}h$, with $d = T/B$ as above.*

*Proof.* For $p = 1$, square (I.4.5) and multiply by the positive denominators. Both sides of the unsquared equation are positive on the specified interval, so the quartic introduces no extra root there. For the determinant, apply the same convexity and stationarity proof to $-\log\det'(L)$. Its derivative replaces the power $q = p + 1$ by $q = 1$. The root equation simplifies to

$$(r-1)t^2 + (s-1)t - (n-r) = 0.$$

Its unique positive root is the formula above. Positivity follows from the preceding proof: at $q = 1$ the strict inequality $v/h > \theta$ remains valid. The spectrum gives the determinant. □

For an explicit non-edge-transitive example take $K_6 - M_2$, $T = 1$. For resistance minimization, $t$ is the unique root in $(1, 2)$ of

$$t^2 = 3(t^2 + 2)(2 - t)^2,$$

and the weights on the paired–paired, paired–unpaired, and unpaired–unpaired edge orbits are approximately

$$(a, b, c_0) = (0.04717362924,\ 0.03545488293,\ 0.02766641956).$$

The exact value is given by the unique-root formulas, not by these rounded numbers. For determinant maximization on the same graph, $t = (\sqrt{17} - 1)/2$. This supplies exact formulas for a family whose full inverse-eigenvalue region need not be known.

### I.4.3 Statistical and stochastic meanings of the optimized quantities

Consider noisy pairwise measurements of a vertex signal modulo its additive constant, with independent Gaussian errors and edge precision $w_{ij}$. The Fisher information on $\mathbf{1}^{\perp}$ is $L$; the efficient estimator's covariance is $L^{+}$. Consequently $J_1$ is total contrast-estimation variance, $J_p$ is the sum of the $p$th powers of the covariance eigenvalues, and maximizing $\det'(L)$ minimizes the volume of the confidence ellipsoid. The preceding theorems give the exact designs under a total precision budget and either one forbidden comparison or an arbitrary set of disjoint forbidden comparisons.

For the projected Ornstein–Uhlenbeck consensus dynamics

$$dX_t = -LX_t\,dt + \sigma H\,dB_t, \qquad X_t \in \mathbf{1}^{\perp}, \qquad H = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n,$$

the stationary covariance is $\Sigma = (\sigma^2/2)L^{+}$, since $L\Sigma + \Sigma L = \sigma^2 H$. Thus the exact minimum stationary mean-square disagreement is $(\sigma^2/2)\min J_1$, and every higher covariance power sum is obtained by multiplying (I.4.1) or (I.4.8) by $(\sigma^2/2)^p$. Finally, summing $(e_i - e_j)^{\mathsf{T}}L^{+}(e_i - e_j)$ over $i < j$ gives $nJ_1$, the total effective resistance. These interpretations use standard identities; the explicit constrained optimizers and their strict-support classification are the content of the preceding results.

## I.5 Arithmetic clocks and action quantization in algebraic Hamiltonian dynamics

The previous manuscripts already established transcendence of periods of regular compact algebraic trajectories. The additional ingredient here is *global holomorphy of the time differential*, including at complex points at infinity. It upgrades transcendence of a period to separation from all algebraic linear combinations of logarithms, then gives a simultaneous arithmetic statement for position and momentum at prescribed times. A superelliptic extension treats nonquadratic kinetic energies and yields an exact algebraic/transcendental dichotomy for action-quantized energy levels. These are transfers of the classical analytic subgroup theorem, not claims of a new transcendence theorem or of a solution of an open quantum spectral problem.

Throughout, arithmetic statements refer to the displayed dimensionless coordinates and coefficients. Rescalings by nonzero algebraic constants preserve them; arbitrary real changes of units need not do so. Put

$$\mathcal{L}_{\mathrm{alg}} = \overline{\mathbb{Q}} + \mathrm{span}_{\overline{\mathbb{Q}}}\{\lambda \in \mathbb{C} : e^{\lambda} \in \overline{\mathbb{Q}}^{\times}\}, \qquad \mathcal{C}_{\mathrm{alg}} = \mathcal{L}_{\mathrm{alg}} \cap \mathbb{R}.$$

All logarithm branches are allowed. This is a vector space, not an assertion about closure under multiplication. It contains every real algebraic number, $\pi$, and the real logarithm of every positive algebraic number. We call its real elements the prescribed arithmetic clock times.

### I.5.1 The separation input and a general clock principle

**Lemma I.5.1** (First-kind integrals and logarithmic separation)**.** *Let $C_j$ be smooth projective curves over $\overline{\mathbb{Q}}$, $\eta_j$ holomorphic differentials defined over $\overline{\mathbb{Q}}$, and $\delta_j$ paths with algebraic endpoints; closed paths are allowed without a restriction on their basepoint. For algebraic coefficients $c_j$, set*

$$S = \sum_{j=1}^{k} c_j \int_{\delta_j} \eta_j.$$

*Then $S \in \mathcal{L}_{\mathrm{alg}}$ implies $S = 0$. In particular, if $S \neq 0$, both $S$ and $S/\pi$ are transcendental.*

*Proof.* Use Abel–Jacobi maps, based at algebraic points, to place the integrals on the product $A$ of the Jacobians. There are an invariant algebraic differential $\eta$ on $A$ and a logarithm $v \in \mathrm{Lie}(A)_{\mathbb{C}}$ such that $\exp_A(v) \in A(\overline{\mathbb{Q}})$ and $S = \eta(v)$. The chosen paths determine the logarithm, including its periods. For a closed path based at a nonalgebraic point, prepend a connector from any algebraic point and append its inverse. The added integrals cancel, and the resulting closed path has algebraic basepoint. If $S \in \mathcal{L}_{\mathrm{alg}}$, write

$$\eta(v) - c_0 - \sum_{j=1}^{s} c_j \lambda_j = 0, \quad c_j \in \overline{\mathbb{Q}}, \quad e^{\lambda_j} \in \overline{\mathbb{Q}}^{\times}.$$

Apply the refined analytic subgroup theorem in [3, Theorem 6.2] to

$$G = A \times \mathbb{G}_a \times \mathbb{G}_m^s, \qquad u = (v, 1, \lambda_1, \dots, \lambda_s).$$

Its exponential is algebraic. The theorem supplies a connected algebraic subgroup $H$ with $u \in \mathrm{Lie}(H)_{\mathbb{C}}$ on which the displayed linear form vanishes identically.

A connected subgroup of a product of an abelian variety and a connected commutative affine group splits between these two types. Indeed, the algebraic-group Goursat argument reduces any mixing to a common connected quotient of an abelian variety and an affine group; such a quotient is both proper and affine and hence trivial. Thus $H = B \times H_0$ for an abelian subvariety $B \subseteq A$ and an affine subgroup $H_0$. Restriction of the linear form to $\mathrm{Lie}(B)$ gives $\eta|_{\mathrm{Lie}(B)} = 0$. Since $v$ lies in this Lie algebra, $S = 0$. Finally, $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}}\pi$ are contained in $\mathcal{L}_{\mathrm{alg}}$. □

**Proposition I.5.2** (Arithmetic clock exclusion)**.** *Let $C$ be a smooth projective curve over the real algebraic numbers, and let $\eta$ be an algebraic holomorphic differential. Suppose a real trajectory $P(t)$, defined for $t$ in an interval, satisfies*

$$\eta(\dot{P}(t)) = 1.$$

*If $P(t_1)$ and $P(t_2)$ are algebraic and $t_1 \neq t_2$, then*

$$t_2 - t_1 \notin \mathcal{C}_{\mathrm{alg}}.$$

*Consequently the times at which the trajectory is algebraic meet every coset of $\mathcal{C}_{\mathrm{alg}}$ in at most one point. Every positive return time $T$ satisfies $T \notin \mathcal{C}_{\mathrm{alg}}$ and $T/\pi$ is transcendental.*

*Proof.* Integrate the time form along the actual oriented trajectory segment:

$$\int_{P|_{[t_1,t_2]}} \eta = t_2 - t_1.$$

This is a nonzero first-kind integral with algebraic endpoints when the endpoints are algebraic, so Lemma I.5.1 applies. For a return time the segment is a closed path; a closed period is covered by the same lemma, without any arithmetic restriction on the chosen starting point. Its integral is the nonzero real number $T$. □

### I.5.2 Polynomial potentials: a global condition and its exact role

Consider the classical Hamiltonian system

$$H(x,p) = \tfrac{1}{2}p^2 + V(x), \qquad \dot{x} = p, \qquad \dot{p} = -V'(x), \tag{I.5.1}$$

with $V \in (\overline{\mathbb{Q}} \cap \mathbb{R})[x]$. An energy $E$ is called algebraically regular if $E - V(x)$ is squarefree over $\mathbb{C}$. This is stronger than absence of real critical points on one selected orbit.

**Theorem I.5.3** (Position and momentum at arithmetic clock times)**.** *Let* $\deg V = d \geq 3$ *and let* $E \in \overline{\mathbb{Q}} \cap \mathbb{R}$ *be algebraically regular. Suppose* (I.5.1) *has a nonconstant bounded real orbit of energy* $E$ *and an algebraic initial phase point* $(x(0), p(0))$*. Then for every* $t \in \mathcal{C}_{\mathrm{alg}} \setminus \{0\}$*,*

$$x(t) \notin \overline{\mathbb{Q}}, \qquad p(t) \notin \overline{\mathbb{Q}}.$$

*The period* $T$ *and angular frequency* $\Omega = 2\pi/T$ *satisfy*

$$T \notin \mathcal{C}_{\mathrm{alg}}, \qquad T/\pi \notin \overline{\mathbb{Q}}, \qquad \Omega \notin \overline{\mathbb{Q}}.$$

*No such orbit is rationally resonant with a harmonic oscillator of positive algebraic angular frequency.*

*Proof.* Let $C_E$ be the smooth projective model of $p^2 = 2(E - V(x))$. Squarefreeness makes the affine curve smooth. The time differential is $\eta = dx/p$. It is regular at a finite point where $p \neq 0$. At a simple root $x_0$ of $E - V$, use $p$ as local parameter: $x - x_0$ has order two, so $dx/p$ is again regular.

We check infinity, since finite regularity is insufficient. If $d = 2g + 1 \geq 3$, the unique point above infinity has $\mathrm{ord}(x) = -2$ and $\mathrm{ord}(p) = -(2g + 1)$. Thus

$$\mathrm{ord}(dx/p) = 2g - 2 \geq 0.$$

If $d = 2g + 2 \geq 4$, the two points above infinity have $\mathrm{ord}(x) = -1$ and $\mathrm{ord}(p) = -(g + 1)$, so $\mathrm{ord}(dx/p) = g - 1 \geq 0$ at both. Therefore $\eta$ is holomorphic on all of $C_E$.

The real orbit is compact, and its vector field never vanishes: a zero would give $p = 0$ and $V'(x) = 0$ at energy $E$, contrary to squarefreeness. Hence it is periodic and exists for all real times. The energy identity extends $\eta(\dot{P}) = 1$ through turning points. Proposition I.5.2 rules out an algebraic phase point at any nonzero prescribed clock time.

On this energy curve either coordinate is algebraic if and only if both are algebraic. In one direction this follows by taking the square root in $p^2 = 2(E - V(x))$; in the other, an algebraic $p$ makes $x$ a root of the nonconstant polynomial $V(X) - E + p^2/2$ over $\overline{\mathbb{Q}}$. Thus both displayed coordinates are transcendental. The period assertion follows from the same proposition. If $\Omega$ were algebraic, $T/\pi = 2/\Omega$ would be algebraic. Finally, a rational relation between $T$ and the harmonic period $2\pi/\omega$, for algebraic $\omega > 0$, would have the same forbidden consequence. □

For example, consider $H = p^2/2 + x^4$, with $x(0) = 0$, $p(0) = \sqrt{2}$. The energy is 1 and $1 - x^4$ is squarefree. Position and momentum are therefore both transcendental at each of the exact times 1, $\pi$, and $\log 2$. This is an arithmetic assertion about those exact values; numerical orbit sampling is not its proof.

**Corollary I.5.4** (The harmonic exception)**.** *Among algebraic polynomial potentials admitting a bounded nonconstant orbit at an algebraically regular algebraic energy, the period is an algebraic multiple of* $\pi$ *if and only if the potential has degree two.*

*Proof.* Degrees zero and one admit no bounded nonconstant orbit. A quadratic potential admitting one has the form $V(x) = a(x - b)^2 + c$ with $a > 0$ and algebraic $a, b, c$; its period is $2\pi/\sqrt{2a}$. Degrees at least three are excluded by Theorem I.5.3. □

**Example I.5.5** (A real regular oval with singular complex energy curve)**.** The squarefreeness assumption cannot be replaced by regularity of the chosen real oval. Take

$$V(x) = -\tfrac{1}{2}(1 - x^2)(1 + x^2)^2, \qquad E = 0.$$

The orbit above $-1 \le x \le 1$ has $p = \pm(1+x^2)\sqrt{1-x^2}$ and is a smooth compact real oval with a nonvanishing vector field. Nevertheless

$$T = 2\int_{-1}^{1} \frac{dx}{(1+x^2)\sqrt{1-x^2}} = \sqrt{2}\,\pi.$$

For the last equality, put $x = \sin\theta$ and then $u = \tan\theta$; the resulting integral is $2\int_{-\infty}^{\infty} du/(1+2u^2)$. The polynomial $2(E-V)$ has repeated roots at $x = \pm i$. On the normalization $u^2 = 1 - x^2$, its time differential is $dx/((1+x^2)u)$, which has complex poles. The loss of global holomorphy explains the algebraic multiple of $\pi$ without contradicting the theorem. In particular, starting from $(0,1)$ returns to the algebraic initial point at an arithmetic clock time.

### I.5.3 Superelliptic kinetic energies and an action-level dichotomy

Let $a, b > 0$ be real algebraic numbers, and let $r, m$ be positive integers. Consider

$$H_{r,m}(x,p) = ap^{2r} + bx^{2m}, \qquad E > 0. \tag{I.5.2}$$

The real energy curve is a smooth compact oval. Its Hamiltonian vector field is

$$(2rap^{2r-1}, -2mbx^{2m-1}),$$

and never vanishes on the oval. Write $T_{r,m}(E)$ for its period and

$$\mathcal{A}_{r,m}(E) = \oint p\,dx > 0$$

for the action, with the orientation of the Hamiltonian flow.

**Lemma I.5.6** (Global time form for the superelliptic family)**.** *For algebraic $E > 0$, the time differential on the smooth projective energy curve is*

$$\eta = \frac{dx}{2rap^{2r-1}}.$$

*It is holomorphic everywhere if $(r,m) \neq (1,1)$. For $(r,m) = (1,1)$ it has simple poles at infinity.*

*Proof.* Set $N = 2r$, $D = 2m$, and $g = \gcd(N, D)$. The curve $p^N = (E - bx^D)/a$ is irreducible because its right side has simple zeros and hence is not a proper power in $\overline{\mathbb{Q}}(x)$. At a finite zero of $p$, $p$ is a local parameter and $x - x_0$ has order $N$. Accordingly $dx/p^{N-1}$ has order zero. There are no other finite poles.

At each of the $g$ points above infinity on the normalization,

$$\operatorname{ord}(x) = -N/g, \qquad \operatorname{ord}(p) = -D/g.$$

These orders follow by balancing $p^N$ and $x^D$, or from the local normalization of $p^N x^{-D} = -b/a + O(x^{-D})$. Thus

$$\operatorname{ord}(\eta) = \frac{D(N-1) - N}{g} - 1 = \frac{2m(2r-1) - 2r}{\gcd(2r, 2m)} - 1. \tag{I.5.3}$$

If $r = 1$ and $m \ge 2$, the quotient before subtraction is $m - 1 \ge 1$; the case $m = 1, r \ge 2$ is symmetric and gives $r - 1 \ge 1$. If $r, m \ge 2$, then $2m(2r-1) - 2r \ge 2\min(r,m) \ge g$. The order is therefore nonnegative in every nonharmonic case. For $r = m = 1$ the numerator is zero, giving order $-1$. □

**Theorem I.5.7** (Arithmetic action quantization)**.** *Fix algebraic $a, b, \hbar, \nu > 0$ and positive integers $r, m$. There is a unique positive solution $E_\nu$ of*

$$\mathcal{A}_{r,m}(E_\nu) = 2\pi\hbar\nu. \tag{I.5.4}$$

*Its arithmetic type is completely determined by*

$$E_\nu \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad (r, m) = (1, 1).$$

*In the harmonic case $E_\nu = 2\hbar\sqrt{ab}\,\nu$. In every other case $E_\nu$ is transcendental. Moreover, at every algebraic $E > 0$ in a nonharmonic case,*

$$T_{r,m}(E) \notin \mathcal{C}_{\mathrm{alg}}, \qquad \mathcal{A}_{r,m}(E)/\pi \notin \overline{\mathbb{Q}}.$$

*An algebraic initial point on that energy oval has transcendental position and momentum at every nonzero time in $\mathcal{C}_{\mathrm{alg}}$.*

*Proof.* The clock statements follow from Lemma I.5.6 and Proposition I.5.2. Algebraicity of either coordinate implies that of the other by the energy equation.

The positive action is

$$\mathcal{A}_{r,m}(E) = 4a^{-1/(2r)} \int_0^{(E/b)^{1/(2m)}} (E - bx^{2m})^{1/(2r)}\, dx.$$

Rescaling $x = (E/b)^{1/(2m)}u$ gives

$$\mathcal{A}_{r,m}(E) = C_{r,m,a,b}E^\sigma, \quad \sigma = \frac{1}{2r} + \frac{1}{2m}, \quad C_{r,m,a,b} = 4a^{-1/(2r)}b^{-1/(2m)} \int_0^1 (1 - u^{2m})^{1/(2r)}\, du > 0. \tag{I.5.5}$$

Consequently the action is continuous and strictly increasing from zero to infinity, proving existence and uniqueness in (I.5.4). Differentiation under the integral gives $\mathcal{A}'_{r,m}(E) = T_{r,m}(E)$. This can be justified after the same fixed-interval substitution; the differentiated turning-point singularity has exponent $1/(2r) - 1 > -1$ and is integrable. It follows that

$$\mathcal{A}_{r,m}(E) = \frac{E}{\sigma} T_{r,m}(E). \tag{I.5.6}$$

For nonharmonic $(r, m)$ and algebraic $E > 0$, Lemma I.5.1 makes $T_{r,m}(E)/\pi$ transcendental. Multiplication by the nonzero algebraic number $E/\sigma$ proves the asserted action statement. If $E_\nu$ in (I.5.4) were algebraic, the same statement would contradict $\mathcal{A}_{r,m}(E_\nu)/\pi = 2\hbar\nu$.

For $r = m = 1$, the energy oval is an ellipse with semiaxes $\sqrt{E/b}$ and $\sqrt{E/a}$. Its action is its enclosed phase-space area, $\pi E/\sqrt{ab}$, giving the stated algebraic solution. □

*Remark* I.5.8 (What is, and is not, quantized here)*.* For the usual quadratic kinetic energy $r = 1$, the choice $\nu = n + 1/2$, $n \geq 0$, is the standard leading Bohr–Sommerfeld rule for a smooth two-turning-point well. Thus all these levels for even power potentials of degree at least four are transcendental when $a, b, \hbar$ are algebraic. Leading Bohr–Sommerfeld levels and exact Schrödinger eigenvalues are different objects; this distinction is explicit in [4]. The theorem makes no arithmetic claim about the exact eigenvalues. For $r > 1$, it classifies the stated action equation and does not assign a universal Maslov correction to every higher-order quantum operator.

*Remark* I.5.9 (No independence claim for the resulting levels)*.* For two positive algebraic labels $\mu, \nu$, equation (I.5.5) gives

$$\frac{E_\nu}{E_\mu} = \left(\frac{\nu}{\mu}\right)^{2rm/(r+m)} \in \overline{\mathbb{Q}}.$$

The infinitely many transcendental levels therefore belong to one algebraic proportionality class. Their individual transcendence is not algebraic independence. No analogous action conclusion for an arbitrary mixed polynomial potential is inferred here: its action differential need not be holomorphic, and the homogeneous identity (I.5.6) is essential to this argument.

## I.6 Discount resonance and arithmetic of CIR stopping boundaries

The transfer in this section changes both the stochastic model and its parameter geometry. The dimension and payoff exponent of a finite-horizon Bessel bridge no longer determine the same Kummer parameter. Instead an independent discount rate determines the upper parameter, while a fixed cost translates the payoff. This produces an arithmetic resonance in the discount-to-mean-reversion ratio. The case of linear payoff is the published CIR liquidation problem of Leung, Li, and Wang [5, Theorem 3.1]; their optimality theorem is credited, not claimed anew. We prove the verification for all positive power payoffs and then establish arithmetic conclusions. We do not claim that the arithmetic question was posed as an open problem in that source, or that historical priority has been established.

Let $Y$ be the nonnegative CIR diffusion

$$dY_t = \kappa(\theta - Y_t)\,dt + \sigma\sqrt{Y_t}\,dW_t, \qquad \kappa, \theta, \sigma > 0. \tag{I.6.1}$$

We use its usual nonexplosive solution, including the instantaneously reflecting boundary when $2\kappa\theta < \sigma^2$ and the entrance interpretation when started at zero in the other cases. Fix $r, m > 0$ and $K \geq 0$. Define

$$V(y) = \sup_{\tau} \mathbb{E}_y\big[e^{-r\tau}(Y_\tau - K)_+^m\big], \qquad y \geq 0, \tag{I.6.2}$$

where infinite stopping times are permitted and have payoff zero. Put

$$s = \frac{2\kappa}{\sigma^2}, \qquad a = \frac{r}{\kappa}, \qquad b = \frac{2\kappa\theta}{\sigma^2}, \qquad k = sK, \qquad f(z) = M(a; b; z). \tag{I.6.3}$$

The generator of $Z = sY$, divided by $\kappa$, is $z\partial_{zz} + (b - z)\partial_z$. Thus

$$zf'' + (b - z)f' - af = 0. \tag{I.6.4}$$

All Taylor coefficients of $f$, and all its derivatives on the nonnegative real axis, are strictly positive.

**Theorem I.6.1** (The global stopping rule, including concave powers)**.** *For every set of real parameters specified above, there is a unique $z_* > k$ satisfying*

$$(z_* - k)f'(z_*) = mf(z_*). \tag{I.6.5}$$

*With $y_* = z_*/s$, the value and an optimal stopping time are*

$$V(y) = \begin{cases} s^{-m}(z_* - k)^m \dfrac{f(sy)}{f(z_*)}, & 0 \leq y < y_*, \\ (y - K)^m, & y \geq y_*, \end{cases} \qquad \tau_* = \inf\{t \geq 0 : Y_t \geq y_*\}. \tag{I.6.6}$$

*The continuation inequality is strict at every $y < y_*$. No Feller inaccessibility condition on zero is needed.*

*Proof.* For $w > 0$ write $f(k + w) = \sum_{j \geq 0} c_j w^j$, where $c_j = f^{(j)}(k)/j! > 0$. Its tilted coefficients $c_j w^j / f(k + w)$ form a probability distribution. Their mean is

$$H(w) = \frac{wf'(k + w)}{f(k + w)}, \qquad wH'(w) = \mathrm{Var}_w(j) > 0.$$

The mean tends to zero at zero. It tends to infinity at infinity: for any fixed $J$, the mass on $0 \leq j \leq J$ tends to zero by comparison with the positive term of degree $J + 1$. Hence there

is exactly one $w_* > 0$ with $H(w_*) = m$, giving $z_* = k + w_*$. Moreover $w^m/f(k+w)$ increases strictly up to $w_*$ and decreases strictly thereafter. Consequently the proposed value majorizes the payoff strictly below the boundary. Value matching and continuous first derivative follow from the contact equation.

It remains to verify superharmonicity in the whole stopping region; this is needed in addition to locating a maximum of a ratio. For $z = k + w > k$ and $g(z) = w^m$,

$$\big[z\partial_{zz} + (b-z)\partial_z - a\big]g(z) = w^{m-2}R(w), \tag{I.6.7}$$

where

$$R(w) = m(m-1)k + m(m+b-1-k)w - (m+a)w^2.$$

At the contact equation, direct use of (I.6.4) gives

$$R(w_*) = -z_* w_* H'(w_*) < 0. \tag{I.6.8}$$

If $m \geq 1$, the concave quadratic $R$ has nonnegative constant term, so it is negative after its possible positive root; (I.6.8) then gives $R(w) < 0$ for $w \geq w_*$. If $k = 0$, the same conclusion follows immediately by factoring out $w$. Suppose $0 < m < 1$ and $k > 0$. If $R$ is never positive for $w > 0$, nothing remains. Otherwise it has two positive roots $w_- < w_+$, and

$$0 < w_- < \frac{m(m+b-1-k)}{2(m+a)} < \frac{m(b-k)}{m+a} < b - k.$$

At $z = k + w_- < b$, positivity of $f''$ and (I.6.4) imply $f'/f < a/(b-z)$. The preceding bound on $w_-$ therefore gives

$$H(w_-) < \frac{aw_-}{b-k-w_-} < m.$$

Thus $w_* > w_-$. In view of (I.6.8), the contact cannot lie in $[w_-, w_+]$, and hence $w_* > w_+$. This proves the required sign everywhere above the boundary, also for concave powers.

The proposed function is nonnegative, continuously differentiable, piecewise twice continuously differentiable, and analytic near zero. The generalized Itô formula has no local-time contribution at the boundary because of smooth fit. Applied after localization, it shows that $e^{-rt}V(Y_t)$ is a nonnegative local supermartingale, hence a supermartingale. Optional sampling and Fatou's lemma give $V(y) \geq \mathbb{E}_y[e^{-r\tau}(Y_\tau - K)_+^m]$ for every stopping time, with the stipulated zero payoff at infinity. This argument also proves finiteness of the supremum.

For $y < y_*$ the stopped process $e^{-r(t\wedge\tau_*)}V(Y_{t\wedge\tau_*})$ is a martingale: its drift is zero and its state remains in the compact interval $[0, y_*]$, where the function and its derivative are bounded. As $t \to \infty$, the term on $\{\tau_* > t\}$ is bounded by $e^{-rt}\max_{[0,y_*]} V$ and tends to zero. The term on $\{\tau_* \leq t\}$ tends, boundedly, to the discounted boundary payoff. Equality in the value follows. For $y \geq y_*$ stop at zero. Analyticity at zero makes the same Itô calculation valid for the usual CIR solution whether zero is attainable or not. □

**Theorem I.6.2** (A discount-resonance arithmetic phase diagram)**.** *Assume $a, b, m \in \mathbb{Q}_{>0}$, $s \in \overline{\mathbb{Q}}_{>0}$, and $K \in \overline{\mathbb{Q}}_{\geq 0}$. Then*

$$y_* \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad a - b \in \mathbb{Z}_{\geq 0}. \tag{I.6.9}$$

*Outside this set the stopping boundary is transcendental, for every algebraic fixed cost $K \geq 0$ and every positive rational payoff power. On the resonance $a - b = N \in \mathbb{Z}_{\geq 0}$, put $P_N(z) = M(-N; b; -z)$. The boundary is the unique root $z_* > k$ of*

$$(z-k)\{P_N'(z) + P_N(z)\} - mP_N(z) = 0, \qquad y_* = z_*/s. \tag{I.6.10}$$

*In this resonant case, for every algebraic starting state $y \geq 0$,*

$$V(y) \text{ is transcendental} \quad \Longleftrightarrow \quad y < y_*; \qquad V(y) \text{ is algebraic} \quad \Longleftrightarrow \quad y \geq y_*. \tag{I.6.11}$$

*Proof.* For rational positive $a, b$, $f$ and $f'$ are $E$-functions. The usual rational Pochhammer-ratio denominator estimate gives the arithmetic coefficient bounds, and (I.6.4) supplies their two-dimensional differential system, whose only finite singularity is zero. We recall precisely why its reducible special solutions are the only exceptions. If the logarithmic derivative of the entire exponential-type function $f$ is rational, removing its finitely many zero factors gives $f = e^{\lambda z}P(z)$ with $P$ a polynomial. Substitution into (I.6.4), at the two highest degrees, forces either $\lambda = 0$, $a = -\deg P$, or $\lambda = 1$, $a - b = \deg P$. The first is impossible because $a > 0$. Conversely Kummer's transformation gives $f = e^z P_N(z)$ when $a - b = N \geq 0$.

If $a - b$ is not a nonnegative integer, $f, f'$ are linearly independent over $\overline{\mathbb{Q}}(z)$. Beukers' ordinary-point specialization theorem [68, Corollary 1.4] makes $f(\xi), f'(\xi)$ linearly independent over $\overline{\mathbb{Q}}$ at every nonzero algebraic $\xi$. An algebraic $z_* > k$ would give the nontrivial algebraic linear relation (I.6.5), a contradiction. Since $s$ is nonzero algebraic, $y_*$ is then transcendental.

On resonance the contact equation is (I.6.10), a nonzero polynomial of degree $N + 1$ with algebraic coefficients. Its specified root is algebraic. For an algebraic $y < y_*$, formula (I.6.6) becomes

$$V(y) = s^{-m}(z_* - k)^m \frac{P_N(sy)}{P_N(z_*)} e^{sy - z_*}.$$

The prefactor is nonzero algebraic: $P_N$ has positive coefficients and $m$ is rational. The exponent is a nonzero algebraic number. Hermite–Lindemann proves transcendence. If $y \geq y_*$, the value $(y - K)^m$ is algebraic, including the boundary case. □

*Remark* I.6.3 (The OU interpretation). If $X$ is a $d$-dimensional Ornstein–Uhlenbeck process satisfying $dX_t = -\lambda X_t\, dt + \eta\, dW_t$, then $Y_t = \|X_t\|^2$ has (I.6.1) with $\kappa = 2\lambda$, $\theta = d\eta^2/(2\lambda)$, and $\sigma = 2\eta$. Consequently $a = r/(2\lambda)$, $b = d/2$, and $s = \lambda/\eta^2$. For rational $r/\lambda > 0$, integer $d \geq 1$, algebraic $s, K$, and rational $m > 0$, the optimal radial boundary for the payoff $(\|X\|^2 - K)_+^m$ is algebraic exactly when

$$\frac{r}{\lambda} - d \in 2\mathbb{Z}_{\geq 0}.$$

For $K = 0$, this covers every positive rational power of $\|X\|$. The resonance depends on discount and dimension, rather than on the reward parity appearing in the bridge classification.

### I.6.1 Transcendence of zero-state values at every integer discount ratio

Boundary transcendence alone does not imply transcendence of the optimized value. The following elimination supplies that extra step for an infinite family including nonresonant parameters and nonzero transaction costs.

**Theorem I.6.4** (Integer discount ratios give transcendental optimized values)**.** *Suppose $a = n \in \mathbb{Z}_{>0}$, $b, m \in \mathbb{Q}_{>0}$, $s \in \overline{\mathbb{Q}}_{>0}$, and $K \in \overline{\mathbb{Q}}_{\geq 0}$. Then $V(0)$ is transcendental. This holds for every positive rational payoff power, not only integer powers.*

*Proof.* The resonant cases $b \in \{1, \dots, n\}$ follow from Theorem I.6.2. Suppose otherwise, so that $z_*$ is transcendental. Put $Y(z) = M(1; b; z)$. There are polynomials $A_n, B_n \in \mathbb{Q}[z]$ such that

$$F_n(z) := M(n; b; z) = A_n(z)Y(z) + B_n(z).$$

They are constructed from $A_1 = 1$, $B_1 = 0$ by

$$A_{j+1} = \frac{zA_j' + (z + j + 1 - b)A_j}{j}, \qquad B_{j+1} = \frac{(b-1)A_j + zB_j' + jB_j}{j}.$$

These follow by coefficient comparison from $zY' = (z + 1 - b)Y + (b - 1)$ and $M(j + 1; b; z) = (z\partial_z + j)M(j; b; z)/j$. In particular $\deg A_n = n - 1$ and its leading coefficient is $1/(n - 1)!$.

For completeness the determinant that prevents hidden division by zero is

$$z(B_nA_n' - A_nB_n') + (z+1-b)A_nB_n - (b-1)A_n^2 = D_n(b) := \frac{\prod_{j=1}^{n}(j-b)}{(n-1)!} \neq 0. \tag{I.6.12}$$

Indeed $F_n$ and $A_n e^z z^{1-b}$ solve the same Kummer equation with upper parameter $n$, by the preceding contiguous operator. Their Wronskian is a constant times $e^z z^{-b}$; direct expansion identifies that constant with the polynomial on the left. At zero, $A_n(0) = \prod_{j=2}^{n}(j-b)/(n-1)!$ and $A_n(0) + B_n(0) = 1$, giving exactly $D_n(b)$.

Write $w = z - k$ and define rational functions

$$U = w\left[A_n' + \left(1 + \frac{1-b}{z}\right)A_n\right] - mA_n, \qquad T = w\left[B_n' + \frac{b-1}{z}A_n\right] - mB_n.$$

Then $(w\partial_z - m)F_n = UY + T$ and $B_nU - A_nT = wD_n(b)/z$. At the boundary, therefore,

$$F_n(z_*)U(z_*) = \frac{(z_*-k)D_n(b)}{z_*} \neq 0.$$

Let

$$Q(z) = zU(z) = (z-k)\{zA_n'(z) + (z+1-b)A_n(z)\} - mzA_n(z). \tag{I.6.13}$$

This is a polynomial of degree $n+1$ with leading coefficient $1/(n-1)!$. Since $f(0) = 1$, the value formula reduces to

$$V(0) = \frac{s^{-m}}{D_n(b)}(z_*-k)^{m-1}Q(z_*). \tag{I.6.14}$$

Write $m = u/v$ in lowest positive terms. If $V(0)$ were algebraic, raising (I.6.14) to the $v$th power would make

$$(z_*-k)^{u-v}Q(z_*)^v$$

algebraic. The rational function $(z-k)^{u-v}Q(z)^v$ has algebraic coefficients and has pole order $vn + u > 0$ at infinity, so it is nonconstant. Equating it to an algebraic number and clearing denominators gives a nonzero polynomial over $\overline{\mathbb{Q}}$ vanishing at $z_*$. This contradicts the transcendence of $z_*$. All denominators used above are nonzero by $z_* > k \geq 0$ and (I.6.12). □

**Example I.6.5** (A concrete fixed-cost liquidation value)**.** Take $dY_t = (2 - Y_t)\,dt + \sqrt{2Y_t}\,dW_t$, discount rate $r = 1$, sale cost $K = 1$, and linear payoff. Then $s = a = m = 1$, $b = 2$, and $f(z) = (e^z - 1)/z$. The optimal stopping level $z_* > 1$ is the unique such solution of

$$(z^2 - 3z + 1)e^z + 2z - 1 = 0.$$

Both the boundary and the optimized zero-state value are transcendental, with the exact algebraic relation

$$V(0) = -z_*^2 + 3z_* - 1.$$

Numerically $z_* \approx 2.45628765889609$ and $V(0) \approx 0.335513913443030$. The decimals illustrate the theorem and are not used in its proof. In contrast, if the mean level is changed from 2 to 1, then $a = b = 1$ is resonant, $y_* = 2$ is algebraic, and $V(y) = e^{y-2}$ for $0 \leq y < 2$. Thus an algebraic stopping boundary can coexist with a transcendental value at every algebraic continuation state.

**Corollary I.6.6** (No algebraic knife edge in the published CIR entry test)**.** *For the linear-payoff CIR liquidation and subsequent entry model of Leung–Li–Wang, assume $a = r/\kappa \in \mathbb{Z}_{>0}$, $b \in \mathbb{Q}_{>0}$, and algebraic $s > 0$ and sale cost $K \geq 0$. For every algebraic entry cost $c \geq 0$, the equality $V(0) = c$ is impossible. Consequently the nontrivial-entry condition $V(0) > c$ of [5, Equation (3.7)] has no algebraic equality case, and it can be decided by a terminating certified interval-refinement calculation from exact algebraic input.*

*Proof.* Take $m = 1$ in Theorem I.6.4. The linear liquidation value with payoff $Y_\tau - K$ equals our positive-part version: the verified optimizer stops only above $K$, and allowing never to stop cannot improve a negative exercise. The primary source gives the stated entry criterion. The unique boundary can be enclosed by bisection of the strictly increasing function $H$; the entire hypergeometric series and its derivative are effectively evaluated with certified tails on compact intervals. Thus $V(0)$ has arbitrarily precise computable intervals. Since $V(0) \neq c$, refinement eventually separates it from the exact algebraic number $c$, deciding the strict inequality. No complexity bound or new entry-boundary formula is asserted. □

**Scope.** The first arithmetic theorem classifies every stopping boundary in the stated rational-parameter family. The second classifies a substantial family of optimized values, but does not classify all nonresonant values at arbitrary starting states. This is a transfer to established diffusion and real-option models, using the classical Siegel–Shidlovskii theorem in its refined specialization form.

## I.7 Unknown temporal persistence turns a two-view inverse into a three-visit experiment

The transfer in this section is from finite-dimensional defects of mixture operators to a hidden Markov experiment. Repeated observations need not share one permanent latent state: the state is retained with an unknown probability and is otherwise refreshed from the unknown population. The conclusions are an exact three-visit identification theorem, a sharp persistence-dependent inverse bound, and an exact minimax rate in a continuous-population example.

**Model and conventions.** Let $\Theta$ and $\mathcal{X}$ be compact metric spaces, and let the known map $\theta \mapsto P_\theta \in \mathcal{P}(\mathcal{X})$ be weakly continuous. For a finite signed measure $\nu$ on $\Theta$, put

$$A_m \nu = \int P_\theta^{\otimes m} \, d\nu(\theta), \qquad K_m = \ker A_m.$$

The hidden process is stationary, with population $G \in \mathcal{P}(\Theta)$ and transition kernel

$$M_{G,\rho}(\theta, d\eta) = \rho \delta_\theta(d\eta) + (1 - \rho) G(d\eta), \qquad 0 < \rho \leq 1. \tag{I.7.1}$$

Conditionally on its latent path, the emissions $X_j$ are independent with law $P_{\theta_j}$. The population $G$ and persistence $\rho$ are unknown. Write $T_{G,\rho}$ for the law of $(X_0, X_1, X_2)$ and $\mu_G = A_1 G$. The total-variation norm of a signed measure is its full variation, denoted $\| \cdot \|_{\mathrm{var}}$; thus the distance of two probabilities is at most two. Constants below do not depend on the sample size or on the persistence lower bound.

**Lemma I.7.1** (Finite defects supply a two-view inverse)**.** *Suppose* $\dim K_1 = d < \infty$ *and* $K_1$ *contains no nonzero finitely supported signed measure. Then* $A_2$ *is injective on finite signed measures. In particular, the component map is injective.*

*Proof.* For $f \in C(\mathcal{X}; \mathbb{R})$ put $Tf(\theta) = \int f \, dP_\theta$, and let $V$ be the uniform closure of the range of $T$ in $C(\Theta; \mathbb{R})$. It contains 1 and has annihilator $K_1$, so its codimension is $d$. Products of functions of the two coordinates are dense in $C(\mathcal{X}^2)$ by Stone–Weierstrass. Hence $\nu \in K_2$ implies $v\nu \in K_1$ for every $v \in V$. A vector-space complement of $V$ of dimension $d$ shows that the multiplication map $g \mapsto g\nu$ has rank at most $2d$. If $|\nu|$ had more than $2d$ support points, continuous functions on disjoint neighborhoods of these points would give more than $2d$ linearly independent measures $g\nu$. Thus $\nu$ is finitely supported. Also $K_2 \subset K_1$, by marginalization, so the hypothesis forces $\nu = 0$. Finally, equality of two component laws would put $\delta_\theta - \delta_\eta$ in $K_1$. □

**Theorem I.7.2** (Blind three-visit recovery)**.** *Assume that $A_2$ is injective on finite signed measures. The law $T_{G,\rho}$ determines $G$ for every $0 < \rho \le 1$. If $G$ is not a point mass, it also determines $\rho$. If $G = \delta_\theta$, the emissions are independent with law $P_\theta$, so $\rho$ is unidentifiable and $G$ remains identifiable. The assumptions of Lemma I.7.1 are sufficient.*

*Proof.* Let $Q_h$ denote the law of $(X_0, X_h)$. The transition powers are $M_{G,\rho}^h = \rho^h I + (1-\rho^h)G$. Therefore

$$Q_h = \mu_G^{\otimes 2} + \rho^h C_G, \qquad C_G = A_2 G - \mu_G^{\otimes 2}. \tag{I.7.2}$$

The triple law supplies $\mu_G, Q_1, Q_2$. If $C_G \neq 0$, set $D_h = Q_h - \mu_G^{\otimes 2}$. Then

$$\rho = \frac{\|D_2\|_{\mathrm{var}}}{\|D_1\|_{\mathrm{var}}}, \qquad A_2 G = \mu_G^{\otimes 2} + \rho^{-1} D_1.$$

Injectivity of $A_2$ recovers $G$.

It remains to characterize the zero-signal case. For every real continuous $f$,

$$\int f(x) f(y)\, dC_G(x,y) = \mathrm{Var}_G\Big(\int f\, dP_\theta\Big).$$

If $C_G = 0$, choose a countable uniform-dense subset of $C(\mathcal{X})$ and intersect the resulting full-$G$-measure sets. All component probabilities on that set coincide, by density and uniqueness of measures. Injectivity of $A_2$ implies injectivity of $\theta \mapsto P_\theta$, so the set contains at most one point and $G$ is a point mass. Its observed marginal identifies the point. Conversely a point mass plainly gives $C_G = 0$ and independent emissions for every $\rho$. □

**Theorem I.7.3** (Uniform inverse bound without a covariance floor)**.** *For any two populations $G, H$ and any $\rho, \sigma \in [r, 1]$, where $0 < r \le 1$, let $\epsilon = \|T_{G,\rho} - T_{H,\sigma}\|_{\mathrm{var}}$. Then*

$$\|A_2 G - A_2 H\|_{\mathrm{var}} \le (2 + 9r^{-2})\epsilon. \tag{I.7.3}$$

*This bound does not assume a positive lower bound on $\|C_G\|$ or $\|C_H\|$. If $A_2$ is injective, define*

$$\omega_2(u) = \sup\{W_1(G,H) : \|A_2 G - A_2 H\|_{\mathrm{var}} \le u\}.$$

*Here $W_1$ uses the given compact metric on $\Theta$. Then $\omega_2(u) \to 0$ as $u \downarrow 0$, and*

$$W_1(G,H) \le \omega_2((2 + 9r^{-2})\epsilon).$$

*Proof.* Marginalization is a contraction in variation norm, and $\|\mu^{\otimes 2} - \nu^{\otimes 2}\|_{\mathrm{var}} \le 2\|\mu - \nu\|_{\mathrm{var}}$. Put $D = \rho C_G$ and $E = \sigma C_H$. Equation (I.7.2) gives

$$\|D - E\|_{\mathrm{var}} \le 3\epsilon, \qquad \|\rho D - \sigma E\|_{\mathrm{var}} \le 3\epsilon.$$

It follows that $|\rho - \sigma|\|D\|_{\mathrm{var}} \le 6\epsilon$. Since $\rho\sigma(C_G - C_H) = \sigma D - \rho E$, the triangle inequality gives $\|C_G - C_H\|_{\mathrm{var}} \le 9\epsilon/(\rho\sigma)$. Adding the product-marginal difference proves the bound.

To establish the modulus assertion, suppose it failed and take populations $G_n, H_n$ with a fixed positive $W_1$ separation and $\|A_2 G_n - A_2 H_n\|_{\mathrm{var}} \to 0$. Compactness gives weakly convergent subsequences $G_n \to G$, $H_n \to H$. Weak continuity of the component map implies weak continuity of $A_2$, so $A_2 G = A_2 H$. Injectivity gives $G = H$. On a compact metric space weak convergence implies $W_1$ convergence, a contradiction. □

### I.7.1 A positive entire kernel for which two visits fail

Let $U$ be the uniform probability on $[0,1]$, let $L_n(x) = P_n(2x-1)$ be the shifted Legendre polynomials, and put

$$v_n(\theta) = \theta^n - \frac{2\theta}{n+1}, \qquad k(x \mid \theta) = 1 + \sum_{n=2}^{\infty} \frac{L_n(x)v_n(\theta)}{8n!}, \quad x,\theta \in [0,1]. \tag{I.7.4}$$

**Proposition I.7.4** (Exact two-versus-three threshold)**.** *The kernel in* (I.7.4) *is jointly entire, is a probability density in $x$, and satisfies $3/4 < k(x \mid \theta) < 5/4$ on the real unit square. Its one-view nullspace is exactly*

$$K_1 = \mathbb{R}(U - \delta_0). \tag{I.7.5}$$

*Thus $A_2$ is injective and its components are distinct. Nevertheless two consecutive visits with unknown persistence do not identify all populations. Three visits do, so the universal threshold for this kernel is exactly three.*

*Proof.* The elementary bounds $|L_n(x)| \le 1$, $|v_n(\theta)| \le 2$ show $|k-1| \le (e-2)/4 < 1/4$. All $L_n$, $n \ge 1$, have integral zero, so the density integrates to one. On a compact complex set the two polynomial factors grow at most exponentially in $n$; the factorial denominator gives locally uniform convergence and hence joint entire extension.

If $A_1\nu = 0$, integration against 1 gives $\nu([0,1]) = 0$ and integration against each $L_n(x)$ gives $\int v_n\, d\nu = 0$, $n \ge 2$. Set $b = 2\int \theta\, d\nu$. Then $\int \theta^n\, d\nu = b/(n+1)$ for every $n \ge 1$. These are precisely the moments of $b(U - \delta_0)$, including the zeroth moment. Polynomial density proves (I.7.5); the converse follows from $v_n(0) = \int v_n\, dU = 0$. The nonzero elements of this nullspace are not finitely supported, so Lemma I.7.1 applies.

For $0 < s \le 1$ define $G_s = (1-s)\delta_0 + sU$. Since $P_0 = U$ and $A_1U = U$, all these populations have $\mu_{G_s} = U$. Set $C = A_2U - U^{\otimes 2} \ne 0$. The inequality follows from the zero-signal argument in Theorem I.7.2, or from $\int v_2^2\, dU > 0$. Their consecutive-pair laws are

$$Q_1(G_s, \rho) = U^{\otimes 2} + s\rho C. \tag{I.7.6}$$

Distinct choices with the same product $s\rho$ are indistinguishable from two visits. Three-visit identification follows from the theorem. □

For later use let $S$ be the signed measure on $[0,1]^3$ which is the sum of $C$ on coordinates $(0,1)$ tensored with $U$ on coordinate 2, and $U$ on coordinate 0 tensored with $C$ on coordinates $(1,2)$. Set $D = A_3U - U^{\otimes 3}$ and $K = D - S$. Conditioning on the two refresh decisions gives the exact identity

$$T_{G_s,\rho} = U^{\otimes 3} + s\rho S + s\rho^2 K. \tag{I.7.7}$$

The $(0,2)$ marginal of $K$ is $C$, so $K \ne 0$. All these measures have bounded densities, and the probability densities of the triple laws are bounded below by $(3/4)^3$.

**Corollary I.7.5** (The inverse persistence exponent is sharp)**.** *The order $r^{-2}$ in* (I.7.3) *cannot be replaced by $o(r^{-2})$, even for the fixed kernel* (I.7.4)*.*

*Proof.* Compare $(s,\rho) = (1,r)$ and $(s,\rho) = (1/2, 2r)$, $0 < r \le 1/2$. Their $A_2$ laws differ by $C/2$, while (I.7.7) shows that their triple laws differ by $r^2K$. Both $C$ and $K$ are fixed nonzero signed measures. □

### I.7.2 An exact statistical price for unknown persistence

The next experiment consists of $N$ independent stationary three-visit records, for example $N$ independent subjects. It is not one long dependent record. This distinction is part of the theorem's assumptions.

**Theorem I.7.6** (Sharp population risk near independent refresh)**.** *Use the fixed kernel* (I.7.4)*. Let $0 < r \le 1/2$, let $s \in [1/2, 1]$ and $\rho \in [r, 2r]$, and observe $N \ge 1$ independent records with law $T_{G_s,\rho}$. The estimator knows the kernel, $r$, and the stated parameter ranges, but does not know $s$ or $\rho$. There are fixed constants $0 < c < C < \infty$ such that*

$$c \min\{1, (r^2\sqrt{N})^{-1}\} \le \inf_{\widehat{G}} \sup_{s,\rho} \mathbb{E}_{s,\rho} W_1(\widehat{G}, G_s) \le C \min\{1, (r^2\sqrt{N})^{-1}\}. \tag{I.7.8}$$

*Consequently, for a sequence $r = r_N$, uniform consistent population recovery in this experiment is possible exactly when $Nr_N^4 \to \infty$.*

*Proof.* Let $\phi(x) = 80L_2(x)$. Legendre orthogonality and the coefficient of $L_2$ in (I.7.4) give $\mathbb{E}_\theta \phi(X) = v_2(\theta)$. Moreover

$$a := \int_0^1 v_2(\theta)^2 \, d\theta = \frac{2}{135}, \qquad \int v_2 \, dG_s = 0, \qquad \int v_2^2 \, dG_s = sa.$$

Thus the two bounded statistics $Z_1 = \phi(X_0)\phi(X_1)/a$ and $Z_2 = \phi(X_0)\phi(X_2)/a$ have expectations

$$v = s\rho, \qquad w = s\rho^2. \tag{I.7.9}$$

Let $(\bar{Z}_1, \bar{Z}_2)$ be their empirical averages. Choose a measurable least-squares minimizer $(\widehat{s}, \widehat{\rho})$ of its distance from $(s\rho, s\rho^2)$ over the compact rectangle $[1/2, 1] \times [r, 2r]$. A deterministic lexicographic tie rule, or any measurable minimum selector, may be used. Because the true parameter is admissible, the fitted moment vector differs from the true vector by at most twice the empirical error. The boundedness of $Z_1, Z_2$ gives expected Euclidean error $O(N^{-1/2})$.

On this rectangle, $w \ge r^2/2$, $0 \le v \le 2r$, and $s = v^2/w$. For two points in its moment image, direct subtraction gives

$$|s - s'| \le \frac{8}{r}|v - v'| + \frac{2}{r^2}|w - w'|.$$

For example, use $|v^2 - v'^2|/w \le 8|v - v'|/r$ and $(v'^2/w')|w - w'|/w \le 2|w - w'|/r^2$. Consequently $\mathbb{E}|\widehat{s} - s| \le C/(r^2\sqrt{N})$, clipped at the diameter of the parameter interval. Since

$$W_1(G_s, G_{s'}) = \frac{1}{2}|s - s'|, \tag{I.7.10}$$

the estimator $\widehat{G} = G_{\widehat{s}}$ proves the upper bound.

For the lower bound, set $s_0 = 3/4$, $\rho_0 = 3r/2$, and, for $0 < h \le 1/8$, put $s_1 = s_0 + h$ and $\rho_1 = \rho_0 s_0/s_1$. Both points lie in the parameter rectangle and $s_0\rho_0 = s_1\rho_1$. Writing $v = s_0\rho_0$, (I.7.7) gives

$$T_{G_{s_1},\rho_1} - T_{G_{s_0},\rho_0} = v^2(s_1^{-1} - s_0^{-1})K.$$

Its density is $O(hr^2)$ uniformly. The common positive lower bound on the probability densities therefore implies

$$\chi^2(T_{G_{s_1},\rho_1} \| T_{G_{s_0},\rho_0}) \le C_0 h^2 r^4.$$

For $N$ independent records the chi-squared divergence is at most $\exp(C_0 N h^2 r^4) - 1$. Choose $h = c_0 \min\{1, (r^2\sqrt{N})^{-1}\}$, with a sufficiently small fixed $c_0$. The two product laws then have total-variation distance bounded away from one in the half-variation convention. The elementary two-point testing lower bound and (I.7.10) give the asserted risk. The necessity and sufficiency of $Nr_N^4 \to \infty$ follow directly. □

**Relation to prior theory and limits of the transfer.** Three-observation identification is already a major theme of hidden Markov theory. In particular, Gassiat, Cleynen, and Robin's finite-state nonparametric HMM theorem treats full-rank transitions and linearly independent emissions. General state-space translation models are treated by Gassiat, Le Corff, and Lehéricy, arXiv:1902.01070. Those results are not being claimed here. The precise contribution proved in this section is a known, possibly one-view-nonidentifiable emission family with an arbitrary Borel population, a restricted unknown refresh transition, an explicit two-visit counterexample with positive entire densities, and its sharp $Nr^4$ statistical boundary. The proof does not identify an arbitrary unknown Markov transition or an unknown emission family, and it does not establish a rate from one long trajectory. Historical priority for this particular combination remains to be assessed through a dedicated literature review.

## I.8 Transfer boundaries and reproducibility

The results answer the target problems defined in this manuscript. The literature comparison identifies their established inputs, but does not turn the absence of a matching search result into a priority claim. In particular, the CIR linear exercise rule and the unweighted Jacobi differential system are credited to their original sources. The additional statements proved here concern arithmetic, exact minimality, explicit optimization, and recovery under an unknown persistence parameter.

**Necessary restrictions.** Several tempting extensions are deliberately excluded because a proof would require an additional ingredient.

1. The recurrence theorem requires separation of multiplicative critical values. The anisotropic Jacobi theorem requires distinct subset sums, and its matching upper bound is proved for even nonnegative Dyson index.

2. The Hamiltonian time form is required to be holomorphic on the *complex* compactified curve. A smooth real orbit alone is insufficient; the explicit singular-complex-curve example above demonstrates the failure.

3. Action quantization defines semiclassical levels. It gives no theorem about the arithmetic of exact quantum eigenvalues.

4. The CIR boundary classification does not automatically classify every optimized value at every nonresonant starting point. The integer-discount zero-state theorem supplies an additional elimination argument for its stated family.

5. Three-visit recovery uses a known emission family and a restricted refresh transition. The sharp risk theorem concerns the explicit population submodel and independent three-observation records, not an arbitrary hidden Markov model or observations from one long chain.

6. A direct extension of the binomial-mixture likelihood-transport proof to arbitrary heterogeneous replication counts is not obtained here. The likelihood score normalizes the aggregate posterior, rather than each observation stratum separately; replacing all replication counts by their minimum does not by itself prove the required inequality.

**Exact checks and their role.** The fourth-order recurrence displayed in Section I.2 was checked as an exact rational matrix identity and against directly computed initial coefficients. The contiguous determinant identity (I.6.12) was expanded symbolically for integer upper parameters through six, retaining the lower parameter as an indeterminate. These checks detect normalization

and sign errors; the general proofs do not rely on a finite sample of parameters. The $K_6 - M_2$ example was also checked by direct construction of its weighted Laplacian and comparison with (I.4.7). The hidden-state moment constant $2/135$ was checked by exact integration of $(\theta^2 - 2\theta/3)^2$ on $[0, 1]$.

The arguments were also read separately for their vulnerable steps: noncancellation at critical fibers, diagonal collision strata in Fourier asymptotics, positivity of the optimized conductances, complex points at infinity of the time differential, concave-power stopping verification, and the two-point statistical lower bound. These checks are useful evidence, not a substitute for external expert review or formal verification.

## Part II

# Sharp Bounds and Connection Formulas Beyond Existing Restrictions

**Abstract**

We give four independent results that remove a stated restriction or determine a previously unmatched boundary in a specified problem. First, the worst ratio between the total effective resistance obtained by normalized infinity Lewis weights and its optimum on an $n$-vertex tree is $\sqrt{n}/8 + 1/2 + O(n^{-1/2})$. Balanced double-brooms attain this bound. They disprove the logarithmic approximation conjecture in Suliman–Weinstein's 2023 preprint, and a rational certificate contradicts its stated constant tree bound. Second, McIntosh's alternating binomial sums have minimum recurrence width $\lfloor r/2 \rfloor$ for every integer $r \geq 2$, removing the prime-or-power-of-two restriction in his minimality theorem. Even a recurrence with arbitrary gaps requires at least $\lfloor r/2 \rfloor + 1$ nonzero summands. Third, we determine the physical-density Frobenius connection coefficients for Jacobi traces in a nonresonant real-parameter range, extending the integer-parameter formula of Forrester–Kumar. Finally, for a specified known-emission refresh model observed along one stationary path, the population minimax risk on $\rho \in [r, 1-q]$ is $\Theta(\min\{1, n^{-1/2}(r^{-2} + q^{-1/2})\})$. The low-persistence lower bound uses a uniform full-history filtering argument; the nearly frozen bound counts fresh latent draws. All constructions and proof assumptions are stated explicitly.

**Integration note.** The alternating-sum theorem is extended in Parts VII and VIII. The tree comparison is followed by broader graph families in Parts III–V, and the nonresonant Jacobi connection is extended in Part IV. The hidden-trajectory argument remains a distinct contribution.

## II.1 Targets, bottlenecks, and proof transfers

The four results have different relationships to the literature. The spectral-design result is a negative answer to an explicit conjecture in a preprint. The binomial-sum result extends a published minimality theorem using that paper's published upper bound. The Jacobi result extends a published connection formula within a stated nonresonant range. The statistical result closes a limitation of the preceding research draft in a fixed observation model; it is not attributed to a general hidden-Markov-model conjecture.

1. **Spectral design.** Suliman and Weinstein [13] conjecture an $O(\log n)$ approximation ratio for normalized infinity Lewis weights relative to optimal total effective resistance. Their Theorem 1.3 states a constant bound for trees. The tree objective reduces to a ratio of two moments of square roots of cut sizes. A sharp interval moment inequality identifies the extremal distribution, and a double-broom realizes it asymptotically. This gives both a counterexample and a sharp replacement theorem for trees.

2. **Alternating binomial sums.** McIntosh [29] constructs a recurrence of width $\lfloor r/2 \rfloor$ for every $r \geq 2$, but proves minimality only for prime $r$ or powers of two. His asymptotic lower bound sees the algebraic degree of one dominant growth rate. A positive torus-moment representation exposes all positive critical values, including those in other cyclotomic orbits. Genuine singularities force the missing lower bound, and Descartes' rule also excludes sparse recurrences with fewer nonzero terms.

3. **Jacobi trace connections.** Forrester and Kumar [84] solve the physical-density connection problem for integer $b, \beta$. Their Fuchsian system supplies the local solution spaces; complete corner Fourier expansions determine the connection coefficients for nonresonant real parameters. Both sides of an interior knot must be included. The continuation convention and the distinction from a full arbitrary-basis connection matrix are essential.

4. **Dependent hidden populations.** The preceding draft [15] treats independent three-observation records. We replace those records with one complete dependent path. A posterior filter contraction proves that the whole history cannot overcome the low-persistence information loss. A fresh-draw oracle gives the other boundary, and overlapping moment statistics attain both costs with one estimator.

These transfers use, respectively, exact moment optimization, positivity of singular fibers, complete endpoint asymptotics, and matching statistical upper and lower bounds. Each following section supplies its own proof and identifies the results it imports. The source-specific conclusions do not rely on interpreting a literature search as proof of historical novelty.

## II.2 A sharp correction to the Lewis-weight approximation problem on trees

Suliman and Weinstein formulate a comparison between the $D$-optimal (infinity-Lewis) design and the $A$-optimal effective-resistance design [13]. Their Conjecture 1.5 asserts an $O(\log n)$ approximation ratio on every $n$-vertex graph, while their Theorem 1.3 asserts the constant bound 3.12 on trees. We show that the tree ratio instead has worst-case growth $\sqrt{n}/8$. More precisely, its maximum over $n$-vertex trees equals

$$\frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}).$$

An explicit 441-vertex tree already contradicts the stated constant bound, with a certificate involving only rational numbers. The target statements here are those of arXiv:2302.05966v1.

### II.2.1 The exact tree design ratio

Let $T = (V, E)$ be a tree with $|V| = n \geq 2$. Assign a positive conductance $w_e$ to each edge, with $\sum_{e \in E} w_e = 1$. Deleting $e$ produces two components of sizes $s_e$ and $n - s_e$; write

$$c_e = s_e(n - s_e).$$

Let $R_{uv}(w)$ denote effective resistance and set

$$K_T(w) = \sum_{\{u,v\} \subset V} R_{uv}(w).$$

The unordered-pair convention is immaterial for approximation ratios: ordered pairs multiply the objective by two, and the standard identity $K_T(w) = n \operatorname{tr} L(w)^\dagger$ gives the same $A$-optimal design. The effective-resistance design problem and its convex formulation are classical; see [48].

**Lemma II.2.1** (Exact comparison on a tree)**.** *The unique $A$-optimal and $D$-optimal designs on $T$ are respectively*

$$w_e^A = \frac{\sqrt{c_e}}{\sum_{f \in E} \sqrt{c_f}}, \qquad w_e^D = \frac{1}{n-1}.$$

*Consequently, if $\alpha(T) = K_T(w^D)/K_T(w^A)$, then*

$$\alpha(T) = \frac{(n-1) \sum_{e \in E} c_e}{\left(\sum_{e \in E} \sqrt{c_e}\right)^2}. \tag{II.2.1}$$

*Proof.* The effective resistance between two vertices of a tree is the sum of $1/w_e$ along their unique connecting path. Exactly $c_e$ unordered vertex pairs have paths traversing $e$, so

$$K_T(w) = \sum_{e \in E} \frac{c_e}{w_e}.$$

Cauchy–Schwarz and the budget constraint give

$$\left(\sum_e \sqrt{c_e}\right)^2 \leq \left(\sum_e \frac{c_e}{w_e}\right)\left(\sum_e w_e\right) = K_T(w),$$

with equality exactly at the displayed $w^A$. For the $D$-optimal design, the product of the $n-1$ nonzero Laplacian eigenvalues equals $n \prod_e w_e$ by the matrix-tree identity. The arithmetic–geometric mean inequality therefore gives the unique maximizer $w_e^D = 1/(n-1)$. Substitution proves (II.2.1). □

The normalized infinity-Lewis weights in [13] coincide with this $D$-optimal design. In particular, (II.2.1) is exactly the ratio in their Claim 4.1 and equation (4.7), rather than a different normalization or another loss function.

### II.2.2 A finite bound and its sharp asymptotic

Define

$$M_n = \max\{\alpha(T) : T \text{ is a tree on } n \text{ vertices}\}, \quad a_n = \sqrt{n-1}, \quad b_n = \sqrt{\lfloor n^2/4 \rfloor}.$$

The maximum exists because there are finitely many tree topologies on $n$ vertices.

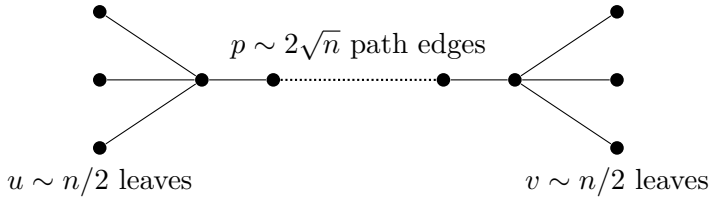



Figure 1: The balanced double-broom. Each fan denotes a larger family of leaves. Leaf-edge cut counts are $n-1$, while central-path cut counts are $n^2/4 + O(n)$. Their proportions realize the extremal moment ratio.

**Theorem II.2.2** (Sharp tree approximation asymptotic)**.** *For every $n \geq 2$ and every $n$-vertex tree $T$,*

$$\alpha(T) \leq \frac{(a_n + b_n)^2}{4a_n b_n}. \tag{II.2.2}$$

*Moreover,*

$$M_n = \frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}) \qquad (n \longrightarrow \infty). \tag{II.2.3}$$

*Balanced double-brooms whose central paths have $\lfloor 2\sqrt{n} \rfloor$ edges attain the asymptotic in (II.2.3).*

*Proof.* We first prove the upper bound. Every tree edge satisfies

$$n - 1 \leq c_e \leq \lfloor n^2/4 \rfloor.$$

Put $x_e = \sqrt{c_e}$ and $\mu = (n-1)^{-1} \sum_e x_e$. The pointwise inequality $(x_e - a_n)(x_e - b_n) \leq 0$ yields

$$\frac{1}{n-1} \sum_e x_e^2 \leq (a_n + b_n)\mu - a_n b_n.$$

By (II.2.1),

$$\alpha(T) = \frac{(n-1)^{-1} \sum_e x_e^2}{\mu^2} \leq \frac{a_n + b_n}{\mu} - \frac{a_n b_n}{\mu^2} \leq \frac{(a_n + b_n)^2}{4a_n b_n}.$$

For the last inequality one may maximize the scalar expression, whose maximum occurs at $\mu = 2a_n b_n/(a_n + b_n)$, or complete the square. In particular,

$$\frac{(a_n + b_n)^2}{4a_n b_n} = \frac{b_n}{4a_n} + \frac{1}{2} + \frac{a_n}{4b_n} = \frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}). \tag{II.2.4}$$

For the lower bound, fix $n \geq 9$ and put

$$p = \lfloor 2\sqrt{n} \rfloor, \qquad L = n - p - 1, \qquad u = \lfloor L/2 \rfloor, \qquad v = L - u.$$

Take a path with $p$ edges, attach $u$ leaves to its left endpoint and $v$ leaves to its right endpoint, and denote the resulting tree by $T_n$. It has exactly $n$ vertices and $n - 1 = L + p$ edges. The $L$ leaf edges have cut count $n - 1$. For the $j$th path edge, $1 \leq j \leq p$, the cut count is

$$c_j = (u + j)(n - u - j) = \frac{n^2}{4} - d_j^2, \qquad d_j = u + j - \frac{n}{2}. \tag{II.2.5}$$

The choice of balanced leaf counts gives $|d_j| \leq p/2$.

Write $r = \sqrt{n}$ and $p = 2r + t$, where $-1 < t \leq 0$. Using $\sum_j d_j^2 = O(p^3)$, we obtain

$$\begin{aligned} S_n := \sum_{e \in E(T_n)} c_e &= L(n-1) + \frac{pn^2}{4} - \sum_{j=1}^{p} d_j^2 \\ &= \frac{r^5}{2} + \left(\frac{t}{4} + 1\right) r^4 + O(r^3). \end{aligned} \tag{II.2.6}$$

For the square-root sum, the exact identity

$$0 \le \frac{n}{2} - \sqrt{\frac{n^2}{4} - d_j^2} = \frac{d_j^2}{n/2 + \sqrt{n^2/4 - d_j^2}} \le \frac{2d_j^2}{n}$$

shows that the total error on the path is $O(p^3/n) = O(r)$. Since $\sqrt{n-1} = r - 1/(2r) + O(r^{-3})$, it follows that

$$\begin{aligned} Q_n := \sum_{e \in E(T_n)} \sqrt{c_e} &= L\sqrt{n-1} + \sum_{j=1}^{p} \sqrt{n^2/4 - d_j^2} \\ &= 2r^3 + \left(\frac{t}{2} - 2\right) r^2 + O(r). \end{aligned} \tag{II.2.7}$$

All remainder constants above are uniform in $t \in [-1, 0]$ and in the parity of $L$. Substituting (II.2.6) and (II.2.7) into (II.2.1) gives

$$\begin{aligned} \alpha(T_n) &= \frac{(r^2-1)\{r^5/2 + (t/4+1)r^4 + O(r^3)\}}{4r^6 + (2t-8)r^5 + O(r^4)} \\ &= \frac{r}{8} + \frac{1}{2} + O(r^{-1}). \end{aligned}$$

The terms depending on $t$ cancel from the constant coefficient. Together with (II.2.4), this proves (II.2.3). □

*Remark* II.2.3 (Why the extremal scale is a square root). The ratio is the empirical second moment of the numbers $\sqrt{c_e}$ divided by their squared empirical mean. Leaf edges have $\sqrt{c_e} \simeq \sqrt{n}$, whereas the central edges of a balanced double-broom have $\sqrt{c_e} \simeq n/2$. About $2\sqrt{n}$ central edges balance these two contributions to the first moment and asymptotically attain the interval moment bound. Thus both the exponent and the leading constant arise from a genuine extremal design calculation.

### II.2.3 A rational certificate and the stated conjecture

**Proposition II.2.4** (An explicit finite counterexample)**.** *Let $T$ consist of a path with 40 edges, with 200 new leaves attached to each endpoint. Then $T$ has 441 vertices and*

$$\alpha(T) > \frac{775676}{247107} > \frac{78}{25} = 3.12.$$

*Proof.* More generally, for an integer $k \ge 1$, take a path with $4k$ edges and attach $2k^2$ leaves to each endpoint. Then $n = (2k+1)^2$ and

$$c_{\text{leaf}} = n - 1, \qquad c_j = (2k^2 + j)(2k^2 + 4k + 1 - j), \quad 1 \le j \le 4k.$$

Summation gives

$$S_k := \sum_e c_e = \frac{4k}{3}(12k^4 + 36k^3 + 26k^2 + 6k + 1).$$

Since $\sqrt{n-1} < 2k+1$ and $\sqrt{c_j} \le n/2$,

$$Q_k := \sum_e \sqrt{c_e} < 4k^2(2k+1) + 2kn = 2k(2k+1)(4k+1).$$

Therefore

$$\alpha(T_k) > \frac{4(k+1)(12k^4 + 36k^3 + 26k^2 + 6k + 1)}{3(2k+1)^2(4k+1)^2}. \tag{II.2.8}$$

At $k = 10$, the right-hand side is $775676/247107$, and

$$\frac{775676}{247107} - \frac{78}{25} = \frac{117554}{6177675} > 0.$$

Thus the certificate requires no floating-point approximation. □

**Corollary II.2.5** (Negative resolution of the logarithmic comparison)**.** *The D-optimal/infinity-Lewis effective-resistance approximation ratio is not $O(\log n)$ on general graphs, or even on trees. In particular, Conjecture 1.5 of [13], as stated in arXiv v1, is false. The stronger Conjecture 4.6 in the same version is false as well.*

*Proof.* Theorem II.2.2 supplies trees with $\alpha(T_n) \sim \sqrt{n}/8$, and $\sqrt{n}/\log n \to \infty$. The source's Theorem 1.4 bounds the actual approximation ratio above by the quantity in its Conjecture 4.6, so that stronger proposed logarithmic bound cannot hold either. □

### II.2.4 The subtree reduction does not preserve the ratio

The discrepancy with the asserted constant tree bound can be traced to a specific monotonicity step. Claim 4.3 of [13] asserts that $T \subset T'$ implies $\alpha(T) \le \alpha(T')$ for trees. Under ordinary subtree containment this assertion is false, as the following elementary argument shows independently of the preceding asymptotic theorem.

**Proposition II.2.6** (Failure of subtree monotonicity)**.** *There are trees $T \subset T'$ with $\alpha(T) > \alpha(T')$.*

*Proof.* Take $T = P_4$, the path on four vertices. Its cut counts are $3, 4, 3$; hence

$$\alpha(P_4) = \frac{30}{(2\sqrt{3}+2)^2} > 1.$$

For $N \ge 1$, attach $N$ new leaves to an endpoint of $P_4$, forming $T_N \supset P_4$. The new leaf edges have cut count $N+3$ and the three original edges have cut counts $3(N+1)$, $2(N+2)$, and $N+3$. Thus

$$\sum_e c_e = N^2 + 9N + 10,$$

$$\sum_e \sqrt{c_e} = (N+1)\sqrt{N+3} + \sqrt{3(N+1)} + \sqrt{2(N+2)} = N^{3/2} + O(N^{1/2}).$$

Since $|E(T_N)| = N + 3$, formula (II.2.1) gives $\alpha(T_N) \to 1$. For all sufficiently large $N$, $\alpha(T_N) < \alpha(P_4)$, proving the assertion. □

In the proposed monotonicity proof, a design on $T$ is treated as feasible for the resistance problem on $T'$. When vertices are added, extending the old design by zero conductances disconnects those new vertices and yields infinite total effective resistance; it does not give the asserted comparison of optimal values. The subsequent reduction from arbitrary bowties to a family with three proportional size parameters therefore does not apply. The extremal double-brooms above have a path of order $\sqrt{n}$ and two leaf populations of order $n$, which is exactly the scale missed by that reduction.

## II.3 Exact recurrence minimality for alternating binomial power sums

For an integer $r \ge 2$, consider McIntosh's alternating binomial sum

$$A_r(n) = \sum_{k=-n}^{n} (-1)^k \binom{2n}{n+k}^r, \qquad n \ge 0. \tag{II.3.1}$$

McIntosh proved that this sequence has a polynomial-coefficient recurrence with consecutive shift width at most $\lfloor r/2 \rfloor$, and proved that this width is minimal when $r$ is prime or a power of two [29, Theorems 1 and 2]. His lower-bound argument uses the algebraic degree of the dominant exponential growth rate. That degree is $\varphi(2r)/2$, which can be strictly smaller than $\lfloor r/2 \rfloor$. Here all the positive critical values of a real moment representation provide the missing lower bounds.

By a recurrence we mean an eventual identity

$$\sum_{j \in J} q_j(n) A_r(n+j) = 0, \qquad q_j \in \mathbb{C}(n) \setminus \{0\}, \tag{II.3.2}$$

where $J \subset \mathbb{Z}$ is finite and nonempty. Its *shift width* is $\max J - \min J$, and its *number of terms* is $|J|$. These are different quantities when the shifts have gaps.

**Theorem II.3.1.** *For every integer $r \geq 2$, put $m = \lfloor r/2 \rfloor$. Every recurrence (II.3.2) has shift width at least $m$ and at least $m+1$ nonzero terms. Both bounds are attained by a recurrence with coefficients in $\mathbb{Q}[n]$. Consequently the minimum shift width is $\lfloor r/2 \rfloor$, and the minimum number of terms, even allowing arbitrary gaps and coefficients in $\mathbb{C}(n)$, is $\lfloor r/2 \rfloor + 1$.*

The new assertion is the lower bound for every $r$, including the stronger bound for sparse recurrences over $\mathbb{C}(n)$. Attainment uses the general upper bound in McIntosh's Theorem 1. Thus the theorem removes the prime-or-power-of-two restriction from his minimality conclusion; no claim about an exhaustive historical priority search is needed for this precise comparison.

### II.3.1 A positive moment representation

Set $d = r - 1$, $X = x_1 \cdots x_d$, and

$$P_r(x) = \prod_{i=1}^{d} (1 + x_i)(1 - X^{-1}). \tag{II.3.3}$$

Here CT denotes the constant term in all $d$ variables. For every integer $\ell \geq 0$, direct expansion gives

$$\mathrm{CT}\, P_r^{\ell} = \sum_{j=0}^{\ell} (-1)^j \binom{\ell}{j}^r. \tag{II.3.4}$$

Indeed, the term indexed by $j$ in the last factor forces the same exponent $j$ in each of the other $d$ factors.

On $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, write $x_i = e^{i\theta_i}$ and $S = \sum_i \theta_i$. The elementary half-angle identities yield

$$P_r(e^{i\theta}) = i\Psi_r(\theta), \qquad \Psi_r(\theta) = 2^r \prod_{i=1}^{d} \cos(\theta_i/2) \sin(S/2). \tag{II.3.5}$$

The half-angle expression is globally periodic: increasing one $\theta_i$ by $2\pi$ changes both its cosine and the final sine by a minus sign. Thus $\Psi_r = -iP_r|_{\mathbb{T}^d}$ is a globally real-analytic real-valued function. Define

$$\Phi_r = \Psi_r^2 = -P_r^2|_{\mathbb{T}^d} \geq 0. \tag{II.3.6}$$

For normalized Haar probability $\mu$ on the torus,

$$A_r(n) = (-1)^n \,\mathrm{CT}\, P_r^{2n} = \int_{\mathbb{T}^d} \Phi_r(\theta)^n \, d\mu(\theta). \tag{II.3.7}$$

The factor $(-1)^n$ comes from replacing $k$ by $j - n$ in (II.3.1). It is also exactly the factor contributed by $P_r^{2n} = (-1)^n \Psi_r^{2n}$.

### II.3.2 The complete positive critical set

**Lemma II.3.2.** *The distinct positive critical values of $\Phi_r$ are*

$$v_j = \left(2\cos\frac{(2j+1)\pi}{2r}\right)^{2r}, \qquad 0 \le j < m, \qquad m = \lfloor r/2 \rfloor. \tag{II.3.8}$$

*Every critical point on each of these fibers is a nondegenerate local maximum. There are no other critical points with $\Phi_r > 0$.*

*Proof.* At a point where $P_r \neq 0$, all its factors are nonzero and

$$x_i \frac{\partial}{\partial x_i} \log P_r = \frac{x_i}{1+x_i} + \frac{1}{X-1}. \tag{II.3.9}$$

Its vanishing is equivalent to $x_i X = -1$. On the torus, $\partial_{\theta_i} P_r = i x_i \partial_{x_i} P_r$; also $P_r = i\Psi_r$. It follows that these equations are precisely the critical equations for $\Psi_r$ away from its zero set. They force $x_1 = \cdots = x_d = \omega$ with $\omega^r = -1$. Conversely each such root with $\omega \neq -1$ gives a critical point. The root $-1$ occurs only for odd $r$ and has $P_r = 0$, so it is excluded. Since $d\Phi_r = 2\Psi_r\, d\Psi_r$, this also exhausts the positive critical set of $\Phi_r$.

At such a point $X = -1/\omega$, hence $P_r = (1+\omega)^r$. With $\omega = e^{i(2j+1)\pi/r}$ this is

$$P_r = i(-1)^j \left(2\cos\frac{(2j+1)\pi}{2r}\right)^r.$$

Conjugate roots produce equal values of $\Phi_r$. After pairing them, the resulting values are exactly (II.3.8). Their angles lie strictly between 0 and $\pi/2$, so they are positive and strictly decreasing. There are $m$ of them.

For the Hessian, let $I_d$ be the identity and $J_d$ the all-ones matrix. Differentiating (II.3.9) gives, at the critical point,

$$\mathrm{Hess}_{\log x} \log P_r = \frac{\omega}{(1+\omega)^2}(I_d + J_d).$$

The diagonal contribution is $\omega/(1+\omega)^2$. Every entry also receives $-X/(X-1)^2$, which has the same value when $X = -1/\omega$. If $\omega = e^{i\alpha}$, then

$$\frac{\omega}{(1+\omega)^2} = \frac{1}{4\cos^2(\alpha/2)} > 0.$$

Angular differentiation contributes a minus sign. On the real torus one may use the real logarithm $\log|\Psi_r|$, whose Hessian agrees with that of a local complex logarithm of $P_r$. Since the gradient vanishes, this gives

$$\mathrm{Hess}_\theta\, \Phi_r = -\frac{\Phi_r}{2\cos^2(\alpha/2)}(I_d + J_d). \tag{II.3.10}$$

The eigenvalues of $I_d + J_d$ are 1, with multiplicity $d-1$, and $d+1 = r$. The Hessian is therefore negative definite at every positive critical point. □

### II.3.3 Visibility of every positive critical value

We record the real-variable argument that prevents the smaller critical values from disappearing under analytic continuation. In particular, the argument includes the integer exponents that occur in even dimensions.

**Lemma II.3.3.** *Let $M$ be a compact real-analytic manifold of dimension $d \geq 1$, let $\mu$ have a positive real-analytic density, and let $\Phi : M \to \mathbb{R}$ be real analytic. Suppose that the critical points in the fiber over $v \neq 0$ form a finite nonempty set and are all nondegenerate local maxima. Regular points in that fiber are allowed. Then the generating germ*

$$F(t) = \int_M \frac{d\mu}{1 - t\Phi}$$

*has a branch continued from $t = 0$ that is singular at $t = 1/v$.*

*Proof.* The analytic Morse lemma gives local coordinates near each critical point in which $\Phi = v - |y|^2$ and $d\mu = g(y)\, dy$, with $g$ analytic and $g(0) > 0$. For $w > 0$ sufficiently small, the local density of $\Phi_*\mu$ at $v - w$ is

$$\frac{1}{2} w^{d/2-1} \int_{S^{d-1}} g(\sqrt{w}\,\omega)\, d\omega.$$

The spherical average is analytic in $w$: odd Taylor terms integrate to zero. Its constant term is positive. Choose smooth cutoffs equal to one near the critical points. After restricting to a sufficiently small interval about $v$, the remaining compact region contains only regular points, so its pushforward has a smooth density. Summing the critical contributions therefore gives

$$\frac{d\Phi_*\mu}{dx}(v + x) = b(x) + \mathbf{1}_{\{x<0\}}(-x)^{d/2-1} h(-x), \tag{II.3.11}$$

where $b$ is smooth, $h$ is analytic near zero, and $h(0) > 0$. There is no cancellation because all critical points contribute from the same side with positive leading coefficients.

The density in (II.3.11) is not smooth at zero. If $d$ is even, a finite derivative has a jump. If $d$ is odd and $d \geq 3$, a noninteger power prevents smoothness; for $d = 1$, the density is locally integrable but unbounded.

Consider the Cauchy transform

$$G(z) = \int_M \frac{d\mu}{z - \Phi}.$$

It is holomorphic in the upper half-plane and satisfies $G(z) = z^{-1} F(z^{-1})$ near infinity. Its boundary distribution is

$$-\frac{1}{\pi} \operatorname{Im} G(x + i0) = \Phi_*\mu,$$

as follows from the Poisson kernel approximation to the identity. If this upper-half-plane branch extended holomorphically through $v$, its boundary imaginary part would be real analytic there. This contradicts (II.3.11). Thus the branch is singular at $v$. The change of variables $t = 1/z$, which is locally biholomorphic at $v \neq 0$, gives the claimed singularity at $1/v$. The continuation from infinity in the upper half-plane corresponds to a continuation of the generating germ from zero, including when $v$ is an interior point of the real range of $\Phi$. □

Applying Lemmas II.3.2 and II.3.3 to (II.3.7) proves that

$$F_r(t) = \sum_{n \geq 0} A_r(n) t^n \tag{II.3.12}$$

has genuine continued-branch singularities at the $m$ distinct positive points $t_j = 1/v_j$. This assertion concerns all these continued singularities, not merely the first singularity on the circle of convergence.

### II.3.4 Singularities force width and support bounds

**Lemma II.3.4.** *Let $a_n$ have a generating function analytic near zero. Suppose its continued germ has singularities at $m$ distinct positive finite points. Then every eventual rational-coefficient recurrence for $a_n$ has shift width at least $m$ and at least $m+1$ nonzero terms, even if its coefficients lie in $\mathbb{C}(n)$ and its shifts have gaps.*

*Proof.* After translating the index, write the recurrence as

$$\sum_{j\in J} q_j(n)a_{n+j} = 0, \qquad \min J = 0, \qquad \max J = L.$$

Clear denominators so that $q_j \in \mathbb{C}[n]$. Set $e = \max_j \deg q_j$ and $c_j = [n^e]q_j$. Not all $c_j$ vanish. Let $F(t) = \sum_{n\geq 0} a_n t^n$ and $\vartheta = t\,d/dt$. Multiplying by $t^{n+L}$ and summing over all sufficiently large $n$ gives

$$\left(\sum_{j\in J} t^{L-j} q_j(\vartheta - j)\right) F(t) = R(t), \tag{II.3.13}$$

where $R$ is a polynomial accounting for finitely many initial terms. The coefficient of the highest ordinary derivative is $t^e H(t)$, where

$$H(t) = \sum_{j\in J} c_j t^{L-j} \neq 0. \tag{II.3.14}$$

Distinct shifts give distinct monomials, so this leading coefficient cannot vanish identically. Every nonzero finite singularity of a continued solution of (II.3.13) must be a zero of $H$: otherwise the equation is an inhomogeneous linear differential equation with an ordinary point and analytic right-hand side. For $e = 0$, the same conclusion follows directly from $H(t)F(t) = R(t)$.

Consequently $H$ has at least $m$ distinct positive roots. Since $\deg H \leq L$, this implies $L \geq m$. For the term bound, choose a nonzero real or imaginary part of $H$. It is a real polynomial with the same $m$ positive roots and no more nonzero monomials than $H$. Descartes' rule of signs implies that a real polynomial with $k$ nonzero monomials has at most $k-1$ positive roots. Thus $H$ has at least $m+1$ nonzero monomials. Formula (II.3.14) now gives $|J| \geq m+1$. □

*Proof of Theorem II.3.1.* The moment representation and the complete critical-point calculation give $m$ distinct positive singularities of $F_r$ by Lemma II.3.3. Lemma II.3.4 proves both lower bounds over $\mathbb{C}(n)$. McIntosh's general existence theorem supplies a recurrence over $\mathbb{Q}[n]$ with shifts in an interval of length $m$ [29, Theorem 1]. It has at most $m+1$ positions; the lower bounds force it to attain both claimed minima. □

**Corollary II.3.5.** *For any minimum-width recurrence, translate the shifts to $0, \ldots, m$ and define its highest-degree symbol $H$ by (II.3.14). Then*

$$H(t) = c\prod_{j=0}^{m-1}(1 - v_j t), \qquad c \neq 0. \tag{II.3.15}$$

*The set of nonzero finite points at which some branch of $F_r$ continued from zero is singular is exactly $\{1/v_j : 0 \leq j < m\}$.*

*Proof.* The polynomial $H$ has degree at most $m$ and already has the $m$ distinct roots $1/v_j$, so (II.3.15) follows. The differential equation (II.3.13) has no other nonzero finite nonordinary points, hence no continued solution branch can acquire a singularity elsewhere. Visibility supplies a singular branch at every displayed point. □

### II.3.5 The first missing composite case

For $r = 6$, the three critical scales are

$$v_0 = 1351 + 780\sqrt{3}, \qquad v_1 = 64, \qquad v_2 = 1351 - 780\sqrt{3}.$$

The dominant scale has algebraic degree two, so the dominant-growth argument alone cannot prove the correct width three. The rational interior scale 64 supplies the third visible singularity. Theorem II.3.1 gives width three and at least four nonzero terms for every sparse recurrence. Moreover,

$$\prod_{j=0}^{2}(X - v_j) = (X - 64)(X^2 - 2702X + 1) = X^3 - 2766X^2 + 172929X - 64.$$

Thus a minimum-width recurrence has highest-degree generating symbol proportional to

$$1 - 2766t + 172929t^2 - 64t^3.$$

For comparison, $r = 2$ and $r = 3$ have the sole critical scales 4 and 27, respectively, in agreement with the familiar hypergeometric formulas $A_2(n) = \binom{2n}{n}$ and $A_3(n) = (3n)!/(n!)^3$.

## II.4 Nonresonant connection coefficients for Jacobi trace densities

### II.4.1 The published connection problem and the new scope

Forrester and Kumar explicitly identify the determination of the scalars in the Frobenius representation of the Jacobi trace density as the connection problem left by Davis. Their Proposition 1.3 and equation (3.55) solve that problem when the endpoint parameter $b$ and the Dyson index $\beta$ are nonnegative integers. Their discussion after equation (3.54) states that the general problem remains; see *Differential recurrences for the distribution of the trace of the $\beta$-Jacobi ensemble*, Physica D **434** (2022), 133220, `arXiv:2011.00787v2`, Section 3.3 [84].

Here the corner Fourier amplitudes determine those physical-density connection scalars for real, noninteger parameters, provided the local exponents are nonintegral. The answer uses an explicitly specified upper half-plane continuation. It generalizes the sign in the integer-parameter formula to a complex phase. The theorem concerns the particular physical density, not the full connection matrix between arbitrary bases. Resonant exponents and numerical stability of long continuation chains are not settled by it.

Fix

$$N \geq 1, \qquad a > -1, \qquad b > 0, \qquad \beta \geq 0. \tag{II.4.1}$$

The stronger restriction $b > 0$ makes every component of the published auxiliary integral system absolutely integrable. Define

$$S_N(a, b, \beta) = \int_{[0,1]^N} |\Delta(u)|^\beta \prod_{i=1}^{N} u_i^a (1 - u_i)^b \, du$$

and let $\rho(t)$ be the density of $\sum_{i=1}^{N} u_i$ under this normalized measure, extended by zero outside $[0, N]$. For $r \geq 0$ put

$$\begin{aligned} W_r(q, \beta) &= \int_{(0,\infty)^r} e^{-\sum x_i} |\Delta(x)|^\beta \prod x_i^q \, dx, \qquad W_0 = 1, \\ &= \prod_{j=0}^{r-1} \frac{\Gamma(q + 1 + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(1 + \beta/2)}. \end{aligned}$$

Only $q = a, b > -1$ will occur. The Laguerre–Selberg and Selberg evaluations give this product and

$$S_N(a,b,\beta) = \prod_{j=0}^{N-1} \frac{\Gamma(a+1+j\beta/2)\Gamma(b+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(a+b+2+(N+j-1)\beta/2)\Gamma(1+\beta/2)}.$$

For $0 \le p \le N$ define

$$\begin{aligned}
\alpha_p &= (N-p)(a+1) + \frac{\beta}{2}(N-p)(N-p-1),\\
\eta_p &= p(b+1) + \frac{\beta}{2}p(p-1),\\
D_p &= \alpha_p + \eta_p, \qquad \gamma_p = D_p - 1, \qquad \text{(II.4.2)}\\
K_p &= \binom{N}{p} \frac{W_{N-p}(a,\beta)W_p(b,\beta)}{S_N(a,b,\beta)}. \qquad \text{(II.4.3)}
\end{aligned}$$

Here $D_p > 0$ and $K_p > 0$. Our nonresonance assumption is

$$\gamma_p \notin \mathbb{Z} \qquad (0 \le p < N). \qquad \text{(II.4.4)}$$

It removes a countable collection of parameter hyperplanes; it does not require $a, b$ or $\beta$ to be rational. The exponent at the final endpoint $p = N$ need not be nonintegral, since no continuation across that endpoint is needed to represent the density on $(0, N)$.

### II.4.2 The normalized Frobenius branches

We specify the branches through the Forrester–Kumar system, so that the normalization of the coefficients below is unambiguous. With indices $j = 0, \dots, N$, set

$$\Lambda = -\operatorname{diag}(N, N-1, \dots, 0), \qquad h = a+b+1+\tfrac{\beta}{2}(2N-1), \qquad k = b+\tfrac{\beta}{2}N,$$

and let $X$ be tridiagonal, with entries

$$\begin{aligned}
X_{j,j-1} &= u_j = -jk + \tfrac{\beta}{2}j^2,\\
X_{j,j} &= v_j = hj - \tfrac{\beta}{2}(N-j)(2j+1) + k(N-2j) - 1,\\
X_{j,j+1} &= w_j = h(j-N) + \tfrac{\beta}{2}(N-j)(N-j-1) + k(N-j).
\end{aligned}$$

Entries outside the index range are omitted. The published system is

$$(tI + \Lambda)G'(t) = XG(t). \qquad \text{(II.4.5)}$$

To relate it to the physical density, form

$$H_j(t) = \binom{N}{j}^{-1} \int_{[0,1]^N} \delta\Big(t - \sum u_i\Big) |\Delta(u)|^\beta \prod u_i^a (1-u_i)^{b-1} e_j(1-u_1, \dots, 1-u_N)\, du.$$

If $P_{jk} = \binom{N-j}{N-k}$, then $G = P^{-1}H$ satisfies (II.4.5). The last row of $P$ is the last coordinate row, so

$$G_N(t) = H_N(t) = S_N(a,b,\beta)\rho(t). \qquad \text{(II.4.6)}$$

These are Proposition 3.9 and equation (3.48) of the cited paper. They also hold at $\beta = 0$, by the same integration identities or by continuity. On every interval between the integers the system is ordinary, so its distributional solution is real analytic there.

**Lemma II.4.1** (A uniquely normalized singular branch)**.** *Fix $p \in \{0, \dots, N\}$ with $\gamma_p \notin \mathbb{Z}$. At $t = p$, the system (II.4.5) has $N$ holomorphic local solutions and one nonintegral Frobenius solution. The last component of the latter, after a unique constant normalization, has the form*

$$F_p(t) = (t-p)^{\gamma_p} U_p(t-p), \qquad U_p(0) = 1, \tag{II.4.7}$$

*where $U_p$ is holomorphic with real Taylor coefficients. Initially $t - p > 0$ and the power is positive real. Any last component of a local solution is the sum of a holomorphic function and a constant multiple of this Frobenius branch, with the branch of the power specified.*

*Proof.* Put $j_0 = N - p$ and $s = t - p$. The residue matrix is the matrix with only row $j_0$ nonzero, that row being row $j_0$ of $X$. It has rank one, with its nonzero eigenvalue

$$v_{j_0} = \gamma_p - p =: v.$$

This identity follows by substitution into the displayed definition of $v_j$. The hypothesis $\gamma_p \notin \mathbb{Z}$ implies that $v$ is nonintegral, so the residue is diagonalizable with eigenvalues 0 and $v$, and no difference between them is a nonzero integer. The ordinary nonresonant Frobenius construction gives $N$ holomorphic solutions and one solution

$$s^v \sum_{\ell \geq 0} g_\ell s^\ell, \qquad g_0 = e_{j_0}.$$

For completeness, substitution gives the recurrence

$$(v + \ell)(pI + \Lambda) g_\ell = (X - (v + \ell - 1)I) g_{\ell - 1}.$$

Tridiagonality shows that its last component vanishes up to $\ell = p - 1$, while

$$(g_p)_N = \frac{\prod_{\ell=1}^{p} u_{N-p+\ell}}{p! \prod_{\ell=1}^{p} (v + \ell)} \neq 0. \tag{II.4.8}$$

Indeed, the first contribution to row $N - p + \ell$ travels through its subdiagonal entry $u_{N-p+\ell}$, and that row of $pI + \Lambda$ has diagonal entry $\ell$. Each numerator is $-j(b + \beta(N - j)/2)$ with $j \geq 1$, hence is nonzero by $b > 0$; the denominators are nonzero by nonresonance. For $p = 0$, the empty product is one. Division by (II.4.8) gives (II.4.7), since $v + p = \gamma_p$. The nonintegral Frobenius line is unique, and all coefficients are real. □

Write $F_p^{\uparrow}(t)$ for its continuation from just to the right of $p$ to a real $t > p$, moving to the right and passing every intervening integer by a small semicircle in the *upper* half-plane. This fixes a homotopy class and hence a branch. In particular, when a germ on the left of a knot is continued to its right in this convention, $(-s)^\gamma$ becomes $e^{-\pi i \gamma} s^\gamma$.

### II.4.3 The connection formula

**Theorem II.4.2** (Physical-density connection coefficients)**.** *Assume (II.4.1) and (II.4.4). For $0 \leq p < N$ set*

$$\boxed{C_p = e^{-\pi i \eta_p} \frac{K_p}{\Gamma(D_p)} = e^{-\pi i \eta_p} \binom{N}{p} \frac{W_{N-p}(a, \beta) W_p(b, \beta)}{S_N(a, b, \beta) \Gamma(D_p)}.} \tag{II.4.9}$$

*For each $p \in \{0, \dots, N-1\}$, let $\rho_{p,-}^{\uparrow}$ be the continuation to the right of $p$ of the density germ on its left, with the upper convention. Let $\rho_{p,+}$ be its physical germ on the right. At $p = 0$ use the zero density on the exterior left interval. Then*

$$\rho_{p,+}(t) - \rho_{p,-}^{\uparrow}(t) = C_p F_p(t) \qquad (t > p \text{ sufficiently close to } p). \tag{II.4.10}$$

*Consequently, for $k = 0, \dots, N-1$ and $k < t < k+1$,*

$$\boxed{\rho(t) = \sum_{p=0}^{k} C_p F_p^{\uparrow}(t).} \tag{II.4.11}$$

*Although the summands can be complex, their sum is the real, positive physical density. The lower-half-plane formula is its complex conjugate. If in addition $\gamma_N \notin \mathbb{Z}$, the local jump (II.4.10) also holds at $p = N$, with the same formula (II.4.9) and zero physical density on the right.*

The distinction between (II.4.10) and a one-sided real coefficient is essential. At noninteger parameters the density generally has nonanalytic terms on *both* sides of an interior knot. The coefficient $C_p$ is the discrepancy after a specified analytic continuation; it is not, in general, the coefficient on the right alone.

**Lemma II.4.3** (The complete local Fourier symbol)**.** *Fix $p \in \{0, \dots, N\}$. Let $\chi$ be smooth, supported in a sufficiently small interval about $p$, and identically one near $p$. Then as $R \to +\infty$,*

$$\int e^{iRt} \chi(t) \rho(t)\, dt \sim e^{ipR} R^{-D_p} \sum_{\ell \geq 0} A_{p,\ell} R^{-\ell}, \qquad A_{p,0} = K_p e^{\frac{\pi i}{2}(\alpha_p - \eta_p)}. \tag{II.4.12}$$

*The expansion is complete to every prescribed finite order. If $D_p$ is nonintegral, there are no additional integer-power terms at this frequency.*

*Proof.* Insert the defining pushforward integral for $\rho$. The relevant corners of the cube have $p$ coordinates equal to one and $N - p$ equal to zero. Near one of these corners use $x_i = u_i$ in the zero cluster and $y_i = 1 - u_i$ in the one cluster. The phase becomes $p + \sum x_i - \sum y_i$. The weight is

$$|\Delta(x)|^{\beta} \prod x_i^{a}\ |\Delta(y)|^{\beta} \prod y_i^{b}\ B(x, y), \qquad B(0,0) = 1,$$

where $B$ is analytic near the origin. The two homogeneous weights, including their measures, have degrees $\alpha_p$ and $\eta_p$. Radial integration with $r = \sum x_i$ and $q = \sum y_i$ reduces their leading Fourier integrals to the one-dimensional endpoint formula

$$\int_0^{\infty} e^{\pm iRr} r^{d-1} \chi_0(r)\, dr \sim \Gamma(d) e^{\pm \pi i d/2} R^{-d}, \quad d > 0,$$

where $\chi_0$ is smooth and equals one near zero. For nonempty clusters the angular constants are $W_{N-p}(a, \beta)/\Gamma(\alpha_p)$ and $W_p(b, \beta)/\Gamma(\eta_p)$, respectively; an empty cluster contributes one without radial integration. The angular integrals are over compact simplexes with integrable weights. Consequently, after choosing smooth cutoffs in $r$ and $q$, angular integration of $B$ produces a smooth amplitude in these two radial variables, with Taylor remainders of every order. Applying the endpoint formula successively in $r$ and $q$ gives a complete expansion. Taylor expansion of $B$ gives only integer increments of $D_p$. Thus one corner contributes the claimed leading coefficient without the binomial factor, and all $\binom{N}{p}$ corners contribute identically.

To justify the complete expansion and exclude other contributions, cover the complement of the corner neighbourhoods by small sets. At every point of this complement, some cluster of coincident coordinates lies strictly between zero and one. Translating all coordinates of that cluster together leaves its internal differences unchanged, is tangent to all relevant collision faces, and has nonzero phase derivative equal to the cluster size. Other factors in the translated coordinates are smooth and nonzero. A smooth partition of unity and repeated integration by parts in this direction give an error $O(R^{-M})$ for every $M$, even when $\beta$ is not integral. Differentiated factors are bounded by integrable endpoint and collision weights. The radial endpoint expansions of the corner pieces have remainders to arbitrary order. This proves the complete expansion and the absence of another power scale in (II.4.12). □

*Proof of Theorem II.4.2.* Fix a knot $p$ with $\gamma_p \notin \mathbb{Z}$, and abbreviate $D = D_p$, $\gamma = \gamma_p$, $\alpha = \alpha_p$, $\eta = \eta_p$, and $K = K_p$. By Lemma II.4.1, on the two real sides of $s = t - p = 0$ the physical density has the form

$$\rho(p+s) = \begin{cases} a_+(s) + c_+ s^\gamma U_p(s), & s > 0, \\ a_-(s) + c_-(-s)^\gamma U_p(s), & s < 0, \end{cases} \tag{II.4.13}$$

where $a_\pm$ are holomorphic near zero and $c_\pm$ are real. At an endpoint the exterior expression is zero. The reality follows from the real equation and the real normalization of $U_p$.

We first prove $a_+ = a_-$. After multiplication by a cutoff, a nonzero first Taylor coefficient of $a_+ - a_-$ of degree $q$ would give a nonzero term of order $R^{-q-1}$ in the Fourier transform at frequency $p$. In contrast, the nonanalytic terms in (II.4.13) give only powers $R^{-D-\ell}$, $\ell \geq 0$. Since $D$ is nonintegral, the two power scales are disjoint. Lemma II.4.3 supplies a complete expansion containing only the second scale. Uniqueness of an asymptotic power expansion forces every Taylor coefficient of $a_+ - a_-$ to vanish. The two holomorphic functions are therefore identical. This step is why leading asymptotics alone would not suffice for the proof.

Comparing the leading Fourier coefficient in (II.4.13) with (II.4.12) gives

$$\Gamma(D)\left(c_+ e^{\pi i D/2} + c_- e^{-\pi i D/2}\right) = K e^{\pi i(\alpha-\eta)/2}.$$

The real and imaginary parts uniquely determine the two real coefficients because $\sin(\pi D) \neq 0$. They are

$$c_+ = \frac{K \sin(\pi\alpha)}{\Gamma(D)\sin(\pi D)}, \qquad c_- = \frac{K \sin(\pi\eta)}{\Gamma(D)\sin(\pi D)}. \tag{II.4.14}$$

These formulas remain meaningful when an individual sine in the numerator vanishes. They also cover the empty-cluster endpoints.

Under upper-half-plane continuation of the left germ, $(-s)^\gamma$ becomes $e^{-\pi i\gamma} s^\gamma$. As the analytic backgrounds agree, the discrepancy is

$$(c_+ - e^{-\pi i\gamma} c_-) F_p(t).$$

Using $D = \alpha + \eta$ and $\gamma = D - 1$,

$$\sin(\pi\alpha) - e^{-\pi i\gamma}\sin(\pi\eta) = e^{-\pi i\eta}\sin(\pi D).$$

Substitution into (II.4.14) gives exactly $C_p = K e^{-\pi i\eta}/\Gamma(D)$ and proves (II.4.10).

At $p = 0$ the density on the left is zero, so the first interval satisfies $\rho = C_0 F_0$. Continue this identity to the next knot through the specified upper path and add the local discrepancy $C_1 F_1$. The identity extends throughout the next interval by analytic continuation of the same differential system. Inductively, this proves (II.4.11). Reversing upper and lower paths conjugates the coefficients and branches. □

### II.4.4 Comparison, checks, and limitations

When $b$ is a positive integer and $\beta$ is a nonnegative integer,

$$\eta_p = p(b+1) + \frac{\beta}{2}p(p-1) \in \mathbb{Z}, \qquad e^{-\pi i\eta_p} = (-1)^{p(b+1)+\beta p(p-1)/2}.$$

If $a$ is irrational, all exponents $\gamma_p$ with $p < N$ are nonintegral. Thus the theorem has a nonempty intersection with the integer-parameter result, and (II.4.9) there is exactly the sign and Laguerre–Selberg constant of Forrester–Kumar (1.12), Proposition 3.4, and (3.55). Their $b = 0$ boundary case is not included in the present auxiliary-integral hypothesis. The new range is the nonresonant

real-parameter range outside those integer restrictions. In that range the coefficient on the left of an interior knot generally does not vanish, and the continuation convention is indispensable.

As an exact consistency test of the additional final-endpoint statement, take $N = 1$ and $a, b \notin \mathbb{Z}$. The density is $t^a(1-t)^b/B(a+1, b+1)$ on $(0, 1)$. At $t = 1$ its upper continuation is $e^{-\pi i b}(t-1)^b t^a/B(a+1, b+1)$. Formula (II.4.9) gives $C_1 = e^{-\pi i(b+1)}/B(a+1, b+1)$, precisely the coefficient that cancels this branch to produce the zero exterior density.

A second, noninteger numerical diagnostic uses the exact $N = 2$ hypergeometric density in Forrester–Kumar (3.36)–(3.37). For $a = 0.2$, $b = 0.35$, $\beta = 0.7$, the predicted coefficient at $p = 1$ is

$$C_1 = -3.7279987849354161471 + 7.3166095798246842339\, i.$$

Evaluating the difference between the physical right density and the upper continuation of the left density, then dividing by $(t-1)^{a+b+1}$, converges to this value as $t \downarrow 1$. This diagnostic does not replace the proof.

*Remark* II.4.4 (Remaining questions). This result does not assert a full connection matrix for arbitrary Frobenius bases, a uniform error bound for numerical continuation, or a treatment of resonant $\gamma_p \in \mathbb{Z}$. Resonance can introduce logarithms and requires a separately normalized limiting basis; simply substituting resonant parameters into (II.4.14) would be invalid. The condition $b > 0$ has also been retained instead of asserting an unproved extension through divergent auxiliary integrals. A broader priority audit of general Selberg connection formulas remains necessary before any claim of historical first solution.

## II.5 One hidden-refresh trajectory: sharp costs at both persistence boundaries

A population that is invisible from one observation can become identifiable through temporal dependence. For a fixed known emission family, the preceding three-visit construction gives a persistence-dependent rate from independent triples. This section removes that independence assumption: all observations now come from one stationary trajectory. Two distinct obstructions must be controlled simultaneously. As persistence tends to zero, the one-view defect reappears. As persistence tends to one, the trajectory contains too few fresh population draws. We obtain matching upper and lower bounds over an interval containing both boundaries, using an estimator that knows neither boundary margin. The proof of the small-persistence lower bound compares full-history filters; a calculation for isolated triples alone would not suffice.

### II.5.1 A fixed analytic emission family and the statistical experiment

Let $U$ be uniform probability on $[0, 1]$, and write $L_j(x) = P_j(2x-1)$ for the shifted Legendre polynomial of degree $j$. Set

$$v_j(\theta) = \theta^j - \frac{2\theta}{j+1}, \qquad k(x \mid \theta) = 1 + \sum_{j=2}^{\infty} \frac{L_j(x) v_j(\theta)}{8 j!}, \quad x, \theta \in [0, 1]. \tag{II.5.1}$$

The family $k$ is known. Factorial decay makes the series normally convergent on every compact subset of $\mathbb{C}^2$, so it defines an entire function. The identities $\int L_j\, dU = 0$ and $\int v_j\, dU = 0$ show that

$$\int k(x \mid \theta)\, dU(x) = 1, \qquad k(x \mid 0) = 1, \qquad \int k(x \mid \theta)\, dU(\theta) = 1. \tag{II.5.2}$$

Since $|L_j| \le 1$ and $|v_j| \le 2$ on the real unit interval,

$$|k - 1| \le \frac{e-2}{4} < \frac{1}{4}, \qquad \frac{3}{4} \le k \le \frac{5}{4}. \tag{II.5.3}$$

In particular $k(\cdot \mid \theta)$ is a strictly positive probability density.

For $s \in [1/2, 1]$ let

$$G_s = (1 - s)\delta_0 + sU. \tag{II.5.4}$$

The latent chain is stationary with population $G_s$ and transition kernel

$$M_{s,\rho}(\theta, d\eta) = \rho\delta_\theta(d\eta) + (1 - \rho)G_s(d\eta), \qquad 0 < \rho < 1. \tag{II.5.5}$$

Conditionally on the latent path, observations are independent with density $k(\cdot \mid \theta_i)$. Both $s$ and $\rho$ are unknown. The data consist of *one complete trajectory* $(X_1, \dots, X_n)$; denote its law by $\mathsf{P}_{s,\rho}^{(n)}$. By (II.5.2), every individual observation has law $U$, independently of $s$ and $\rho$.

We estimate the population in Wasserstein distance $W_1$ for the usual metric on $[0, 1]$. The one-dimensional distribution-function formula gives

$$W_1(G_s, G_t) = \frac{|s - t|}{2}. \tag{II.5.6}$$

The infima below allow arbitrary measurable probability-valued estimators, not just estimates within the family (II.5.4).

**Theorem II.5.1** (Two-boundary minimax rate from one path)**.** *For $0 < r \le 1/8$, $0 < q \le 1/4$ and $n \ge 1$, define*

$$R_n(r, q) = \inf_{\widehat{G}} \sup_{\substack{s\in[1/2,1]\\ \rho\in[r,1-q]}} \mathbb{E}_{s,\rho} W_1(\widehat{G}, G_s).$$

*There are constants $0 < c < C < \infty$, depending only on the fixed emission family, such that*

$$c \min\left\{1, \frac{r^{-2} + q^{-1/2}}{\sqrt{n}}\right\} \le R_n(r, q) \le C \min\left\{1, \frac{r^{-2} + q^{-1/2}}{\sqrt{n}}\right\}. \tag{II.5.7}$$

*A single estimator $\widehat{G}_n$, using neither $r$ nor $q$, satisfies the stronger pointwise upper bound*

$$\mathbb{E}_{s,\rho} W_1(\widehat{G}_n, G_s) \le C \min\left\{1, \frac{1}{\sqrt{n}\,\rho^2\sqrt{1 - \rho}}\right\}, \qquad s \in [1/2, 1], \quad 0 < \rho < 1. \tag{II.5.8}$$

The rate in (II.5.7) contains the sum, equivalently the maximum, of two boundary costs. A product $r^{-2}q^{-1/2}$ would lose this information: one value of $\rho$ cannot lie close to both zero and one. The restrictions on $r, q$ guarantee that both lower-bound experiments lie in the parameter interval; the upper bound applies throughout that interval, including its interior.

**Corollary II.5.2** (Exact uniform-consistency criterion)**.** *For sequences $r_n \in (0, 1/8]$ and $q_n \in (0, 1/4]$, uniform consistent population estimation over the corresponding classes is possible if and only if*

$$nr_n^4 \longrightarrow \infty \qquad \text{and} \qquad nq_n \longrightarrow \infty.$$

*Proof.* Apply both sides of (II.5.7). A sum of two nonnegative sequences tends to zero exactly when both summands do. □

### II.5.2 An estimator that needs neither persistence margin

Let $\phi = 80L_2$ and $a = 2/135$. Legendre orthogonality gives

$$\int \phi(x)k(x \mid \theta)\, dU(x) = v_2(\theta), \qquad \int v_2(\theta)^2\, dU(\theta) = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = a.$$

Also $v_2(0) = 0$ and $\int v_2\, dU = 0$. The refresh transition has powers $M_{s,\rho}^h = \rho^h I + (1-\rho^h)G_s$. Conditional independence of emissions therefore yields, for every integer $h \geq 1$,

$$\mathbb{E}_{s,\rho}\frac{\phi(X_i)\phi(X_{i+h})}{a} = s\rho^h. \tag{II.5.9}$$

For $n \geq 4$, put $m = n-2$ and define the overlapping-window averages

$$\widehat{V} = \frac{1}{m}\sum_{i=1}^{m}\frac{\phi(X_i)\phi(X_{i+1})}{a}, \qquad \widehat{W} = \frac{1}{m}\sum_{i=1}^{m}\frac{\phi(X_i)\phi(X_{i+2})}{a}.$$

Their mean vector is $(v, w) = (s\rho, s\rho^2)$.

**Lemma II.5.3** (Variance of overlapping windows)**.** *There is a fixed constant $C$ such that*

$$\mathbb{E}_{s,\rho}[(\widehat{V}-v)^2 + (\widehat{W}-w)^2] \leq \frac{C}{n(1-\rho)}. \tag{II.5.10}$$

*Proof.* Each summand is a bounded function of the block $(X_i, X_{i+1}, X_{i+2})$. For two such blocks with starting positions separated by $\ell \geq 3$, condition on the latent state at the right endpoint of the first block and the state at the left endpoint of the second. Across the intervening gap, the transition is

$$M_{s,\rho}^{\ell-2} = \rho^{\ell-2}I + (1-\rho^{\ell-2})G_s.$$

The second term produces the product of the two expectations. It follows that the absolute covariance is at most $C\rho^{\ell-2}$. Overlaps with $\ell \leq 2$ have covariance at most $C$. Summing the covariance series, and using $m \geq n/2$, proves (II.5.10). □

Choose a measurable minimizer $(\widehat{s}, \widehat{\rho})$ of

$$(\widehat{V} - s'\rho')^2 + (\widehat{W} - s'\rho'^2)^2 \quad \text{over} \quad (s', \rho') \in [1/2, 1] \times [0, 1]. \tag{II.5.11}$$

Existence follows from compactness and continuity; a fixed measurable tie-breaking rule supplies measurability. Set $\widehat{G}_n = G_{\widehat{s}}$. For $n < 4$, use a fixed member of the population family.

**Lemma II.5.4** (A global inverse with the true persistence in its denominator)**.** *Let $s, s' \in [1/2, 1]$, $0 < \rho \leq 1$, $0 \leq \rho' \leq 1$, and put $(v, w) = (s\rho, s\rho^2)$ and $(v', w') = (s'\rho', s'\rho'^2)$. Then*

$$|s - s'| \leq \frac{4}{\rho^2}(|v - v'| + |w - w'|). \tag{II.5.12}$$

*The candidate persistence $\rho'$ may be zero.*

*Proof.* The polynomial identities $v^2 = sw$ and $v'^2 = s'w'$ hold even at $\rho' = 0$. Consequently

$$w(s - s') = v^2 - v'^2 + s'(w' - w).$$

Since $w \geq \rho^2/2$, $v, v' \in [0, 1]$ and $s' \leq 1$, the claimed bound follows. □

*Proof of the upper bounds in Theorem II.5.1.* Least-squares projection makes the Euclidean error of the fitted moment vector at most twice the empirical error from the true moment vector. Lemma II.5.4, Cauchy–Schwarz and Lemma II.5.3 give

$$\mathbb{E}_{s,\rho}|\widehat{s} - s| \leq \frac{C}{\sqrt{n}\,\rho^2\sqrt{1-\rho}}.$$

The error is also bounded by $1/2$. Equation (II.5.6) proves (II.5.8), after increasing the constant to include $n < 4$. Crucially, the persistence in this bound is the true parameter, not a worst-case lower bound substituted before the variance calculation.

For $\rho \leq 1/2$, the right-hand side before clipping is at most $C/(\sqrt{n}\,r^2)$. For $\rho \geq 1/2$, it is at most $C/\sqrt{nq}$. Taking the supremum over $[r, 1-q]$, then using the bounded loss, proves the upper half of (II.5.7). □

### II.5.3 Small persistence: a full-path information bound

The moment map has first coordinate $s\rho$. We compare two parameter points along a curve on which this coordinate is constant. The next lemma shows that the entire observed trajectory, including all its long-range dependence, has only fourth-order information in small persistence along that curve. Write $D(P\|Q)$ for relative entropy.

**Lemma II.5.5** (Uniform filter comparison along $s\rho$ = constant)**.** *Suppose $s, s' \in [1/2, 1]$, $0 < \rho, \rho' \le 1/4$ and $\alpha = s\rho = s'\rho'$. Then, for every $n \ge 1$,*

$$D(\mathsf{P}_{s,\rho}^{(n)} \| \mathsf{P}_{s',\rho'}^{(n)}) \le Cn\alpha^2(\rho - \rho')^2. \tag{II.5.13}$$

*The constant is independent of the time horizon and both parameters.*

*Proof.* Put $q_x(\theta) = k(x \mid \theta) - 1$. For a fixed observed history $(x_1, \dots, x_t)$, let $f_t^{s,\rho}$ be the posterior density of $\theta_t$ relative to $G_s$. Use the recursive formula below to extend this density to a function on all of $[0, 1]$. With $f_0 = 1$, Bayes' formula gives

$$f_t^{s,\rho}(\theta) = \frac{k(x_t \mid \theta)[1 - \rho + \rho f_{t-1}^{s,\rho}(\theta)]}{p_t^{s,\rho}(x_t)}, \tag{II.5.14}$$

$$p_t^{s,\rho}(x) = 1 + s\rho \int q_x(\theta) f_{t-1}^{s,\rho}(\theta)\, dU(\theta). \tag{II.5.15}$$

Here $p_t^{s,\rho}$ is the conditional density of $X_t$ given the preceding history. To check the second formula, integrate the numerator of the first before observing $x_t$. The constant part integrates to one because $\int f_{t-1}^{s,\rho}\, dG_s = 1$; the remaining part uses $\int q_x\, dG_s = 0$ and $q_x(0) = 0$. For $t = 1$, the convention $f_0 = 1$ gives the stationary initial law and $p_1 = 1$.

Every predictive distribution is a mixture of emission densities, so

$$\frac{3}{4} \le p_t^{s,\rho} \le \frac{5}{4} \tag{II.5.16}$$

for every history, including histories of probability zero. Indeed all formulas are pointwise defined with positive denominators. An induction in (II.5.14) gives

$$\|f_t^{s,\rho}\|_\infty \le 3. \tag{II.5.17}$$

If the preceding norm is at most three, the next one is at most $(5/3)(1 - \rho + 3\rho) \le 5/2 < 3$; the induction starts at $f_0 = 1$.

Compare both filters on the same arbitrary history, and write $\Delta_t = \|f_t^{s,\rho} - f_t^{s',\rho'}\|_\infty$. Equation (II.5.15), the fixed value of $\alpha$, and $\|q_x\|_\infty \le 1/4$ imply

$$\sup_x |p_t^{s,\rho}(x) - p_t^{s',\rho'}(x)| \le \frac{\alpha}{4}\Delta_{t-1}. \tag{II.5.18}$$

If $N, N'$ denote the numerators in (II.5.14), then

$$\|N - N'\|_\infty \le \frac{5}{4}(\rho\Delta_{t-1} + 4|\rho - \rho'|).$$

This follows by adding and subtracting $\rho f_{t-1}^{s',\rho'}$ and using (II.5.17). Using $N'/p' = f_t^{s',\rho'}$, whose norm is at most three, and (II.5.16), we obtain

$$\Delta_t \le \left(\frac{5}{3}\rho + \alpha\right)\Delta_{t-1} + \frac{20}{3}|\rho - \rho'| \le \frac{2}{3}\Delta_{t-1} + \frac{20}{3}|\rho - \rho'|.$$

The last inequality uses $\alpha \le \rho \le 1/4$. Since $\Delta_0 = 0$, geometric summation proves $\Delta_t \le 20|\rho - \rho'|$ uniformly in time and history. Therefore

$$\sup_x |p_t^{s,\rho}(x) - p_t^{s',\rho'}(x)| \le 5\alpha|\rho - \rho'|. \tag{II.5.19}$$

For two densities $p, p'$ relative to the probability measure $U$, $D(p\|p') \le \int (p - p')^2/p'\, dU$. Equations (II.5.16) and (II.5.19) thus bound every conditional relative entropy by $C\alpha^2(\rho - \rho')^2$, uniformly in the preceding history. Summing by the chain rule for relative entropy gives (II.5.13). □

**Proposition II.5.6** (The small-persistence obstruction)**.** *On the parameter class of Theorem II.5.1,*

$$R_n(r, q) \ge c \min\{1, (\sqrt{n}\, r^2)^{-1}\}. \tag{II.5.20}$$

*Proof.* Take

$$s_0 = \frac{3}{4}, \quad \rho_0 = \frac{3r}{2}, \qquad s_1 = s_0 + h, \quad \rho_1 = \rho_0 \frac{s_0}{s_1}, \qquad 0 < h \le \frac{1}{8}.$$

Both persistence values belong to $[r, 1 - q]$ and are at most $1/4$. Their products $s_i\rho_i$ agree, and $|\rho_0 - \rho_1| \le Crh$. Lemma II.5.5 gives

$$D(\mathsf{P}^{(n)}_{s_0,\rho_0} \| \mathsf{P}^{(n)}_{s_1,\rho_1}) \le Cnh^2r^4.$$

Choose $h = c_0 \min\{1, (\sqrt{n}\, r^2)^{-1}\}$ with a fixed sufficiently small $c_0 > 0$. Then the total variation distance of the two observed-path laws is at most $1/2$, by Pinsker's inequality. Their population separation is $h/2$ by (II.5.6). For any probability-valued estimator, the nearest-population test and the triangle inequality give a maximal expected loss at least one quarter of this separation times one minus the total variation distance. This proves (II.5.20). □

### II.5.4 Nearly frozen paths: only fresh latent draws carry population information

**Proposition II.5.7** (The nearly frozen obstruction)**.** *On the parameter class of Theorem II.5.1,*

$$R_n(r, q) \ge \frac{c}{\sqrt{1 + nq}} \ge c' \min\{1, (nq)^{-1/2}\}. \tag{II.5.21}$$

*Proof.* Fix $\rho = 1 - q$. Consider an oracle that reveals all refresh indicators, the initial latent value, and every newly drawn latent value. The oracle can simulate the whole observed path using a parameter-independent Markov kernel, so data processing bounds observed-path relative entropy by oracle relative entropy. The number of independent population draws revealed is

$$M = 1 + \mathrm{Binomial}(n - 1, q), \qquad \mathbb{E}M = 1 + (n - 1)q.$$

Refresh indicators have the same law at all values of $s$. Because $\delta_0$ and $U$ are mutually singular,

$$D(G_s\|G_t) = s \log \frac{s}{t} + (1 - s) \log \frac{1 - s}{1 - t}.$$

For $s_0 = 3/4$ and $s_1 = s_0 + h$, $0 < h \le 1/8$, this is at most $Ch^2$. Conditioning on the indicators and averaging the product-law relative entropies therefore gives

$$D(\mathsf{P}^{(n)}_{s_0,1-q} \| \mathsf{P}^{(n)}_{s_1,1-q}) \le C(1 + nq)h^2.$$

Choose $h = c_0/\sqrt{1 + nq}$ and apply the same two-point argument as before. The resulting bound is $c/\sqrt{1 + nq}$. This is comparable, with absolute constants, to $\min\{1, (nq)^{-1/2}\}$. □

*Completion of Theorem II.5.1.* The upper bound has already been proved. The maximum of the lower bounds in Propositions II.5.6 and II.5.7 is, within an absolute constant, their clipped sum. Explicitly, for $A, B \geq 0$,

$$\max\{\min(1, A), \min(1, B)\} \geq \tfrac{1}{2} \min(1, A + B).$$

Taking $A = (\sqrt{n}\, r^2)^{-1}$ and $B = (nq)^{-1/2}$ proves the lower half of (II.5.7). □

*Remark* II.5.8 (The two endpoints are different impossibility mechanisms). At $\rho = 0$, all observed paths are independent uniform samples for every $s$, so the population is unidentifiable. At $\rho = 1$, the oracle above reveals just one population draw, regardless of the length of the observed path. The same two-point argument then gives a positive constant minimax risk for every $n$. Thus the obstruction at one is lack of fresh population draws, even though the emission law of the retained latent state may be learned accurately from its long run.

### II.5.5 Transfer conditions and precise relation to earlier work

**Proposition II.5.9** (The same rate for other fixed emission families)**.** *Let $\Theta$ be a compact metric space, let $\theta_0 \in \Theta$, and let $H$ be a probability measure with $H(\{\theta_0\}) = 0$. Let $\mu$ be a probability measure on the observation space. Suppose the known emission densities satisfy, pointwise,*

$$|k(x \mid \theta) - 1| \leq \frac{1}{4}, \qquad k(x \mid \theta_0) = 1, \qquad \int k(x \mid \theta)\, dH(\theta) = 1,$$
$$\int k(x \mid \theta)\, d\mu(x) = 1.$$

*Assume there is a bounded measurable real function $\phi$ with $\int \phi\, d\mu = 0$ such that*

$$b(\theta) = \int \phi(x) k(x \mid \theta)\, d\mu(x), \qquad a = \int b(\theta)^2\, dH(\theta) > 0.$$

*Replace $U$ by $H$ and $0$ by $\theta_0$ in the population (II.5.4), and use the same refresh transition. Then Theorem II.5.1 and Corollary II.5.2 hold, with constants depending on this fixed experiment and its metric.*

*Proof.* The assumptions give $b(\theta_0) = 0$ and $\int b\, dH = 0$. For $h = 1, 2$, the bounded statistic $\phi(X_i)\phi(X_{i+h})/a$ again has expectation $s\rho^h$. The covariance bound and moment inverse are unchanged. Also

$$W_1(G_s, G_t) = |s - t| \int d(\theta_0, \theta)\, dH(\theta).$$

The upper bound follows by coupling the excess mass between $\delta_{\theta_0}$ and $H$, and the reverse inequality follows from the 1-Lipschitz function $d(\theta_0, \cdot)$. The displayed multiplier is strictly positive.

For the small-persistence lower bound, the filter proof is identical with $H$ replacing $U$ in latent integrals and $\mu$ replacing $U$ in predictive relative entropies. Every numerical bound in that proof used only the stated bound $|k - 1| \leq 1/4$ and the two cancellation identities. Finally, $H(\{\theta_0\}) = 0$ gives the same Bernoulli relative entropy for population draws, proving the nearly frozen lower bound. This verifies all assertions. □

The direct advance over the independent-three-visit result is the matching rate for a single dependent path, including both persistence boundaries and an estimator that does not require their margins. The small-persistence lower bound required a separate full-path argument; it cannot be obtained by declaring overlapping triples independent or by substituting an informal effective sample size.

For comparison, Abraham, Gassiat and Naulet study minimax frontiers for nonparametric hidden Markov models with unknown emission densities [14]. Their Section 1.2 distinguishes the independence boundary from an absolute spectral-gap condition, and their informal Theorem 1 allows constants to depend on that gap. Section 4 states that exact dependence on the gap was not its focus, and poses a different adaptation question concerning emission densities of different smoothness. The present theorem does not resolve that question: the emission family is fixed and known, the population submodel is one-dimensional, and transitions have the restricted refresh form (II.5.5). Within this specified experiment, both boundary dependences in (II.5.7) are proved sharply.

Although the population has only one scalar unknown coefficient, the latent state is continuous. A value drawn from the continuous part retains its own emission law throughout a persistence run. Collapsing all such values into one of two state labels and assigning that label a fixed emission law would remove the within-run dependence and would change the experiment.

## II.6 What has and has not been resolved

The double-broom construction is an explicit obstruction to the logarithmic Lewis-weight claim under the source's own normalization. The result determines the worst tree asymptotic, including its leading coefficient and constant term. It does not determine the worst asymptotic over all connected graphs.

For alternating binomial sums, the lower bound holds for every integer exponent and for coefficients in $\mathbb{C}(n)$, with no coefficient-degree restriction. The matching upper bound is credited to McIntosh. We remove the restriction in his theorem; we do not assert that an exhaustive survey has certified the absence of every subsequent proof.

For Jacobi traces, the conclusion gives the scalars for the physical density in the stated nonresonant range. Resonance, the remaining endpoint parameter range, and stable numerical continuation remain separate tasks. For the hidden-population model, both boundary rates concern the explicitly defined known emission family and population class. Unknown emissions and general transition kernels are outside the theorem.

The heterogeneous binomial-mixture NPMLE was also examined during target selection. A posterior-coupling bound works for comparable observation depths, but a matching all-profile, all-range theorem was not obtained. It is therefore not counted among the four completed results here.

## Part III

# Entropy, Likelihood Geometry, and Sharp Proof Transfers

**Abstract**

We transfer entropy–transport and singularity methods to five specified problems. For binomial population recovery, every ordinary unrestricted nonparametric maximum-likelihood estimator attains the sharp Wasserstein risk under an unknown fraction of adversarially replaced reports; the estimator requires no modification and no contamination parameter. A second theorem proves uniform minimax optimality for a joint likelihood with arbitrarily many depth-one observations and any number of observations at a common deeper level. Its key identity is an affine posterior tilt, which repairs the failure of stratumwise normalization. For exponential families, likelihood contact gives a conditional moment-generating-function bound and an entropy bound for posterior barycenter movement. Gaussian and Poisson applications give martingale couplings and high-information minimax regimes, with explicit boundary and approximation limitations. Beyond statistics, the worst D-optimal/A-optimal design ratio over all connected graphs is of order $\sqrt{n}$: an electrical maximum-principle argument yields a universal upper bound $\sqrt{n/8} + 1/2$, and complete-graph barbells improve the preceding tree lower constant. Finally, the minimum recurrence width of phase-weighted central binomial power sums is classified on the entire unit circle: it is $r$ at every nonreal phase and drops at the two real phases. A polynomial moment reduction supplies the matching upper bound. All five developments distinguish previously proved ingredients from new arguments and state the unresolved extensions explicitly.

**Integration note.** The robust binomial, unequal-depth, and posterior-contact results retain their own hypotheses. The phase-weight recurrence result is extended in Parts VII and VIII. The all-connected-graph result complements the more specialized extremal theorems in Parts II, IV, and V.

## III.1 Questions closed and methods transferred

The statistical starting point is the entropy–transport argument in the supplied observation compilation. Likelihood scores define a posterior coupling; a relative-entropy decomposition then converts componentwise concentration into control of the fitted distribution. This transfer does not assert that an arbitrary mixture inherits its components' concentration. The new arguments address three obstructions to a direct application: adversarial selection of observations, failure of normalization within an observation stratum, and infinite relative entropy between an empirical measure and a continuous density.

1. **Ordinary likelihood under report corruption.** The supplied compilation already establishes a sharp robust rate using different estimators, but expressly leaves ordinary NPMLE robustness unproved. We establish that property, including strong, data-dependent replacements and adaptation to their unknown budget. A bounded retained-sample score gives a linear posterior repair cost. Uniform likelihood events over all retained subsets handle the adversary without an invalid iid assumption.

2. **Arbitrary imbalance between two observation depths.** For depths 1 and $T$, the shallow likelihood forces the deep posterior update to be an affine tilt of the fitted mixing law. Its transport cost is exactly its change of mean. This restores a uniform predictive bound and matches lower bounds whose alternatives have identical means. The number of shallow observations may be arbitrarily large.

3. **Likelihood contact in continuous and count families.** A natural-parameter perturbation turns the global likelihood score condition into a conditional moment-generating-function inequality. Entropy is compared between two weightings of the same observed sample, so no empirical-to-density KL divergence is used. Boundary-constrained fits require a separate clipping argument.

4. **All connected graph topologies.** The previous paper settled the tree asymptotic and refuted the logarithmic conjecture stated by Suliman and Weinstein. The present extension determines the correct worst-case growth order over all connected graphs. The exact leading constant in that larger class is still undetermined.

5. **The complete unit-circle phase family.** The Franel singularity method and the preceding alternating-sum calculation extend to all complex phases of modulus one. A new finite-dimensional moment reduction supplies the generic upper recurrence. This is a self-posed parameter classification, not an asserted solution of a named external open problem.

These are five related developments, not five independent claims to have settled famous conjectures. Their precise statements and proofs follow. The binomial approximation and likelihood-cover tools are reproduced in Section III.A for independent checking; those supporting lemmas are credited to the preceding compilation and are not new claims.

## III.2 The ordinary mixture MLE withstands unknown adversarial report replacement

The sharp robust population-recovery rate was previously attained by minimum-distance estimators. The preceding compilation explicitly left open whether the ordinary, unmodified nonparametric maximum likelihood estimator attains that rate [20, Parts VII–VIII]. We answer that question affirmatively, including replacement locations chosen after inspecting the entire clean sample. The contamination budget is not supplied to the estimator. The rate itself and its

lower bound are prior results; the new conclusion concerns the ordinary MLE and a certificate sufficient for its numerical approximation.

Write

$$K_\theta(j) = \binom{t}{j}\theta^j(1-\theta)^{t-j}, \qquad p_G(j) = \int_0^1 K_\theta(j)\,dG(\theta), \quad 0 \le j \le t.$$

On the count alphabet, $W_{1,t}$ uses cost $|i-j|/t$; population $W_1$ uses the ordinary distance on $[0,1]$. Total variation is one half of the $\ell_1$ distance, and $h(p,q)^2 = \sum_j(\sqrt{p_j} - \sqrt{q_j})^2$. Set $\log(e/0) = +\infty$ and

$$r_{N,t}(\varepsilon) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + \varepsilon + \frac{1}{\sqrt{t\log(e/\varepsilon)}}\right\}. \tag{III.2.1}$$

**Theorem III.2.1** (Sharp robustness of the unmodified NPMLE)**.** *Let $J_1, \dots, J_N$ be independent with common law $p_G$, where $G$ is an arbitrary probability law on $[0,1]$. A possibly randomized adversary sees the entire clean sample and replaces at most $\lfloor \varepsilon N \rfloor$ reports by arbitrary values in $\{0, \dots, t\}$, where $0 \le \varepsilon \le 1/4$. Every measurable choice $\widehat{G}$ of an unrestricted mixing MLE of the corrupted reports satisfies*

$$\sup_G \sup_A \mathbb{E} W_1(\widehat{G}, G) \le C r_{N,t}(\varepsilon). \tag{III.2.2}$$

*The corresponding minimax risk over all estimators is at least $c r_{N,t}(\varepsilon)$. The constants are absolute, uniformly in $N, t \ge 1$. Thus the ordinary MLE adapts to the unknown replacement budget and attains the sharp robust rate.*

*The same upper bound holds for any fitted law $H$ whose corrupted-data score satisfies*

$$\sup_{\theta\in[0,1]} \sum_{j:q(j)>0} \frac{q(j)K_\theta(j)}{p_H(j)} \le 1 + \frac{1}{N}, \tag{III.2.3}$$

*with $p_H > 0$ on the observed support. Here $q$ is the empirical law of the corrupted reports.*

This result concerns replacement of whole reports, not a budget of individual Bernoulli-bit changes. Unrestricted mixing is essential to the score condition; no claim is made for an MLE with a prescribed number of mixture components. The clean population problem was studied by Vinayak, Kong, Valiant and Kakade [70], with subsequent Chebyshev-moment improvements by Musco, Musco, Rosenblatt and Singh [16]. General contamination lower bounds based on total variation moduli are classical [17]. We do not attribute the estimator-specific question here to a named conjecture in those papers.

### III.2.1 A retained-sample score and entropy–transport repair

For a probability vector $a$ on the count alphabet, put $\ell_a(H) = \sum_j a(j)\log p_H(j)$, with zero terms omitted. The following finite-space lemma isolates the mechanism. Its likelihood geometry and entropy–transport duality are classical [18, 19]; the use of a retained subsample and a linear repair cost is what is needed below.

**Lemma III.2.2** (Score-controlled posterior repair)**.** *Let $(\mathcal{X}, d)$ be a finite metric space of diameter one, and suppose the component kernels satisfy*

$$W_{1,d}(R, K_\theta)^2 \le 2cD(R\|K_\theta).$$

*Let $H$ be a mixing law with $p_H > 0$ on the support of $a$, and suppose*

$$S_a(\theta) = \sum_{x:a(x)>0} \frac{a(x)K_\theta(x)}{p_H(x)} \le \sigma, \qquad \sigma \ge 1.$$

*Then*

$$\sup_F \ell_a(F) - \ell_a(H) \le \log \sigma, \tag{III.2.4}$$

$$W_{1,d}(a, p_H) \le \sqrt{2cD(a\|p_H)} + 1 - \sigma^{-1}. \tag{III.2.5}$$

*Proof.* Since $\int S_a \, dH = 1$, the measure $dH^+ = S_a \, dH$ is a probability law. For $S_a(\theta) > 0$, define

$$R_\theta(x) = \frac{a(x)K_\theta(x)}{p_H(x)S_a(\theta)}.$$

It is a probability vector and $\int R_\theta \, dH^+ = a$. Direct expansion of the logarithms gives

$$D(a\|p_H) = D(H^+\|H) + \int D(R_\theta\|K_\theta) \, dH^+(\theta). \tag{III.2.6}$$

Zero-density terms are omitted. Convexity of transportation cost under mixtures, the component inequality, and Cauchy–Schwarz give $W_{1,d}(a, p_{H^+}) \le \sqrt{2cD(a\|p_H)}$. Also, $(u-1)_+ \le (1-\sigma^{-1})u$ for $0 \le u \le \sigma$, so

$$\|H^+ - H\|_{\mathrm{TV}} = \int (S_a - 1)_+ \, dH \le 1 - \sigma^{-1}.$$

Coupling their common mixing mass costs at most this amount because the observation diameter is one. This proves (III.2.5). Finally, Jensen's inequality gives, for every $F$,

$$\ell_a(F) - \ell_a(H) \le \log \sum_x a(x) \frac{p_F(x)}{p_H(x)} = \log \int S_a \, dF \le \log \sigma.$$

□

For the normalized binomial count metric the component constant is $c = 1/(4t)$. Indeed, a function with count-grid Lipschitz constant one changes by at most $1/t$ when one of the $t$ underlying Bernoulli bits is changed. Conditional Hoeffding inequalities for the revealing martingale imply

$$\log \mathbb{E}_{K_\theta} e^{\lambda(f - \mathbb{E}_{K_\theta} f)} \le \frac{\lambda^2}{8t}.$$

The entropy variational inequality, optimized over $\lambda > 0$, and Wasserstein duality therefore give

$$W_{1,t}(R, K_\theta)^2 \le \frac{D(R\|K_\theta)}{2t}. \tag{III.2.7}$$

This argument also covers the degenerate endpoint kernels whenever the relative entropy is finite.

Now put $k = \lfloor \varepsilon N \rfloor$, $\delta = k/N$, and $n = N - k$. Pad the actual changed set to exactly $k$ indices. Let $I$ be its complement, $q_I$ the empirical law of the retained clean observations, $q_J$ that of the $k$ removed observations, and $q^0$ the full clean empirical law. When $k = 0$, terms involving $q_J$ are absent. Then

$$q \ge (1-\delta)q_I \quad \text{coordinatewise}, \qquad q^0 = (1-\delta)q_I + \delta q_J. \tag{III.2.8}$$

No independence of the data-selected retained sample is assumed.

The MLE exists: the binomial moment body is compact and contains a vector positive at every cell, so the likelihood maximum is finite and attained. Its directional score satisfies $S_q(\theta) \le 1$. More generally, suppose $S_q \le 1 + \zeta$. Equation (III.2.8) implies

$$S_{q_I} \le \sigma_\delta := \frac{1+\zeta}{1-\delta}, \qquad e_\delta := \log \sigma_\delta, \qquad a_\delta := 1 - \sigma_\delta^{-1} = \frac{\delta + \zeta}{1+\zeta}. \tag{III.2.9}$$

Consequently the fit is within $e_\delta$ of the optimal retained normalized likelihood, and its transportation repair costs at most $a_\delta$, not a square root of the contamination fraction.

**Lemma III.2.3** (Uniform predictive transportation under deletions)**.** *Let $0 \le \delta \le 1/4$ and $0 \le \zeta \le 1/N$. Every fit satisfying $S_q \le 1+\zeta$ obeys, uniformly over the replacement adversary,*

$$\mathbb{E}W_{1,t}(p_H, p_G) \le C\left\{N^{-1/2} + \delta + \sqrt{\frac{b(\delta)}{t}}\right\}, \qquad b(u) = -u\log u - (1-u)\log(1-u). \tag{III.2.10}$$

*Here $b(0) = 0$.*

*Proof.* Introduce the binary label specifying which part of (III.2.8) generated a count. Its mutual information with the count is at most $b(\delta)$. The entropy chain rule gives

$$(1-\delta)D(q_I\|p_G) + \delta D(q_J\|p_G) \le D(q^0\|p_G) + b(\delta). \tag{III.2.11}$$

All clean observations lie in the support of $p_G$ almost surely, so this identity requires no lower bound on its nonzero cell probabilities. By (III.2.4),

$$D(q_I\|p_H) \le D(q_I\|p_G) + e_\delta.$$

Apply (III.2.5) with (III.2.7). Since $W_{1,t}(q_I, q^0) \le \delta$, it follows samplewise that

$$\begin{aligned} W_{1,t}(p_H, p_G) \le{}& W_{1,t}(q^0, p_G) + \delta + a_\delta \\ &+ \sqrt{\frac{1}{2t}\left\{\frac{D(q^0\|p_G) + b(\delta)}{1-\delta} + e_\delta\right\}}. \end{aligned} \tag{III.2.12}$$

The full clean empirical law satisfies

$$\mathbb{E}D(q^0\|p_G) \le \mathbb{E}\chi^2(q^0, p_G) \le t/N, \qquad \mathbb{E}W_{1,t}(q^0, p_G) \le 1/(2\sqrt{N}).$$

The second bound follows by integrating empirical-CDF standard deviations on the normalized interval. Jensen's inequality, $e_\delta \le C\delta + \zeta$, $a_\delta \le \delta + \zeta$, and $\delta \le Cb(\delta)$ prove (III.2.10). Crucially, (III.2.12) holds for every retained subset on the same clean sample; it is not an expectation conditional on an adaptively chosen subset. □

### III.2.2 Likelihood envelopes uniform over the deletion set

We record the approximation inputs from the earlier clean likelihood analysis [20, Parts VI–VIII]. Their full proofs are reproduced in Section III.A; they concern only the binomial family, not the corruption mechanism, and are not new claims.

**Lemma III.2.4** (Binomial cover and two-scale inversion)**.** *For $t \le N$, $u = (eN)^{-2}$, and $d = t+1$, the binomial mixture moment body has an $\ell_1$ cover of radius $u/d$, with centers in that body and cardinality satisfying*

$$\log M \le C\sqrt{t}\,\log(eN)^{3/2}. \tag{III.2.13}$$

*For arbitrary mixing laws $F, G$ and integers $1 \le S \le K \le t/2$,*

$$W_1(F,G) \le \frac{C}{K} + Ce^{4S^2/t}W_{1,t}(p_F, p_G) + \frac{Ce^{2K^2/t}}{S}h(p_F, p_G). \tag{III.2.14}$$

*Proof.* For (III.2.13), take $\beta = u/(t+1)$ in Lemma III.A.1; its degree cutoff is $O(\sqrt{t\log(eN)})$ when $t \le N$. Equation (III.2.14) is Proposition III.A.4, whose proof uses the derivative-stable approximation and coefficient estimates in Lemma III.A.3. The weighted Chebyshev-moment approach is also developed in [16]. □

**Lemma III.2.5** (Hellinger control for all retained samples)**.** *Under the hypotheses of Lemma III.2.3, if $t \le N$, then*

$$\mathbb{E}h(p_G, p_H) \le C\left\{\frac{t^{1/4}\log(eN)^{3/4}}{\sqrt{N}} + \sqrt{b(\delta)}\right\}. \tag{III.2.15}$$

*The bound is uniform over every data-dependent choice of the changed indices and every corresponding score-certified fit.*

*Proof.* Let $p_1, \dots, p_M$ be the cover from Lemma III.2.4, and set $u_a(j) = p_a(j) + u/d$. Every model law is dominated coordinatewise by at least one $u_a$, and $\sum_j u_a(j) = 1 + u$. If $p_H \le u_a$, then $h(p_H, u_a)^2 \le u$.

Fix a deterministic index set $I$ of size $n = N - k$. For every fitted law satisfying $\ell_{q_I}(H) \ge \ell_{q_I}(G) - e_\delta$, domination implies

$$\prod_{i\in I} \frac{u_a(J_i)}{p_G(J_i)} \ge e^{-ne_\delta}.$$

Under the original clean experiment these $n$ observations are iid. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}\mathbb{P}\left\{\prod_{i\in I} \frac{u_a(J_i)}{p_G(J_i)} \ge e^{-ne_\delta}\right\} &\le e^{ne_\delta/2}\left(\sum_j \sqrt{p_G(j)u_a(j)}\right)^n \\ &\le \exp\left\{\frac{n}{2}(e_\delta + u - h(p_G, u_a)^2)\right\}.\end{aligned}$$

Ratios need be defined only on the support of $p_G$. Take a union over the $M$ envelopes and all $\binom{N}{k}$ index sets. Except on an event of probability $e^{-x}$, simultaneously for all these subsets and all such fits,

$$h(p_G, p_H) \le \sqrt{u} + \sqrt{e_\delta + u + \frac{2\{\log M + \log\binom{N}{k} + x\}}{N-k}}. \tag{III.2.16}$$

This is the step allowing adversarial selection of the retained set. There is no assertion that it remains iid after that selection.

Since $\log\binom{N}{k} \le Nb(\delta)$, $N - k \ge 3N/4$, and $e_\delta \le Cb(\delta) + \zeta$, integrating (III.2.16) and using (III.2.13) proves (III.2.15). The additional terms $u$, $1/N$, and $\zeta \le 1/N$ are absorbed by its sampling term. □

### III.2.3 Proof of the sharp upper bound and matching lower bound

*Upper bound in Theorem III.2.1.* First let $t \le N$. Bounded $t$ is covered by increasing the constant in (III.2.2). For larger $t$ put

$$L_N = \log(eN), \quad L_\delta = \log(e/\delta), \quad L = \min(L_N, L_\delta), \quad a = 1/64,$$

and choose

$$K = \left\lfloor \min\{t/2, \sqrt{atL}\}\right\rfloor, \qquad S = \lfloor\sqrt{t}/16\rfloor.$$

For sufficiently large absolute $t$ these satisfy $1 \le S \le K \le t/2$, $S \asymp \sqrt{t}$, and $e^{4S^2/t} \le C$. Also

$$\frac{1}{K} \le C\left\{\frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{tL_\delta}}\right\}. \tag{III.2.17}$$

Apply (III.2.14), then Lemmas III.2.3 and III.2.5. The high-frequency sampling term is at most

$$\frac{C(eN)^{2a}L_N^{3/4}}{\sqrt{N}\,t^{1/4}} \le \frac{C}{\sqrt{tL_N}}.$$

Indeed, its ratio to the last expression is bounded by $C(eN)^{2a}N^{-1/4}L_N^{5/4}$ because $t \le N$, and this is uniformly bounded as $2a < 1/4$.

For $0 < \delta \le 1/4$ we have $b(\delta) \le C\delta L_\delta$. Consequently

$$\sqrt{\frac{b(\delta)}{t}} \le \frac{C}{\sqrt{tL_\delta}}, \qquad \frac{e^{2aL_\delta}\sqrt{b(\delta)}}{\sqrt{t}} \le \frac{C}{\sqrt{tL_\delta}}. \tag{III.2.18}$$

After multiplying by $\sqrt{tL_\delta}$, the two left sides are bounded respectively by $C\sqrt{\delta}L_\delta$ and $C\delta^{1/2-2a}L_\delta$, uniformly in this interval. At $\delta = 0$ the entropy contributions vanish. Thus (III.2.17)–(III.2.18) prove the upper bound with $\delta$ in place of $\varepsilon$.

If $t > N$, the elementary component coupling gives

$$W_1(H, G) \le \frac{1}{\sqrt{t}} + W_{1,t}(p_H, p_G).$$

Here the normalized count law of each mixture is coupled to its latent parameter at expected cost at most $1/(2\sqrt{t})$. The boundedness of $b(\delta)$ and (III.2.10) now give $\mathbb{E}W_1(H, G) \le C(N^{-1/2} + \delta)$. Finally, $\delta \le \varepsilon$ and all terms in (III.2.1) are nondecreasing in the contamination argument. The loss is at most one, so clipping gives the asserted bound. The proof allowed $\zeta \le 1/N$ throughout, establishing the certificate assertion as well. □

*Matching lower bound.* We recall the existing lower-bound construction to make the matching claim explicit. The clean experiment already gives

$$c \min\left\{1, t^{-1} + [t\log(eN)]^{-1/2} + N^{-1/2}\right\}.$$

The first term follows from mixing laws matching moments through degree $t$. For the second, let $\pi(dx) = dx/(\pi\sqrt{x(1-x)})$ and $G_m^{\pm} = (1 \pm T_m(2x-1)/2)\pi$. Their population distance is comparable to $1/m$. The beta–binomial conditional-expectation operator has squared singular value

$$\lambda_{t,m} = \prod_{j=0}^{m-1} \frac{t-j}{t+1+j} \le e^{-m^2/(2t)}$$

on this Chebyshev mode, and zero when $m > t$. Choosing $m$ a sufficiently large multiple of $\sqrt{t\log(eN)}$, or using exact moment matching if it exceeds $t$, makes the product experiments close and proves the inverse sampling term. Endpoint mixing laws with masses $1/2 \pm c/\sqrt{N}$ prove the parametric term.

The same witnesses, together with an endpoint mass change of size $\eta$, prove the sharp total-variation inverse lower bound

$$\sup_{\|p_F - p_G\|_{\mathrm{TV}} \le \eta} W_1(F, G) \ge c \min\left\{1, t^{-1} + [t\log(e/\eta)]^{-1/2} + \eta\right\}. \tag{III.2.19}$$

Indeed the observed TV of the Chebyshev witness is at most $C\sqrt{\lambda_{t,m}}$, so $m \asymp \sqrt{t\log(e/\eta)}$ suffices. These are the lower witnesses in [20, Parts VII–VIII]; no estimator-specific assumption enters them.

If $k \ge 1$, use (III.2.19) with $\eta = k/(16N)$. Independently maximally couple $N$ observations from the resulting $p_F, p_G$. The expected number of unequal reports is at most $k/16$, and its probability of exceeding $k$ is at most $1/16$. In the $F$ experiment an adversary draws the conditionally coupled $G$ sample, makes all replacements if their number is at most $k$, and otherwise leaves the sample unchanged. Its output law is within TV $1/16$ of the clean $G$ experiment. This adversary always respects the hard budget. Le Cam's two-point inequality proves the contamination lower bound, since $k/N \ge \varepsilon/2$ whenever $k = \lfloor \varepsilon N \rfloor \ge 1$. If $k = 0$, then $\varepsilon < 1/N$ and both contamination terms in (III.2.1) are bounded by clean terms. Combining the lower bounds and taking a clipped maximum, equivalent to the clipped sum, completes the proof. □

### III.2.4 Independent Huber contamination and the meaning of the result

**Corollary III.2.6** (The same MLE adapts to Huber contamination)**.** *Suppose the reports are independent with common law $(1-\varepsilon)p_G+\varepsilon Q$, where $Q$ is any law on the count alphabet and $0\le\varepsilon\le 1/4$. Every ordinary NPMLE, and every fit satisfying* (III.2.3)*, has worst-case expected population loss at most $Cr_{N,t}(\varepsilon)$. The minimax risk is at least $cr_{N,t}(\varepsilon)$.*

*Proof.* Condition on the contamination indicators. Their count $K$ is binomial with parameters $N,\varepsilon$. Fill the corrupt positions with independent unobserved clean reports; conditional on the indicator set, the experiment is a replacement experiment with budget $K$. Theorem III.2.1 applies when $K\le N/4$. Put $\Delta=K/N$, $\Delta'=\min(\Delta,1/4)$ and $g(x)=[t\log(e/x)]^{-1/2}$, with $g(0)=0$. This function is increasing and concave on $[0,1/4]$: its second derivative has the sign of $3/2-\log(e/x)$, which is negative there. Therefore

$$\mathbb{E}g(\Delta')\le g(\mathbb{E}\Delta')\le g(\varepsilon),\qquad \mathbb{E}\Delta=\varepsilon,\qquad \mathbb{P}\{\Delta>1/4\}\le 4\varepsilon.$$

Bound the loss by one on the last event and average the theorem on its complement. This proves the upper bound without a logarithmic loss from random-budget control.

For the lower bound, the clean experiment is a subcase. Huber neighborhoods of two laws intersect if their TV distance is at most $\varepsilon/(1-\varepsilon)$: the vector $(1-\varepsilon)\max(p_F,p_G)$ has mass at most one and can be completed to a common probability law. Apply (III.2.19) and the triangle inequality for population loss. □

The proof does not make arbitrary mixture MLEs robust in arbitrary models. It uses unrestricted mixing, bounded count geometry, component concentration at scale $t^{-1/2}$, and a two-scale inverse bound specific to the binomial operator. The entropy–transport step is substantive: a likelihood-deficit bound alone repairs the mixing law at order $\sqrt{\varepsilon}$, whereas the retained-score bound repairs it at order $\varepsilon$. The remaining binary-entropy cost is carried by the high-frequency inverse term and is absorbed in the sharp contamination modulus.

Thus the advance is an estimator theorem closing the explicitly recorded unmodified-MLE question. It is not a rederivation of the already known robust minimax curve, and it is not claimed as a resolution of a general external robust-mixture conjecture. A bounded check of the primary sources above did not identify this precise strong-replacement theorem; that check is not a proof of historical priority.

## III.3 A sharp two-depth binomial NPMLE theorem from affine posterior repair

The homogeneous binomial-mixture likelihood problem asks whether an unrestricted mixing-measure maximum likelihood estimator attains the optimal population-recovery rate at every replication depth. The preceding integrated manuscript proves that assertion using likelihood transportation and a two-scale inverse inequality. Heterogeneous replication introduces a separate obstruction: likelihood optimality normalizes an aggregate posterior score, but generally does not normalize the score in any one replication stratum.

We resolve a nontrivial heterogeneous case with arbitrary imbalance. There may be any number of depth-one observations, together with $n$ observations of a common, unrestricted depth $T$. Every maximizer of their joint likelihood attains the same worst-case order as the optimal estimator based on the $n$ depth-$T$ observations alone. The new argument identifies an affine posterior tilt and controls its Wasserstein cost exactly by its change in mean.

### III.3.1 Experiment and result

For a mixing law $F \in \mathcal{P}([0,1])$, define

$$\mu_F = \int \theta \, dF(\theta), \qquad p_F(j) = \int B_j^T(\theta) \, dF(\theta), \qquad B_j^T(\theta) = \binom{T}{j} \theta^j (1-\theta)^{T-j}.$$

Observe $m \geq 0$ depth-one variables $Y_1, \ldots, Y_m$ and $n \geq 1$ depth-$T$ counts $X_1, \ldots, X_n$, where $T \geq 1$. Every observation has its own independent latent parameter drawn from the same unknown law $G$. Thus the $Y_i$ are independent Bernoulli variables of parameter $\mu_G$, the $X_i$ are independent with law $p_G$, and the two samples are independent. The joint log likelihood is

$$L(F) = \sum_{i=1}^{n} \log p_F(X_i) + \sum_{i=1}^{m} \{Y_i \log \mu_F + (1 - Y_i) \log(1 - \mu_F)\}. \tag{III.3.1}$$

Let $H$ be any measurable choice of an unrestricted maximizer. Existence follows by maximizing the continuous likelihood on the compact space of mixing laws; the competitor $\delta_{1/2}$ gives positive likelihood, so all fitted probabilities at observed cells are positive.

Write $W_{1,T}$ for Wasserstein distance between count laws with cost $|j - l|/T$, and use $W_1$ for the usual cost on $[0,1]$. Our Hellinger convention is $h(p,v)^2 = \sum_j (\sqrt{p(j)} - \sqrt{v(j)})^2$, including nonnormalized nonnegative $v$. Relative entropy is denoted by $D$; $D_B$ denotes its Bernoulli specialization.

**Theorem III.3.1** (All-depth optimality under arbitrary two-stratum imbalance)**.** *Put*

$$r_{n,T} = \min \left\{ 1, \frac{1}{T} + \frac{1}{\sqrt{T \log(en)}} + \frac{1}{\sqrt{n}} \right\}.$$

*There are absolute constants $0 < c < C < \infty$ such that, for every $m \geq 0$, $n, T \geq 1$,*

$$c r_{n,T} \leq \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G} \mathbb{E}_G W_1(H, G) \leq C r_{n,T}. \tag{III.3.2}$$

*The infimum allows all estimators using both samples. The upper bound holds for every measurable joint NPMLE, with constants independent of the number $m$ of depth-one observations. In particular, it covers $m \gg n$, $T \gg n$, and the transition $T \asymp n/\log(en)$.*

The number of depth-one observations can substantially improve mean estimation. The theorem says that this improvement does not change the worst-case order for recovering the whole mixing law; all lower-bound pairs below have exactly the same mean. The upper bound concerns the actual joint maximizer, rather than an estimator that discards the shallow sample.

### III.3.2 The likelihood advantage of the shallow sample

Let $q$ be the empirical law of the depth-$T$ sample, set $\mu_0 = \mu_G$, and, for $m > 0$, write $z = m^{-1} \sum_i Y_i$. Define

$$A = m D_B(z \| \mu_0),$$

with $A = 0$ if $m = 0$ and all expressions involving $z$ omitted in that case. This is precisely the maximal possible shallow-sample log-likelihood gain over its value at the truth. Joint optimality therefore implies

$$n \sum_j q_j \log p_H(j) \geq n \sum_j q_j \log p_G(j) - A, \qquad D(q \| p_H) \leq D(q \| p_G) + A/n. \tag{III.3.3}$$

The elementary inequality $D \le \chi^2$ and the multinomial variance identity give

$$\mathbb{E}_G A \le 1, \qquad \mathbb{E}_G D(q\|p_G) \le T/n, \qquad \mathbb{E}_G D(q\|p_H) \le (T+1)/n. \tag{III.3.4}$$

For example, if $0 < \mu_0 < 1$ and $m > 0$, $A \le m(z-\mu_0)^2/[\mu_0(1-\mu_0)]$, whose expectation is one. True zero-probability cells are omitted from the multinomial identity. At $\mu_0 \in \{0,1\}$ the sample is deterministic and the fitted law is the same endpoint point mass, so the risk is zero. Hence this boundary case also causes no exception.

### III.3.3 The affine posterior tilt

Define the depth-$T$ score and its posterior update by

$$S_T(\theta) = \sum_{j:q_j>0} \frac{q_j B_j^T(\theta)}{p_H(j)}, \qquad dH^+(\theta) = S_T(\theta)\, dH(\theta). \tag{III.3.5}$$

The measure $H^+$ is always a probability measure, since $\int S_T\, dH = 1$. It need not equal $H$.

**Lemma III.3.2** (Exact cost of repairing the deep stratum)**.** *Let $\mu = \mu_H$ and $\mu^+ = \mu_{H^+}$. If $0 < \mu < 1$ and $m > 0$, then*

$$dH^+(\theta) = \{1 - b(\theta-\mu)\}\, dH(\theta), \qquad b = \frac{m}{n}\frac{z-\mu}{\mu(1-\mu)}. \tag{III.3.6}$$

*Consequently*

$$W_1(H, H^+) = |\mu - \mu^+| \le |\mu^+ - \mu_0| + w|z-\mu_0|, \qquad w = \min\{1, m/n\}. \tag{III.3.7}$$

*When $m = 0$, take $w = 0$ and omit $z$; then $H^+ = H$. If the fitted mean is an endpoint, $H^+ = H$ also holds directly.*

*Proof.* The depth-one score is

$$S_1(\theta) = \frac{z\theta}{\mu} + \frac{(1-z)(1-\theta)}{1-\mu} = 1 + \frac{z-\mu}{\mu(1-\mu)}(\theta-\mu).$$

The likelihood directional derivative in the direction $\delta_\theta - H$ gives

$$nS_T(\theta) + mS_1(\theta) \le n + m.$$

Its integral against $H$ is $n+m$, so equality holds $H$-almost surely. Solving for $S_T$ on this full-measure set proves (III.3.6). No individual-stratum score normalization is assumed.

For every $x \in [0,1]$,

$$F_{H^+}(x) - F_H(x) = -b \int_{[0,x]} (\theta-\mu)\, dH(\theta).$$

The integral on the right is nonpositive: it decreases up to $\mu$ and then increases to its full value zero. Hence the distribution-function difference has a constant sign. The CDF formula for $W_1$ proves $W_1(H, H^+) = |\mu - \mu^+|$.

Integrating (III.3.6) against $\theta$ gives

$$\mu^+ = \mu - a(z-\mu), \qquad a = \frac{m}{n}\frac{\mathrm{Var}_H(\theta)}{\mu(1-\mu)} \in [0, m/n].$$

Thus $\mu = (\mu^+ + az)/(1+a)$, and

$$|\mu - \mu^+| = \frac{a}{1+a}|z-\mu^+| \le w|z-\mu_0| + |\mu^+ - \mu_0|.$$

For $m = 0$, the joint score condition is $S_T = 1$ $H$-almost surely. If $\mu$ is an endpoint, $H$ is the corresponding point mass; finite likelihood forces all depth-$T$ observations to be that endpoint count, and $S_T = 1$ on the support of $H$. These cases establish the remaining assertions without division by zero. □

### III.3.4 Entropy–transport after the repair

**Lemma III.3.3** (Posterior transportation for an arbitrary fitted law)**.** *For the update in* (III.3.5)*, whether or not $H$ maximizes a likelihood,*

$$W_{1,T}(q, p_{H^+}) \le \sqrt{\frac{D(q\|p_H)}{2T}}. \tag{III.3.8}$$

*Proof.* For $S_T(\theta) > 0$, set

$$R_\theta(j) = \frac{q_j B_j^T(\theta)}{p_H(j) S_T(\theta)}.$$

This is a probability vector, and $\int R_\theta \, dH^+(\theta) = q$. Direct expansion gives the entropy identity

$$\int D(R_\theta \| B_\theta^T) \, dH^+(\theta) = D(q\|p_H) - D(H^+\|H) \le D(q\|p_H). \tag{III.3.9}$$

Terms with $S_T = 0$ have zero $H^+$ mass.

For completeness, every binomial kernel satisfies

$$W_{1,T}(r, B_\theta^T)^2 \le \frac{D(r\|B_\theta^T)}{2T}.$$

Indeed a 1-Lipschitz function of the normalized sum of $T$ independent Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8T)$ by the bounded-differences martingale argument. The entropy variational inequality bounds its expectation difference by $D/\lambda + \lambda/(8T)$; optimizing $\lambda > 0$ and using Wasserstein duality proves the displayed inequality. Mix the corresponding couplings over $H^+$ and use Cauchy–Schwarz and (III.3.9) to obtain (III.3.8). □

**Proposition III.3.4** (Predictive transport uniform in the shallow sample size)**.** *Every joint maximizer satisfies*

$$\mathbb{E}_G W_{1,T}(p_H, p_G) \le C n^{-1/2}, \tag{III.3.10}$$

*with an absolute constant independent of $m, T, G$.*

*Proof.* Coupling binomial sums with the same uniforms at two parameters shows

$$W_{1,T}(p_F, p_J) \le W_1(F, J)$$

for any two mixing laws $F, J$. Also the normalized predictive mean is the mixing mean. Combining Lemmas III.3.2 and III.3.3 therefore gives the deterministic bound

$$\begin{aligned} W_{1,T}(p_H, p_G) &\le 2W_{1,T}(p_{H^+}, p_G) + w|z - \mu_0| \\ &\le 2\sqrt{\frac{D(q\|p_H)}{2T}} + 2W_{1,T}(q, p_G) + w|z - \mu_0|. \end{aligned} \tag{III.3.11}$$

At a fitted endpoint the same bound holds since $H^+ = H$. The empirical CDF identity gives $\mathbb{E}W_{1,T}(q, p_G) \le 1/(2\sqrt{n})$. If $m > 0$,

$$w\mathbb{E}|z - \mu_0| \le \frac{\min(1, m/n)}{2\sqrt{m}} \le \frac{1}{2\sqrt{n}}.$$

Finally (III.3.4) and Jensen's inequality bound the expected square-root term by $1/\sqrt{n}$. This proves (III.3.10). □

This is the step absent from the homogeneous argument. Directly asserting $S_T = 1$ would be wrong. Bounding an arbitrary posterior correction only by total variation would also lose the uniformity in $m/n$ supplied by the affine structure.

### III.3.5 Predictive Hellinger control with a conditional likelihood deficit

We record the finite-cover and likelihood-envelope ingredients to make their application to the joint estimator explicit. These two ingredients, and the polynomial inverse used below, are inherited from Section 6.3 of the preceding integrated manuscript; the conditional application here is new.

**Lemma III.3.5** (A cover of the binomial moment body)**.** *For $0 < \beta < 1$, the set of laws $\{p_F : F \in \mathcal{P}([0,1])\}$ has an $\ell_1$ cover of radius $\beta$, with centers in that set and cardinality $M$ satisfying*

$$\log M \le d \log \frac{C(d+1)}{\beta}, \qquad d = \min\left\{T, \left\lceil \sqrt{2T \log(16T/\beta)} \right\rceil \right\}. \tag{III.3.12}$$

*Proof.* This is Lemma III.A.1, whose Fourier contour and moment-cell proof is included in the supporting appendix. □

**Lemma III.3.6** (Conditional envelopes)**.** *For $T \le n$, every joint maximizer satisfies*

$$\mathbb{E}_G h(p_H, p_G)^2 \le \frac{C\{\sqrt{T}\log(en)^{3/2} + 1\}}{n}, \qquad \mathbb{E}_G h(p_H, p_G) \le \frac{CT^{1/4}\log(en)^{3/4}}{\sqrt{n}}. \tag{III.3.13}$$

*Proof.* Condition on the shallow sample. Then $A$ in (III.3.3) is fixed, while the deep observations retain their independent product law. Let $\eta = (en)^{-2}$ and take the cover in Lemma III.3.5 with $\beta = \eta/(T+1)$. For each center $p_a$, put $u_a(j) = p_a(j) + \eta/(T+1)$. Every model law is bounded coordinatewise by some $u_a$, each $u_a$ has mass $1+\eta$, and

$$\widetilde{p} \le u_a \quad \Longrightarrow \quad h(u_a, \widetilde{p})^2 \le \eta.$$

For a fixed envelope, Markov's inequality at exponent $1/2$ gives

$$\begin{aligned}\Pr\left\{ \prod_{i=1}^n \frac{u_a(X_i)}{p_G(X_i)} \ge e^{-A} \,\middle|\, Y_1, \dots, Y_m \right\} &\le e^{A/2} \left( \sum_j \sqrt{p_G(j) u_a(j)} \right)^n \\ &= e^{A/2} \left(1 + \eta/2 - h(p_G, u_a)^2/2\right)^n.\end{aligned}$$

Ratios need be defined only on the true support, where the sample lies almost surely. The triangle inequality and a union over $M$ envelopes now show that, conditionally with probability at least $1 - e^{-x}$, simultaneously for every $\widetilde{p}$ obeying the likelihood comparison in (III.3.3),

$$h(p_G, \widetilde{p}) \le \sqrt{\eta} + \sqrt{A/n + \eta + 2(\log M + x)/n}.$$

Thus the conclusion applies to the joint maximizer even though it uses both samples. Integrating this conditional tail and then averaging over the shallow sample gives

$$\mathbb{E} h(p_G, p_H)^2 \le C\{\eta + (\mathbb{E}A + \log M + 1)/n\}.$$

For $T \le n$, (III.3.12) gives $\log M \le C\sqrt{T}\log(en)^{3/2}$. Use $\mathbb{E}A \le 1$ and Jensen's inequality to finish the proof. □

### III.3.6 The inherited inverse inequality and the complete upper bound

To state the required analytic input explicitly, for every 1-Lipschitz $f : [0,1] \to \mathbb{R}$ and integer $k \geq 1$ there is a polynomial $P_k$ of degree at most $k$ such that

$$\|f - P_k\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq 2. \tag{III.3.14}$$

Also, if $P = \sum_{j=0}^{T} b_j B_j^T$ has degree $k \leq T/2$, then

$$\max_j |b_j| \leq Ce^{2k^2/T}\|P\|_\infty, \qquad \max_{j<T} T|b_{j+1} - b_j| \leq Ce^{4k^2/T} \quad \text{when } P = P_k. \tag{III.3.15}$$

The proofs of (III.3.14) and (III.3.15) are given in Lemma III.A.3 of the supporting appendix. They are inherited ingredients. The following form of Proposition III.A.4 states the exact inverse used here.

**Lemma III.3.7** (Two-scale inverse inequality)**.** *For integers $1 \leq s \leq k \leq T/2$ and any two mixing laws $F, J$,*

$$W_1(F, J) \leq \frac{C}{k} + Ce^{4s^2/T} W_{1,T}(p_F, p_J) + \frac{Ce^{2k^2/T}}{s} h(p_F, p_J). \tag{III.3.16}$$

*Proof.* For a 1-Lipschitz test $f$, approximate it by $P_k$ and $P_s$ as in (III.3.14). Replacing $f$ by $P_k$ costs $C/k$. The Bernstein coefficients of $P_s$ have normalized-grid Lipschitz constant at most $Ce^{4s^2/T}$, so their integral against $p_F - p_J$ is bounded by the middle term of (III.3.16). Since $\|P_k - P_s\|_\infty \leq C/s$, the remaining coefficient vector has supremum norm at most $Ce^{2k^2/T}/s$. Use $\|p_F - p_J\|_1 \leq 2h(p_F, p_J)$ to bound its integral. Take the Wasserstein dual supremum over $f$. □

*Upper bound in Theorem III.3.1.* First suppose $T \leq n$. Bounded $T$ or $n$ are absorbed by the loss bound one. Write $L_n = \log(en)$, fix $a = 1/64$, and take

$$k = \left\lfloor \min\{T/2, \sqrt{aTL_n}\} \right\rfloor, \qquad s = \lfloor \sqrt{T}/4 \rfloor.$$

In the remaining range these satisfy $1 \leq s \leq k \leq T/2$, $s \asymp \sqrt{T}$, $e^{4s^2/T} \leq C$, and $k^{-1} \leq C\{T^{-1} + (TL_n)^{-1/2}\}$. Proposition III.3.4, Lemma III.3.6, and (III.3.16) give

$$\mathbb{E}W_1(H, G) \leq \frac{C}{k} + \frac{C}{\sqrt{n}} + \frac{Ce^{2k^2/T} L_n^{3/4}}{\sqrt{n}\, T^{1/4}}.$$

Multiplying the last term by $\sqrt{TL_n}$ bounds it by

$$C(en)^{2a} n^{-1/2} T^{1/4} L_n^{5/4} \leq C(en)^{2a} n^{-1/4} L_n^{5/4} \leq C,$$

because $2a < 1/4$. This proves the required rate for $T \leq n$.

For $T \geq n$, coupling a mixing parameter with its own normalized binomial count gives $W_1(F, p_F) \leq 1/(2\sqrt{T})$, after viewing $p_F$ as a law on the normalized grid. Therefore

$$W_1(H, G) \leq T^{-1/2} + W_{1,T}(p_H, p_G).$$

Proposition III.3.4 yields the root-$n$ bound. Clipping at the diameter one completes the upper bound in every range. □

### III.3.7 Matching lower bounds with identical shallow-data laws

*Lower bound in Theorem III.3.1.* Every pair below has common mean $1/2$. Thus the entire depth-one sample has exactly the same product Bernoulli law at the two alternatives, regardless of $m$.

For the root-$n$ term, consider

$$G_u = \frac{u}{2}\delta_0 + \frac{u}{2}\delta_1 + (1-u)\delta_{1/2}, \qquad u_0 = \frac{1}{2}, \quad u_1 = \frac{1}{2} + h, \quad h = \frac{1}{16\sqrt{n}}.$$

Their Wasserstein distance is $h/2$. An oracle revealing the $n$ deep latent parameters has relative entropy $nD_B(u_0\|u_1) \le 5nh^2 = 5/256$ between these alternatives. The observed deep counts are generated by a parameter-independent kernel, so data processing, Pinsker's inequality and the two-point metric-risk bound give a universal lower bound $c/\sqrt{n}$.

For the other terms, let

$$\pi(d\theta) = \frac{d\theta}{\pi\sqrt{\theta(1-\theta)}}, \qquad g_k(\theta) = \mathcal{T}_k(2\theta - 1), \qquad G_k^{\pm} = (1 \pm g_k/2)\pi, \quad k \ge 2.$$

Orthogonality makes these probability measures of common mean $1/2$. Direct integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(ku)| \sin u \, du \asymp k^{-1}. \tag{III.3.17}$$

The lower constant is uniform: restrict the integral to $[\pi/4, 3\pi/4]$ and use the period of $|\sin(ku)|$. If $k = T+1$, orthogonality to all degree-$T$ polynomials makes the two deep predictive laws identical. This gives the lower bound $c/T$.

For $k \le T$, draw $\theta \sim \pi$ and $J \mid \theta \sim \mathrm{Binomial}(T, \theta)$, and define $Qf(j) = \mathbb{E}[f(\theta) \mid J = j]$. The posterior is $\mathrm{Beta}(j + 1/2, T - j + 1/2)$. The self-adjoint operator $Q^*Q$ preserves degree-$k$ polynomial spaces and has leading coefficient

$$\lambda_{T,k} = \prod_{i=0}^{k-1} \frac{T-i}{T+1+i}.$$

Indeed the degree-$k$ coefficient in the posterior moment comes from $(J+1/2)\cdots(J+k-1/2)/[(T+1)\cdots(T+k)]$; taking its expectation under $\mathrm{Binomial}(T,\theta)$ contributes the falling factorial $(T)_k$. Self-adjointness and orthogonality therefore imply $Q^*Qg_k = \lambda_{T,k} g_k$. Since $\int g_k^2 \, d\pi = 1/2$, $\mathbb{E}(Qg_k(J))^2 = \lambda_{T,k}/2$.

The predictive laws are $p_\pi(1 \pm Qg_k/2)$, with $|Qg_k| \le 1$. Consequently

$$\chi^2(p_{G_k^+}\|p_{G_k^-}) \le 2\mathbb{E}(Qg_k(J))^2 = \lambda_{T,k} \le e^{-k^2/(2T)}. \tag{III.3.18}$$

For the last inequality, apply $\log(1-x) \le -x$ to each factor and use $T + 1 + i \le 2T$.

Take $k = \lceil\sqrt{8T\log(8n)}\rceil$ whenever this is at most $T$. The deep product chi-squared divergence is at most $\exp\{n(8n)^{-4}\} - 1 < 10^{-3}$, by tensorization. Two-point testing and (III.3.17) give $c/\sqrt{T\log(en)}$. If this choice exceeds $T$, then $T < 8\log(8n) \le C\log(en)$, so $1/T$ already dominates the desired logarithmic term up to an absolute constant. All degrees used are at least two, preserving equality of means. Finally the maximum of the three lower bounds is comparable to their sum clipped at one, proving the result. □

**Corollary III.3.8** (Equal total budgets can have different recovery orders)**.** *For $N \ge 2$, consider two profiles with exactly $N$ independent individuals and exactly $N^2$ Bernoulli trials. If all depths equal $N$, the minimax population risk is $\Theta(N^{-1/2})$. If $N-1$ depths equal one and the last equals $N^2 - N + 1$, the minimax risk is $\Theta(1)$. Every joint NPMLE has the corresponding upper order.*

*Proof.* Apply Theorem III.3.1 with $(m, n, T) = (0, N, N)$ and $(N-1, 1, N^2 - N + 1)$ respectively. □

### III.3.8 Scope, provenance, and the unresolved general profile

The homogeneous all-depth theorem, finite moment-body covers, posterior entropy decomposition, and two-scale polynomial inverse are existing ingredients in the supplied integrated manuscript. We have reproduced the needed statements; proofs are supplied here and in the supporting appendix to specify all hypotheses. The new ingredient is the affine score identity (III.3.6), which repairs the deep stratum and restores the predictive root-$n$ transportation bound uniformly over all shallow sample sizes. Conditional envelopes then preserve the full replication transition.

Vinayak, Kong, Valiant and Kakade study homogeneous mixture likelihood [70]; the Chebyshev-moment framework of Musco, Musco, Rosenblatt and Singh is also homogeneous in the relevant binomial application [16]. Recent heterogeneous-trial papers address different losses and parameter spaces. Lee, Baćak and Kennedy analyze smooth-density estimators [21]. Lee and Kennedy's October 2026 manuscript treats pointwise estimation of a Hölder-smooth mixing density, with profile-dependent effective sample sizes and a regularized series estimator [22]. The present theorem concerns arbitrary mixing laws, Wasserstein loss, and every unrestricted joint likelihood maximizer. It is not asserted to resolve a question posed in those smooth-density papers, nor is this comparison a comprehensive historical-priority assessment.

An arbitrary replication profile remains unresolved here. At shallow depth $d \geq 2$, the corresponding score is generally a polynomial of degree $d$. The posterior correction can have several sign changes, and its Wasserstein cost is no longer determined by its mean shift. Neither aggregate posterior normalization nor the present affine argument supplies the missing control for that general case.

## III.4 Likelihood contact and empirical-posterior transportation

An empirical measure on Euclidean space is singular with respect to a Gaussian mixture density. Consequently its relative entropy to that density is infinite, and a finite-space empirical-entropy argument cannot be transferred directly. We use a different consequence of mixture likelihood optimality: translating the parameter at a likelihood-score contact point controls the entire moment generating function of an *empirical posterior*. This gives a sharp entropy bound for one EM update after changing sample weights, exact Gaussian and Poisson transport bounds, and minimax bounds in a specified high-precision regime.

The score contact condition is classical; see [47, Proposition 1.1] and its attribution to Lindsay. Rigollet–Weed's identification of maximum-likelihood deconvolution with entropic transport [23, Theorem 4.2] is closely related prior work. Posterior reweighting also underlies the information inequalities in [24, Theorems 3.1 and 3.3]. The results below concern finite empirical-posterior kernels and sample-weight entropy. They do not assert a new proof of those variational or functional-inequality theorems, nor historical priority for every consequence of likelihood stationarity.

### III.4.1 Moment domination at a likelihood contact point

Let $x_1, \dots, x_N$ be fixed observations and let $q_i > 0$ satisfy $\sum_i q_i = 1$. Repeated observation values are allowed: $i$ is an observation label. Consider a regular natural exponential family

$$k_\eta(x) = h(x) \exp\{\langle \eta, T(x) \rangle - A(\eta)\}, \qquad \eta \in U \subset \mathbb{R}^d,$$

where $U$ is open and convex, $A$ is twice continuously differentiable, and $0 < h(x_i) < \infty$, $T(x_i) \in \mathbb{R}^d$ for every observed label. For a mixing law $H$ on $U$, put

$$f_H(x_i) = \int k_\eta(x_i)\, dH(\eta), \qquad \ell_q(H) = \sum_i q_i \log f_H(x_i).$$

The following theorem assumes that a maximizer exists and that its fitted values at the observations are finite and positive. These conditions will be verified in the applications.

**Theorem III.4.1** (Empirical-posterior moment domination)**.** *Let $H$ be any maximizer of $\ell_q$. For $H$-almost every $\eta$,*

$$R_\eta(i) = \frac{q_i k_\eta(x_i)}{f_H(x_i)} \tag{III.4.1}$$

*is a probability vector. The joint law $\Gamma(d\eta, i) = H(d\eta)R_\eta(i)$ has marginals $H$ and $q$. For every $u$ such that $\eta + u \in U$,*

$$\mathbb{E}_{R_\eta} e^{\langle u, T(x_i)\rangle} \le e^{A(\eta+u)-A(\eta)}. \tag{III.4.2}$$

*In particular,*

$$\mathbb{E}_{R_\eta} T(x_i) = \nabla A(\eta), \qquad \mathrm{Cov}_{R_\eta}(T(x_i)) \preceq \nabla^2 A(\eta). \tag{III.4.3}$$

*Thus the law $(\nabla A)_{\#}H$ is dominated in convex order by $\sum_i q_i \delta_{T(x_i)}$.*

*Proof.* Taking the directional derivative toward a point mass gives

$$S(\zeta) := \sum_i \frac{q_i k_\zeta(x_i)}{f_H(x_i)} \le 1 \quad (\zeta \in U).$$

Since $\int S\, dH = 1$, equality holds $H$-almost surely. This proves normalization in (III.4.1); summing over the parameter gives the label marginal $q_i$. At a point with $S(\eta) = 1$, direct substitution shows

$$S(\eta + u) = e^{A(\eta)-A(\eta+u)} \mathbb{E}_{R_\eta} e^{\langle u, T(x_i)\rangle}.$$

This proves (III.4.2). The difference between $A(\eta+u) - A(\eta)$ and the logarithm of the displayed moment generating function has a minimum of zero at $u = 0$. Its gradient vanishes and its Hessian is positive semidefinite, giving (III.4.3). Finally Jensen's inequality, conditional on $\eta$, gives

$$\int \varphi(\nabla A(\eta))\, dH(\eta) \le \sum_i q_i \varphi(T(x_i))$$

for every convex function $\varphi$ for which the expression is defined. □

The theorem concerns an unrestricted natural-parameter domain. If the optimization is restricted to a closed set, only increments staying inside that set are available. Interior support points retain (III.4.3); boundary support points need a separate argument.

### III.4.2 A sharp entropy bound for reweighted EM updates

For a contact point $\eta$, define

$$\psi_\eta(u) = A(\eta + u) - A(\eta) - \langle u, \nabla A(\eta)\rangle, \qquad \eta + u \in U,$$

and let

$$\psi_\eta^*(z) = \sup_{u:\eta+u\in U} \{\langle u, z\rangle - \psi_\eta(u)\}.$$

Now change the observation weights from $q$ to an arbitrary probability vector $q'$. Keep the old posterior responsibilities, and set

$$S'(\eta) = \sum_i \frac{q_i' k_\eta(x_i)}{f_H(x_i)}, \qquad dH'(\eta) = S'(\eta)\, dH(\eta), \tag{III.4.4}$$

$$R_\eta'(i) = \frac{q_i' k_\eta(x_i)}{f_H(x_i) S'(\eta)}, \qquad m_\eta = \sum_i R_\eta'(i) T(x_i).$$

The last line is needed only where $S' > 0$.

**Theorem III.4.2** (Entropy budget for the location update)**.** *Under the assumptions of Theorem III.4.1, $H'$ is a probability measure and*

$$\int \psi_\eta^*(m_\eta - \nabla A(\eta))\, dH'(\eta) \le D(q'\|q) - D(H'\|H). \tag{III.4.5}$$

*All relative entropies on the right are finite. In particular, for a Gaussian location family with common positive definite covariance $\Sigma$, using the mean $\theta$ as the parameter,*

$$W_{2,\Sigma^{-1}}(H', B)^2 \le \int \|m_\theta - \theta\|_{\Sigma^{-1}}^2\, dH'(\theta) \le 2\{D(q'\|q) - D(H'\|H)\}, \qquad B = (m_\cdot)_\# H'. \tag{III.4.6}$$

*Here $B$ is one EM update of both weights and component locations using the old responsibilities. The bound does not assert stability of an arbitrary fully refitted NPMLE. The factor 2 is best possible. The notation $W_{2,\Sigma^{-1}}$ denotes quadratic Wasserstein distance with cost $(x-y)^\mathsf{T}\Sigma^{-1}(x-y)$.*

*Proof.* The joint measure $\Gamma'(d\eta, i) = H'(d\eta)R'_\eta(i)$ is obtained from $\Gamma$ by multiplying its density on label $i$ by $q'_i/q_i$. Consequently its label marginal is $q'$, its parameter marginal is $H'$, and the relative-entropy chain rule gives

$$D(q'\|q) = D(\Gamma'\|\Gamma) = D(H'\|H) + \int D(R'_\eta\|R_\eta)\, dH'(\eta). \tag{III.4.7}$$

Finiteness follows, for example, from $S' \le \max_i(q'_i/q_i) < \infty$ and the finite label space. The entropy variational inequality gives, for every admissible $u$,

$$\langle u, m_\eta - \nabla A(\eta)\rangle \le D(R'_\eta\|R_\eta) + \log \mathbb{E}_{R_\eta} e^{\langle u, T(x_i) - \nabla A(\eta)\rangle} \le D(R'_\eta\|R_\eta) + \psi_\eta(u).$$

Take the supremum over $u$ and integrate, then use (III.4.7). For the Gaussian family, $\psi_\theta^*(z) = \frac{1}{2} z^\mathsf{T}\Sigma^{-1}z$. The coupling $\theta \mapsto m_\theta$ proves the first inequality in (III.4.6).

For sharpness it suffices to take one dimension, observations $-\sigma, \sigma$, and $q = (1/2, 1/2)$. The unique NPMLE is $H = \delta_0$: its score is $e^{-\theta^2/(2\sigma^2)}\cosh(\theta/\sigma) \le 1$, with equality only at zero. For $q' = (1/2 + \varepsilon, 1/2 - \varepsilon)$, $H' = \delta_0$ and the updated mean has magnitude $2\sigma\varepsilon$. The squared normalized transport is $4\varepsilon^2$, whereas $D(q'\|q) = 2\varepsilon^2 + O(\varepsilon^4)$. No factor smaller than 2 can therefore hold uniformly. □

For Poisson observations of mean $\lambda = t\theta$, the conjugate in (III.4.5) is the Poisson mean divergence

$$\psi_\lambda^*(m - \lambda) = m\log(m/\lambda) - m + \lambda. \tag{III.4.8}$$

Thus the same theorem controls the component-mean update in a nonquadratic cost adapted to counts. The multiplicative constant 1 in (III.4.5) is sharp even in this family. Indeed, an empirical Bernoulli law with mass $p$ at one has Poisson-mixture NPMLE $\delta_p$ in the mean parameter: its score is $e^{p-\lambda}(1 - p + \lambda) \le 1$. Reweighting $p$ to $p + h$ gives

$$\frac{(p+h)\log((p+h)/p) - h}{D(\mathrm{Bernoulli}(p+h)\,\|\,\mathrm{Bernoulli}(p))} \longrightarrow 1 - p \quad (h \longrightarrow 0).$$

Letting $p$ tend to zero proves sharpness. At $\lambda = 0$ the old posterior is concentrated on zero counts, and the corresponding term is zero whenever its reweighted mass is nonzero.

### III.4.3 Exact Gaussian and Poisson coupling bounds

**Theorem III.4.3** (A sharp Gaussian martingale coupling)**.** *For arbitrary $x_i \in \mathbb{R}^d$ and positive $q_i$, let $H$ be any NPMLE for Gaussian locations with common covariance $\Sigma \succ 0$, optimized over all probability measures on $\mathbb{R}^d$. There is a coupling $(\Theta, X)$ of $H$ and $Q = \sum_i q_i\delta_{x_i}$ such that*

$$\mathbb{E}[X \mid \Theta] = \Theta, \qquad \mathbb{E}[e^{\langle v, X - \Theta\rangle} \mid \Theta] \le e^{v^\mathsf{T}\Sigma v/2} \quad (v \in \mathbb{R}^d). \tag{III.4.9}$$

*Consequently*

$$W_2(H,Q)^2 \le \operatorname{tr}\Sigma, \qquad \operatorname{Cov}(Q)-\Sigma \preceq \operatorname{Cov}(H) \preceq \operatorname{Cov}(Q). \tag{III.4.10}$$

*The constant* 1 *multiplying* $\operatorname{tr}\Sigma$ *is sharp.*

*Proof.* Projection onto the convex hull of the observations in the $\Sigma^{-1}$ metric decreases every squared distance to an observation, strictly for a point outside the hull. Therefore the maximization may be restricted to that compact hull, where a maximizer exists; every maximizer is supported there. All fitted Gaussian likelihoods are positive. Apply Theorem III.4.1 in natural coordinates $\eta = \Sigma^{-1}\theta$ to obtain (III.4.9) and $\operatorname{Cov}(X \mid \Theta) \preceq \Sigma$. The coupling cost is at most $\operatorname{tr}\Sigma$. The total-covariance identity gives the sandwich in (III.4.10).

For equality, take the $2^d$ points $\Sigma^{1/2}s$, uniformly weighted, where $s \in \{-1,1\}^d$. The score of $\delta_0$ is

$$e^{-\|z\|^2/2}\prod_{j=1}^{d}\cosh z_j \le 1, \qquad z=\Sigma^{-1/2}\theta,$$

with equality only at $z=0$. The contact criterion implies that $\delta_0$ is the unique NPMLE. Its squared transport distance to the empirical law is exactly $\operatorname{tr}\Sigma$. □

**Theorem III.4.4** (A sharp Poisson martingale coupling)**.** *Fix $t>0$ and nonnegative integer observations $x_i$. Let $H$ be any NPMLE for* Poisson$(t\theta)$ *mixtures optimized over $\theta \in [0,\infty)$, with sample weights $q_i>0$. Put $\bar{x}=\sum_i q_i x_i$ and $Q_t=\sum_i q_i\delta_{x_i/t}$. There is a coupling of $\Theta \sim H$ and the empirical count $X$ such that*

$$\mathbb{E}[X \mid \Theta] = t\Theta, \qquad \mathbb{E}[e^{s(X-t\Theta)} \mid \Theta] \le \exp\{t\Theta(e^s-1-s)\} \quad (s\in\mathbb{R}). \tag{III.4.11}$$

*In particular,*

$$W_2(H,Q_t)^2 \le \frac{\bar{x}}{t^2}. \tag{III.4.12}$$

*The constant* 1 *is best possible uniformly over finite datasets.*

*Proof.* If $x_{\max}>0$, replacing any parameter greater than $x_{\max}/t$ by $x_{\max}/t$ strictly increases every observed kernel likelihood. Thus maximizers exist on a compact interval and are supported there. If all counts are zero, the unique maximizer is $\delta_0$. For positive support parameters, Theorem III.4.1 with natural parameter $\eta=\log(t\theta)$ proves (III.4.11), including $\operatorname{Var}(X \mid \Theta) \le t\Theta$. At $\theta=0$ the empirical posterior can put mass only on zero counts, so the same formulas hold with both sides degenerate. Integrating the conditional mean gives $t\mathbb{E}_H\Theta=\bar{x}$. The squared cost of the coupling $(\Theta, X/t)$ is therefore at most $\mathbb{E}_H\Theta/t=\bar{x}/t^2$.

For an empirical Bernoulli law with mass $p\in(0,1)$ at one, the unique maximizing intensity is $p$, by the score calculation following (III.4.8). Hence $H=\delta_{p/t}$ and the actual squared transport distance is $p(1-p)/t^2$. Taking rational $p$ tending to zero realizes finite equally weighted datasets and proves sharpness. □

### III.4.4 Boundary repair and the high-precision statistical regime

The preceding martingale conclusions cannot simply be asserted for a constrained parameter space. For example, a Poisson fit over $[0,M]$ to a single count $x>tM$ is $\delta_M$, and $(x/t-M)^2$ can exceed $x/t^2$ arbitrarily. The correct comparison for a bounded fit uses clipped observations.

**Theorem III.4.5** (Clipped-data transport for bounded NPMLEs)**.** *Let* $C_{a,b}(y) = \min(b,\max(a,y))$.

*(i) For every NPMLE $H$ of* $\mathrm{Poisson}(t\theta)$ *mixtures over* $[0, M]$*, where* $t, M > 0$*,*

$$W_2\left(H, \sum_i q_i \delta_{C_{0,M}(x_i/t)}\right)^2 \le \frac{2M}{t}. \tag{III.4.13}$$

*(ii) For every NPMLE $H$ of one-dimensional* $\mathcal{N}(\theta, \sigma^2)$ *mixtures over* $[a, b]$*, where* $a < b$*,*

$$W_2\left(H, \sum_i q_i \delta_{C_{a,b}(x_i)}\right)^2 \le 2\sigma^2. \tag{III.4.14}$$

*Proof.* Compactness gives existence, and the score contact condition again normalizes the empirical posterior kernels. At an interior Poisson parameter, the conditional mean and variance identities give

$$\mathbb{E}[(C_{0,M}(X/t) - \theta)^2 \mid \theta] \le \mathbb{E}[(X/t - \theta)^2 \mid \theta] \le M/t.$$

At zero the conditional count is zero. At $\theta = M$, natural increments $s \le 0$ are admissible, giving

$$\mathbb{E}[e^{s(X-tM)} \mid \theta = M] \le e^{tM(e^s - 1 - s)} \le e^{tMs^2/2}.$$

Chernoff's bound with $s = -r/(tM)$ yields $\mathbb{P}(tM - X > r \mid \theta = M) \le e^{-r^2/(2tM)}$. Integrating this bound gives

$$\mathbb{E}[(tM - X)_+^2 \mid \theta = M] = \int_0^\infty 2r\mathbb{P}(tM - X > r \mid \theta = M)\, dr \le 2tM.$$

This is exactly the required conditional cost after clipping at $M$. Integration over $H$ proves (III.4.13).

For Gaussian locations, interior parameters have conditional variance at most $\sigma^2$. At $\theta = a$, a shift $v \in [0, b - a]$ gives

$$\mathbb{E}[e^{v(X-a)/\sigma^2} \mid \theta = a] \le e^{v^2/(2\sigma^2)}.$$

For a tail level $0 < r < b - a$ take the admissible shift $v = r$. Then

$$\mathbb{P}(C_{a,b}(X) - a > r \mid \theta = a) \le e^{-r^2/(2\sigma^2)}.$$

Integration up to $b - a$ bounds the conditional squared cost by $2\sigma^2$. The endpoint $b$ is identical with negative shifts. Integrating over $H$ proves (III.4.14). □

**Corollary III.4.6** (Minimax optimality at high exposure or small noise)**.** *Suppose* $\Theta_1, \dots, \Theta_N$ *are independent with common law* $G$*.*

*(i) If $G$ is supported on* $[0, M]$ *and* $X_i \mid \Theta_i \sim \mathrm{Poisson}(t\Theta_i)$ *independently, every measurable NPMLE $H$ over* $[0, M]$ *satisfies*

$$\sup_G \mathbb{E}_G W_1(H, G) \le (1 + \sqrt{2})\sqrt{M/t} + \frac{M}{2\sqrt{N}}. \tag{III.4.15}$$

*The unrestricted NPMLE over* $[0, \infty)$ *satisfies the same bound with* $1 + \sqrt{2}$ *replaced by* $2$*. There are absolute constants* $c, C > 0$ *such that, whenever* $tM \ge N$*,*

$$c\frac{M}{\sqrt{N}} \le \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,M])} \mathbb{E}_G W_1(\widetilde{G}, G) \le \sup_G \mathbb{E}_G W_1(H, G) \le C\frac{M}{\sqrt{N}}.$$

*(ii) If $G$ is supported on* $[a, b]$ *and* $X_i = \Theta_i + \sigma Z_i$ *for independent standard Gaussian* $Z_i$*, every NPMLE over* $[a, b]$ *satisfies*

$$\sup_G \mathbb{E}_G W_1(H, G) \le (\sqrt{2} + \sqrt{2/\pi})\sigma + \frac{b - a}{2\sqrt{N}}. \tag{III.4.16}$$

*The minimax risk is of order* $(b - a)/\sqrt{N}$ *uniformly when* $\sigma \le (b - a)/\sqrt{N}$*.*

*Proof.* Write $Q_\Theta = N^{-1}\sum_i \delta_{\Theta_i}$. For a law supported on an interval of length $L$, the one-dimensional CDF formula gives

$$\mathbb{E}W_1(Q_\Theta, G) \le \int \sqrt{F_G(x)(1-F_G(x))/N}\,dx \le \frac{L}{2\sqrt{N}}.$$

In the Poisson model, clipping cannot increase the distance to $\Theta_i \in [0, M]$, and

$$\mathbb{E}|C_{0,M}(X_i/t) - \Theta_i| \le \mathbb{E}|X_i/t - \Theta_i| \le \mathbb{E}\sqrt{\Theta_i/t} \le \sqrt{M/t}.$$

The triangle inequality, (III.4.13), and $W_1 \le W_2$ prove (III.4.15). For the unrestricted estimator use (III.4.12) and $\mathbb{E}\bar{X} = t\mathbb{E}\Theta_i \le tM$ instead. In the Gaussian case the corresponding noise term is $\sigma\mathbb{E}|Z_i| = \sigma\sqrt{2/\pi}$, and (III.4.14) proves (III.4.16).

For the lower bounds, use mixing laws on the two endpoints of the parameter interval, with upper-endpoint probabilities $1/2 \pm h$, where $h = 1/(16\sqrt{N})$. Their $W_1$ separation is $2Lh$. The relative entropy between their latent $N$-sample experiments is bounded by an absolute constant smaller than one, using $D(\text{Bernoulli}(p)\|\,\text{Bernoulli}(q)) \le (p-q)^2/[q(1-q)]$. Passing the latent variables through either observation kernel cannot increase total variation. The two-point testing bound therefore gives $cL/\sqrt{N}$ for every estimator. Substitute $L = M$ or $L = b - a$ and impose the respective high-precision condition to match the upper bound. □

In the rate-one Poisson parameterization, with support $[0, B]$, the high-exposure condition is $B \ge N$ and the risk is $\Theta(B/\sqrt{N})$. This supplies a complete statement for that parameter regime. Theorem 6.1 of the version of [78] inspected here covers smaller support scales, up to $N^{1/3-\varepsilon}$. We do not claim that Corollary III.4.6 closes the intervening range, determines all noise-dependent minimax rates, or resolves a named conjecture from that paper.

### III.4.5 Two limits of the transfer

**Proposition III.4.7** (Heteroscedastic contact is precision weighted)**.** *For Gaussian observations with known positive definite covariances $\Sigma_i$, let $H$ maximize $\sum_i q_i \log \int \phi_{\Sigma_i}(x_i - \theta)\,dH(\theta)$ over unrestricted locations. At a contact point, the empirical posterior satisfies*

$$\mathbb{E}_{R_\theta} \exp\{v^\mathsf{T}\Sigma_i^{-1}(x_i - \theta) - \tfrac{1}{2}v^\mathsf{T}\Sigma_i^{-1}v\} \le 1. \tag{III.4.17}$$

*In particular,*

$$\mathbb{E}_{R_\theta}[\Sigma_i^{-1}(x_i - \theta)] = 0.$$

*The ordinary martingale identity $\mathbb{E}_{R_\theta} x_i = \theta$ can fail.*

*Proof.* The likelihood score at $\theta + v$ is exactly the left side of (III.4.17), which proves the inequality. Its derivative at zero vanishes. For a counterexample to the unweighted identity, take equal weights, observations $x_1 = 0, x_2 = 1$, and variances $1, 4$. For $H = \delta_{1/5}$ the score is

$$S(\theta) = \tfrac{1}{2}e^{((1/5)^2 - \theta^2)/2} + \tfrac{1}{2}e^{((4/5)^2 - (1-\theta)^2)/8}.$$

It is strictly concave on $[0, 1]$, has derivative zero at $1/5$, and has positive derivative for $\theta < 0$ and negative derivative for $\theta > 1$. Its unique maximum is $S(1/5) = 1$. The score criterion therefore makes $\delta_{1/5}$ the unique NPMLE, while its conditional empirical mean is $1/2$. □

Thus the common-covariance convex-order and transport conclusions cannot be transferred to arbitrary heteroscedastic models by merely replacing $\Sigma$ with an average covariance. No sharp heteroscedastic minimax conclusion is asserted here.

**Proposition III.4.8** (Likelihood accuracy alone does not control unbounded latent transport)**.** *In the unrestricted Gaussian location model, no uniform bound on latent $W_2$ error follows from a fixed positive likelihood tolerance, even if the fit also has a uniformly bounded likelihood-score certificate.*

*Proof.* For the single observation zero, the exact NPMLE is $\delta_0$. Fix $0 < \varepsilon < 1$ and set $H_R = (1-\varepsilon)\delta_0 + \varepsilon\delta_R$. Its normalized likelihood deficit is

$$-\log\{1 - \varepsilon + \varepsilon e^{-R^2/(2\sigma^2)}\} \le -\log(1-\varepsilon).$$

Its score supremum is at most $1/(1-\varepsilon)$, independently of $R$, whereas $W_2(H_R, \delta_0) = \sqrt{\varepsilon}R \to \infty$. For any specified positive deficit or score tolerance, choose a sufficiently small fixed $\varepsilon$ and then let $R$ increase. □

The exact posterior results and the reweighted EM bound therefore address a different question from unrestricted stability of all approximate likelihood fits. This distinction is necessary, not an artifact of using empirical entropy.

## III.5 The square-root order of the design gap on all connected graphs

The comparison between $D$-optimal and $A$-optimal graph designs asks how much total effective resistance is lost by optimizing determinant instead of inverse trace. Suliman and Weinstein give graph-dependent bounds and propose an $O(\log n)$ bound in Conjecture 1.5 of [13]. The preceding tree counterexample rules out that conjecture. Here we determine the correct worst-case order on *all* connected graphs: it is $\Theta(\sqrt{n})$. We also give an exact solution on complete-graph barbells, whose asymptotic ratio exceeds the worst-tree ratio by a factor of $\sqrt{2}$.

### III.5.1 A universal bound from electrical potentials

Let $G$ be a connected undirected graph on $n \ge 2$ vertices. For each edge choose an orientation and write $b_e = e_u - e_v$. A design assigns nonnegative conductances $w_e$ with $\sum_e w_e = 1$; its Laplacian is

$$L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}.$$

Only connected supports have finite objective. On $H = \mathbf{1}^{\perp}$, of dimension $d = n - 1$, let

$$A = L(w^D)|_H, \qquad B = L(w^A)|_H$$

be respectively a determinant-maximizing and an inverse-trace-minimizing design. Define

$$t = \operatorname{tr} A^{-1}, \qquad \tau = \operatorname{tr} B^{-1}, \qquad T = \frac{t}{d^2}, \qquad \alpha(G) = \frac{t}{\tau}.$$

Both matrices have trace two. The unordered-pair total resistance is $n$ times the inverse trace, so this is precisely the infinity-Lewis/ERMP approximation ratio of [13].

**Lemma III.5.1** (Edge potentials at a determinant-optimal design)**.** *For every candidate edge $e$, including an edge assigned zero weight,*

$$R_e = b_e^{\mathsf{T}} A^{-1} b_e \le d, \qquad \|L(w^D)^{\dagger} b_e\|^2 \le \frac{n R_e^2}{4}.$$

*Consequently,*

$$\frac{1}{2} \le T \le \frac{n}{4}. \tag{III.5.1}$$

*Proof.* The derivative of $\log\det A$ in the edge coordinate is $R_e$. The simplex optimality conditions give $R_e \le d$, with equality on positive-weight edges: the common multiplier equals $\sum_e w_e^D R_e = \operatorname{tr}(A^{-1}A) = d$.

Write $v = L(w^D)^\dagger b_e$, considered as a zero-mean vector on the $n$ vertices. It is harmonic away from the source and sink of $b_e$. The maximum principle on the connected support puts its maximum at the source and its minimum at the sink. Thus its range is $v_u - v_v = R_e$, even when $e$ itself has weight zero. A zero-mean vector of range $R_e$ has squared norm at most $nR_e^2/4$: if its minimum and maximum are $a \le 0 \le b$, then averaging $(v_i - a)(v_i - b) \le 0$ gives $n^{-1}\sum_i v_i^2 \le -ab \le (b-a)^2/4$. This proves the potential bound. Finally,

$$t = \sum_e w_e^D \|A^{-1}b_e\|^2 \le \frac{nd^2}{4}.$$

The opposite bound $t \ge d^2/2$ follows from $\operatorname{tr} A = 2$ and Cauchy–Schwarz on its eigenvalues. □

**Theorem III.5.2** (Universal square-root upper bound)**.** *For $n > 2$, every connected $n$-vertex graph satisfies*

$$\alpha(G) \le \frac{2T(n-2)}{8T^2 - 8T + n} \le \sqrt{\frac{n}{8}} + \frac{1}{2}. \tag{III.5.2}$$

*For $n = 2$, the ratio is one and the last inequality remains valid.*

*Proof.* For every symmetric operator $Y$ on $H$, matrix Cauchy–Schwarz gives

$$(\operatorname{tr} Y)^2 \le \operatorname{tr}(B^{-1})\operatorname{tr}(Y^2 B).$$

Choose $Y = A^{-1} + \eta I$ with $\eta \ge 0$. Because $B = \sum_e w_e^A b_e b_e^{\mathsf{T}}$, the preceding lemma gives

$$\begin{aligned}\operatorname{tr}(Y^2 B) &= \sum_e w_e^A(\|A^{-1}b_e\|^2 + 2\eta R_e + 2\eta^2)\\ &\le \frac{nd^2}{4} + 2\eta d + 2\eta^2.\end{aligned}$$

Put $\eta = dz$. It follows that, for every $z \ge 0$,

$$\alpha(G) \le \frac{T(n/4 + 2z + 2z^2)}{(T+z)^2}. \tag{III.5.3}$$

If $T = 1/2$, equality in the eigenvalue Cauchy–Schwarz bound forces $A = (2/d)I$; this already minimizes inverse trace, so $\alpha(G) = 1$. If $T > 1/2$, choose

$$z = \frac{n/4 - T}{2T - 1} \ge 0,$$

where nonnegativity follows from (III.5.1). Substitution into (III.5.3) yields the first inequality of (III.5.2). The derivative of that rational function of $T$ has the sign of $n - 8T^2$. Its maximum on $[1/2, n/4]$ therefore occurs at $T = \sqrt{n/8}$ and equals $\sqrt{n/8} + 1/2$. For $n = 2$, the normalized Laplacian on $H$ is the scalar two, so both designs coincide. □

*Remark* III.5.3 (An elementary first bound). Taking $z = 0$ above gives $\alpha(G) \le n/(4T)$. The bound $\tau \ge d^2/2$ also gives $\alpha(G) \le 2T$. Thus already

$$\alpha(G) \le \min\{2T, n/(4T)\} \le \sqrt{n/2}.$$

Optimizing the nonnegative shift improves its leading constant by a factor of two. The graph-specific input is the maximum principle for electrical potentials; a general positive semidefinite design need not satisfy that input.

### III.5.2 Exact designs on complete-graph barbells

Let $G_{m,p}$ consist of two disjoint copies of $K_m$, $m \geq 2$, with a distinguished root in each, joined by a path of $p \geq 1$ edges. Its $p-1$ interior vertices are new. Set

$$n = 2m + p - 1, \qquad d = n - 1, \qquad h = n - m + 1,$$

$$c_j = (m + j - 1)(m + p - j) \quad (1 \leq j \leq p), \qquad q = \sqrt{\frac{mh}{2}} + (m-2)\sqrt{\frac{n}{2}},$$

$$Q = 2q + \sum_{j=1}^{p} \sqrt{c_j}.$$

Here $c_j$ is the cut count of the $j$th bridge on the joining path.

**Theorem III.5.4** (Exact barbell objectives and optimal conductances)**.** *For the unordered-pair resistance objective $K$, the optimal values and the $D$-design's resistance are*

$$K(w^A) = Q^2, \qquad K(w^D) = d\left((m-1)(2n-m) + \sum_{j=1}^{p} c_j\right). \tag{III.5.4}$$

*The $D$-optimal design assigns $1/d$ to each joining-path edge and $2/(md)$ to every clique edge. The $A$-optimal design assigns $\sqrt{c_j}/Q$ to the $j$th joining-path edge and total conductance $s = q/Q$ to each clique. Within either clique, root-to-other conductances are*

$$a = \frac{s\sqrt{2h/m}}{q},$$

*and, for $m \geq 3$, other-to-other conductances are*

$$b = \frac{1}{m-1}\left(\frac{s\sqrt{2n}}{q} - a\right).$$

*Proof.* The displayed $D$-weights sum to one. Each bridge has effective resistance $d$, and the uniform clique weights likewise give resistance $2/(m \cdot 2/(md)) = d$ between every clique pair. Attachments at a single root do not change resistances inside a clique. Thus every candidate edge has resistance $d$, which verifies the determinant optimality conditions.

For the $A$-optimal objective, resistances add across the graph's blocks. Each clique contributes a weighted resistance sum with mass $h$ at its root and mass one at each other vertex: all vertices outside the clique project to the root. For a fixed clique budget $s$, symmetry and convexity allow equal weights $a$ on root-to-other edges and $b$ on other-to-other edges. For $m \geq 3$ put $\lambda = a + (m-1)b$. The root contrast eigenvalue is $ma$, while the peripheral zero-sum eigenspace has eigenvalue $\lambda$ and dimension $m-2$. Consequently

$$R_{\mathrm{root},i} = \frac{1}{(m-1)a} + \frac{m-2}{(m-1)\lambda}, \qquad R_{i,j} = \frac{2}{\lambda}$$

for distinct peripheral vertices $i, j$. The weighted contribution and budget are therefore

$$H(a,\lambda) = \frac{h}{a} + \frac{n(m-2)}{\lambda}, \qquad s = \frac{m}{2}a + \frac{m-2}{2}\lambda.$$

Cauchy–Schwarz gives $H \geq q^2/s$, with equality at

$$a = \frac{s\sqrt{2h/m}}{q}, \qquad \lambda = \frac{s\sqrt{2n}}{q}.$$

Since $nm \geq h$, these weights satisfy $\lambda \geq a$ and hence $b \geq 0$. When $m = 2$, the clique is a single edge with contribution $h/a$ and budget $a = s$, and the same formula follows directly from $q = \sqrt{h}$.

The joining-path contribution is $\sum_j c_j/w_j$. Allocating the unit total budget among the two clique blocks and the $p$ bridges yields

$$\min\left\{\frac{q^2}{s_1} + \frac{q^2}{s_2} + \sum_j \frac{c_j}{w_j} : s_1 + s_2 + \sum_j w_j = 1\right\} = Q^2.$$

Equality gives exactly the asserted conductances. This proves global optimality, not just stationarity.

At the $D$-weights, every clique pair has resistance $d$. The weighted-pair coefficient of one clique is

$$h(m-1) + \binom{m-1}{2} = (m-1)(n - m/2).$$

Summing over the two cliques and adding the bridge contributions $d\sum_j c_j$ proves the second formula in (III.5.4). □

### III.5.3 The general-graph order and the remaining constant gap

**Theorem III.5.5** (Square-root worst-case order and a denser obstruction)**.** *Let $\mathcal{R}_n$ be the maximum of $\alpha(G)$ over connected simple graphs on $n$ vertices. Then*

$$\left(\frac{1}{4\sqrt{2}} + o(1)\right)\sqrt{n} \leq \mathcal{R}_n \leq \frac{\sqrt{n}}{2\sqrt{2}} + \frac{1}{2}. \tag{III.5.5}$$

*In particular, $\mathcal{R}_n = \Theta(\sqrt{n})$. The lower-bound graphs have an asymptotic leading constant $\sqrt{2}$ times the worst-tree leading constant $1/8$.*

*Proof.* The upper bound is Theorem III.5.2. For each sufficiently large $n$, choose an integer $p = \sqrt{2n} + O(1)$ with $p \equiv n + 1 \pmod 2$, and set $m = (n - p + 1)/2$. Then $G_{m,p}$ has exactly $n$ vertices, $m = n/2 + O(\sqrt{n})$, and uniformly along the joining path

$$c_j = n^2/4 + O(n).$$

The quantities in Theorem III.5.4 satisfy

$$2q = \frac{n^{3/2}}{\sqrt{2}} + O(n), \qquad \sum_j \sqrt{c_j} = \frac{n^{3/2}}{\sqrt{2}} + O(n),$$

$$\sum_j c_j = \frac{n^{5/2}}{2\sqrt{2}} + O(n^2).$$

The clique term $(m-1)(2n-m)$ is $O(n^2)$. It follows that

$$K(w^D) = \frac{n^{7/2}}{2\sqrt{2}} + O(n^3), \qquad K(w^A) = 2n^3 + O(n^{5/2}),$$

and hence

$$\alpha(G_{m,p}) = \frac{\sqrt{n}}{4\sqrt{2}} + O(1).$$

This proves the lower bound and the claimed order. Comparison with the previously established tree leading constant $1/8$ gives the final assertion. □

The exact asymptotic leading constant for all connected graphs is not determined by these bounds. It lies between $1/(4\sqrt{2})$ and $1/(2\sqrt{2})$ if a limiting constant exists; without assuming such existence, the same numbers bound the liminf and limsup. Neither endpoint is asserted sharp here.

## III.6 A phase classification for weighted binomial power sums

For $r \geq 2$ and $\phi \in \mathbb{R}$, define

$$B_{r,\phi}(n) = \sum_{k=-n}^{n} \binom{2n}{n+k}^{r} e^{ik\phi}, \qquad n \geq 0. \tag{III.6.1}$$

The sum is real by pairing $k$ with $-k$. Weighted binomial powers belong to the hypergeometric polynomial families considered, for example, by Matala-Aho for powers three and four [26] and by Simsek for general powers [27]. The theorem below is a classification of this phase family; it is not presented as the resolution of a named external conjecture. Its new content relative to the preceding alternating-sum result is the complete nonreal phase range. The real phases are included to identify exactly where the recurrence width drops.

A recurrence is an eventual identity $\sum_{j\in J} q_j(n) B_{r,\phi}(n+j) = 0$, with finite nonempty $J \subset \mathbb{Z}$ and nonzero $q_j \in \mathbb{C}(n)$. Its width is $\max J - \min J$ and its number of terms is $|J|$.

**Theorem III.6.1.** *Put*

$$m(r,\phi) = \begin{cases} \lceil r/2 \rceil, & \phi \equiv 0 \pmod{2\pi}, \\ \lfloor r/2 \rfloor, & \phi \equiv \pi \pmod{2\pi}, \\ r, & \phi \notin \pi\mathbb{Z}. \end{cases} \tag{III.6.2}$$

*The minimum recurrence width of $B_{r,\phi}$ over $\mathbb{C}(n)$ is $m(r,\phi)$. The minimum number of nonzero terms, even allowing arbitrary gaps between shifts, is $m(r,\phi)+1$. A recurrence attaining both minima can be chosen with coefficients in $\mathbb{Q}(2\cos\phi)[n]$.*

In particular, every nonreal root of unity gives width $r$, independently of its order. The widths at 1 and $-1$ are roughly half as large. This distinction will follow from collisions of positive critical values, not from an asymptotic comparison involving only the largest one. The theorem makes no assertion for weights off the unit circle or about the minimum differential order of a generating function.

### III.6.1 Positive moments and critical-value collisions

Write $\lambda = e^{i\phi}$, $d = r-1$, $X = x_1 \cdots x_d$, and

$$P_{r,\lambda}(x) = \prod_{i=1}^{d} (1 + x_i)(1 + \lambda/X).$$

Direct expansion gives

$$\mathrm{CT}\, P_{r,\lambda}^{\ell} = \sum_{j=0}^{\ell} \binom{\ell}{j}^{r} \lambda^{j}. \tag{III.6.3}$$

On $x_i = e^{i\theta_i}$, with $S = \sum_i \theta_i$, we have

$$P_{r,\lambda}(e^{i\theta}) = e^{i\phi/2} \Psi_{r,\phi}(\theta), \qquad \Psi_{r,\phi}(\theta) = 2^{r} \prod_{i=1}^{d} \cos(\theta_i/2) \cos((\phi - S)/2). \tag{III.6.4}$$

The real function $\Psi_{r,\phi}$ is globally periodic and analytic: shifting a single angle by $2\pi$ changes two factors by a minus sign. Thus $\Phi_{r,\phi} = \Psi_{r,\phi}^2 \geq 0$ is a real-analytic function on $\mathbb{T}^d$. For normalized Haar measure $\mu$,

$$B_{r,\phi}(n) = \lambda^{-n}\,\mathrm{CT}\,P_{r,\lambda}^{2n} = \int_{\mathbb{T}^d} \Phi_{r,\phi}^n \, d\mu. \tag{III.6.5}$$

The first equality is the change of index $j = n + k$ in (III.6.1). The phase in the second equality cancels because $(e^{i\phi/2})^{2n} = \lambda^n$. In particular these real sums are strictly positive.

**Lemma III.6.2.** *The function $\Phi_{r,\phi}$ has exactly $m(r,\phi)$ distinct positive critical values. Every critical point on every positive critical fiber is a nondegenerate local maximum. For $0 \leq \phi < 2\pi$, the values are the distinct nonzero members of*

$$v_j = \left(2\cos\frac{\phi + 2\pi j}{2r}\right)^{2r}, \qquad j = 0, \ldots, r-1. \tag{III.6.6}$$

*Proof.* At $P_{r,\lambda} \neq 0$, the logarithmic critical equations are

$$\frac{x_i}{1+x_i} - \frac{\lambda}{X+\lambda} = 0 \quad \Longleftrightarrow \quad x_i X = \lambda.$$

They force $x_1 = \cdots = x_d = \omega$, where $\omega^r = \lambda$. Every such root is on the unit circle, and it gives a critical point unless $\omega = -1$, when $P = 0$. These are all positive critical points of $\Phi$: angular differentiation of $P = e^{i\phi/2}\Psi$ is equivalent to differentiation of the real $\Psi$, and away from zero $d\Phi = 2\Psi\, d\Psi$. At a critical point $P = (1+\omega)^r$, giving (III.6.6).

For completeness, differentiation once more gives

$$\mathrm{Hess}_{\log x} \log P = \frac{\omega}{(1+\omega)^2}(I_d + J_d),$$

where $J_d$ is the all-ones matrix. Indeed the derivative of $-\lambda/(X+\lambda)$ in each logarithmic direction is $\lambda X/(X+\lambda)^2 = \omega/(1+\omega)^2$. For $\omega = e^{i\alpha} \neq -1$, the common factor is $1/(4\cos^2(\alpha/2)) > 0$. Angular differentiation introduces a minus sign; using the local real logarithm $\log|\Psi|$ yields

$$\mathrm{Hess}_\theta\, \Phi = -\frac{\Phi}{2\cos^2(\alpha/2)}(I_d + J_d). \tag{III.6.7}$$

The eigenvalues of $I_d + J_d$ are 1 and $d+1 = r$, so this Hessian is negative definite.

It remains to count distinct values. The half-angles $\theta_j = (\phi + 2\pi j)/(2r)$ lie in $[0, \pi)$. For distinct indices, equality of the nonzero values in (III.6.6) is equivalent to $\theta_j + \theta_k = \pi$, hence to

$$\phi/\pi = r - j - k.$$

If $\phi \notin \pi\mathbb{Z}$, no collision is possible and no $\theta_j$ equals $\pi/2$, giving $r$ positive values. At $\phi = 0$, pairing $j$ with $r - j$ leaves exactly $\lceil r/2 \rceil$ positive values, represented by $0 \leq j < \lceil r/2 \rceil$; the root $-1$, if present, contributes zero and is excluded. At $\phi = \pi$, pairing $j$ with $r - 1 - j$ leaves $\lfloor r/2 \rfloor$ positive values, represented by $0 \leq j < \lfloor r/2 \rfloor$. □

### III.6.2 Visibility and the lower bounds

We recall the precise real-variable mechanism, including even-dimensional integer exponents. If a real-analytic function on a compact manifold has finitely many critical points on a positive fiber $v$, all nondegenerate maxima, the analytic Morse lemma gives local coordinates $\Phi = v - |y|^2$. A positive analytic measure has the form $g(y)\, dy$, $g(0) > 0$. Its local pushforward density at $v - w$, $w > 0$, is

$$\frac{1}{2} w^{d/2-1} \int_{S^{d-1}} g(\sqrt{w}\,\omega)\, d\omega.$$

The spherical average is analytic in $w$ and has positive constant term. Smooth cutoffs leave a compact submersion region, whose pushforward is smooth. Consequently the complete density near $v$ has the form

$$b(x) + \mathbf{1}_{\{x<0\}}(-x)^{d/2-1}h(-x), \qquad x = t - v, \tag{III.6.8}$$

with $b$ smooth, $h$ analytic, and $h(0) > 0$. This is not smooth: in even dimension a finite derivative jumps, whereas in odd dimension there is a noninteger power (an integrable unbounded density when $d = 1$). All critical contributions have the same side and a positive leading coefficient.

The Cauchy transform $G(z) = \int (z - \Phi)^{-1}\, d\mu$ is holomorphic in the upper half-plane, and its boundary distribution satisfies $-\operatorname{Im} G(x + i0)/\pi = \Phi_* \mu$. A holomorphic extension through $v$ would give a real-analytic boundary density, contradicting (III.6.8). Since $G(z) = z^{-1}F(z^{-1})$ near infinity for $F(t) = \sum_n (\int \Phi^n\, d\mu) t^n$, the generating germ has a genuine continued-branch singularity at $1/v$. This argument applies also to critical values inside the range of $\Phi$.

By Lemma III.6.2, the generating function of $B_{r,\phi}$ therefore has $m = m(r, \phi)$ distinct positive finite singularities. To see their implications for an arbitrary recurrence, translate its shifts to $\min J = 0$, $\max J = L$ and clear denominators in $q_j(n)$. Put $e = \max_j \deg q_j$ and $c_j = [n^e]q_j$. With $\vartheta = t\, d/dt$, summation gives

$$\left( \sum_{j \in J} t^{L-j} q_j(\vartheta - j) \right) F(t) = R(t),$$

where $R$ is a polynomial of initial terms. Its highest ordinary derivative coefficient is $t^e H(t)$, where

$$H(t) = \sum_{j \in J} c_j t^{L-j} \neq 0.$$

Every nonzero finite singularity of a continued solution must be a zero of $H$, by ordinary-point theory for linear differential equations; for $e = 0$ this follows directly from $HF = R$. Thus $H$ has at least $m$ positive roots. Its degree is at most $L$, giving $L \geq m$. Moreover, a nonzero real or imaginary part of $H$ has the same positive roots and no more monomials. Descartes' rule of signs gives at least $m + 1$ nonzero monomials, hence $|J| \geq m + 1$. Both lower bounds hold over $\mathbb{C}(n)$ and allow gaps in $J$.

### III.6.3 An independent upper bound for every nonreal phase

The matching upper bound uses an explicit $r$-dimensional state. It requires neither a general weighted-recurrence theorem nor an invertibility assumption about a moment matrix.

For an integer $N \geq 0$, define a polynomial functional

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k.$$

For every $h \geq 0$, shifting the summation index gives

$$L_N[R_h] = 0, \qquad R_h(k) = k^{r+h} - \lambda (N - k)^r (k + 1)^h. \tag{III.6.9}$$

Indeed $k^r \binom{N}{k}^r \lambda^k = \lambda (N - k + 1)^r \binom{N}{k-1}^r \lambda^{k-1}$, and the boundary terms vanish. The leading coefficient of every $R_h$ is

$$1 - (-1)^r \lambda. \tag{III.6.10}$$

It is nonzero if $\lambda$ is nonreal. Successive subtraction of multiples of $R_h$ reduces any polynomial in $k$ to degree less than $r$, without changing $L_N$. Denote this triangular reduction by $\mathcal{R}_N$. All reduction coefficients belong to $\mathbb{Q}(\lambda, N)$.

Form the column vector

$$M_N = (L_N[1], L_N[k], \dots, L_N[k^{r-1}])^{\mathsf{T}}.$$

For $N \geq 1$ and $0 \leq j < r$, the binomial identity implies

$$(M_{N-1})_j = N^{-r} L_N[k^j (N-k)^r].$$

Define the rational matrix $T_r(N, \lambda)$ by taking its $j$th row to be the coefficients of $N^{-r}\mathcal{R}_N[k^j(N-k)^r]$ in the basis $1, k, \dots, k^{r-1}$. Then

$$M_{N-1} = T_r(N, \lambda) M_N. \tag{III.6.11}$$

The index $N$ here is the upper binomial index and is distinct from the index $n$ of (III.6.1).

Set $V_n = \lambda^{-n} M_{2n}$. Equations (III.6.3) and (III.6.11) give

$$\begin{aligned} V_{n-1} &= \lambda T_r(2n-1, \lambda) T_r(2n, \lambda) V_n, \\ B_{r,\phi}(n) &= (V_n)_0. \end{aligned} \tag{III.6.12}$$

Thus $B_{r,\phi}(n), \dots, B_{r,\phi}(n-r)$ are $r+1$ row functionals of the same $r$-dimensional vector $V_n$. Their rows have entries in $\mathbb{Q}(\lambda)(n)$, so they are linearly dependent over that field. This is a nonzero eventual recurrence of width at most $r$. No matrix inverse is involved, and possible special initial indices are immaterial to an eventual recurrence. The lower bounds already proved force its width to be $r$ and all $r+1$ positions to be nonzero.

### III.6.4 Exceptional phases and the coefficient field

At $\phi = \pi$, the sequence is exactly McIntosh's alternating sum. His general existence theorem gives width at most $\lfloor r/2 \rfloor$ [29, Theorem 1]. This matches the critical-value lower bound above. This boundary assertion agrees with the previously established alternating minimality result and is not counted as a second new application.

At $\phi = 0$, write $f_r(N) = \sum_{k=0}^{N} \binom{N}{k}^r$, so $B_{r,0}(n) = f_r(2n)$. Stoll's existence theorem gives $f_r$ a recurrence of width at most $\lceil r/2 \rceil$ [28]. To pass to the even subsequence, solve such a recurrence for its highest shift over $\mathbb{Q}(N)$ and use its usual companion state. Composing two consecutive transitions produces a state for the even subsequence of the same dimension. Linear dependence of one more consecutive row functional therefore gives width at most $\lceil r/2 \rceil$, not twice that quantity. The lower bound proves equality.

Finally, in the nonreal case the construction provides a recurrence over $K(n)$, where $K = \mathbb{Q}(\lambda)$. Translate its shifts to $0, \dots, r$ and normalize the coefficient of shift $r$ to one. Such a monic recurrence on this fixed shift interval is unique: the difference of two would be a nonzero recurrence of smaller width. The sequence is real, so complex conjugation gives the same normalized recurrence. Consequently its coefficients lie in the conjugation-fixed field

$$K^{\mathrm{conj}}(n) = \mathbb{Q}(\lambda + \lambda^{-1})(n) = \mathbb{Q}(2\cos\phi)(n).$$

For clarity, the fixed-field equality also holds for transcendental $\lambda$: conjugation sends $\lambda$ to $\lambda^{-1}$, and $\lambda$ satisfies a quadratic equation over $\mathbb{Q}(\lambda + \lambda^{-1})$; as $\lambda$ is nonreal, the extension has degree two. Clearing denominators gives coefficients in $\mathbb{Q}(2\cos\phi)[n]$. At the real phases the cited upper recurrences already have rational coefficients. This completes the proof of Theorem III.6.1.

### III.6.5 A concrete nonreal phase

For $r = 3$ and $\phi = \pi/2$, the sequence is integer-valued:

$$B(n) = \sum_{\substack{-n \le k \le n \\ k \text{ even}}} (-1)^{k/2} \binom{2n}{n+k}^3.$$

Its critical scales are $26 + 15\sqrt{3}$, $8$, and $26 - 15\sqrt{3}$, with scale polynomial

$$(X - 8)(X^2 - 52X + 1) = X^3 - 60X^2 + 417X - 8.$$

Therefore the minimum width is three and the minimum number of terms is four, over $\mathbb{C}(n)$ as well as $\mathbb{Q}(n)$. Every minimum-width recurrence has highest-degree generating symbol proportional to

$$1 - 60t + 417t^2 - 8t^3.$$

Indeed the symbol has degree at most three and must vanish at the reciprocals of all three visible scales. In contrast, the same power $r = 3$ has width two at phase zero and width one at phase $\pi$.

The phase classification consequently describes an exact change in recurrence complexity under the weight, while keeping the power fixed. It supplies no uniform bound on coefficient sizes near the exceptional phases, where the critical values collide.

## III.7 What is closed, and what remains open

The main statistical closure is estimator-specific. The robust minimax curve was already established in the supplied compilation; Theorem III.2.1 proves that the ordinary NPMLE itself attains it without knowing the contamination fraction. Theorem III.3.1 closes the full parameter range for depths 1 and $T$, but not for arbitrary replication profiles. The affine identity behind that theorem is special to a depth-one stratum.

For continuous families, Theorem III.4.2 gives an exact entropy budget for one reweighted posterior update, with optimal Gaussian constant. Corollary III.4.6 supplies matching statistical rates in the stated high-precision regimes. It does not settle all intermediate Poisson exposure or Gaussian noise regimes. The counterexamples in Section III.4 explain why unrestricted approximate fits and heteroscedastic observations require additional ideas.

For graph design, Theorem III.5.5 determines the worst growth order over all connected graphs. The leading constants in its universal upper and constructed lower bounds still differ by a factor of two. Determining the exact constant and the extremal graph families remains open here. For weighted binomial sums, Theorem III.6.1 classifies the full unit circle; it does not classify arbitrary complex weights or identify recurrence width with differential-equation order.

The supporting analytic tools and the preceding real-phase and tree results are credited to the earlier manuscripts [20, 25]. The proofs here establish the specified assertions subject to their assumptions. Internal proof checks are not a substitute for external expert review or a comprehensive assessment of historical priority.

## III.A Supporting binomial approximation and likelihood lemmas

The following tools were established in the supplied observation compilation. They are reproduced with proofs so that the new statistical arguments do not require an external supplement. They are not counted as new results. Write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and $p_G(j) = \int B_j^t \, dG$. The distance $W_{1,t}$ uses ground cost $|j - k|/t$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$. The same formula defines $h$ for nonnegative vectors of unequal mass. All constants below are absolute.

**Lemma III.A.1** (Small covers of the binomial moment body)**.** *For $0 < \beta < 1$ the set $\{p_G : G \in \mathcal{P}([0,1])\}$ has a finite $\ell_1$ cover of radius $\beta$, with centers in that same set, whose cardinality $M$ satisfies*

$$\log M \le m \log \frac{C(m+1)}{\beta}, \qquad m = \min\left\{t, \left\lceil \sqrt{2t \log(16t/\beta)} \right\rceil \right\}. \tag{III.A.1}$$

*Proof.* View $K_x = (B_0^t(x), \dots, B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x = \sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k \ge 1$. Indeed, the even periodic function $F(z) = K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z = \theta + iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right| + \left|\frac{1-\cos z}{2}\right| = \cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1 = (\cosh u)^t \le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u = k/t$. This proves the claim. In particular, truncating after degree $m < t$ has vector $\ell_1$ error at most $2te^{-(m+1)^2/(2t)} \le \beta/8$, uniformly in $x$. For $m = t$ the error is zero.

Each mixing law contributes its first $m$ shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If $G, H$ lie in one cell, their truncated transforms differ in $\ell_1$ norm by at most $2m\beta/[8(m+1)] \le \beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most $\beta$. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

**Lemma III.A.2** (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let $p$ be the true law on a finite alphabet with $d$ cells, and suppose a model admits an $\ell_1$ cover of radius $\eta/d$, with $M$ probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

*For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most $e^{-x}$,*

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{III.A.2}$$

*Consequently*

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{III.A.3}$$

*Proof.* For each center $p_a$, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one $u_a$, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of $p$, where all sampled

observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\ &= e^{N\varepsilon/2}\left(1+\frac{\eta}{2}-\frac{h(p,u_a)^2}{2}\right)^N \\ &\le \exp\{N(\varepsilon+\eta-r^2)/2\}\end{aligned}$$

when $h(p,u_a) > r$. Taking the union over the $M$ envelopes proves (III.A.2). Integrating that tail, and using $(a+b)^2 \le 2a^2+2b^2$, proves (III.A.3). □

**Lemma III.A.3** (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute $C$ with the following properties. For every 1-Lipschitz $f:[0,1]\to\mathbb{R}$ and integer $k\ge 1$, there is a polynomial $P_k$ of degree at most $k$ such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

*If $P=\sum_{j=0}^t b_j B_j^t$ has degree $k\le t/2$, then*

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{III.A.4}$$

*For the derivative-stable $P_k$, its coefficient vector also satisfies*

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{III.A.5}$$

*Proof.* Subtract $f(0)$, reflect $f$ on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial $Q_s$ has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most $e^{-100s}$. Indeed $|\pi l(x-1/2)| \le \pi s$ and the factorial inequality $M! \ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s=\lfloor k/100\rfloor$ for $k\ge 100$; use a constant polynomial and enlarge $C$ for smaller $k$.

For (III.A.4), put $M=\|P\|_\infty$ and $F(z)=(1+z)^tP(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w|\ge 1$. For $0<j<t$, set $r=j/(t-j)$, $v=r/(1+r)^2$, and $u=2\sqrt{v}\,|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k\le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c/\sqrt{1+tv}$, proves (III.A.4). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (III.A.4) at degree $k-1\le (t-1)/2$ and use $2(k-1)^2/(t-1)\le 4k^2/t$. □

**Proposition III.A.4** (Two-scale deterministic inversion)**.** *For integers $1\le s\le k\le t/2$ and any two mixing laws $G,H$,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t}W_{1,t}(p_G,p_H) + \frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{III.A.6}$$

*Proof.* For any 1-Lipschitz test $f$, take the polynomials $P_k, P_s$ from Lemma III.A.3. The approximation error in integrating $P_k$ instead of $f$ against $G - H$ is at most $C/k$. The Bernstein coefficients of $P_s$ have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test $P_s$ by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G, p_H)$. All constants are uniform over $f$, so taking the Wasserstein dual supremum proves the result. □

The small amplitude $1/s$ is essential. Applying only the Hellinger bound to the entire polynomial loses the low-frequency empirical $N^{-1/2}$ scale; applying only transportation loses the critical transition. Proposition III.A.4 retains both.

# Part IV

# Global Parameter Theorems from Entropy, Transport, and Spectral Structure

**Abstract**

We transfer likelihood entropy, two-scale inversion, canonical singular germs, and spectral optimization to four precisely specified problems. The main statistical theorem determines the Wasserstein minimax risk for Poisson population recovery for every sample size $N \geq 3$ and every known support radius $B > 0$, and proves that every ordinary unrestricted Poisson mixture maximum-likelihood estimator attains it. An exact occupancy channel converts Poisson kernels into binomial kernels while preserving the posterior entropy argument for the original estimator. A second global theorem determines the physical-density connection coefficients for the Jacobi trace for every $N \geq 1$, $a, b > -1$, and $\beta \geq 0$, including all resonances. Convergent corner integrals and a shifted auxiliary differential system give parameter-continuous continuation without singular normalization factors. For every connected block graph, we compute both optimal designs and prove that the worst $D$-optimal/$A$-optimal resistance ratio has leading constant $1/(4\sqrt{2})$ times $\sqrt{n}$. Finally, in a known two-Gaussian dictionary, we prove that ordinary likelihood incurs contamination error of order $\sqrt{\varepsilon}$, whereas the minimax error is of order $\varepsilon$, with matching finite-sample bounds and an exact population bias equation. The supporting binomial approximation arguments are reproduced, and the unresolved extensions are stated explicitly.

**Integration note.** The all-support Poisson theorem is incorporated again in Part VIII with a complete supporting argument. The all-real-parameter Jacobi connection, block-graph extremal constant, and Gaussian corruption obstruction are retained as separate results.

## IV.1 The global questions and their precise scope

The first two theorems remove parameter exclusions from specific likelihood and connection problems. The third solves an exact extremal problem throughout a graph class. The fourth identifies a sharp obstruction to extending the same statistical method without additional assumptions. Their conclusions are summarized below; each has a complete proof in its indicated section.

| Problem | Scope of the theorem | What is determined |
|---|---|---|
| Poisson recovery | Every $N \geq 3$, every known $B > 0$, all mixing laws on $[0, B]$ | The minimax $W_1$ rate and optimality, up to absolute factors, of the original NPMLE; Theorem IV.2.1. |
| Jacobi trace | Every $N \geq 1$, $a, b > -1$, $\beta \geq 0$, including resonances | All connection discrepancies of the physical density in a canonical normalization; Theorem IV.3.1. |
| Graph design | Every connected block graph | Exact optimal weights and the sharp asymptotic extremal constant; Theorems IV.4.2 and IV.4.4. |
| Likelihood robustness | Two fixed Gaussian components, every $N \geq 1$, $0 \leq \varepsilon \leq 1/4$ | Matching MLE and minimax risks under both strong replacement and Huber contamination; Theorem IV.5.1. |

**Provenance of the methods.** The entropy–transport and two-scale binomial tools come from the supplied manuscript *Observation and Recovery: Geometry, Mixtures, and Spectral Inversion*, especially its complete binomial-mixture likelihood transition [30]. The preceding proof-transfer drafts developed their robustness consequences and the nonresonant Jacobi and graph comparisons [25, 31]. The new Poisson argument requires an exact channel identity, an infinite-alphabet entropy estimate, and a Poisson analytic cover. The new Jacobi argument requires a full vector of convergent corner integrals; its scalar leading term alone would not justify continuation through resonance. The graph argument requires feasibility of a matrix square-root design on every complete block. The Gaussian result uses a bounded likelihood contrast and an explicit finite outlier attack to determine the limitation of the transfer.

**Relation to external questions.** The Poisson comparison is with the support-growth restriction in Miao et al.'s 2021 author version, Theorem 6.1(b) [78]. The Jacobi comparison is with the integer-parameter physical connection formula of Forrester–Kumar [84]. These are concrete theorem extensions. The block-graph extremal question and the precise Gaussian separation are formulated and answered here; they are not presented as named conjectures from those sources. These comparisons do not establish priority over all subsequent or related work.

All constants denoted $c, C$ are positive and may change between displays. Unless a dependency is stated, constants in statistical rate comparisons are numerical. $W_1$ denotes transportation with the usual absolute-distance cost on the underlying real parameter space.

## IV.2 Poisson mixture likelihood across all support regimes

Consider the experiment

$$p_G(j) = \int_0^B e^{-\lambda} \frac{\lambda^j}{j!}\, dG(\lambda), \qquad X_1, \dots, X_N \overset{\text{iid}}{\sim} p_G,$$

where the support radius $B > 0$ is known. Let $\widehat{G}$ be any measurable maximizer of the original Poisson mixture likelihood over all probability laws on $[0, B]$. There is no restriction on the number of mixing atoms. Set $L = \log(eN)$ and, for $N \geq 3$, define

$$r(N, B) = \begin{cases} B \min\{1, L^{-1} \log(e + L/B)\}, & 0 < B \leq L, \\ \max\{\sqrt{B/L}, B/\sqrt{N}\}, & B > L. \end{cases} \tag{IV.2.1}$$

The two expressions at the breakpoint agree up to absolute constants.

**Theorem IV.2.1** (Global Poisson population recovery)**.** *There exist absolute constants $0 < c < C < \infty$ such that, for every $N \geq 3$ and every $B > 0$,*

$$c\, r(N, B) \leq \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\, r(N, B).$$

*In particular, the ordinary NPMLE is minimax up to absolute factors throughout the entire support/sample-size plane. For observations $X_i \mid \Theta_i \sim \mathrm{Poi}(t\Theta_i)$ with $G \in \mathcal{P}([0, M])$, the corresponding rate is $r(N, tM)/t$.*

The small-support rate was established for the NPMLE by Miao et al. [78, Theorem 6.1(a)]. Their Theorem 6.1(b) gives the growing-support rate $\sqrt{B/\log N}$ under $B \leq C_0 N^{1/3-\epsilon_0}$. The point addressed here is the unrestricted growing-support range for the same estimator. We also include the necessary diameter clipping as $B \downarrow 0$, and give direct lower bounds rather than extending a displayed lower-rate formula beyond its meaningful range. The proof uses the two-scale binomial inverse inequality proved in Section IV.A; all additional ingredients are proved below. No claim about the historical priority of every ingredient is intended.

### IV.2.1 A parameter-free occupancy channel

For an integer $m \geq 1$, define a Markov kernel $A_m$ as follows. Given a count $x$, independently assign $x$ balls uniformly to $m$ boxes, and output the number of occupied boxes. Poisson splitting proves

$$A_m \,\mathrm{Poi}(\lambda) = \mathrm{Bin}(m, \phi_m(\lambda)), \qquad \phi_m(\lambda) = 1 - e^{-\lambda/m}. \tag{IV.2.2}$$

Indeed the box counts are independent Poisson random variables of mean $\lambda/m$, so their nonempty indicators are independent Bernoulli variables. In particular, if $F = \phi_m \# G$, then $A_m p_G$ is exactly the binomial mixture with mixing law $F$. On $[0, \phi_m(B)]$ the inverse map has derivative at most $me^{B/m}$, and consequently

$$W_1(G, H) \leq me^{B/m} W_1(\phi_m \# G, \phi_m \# H). \tag{IV.2.3}$$

This channel is a proof device. We do not transform the data before fitting, or assert that a transformed MLE remains an MLE.

We use $W_{1,m}$ for transportation on $\{0, \dots, m\}$ with cost $|i - j|/m$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

**Lemma IV.2.2** (Empirical entropy on the infinite alphabet)**.** *Let $q$ be the empirical law of $N$ observations from $p_G$. Uniformly over $G \in \mathcal{P}([0, B])$,*

$$\mathbb{E} D(q \| p_G) \leq C(B + L)/N. \tag{IV.2.4}$$

*Proof.* For every coordinate with probability $p_j > 0$, convexity and $\log u \le u - 1$ give

$$0 \le \mathbb{E}\{q_j \log(q_j/p_j) - q_j + p_j\} \le \min\{(1-p_j)/N,\ p_j \log(1/p_j)\}.$$

The second bound also uses $q_j \log q_j \le 0$. The expression inside braces is nonnegative, so its infinite sum can be averaged by Tonelli's theorem. Coordinates with $p_j = 0$ contribute zero almost surely. For $j \ge B$, the Poisson mass is increasing in its mean throughout $[0, B]$, and hence

$$p_G(j) \le e^{-B} B^j / j! \le (eB/j)^j \le e^{-j} \quad (j \ge e^2 B).$$

Take $J = \lceil e^2 B + 3L \rceil$. The first $J+1$ coordinates contribute at most $(J+1)/N$. Since $u \log(1/u)$ is increasing on $(0, e^{-1}]$, the remaining contribution is bounded by $\sum_{j>J} j e^{-j} \le C(J+1)e^{-J} \le C/N$. This proves (IV.2.4). Notice that the expected empirical chi-square divergence on the full infinite alphabet would not give this conclusion. □

**Lemma IV.2.3** (Root-$N$ prediction after occupancy)**.** *For $m = \lceil \max(B, L) \rceil$ and every Poisson NPMLE $\widehat{G}$,*

$$\mathbb{E} W_{1,m}(A_m p_{\widehat{G}}, A_m p_G) \le C/\sqrt{N}. \tag{IV.2.5}$$

*Proof.* Existence follows from compactness of the mixing parameter space and upper semicontinuity of the likelihood; the admissible fit $\delta_B$ is strictly positive at every observed count. Write $H = \widehat{G}$. The likelihood directional derivative gives

$$S(\lambda) = \sum_{j: q_j > 0} \frac{q_j K_\lambda(j)}{p_H(j)} \le 1 \quad (0 \le \lambda \le B).$$

Since $\int S\, dH = 1$, equality holds $H$-almost everywhere. Thus $R_\lambda(j) = q_j K_\lambda(j)/p_H(j)$ is a probability kernel $H$-almost everywhere and $\int R_\lambda\, dH = q$. Its entropy chain rule and the likelihood comparison yield

$$\int D(R_\lambda \| K_\lambda)\, dH(\lambda) = D(q \| p_H) \le D(q \| p_G). \tag{IV.2.6}$$

The binomial transport-entropy inequality is

$$W_{1,m}(R, \mathrm{Bin}(m,p))^2 \le D(R \| \mathrm{Bin}(m,p))/(2m).$$

For completeness, a normalized-count 1-Lipschitz function changes by at most $1/m$ on changing one of the $m$ Bernoulli bits. The conditional Hoeffding bound gives centered log-mgf at most $u^2/(8m)$; entropy duality and Wasserstein duality give the displayed inequality, including degenerate endpoints whenever the entropy is finite. Apply it to $A_m R_\lambda$, use entropy data processing, and integrate (IV.2.6). Convexity of transportation under mixtures and Cauchy–Schwarz give

$$W_{1,m}(A_m q, A_m p_H) \le \sqrt{D(q \| p_G)/(2m)}.$$

For each count threshold, the CDF of $A_m q$ is an average of independent $[0,1]$-valued functions of the $X_i$. Its expected absolute error is at most $1/(2\sqrt{N})$. The CDF formula for $W_{1,m}$ therefore gives $\mathbb{E} W_{1,m}(A_m q, A_m p_G) \le 1/(2\sqrt{N})$. Now use Lemma IV.2.2 and $(B+L)/m \le 2$. □

### IV.2.2 A growing-support Hellinger bound

**Lemma IV.2.4** (Poisson mixture entropy and likelihood control)**.** *For every fixed absolute $b_0 > 0$, uniformly over $b_0 L \le B \le N$,*

$$\mathbb{E} h(p_{\widehat{G}}, p_G) \le C_{b_0} B^{1/4} L^{3/4} / \sqrt{N}. \tag{IV.2.7}$$

*Proof.* For complex $z$ the vector $K_z = (e^{-z} z^j/j!)_{j\ge 0}$ belongs to $\ell_1$ and satisfies $\|K_z\|_1 = e^{|z|-\operatorname{Re} z}$. On the ellipse $z = B(1+\cos(\theta+iu))/2$, direct calculation gives

$$|z| - \operatorname{Re} z = \tfrac{B}{2}(\cosh u - 1)(1-\cos\theta) \le B(\cosh u - 1).$$

The $k$th vector Chebyshev coefficient $v_k$ of $x \mapsto K_{Bx}$ therefore satisfies, by contour shifting,

$$\|v_k\|_1 \le 2\exp\{B(\cosh u - 1) - ku\} \le 2\exp\{-c\min(k^2/B, k)\}.$$

For the second inequality take $u = k/(2B)$ if $k \le 2B$, using $\cosh u - 1 \le u^2$ for $0 \le u \le 1$, and take $u = 1$ otherwise. It follows by summing the Gaussian and exponential tails that truncating the expansion at

$$d = \left\lceil C\left\{\sqrt{B\log(C(B+1)/\beta)} + \log(C(B+1)/\beta)\right\}\right\rceil$$

has uniform $\ell_1$ error at most $\beta/8$. Every Chebyshev moment of a probability law on $[0,B]$ belongs to $[-1,1]$, and $\|v_k\|_1 \le 2$ also follows directly from its real integral formula. Grid the first $d$ moments with mesh $\beta/(16(d+1))$. Choose one actual mixture in each occupied grid box. This gives an $\ell_1$ cover of radius $\beta$, with centers in the mixture class and cardinality $M$ satisfying

$$\log M \le Cd\log(C(d+1)/\beta). \tag{IV.2.8}$$

Take $J = \lceil e^2 B + 6L\rceil$ and $\beta = N^{-3}/(J+1)$. In the indicated range, $d \le C_{b_0}\sqrt{BL}$ and $\log M \le C_{b_0}\sqrt{B}L^{3/2}$. For every cover center $p_a$, define the pointwise envelope

$$u_a(j) = \begin{cases} p_a(j) + \beta, & j \le J, \\ e^{-j}, & j > J. \end{cases}$$

Every mixture is bounded coordinatewise by its associated envelope, whose total mass $U_a$ lies in $[1, 1+\eta]$ for $\eta = CN^{-3}$, whenever that envelope is used for at least one mixture. Unused envelopes can be discarded. Normalize $r_a = u_a/U_a$. An associated mixture $p$ satisfies $h(p, r_a) \le \sqrt{2\eta}$.

Here is the elementary likelihood argument, to specify the treatment of the infinite alphabet. If $\widehat{p}$ is a likelihood maximizer and $u_a \ge \widehat{p}$, then

$$\prod_{i=1}^{N} \frac{r_a(X_i)}{p_G(X_i)} \ge e^{-N\eta}.$$

Under $p_G$, the expectation of the square root of this product is the $N$th power of the affinity, at most $\exp\{-Nh(p_G, r_a)^2/2\}$. Markov's inequality followed by a union bound shows, with probability at least $1 - e^{-x}$,

$$h(p_G, \widehat{p}) \le \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}.$$

Integration gives $\mathbb{E}h(p_G, \widehat{p})^2 \le C\eta + C(\log M + 1)/N$, which proves (IV.2.7). □

### IV.2.3 The two-scale argument closes every growing-support regime

Recall the already proved binomial inverse inequality: if $F, H$ are mixing laws on $[0,1]$, $b_F, b_H$ their $m$-trial binomial mixtures, and $1 \le S \le K \le m/2$, then

$$W_1(F,H) \le C/K + Ce^{4S^2/m} W_{1,m}(b_F, b_H) + Ce^{2K^2/m} h(b_F, b_H)/S. \tag{IV.2.9}$$

Its small high-frequency amplitude $1/S$ is indispensable. The proof approximates a Lipschitz test function at two polynomial degrees, preserves a bounded derivative at the lower degree, and uses

the degree-elevated Bernstein coefficient bounds. This is Proposition IV.A.2, not an additional unproved input.

Suppose $b_0 L \le B \le N$, with $b_0 > 0$ an absolute constant to be fixed below. Set $m = \lceil \max(B, L) \rceil$, so $m \asymp_{b_0} B$. For sufficiently large absolute $N$, take

$$S = \lfloor \sqrt{m}/16 \rfloor, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These are admissible, $S \asymp \sqrt{m}$, $e^{4S^2/m} \le C$, and $m/K \le C\sqrt{m/L}$ because $m \ge L$. Apply (IV.2.9) to the pushforwards under $\phi_m$, then use (IV.2.3), Lemmas IV.2.3 and IV.2.4, and Hellinger data processing. This gives

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \left\{ \sqrt{B/L} + \frac{B}{\sqrt{N}} + \frac{e^{2aL} B^{3/4} L^{3/4}}{\sqrt{N}} \right\}.$$

The ratio of the last term to $\sqrt{B/L}$ is at most $Ce^{2aL} N^{-1/4} L^{5/4}$, which is uniformly bounded since $B \le N$ and $2a < 1/4$. We have proved

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \{ \sqrt{B/L} + B/\sqrt{N} \}. \tag{IV.2.10}$$

Bounded $N$ is absorbed in the absolute constant by the diameter bound; this is uniform in $B$ because $B \ge b_0 L$ in this step.

If $B > N$, no Hellinger estimate or approximation is necessary. For any binomial mixing law, coupling its parameter $p$ with its normalized count gives expected cost at most $1/(2\sqrt{m})$. Thus

$$W_1(F, H) \le m^{-1/2} + W_{1,m}(b_F, b_H).$$

Together with (IV.2.3) and Lemma IV.2.3, this gives $\mathbb{E}W_1(\widehat{G}, G) \le C(\sqrt{B} + B/\sqrt{N}) \le CB/\sqrt{N}$. Every $B \ge L$ is now covered.

### IV.2.4 The small-support regime, including arbitrarily small radii

We include a direct proof to ensure uniformity as $B \downarrow 0$. Let $f$ be 1-Lipschitz on $[0, B]$ and normalize $f(0) = 0$. Jackson approximation gives a degree-$k$ polynomial $P(\lambda) = \sum_{l=0}^{k} a_l \lambda^l$ such that

$$\|f - P\|_{[0,B]} \le CB/k, \qquad \|P\|_{[0,B]} \le CB.$$

Its monomial coefficients satisfy

$$|a_l| \le BC_1^k B^{-l}. \tag{IV.2.11}$$

Indeed expand $P(Bx)$ in shifted Chebyshev polynomials: each Chebyshev coefficient has magnitude at most $2\|P\|$, and the sum of absolute monomial coefficients of $T_j(2x-1)$ is at most $7^j$ by its three-term recurrence. Summing the finite geometric series proves (IV.2.11) after enlarging $C_1$.

Take $k \ge 2eB$ and put $Q_k(\lambda) = \sum_{r=0}^{k} \lambda^r / r!$. The exponential remainder implies

$$\|e^{-\lambda} P(\lambda) Q_k(\lambda) - f(\lambda)\|_{[0,B]} \le CB/k.$$

For example, $0 \le 1 - e^{-\lambda} Q_k(\lambda) \le C(eB/(k+1))^{k+1} \le C2^{-k}$, which suffices. Write this weighted polynomial as $\sum_{j=0}^{2k} b_j K_\lambda(j)$. Since

$$b_j = \sum_{\substack{0 \le l \le k \\ 0 \le j-l \le k}} a_l \frac{j!}{(j-l)!},$$

equation (IV.2.11) gives

$$\max_{0\le j\le 2k}|b_j| \le B(C_2k/B)^k. \tag{IV.2.12}$$

The factors $k+1$ and $C_1^k$ have been absorbed into $C_2^k$; $2k/B \ge 1$ by hypothesis. Wasserstein duality consequently gives the deterministic inequality

$$W_1(F,H) \le CB/k + B(C_2k/B)^k\|p_F - p_H\|_1. \tag{IV.2.13}$$

For the MLE, likelihood comparison, $h^2 \le D$, the Hellinger triangle inequality and $\|p-q\|_1 \le 2h(p,q)$ imply

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_1 \le 4\sqrt{\mathbb{E}D(q\|p_G)} \le C\sqrt{L/N} \qquad (B \le L).$$

Put $d_0 = \log(e + L/B)$ and $k = \lfloor \gamma L/d_0 \rfloor$, where $\gamma > 0$ is a sufficiently small absolute constant. If $\gamma L/d_0 < 2$, the diameter bound $W_1 \le B$ is already at most a constant times $B\min(1, d_0/L)$. Otherwise $k \ge \gamma L/(2d_0)$. Choose a fixed $b_0 > 0$ sufficiently small that $B \le b_0 L$ implies $\gamma(L/B)/(2\log(e+L/B)) \ge 2e$; then $k \ge 2eB$. Furthermore

$$k\log(C_2k/B) \le C_3\gamma L \le L/8$$

after decreasing $\gamma$ once, since $\log(C_2\gamma(L/B)/d_0) \le C_3 d_0$. The stochastic term in (IV.2.13) is therefore at most $CBe^{L/8}\sqrt{L/N} \le CB/k$: here $k \le \gamma L$ and $e^{-3L/8}L^{3/2}$ is bounded. This proves the required rate for $B \le b_0 L$. On $b_0 L \le B \le L$, equation (IV.2.10) is bounded by an absolute constant, whereas $B\min(1, d_0/L)$ is bounded below by a positive absolute constant. This proves the upper bound in Theorem IV.2.1 for every $B$.

### IV.2.5 Matching lower bounds without support-range restrictions

We first record a convenient moment-matching pair. On an interval $[a-M, a+M]$, push forward the probability measures

$$(1 \pm \tfrac{1}{2}T_k(x))\frac{dx}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

under $x \mapsto a + Mx$. Denote them by $F_k^+, F_k^-$. They match all moments of degree less than $k$. Their CDF difference is, up to orientation, $\sin(k\theta)/(\pi k)$ for $x = \cos\theta$. Consequently

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\, d\theta \ge \frac{M}{\pi k}. \tag{IV.2.14}$$

The last inequality follows from $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1}$.

For $B \le L$, use this pair on $[0, B]$. Expanding $e^{-\lambda}$ and using the matched moments gives the elementary bound

$$\|p_{F_k^+} - p_{F_k^-}\|_1 \le 2\sum_{r\ge k}\frac{(2B)^r}{r!}. \tag{IV.2.15}$$

Indeed the term with observed count $j$ and exponential-expansion index $l$ vanishes if $j + l < k$; summing the absolute values of all remaining terms gives (IV.2.15). Set $d_0 = \log(e + L/B)$ and $k = \lceil 100L/d_0 \rceil$. Writing $t = L/B \ge 1$, we have $t = e^{d_0} - e \ge e^{d_0}/(e+1)$ and $e^{d_0/2}/d_0 \ge e/2$. These inequalities show

$$\log\frac{k}{2eB} \ge \frac{d_0}{2}, \qquad k\log\frac{k}{2eB} \ge 50L.$$

They also ensure $k \ge 4B$. Thus the tail in (IV.2.15) is at most $4(2eB/k)^k \le 4e^{-50L}$. The two $N$-sample laws have TV at most $2Ne^{-50L}$, whereas (IV.2.14) is at least $cB\min\{1, d_0/L\}$. Le Cam's two-point inequality proves the small-support lower bound, including arbitrarily tiny $B$.

For $B \ge L$, set $a = B/2$, $M = \sqrt{BL}/8$, and $k = \lceil 20L \rceil$. The moment-matching interval lies inside $[0, B]$. The exact Poisson identity

$$\sum_{x=0}^{\infty} \frac{K_\lambda(x) K_\mu(x)}{K_a(x)} = \exp\{(\lambda - a)(\mu - a)/a\}$$

shows, upon integrating against the signed difference $F_k^+ - F_k^-$, that

$$\sum_x \frac{(p_{F_k^+}(x) - p_{F_k^-}(x))^2}{K_a(x)} \le 4 \sum_{r \ge k} \frac{(M^2/a)^r}{r!} = 4 \sum_{r \ge k} \frac{(L/32)^r}{r!}.$$

All exchanges are justified by boundedness of $\lambda, \mu$ on the integration interval and the absolutely convergent exponential series. The last tail is at most $Ce^{-40L}$, for instance by $r! \ge (r/e)^r$ and a geometric-tail bound. Cauchy–Schwarz bounds the predictive TV by half the square root of this quantity. Le Cam's inequality and (IV.2.14) therefore give the lower rate $c\sqrt{B/L}$.

Finally consider mixing laws on $\{0, B\}$ with respective weights $1/2 \pm c_0/\sqrt{N}$ at $B$, for a sufficiently small absolute $c_0 > 0$. Their latent $N$-sample laws have TV bounded away from one, by the Bernoulli KL formula and Pinsker's inequality. The Poisson sample is a Markov image of that latent sample, so its TV is no larger. The mixing-law distance is $2c_0 B/\sqrt{N}$. Le Cam's inequality yields the remaining lower rate $cB/\sqrt{N}$. Taking the maximum completes the proof of Theorem IV.2.1.

The proof uses a common known exposure and a known compact support. It makes no assertion for heterogeneous exposures, unknown support radii, or an MLE constrained to a prescribed number of components. The gain over a direct one-scale inverse estimate is structural: the occupancy channel imports a binomial concentration inequality, likelihood contact supplies posterior entropy, and the two-scale inverse places predictive transportation and high-frequency Hellinger error in different terms.

## IV.3 Jacobi trace connections for all real parameters

Forrester and Kumar formulate the physical-density connection problem for the trace of the $\beta$-Jacobi ensemble and give its coefficients when $b$ and $\beta$ are nonnegative integers [84, Proposition 1.3 and Section 3.3]. The preceding research draft extended the formula to nonintegral local exponents, retaining $b > 0$. Here we remove both restrictions. The normalization at a resonant exponent is specified by a convergent simplex integral, and continuation uses an auxiliary system whose coefficients have no parameter poles. The result concerns the physical density in this canonical normalization, not a connection matrix between arbitrary choices of local bases.

### IV.3.1 Canonical corner functions

Fix $N \ge 1$, $a, b > -1$, and $\beta \ge 0$. Write

$$S_N = \int_{[0,1]^N} |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx, \qquad \rho(t) = \frac{1}{S_N} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx.$$

The density is zero outside $[0, N]$. At an integer it can diverge; statements about values there are understood by one-sided limits or as identities of locally integrable distributions. Define the Laguerre integrals

$$W_r(q) = \int_{(0,\infty)^r} e^{-\sum x_i} |\Delta(x)|^\beta \prod_i x_i^q \, dx = \prod_{j=0}^{r-1} \frac{\Gamma(q + 1 + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(1 + \beta/2)}, \qquad W_0 = 1.$$

The normalizing integral is

$$S_N = \prod_{j=0}^{N-1} \frac{\Gamma(a+1+j\beta/2)\Gamma(b+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(a+b+2+(N+j-1)\beta/2)\Gamma(1+\beta/2)}.$$

For $0 \le p \le N$, put $m = N-p$ and

$$\alpha_p = m(a+1) + \tfrac{\beta}{2}m(m-1), \qquad \eta_p = p(b+1) + \tfrac{\beta}{2}p(p-1),$$
$$D_p = \alpha_p + \eta_p, \qquad \gamma_p = D_p - 1,$$
$$Z_p = \frac{W_m(a)W_p(b)}{\Gamma(D_p)}.$$

In particular $D_p > 0$ and $Z_p > 0$.

On the simplex

$$\mathcal{D}_p = \{(u,v) \in [0,\infty)^m \times [0,\infty)^p : \sum_i u_i + \sum_j v_j = 1\},$$

use the measure $d\sigma$ obtained by eliminating the last coordinate with Jacobian one, equivalently the Dirac-delta simplex measure. For $N = 1$ it is the unit mass on the single point. Set

$$w_p(u,v) = |\Delta(u)|^\beta |\Delta(v)|^\beta \prod_i u_i^a \prod_j v_j^b,$$
$$A_p(s;u,v) = \prod_i (1-su_i)^b \prod_j (1+sv_j)^a \prod_{i,j} (1+s(v_j-u_i))^\beta,$$
$$U_p(s) = Z_p^{-1} \int_{\mathcal{D}_p} w_p(u,v) A_p(s;u,v)\, d\sigma, \qquad |s| < 1,$$
$$F_p(p+s) = s^{\gamma_p} U_p(s), \qquad 0 < s < 1.$$

All analytic factors in $A_p$ use the branches equal to one at $s = 0$. Empty products are one. Radial integration of the two Laguerre clusters gives $\int_{\mathcal{D}_p} w_p\, d\sigma = Z_p$. Consequently $U_p(0) = 1$. The integral converges for every stated parameter and is holomorphic in $s$ on $|s| < 1$. On compact subsets of the parameter domain it and all its $s$ derivatives depend continuously on the parameters. No division by $\sin(\pi D_p)$ occurs.

**Theorem IV.3.1** (All-parameter physical connection formula)**.** *For every $N \ge 1$, $a, b > -1$, and $\beta \ge 0$, each canonical $F_p$ has analytic continuation along paths avoiding $0, 1, \dots, N$. Let $F_p^{\uparrow}(t)$ denote continuation to the right with every intervening integer passed in the upper half-plane. Define*

$$C_p = e^{-\pi i \eta_p} \binom{N}{p} \frac{Z_p}{S_N}. \tag{IV.3.1}$$

*Then, for $k = 0, \dots, N-1$ and $k < t < k+1$,*

$$\boxed{\rho(t) = \sum_{p=0}^{k} C_p F_p^{\uparrow}(t).} \tag{IV.3.2}$$

*Equivalently, at every knot $p$, the physical right germ minus the upper continuation of the physical left germ is $C_p F_p$. This includes $p = N$, with zero physical density on its right. The lower-half-plane formula is the complex conjugate. All assertions include resonances $\gamma_p \in \mathbb{Z}$ and the entire range $-1 < b \le 0$.*

### IV.3.2 An auxiliary system that remains regular in the parameters

The following construction supplies both continuation and the normalization needed at resonance. Let $\widetilde{b} = b + 1 > 0$ and

$$\Lambda = -\operatorname{diag}(N, N-1, \dots, 0), \qquad h = a + \widetilde{b} + 1 + \tfrac{\beta}{2}(2N-1), \qquad k = \widetilde{b} + \tfrac{\beta}{2}N.$$

For indices $j = 0, \dots, N$, define the tridiagonal matrix $X$ by

$$\begin{aligned} X_{j,j-1} &= -jk + \tfrac{\beta}{2}j^2, \\ X_{j,j} &= hj - \tfrac{\beta}{2}(N-j)(2j+1) + k(N-2j) - 1, \\ X_{j,j+1} &= h(j-N) + \tfrac{\beta}{2}(N-j)(N-j-1) + k(N-j). \end{aligned}$$

Missing entries are zero. Write $P_{j\ell} = \binom{N-j}{N-\ell}$, with a binomial coefficient zero outside its usual range. The Forrester–Kumar system is

$$(tI + \Lambda)G'(t) = XG(t), \qquad H = PG. \tag{IV.3.3}$$

With the shifted parameter, its physical auxiliary vector is

$$H_j(t) = \binom{N}{j}^{-1} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b e_j(1-x_1, \dots, 1-x_N)\, dx.$$

Thus $H_0 = S_N \rho$, and every defining integral is absolutely convergent throughout the parameter range of the theorem.

**Lemma IV.3.2** (Explicit corner vectors and parameter-continuous transport)**.** *For a fixed $p$, define the vector germ*

$$\begin{aligned} H_{p,j}(p+s) = \frac{s^{\gamma_p}}{\binom{N}{j}} \int_{\mathcal{D}_p} & w_p A_p(s; u, v) \\ & \cdot e_j(1 - su_1, \dots, 1 - su_m, -sv_1, \dots, -sv_p)\, d\sigma. \end{aligned} \tag{IV.3.4}$$

*Then $G_p = P^{-1}H_p$ solves (IV.3.3), and $H_{p,0} = Z_p F_p$. Its continuation along any fixed path avoiding the integers is continuous in all real parameters $a, b > -1$, $\beta \geq 0$.*

*Proof.* First take $a, b, \beta$ sufficiently large and positive. The vector integral is the slice integral on the cone having $m$ coordinates $x_i \geq 0$ and $p$ coordinates $x_j \geq 1$, with $x_i = su_i$ and $x_j = 1 + sv_j$. Restrict to $0 < s < 1$. The slice is compact, the lower coordinates stay below one, and the upper coordinates exceed one. Cross-cluster differences are $1 + s(v_j - u_i) > 0$. The upper factors $(x_j - 1)^b$ differ from the corresponding analytic Jacobi powers by a constant phase, so their logarithmic derivatives are the same. Sum over the choices of the upper coordinates to make the domain symmetric; this multiplies every vector component by the same factor $\binom{N}{p}$.

The integration-by-parts identities giving (IV.3.3) therefore apply on this cone as well. Endpoint and within-cluster collision boundary terms vanish for the chosen large parameters; the compact slice has no boundary at infinity. Equivalently these identities may be tested against a smooth compactly supported function of $s \in (0, 1)$ before integration by parts. The change of variables gives exactly (IV.3.4), including its factor $s^{D_p - 1}$ and its elementary symmetric polynomials.

For fixed $0 < s < 1$, the vector integral and its derivative are holomorphic for $\operatorname{Re} a, \operatorname{Re} b > -1$ and $\operatorname{Re} \beta > 0$. On compact subsets the endpoint factors have a common integrable majorant; the collision powers only improve integrability. The identity theorem continues the differential

identity from the large-parameter set to this connected domain. Dominated convergence includes $\beta = 0$. The zeroth coordinate is $Z_p F_p$ by definition. More explicitly, the leading coefficients are

$$H_{p,j}(p+s) = Z_p s^{\gamma_p} \left( \frac{\binom{N-p}{j}}{\binom{N}{j}} + O(s) \right), \qquad G_p(p+s) = \frac{Z_p}{\binom{N}{p}} s^{\gamma_p} \left( e_{N-p} + O(s) \right),$$

because $\binom{N-p}{j}/\binom{N}{j} = P_{j,N-p}/\binom{N}{p}$.

Finally, away from the integers, the coefficient matrix $(tI + \Lambda)^{-1}X$ is holomorphic in $t$ and polynomial in the parameters. Its fundamental matrix along a fixed compact path depends continuously, indeed locally holomorphically, on these parameters. Start at any fixed $s_0 \in (0,1)$ with the explicit vector integral. This proves the claimed continuation and parameter dependence without dividing by resonant exponent differences. □

### IV.3.3 Determination of the coefficients off resonance

We give the local argument in order to distinguish the complex connection coefficient from a one-sided real singular coefficient. At $t = p$, the residue of the shifted system has only row $N - p$ nonzero, and its nonzero eigenvalue is $\gamma_p$. If $\gamma_p$ is nonintegral, the residue is diagonalizable with eigenvalues 0 and $\gamma_p$. The regular-singular recurrence gives $N$ holomorphic solutions and one nonintegral solution. The zeroth $H$ coordinate of its leading vector is nonzero because $P_{0,N-p} = \binom{N}{p} \neq 0$. Lemma IV.3.2 therefore identifies its normalized zeroth coordinate with $F_p$.

For a smooth cutoff $\chi$ equal to one near $p$, the complete local Fourier expansion of the physical density is

$$\int e^{iRt} \chi(t) \rho(t)\, dt \sim e^{ipR} R^{-D_p} \sum_{\ell \geq 0} A_{p,\ell} R^{-\ell}, \quad A_{p,0} = K_p e^{\pi i(\alpha_p - \eta_p)/2}, \quad K_p = \binom{N}{p} \frac{W_m(a) W_p(b)}{S_N}. \tag{IV.3.5}$$

Here and below completeness means a remainder to any prescribed finite order. To verify it, near a corner write the lower coordinates as $x$ and the upper coordinates as $1 - y$. The phase is $p + \sum x - \sum y$, and the weight is

$$|\Delta(x)|^{\beta} |\Delta(y)|^{\beta} \prod x_i^a \prod y_j^b \prod_i (1 - x_i)^b \prod_j (1 - y_j)^a \prod_{i,j} (1 - x_i - y_j)^{\beta}.$$

Its final three factors are analytic near the corner. Radial variables $r = \sum x_i$ and $q = \sum y_j$ have powers $r^{\alpha_p - 1}$ and $q^{\eta_p - 1}$. Taylor expansion of the analytic factor and successive one-dimensional endpoint Fourier integrals give only the powers $R^{-D_p - \ell}$. The angular constants are the two Laguerre integrals divided by $\Gamma(\alpha_p)$ and $\Gamma(\eta_p)$; empty clusters are omitted. This proves the leading coefficient in (IV.3.5) and the entire power scale.

Away from these corners, a cluster of coincident coordinates lies strictly inside $(0,1)$. Translating that whole cluster leaves its internal differences fixed and gives a nonzero phase derivative. It also stays tangent to any other fixed boundary faces. A local partition of unity and repeated integration by parts in that direction give arbitrarily rapid decay. All differentiated noncluster factors are smooth on the selected neighborhood, while the remaining endpoint and collision powers are integrable. Thus there are no other power contributions at frequency $p$.

Suppose now that $D_p$ is nonintegral. On the two sides of $s = 0$, regular-singular theory gives

$$\rho(p+s) = \begin{cases} a_+(s) + c_+ s^{\gamma_p} U_p(s), & s > 0, \\ a_-(s) + c_- (-s)^{\gamma_p} U_p(s), & s < 0, \end{cases}$$

where $a_\pm$ are holomorphic and $c_\pm$ are real. If the analytic backgrounds first differ at Taylor degree $q$, their cutoff Fourier transform has a nonzero $R^{-q-1}$ term. This contradicts (IV.3.5),

whose powers are all nonintegral. Hence $a_+ = a_-$. Comparing the leading coefficient yields

$$c_+ = \frac{K_p \sin(\pi\alpha_p)}{\Gamma(D_p)\sin(\pi D_p)}, \qquad c_- = \frac{K_p \sin(\pi\eta_p)}{\Gamma(D_p)\sin(\pi D_p)}.$$

Upper continuation changes $(-s)^{\gamma_p}$ into $e^{-\pi i\gamma_p} s^{\gamma_p}$. Therefore the discrepancy is

$$(c_+ - e^{-\pi i\gamma_p} c_-)F_p = e^{-\pi i\eta_p} \frac{K_p}{\Gamma(D_p)} F_p = C_p F_p.$$

The sine identity uses $D_p = \alpha_p + \eta_p$. It includes both endpoints with the exterior density zero. Starting from the first interval and successively adding these discrepancies proves (IV.3.2) whenever all relevant $\gamma_p$ are nonintegral.

### IV.3.4 Crossing every resonant hyperplane

*Completion of Theorem IV.3.1.* Fix any admissible parameters and a noninteger $t$. Approximate the parameters by admissible real triples with $\beta > 0$ and all $\gamma_p$ nonintegral; the excluded affine hyperplanes are locally finite. Each $C_p$ has a finite continuous limit. By Lemma IV.3.2, so does each continued $F_p^{\uparrow}(t)$, using the same paths and starting points throughout.

The physical slice integral is continuous in the parameters at this fixed $t$. To see a common integrable bound, note that, since $t$ is not an integer, at least one coordinate of every point of the slice is bounded away from both zero and one. A finite cover selects such a coordinate and eliminates it with Jacobian one. Its endpoint factors are bounded, the other endpoint factors have a uniform integrable majorant with powers greater than $-1$, and $|\Delta|^\beta \le 1$ for real $\beta \ge 0$ on the cube. Dominated convergence now applies on each patch. The normalizer is positive and continuous as well.

Pass to the limit in the nonresonant identity to obtain (IV.3.2) at the chosen parameters. The same argument applies to the local discrepancy, using ODE continuation for the left germ and zero exterior density at the endpoints. It also includes $\beta = 0$. No one-sided formula containing $1/\sin(\pi D_p)$ has been evaluated at a pole: the continued canonical functions and their connection coefficients are separately finite before the limit is taken. □

**Example IV.3.3** (Complete resonance and logarithmic resonance)**.** If $a = b = \beta = 0$, then $U_p = 1$ and $Z_p = 1/(N-1)!$. The theorem gives the familiar identity

$$\rho(t) = \frac{1}{(N-1)!} \sum_{p=0}^{\lfloor t \rfloor} (-1)^p \binom{N}{p} (t-p)^{N-1}, \qquad 0 < t < N,$$

with the sum taken away from integer $t$. Every exponent is resonant. For $N = 1$ it gives the beta density directly. For $N = 2$, $a = b = -1/2$, $\beta = 0$, the physical density has a logarithmic singularity at $t = 1$, while $\gamma_1 = 0$ and $U_1$ is the convergent analytic simplex integral above. The logarithm appears in the continued earlier branch; it does not force a pole in $C_1$ or in the canonical $U_1$.

This removes the nonresonance and auxiliary-integrability restrictions of the preceding connection theorem for every integrable real Jacobi parameter with $\beta \ge 0$. The system itself and its classical Selberg constants are inherited from the cited literature. No claim is made about stable floating-point evaluation for large $N$, arbitrary-basis connection matrices, or a complete historical-priority assessment of related Selberg connection formulas.

## IV.4 Exact designs and a sharp extremal constant on block graphs

The universal connected-graph comparison leaves a factor of two between its upper and lower asymptotic constants. We close that gap for the entire class of connected *block graphs*, namely graphs whose blocks are complete graphs, with every bridge regarded as a block $K_2$. This class contains all trees, all complete graphs, and arbitrary configurations of cliques joined at articulation vertices. In addition to its sharp leading constant, it admits an exact finite-graph formula for every optimal conductance. We do not claim that the remaining constant gap on arbitrary connected graphs is closed.

For clarity, a design consists of conductances $w_e \geq 0$ with $\sum_{e\in E} w_e = 1$ and connected positive-weight support. Choose an orientation of each edge and let $b_e = e_u - e_v$; its Laplacian is $L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}$. The $D$-optimal design $w^D$ maximizes $\det(L(w)|_{\mathbf{1}^\perp})$, while the $A$-optimal design $w^A$ minimizes the unordered-pair total effective resistance

$$K_G(w) = n \operatorname{tr} L(w)^\dagger.$$

Throughout this section, $\alpha(G) = K_G(w^D)/K_G(w^A)$.

### IV.4.1 The demand matrix of a block

Let $G$ be a connected block graph with $n \geq 2$ vertices and $d = n - 1$. Write $\mathcal{B}$ for its blocks. If $B \in \mathcal{B}$ has vertex set $\{1, \dots, m_B\}$, delete all edges of $B$ from $G$. The resulting components contain exactly one vertex of $B$ each. Let their sizes be

$$a_{B,1}, \dots, a_{B,m_B}, \qquad a_{B,i} \geq 1, \qquad \sum_i a_{B,i} = n.$$

Define the block demand Laplacian

$$C_B = n \operatorname{diag}(a_B) - a_B a_B^{\mathsf{T}}. \tag{IV.4.1}$$

It is the complete-graph Laplacian with conductance $a_{B,i}a_{B,j}$ on the pair $\{i, j\}$. Put

$$q_B = \operatorname{tr} C_B^{1/2}, \qquad Q = \sum_{B\in\mathcal{B}} q_B.$$

All square roots here are the positive semidefinite matrix square roots.

The unordered-pair total effective resistance decomposes by blocks:

$$K_G(w) = \sum_{B\in\mathcal{B}} \sum_{i<j} a_{B,i}a_{B,j} R_{ij}(L_B) = \sum_{B\in\mathcal{B}} \operatorname{tr}(C_B L_B^\dagger). \tag{IV.4.2}$$

Indeed, the block-cut tree specifies the blocks traversed by each pair of vertices. Resistances add along that chain, and precisely $a_{B,i}a_{B,j}$ pairs project to distinct vertices $i, j$ of $B$.

**Lemma IV.4.1** (Feasibility of the square-root block design)**.** *For each block $B$, the matrix $C_B^{1/2}$ has zero row sums and strictly negative off-diagonal entries. It is therefore a feasible positive-conductance Laplacian on the complete block $B$.*

*Proof.* Zero row sums follow from $C_B\mathbf{1} = 0$ and spectral calculus. For a positive semidefinite matrix $C$, the scalar integral for the square root gives

$$C^{1/2} = \frac{1}{\pi} \int_0^\infty t^{-1/2} C(tI + C)^{-1}\, dt.$$

For the special matrix in (IV.4.1), put $D_t = tI + n\,\mathrm{diag}(a_B)$. Sherman–Morrison gives

$$(tI + C_B)^{-1} = D_t^{-1} + \frac{D_t^{-1} a_B a_B^{\mathsf{T}} D_t^{-1}}{1 - a_B^{\mathsf{T}} D_t^{-1} a_B}.$$

The denominator is strictly positive for $t > 0$, since

$$\sum_i \frac{a_{B,i}^2}{t + n a_{B,i}} < \sum_i \frac{a_{B,i}}{n} = 1.$$

Every off-diagonal resolvent entry is therefore positive. For $i \neq j$,

$$\big[C_B(tI + C_B)^{-1}\big]_{ij} = -t\big[(tI + C_B)^{-1}\big]_{ij} < 0.$$

Integration proves the claimed strict sign. □

### IV.4.2 Exact optimal weights and a spectral moment formula

**Theorem IV.4.2** (Exact designs on every connected block graph)**.** *The $A$-optimal Laplacian on each block is*

$$L_B^A = \frac{2}{Q} C_B^{1/2}, \qquad w_{ij}^A = -\frac{2}{Q}(C_B^{1/2})_{ij} \quad (i \neq j). \tag{IV.4.3}$$

*The total conductance assigned to $B$ is $q_B/Q$, and*

$$K_G(w^A) = \frac{Q^2}{2}.$$

*The $D$-optimal conductance on every edge of a block of size $m_B$ is*

$$w_e^D = \frac{2}{m_B d}, \qquad K_G(w^D) = \frac{d}{2} \sum_B \operatorname{tr} C_B.$$

*If $\lambda_{B,1}, \dots, \lambda_{B,m_B-1}$ are the nonzero eigenvalues of $C_B$, then the approximation ratio is exactly*

$$\alpha(G) = \frac{d \sum_{B,j} \lambda_{B,j}}{\left(\sum_{B,j} \sqrt{\lambda_{B,j}}\right)^2}. \tag{IV.4.4}$$

*There are exactly $d$ eigenvalues in these sums.*

*Proof.* Fix a positive budget $s_B$ for block $B$, so $\operatorname{tr} L_B = 2s_B$. On its subspace $\mathbf{1}^\perp$, matrix Cauchy–Schwarz gives

$$(\operatorname{tr} C_B^{1/2})^2 \le \operatorname{tr}(C_B L_B^{-1}) \operatorname{tr} L_B.$$

Equality holds precisely when $L_B$ is proportional to $C_B^{1/2}$. By Lemma IV.4.1, that choice is feasible; hence the block minimum is $q_B^2/(2s_B)$, attained at $L_B = (2s_B/q_B) C_B^{1/2}$. Allocating the total unit budget among the blocks yields

$$\min_{s_B>0,\ \sum_B s_B = 1} \sum_B \frac{q_B^2}{2s_B} = \frac{1}{2}\left(\sum_B q_B\right)^2, \qquad s_B = \frac{q_B}{Q}.$$

This proves (IV.4.3) and its global optimality.

The block-cut tree identity is $\sum_B (m_B - 1) = n - 1 = d$. Consequently the stated $D$-weights sum to one. Within a uniformly weighted $K_{m_B}$, every pair has resistance $2/(m_B w_e^D) = d$; attachments at a single articulation vertex do not alter that resistance. Thus every candidate edge has effective resistance $d$, verifying the determinant-optimality conditions. Using (IV.4.2) now gives

$$K_G(w^D) = d \sum_B \sum_{i<j} a_{B,i} a_{B,j} = \frac{d}{2} \sum_B \operatorname{tr} C_B.$$

Dividing by the $A$-optimal value proves (IV.4.4). The same block-cut identity counts the nonzero eigenvalues. □

### IV.4.3 A common spectral interval and the sharp constant

**Lemma IV.4.3.** *Every nonzero eigenvalue of every $C_B$ lies in*

$$n \le \lambda \le n^2/2.$$

*Proof.* Suppress the block subscript and put $D = \operatorname{diag}(a)$, $u = (\sqrt{a_i})_i$, and $P = I - uu^{\mathsf{T}}/n$. Since $\|u\|^2 = n$, $P$ is an orthogonal projection and

$$C = nD^{1/2}PD^{1/2}.$$

The nonzero eigenvalues of $D^{1/2}PD^{1/2}$ equal those of $PDP$ on range $P$. There the quadratic form is at least $\min_i a_i \ge 1$, which proves the lower bound. For the upper bound, the diagonal entry in row $i$ of $C$ is $a_i(n - a_i)$, and the sum of the absolute off-diagonal entries is the same. Gershgorin's bound therefore gives

$$\lambda \le 2 \max_i a_i(n - a_i) \le n^2/2.$$

□

**Theorem IV.4.4** (Sharp leading constant for all block graphs)**.** *Every connected $n$-vertex block graph satisfies*

$$\alpha(G) \le \frac{\sqrt{n}}{4\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2n}}. \tag{IV.4.5}$$

*If $\mathcal{B}_n$ is the largest ratio among all such graphs, then*

$$\lim_{n\to\infty} \frac{\mathcal{B}_n}{\sqrt{n}} = \frac{1}{4\sqrt{2}}. \tag{IV.4.6}$$

*Proof.* Set $x_{B,j} = \sqrt{\lambda_{B,j}}$, $a = \sqrt{n}$, and $b = n/\sqrt{2}$. There are $d$ such numbers and each lies in $[a, b]$. By (IV.4.4), the ratio is their empirical second moment divided by the squared empirical mean. The pointwise inequality $(x - a)(x - b) \le 0$ gives

$$\alpha(G) \le \frac{(a+b)^2}{4ab} = \frac{\sqrt{n}}{4\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2n}},$$

proving the upper bound.

For the lower bound, take two $K_m$ blocks joined at distinguished roots by a path with $p$ edges, with

$$n = 2m + p - 1, \qquad p = \sqrt{2n} + O(1).$$

An appropriate parity choice of $p$ gives such a graph for every sufficiently large $n$. Put $h = n-m+1$,

$$q = \sqrt{mh/2} + (m-2)\sqrt{n/2}, \qquad c_j = (m + j - 1)(m + p - j).$$

The exact design formulas reduce to

$$K_G(w^A) = \left(2q + \sum_{j=1}^{p} \sqrt{c_j}\right)^2, \qquad K_G(w^D) = d\left((m-1)(2n-m) + \sum_{j=1}^{p} c_j\right).$$

For example, the nonzero eigenvalues of an end-clique demand matrix are $n$ with multiplicity $m - 2$ and $mh$ once, which gives $q$; a bridge contributes $\sqrt{c_j}$ to the square-root budget. Since $m = n/2 + O(\sqrt{n})$,

$$2q = \frac{n^{3/2}}{\sqrt{2}} + O(n), \qquad \sum_j \sqrt{c_j} = \frac{n^{3/2}}{\sqrt{2}} + O(n),$$

$$K_G(w^D) = \frac{n^{7/2}}{2\sqrt{2}} + O(n^3), \qquad K_G(w^A) = 2n^3 + O(n^{5/2}).$$

Thus $\alpha(G) = \sqrt{n}/(4\sqrt{2}) + O(1)$. These graphs belong to the block-graph class, so the upper and lower bounds prove (IV.4.6). □

**Proposition IV.4.5** (Exact spectral deficit identity)**.** *Let $x$ range uniformly over the $d$ numbers in the preceding proof, and define*

$$\mu = \mathbb{E}x, \qquad \delta = \mathbb{E}[(x-a)(b-x)].$$

*Then*

$$\frac{(a+b)^2}{4ab} - \alpha(G) = \frac{((a+b)\mu - 2ab)^2}{4ab\mu^2} + \frac{\delta}{\mu^2}. \tag{IV.4.7}$$

*In particular, equality in the finite bound requires and is implied by all $x \in \{a, b\}$ and $\mu = 2ab/(a+b)$. If a sequence of block graphs attains the leading constant in (IV.4.6), then*

$$\frac{\mu}{2\sqrt{n}} \longrightarrow 1, \qquad \delta = o(n^{3/2}).$$

*Proof.* The identity $\mathbb{E}x^2 = (a+b)\mu - ab - \delta$ gives (IV.4.7) by completing the square. Both terms on its right are nonnegative. Dividing the first term by $(a+b)^2/(4ab)$ shows that a vanishing relative deficit forces $(a+b)\mu/(2ab) \to 1$. Since $a/b \to 0$, this gives $\mu/(2\sqrt{n}) \to 1$. The second term then gives $\delta = o(n^{3/2})$. □

The stability statement concerns the block demand spectra; it does not assert that all nearly extremal graphs are barbells. For a noncomplete block, the matrix $C_B^{1/2}$ may require edges absent from the graph, so its trace-optimal design need not be feasible. Completing such blocks is not a valid monotonicity argument for the ratio. Accordingly, the exact leading constant on all connected graphs, and a full combinatorial characterization of their extremizers, remain open in this manuscript.

## IV.5 A sharp limit to universal robustness of mixture likelihood

The component concentration needed for entropy–transport does not, by itself, make an ordinary mixture maximum-likelihood estimator optimal under contamination. We prove this in a fixed, identifiable, one-dimensional model. Both Gaussian components are known, their variances equal one, and only their mixing weight is estimated. The comparison is uniform in the sample size and contamination level. An exact population bias equation identifies the mechanism behind the finite-sample separation.

### IV.5.1 Model and the two risk laws

Let $f_j$ denote the density of $N(j, 1)$, $j = 0, 1$, and put

$$p_w = (1-w)f_0 + wf_1, \qquad G_w = (1-w)\delta_0 + w\delta_1, \qquad 0 \le w \le 1.$$

Write $P_w$ for the observed distribution with density $p_w$. Then $W_1(G_w, G_v) = |w - v|$. Throughout this section, $\mathrm{TV}(P, Q) = \frac{1}{2}\int |p - q|$.

For the strong contamination experiment, start with independent $X_1, \dots, X_N \sim P_w$. An adversary, possibly randomized and depending on the whole clean sample, produces $Y_1, \dots, Y_N$ by replacing at most $k = \lfloor \varepsilon N \rfloor$ coordinates. Denote the class of these transformations by $\mathcal{A}_{N,k}$.

In the Huber experiment the $Y_i$ are independent with common law $(1-\varepsilon)P_w + \varepsilon Q$, where $Q$ is any probability measure on $\mathbb{R}$.

An ordinary MLE is any measurable choice

$$\widehat{w} \in \operatorname*{argmax}_{0\le v\le 1} \sum_{i=1}^{N} \log p_v(Y_i).$$

Such choices exist: the objective is finite and continuous on a compact interval for every finite real sample. Bounds below hold for every measurable choice, including possible nonunique maximizers.

**Theorem IV.5.1** (Sharp finite-sample separation)**.** *There are numerical constants $0 < c < C < \infty$ such that, for every $N \ge 1$ and $0 \le \varepsilon \le 1/4$,*

$$c\min\{1, N^{-1/2} + \sqrt{\varepsilon}\} \le \sup_{w\in[0,1]} \sup_{A\in\mathcal{A}_{N,\lfloor \varepsilon N\rfloor}} \mathbb{E}_w|\widehat{w}(A(X)) - w| \le C\min\{1, N^{-1/2} + \sqrt{\varepsilon}\}, \quad \text{(IV.5.1)}$$

$$c\min\{1, N^{-1/2} + \varepsilon\} \le \inf_{\widetilde{w}} \sup_{w\in[0,1]} \sup_{A\in\mathcal{A}_{N,\lfloor \varepsilon N\rfloor}} \mathbb{E}_w|\widetilde{w}(A(X)) - w| \le C\min\{1, N^{-1/2} + \varepsilon\}. \quad \text{(IV.5.2)}$$

*The infimum is over measurable estimators with values in $[0,1]$. The same two inequalities hold when the adversarial supremum is replaced by the supremum over $Q$ in the Huber experiment. An estimator attaining the minimax upper bound does not need to know $\varepsilon$.*

Thus, along $\varepsilon \to 0$ and $N\varepsilon \to \infty$, the ratio between the worst MLE risk and the minimax risk tends to infinity. This is a statement about the two-component dictionary just specified. It does not by itself prove a lower bound for the unrestricted Gaussian location-mixture NPMLE with a continuum of allowed components.

### IV.5.2 Uniform upper bound for every likelihood maximizer

Set

$$r(x) = \frac{f_1(x)}{f_0(x)} = e^{x-1/2}, \qquad U(x) = r(x) - 1, \qquad a = \int_{\mathbb{R}} \frac{(f_1 - f_0)^2}{f_0 + f_1}\, dx > 0.$$

Let $P_N$ be the clean empirical measure, let $q$ be the corrupted empirical measure, and suppose the true parameter is $w_0$. The likelihood directional condition towards $p_{w_0}$ gives

$$q\left(\frac{p_{w_0}}{p_{\widehat{w}}}\right) \le 1. \quad \text{(IV.5.3)}$$

This condition remains valid at an endpoint maximizer because the direction towards $w_0$ is feasible. The function $g(z) = (1-z)/(1+z)$ is convex and decreasing for $z \ge 0$. Jensen's inequality applied to (IV.5.3) therefore implies

$$q\phi \ge 0, \qquad \phi = \frac{p_{\widehat{w}} - p_{w_0}}{p_{\widehat{w}} + p_{w_0}}.$$

Put $\Delta = \widehat{w} - w_0$ and $v = (\widehat{w} + w_0)/2$. Then

$$\phi = \Delta\psi_v, \qquad \psi_v(x) = \frac{U(x)}{2(1 + vU(x))}.$$

The equality of the total masses of $p_{\widehat{w}}$ and $p_{w_0}$ gives the exact identity

$$P_{w_0}\phi = -\frac{1}{2}\int_{\mathbb{R}} \frac{(p_{\widehat{w}} - p_{w_0})^2}{p_{\widehat{w}} + p_{w_0}}\, dx \le -\frac{a}{4}\Delta^2,$$

where the inequality uses $p_{\widehat{w}} + p_{w_0} \le 2(f_0 + f_1)$. Since $|\phi| \le 1$, replacing at most $k$ observations changes its empirical integral by at most $2k/N$, even for this data-dependent function. Consequently, with $\delta = k/N$,

$$\frac{a}{4}\Delta^2 \le |\Delta| Z_N + 2\delta, \qquad Z_N = \sup_{0 \le v \le 1} |(P_N - P_{w_0})\psi_v|. \tag{IV.5.4}$$

It remains to control $Z_N$ without a logarithmic loss. Direct differentiation yields

$$0 \le \psi_v'(x) = \frac{r(x)}{2(1 + v(r(x) - 1))^2} \le \frac{1}{2} e^{|x - 1/2|}.$$

Let $F_N, F_{w_0}$ be the corresponding empirical and true distribution functions. Integration by parts gives

$$Z_N \le \frac{1}{2} \int_{\mathbb{R}} e^{|x-1/2|} |F_N(x) - F_{w_0}(x)| \, dx.$$

The boundary terms vanish also for $v = 0, 1$: $\psi_v$ grows at most exponentially, the true tails are Gaussian, and an empirical sample has finite extrema. The variance of the empirical distribution function gives

$$\mathbb{E} Z_N \le \frac{1}{2\sqrt{N}} \int_{\mathbb{R}} e^{|x-1/2|} \sqrt{F_{w_0}(x)(1 - F_{w_0}(x))} \, dx \le \frac{C_0}{\sqrt{N}}. \tag{IV.5.5}$$

The last constant is uniform over $w_0 \in [0, 1]$. For example, on the left tail $F_{w_0} \le \Phi(x)$, and on the right tail $1 - F_{w_0} \le 1 - \Phi(x - 1)$; these bounds have Gaussian decay. Solving (IV.5.4) gives

$$|\Delta| \le \frac{4}{a} Z_N + \sqrt{\frac{8\delta}{a}}. \tag{IV.5.6}$$

Together with $|\Delta| \le 1$, this proves the strong-contamination upper bound in (IV.5.1). In particular, the argument requires no moment condition on the replacement values.

### IV.5.3 A finite outlier attack attaining the square-root term

We first establish a clean-sample event whose probability is uniformly positive. Under $P_0$, let $U = e^{X - 1/2} - 1$, so

$$\mathbb{E}_0 U = 0, \qquad A := \mathbb{E}_0 U^2 = e - 1, \qquad B := \mathbb{E}_0 U^4 = e^6 - 4e^3 + 6e - 3.$$

For $n \ge 1$ independent copies, put $S = n^{-1} \sum_{i=1}^n U_i$ and $V = n^{-1} \sum_{i=1}^n U_i^2$. Then

$$\mathbb{E} S^2 = A/n, \qquad \mathbb{E} S^4 \le (B + 3A^2)/n^2.$$

Interpolation between the first, second, and fourth moments gives

$$\mathbb{E}|S| \ge \frac{(\mathbb{E} S^2)^{3/2}}{(\mathbb{E} S^4)^{1/2}} \ge \frac{A^{3/2}}{\sqrt{B + 3A^2}\sqrt{n}}.$$

Since $\mathbb{E} S_+ = \mathbb{E}|S|/2$, Cauchy–Schwarz implies

$$\mathbb{P}(S > 0) \ge p_* := \frac{A^2}{4(B + 3A^2)} > 0.$$

Choose $M = 2A/p_*$. Markov's inequality, using $\mathbb{E} V = A$, gives

$$\mathbb{P}(S \ge 0, \ V \le M) \ge p_*/2. \tag{IV.5.7}$$

Suppose $1 \le k \le N/4$, set $\delta = k/N$, and take true parameter $w_0 = 0$. Replace a predetermined set of $k$ observations by a common value $z$, leaving $n = N - k$ independent $P_0$ observations. Choose a numerical $c_0 > 0$ such that $c_0 \le 1/2$ and $c_0^2 \le 1/(16M)$, and put

$$w_* = c_0\sqrt{\delta}, \qquad z = \frac{1}{2} + \log(1 + 1/w_*). \tag{IV.5.8}$$

In particular, this is a finite replacement value. On the event in (IV.5.7), the average clean likelihood score at $w_*$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n} \frac{U_i}{1 + w_* U_i} = S - \frac{w_*}{n}\sum_{i=1}^{n} \frac{U_i^2}{1 + w_* U_i} \ge -2M w_*,$$

because $U_i \ge -1$ and $w_* \le 1/2$. The choice of $z$ gives $U(z) = 1/w_*$, so its score equals $1/(2w_*)$. The normalized corrupted log likelihood has derivative at least

$$-2M w_* + \frac{\delta}{2w_*} > 0.$$

By concavity every maximizer therefore exceeds $w_*$. It follows that every measurable MLE has expected error at least $(p_* c_0/2)\sqrt{k/N}$. If $k = \lfloor \varepsilon N \rfloor \ge 1$, then $k/N \ge \varepsilon/2$. If $k = 0$, then $\sqrt{\varepsilon} \le N^{-1/2}$; the clean lower bound proved next already covers this case.

### IV.5.4 The minimax rate and the clean lower bound

Let

$$b = \Phi(-1/2), \qquad d = 2\Phi(1/2) - 1 > 0.$$

The threshold probability is affine in the parameter: $P_w(X > 1/2) = b + dw$. Hence the estimator

$$\widetilde{w} = \Pi_{[0,1]}\left\{ \frac{N^{-1}\sum_{i=1}^{N} \mathbf{1}_{\{Y_i > 1/2\}} - b}{d} \right\} \tag{IV.5.9}$$

satisfies

$$\mathbb{E}|\widetilde{w} - w| \le \frac{1}{d}\left( \frac{1}{2\sqrt{N}} + \frac{k}{N} \right)$$

under strong contamination. Here $\Pi_{[0,1]}$ denotes projection onto the interval, which cannot increase the distance to $w$. This proves the minimax upper bound without knowing the corruption budget.

For clarity, a clean lower bound of order $N^{-1/2}$ holds uniformly for every estimator. Choose $w_\pm = 1/2 \pm h/\sqrt{N}$, with a fixed sufficiently small $h > 0$. Both parameters lie in $[1/4, 3/4]$, and

$$\chi^2(P_{w_+}, P_{w_-}) = \frac{4h^2}{N}\int \frac{(f_1 - f_0)^2}{p_{w_-}} \le \frac{16ah^2}{N}.$$

The product chi-squared divergence is at most $\exp(16ah^2) - 1$. Taking $h$ so this is at most $1/4$ bounds the product total variation by $1/4$. The two-point testing inequality then yields expected absolute error at least $c/\sqrt{N}$ for one of the two parameters. This applies in both contamination experiments: the strong adversary may leave all reports unchanged, and in the Huber experiment one may choose $Q = P_w$.

To obtain the strong-contamination lower bound, note the exact identity

$$\mathrm{TV}(P_w, P_v) = d|w - v|. \tag{IV.5.10}$$

For $k \ge 1$, choose $w = 1/4$ and $v = w + k/(16Nd)$; these parameters lie in $[0, 1]$. Couple independent pairs $(X_i, Z_i)$ maximally with marginals $P_w, P_v$. The expected number $J$ of mismatches is

$k/16$. Conditional on $X$, an adversary can sample this coupling and output $Z$ when $J \le k$, and $X$ otherwise. It makes at most $k$ replacements in every realization, and its output law is within total variation $1/16$ of $P_v^{\otimes N}$ because $\mathbb{P}(J > k) \le 1/16$. The experiment at parameter $v$ may use the identity adversary. Two-point testing therefore gives a lower bound $ck/N$. Randomization is harmless: for any estimator a randomized attack's risk is an average of risks of deterministic attacks. For $k \ge 1$ this is at least $c'\varepsilon$; for $k = 0$ the term $\varepsilon$ is absorbed by $N^{-1/2}$. Combining the clean and contamination lower bounds proves (IV.5.2); combining the clean bound with (IV.5.8) proves the lower bound in (IV.5.1). The sum and maximum of the two terms differ by at most a factor two.

### IV.5.5 Huber contamination

Represent the observations by independent contamination indicators $B_i \sim \text{Bernoulli}(\varepsilon)$, clean $X_i \sim P_w$, and outliers $Z_i \sim Q$, all mutually independent, with $Y_i = X_i$ when $B_i = 0$ and $Y_i = Z_i$ otherwise. Set $K = \sum B_i$. On $K \le N/4$, the pathwise inequality (IV.5.6) applies with $\delta = K/N$ to the complete latent clean sample. Therefore

$$\mathbb{E}|\widehat{w} - w| \le C\mathbb{E}Z_N + C\mathbb{E}\sqrt{K/N} + \mathbb{P}(K > N/4) \le C'(N^{-1/2} + \sqrt{\varepsilon}).$$

We used Jensen's inequality and $\mathbb{P}(K > N/4) \le 4\varepsilon$. For (IV.5.9), replacement changes the threshold average by at most $K/N$, giving the minimax upper bound $d^{-1}(1/(2\sqrt{N}) + \varepsilon)$.

For the minimax Huber lower bound, take $w = 1/4$ and $v - w = \varepsilon/(2d)$. These parameters remain in $[0, 1]$. By (IV.5.10), $\text{TV}(P_w, P_v) = \varepsilon/2 \le \varepsilon/(1 - \varepsilon)$. Their Huber neighborhoods intersect. Indeed, the measure with density $(1 - \varepsilon)\max(p_w, p_v)$ has total mass at most one; adding arbitrary nonnegative mass to make it a probability gives a law of the form $(1-\varepsilon)P_w+\varepsilon Q_w$ and also $(1 - \varepsilon)P_v + \varepsilon Q_v$. Identical observed laws force absolute-error risk at least $|v - w|/2$ for one parameter. The clean lower bound handles $\varepsilon = 0$ and supplies the $N^{-1/2}$ term.

Finally, take $w_0 = 0$ to prove the MLE lower bound. If $N\varepsilon < 16$, the clean lower bound absorbs $\sqrt{\varepsilon} \le 4N^{-1/2}$. Otherwise choose $w_* = c_0\sqrt{\varepsilon}$ and a single point-mass contamination at $z = 1/2 + \log(1 + 1/w_*)$, with $c_0$ as above. Chebyshev's inequality gives

$$\mathbb{P}\{N\varepsilon/2 \le K \le 2N\varepsilon\} \ge 1 - \frac{5}{N\varepsilon} \ge \frac{11}{16}.$$

On this event $K \le N/2$. Conditional on the contamination indices, the retained $N - K \ge 1$ clean observations satisfy (IV.5.7) with probability at least $p_*/2$. At $w_*$ the normalized score is at least

$$-2Mw_* + \frac{\varepsilon}{4w_*} > 0.$$

Thus every MLE exceeds $w_*$ on an event of fixed positive probability, proving the lower bound $c\sqrt{\varepsilon}$. This completes the proof of Theorem IV.5.1.

### IV.5.6 Exact population bias

The preceding attack is not merely a finite-sample artifact. For $0 < \varepsilon \le 1/4$ and finite $z \in \mathbb{R}$, let $w_z(\varepsilon)$ maximize

$$L_z(w) = (1 - \varepsilon)\mathbb{E}_0 \log(1 + wU) + \varepsilon \log(1 + wU(z)), \qquad 0 \le w \le 1.$$

Subtracting the fixed $f_0$ log likelihood gives this finite contrast. It is strictly concave, since the clean component assigns positive probability to $U \ne 0$, so the maximizer is unique.

**Theorem IV.5.2** (Exact worst positive point-contamination bias)**.** *Define*

$$A(w) = \mathbb{E}_0 \frac{U^2}{1+wU}, \qquad 0 \le w \le 1.$$

*For each* $0 < \varepsilon \le 1/4$ *there is a unique* $w_\varepsilon \in (0,1)$ *solving*

$$w_\varepsilon^2 A(w_\varepsilon) = \frac{\varepsilon}{1-\varepsilon}. \tag{IV.5.11}$$

*It satisfies*

$$\sup_{z\in\mathbb{R}} w_z(\varepsilon) = \lim_{z\to+\infty} w_z(\varepsilon) = w_\varepsilon, \qquad w_\varepsilon \sim \sqrt{\frac{\varepsilon}{e-1}} \quad (\varepsilon \downarrow 0). \tag{IV.5.12}$$

*Set* $w_0 = 0$ *when* $\varepsilon = 0$.

*Proof.* All positive and negative moments of $r = e^{X-1/2}$ are finite. For $w \in [0,1]$, the integrand in $A(w)$ is dominated by $U^2 + U^2/r$, an integrable function. Thus $A$ is finite and continuous on this interval. Since $\mathbb{E}_0 U = 0$,

$$\mathbb{E}_0 \frac{U}{1+wU} = -wA(w).$$

For every nonzero $u > -1$, the function $w^2u^2/(1+wu)$ is strictly increasing on $[0,1]$; for $w > 0$ its derivative is $wu^2(2+wu)/(1+wu)^2 > 0$. Consequently $w^2A(w)$ is continuous and strictly increasing, starts at zero, and ends at

$$A(1) = \mathbb{E}_0(r - 2 + r^{-1}) = e - 1.$$

Since $\varepsilon/(1-\varepsilon) \le 1/3 < e-1$, (IV.5.11) has exactly one interior solution.

For $w > 0$ the outlier score is $U(z)/(1+wU(z)) \le 1/w$. Hence

$$L_z'(w) \le -(1-\varepsilon)wA(w) + \varepsilon/w.$$

The right side is negative for $w > w_\varepsilon$, so $w_z(\varepsilon) \le w_\varepsilon$. As $z \to +\infty$, the outlier score converges locally uniformly for $w > 0$ to $1/w$. At any fixed $0 < u < w_\varepsilon$ the limiting derivative is positive, so eventually $L_z'(u) > 0$ and $w_z(\varepsilon) > u$. Letting $u \uparrow w_\varepsilon$ establishes the limit and supremum. Finally, strict monotonicity shows $w_\varepsilon \to 0$, and $A(0) = e - 1$ in (IV.5.11) proves the asymptotic formula. □

### IV.5.7 What entropy–transport does and does not imply

The following posterior argument works even when the observation space is unbounded. It only needs a bound on the distances between the component distributions.

**Lemma IV.5.3** (Posterior repair with finite component diameter)**.** *Let* $K_1, \dots, K_m$ *be probability measures with finite first moments on a metric space, with densities* $k_j$ *relative to a common measure. Suppose*

$$W_1(R, K_j) \le \sqrt{2c\, D(R\|K_j)} \quad \text{for every } R, \qquad \max_{i,j} W_1(K_i, K_j) \le D_K.$$

*Let* $H = (H_j)_{j=1}^m$ *be a probability vector and* $p_H = \sum_j H_j k_j$. *Let* $a$ *be a probability density with* $D(a\|p_H) < \infty$. *Suppose, for all* $j$ *with* $H_j > 0$,

$$S_j := \int \frac{ak_j}{p_H} \le \sigma, \qquad \sigma \ge 1.$$

*Then*

$$W_1(a, p_H) \le \sqrt{2cD(a\|p_H)} + D_K(1 - \sigma^{-1}). \tag{IV.5.13}$$

*Proof.* Set $H_j^+ = H_j S_j$; these nonnegative numbers sum to one. For $H_j^+ > 0$, let $R_j$ have density $ak_j/(S_j p_H)$; zero-mass terms may use an arbitrary conditional distribution. Then $\sum_j H_j^+ R_j = a$. Apply the chain rule for relative entropy to the joint law with $(j, x)$ density $H_j a(x) k_j(x)/p_H(x)$ and reference law $H_j k_j(x)$. It gives

$$D(a\|p_H) = D(H^+\|H) + \sum_j H_j^+ D(R_j\|K_j).$$

Convexity of transport under mixing and Cauchy–Schwarz therefore give

$$W_1(a, p_{H^+}) \le \sum_j H_j^+ W_1(R_j, K_j) \le \sqrt{2cD(a\|p_H)}.$$

The domination $H^+ \le \sigma H$ yields $\sum_j \min(H_j, H_j^+) \ge 1/\sigma$, hence $\mathrm{TV}(H^+, H) \le 1 - 1/\sigma$. Couple the common mixing mass identically, and couple the residual component laws at cost at most $D_K$ per unit mass. This gives $W_1(p_{H^+}, p_H) \le D_K(1 - 1/\sigma)$. The triangle inequality proves (IV.5.13). □

For the Gaussian pair, each component satisfies the classical $T_1(1)$ inequality, and $D_K = 1$. Consider a population MLE $p_w$ under $(1-\varepsilon)P_0 + \varepsilon\delta_z$. The directional likelihood conditions towards either component give

$$(1-\varepsilon)\int f_0 \frac{f_j}{p_w} + \varepsilon \frac{f_j(z)}{p_w(z)} \le 1, \qquad j = 0, 1.$$

In particular $S_j \le \sigma = (1-\varepsilon)^{-1}$. Jensen's inequality also gives the clean likelihood deficit bound

$$D(f_0\|p_w) \le \log \int f_0 \frac{f_0}{p_w} \le \log \frac{1}{1-\varepsilon}.$$

Applying Lemma IV.5.3 yields

$$W_1(P_0, P_w) \le \sqrt{2 \log \frac{1}{1-\varepsilon}} + \varepsilon. \tag{IV.5.14}$$

Finally, $W_1(P_0, P_w) = w$: coupling $X \sim N(0,1)$ with $X + B$, where $B \sim \mathrm{Bernoulli}(w)$, gives the upper bound $w$, and the difference of means gives the lower bound. Theorem IV.5.2 shows that the square-root order in (IV.5.14) cannot generally be improved to a linear contamination term for the ordinary likelihood maximizer. Thus Gaussian component concentration, finite component diameter, convexity, and a well-conditioned scalar mixing parameter together do not imply contamination-minimax ordinary MLE risk.

### IV.5.8 Relation to existing robustness results

The nonrobustness of ordinary likelihood in normal mixture problems is established motivation for robust alternatives; see, for example, Fujisawa–Eguchi [33]. Baraud–Birgé [32, Section 6] show that MLEs on suitable convex separable density models are $\rho$-estimators. Lecestre [34] develops nonasymptotic Hellinger robustness for finite mixtures, including contamination and outlier guarantees. These facts are consistent with the present result: a squared Hellinger error of order $\varepsilon$ permits parameter error of order $\sqrt{\varepsilon}$ in this model, while a bounded threshold statistic attains parameter error of order $\varepsilon$.

The role of this section is a precise limitation theorem for a proposed universal entropy–transport extension, with matching finite-sample bounds for two contamination experiments and an exact population worst-bias identity. It makes no historical-priority claim and does not assert resolution of a named external conjecture.

## IV.6 Consequences and remaining boundaries

Theorem IV.2.1 supplies a single risk curve for the entire known-support Poisson experiment. With $L = \log(eN)$, its growing-support part changes from $\sqrt{B/L}$ to $B/\sqrt{N}$ at $B \asymp N/L$. The first term expresses the difficulty of inversion; the second is already forced by uncertainty in the weights of two endpoint atoms. For fixed positive support radius, the small-support expression has the familiar order $(\log\log N)/\log N$. The new proof covers the intermediate and unbounded growing-support ranges for the same original likelihood estimator. It does not settle unknown-support adaptation, heterogeneous exposures, or an imposed finite-component constraint.

Theorem IV.3.1 gives a single canonical connection formula throughout the real integrable Jacobi parameter region considered here. Its separate summands may be complex, and transported branches may contain logarithms at resonance; their prescribed sum is the physical density. The theorem resolves that physical combination, not every connection matrix for arbitrary local bases or a stable large-$N$ numerical algorithm.

For block graphs, the exact ratio is a spectral second moment divided by a squared spectral first moment. A common spectral interval then determines the sharp leading extremal constant. The feasibility argument depends on each block being complete. The exact constant for all connected graphs remains unresolved here, as does a combinatorial classification of all nearly extremal graphs.

The Gaussian separation closes a universal-transfer question negatively: Gaussian component concentration and a finite component transportation diameter do not suffice for contamination-minimax performance of ordinary mixture likelihood. The square-root contamination cost predicted by the entropy bound is attained in the specified two-component model. This lower bound has not been extended here to an unrestricted continuum of Gaussian locations.

The global statements in this manuscript are therefore global in their explicitly stated model and parameter classes. Arbitrary heterogeneous observation profiles and arbitrary complex weights in the binomial-power recurrence problem are also outside the completed results. Independent internal checks found no substantive gaps in the four arguments; external expert review, broader novelty assessment, and publication decisions remain separate steps.

## IV.A Supporting two-scale binomial inversion

These supporting tools are reproduced from the complete binomial-mixture likelihood argument in the supplied manuscript [30], also used in [31]. They are included for independent verification and are not counted as new results here. Write $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and $p_G(j) = \int B_j^t\,dG$. The distance $W_{1,t}$ uses ground cost $|j-k|/t$, and $h(p,q)^2 = \sum_j(\sqrt{p_j}-\sqrt{q_j})^2$. All constants below are absolute.

**Lemma IV.A.1** (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute $C$ with the following properties. For every 1-Lipschitz $f:[0,1]\to\mathbb{R}$ and integer $k\ge 1$, there is a polynomial $P_k$ of degree at most $k$ such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

*If $P = \sum_{j=0}^t b_j B_j^t$ has degree $k \le t/2$, then*

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{IV.A.1}$$

*For the derivative-stable $P_k$, its coefficient vector also satisfies*

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{IV.A.2}$$

*Proof.* Subtract $f(0)$, reflect $f$ on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial $Q_s$ has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most $e^{-100s}$. Indeed $|\pi l(x-1/2)| \le \pi s$ and the factorial inequality $M! \ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s = \lfloor k/100\rfloor$ for $k \ge 100$; use a constant polynomial and enlarge $C$ for smaller $k$.

For (IV.A.1), put $M = \|P\|_\infty$ and $F(z) = (1+z)^t P(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w| \ge 1$. For $0 < j < t$, set $r = j/(t-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k \le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c/\sqrt{1+tv}$, proves (IV.A.1). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (IV.A.1) at degree $k-1 \le (t-1)/2$ and use $2(k-1)^2/(t-1) \le 4k^2/t$. □

**Proposition IV.A.2** (Two-scale deterministic inversion)**.** *For integers $1 \le s \le k \le t/2$ and any two mixing laws $G, H$,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t}W_{1,t}(p_G,p_H) + \frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{IV.A.3}$$

*Proof.* For any 1-Lipschitz test $f$, take the polynomials $P_k, P_s$ from Lemma IV.A.1. The approximation error in integrating $P_k$ instead of $f$ against $G-H$ is at most $C/k$. The Bernstein coefficients of $P_s$ have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test $P_s$ by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G, p_H)$. All constants are uniform over $f$, so taking the Wasserstein dual supremum proves the result. □

The small amplitude $1/s$ is essential. Applying only the Hellinger bound to the entire polynomial loses the low-frequency empirical $N^{-1/2}$ scale; applying only transportation loses the critical transition. Proposition IV.A.2 retains both.

# Part V
# Global Algebraicity and Exact Complexity

**Abstract**

We establish three complete classifications using methods from coefficient asymptotics, differential modules, and arithmetic inverse periods. First, for every integer $r \geq 1$ and every complex weight $\lambda$, the weighted Franel generating function $\sum_{n\geq 0} \sum_{k=0}^{n} \binom{n}{k}^{r} \lambda^{k} z^{n}$, and its even section, are algebraic precisely when $r \leq 2$ or $\lambda = 0$. A global torus maximum inequality resolves complex cancellation, and an arithmetic Puiseux obstruction finishes the classification. Second, the exact minimal differential order of every anisotropic hypercubic lattice Green function is determined solely by the multiplicities of its nonzero squared weights. The formula holds in every dimension and at every complex specialization, including colliding thresholds. Irreducibility and local monodromy imply that the function is algebraic exactly when at most one weight is nonzero. Third, all algebraic quantiles at algebraic levels for rational polynomial tilts of the arcsine law are classified by cyclotomic factors. Every allowable finite exceptional set is realized, with minimum polynomial degree one plus its cardinality. Two further transfers prove sharp support-free Gaussian likelihood recovery throughout a specified growing-support regime and an exact design-comparison constant for dense inflations of every fixed graph skeleton. The arguments are given in full, with classical inputs, previously known cases, and remaining limitations stated explicitly.

**Integration note.** The rank and algebraicity statements are the foundation for the Galois-group refinements in Parts VI and VII. Its support-free Gaussian theorem differs from the known-support all-scale theorem of Part VI; both estimation statements are retained. The graph-inflation comparison and quantile constructions remain included.

## V.1 What is global, and where the methods come from

The central questions concern entire families, rather than finitely many examples. The first three results classify every parameter in their stated domains. Their conclusions have different meanings: nonalgebraicity of a generating function, minimum order of a differential equation, and arithmetic of inverse distribution values must not be conflated.

| Family | Complete domain | Conclusion |
|---|---|---|
| Weighted Franel series | Every integer $r \geq 1$ and every $\lambda \in \mathbb{C}$ | Algebraic exactly for $r \leq 2$ or $\lambda = 0$; also for the even section. |
| Hypercubic lattice periods | Every dimension and every complex weight tuple | Exact minimal differential order, irreducibility, and an algebraicity dichotomy. |
| Polynomial arcsine tilts | Every positive rational polynomial weight | Exact exceptional quantile sets, inverse realization, and minimum degree. |
| Gaussian likelihood | Every $N \geq 3$ and $A \geq \sigma\sqrt{\log(eN)}$ | Uniform minimax $W_1$ risk without supplying the support bound. |
| Graph inflation | Every fixed connected simple skeleton, positive masses and lengths | Exact asymptotic ratio and the sharp optimized constant within this class. |

The method transfers are as follows.

1. **From Franel singularities to all complex weights.** The coefficient-asymptotic obstruction is already present in the unweighted literature. A pair-averaging inequality identifies every dominant saddle for arbitrary complex weights. The two competing saddles on the negative real axis are treated explicitly, including their collision after taking the even section. The result is an algebraicity classification, not a new claim about every minimum recurrence width.

2. **From Bessel periods to all anisotropic hypercubic lattices.** The supplied manuscript [49] uses Bessel differential modules, diagonal $\mathrm{SL}_2$ decompositions, and Fourier transforms for exact order problems. Here a single Laplace transform in the Bessel variable produces a different period family. An upper bound in middle de Rham cohomology and a lower bound from a visible simple Fourier constituent agree. This also removes the possible extra rank from the unilateral integration endpoint.

3. **From arithmetic inverse periods to a complete design problem.** The supplied inverse-period work [50] motivates reducing a distribution function to a logarithmic period and an algebraic correction. For polynomial arcsine tilts the correction is an explicit rational polynomial. Its cyclotomic factors determine every exception, and the same operator constructs every possible exceptional set with optimal degree. Baker's and Gelfond–Schneider's theorems remain classical inputs.

4. **From likelihood contact to a continuous-data bound.** The posterior coupling method in [30, 31] gives an empirical-posterior martingale for an unrestricted Gaussian fit. A second-order Taylor cancellation and an exact Gaussian-to-binomial channel replace an inapplicable empirical-to-density entropy argument. The theorem is uniform on the stated growing-support range; the small-support problem is not claimed to be settled.

5. **From electrical maximum principles to all fixed skeletons.** A flow norm inequality has an exact equality classification. Combined with series reduction and clique budget allocation, it controls every fixed cyclic skeleton as well as trees. It does not provide the sharp upper constant for arbitrary graph sequences.

## V.2 Algebraicity of weighted Franel series for every complex weight

For an integer $r \geq 1$ and $\lambda \in \mathbb{C}$, define

$$a_n^{(r)}(\lambda) = \sum_{k=0}^{n} \binom{n}{k}^r \lambda^k, \qquad A_{r,\lambda}(z) = \sum_{n\geq 0} a_n^{(r)}(\lambda) z^n, \qquad E_{r,\lambda}(z) = \sum_{n\geq 0} a_{2n}^{(r)}(\lambda) z^n.$$

All these series have positive radii of convergence. For nonzero $\lambda$, the normalized central sequence considered previously has generating function $E_{r,\lambda}(z/\lambda)$.

The unweighted nonalgebraicity result for $r \geq 3$ is already established by Banderier and Drmota [35, Proposition 7], using the exponent and the power of $\pi$ in the coefficient asymptotics. Algebraic independence results for the unweighted Franel family also appear in Adamczewski, Bell, and Delaygue [36, Theorem 9.8]. The point here is the complete complex-weight classification, including algebraic weights off the unit circle and the cancellation boundary $\lambda < 0$. We do not identify that full classification as an explicitly stated open conjecture in those sources.

**Theorem V.2.1** (Complete algebraicity classification)**.** *For every integer $r \geq 1$ and every $\lambda \in \mathbb{C}$, each of $A_{r,\lambda}$ and $E_{r,\lambda}$ is algebraic over $\mathbb{C}(z)$ if and only if $r \leq 2$ or $\lambda = 0$. Consequently, for every $r \geq 3$ and every nonzero complex $\lambda$, the normalized central generating function $E_{r,\lambda}(z/\lambda)$ is transcendental over $\mathbb{C}(z)$.*

The proof does not require the visibility of every complex critical value, nor a classification of minimum recurrence widths. It finds all dominant saddles and proves that their relevant amplitudes cannot cancel.

### V.2.1 A global maximum inequality on the coefficient torus

**Lemma V.2.2** (Pair averaging)**.** *Fix $\rho > 0$. Among $r$ unit complex numbers with prescribed product, every maximizer of $\prod_{j=1}^{r} |1 + \rho e^{it_j}|$ has all $e^{it_j}$ equal. The maximizing common angles are precisely those $r$th roots of the prescribed product having the largest real part.*

*Proof.* For two angles choose a midpoint $a$ modulo $\pi$ with $\cos a \geq 0$, and choose $b$ so that $e^{it} = e^{i(a+b)}$ and $e^{iu} = e^{i(a-b)}$. Replacing both angles by $a$ preserves their product. Direct expansion gives

$$|1 + \rho e^{ia}|^4 - |(1 + \rho e^{it})(1 + \rho e^{iu})|^2 = 4\rho(1 - \cos b)\big((1 + \rho^2)\cos a + \rho(1 + \cos b)\big) \geq 0.$$

Equality requires either $\cos b = 1$, or $\cos a = 0$ and $\cos b = -1$; both cases imply $e^{it} = e^{iu}$. The maximum of the full product is positive. Thus every pair at a maximizer must have equal phases. The prescribed product then forces the common phase to be an $r$th root, and comparison of $|1 + \rho e^{it}|^2 = 1 + \rho^2 + 2\rho\cos t$ proves the last assertion. □

For $\lambda \neq 0$, let $\rho = |\lambda|^{1/r}$ and let $\Omega_\lambda$ consist of the roots of $\omega^r = \lambda$ with largest real part. If $\lambda \notin (-\infty, 0)$, this is the single root with $\arg\omega = (\operatorname{Arg}\lambda)/r$, where $-\pi < \operatorname{Arg}\lambda < \pi$. If $\lambda < 0$, it consists of the two roots $\rho e^{\pm\pi i/r}$. Put

$$d = r - 1, \qquad \nu = d/2, \qquad \sigma_\omega = (1 + \omega)^r, \qquad V_\omega = \frac{\omega}{(1 + \omega)^2}, \qquad \kappa_\omega = \frac{(2\pi)^{-\nu}}{\sqrt{r}\, V_\omega^\nu}.$$

For $r \geq 3$ every root in $\Omega_\lambda$ has positive real part, and

$$\operatorname{Re} V_\omega = \frac{(1 + |\omega|^2)\operatorname{Re}\omega + 2|\omega|^2}{|1 + \omega|^4} > 0.$$

The power of $V_\omega$ in $\kappa_\omega$ uses the logarithm on the right half-plane.

**Lemma V.2.3** (Dominant asymptotics at every complex weight)**.** *Fix $r \geq 3$ and $\lambda \neq 0$, and write $M = |\sigma_\omega|$ for $\omega \in \Omega_\lambda$. Then*

$$a_n^{(r)}(\lambda) = n^{-\nu} \sum_{\omega \in \Omega_\lambda} \kappa_\omega \sigma_\omega^n + O(M^n n^{-\nu-1}). \tag{V.2.1}$$

*For the ordinary coefficients the displayed exponential bases are distinct. For the even coefficients their squares are distinct except when $\lambda = -1$. In that exceptional case the two amplitudes are equal positive real numbers and therefore add without cancellation.*

*Proof.* With $X = x_1 \cdots x_d$, define the Laurent polynomial

$$P_\lambda(x) = \prod_{j=1}^{d} (1 + x_j)(1 + \lambda/X).$$

Coefficient extraction gives $a_n^{(r)}(\lambda) = \mathrm{CT}\, P_\lambda^n$. Evaluate this constant term on the torus $|x_j| = \rho$. Its $r$ factors have arguments $t_1, \ldots, t_r$ satisfying $\sum_j t_j = \operatorname{Arg} \lambda$ modulo $2\pi$. Lemma V.2.2 identifies the complete set of maxima of $|P_\lambda|$: they are $x_1 = \cdots = x_d = \omega$ with $\omega \in \Omega_\lambda$.

In local angular coordinates $x_j = \omega e^{i\theta_j}$, the last factor is $1 + \omega e^{-i\sum_j \theta_j}$. The first derivatives of $\log P_\lambda$ vanish and its Hessian is

$$-V_\omega(I_d + J_d), \qquad \det(I_d + J_d) = r,$$

where $J_d$ has all entries one. Its real part is negative definite. Outside fixed small neighborhoods of these finitely many points, $|P_\lambda| \leq M - \epsilon$. Inside a neighborhood, Taylor expansion, the scaling $\theta = n^{-1/2} y$, and dominated Gaussian integration give the contribution

$$\sigma_\omega^n (2\pi)^{-d} \frac{(2\pi/n)^{d/2}}{\sqrt{r}\, V_\omega^{d/2}} (1 + O(n^{-1})).$$

The Gaussian branch is determined by $\operatorname{Re} V_\omega > 0$; odd Taylor terms integrate to zero on a symmetric local neighborhood. This proves (V.2.1).

Only the case $\lambda < 0$ has two contributions. Write $\omega = \rho e^{i\pi/r}$ and $\alpha = \arg(1 + \omega)$, so $0 < \alpha < \pi/r$. Their ordinary bases have arguments $\pm r\alpha \in (-\pi, \pi)$ and are distinct. Their squares coincide if and only if $2r\alpha = \pi$, equivalently $\alpha = \pi/(2r)$. Since the argument of $1 + \rho e^{i\pi/r}$ is strictly increasing in $\rho$, this occurs exactly at $\rho = 1$, that is, $\lambda = -1$. At that value

$$V_\omega = \frac{1}{4\cos^2(\pi/(2r))} > 0$$

for both roots, so their $\kappa_\omega$ are the same positive number. □

### V.2.2 Algebraic-series asymptotics with several dominant phases

**Lemma V.2.4** (Arithmetic amplitude obstruction)**.** *Let $f(z) = \sum_{n \geq 0} c_n z^n$ have algebraic coefficients. Suppose that, for distinct nonzero algebraic numbers $\tau_1, \ldots, \tau_s$ of the same modulus $T > 0$, nonzero constants $b_j$, and $\nu > 0$,*

$$c_n = n^{-\nu} \sum_{j=1}^{s} b_j \tau_j^n + o(T^n n^{-\nu}).$$

*If $f$ is algebraic over $\mathbb{C}(z)$, then $\nu$ is not a positive integer. If $\nu$ is a positive half-integer that is not an integer, every $b_j$ belongs to $\overline{\mathbb{Q}}/\sqrt{\pi}$.*

*Proof.* First the algebraic relation descends to $\overline{\mathbb{Q}}(z)$. Indeed, fix the finitely many monomials $z^i f^j$ in any polynomial relation. Their coefficients form an infinite linear system with finitely many columns over $\overline{\mathbb{Q}}$. Its rank is attained by finitely many rows, so a nonzero complex kernel implies a nonzero kernel over $\overline{\mathbb{Q}}$.

The given expansion implies a radius of convergence $T^{-1}$. For the lower bound on coefficient growth, use the elementary fact that a nonzero finite sum $\sum_j b_j \zeta_j^n$ with distinct $|\zeta_j| = 1$ cannot tend to zero: its Cesàro squared mean tends to $\sum_j |b_j|^2 > 0$.

An algebraic function has finitely many singularities on this circle and convergent Puiseux expansions there, with rational exponents and algebraic coefficients. A nonintegral local term $a(1 - z/\zeta)^q$ contributes

$$\frac{a}{\Gamma(-q)} \zeta^{-n} n^{-q-1} (1 + O(n^{-1})).$$

This follows from the exact binomial coefficient formula and Stirling's formula. Nonnegative integral powers are locally analytic and make no such contribution; poles give nonnegative powers of $n$. Truncating the finitely many Puiseux expansions and deforming the Cauchy contour gives their coefficient expansions to any prescribed order.

Distinct phases in these expansions cannot cancel each other. More explicitly, if $\sum_j d_j \zeta_j^n = o(1)$, multiply by $\overline{\zeta_k}^n$ and take Cesàro means to obtain $d_k = 0$. Apply this first to any powers of $n$ larger than $n^{-\nu}$, then at $n^{-\nu}$. Thus each specified phase $\tau_j^n$ must have a nonzero leading Puiseux contribution with exponent $q = \nu - 1$. When $\nu$ is a positive integer, this exponent is a nonnegative integer, which is impossible. When $\nu$ is a nonintegral half-integer, $\Gamma(1 - \nu)$ is a nonzero rational multiple of $\sqrt{\pi}$, so $b_j \in \overline{\mathbb{Q}}/\sqrt{\pi}$. □

### V.2.3 Proof of the classification

*Proof of Theorem V.2.1.* First suppose $r \geq 3$ and $\lambda \neq 0$ is algebraic. Every dominant $\omega$, $\sigma_\omega$, and $V_\omega$ is algebraic, and

$$\kappa_\omega \in \overline{\mathbb{Q}}^\times \pi^{-\nu}.$$

For $A_{r,\lambda}$, Lemma V.2.3 supplies distinct dominant bases and nonzero amplitudes. For $E_{r,\lambda}$ use the same formula with $n$ replaced by $2n$. The only collision of squared bases occurs at $\lambda = -1$ and adds two equal nonzero amplitudes; the extra factor $2^{-\nu}$ is algebraic. Thus in both cases every combined amplitude is still in $\overline{\mathbb{Q}}^\times \pi^{-\nu}$.

If $r$ is odd, $\nu = (r-1)/2$ is a positive integer, contradicting Lemma V.2.4. If $r$ is even, then $r \geq 4$ and $\nu$ is a nonintegral half-integer. Algebraicity would force a nonzero amplitude to belong simultaneously to $\overline{\mathbb{Q}}^\times \pi^{-\nu}$ and $\overline{\mathbb{Q}}^\times \pi^{-1/2}$. This would make $\pi^{\nu - 1/2} = \pi^{(r-2)/2}$ algebraic, contradicting the transcendence of $\pi$.

Now suppose $\lambda$ is transcendental and one of the two series has a polynomial relation over $\mathbb{C}(z)$. Its coefficients lie in $K = \mathbb{Q}(\lambda)$. The finite-column linear-system argument in Lemma V.2.4 descends a nonzero polynomial relation to $K[z, Y]$. Replace the transcendental $\lambda$ by an indeterminate $w$. Each coefficient identity becomes an identity in $\mathbb{Q}(w)$, so the corresponding symbolic weighted series is algebraic over $\mathbb{Q}(w, z)$. After clearing coefficient denominators, specialize $w$ to a positive rational number for which the relation does not become the zero polynomial. There are only finitely many excluded values. This contradicts the algebraic-weight case already proved. Therefore both series are nonalgebraic at every nonzero complex weight.

Finally, if $\lambda = 0$, then $a_n^{(r)}(0) = 1$ and both series are $(1 - z)^{-1}$. For $r = 1$,

$$A_{1,\lambda}(z) = \frac{1}{1 - (1 + \lambda)z}, \qquad E_{1,\lambda}(z) = \frac{1}{1 - (1 + \lambda)^2 z}.$$

For $r = 2$, constant-term extraction of a quadratic Laurent polynomial gives

$$A_{2,\lambda}(z) = \left(1 - 2(1 + \lambda)z + (1 - \lambda)^2 z^2\right)^{-1/2},$$

with value one at zero. Its even section

$$E_{2,\lambda}(z) = \tfrac{1}{2}\big(A_{2,\lambda}(\sqrt{z}) + A_{2,\lambda}(-\sqrt{z})\big)$$

is algebraic as well. These cases exhaust the classification. □

*Remark* V.2.5 (What has and has not been classified). The theorem includes all algebraic collision parameters, such as $\lambda = 8$ and $1/8$ for the cubic central family. It does not determine the minimum recurrence width at those parameters. The mechanism is a global maximum inequality followed by arithmetic asymptotics, rather than an assertion that all complex critical values are visible in the physical period. The unweighted case and the general Puiseux obstruction are inherited ideas; the complex-weight maximum analysis and its cancellation check supply the extension.

## V.3 Exact order and algebraicity for every anisotropic hypercubic period

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{V.3.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{V.3.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the $n$-dimensional hypercubic lattice when each step in either coordinate direction $i$ has weight $m_i$. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of $\mathrm{SL}_2$, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{V.3.3}$$

The decomposition in the middle uses the diagonal $\mathrm{SL}_2$; the tensor product itself is irreducible for the product group $\mathrm{SL}_2^q$.

**Theorem V.3.1** (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle \partial_z \rangle$ annihilating $F_m$ is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{V.3.4}$$

*The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.*

*More precisely, define*

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

*Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size $d$ for each of the $a_d$ summands with $d \geq 1$ in* (V.3.3).

**Corollary V.3.2** (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is $1$; with one nonzero weight $m$, it is $(1-4m^2z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (V.3.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2,1]$, $[1,1,1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem V.3.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

### V.3.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection $M$ on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy $T$, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size $d$, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if $M$ has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank} \mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{V.3.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection $N$ on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero $H^0$ for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

### V.3.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let $B_\mu$ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{V.3.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is $\mathrm{SL}_2$; this is the classical non-half-integral Bessel case [42, 43]. At zero, $I_0$ is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ.}$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

**Lemma V.3.3.** *If the $\mu_j^2$ are pairwise distinct and nonzero, the direct sum of the $B_{\mu_j}$ has differential Galois group $\mathrm{SL}_2^q$. Consequently*

$$M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$$

*is irreducible of rank $Q$, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities $a_d$ from (V.3.3); its slopes at infinity are at most one.*

*Proof.* The projections of the Galois group, and of its identity component, onto each $\mathrm{SL}_2$ factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the $I_0$ solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by $g$. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator $e$ on $W$. On $\mathrm{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j - 2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (V.3.3) has the parity of $n$ and contains exactly one weight-zero vector if $n$ is even, or exactly one weight-one vector if $n$ is odd. Thus $b$ is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (V.3.4). In particular $\mathrm{Sym}^n V$ occurs once, so $r \geq n$.

### V.3.3 An upper bound that eliminates the endpoint ambiguity

For $\operatorname{Re} s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\,dx = \sum_{k\ge 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{V.3.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist $M$ by $e^{-sx}$; call the resulting connection $N$. On the cyclic vector $\eta$ evaluating to $g$, its $x$-connection is $\nabla_x^M - s$ and its commuting $s$-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice $N_0$ stable under $\theta_x$. Its residue at zero is nilpotent with the same $b$ Jordan blocks as $M$.

The connection $N$ is irreducible of rank $Q \ge 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since $s$ is transcendental over the mass constants, every formal exponential factor at infinity is nonzero and linear in $x$ after the twist. Thus $\operatorname{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension $b$. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{V.3.8}$$

Here $\omega = [\eta\,dx]$ belongs to this kernel. To check this without conflating $dx$ and $dx/x$, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator $\theta_x$ acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \ge 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\,dx$ is formally exact at zero. The commuting connection $\nabla_s$ preserves $N_0$, acts on both global and local cohomology, and hence preserves the kernel in (V.3.8).

It remains to verify that (V.3.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of $M$, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to $\varepsilon$, circle zero once, and return to infinity on the continued branch. With the factor $e^{-sx}$ this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of $\omega$ along its small circle is $O(\varepsilon(1 + |\log\varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of $K_0$ have at most exponential growth, which is dominated by $e^{-sx}$ in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over $K$ in (V.3.8), period integration proves

$$\operatorname{ord}_{\mathbb{C}(s)} H_m \le r. \tag{V.3.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

### V.3.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine $x$-line. Lemma V.3.3 and (V.3.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank $r$. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes $d$ stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is $M$; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module $\delta_0$. For an entire function $f$ in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\,dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an $A_s$-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{V.3.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (V.3.10) is an actual isomorphism. Since Fourier sends $\delta_0$ to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{F}\mathcal{K}$ (at most once) or the trivial affine-line connection.

If the former were absent, $H_m$ would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{F}\mathcal{K}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (V.3.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by $H_m$ is isomorphic to the irreducible generic module of $\mathcal{F}\mathcal{K}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

### V.3.5 Descent and the algebraicity dichotomy

Multiplication by $s$ does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order $r$ over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to $z$ are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem V.3.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in $W$ supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus $H_m$, and consequently $F_m$, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary V.3.2.

The argument proves a statement about a function of the variable $z$. It does not assert transcendence of every special value at algebraic $z$, nor does it prove a general formula for the differential Galois group at all weight specializations. Those are distinct arithmetic and differential-algebraic questions.

## V.4 Cyclotomic classification and design of algebraic quantiles

We classify all algebraic quantiles at algebraic probability levels for every rational polynomial weighting of the arcsine distribution. The exceptional levels form finite sets determined exactly by cyclotomic factors of an explicit polynomial. Conversely, every allowable finite set can be realized, and its minimum polynomial degree is determined. A real-variable root count reduces the entire two-parameter family of strict half-integer beta shapes to three possible levels. The classical transcendence inputs are Baker's theorem on linear forms in logarithms and the Gelfond–Schneider theorem; the classification and construction below are proved from those inputs.

### V.4.1 The correction polynomial

Let $w \in \mathbb{Q}[X]$ satisfy $w(x) > 0$ for $0 < x < 1$. Zeros at the two endpoints are permitted. Define

$$c_w = \frac{1}{\pi}\int_0^1 \frac{w(t)}{\sqrt{t(1-t)}}\,dt > 0, \qquad F_w(x) = \frac{1}{\pi c_w}\int_0^x \frac{w(t)}{\sqrt{t(1-t)}}\,dt. \tag{V.4.1}$$

This is a continuous strictly increasing distribution function. Write

$$\mathcal{E}_w = \{r \in \overline{\mathbb{Q}} \cap (0,1) : F_w^{-1}(r) \in \overline{\mathbb{Q}}\}$$

for its set of algebraic probability levels with algebraic quantiles. Only the untruncated distribution on $[0,1]$ is considered here.

**Lemma V.4.1** (Exact polynomial reduction)**.** *There is a unique polynomial $R_w \in \mathbb{Q}[X]$ such that*

$$w(X) = c_w + X(1-X)R_w'(X) + (1/2 - X)R_w(X). \tag{V.4.2}$$

*If $w(X) = \sum_j w_j X^j$, then*

$$c_w = \sum_j w_j\, 4^{-j}\binom{2j}{j} \in \mathbb{Q}_{>0}.$$

*For $0 \le x \le 1$, with the positive square root and $\arcsin\sqrt{x} \in [0, \pi/2]$,*

$$F_w(x) = \frac{2}{\pi}\arcsin\sqrt{x} + \frac{\sqrt{x(1-x)}}{\pi c_w}R_w(x). \tag{V.4.3}$$

*If $w$ is nonconstant of degree $d$, then $R_w$ is nonzero of degree $d-1$. If $w$ is constant, then $R_w = 0$.*

*Proof.* For the linear operator $\mathcal{L}R = X(1-X)R' + (1/2 - X)R$,

$$\mathcal{L}(X^k) = (k + 1/2)X^k - (k+1)X^{k+1}.$$

Starting with the highest degree, this triangular identity gives unique rational coefficients for a polynomial $R$ such that $w - \mathcal{L}R$ is constant. A nonzero polynomial $R$ of degree $k$ has $\mathcal{L}R$ of degree $k+1$, proving uniqueness and the degree assertion.

Since

$$\frac{d}{dx}\{\sqrt{x(1-x)}R(x)\} = \frac{\mathcal{L}R(x)}{\sqrt{x(1-x)}},$$

the integral of $\mathcal{L}R/\sqrt{x(1-x)}$ over $[0,1]$ vanishes. Hence the constant just obtained is $c_w$. The beta integral for each monomial gives $\int_0^1 x^{j-1/2}(1-x)^{-1/2}\,dx = \pi 4^{-j}\binom{2j}{j}$; alternatively this follows by $x = \sin^2\theta$ and the elementary reduction of even powers of sine. Integrating (V.4.2) from zero to $x$ proves (V.4.3). □

**Theorem V.4.2** (Arithmetic reduction)**.** *For every algebraic $x \in (0,1)$,*

$$F_w(x) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad R_w(x) = 0 \ \text{ and } \ \frac{2}{\pi}\arcsin\sqrt{x} \in \mathbb{Q}. \tag{V.4.4}$$

*Whenever these conditions hold, $F_w(x) = 2\arcsin\sqrt{x}/\pi$ is rational. In particular, every algebraic irrational level has a transcendental quantile for every $w$ in* (V.4.1)*.*

*Proof.* Suppose $x$ and $r = F_w(x)$ are algebraic, and put $\theta = \arcsin\sqrt{x} \in (0, \pi/2)$. Equation (V.4.3) shows that

$$2\theta - \pi r = -\frac{\sqrt{x(1-x)}}{c_w} R_w(x) \in \overline{\mathbb{Q}}.$$

The number $\zeta = e^{2i\theta} = 1 - 2x + 2i\sqrt{x(1-x)}$ is algebraic. Choose $\log\zeta = 2i\theta$ and $\log(-1) = i\pi$. Then

$$\log\zeta - r\log(-1) = i(2\theta - \pi r)$$

is algebraic. Baker's theorem says that an algebraic-coefficient linear form in logarithms of algebraic numbers is either zero or transcendental [46]. It must therefore vanish. Consequently $R_w(x) = 0$ and $\zeta = e^{i\pi r}$. If the algebraic number $r$ were irrational, the Gelfond–Schneider theorem, applied to this determination of $(-1)^r$, would make $\zeta$ transcendental. Thus $r \in \mathbb{Q}$. The converse follows immediately from (V.4.3). □

### V.4.2 All exceptional sets and a finite exact algorithm

For each integer $d \geq 3$, put

$$\mathcal{Q}_d = \left\{ \frac{2j}{d} : 1 \leq j < d/2,\ \gcd(j,d) = 1 \right\}, \tag{V.4.5}$$

$$P_d(X) = \prod_{\substack{1 \leq j < d/2 \\ (j,d)=1}} \left( X - \sin^2\frac{\pi j}{d} \right). \tag{V.4.6}$$

We call $\mathcal{Q}_d$ a cyclotomic packet of probability levels. The sets $\mathcal{Q}_d$ are pairwise disjoint: $d$ is the exact order of $e^{i\pi r}$ for every $r \in \mathcal{Q}_d$.

**Lemma V.4.3.** *The polynomial $P_d$ is monic, rational, and irreducible, of degree $\varphi(d)/2$. Its roots are distinct and belong to $(0,1)$. The polynomials $P_d$, $d \geq 3$, are pairwise distinct.*

*Proof.* For a primitive $d$th root $\zeta_d$, write

$$\sin^2\frac{\pi j}{d} = \frac{2 - \zeta_d^j - \zeta_d^{-j}}{4}.$$

The Galois group of $\mathbb{Q}(\zeta_d)$ acts transitively on these numbers. Two such numbers agree exactly when their indices agree up to sign modulo $d$: the sum $z + z^{-1}$ determines the unordered pair $\{z, z^{-1}\}$. Thus the orbit has $\varphi(d)/2$ distinct elements, proving rationality and irreducibility. None of these roots is an endpoint because $d \geq 3$ and the root of unity is primitive. The same unordered-pair observation shows that a root cannot belong to both $P_d$ and $P_e$ unless $d = e$. □

**Theorem V.4.4** (Complete packet classification)**.** *If $w$ is nonconstant, its entire exceptional set is*

$$\mathcal{E}_w = \coprod_{\substack{d \geq 3 \\ P_d \mid R_w}} \mathcal{Q}_d. \tag{V.4.7}$$

*For $r$ in this set the quantile is exactly $\sin^2(\pi r/2)$. If $\nu_w$ is the number of distinct roots of $R_w$ in $(0,1)$, every packet appearing in (V.4.7) satisfies*

$$\frac{\varphi(d)}{2} \le \nu_w, \qquad d \le 8\nu_w^2 \le 8(\deg w - 1)^2. \tag{V.4.8}$$

*Moreover $|\mathcal{E}_w| \le \nu_w$. Thus the whole exceptional set is determined by finitely many exact divisions in $\mathbb{Q}[X]$. For a positive constant $w$, instead, $\mathcal{E}_w = \mathbb{Q} \cap (0,1)$.*

*Proof.* Theorem V.4.2 makes every exceptional level rational. If $r$ is such a level, $e^{i\pi r}$ has an exact order $d \ge 3$, so $r = 2j/d$ with the index conditions in (V.4.5). Its quantile $x = \sin^2(\pi j/d)$ is a root of $R_w$. Lemma V.4.3 shows that $P_d \mid R_w$. Conversely this divisibility makes every root of $P_d$ a root of $R_w$, and (V.4.3) gives precisely the quantiles and levels in $\mathcal{Q}_d$.

All $\varphi(d)/2$ roots of a dividing $P_d$ belong to $(0,1)$, and different packets give different roots. This proves the root count and cardinality bounds. For completeness, the elementary inequality $\varphi(d) \ge \sqrt{d/2}$ follows from

$$\frac{\varphi(d)^2}{d} = \prod_{p^a \| d} p^{a-2}(p-1)^2 \ge \frac{1}{2}.$$

Every odd-prime factor on the right is at least one; the only possible factor below one is $p = 2, a = 1$, equal to $1/2$. This proves the finite bound in (V.4.8). For constant $w$, use $R_w = 0$ in Theorem V.4.2; rational angles have algebraic sine. □

An exact algorithm is now explicit. Compute $c_w, R_w$ by the triangular identity in Lemma V.4.1; for all $3 \le d \le 8(\deg w - 1)^2$, form $P_d$ and test $P_d \mid R_w$ over $\mathbb{Q}$. Output the packets for the successful tests. A Sturm count for $R_w$ can replace $\deg w - 1$ by $\nu_w$ and shorten the list. The polynomial $P_d$ itself can be obtained from the usual cyclotomic polynomial by the substitution $4X = 2 - Z - Z^{-1}$ or by its minimal-polynomial description. No test of numerical equality between transcendental quantities is required. When $\deg w = 1$, $R_w$ is a nonzero constant and the output is empty.

### V.4.3 Converse design and the minimum degree

**Theorem V.4.5** (Realization of every possible finite exceptional set)**.** *A finite set $\mathcal{E} \subset \overline{\mathbb{Q}} \cap (0,1)$ equals $\mathcal{E}_w$ for some rational polynomial $w$ strictly positive on $[0,1]$ if and only if*

$$\mathcal{E} = \coprod_{d \in D} \mathcal{Q}_d$$

*for a finite set $D \subset \{3, 4, \dots\}$. For this prescribed set, the minimum possible degree of $w$ is*

$$1 + \sum_{d \in D} \frac{\varphi(d)}{2} = 1 + |\mathcal{E}|. \tag{V.4.9}$$

*This includes the empty set, whose minimum degree is one.*

*Proof.* Necessity follows from Theorem V.4.4; a constant positive weight has infinitely many exceptions. For sufficiency put $R = \prod_{d \in D} P_d$, with $R = 1$ if $D$ is empty. Choose a positive rational number $c > \max_{x \in [0,1]} |\mathcal{L}R(x)|$ and set

$$w = c + \mathcal{L}R.$$

Then $w > 0$ on $[0,1]$. The integral of $\mathcal{L}R/\sqrt{x(1-x)}$ is zero, so $c_w = c$ and $R_w = R$. Pairwise distinct irreducible factors in Lemma V.4.3 and (V.4.7) show that the exceptional set is exactly the prescribed one. The construction has degree $1 + \deg R = 1 + \sum_{d \in D} \varphi(d)/2$.

Any polynomial weight realizing this finite set is nonconstant, and its correction polynomial must be divisible by every $P_d$, $d \in D$. Those polynomials are pairwise coprime, so its degree is at least the displayed sum. Lemma V.4.1 then proves the lower bound in (V.4.9). When $D$ is empty, the same construction is a positive linear weight and constants are excluded, proving that case too. □

**Example V.4.6** (A single exceptional probability)**.** Choose $R = X - 1/4$ and $c = 1$. Then

$$w(X) = \frac{7}{8} + \frac{7}{4}X - 2X^2$$

is positive on $[0, 1]$: it is concave and its endpoint values are $7/8$ and $5/8$. Its entire exceptional set is $\mathcal{Q}_6 = \{1/3\}$ and $F_w^{-1}(1/3) = 1/4$. Every other algebraic internal level has a transcendental quantile. Degree two is minimal.

**Example V.4.7** (An algebraic quantile forces its conjugate packet)**.** The packet $\mathcal{Q}_{10}$ is $\{1/5, 3/5\}$ and

$$P_{10}(X) = X^2 - \frac{3}{4}X + \frac{1}{16}.$$

Therefore a rational polynomial weighting cannot have $1/5$ as its only algebraic level with an algebraic quantile: $3/5$ must occur too. The two quantiles are $(3 - \sqrt{5})/8$ and $(3 + \sqrt{5})/8$. Theorem V.4.5 realizes exactly this pair with a positive polynomial of degree three, and this degree is minimal.

### V.4.4 Real-variable shape bounds

Put $g(\theta) = w(\sin^2 \theta)$ for $0 \le \theta \le \pi/2$. Its average is $c_w$. Define

$$h(\theta) = \int_0^\theta (g(u) - c_w)\, du.$$

Formula (V.4.3) gives, with $x = \sin^2 \theta$,

$$h(\theta) = \tfrac{1}{2}\sqrt{x(1-x)}R_w(x), \qquad h(0) = h(\pi/2) = 0. \tag{V.4.10}$$

**Proposition V.4.8** (Mean crossings bound arithmetic complexity)**.** *Suppose $w$ is nonconstant and $g - c_w$ has $s$ distinct zeros in $(0, \pi/2)$. Then $\nu_w \le s - 1$. In particular:*

1. *If $g$ is strictly monotone, then $\nu_w = 0$ and $\mathcal{E}_w$ is empty.*
2. *Suppose $g$ is strictly increasing up to one interior maximum and strictly decreasing thereafter, and $g(0), g(\pi/2) < c_w$. Then $\nu_w = 1$. Consequently $\mathcal{E}_w$ has at most one member, and*

$$\mathcal{E}_w \subset \{1/3, 1/2, 2/3\}.$$

*Proof.* If $h$ has $\nu_w$ distinct interior zeros, its two endpoint zeros and Rolle's theorem force at least $\nu_w + 1$ distinct interior zeros of $h' = g - c_w$. This gives the first bound. Strict monotonicity gives exactly one mean crossing.

Under the single-maximum assumptions there are exactly two mean crossings, say $\theta_- < \theta_+$. The function $h$ strictly decreases before $\theta_-$, strictly increases between the crossings, and strictly decreases after $\theta_+$. It is negative near zero and positive just before $\pi/2$, because its derivative is negative there and its final value is zero. It therefore has exactly one interior zero, located between the two mean crossings. Every possible packet now has $\varphi(d) \le 2$. The integers $d \ge 3$ with this property are exactly $3, 4, 6$, whose packets are respectively $\{2/3\}, \{1/2\}, \{1/3\}$. □

### V.4.5 The full half-integer beta lattice

The normalized incomplete beta function is $I_x(\alpha,\beta) = B_x(\alpha,\beta)/B(\alpha,\beta)$; we use its usual strictly increasing inverse on $(0,1)$ [57]. First take strict half-integer shapes

$$\alpha = m + \tfrac{1}{2}, \qquad \beta = n + \tfrac{1}{2}, \qquad m,n \in \mathbb{Z}_{\geq 0}.$$

This is (V.4.1) with $w_{m,n}(X) = X^m(1-X)^n$. Let $R_{m,n}$ be its correction polynomial and let $\mathcal{E}_{m,n}$ denote its exceptional set.

**Theorem V.4.9** (Three-level theorem and exact decision for every pair)**.** *For all nonnegative integers $m,n$ the following statements hold.*

1. *If $m = n = 0$, then $\mathcal{E}_{0,0} = \mathbb{Q} \cap (0,1)$.*
2. *If exactly one of $m,n$ is zero, then $\mathcal{E}_{m,n}$ is empty.*
3. *If $m,n \geq 1$, then $\mathcal{E}_{m,n}$ has at most one member, and is exactly the subset of the following three candidates passing the indicated rational test:*

| *level* | *quantile* | *necessary and sufficient test* |
|---|---|---|
| 1/3 | 1/4 | $R_{m,n}(1/4) = 0$ |
| 1/2 | 1/2 | $m = n$ |
| 2/3 | 3/4 | $R_{m,n}(3/4) = 0.$ |

*In particular, $\mathcal{E}_{m,m} = \{1/2\}$ for $m \geq 1$.*

*In every case, every algebraic irrational level has a transcendental quantile. The tests are finite rational arithmetic, for example by the recurrences*

$$\begin{aligned} R_{0,0}(X) &= 0, \\ R_{0,n}(X) &= \frac{2n-1}{2n} R_{0,n-1}(X) + \frac{(1-X)^{n-1}}{n}, \qquad n \geq 1, \\ R_{m,n}(X) &= \frac{2m-1}{2(m+n)} R_{m-1,n}(X) - \frac{X^{m-1}(1-X)^n}{m+n}, \qquad m \geq 1. \end{aligned} \tag{V.4.11}$$

*Proof.* The arcsine case follows from Theorem V.4.4. With exactly one positive exponent, $w_{m,n}$ is strictly monotone. With both exponents positive, it vanishes at the endpoints and has exactly one strict maximum, because its logarithmic derivative is $m/X - n/(1-X)$. Proposition V.4.8 gives all the candidate restrictions. For each candidate the corresponding $P_d$ is linear, so Theorem V.4.4 gives precisely the stated evaluation test.

It remains to simplify the middle test. If $m = n$, symmetry gives $I_{1/2} = 1/2$, so that test passes. If $m > n$, then for $0 < x < 1/2$,

$$\frac{x^{m-1/2}(1-x)^{n-1/2}}{(1-x)^{m-1/2}x^{n-1/2}} = \left(\frac{x}{1-x}\right)^{m-n} < 1.$$

The probability below $1/2$ is strictly less than $1/2$. For $m < n$ it is strictly greater. Hence the middle test passes if and only if $m = n$.

For completeness, put

$$c_{m,n} = \frac{(2m)!(2n)!}{4^{m+n} m!\, n!\, (m+n)!}.$$

The primitive is $2c_{m,n} \arcsin\sqrt{x} + \sqrt{x(1-x)} R_{m,n}(x)$. Integration by parts, or differentiation of $x^{m-1/2}(1-x)^{n+1/2}$, gives for $m \geq 1$

$$B_x(m+\tfrac{1}{2}, n+\tfrac{1}{2}) = \frac{2m-1}{2(m+n)} B_x(m-\tfrac{1}{2}, n+\tfrac{1}{2}) - \frac{x^{m-1/2}(1-x)^{n+1/2}}{m+n}.$$

The analogous reduction in $n$, starting with $m = 0$, has positive boundary term $x^{1/2}(1-x)^{n-1/2}/n$. Substitution of the primitive proves (V.4.11). □

**Example V.4.10** (A genuine asymmetric exception)**.** For $(m, n) = (1, 2)$,

$$c_{1,2} = \frac{1}{16}, \qquad R_{1,2}(X) = -\frac{(2X-3)(4X-1)}{24}.$$

The unique interior root is $1/4$. Therefore

$$I_{1/4}(3/2, 5/2) = 1/3, \qquad \mathcal{E}_{1,2} = \{1/3\}.$$

By reflection $\mathcal{E}_{2,1} = \{2/3\}$, with quantile $3/4$. Every other algebraic internal probability level has a transcendental quantile for these two distributions.

**Corollary V.4.11** (All positive half-integer shapes)**.** *Let $\alpha, \beta \in \frac{1}{2}\mathbb{Z}_{>0}$. If at least one of $\alpha, \beta$ is an integer, every algebraic level in $(0, 1)$ has an algebraic beta quantile. If neither is an integer, Theorem V.4.9 gives the exact finite tests and all the possible exceptional levels.*

*Proof.* When both shapes are integers, the incomplete beta integral is a polynomial with rational coefficients. Suppose one shape is integral and the other is a strict half-integer. If $\alpha$ is integral, the substitution $t = 1 - y^2$ turns the integral into a polynomial primitive in $y = \sqrt{1-x}$ with rational coefficients; if $\beta$ is integral, use $t = y^2$. The complete integral is then a nonzero algebraic number. Equating the incomplete integral to an algebraic multiple of the complete one is a nonconstant polynomial equation for $y$ with algebraic coefficients. Thus $y$, and hence the unique quantile $x$, is algebraic. The remaining shapes are exactly those in Theorem V.4.9. □

**Corollary V.4.12** (Quantiles of the binomial Jeffreys posterior)**.** *After $m$ successes and $n$ failures, the prior with density proportional to $[x(1-x)]^{-1/2}$ gives the posterior* Beta$(m + 1/2, n + 1/2)$. *If $m + n \geq 1$, every algebraic level outside $\{1/3, 1/2, 2/3\}$ has a transcendental posterior quantile. In particular, both endpoints of the equal-tailed 90%, 95%, and 99% posterior intervals are transcendental for every nonempty data set.*

*Proof.* Multiply the prior density by the binomial likelihood $x^m(1-x)^n$ and apply Theorem V.4.9. The lower-tail probabilities of these intervals are respectively $1/20$, $1/40$, and $1/200$, and the upper-tail probabilities are their complements. None is one of the three candidates. □

**Exact scope of the classification.** Theorem V.4.4 classifies all exceptional sets for rational polynomial arcsine weightings, and Theorem V.4.5 proves both converse realization and optimal degree. Theorem V.4.9 supplies an exact finite decision for every half-integer beta pair. It does not assert that the only asymmetric pairs passing a test are $(1, 2)$ and $(2, 1)$: a closed list of all integer pairs satisfying $R_{m,n}(1/4) = 0$ is not proved here. The restriction to full-interval normalization is also essential to the proof; no statement for arbitrary algebraic truncation is inferred.

**Relation to the literature.** Baker's logarithmic theorem is an established input, not a new transcendence principle. The beta identities are classical [57]. Dhillon–Saha study arithmetic values of incomplete beta functions [55]. Their publisher abstract and bibliographic record were available in preparing this draft, but their full article was not obtained; consequently no claim is made that all potential overlaps with that paper have been excluded. The independently proved claims here are the explicit packet criterion, the converse design with its sharp degree, and the resulting uniform restrictions. No historical-priority or named-open-problem claim is made.

## V.5 Support-free Gaussian likelihood in the complete growing-support regime

Let $X_i = \Theta_i + \sigma Z_i$, where the $\Theta_i$ are independent with law $G$, the $Z_i$ are independent standard normal variables, and $\sigma > 0$ is known. In this section $\widehat{G}$ maximizes the Gaussian mixture likelihood over *all* probability measures on $\mathbb{R}$. Neither a support center nor a support radius is supplied to the estimator.

**Theorem V.5.1** (Uniform growing-support minimaxity)**.** *There exist absolute constants $c, C > 0$ such that the following holds. For $N \geq 3$, put $L = \log(eN)$. If $A \geq \sigma\sqrt{L}$, then*

$$\inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([\mu-A,\mu+A])} \mathbb{E}_G W_1(\widetilde{G}, G) \geq c\left\{\frac{\sigma}{\sqrt{L}} + \frac{A}{\sqrt{N}}\right\},$$
$$\sup_{G \in \mathcal{P}([\mu-A,\mu+A])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\left\{\frac{\sigma}{\sqrt{L}} + \frac{A}{\sqrt{N}}\right\}.$$

*The infimum can even be taken over estimators that know $A$ and $\mu$. Every measurable choice of the unrestricted NPMLE satisfies the upper bound, uniformly in $A, \mu, \sigma, N$ in this range.*

The result includes every intermediate support/noise scale between $A \asymp \sigma\sqrt{\log N}$ and $A \asymp \sigma\sqrt{N}$, as well as the larger-support range. It does not assert the entire known-support minimax curve when $A < \sigma\sqrt{\log N}$. Soloff, Guntuboyina and Sen [64, Theorem 11] prove a deconvolution bound for a fixed compact support once the sample size exceeds a support-dependent threshold. The issue here is a bound with absolute constants throughout the stated growing-radius range. The classical likelihood contact condition is used explicitly [47, Proposition 1.1]; that condition itself is not new.

Translation and scaling reduce the proof to $\mu = 0$, $\sigma = 1$. Write $\varphi$ and $\Phi$ for the standard normal density and CDF, $p_H = H * \varphi$, and $q = N^{-1}\sum_i \delta_{X_i}$. For probability densities $p, r$, use the Hellinger convention $h(p,r)^2 = \int_{\mathbb{R}} (\sqrt{p} - \sqrt{r})^2\, dx$.

### V.5.1 An exact Gaussian-to-binomial experiment

For an integer $m \geq 1$, let $\varepsilon$ be independent of $X$, with centered Gaussian law on $\mathbb{R}^m$ and covariance $mI_m - \mathbf{1}\mathbf{1}^{\mathsf{T}}$. This matrix is positive semidefinite; its zero eigenvalue causes no difficulty. Given $X$, set $Y_j = X + \varepsilon_j$ and retain $Z = \sum_{j=1}^m \mathbf{1}_{\{Y_j \geq 0\}}$. This defines a parameter-free Markov channel $A_m$. If $X \sim N(\theta, 1)$, then the joint Gaussian vector $Y$ has mean $\theta\mathbf{1}$ and covariance $mI_m$. Thus its coordinates are independent and

$$A_m N(\theta, 1) = \mathrm{Bin}(m, \phi_m(\theta)), \qquad \phi_m(\theta) = \Phi(\theta/\sqrt{m}). \tag{V.5.1}$$

For $m = \lceil D^2 \rceil$, $D \geq 1$, the inverse of $\phi_m$ on $[-D, D]$ has Lipschitz constant at most $CD$. In particular, Hellinger data processing transfers predictive Hellinger accuracy to the binomial experiment, while latent distances can be recovered with a factor $CD$.

This exact channel does not by itself justify a Poisson-style entropy transfer: $D(q\|p_H)$ is infinite for continuous Gaussian observations. We instead use a second-order cancellation.

### V.5.2 Likelihood contact supplies the second-order cancellation

Every unrestricted Gaussian likelihood maximizer is supported in $[\min_i X_i, \max_i X_i]$: moving a parameter outside that interval to its nearest endpoint strictly increases every component likelihood. Compactness then proves existence. Its score satisfies

$$S_H(\theta) = \frac{1}{N}\sum_i \frac{\varphi(X_i - \theta)}{p_H(X_i)} \leq 1 \quad (\theta \in \mathbb{R}), \qquad S_H = 1 \quad H\text{-almost everywhere.}$$

At each contact parameter the empirical-posterior weights $R_\theta(i) = N^{-1}\varphi(X_i-\theta)/p_H(X_i)$ sum to one. Evaluating the score at $\theta+u$ gives

$$\mathbb{E}_{R_\theta} e^{u(X-\theta)} = e^{u^2/2} S_H(\theta+u) \le e^{u^2/2}.$$

Consequently $\mathbb{E}_{R_\theta} X = \theta$ and $\mathbb{E}_{R_\theta}(X-\theta)^2 \le 1$, and the joint law $H(d\theta)R_\theta(i)$ has observation marginal $q$. For every globally twice continuously differentiable $F$ with bounded second derivative, Taylor's theorem therefore yields

$$\left| \int F\, d(H-q) \right| \le \tfrac{1}{2}\|F''\|_\infty. \tag{V.5.2}$$

The true Gaussian coupling similarly gives $|\int F\, d(G-p_G)| \le \|F''\|_\infty/2$. Hence

$$\left| \int F\, d(H-G) \right| \le \mathrm{Lip}(F) W_1(q,p_G) + \|F''\|_\infty. \tag{V.5.3}$$

Unlike a transport bound applied to an arbitrary Lipschitz test, this estimate is small when the retained test has small curvature.

**Lemma V.5.2** (Polynomial approximation with two controlled derivatives)**.** *For every 1-Lipschitz $g : [0,1] \to \mathbb{R}$ and every integer $k \ge 1$, there is a real polynomial $P_k$ of degree at most $k$ such that*

$$\|g - P_k\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le C, \qquad \|P_k''\|_\infty \le Ck.$$

*All norms in this display are on* $[0,1]$.

*Proof.* Subtract $g(0)$ and reflect $g$ to a 2-periodic 1-Lipschitz function. Convolve with a positive normalized Jackson trigonometric kernel of degree $2s$. The resulting trigonometric polynomial $Q_s$ approximates $g$ within $C/s$, and $\|Q_s'\|_\infty \le 1$, because convolution preserves the Lipschitz constant. The trigonometric Bernstein inequality applied to $Q_s'$ gives $\|Q_s''\|_\infty \le Cs$.

To replace $Q_s$ by an ordinary polynomial, expand its finitely many exponentials $e^{i\pi j(x-1/2)}$, $|j| \le 2s$, through Taylor degree $100s$. Each Fourier coefficient has magnitude at most the bounded supremum norm of the reflected function. For $|x-1/2| \le 1/2$, the exponential-series remainder and its first two derivatives are at most a polynomial in $s$ times $e^{-cs}$, since $|\pi j(x-1/2)| \le \pi s$ while the truncation degree is $100s$. Their sum over $O(s)$ modes is still exponentially small. Take the real part and add back $g(0)$. With $s = \lfloor k/100 \rfloor$, this proves the assertions for large $k$; bounded $k$ is handled by a constant polynomial and an enlarged absolute constant. □

**Lemma V.5.3** (A two-scale likelihood inverse)**.** *Suppose the true mixing law $G$ and an unrestricted Gaussian MLE $H$ are supported in $[-D,D]$, where $D \ge 1$, and put $m = \lceil D^2 \rceil$. For $1 \le S \le K \le m/2$,*

$$W_1(H,G) \le C\left\{ \frac{D}{K} + \frac{S}{D} + W_1(q,p_G) + \frac{De^{2K^2/m}}{S} h(p_H,p_G) \right\}. \tag{V.5.4}$$

*Proof.* For a 1-Lipschitz test $f$ on $[-D,D]$, define $g(x) = f(\phi_m^{-1}(x))/D$ on $\phi_m([-D,D])$. It has an absolute Lipschitz constant, so it extends to $[0,1]$ with the same constant. Apply Lemma V.5.2, with constants rescaled if necessary, at degrees $S$ and $K$.

The global function $F_S(\theta) = DP_S(\phi_m(\theta))$ satisfies $\mathrm{Lip}(F_S) \le C$. Writing $z = \theta/\sqrt{m}$,

$$F_S''(\theta) = \frac{D}{m}\{P_S''(\Phi(z))\varphi(z)^2 - P_S'(\Phi(z))z\varphi(z)\},$$

so $\|F_S''\|_\infty \le CS/D$. Equation (V.5.3) controls this part by $C\{W_1(q,p_G) + S/D\}$.

The residual polynomial $P_K - P_S$ has norm at most $C/S$. The binomial Bernstein coefficient bound proved in Lemma V.A.1 expands it at degree $m$ with coefficients of magnitude at most $Ce^{2K^2/m}/S$. Its expectation difference under the pushforward mixing laws is therefore bounded by this quantity times $\|A_m p_H - A_m p_G\|_1 \le 2h(p_H,p_G)$. Finally the approximation of $f$ by $DP_K \circ \phi_m$ costs $CD/K$. Wasserstein duality proves (V.5.4). □

### V.5.3 Uniform likelihood accuracy on the random support range

Let $T = \sqrt{40L}$, $D = A + T$, and $E = \{\max_i |Z_i| \le T\}$. Every unrestricted MLE is supported in $[-D, D]$ on $E$. Also $\mathbb{P}(E^c) \le 2Ne^{-20L}$ and

$$\mathbb{E}[\mathbf{1}_{E^c} W_1(\widehat{G}, G)] \le \mathbb{E}[\mathbf{1}_{E^c}(2A + \max_i |Z_i|)] \le C(A+T)N^{-19}. \tag{V.5.5}$$

Since $A \ge \sqrt{L}$, $D \asymp A$ with absolute constants.

We give the density bound used on this event explicitly. If $A \le \sqrt{N}$, then

$$\mathbb{E}[\mathbf{1}_E h(p_{\widehat{G}}, p_G)] \le C\sqrt{D}\, L^{3/4}/\sqrt{N}. \tag{V.5.6}$$

To verify it, consider Gaussian mixtures with means in $[-D, D]$. For complex $z$, $\sup_x |\varphi(x - z)| \le (2\pi)^{-1/2} e^{(\operatorname{Im} z)^2/2}$. On $z = D\cos(\theta + iu)$ this is at most $C\exp(D^2 \sinh^2 u/2)$. Contour shifting, with $u$ a small multiple of $k/D^2$ when $k \le D^2$ and a fixed small constant otherwise, bounds the $k$th Chebyshev coefficient in supremum norm by

$$C\exp\{-c\min(k^2/D^2, k)\}.$$

Truncating at degree $d = C\{D\sqrt{\log(C(D+1)/\beta)} + \log(C(D+1)/\beta)\}$ and gridding the Chebyshev moments gives a uniform density cover of radius $\beta$, with centers that are actual mixtures and $\log M \le Cd\log(C(d+1)/\beta)$.

Take $U = D + \sqrt{12L}$ and $\beta = N^{-3}/(2U)$. As $\sqrt{L} \le D \le C\sqrt{N}$, this gives $\log M \le CDL^{3/2}$. Replace each cover center $p_a$ by the envelope

$$u_a(x) = \begin{cases} p_a(x) + \beta, & |x| \le U, \\ \varphi(|x| - D), & |x| > U. \end{cases}$$

It dominates every associated mixture and has total mass in $[1, 1 + CN^{-3}]$ whenever used. Normalizing the envelope changes an associated density by Hellinger distance at most $CN^{-3/2}$. The square-root likelihood-ratio union bound in Lemma V.A.2 now gives

$$\mathbb{E}[\mathbf{1}_E h(p_{\widehat{G}}, p_G)^2] \le C(\log M + 1)/N + CN^{-3}.$$

The argument applies to every fit in the covered class with likelihood at least that of $G$. Consequently it applies on $E$ without conditioning on $E$ or pretending that the conditional observations remain independent. Cauchy–Schwarz proves (V.5.6).

The ordinary empirical distribution also satisfies

$$\mathbb{E}W_1(q, p_G) \le C(A+1)/\sqrt{N}. \tag{V.5.7}$$

Integrate empirical-CDF standard deviations: on $[-A, A]$ use $\operatorname{Var}(\mathbf{1}_{X \le x}) \le 1/4$, and on the two tails use $\mathbb{P}(|X| > A + u) \le 2\mathbb{P}(Z > u)$ and the finiteness of $\int_0^\infty \sqrt{\mathbb{P}(Z > u)}\, du$.

### V.5.4 Proof of the uniform upper and lower bounds

Assume first $\sqrt{L} \le A \le \sqrt{N}$. For sufficiently large absolute $N$, put $m = \lceil D^2 \rceil$ and choose

$$S = \left\lfloor \frac{D}{16\sqrt{L}} \right\rfloor \vee 1, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These choices satisfy $1 \le S \le K \le m/2$, $D/K \le C/\sqrt{L}$, $S/D \le C/\sqrt{L}$, and $D/S \le C\sqrt{L}$. Apply Lemma V.5.3 on $E$, then equations (V.5.5)–(V.5.7). The only term needing absorption is

$$C\sqrt{L}\, e^{2aL} \frac{\sqrt{D} L^{3/4}}{\sqrt{N}}.$$

Its ratio to $L^{-1/2}$ is bounded by $Ce^{2aL}N^{-1/4}L^{7/4}$, because $D \le C\sqrt{N}$. This is uniformly bounded as $2a < 1/4$. Thus $\mathbb{E}W_1(\widehat{G}, G) \le C\{L^{-1/2} + A/\sqrt{N}\}$. Bounded $N$ is absorbed by the elementary bound in the next paragraph.

If $A > \sqrt{N}$, the empirical-posterior coupling gives $W_1(\widehat{G}, q) \le 1$. Also coupling the true mixing law to its Gaussian observation law costs at most $\mathbb{E}|Z|$. Equation (V.5.7) therefore yields

$$\mathbb{E}W_1(\widehat{G}, G) \le 1 + \mathbb{E}|Z| + C(A+1)/\sqrt{N} \le CA/\sqrt{N}.$$

This proves the upper bound throughout the claimed range.

For the nonparametric lower bound, use the Chebyshev-arcsine pair from Lemma V.A.3 on $[-M, M]$, with $M = \sqrt{L}/8$ and $k = \lceil 20L \rceil$. Its support lies in $[-A, A]$, its moments through degree $k-1$ agree, and its $W_1$ separation is at least $M/(\pi k) \ge c/\sqrt{L}$. The Gaussian identity

$$\int_{\mathbb{R}} \frac{\varphi(x-\lambda)\varphi(x-\mu)}{\varphi(x)}\, dx = e^{\lambda\mu}$$

gives

$$\int \frac{(p_{F_k^+} - p_{F_k^-})^2}{\varphi} \le 4\sum_{j\ge k} \frac{M^{2j}}{j!} = 4\sum_{j\ge k} \frac{(L/64)^j}{j!} \le Ce^{-40L}.$$

Predictive TV and hence the TV of the $N$-sample experiments are bounded away from one. Le Cam's inequality proves the $c/\sqrt{L}$ lower bound. Finally, endpoint mixing laws on $\{-A, A\}$ whose weights differ by $c/\sqrt{N}$ give the lower bound $cA/\sqrt{N}$, by the Bernoulli KL calculation and data processing from the latent experiment. Taking the maximum, equivalent to the displayed sum up to a factor two, completes the proof of Theorem V.5.1.

The distinction between constrained and unrestricted fits is essential to this proof. At a boundary atom of a constrained fit, posterior mean equality can fail, so equation (V.5.2) cannot simply be asserted. The result above genuinely concerns the unrestricted estimator and does not settle that separate small-support problem. Its mechanism is the transfer of the likelihood contact principle through a second-order Taylor cancellation, followed by a binomial inverse at a different scale.

## V.6 A sharp comparison for fixed skeletons and their dense inflations

This section extends the sharp asymptotic comparison on block graphs to dense inflations of arbitrary fixed connected skeletons, including skeletons with cycles. It does not settle the sharp constant for all connected graphs. Both the finite skeleton inequality and the inflation limit are proved below without invoking the earlier block-graph classification. The external motivating problem is the comparison of determinant-optimal and total-resistance-optimal graph designs in [13]; the constant studied here is a continuation of that problem, not a constant explicitly conjectured in that paper. The convex effective-resistance design framework is developed in [48].

A design on a connected simple graph $G$ with $n$ vertices consists of conductances $w_e \ge 0$ with $\sum_e w_e = 1$ and connected positive support. For an arbitrary orientation, write $b_e = e_u - e_v$ and $L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}$. The $D$-optimal design maximizes $\det(L(w)|_{\mathbf{1}^\perp})$, and the $A$-optimal design minimizes

$$K_G(w) = \sum_{u<v} R_{uv}(w) = n \operatorname{tr} L(w)^\dagger.$$

Write $\mathfrak{a}(G) = K_G(w^D)/K_G(w^A)$. Disconnected supports have infinite resistance objective.

### V.6.1 A weighted flow inequality with an exact equality case

Fix a connected simple graph $H = (V, E)$ on $h \geq 2$ vertices, masses $\alpha_i > 0$ with $\sum_i \alpha_i = 1$, and lengths $\ell_e > 0$. For conductances $c$ define

$$\mathcal{K}_\alpha(c) = \sum_{i<j} \alpha_i \alpha_j R_{ij}(c), \qquad K_\ell = \mathcal{K}_\alpha((1/\ell_e)_{e\in E}),$$

and set

$$A_\ell = \min_{\substack{c_e \geq 0 \\ \sum_e \ell_e^2 c_e = 1}} \mathcal{K}_\alpha(c). \tag{V.6.1}$$

**Theorem V.6.1** (Sharp skeleton inequality)**.** *For every such $H$, $\alpha$, and $\ell$,*

$$K_\ell \leq \frac{1}{2}\sqrt{A_\ell}. \tag{V.6.2}$$

*Equality holds if and only if $H = K_2$ and its two masses are both $1/2$. The length of the single edge is unrestricted.*

*Proof.* Orient $H$ and let $B$ be its vertex-edge incidence matrix. Index commodities by unordered pairs $i < j$. A feasible simultaneous flow $F = (F_e^{ij})$ satisfies

$$BF^{ij} = \sqrt{\alpha_i \alpha_j}(e_i - e_j).$$

The row $F_e$ is a vector in $\mathbb{R}^{\binom{h}{2}}$. Thomson's principle, summed over commodities, gives

$$\mathcal{K}_\alpha(c) = \min_{F \text{ feasible}} \sum_e \frac{\|F_e\|^2}{c_e}, \tag{V.6.3}$$

where a zero conductance requires a zero flow on that edge.

Let $F^0$ be the electrical flow for conductances $1/\ell_e$. For a fixed edge $e = (u, v)$ put $z = L(1/\ell)^\dagger(e_u - e_v)$. Reciprocity yields

$$F_e^{0,ij} = \frac{\sqrt{\alpha_i \alpha_j}}{\ell_e}(z_i - z_j).$$

The weighted variance identity therefore gives

$$\|F_e^0\|^2 = \frac{1}{\ell_e^2}\left(\sum_i \alpha_i z_i^2 - \left(\sum_i \alpha_i z_i\right)^2\right). \tag{V.6.4}$$

The maximum principle places every $z_i$ between $z_u$ and $z_v$; their difference is the effective resistance $R_e$. For a probability distribution on an interval of length $R_e$, the variance is at most $R_e^2/4$. For completeness, after translating the interval to $[0, R_e]$, the inequality $x^2 \leq R_e x$ bounds the variance by $\mu(R_e - \mu) \leq R_e^2/4$. The direct edge has resistance $\ell_e$, so $R_e \leq \ell_e$. Consequently

$$\|F_e^0\| \leq \frac{R_e}{2\ell_e} \leq \frac{1}{2}. \tag{V.6.5}$$

Moreover $R_e < \ell_e$ when $e$ is not a bridge, since the graph with that edge removed then supplies a positive parallel conductance.

For every feasible $F$, each column of $F - F^0$ is a circulation. Its scalar product with an electrical potential gradient is zero. Thus Thomson orthogonality, followed by (V.6.5), gives

$$K_\ell = \sum_e \ell_e \langle F_e^0, F_e\rangle \leq \frac{1}{2}\sum_e \ell_e \|F_e\|. \tag{V.6.6}$$

For a fixed feasible $F$, Cauchy–Schwarz gives

$$\left(\sum_e \ell_e \|F_e\|\right)^2 \le \left(\sum_e \frac{\|F_e\|^2}{c_e}\right)\left(\sum_e \ell_e^2 c_e\right).$$

Equality is attained by $c_e = \|F_e\|/(\ell_e \sum_f \ell_f \|F_f\|)$, allowing $c_e = 0$ when $F_e = 0$. Hence

$$\sqrt{A_\ell} = \min_{F \text{ feasible}} \sum_e \ell_e \|F_e\|. \tag{V.6.7}$$

This minimum is attained: the feasible set is a nonempty closed affine set, and the displayed positive weighted norm has compact sublevel sets. The support of any feasible flow is connected, since all pair demands are positive. In particular, the minimizing conductances above are an admissible connected design. Taking the minimum in (V.6.6) proves the inequality.

Suppose equality holds and let $F^*$ minimize (V.6.7). Every edge with $F_e^* \neq 0$ must have $\|F_e^0\| = 1/2$; otherwise (V.6.6) is strict. Each used edge is therefore a bridge of $H$. Since these edges form a connected spanning support, $H$ must itself be a tree. For a tree edge the potential $z$ is constant on either side of its cut, so (V.6.4) says $\|F_e^0\|^2 = a_e(1-a_e)$, where $a_e$ is the mass on one side. Every edge is used and thus every cut has mass $1/2$ on either side. If a tree has at least three vertices, two of its leaves would already have mass $1/2$ each, contradicting the strictly positive mass on a third vertex. Therefore $H = K_2$, with equal masses. Conversely, for one edge of length $\ell$ and equal masses, $K_\ell = \ell/4$ and $A_\ell = \ell^2/4$. □

### V.6.2 Inflating a fixed skeleton

Fix $H$, $\alpha$, and $\ell$ as above, and a scale $s > 0$. Construct $G_n$ by replacing vertex $i$ of $H$ with a clique $K_{m_i(n)}$, designating one of its vertices as the root, and replacing each skeleton edge $e$ with an internally disjoint path of $p_e(n)$ edges between the corresponding roots. Assume

$$n = \sum_i m_i(n) + \sum_e (p_e(n) - 1), \qquad m_i(n) = \alpha_i n + O(\sqrt{n}), \qquad p_e(n) = s\ell_e\sqrt{n} + O(1). \tag{V.6.8}$$

Such integer choices exist for all sufficiently large $n$, by rounding the path lengths and distributing their $O(\sqrt{n})$ interior vertices among the clique-size corrections.

**Theorem V.6.2** (Exact inflation limit)**.** *For every family satisfying* (V.6.8)*,*

$$K_{G_n}(w^D) = sK_\ell n^{7/2} + o(n^{7/2}), \tag{V.6.9}$$

$$K_{G_n}(w^A) = n^3\left(\frac{1}{\sqrt{2}} + s\sqrt{A_\ell}\right)^2 + o(n^3). \tag{V.6.10}$$

*In particular,*

$$\lim_{n\to\infty} \frac{\mathfrak{a}(G_n)}{\sqrt{n}} = \frac{sK_\ell}{(1/\sqrt{2} + s\sqrt{A_\ell})^2}. \tag{V.6.11}$$

*The maximum over $s > 0$ equals*

$$\frac{K_\ell}{2\sqrt{2}\sqrt{A_\ell}} \le \frac{1}{4\sqrt{2}}, \tag{V.6.12}$$

*and is attained uniquely at $s = (2A_\ell)^{-1/2}$. Equality in the final bound holds precisely for the balanced two-vertex skeleton. In that case the joining path has $p(n) \sim \sqrt{2n}$ edges.*

*Proof.* Write $d = n - 1$, and suppress $n$ from $m_i$ and $p_e$. Call the graph of roots and subdivided paths the central graph.

**The determinant-optimal design.** The spanning-tree polynomial factors across the cliques attached at roots and the central graph. The determinant is $n$ times this polynomial. Its central factor has the identity

$$\tau_{\mathrm{cen}}(w) = \left(\prod_{e\in E}\prod_{j=1}^{p_e} w_{ej}\right) \sum_{T\in\mathcal{T}(H)} \prod_{e\notin T} \left(\sum_{j=1}^{p_e} w_{ej}^{-1}\right). \tag{V.6.13}$$

Indeed, a spanning tree contains every segment of each path corresponding to an edge of some $T \in \mathcal{T}(H)$, and deletes exactly one segment from every other path. The identity is first written for positive weights and extends as a polynomial identity.

The logarithmic determinant is strictly concave as a function of $L|_{\mathbf{1}^\perp}$. For a simple graph the edge weights are determined by the off-diagonal entries of $L$, so the maximizing weights are unique. Formula (V.6.13) is invariant under arbitrary permutations of the segments of any one path. Therefore its $D$-optimal segments all have the same weight $w_e^D$. This common weight is positive, since $p_e \geq 2$ for large $n$ and deleting the entire path would isolate its interior vertices. The same uniqueness and the complete-graph factor show that all edges of each attached clique have equal positive weight.

The derivative of the logarithmic determinant with respect to $w_f$ is $R_f$. The budget multiplier is $d$ by homogeneity; hence $R_f = d$ on every positive edge. A clique $K_m$ with constant edge conductance $a$ has pair resistance $2/(ma)$. Since it attaches at one vertex, other parts of the graph do not change this internal resistance. Thus its edge weights are exactly $2/(md)$ and its pair resistances are $d$.

In a weighted random spanning tree, the inclusion probability of an edge $f$ is $w_f R_f = dw_f$. A tree omits at most one segment of any central path, and all its segments are interchangeable under (V.6.13). Consequently

$$\frac{1-1/p_e}{d} \leq w_e^D \leq \frac{1}{d}. \tag{V.6.14}$$

After series reduction, the root network has conductances $w_e^D/p_e = (1+o(1))/(sd\sqrt{n}\,\ell_e)$, uniformly over the fixed edge set. Its root resistances therefore satisfy

$$R_{ij}^{\mathrm{cen}}(w^D) = sd\sqrt{n}\,[R_{ij}(1/\ell) + o(1)].$$

Pairs in distinct cliques contribute a central-resistance term

$$\sum_{i<j} m_i m_j R_{ij}^{\mathrm{cen}}(w^D) = sK_\ell n^{7/2} + o(n^{7/2}).$$

All access resistances inside cliques are $O(n)$, so their total contribution is $O(n^3)$. There are $O(\sqrt{n})$ central interior vertices and hence $O(n^{3/2})$ pairs involving at least one such vertex. By (V.6.14), the resistance of a segment is at most $2d$ for large $n$. Connecting vertices along the fixed skeleton uses $O(\sqrt{n})$ segments, so every relevant pair resistance is $O(n^{3/2})$. These remaining contributions are also $O(n^3)$. This proves (V.6.9).

**The resistance-optimal design.** Resistance adds across a graph joined at a single vertex. In the objective of an attached clique $K_m$, its root has projected mass $b = n - m + 1$, and its other $m - 1$ vertices each have mass one. Permutation averaging over the latter vertices does not increase the objective, by convexity of effective resistance in conductances. Thus write its root-to-other edge weights as $a$ and its other edge weights as $b'$; put $\lambda = a + (m-1)b'$. The two nonzero Laplacian modes give the objective and budget

$$\frac{b}{a} + \frac{(m-2)n}{\lambda}, \qquad \frac{ma + (m-2)\lambda}{2} = t.$$

Cauchy–Schwarz shows that the minimum is $q_m^2/t$, where

$$q_m = \frac{\sqrt{m(n-m+1)} + (m-2)\sqrt{n}}{\sqrt{2}}. \tag{V.6.15}$$

The equality allocation has $\lambda/a = \sqrt{mn/(n-m+1)} \geq 1$, so $b' \geq 0$ and the minimum is feasible. All $m_i \geq 3$ for large $n$; the same formula also holds for $m = 2$ by the single-edge calculation. From (V.6.8),

$$\sum_i q_{m_i} = \frac{n^{3/2}}{\sqrt{2}} + O(n). \tag{V.6.16}$$

Let $J_n$ be the minimum central resistance objective with unit central budget, assigning mass $m_i$ to root $i$ and mass one to every central interior vertex. The articulation decomposition and an optimal allocation of the total budget give the exact identity

$$K_{G_n}(w^A) = \left(\sum_i q_{m_i} + \sqrt{J_n}\right)^2. \tag{V.6.17}$$

Here one uses $\min_{t_j \geq 0, \sum t_j = 1} \sum_j a_j^2/t_j = (\sum_j a_j)^2$ for the clique components and the central component. It remains to prove

$$J_n/n^3 \longrightarrow s^2 A_\ell. \tag{V.6.18}$$

For a lower bound, discard pairs involving central interior vertices. If a path has segment weights $w_{ej}$ and total budget $t_e$, its equivalent conductance is at most $t_e/p_e^2$, by the harmonic-arithmetic mean inequality, with equality for equal weights. Reducing all paths therefore leaves skeleton conductances $\widehat{c}_e$ with $\sum_e p_e^2 \widehat{c}_e \leq 1$. Put $c_e = n\widehat{c}_e$ and $\beta_i = m_i/n$. The root-only objective divided by $n^3$ is $\mathcal{K}_\beta(c)$, and

$$\sum_e (p_e/\sqrt{n})^2 c_e \leq 1.$$

The optimum of this finite skeleton problem converges to $s^2 A_\ell$. To see this without assuming an interior optimizer, observe that the pair coefficients $\beta_i\beta_j/(\alpha_i\alpha_j)$ and the cost coefficients $(p_e/\sqrt{n})^2/(s^2\ell_e^2)$ all converge to one uniformly over the fixed finite sets. Multiplicative comparison of objectives and budget sets, followed by homogeneity, squeezes their optima. This proves the required lower limit.

For the upper limit, fix any positive skeleton design $c$ with $\sum_e \ell_e^2 c_e = 1$. Give path $e$ budget $t_e = \ell_e^2 c_e$ and distribute it equally over its segments. Then its equivalent conductance is $t_e/p_e^2$ and the root-only objective divided by $n^3$ converges to $s^2\mathcal{K}_\alpha(c)$. Every central pair resistance is $O_c(n)$: a whole path has resistance $p_e^2/t_e = O_c(n)$ and the skeleton is fixed and connected. The weighted number of pairs involving an interior vertex is $O(n^{3/2})$, so their contribution is $O_c(n^{5/2}) = o(n^3)$. Finally, positive designs approach the optimum in (V.6.1): mix an optimal connected design with any fixed positive design and let the mixture parameter tend to zero. Effective resistances are continuous on connected supports. Taking first $n \to \infty$ and then this parameter to zero proves the upper limit in (V.6.18). Equations (V.6.16)–(V.6.18) prove (V.6.10).

The ratio limit follows. For $a, b > 0$, the function $s/(a+bs)^2$ has its unique maximum $1/(4ab)$ at $s = a/b$. Use $a = 1/\sqrt{2}$, $b = \sqrt{A_\ell}$, and Theorem V.6.1. In its equality case, $A_\ell = \ell^2/4$ and $s\ell = \sqrt{2}$. □

*Remark* V.6.3 (What is and is not closed). For every fixed simple skeleton with strictly positive fixed vertex masses, the optimized leading coefficient in (V.6.11) is at most $1/(4\sqrt{2})$. It is strictly smaller unless the skeleton is the balanced single edge. This includes cyclic skeletons whose dense inflations are not block graphs. The strictness is for each fixed choice; it is not a uniform gap when some masses tend to zero. This result does not cover a skeleton whose size grows with $n$,

arbitrary non-clique dense components, or general multiscale graph sequences. In particular it does not improve the previously obtained universal upper bound $\mathfrak{a}(G) \le \sqrt{n/8} + 1/2$, and the factor of two between that bound and the sharp constructed lower constant $1/(4\sqrt{2})$ remains open for arbitrary connected graphs. The quoted universal upper bound is background from the earlier general-graph comparison; it is not used in either proof here.

## V.7 Boundaries of the results and next questions

The three algebraicity classifications are complete within their stated domains. They do not imply that every special value of a transcendental function is transcendental. In particular, the hypercubic theorem concerns the minimal differential equation in the energy variable, and does not classify all differential Galois groups or all arithmetic special values. The Franel theorem does not resolve the minimum recurrence width at every weight. For beta quantiles the finite rational tests are exact for every half-integer pair, but a closed list of all asymmetric pairs passing those tests is not proved.

The two additional transfers also have exact boundaries. The Gaussian theorem covers $A \ge \sigma\sqrt{\log(eN)}$ with absolute constants, not the whole small-support parameter plane. The graph theorem covers fixed skeletons with positive fixed masses and lengths; a growing skeleton or a different dense component can require new arguments. The universal all-graph leading constant remains undetermined.

Independent internal readers examined the principal arguments, including the Franel amplitude obstruction, the Fourier endpoint issue, the cyclotomic converse construction, the Gaussian inverse inequality, and the skeleton equality case. Exact symbolic checks and compilation checks were used where useful. These checks support the draft but do not replace external mathematical review. Related literature and known special cases are identified in each section; in particular, the full text of the recent incomplete-beta arithmetic paper was not available for a complete overlap assessment. The manuscript is therefore presented as a set of explicit proofs for scrutiny, without a claim of having resolved a verified famous open conjecture or attained a specified publication tier.

## V.A Supporting estimates for the Gaussian theorem

The following estimates make the Gaussian proof independent of the preceding Poisson manuscript.

**Lemma V.A.1** (Degree-elevated Bernstein coefficients)**.** *If $P$ is a polynomial of degree $k \le m/2$ and*

$$P(x) = \sum_{j=0}^{m} b_j \binom{m}{j} x^j (1-x)^{m-j},$$

*then*

$$\max_j |b_j| \le C e^{2k^2/m} \|P\|_{[0,1]}.$$

*Proof.* Put $M = \|P\|_{[0,1]}$ and $F(z) = (1+z)^m P(z/(1+z))$; its $j$th coefficient is $\binom{m}{j} b_j$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w| \ge 1$. For $0 < j < m$ take $r = j/(m-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}\,|\sin(\phi/2)|$. Direct inversion of the Joukowski map shows that the larger modulus associated with $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Thus

$$\frac{|F(re^{i\phi})|}{M(1+r)^m} \le (1-u^2)^{(m-k)/2}(1+u)^k \le \exp(2k^2/m)\exp(-mu^2/8).$$

For the final inequality, use $k \le m/2$, $\log(1+u) \le u$, $\log(1-u^2) \le -u^2$, and complete the square in $ku - mu^2/8$. The limiting point $u = 1$ is interpreted by continuity. The average of $\exp(-mu^2/8)$ over $\phi$ is at most $C/\sqrt{1+mv}$, by comparison with a Gaussian integral near zero. Stirling's factorial bounds give

$$\binom{m}{j}\left(\frac{j}{m}\right)^j\left(1-\frac{j}{m}\right)^{m-j} \ge \frac{c}{\sqrt{1+mv}}.$$

Cauchy's coefficient formula cancels these two square-root factors and proves the claim. The endpoint coefficients are $P(0)$ and $P(1)$. □

**Lemma V.A.2** (A finite likelihood-envelope bound)**.** *Let $p$ be a probability density and let $u_1, \dots, u_M$ be nonnegative envelopes with total masses $U_a \in [1, 1+\eta]$. Suppose that, on an event $E$, a fitted density $\widehat{p}$ is dominated by some $u_a$ and satisfies $\prod_i \widehat{p}(X_i) \ge \prod_i p(X_i)$, where the $X_i$ are independent with density $p$. Then, for $x \ge 0$,*

$$\mathbb{P}\left(E \cap \left\{h(p,\widehat{p}) > \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}\right\}\right) \le e^{-x},$$

*and $\mathbb{E}[\mathbf{1}_E h(p,\widehat{p})^2] \le C\eta + C(\log M + 1)/N$.*

*Proof.* Put $r_a = u_a/U_a$. If $u_a \ge \widehat{p}$, then $\|r_a - \widehat{p}\|_1 \le 2\eta$, so $h(r_a, \widehat{p}) \le \sqrt{2\eta}$, and the likelihood comparison implies $\prod_i r_a(X_i)/p(X_i) \ge e^{-N\eta}$. The expectation of the square root of this product is the $N$th power of the affinity, at most $e^{-Nh(p,r_a)^2/2}$. Markov's inequality and a union bound over the $M$ deterministic envelopes prove the tail inequality. Integrating it proves the second assertion. The event $E$ is only intersected with the bad event; no conditioning or independence assertion on $E$ is used. □

**Lemma V.A.3** (A quantitative moment-matching pair)**.** *For every interval $[a-M, a+M]$ with $M > 0$ and integer $k \ge 1$, there are probability laws $F_k^+, F_k^-$ on that interval which match moments of every degree less than $k$ and satisfy*

$$W_1(F_k^+, F_k^-) \ge \frac{M}{\pi k}.$$

*Proof.* Push the measures $(1 \pm \frac{1}{2}T_k(x))\,dx/(\pi\sqrt{1-x^2})$ on $[-1,1]$ forward by $x \mapsto a + Mx$. Positivity, normalization and moment matching follow from $|T_k| \le 1$ and Chebyshev orthogonality. With $x = \cos\theta$, their CDF difference is $\pm\sin(k\theta)/(\pi k)$, with a sign depending only on the orientation. Therefore

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\,d\theta \ge \frac{M}{\pi k},$$

because $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\,d\theta = 1 + (4k^2-1)^{-1}$. □

# Part VI
# Global Galois Groups and Complete Parameter Classifications

**Abstract**

We develop six extensions of earlier proof methods, with explicit attention to exceptional parameters. For anisotropic hypercubic periods, the connected differential Galois group in dimension three is exactly $SO_3$, $SO_4$, or $SO_5$, according to the multiplicities of the squared weights. In every even dimension the full group is symplectic for every nonzero complex weight tuple. A unique extremal exponential factor supplies a transvection even when other thresholds collide; the same argument removes the threshold restriction from the odd-propagator banana Galois theorem, using the previously established all-mass order theorem. In arithmetic, we give a closed classification of algebraic quantiles at algebraic probability levels for every strict half-integer beta distribution. A prime-denominator obstruction eliminates all but finitely many shape rows. For cubic weighted binomial sums, local monodromy rank retains the multiplicity of colliding singularities and proves the minimum recurrence width for every nonzero complex weight. Finally, the ordinary constrained Gaussian mixture NPMLE is minimax, up to universal constants, simultaneously for all sample sizes, support radii, and noise scales. A boundary posterior drift estimate joins the small-support Hermite regime to the large-support transport regime. Proofs, supporting estimates, and the precise dependencies on preceding work are included.

**Integration note.** The remaining odd hypercubic and even-propagator Galois cases are treated in Part VII. The cubic-weight recurrence theorem is generalized there and then to all arithmetic subsequences in Part VIII. The all-scale Gaussian and exceptional half-integer quantile results remain distinct.

## VI.1 The questions and the changes in scope

The organizing question is whether a method that succeeds for generic parameters can control its exceptional set. The extensions here address three different obstructions: loss of information when singularities collide, an arithmetic test that remains indexed by infinitely many integer pairs, and a likelihood argument that fails at a support boundary. They are related by a proof strategy, not by a common underlying model.

| Question | Earlier scope or remaining restriction | Extension proved here |
|---|---|---|
| Three-dimensional lattice periods | A public announcement gives the connected $\mathrm{SO}_5$ result for very general directions. | Exact connected groups for every nonzero complex triple; all distinct squared weights give $\mathrm{SO}_5$. |
| Even-dimensional lattice periods | The preceding all-weight theorem determines order and algebraicity, but not the full group. | The full group is $\mathrm{Sp}_R$ for every even dimension and every nonzero complex tuple. |
| Odd-propagator banana periods | Manuscript [032] assumes distinct, nonzero squared signed thresholds for its group theorem. | The full symplectic group holds without either threshold condition. |
| Half-integer beta quantiles | Arithmetic reduction leaves a test depending on each integer shape pair. | A closed list of all exceptional pairs, with no unbounded search. |
| Cubic weighted sums | Distinct-singularity counting fails at $\lambda = 8, 1/8$. | Exact minimum width for every $\lambda \neq 0$, including both collisions. |
| Gaussian mixing recovery | Small support, growing support, and boundary-constrained fits require different arguments. | One uniform minimax curve for the actual constrained NPMLE over all $(N, A/\sigma)$. |

**The principal differential-algebraic advance.** The Bessel–Fourier method from the supplied period manuscript [49] and the preceding hypercubic work is combined with reflected Poincaré duality. In even dimension, all relevant local monodromies are unipotent. An extremal signed sum is unique regardless of collisions elsewhere, and gives one transvection. Connectedness, irreducibility, and that transvection force the full symplectic group. In dimension three, the monodromy block $[3, 1, 1]$ rules out every proper connected orthogonal subgroup acting irreducibly in dimension five. The resulting obstruction to elliptic expressions is functional; it says nothing about reduction of an isolated numerical value.

**The arithmetic and recurrence advances.** The beta classification combines the earlier logarithmic-transcendence reduction with a new uniform exclusion argument. A single prime in a specified interval produces a unique denominator of negative $p$-adic valuation. Nagura's theorem and a one-sided variance bound cover all large first exponents, and exact rational signs handle the remaining rows. For cubic sums, the replacement for distinct-singularity counting is the rank of local monodromy: a collided rank-two action requires at least a double zero of every scalar leading coefficient.

**The statistical advance.** The likelihood-contact and transport method from the preceding mixture work [30, 31] is combined with Hermite inversion. At an endpoint of the known support interval, posterior centering can fail. Monotonicity of the endpoint posterior weight bounds its total drift, so the same likelihood estimator can be controlled across the whole support-to-noise range. Matching testing lower bounds are uniform in that range. The interval center and radius are known; the statement does not assert unrestricted adaptation to an unknown tiny interval.

**Dependencies and unresolved questions.** Appendix VI.A reproduces the preceding hypercubic rank proof; it is supporting material, not a second new claim. The banana extension uses the precise irreducibility, rank, pairing, and infinity-monodromy inputs of [032], listed in Section VI.4; it does not reprove that manuscript in full. The other new classifications and the Gaussian rate are proved here with their stated classical inputs. The public Watson summary is an announcement, not an available complete proof against which historical priority can be settled. The beta and cubic classifications close explicit parameter questions, not identified named conjectures from the external literature. In particular this draft does not claim to resolve the Schneider-type question about general incomplete beta integrals.

## VI.2 The connected Galois group in dimension three, without exceptional directions

Let $m_1, m_2, m_3 \in \mathbb{C}^{\times}$. Put

$$F_m(z) = \sum_{k\geq 0} (2k)! \left( \sum_{a_1+a_2+a_3=k} \prod_{i=1}^{3} \frac{m_i^{2a_i}}{a_i!^2} \right) z^k, \qquad H_m(s) = s^{-1} F_m(s^{-2}).$$

For $\operatorname{Re} s > 2\sum_i |m_i|$, the latter also equals $\int_0^\infty e^{-sx} \prod_i I_0(2m_i x)\, dx$. The differential Galois group of a function means that of its minimal scalar differential equation, acting on its solution space.

**Theorem VI.2.1** (Complete three-dimensional connected-group classification)**.** *Set $r = 3, 4, 5$ according as the number of distinct values among $m_1^2, m_2^2, m_3^2$ is $1, 2, 3$. The differential Galois groups of $H_m$ over $\mathbb{C}(s)$ and $F_m$ over $\mathbb{C}(z)$ have identity component $\mathrm{SO}_r(\mathbb{C})$ in its standard representation. In particular, for every pairwise-distinct triple of nonzero squared complex weights the connected group is $\mathrm{SO}_5$. No condition on collisions or vanishing of signed mass sums is imposed. The minimal order remains $r$ after every finite algebraic extension of the respective coefficient field.*

The public announcement of Elkies, Grimm and Schwartz [39] states the connected $\mathrm{SO}_5$ result for very general directions, outside the equal-rate walls and a further countable union of algebraic curves. The theorem above removes this exceptional-direction restriction using the all-weight rank and local-monodromy theorem. It does not claim priority for the generic result, a new closed formula for the Watson integral, or information about isolated special values of the energy variable.

### VI.2.1 The precise previously established input

We use the following specialization of the all-weight hypercubic rank theorem proved in the preceding work. Its hypotheses and its geometric realization are recorded because irreducibility of a fifth-order operator alone would not suffice here. Let $B_\mu$ be the rank-two Bessel connection

$$(\theta_x^2 - 4\mu^2 x^2) y = 0, \qquad \theta_x = x\partial_x.$$

If the distinct squared masses have multiplicities $n_j$, put $M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$ and, over $K = \mathbb{C}(s)$, put $N_s = M \otimes e^{-sx}$. The connection in $s$ on this last module is $\partial_s - x$. The cyclic differential module of $H_m$ is the middle cohomology

$$\mathcal{P}_m = H^1_{\mathrm{mid}}(\mathbb{G}_{m,K}, N_s) = \operatorname{im}\{H^1_c(N_s) \longrightarrow H^1(N_s)\}. \tag{R}$$

It is irreducible over $K$. For three masses its rank and its unipotent local monodromy $T_\infty$ at $s = \infty$ are

| squared-mass multiplicities | $\bigotimes_j \mathrm{Sym}^{n_j} \mathbb{C}^2$ | $\operatorname{rank} \mathcal{P}_m$ | Jordan blocks of $T_\infty$ |
|---|---|---|---|
| $[3]$ | $\mathrm{Sym}^3$ | 3 | $[3]$ |
| $[2,1]$ | $\mathrm{Sym}^3 \oplus \mathrm{Sym}^1$ | 4 | $[3,1]$ |
| $[1,1,1]$ | $\mathrm{Sym}^3 \oplus 2\,\mathrm{Sym}^1$ | 5 | $[3,1,1]$. |

(R3)

The middle term is a decomposition for the diagonal $\mathrm{SL}_2$. The rank theorem follows from the product Bessel Galois group, intermediate extension, and Fourier stationary phase; in particular a block of size $d+1$ at the original zero contributes a block of size $d$ at the transformed infinity. The equality in (R), including visibility of this particular unilateral Laplace integral and removal of its endpoint terms, is part of that theorem. Thus it is not an assumption about a putative generic Picard–Fuchs equation.

### VI.2.2 A reflected Poincaré pairing

The next construction works in every dimension and for every squared-mass multiplicity pattern. We give it directly rather than infer the sign of a form from a Fourier-transform convention.

**Lemma VI.2.2** (A form for every choice of weights)**.** *For $n$ nonzero complex weights, the middle connection $\mathcal{P}_m$ has a nondegenerate horizontal bilinear form of parity $(-1)^{n+1}$. Here parity $+1$ means symmetric. Thus its Galois group is contained in an orthogonal group for odd $n$ and a symplectic group for even $n$.*

*Proof.* In the Bessel basis $b_\mu, \theta_x b_\mu$, the form $B_\mu(b_\mu, \theta_x b_\mu) = 1$ is alternating, nondegenerate and horizontal. Indeed, $\theta_x^2 b_\mu = 4\mu^2 x^2 b_\mu$ makes its connection matrix infinitesimally symplectic. Symmetric powers and tensor products give $M$ a horizontal nondegenerate form $B$ of parity $\epsilon = (-1)^n$.

Let $\iota : x \mapsto -x$. Its canonical action fixes each Bessel basis vector $b_\mu, \theta_x b_\mu$; the defining matrix is even in $x$. This identifies $\iota^* M$ with $M$ and preserves $B$. Consequently

$$\iota^* N_s = M \otimes e^{sx} \simeq N_s^\vee,$$

compatibly both with the $x$-connection and with the $s$-connections $\partial_s + x$ on the left and the dual of $\partial_s - x$ on the right. Curve Poincaré duality therefore supplies a perfect pairing of $\mathcal{P}_m$ with itself. The use of the middle image makes this pairing nondegenerate: the annihilator of the image of $H^1_c \to H^1$ is the kernel of the dual map, so the perfect compact-support pairing descends to the two images. This is the same duality-and-involution construction used for exponential connections in [51, §2, especially (2.12) and Corollary 2.15].

For completeness, its sign can be checked by residues. Represent $a, b \in \mathcal{P}_m$ by $N_s$-valued one-forms $\alpha, \beta$, with formal primitives $f_a, f_b$ at 0 and $\infty$. The sum of local residues gives

$$P(a,b) = \sum_{v=0,\infty} \operatorname{Res}_v B(f_a, \iota^* \beta). \tag{P}$$

Local exactness at both points is precisely the middle condition; at infinity the twisted connection has no local cohomology. Formula (P) is independent of the choices by the compact-support duality just described. Pullback by $\iota$ preserves the sum of residues, and $\iota$ is an isometry for $B$. Hence

$$P(b,a) = \sum_v \operatorname{Res}_v B(\iota^* f_b, \alpha) = \epsilon \sum_v \operatorname{Res}_v B(\alpha, \iota^* f_b).$$

The one-form

$$dB(f_a, \iota^* f_b) = B(\alpha, \iota^* f_b) + B(f_a, \iota^* \beta)$$

has zero residues. It follows that $P(b,a) = -\epsilon P(a,b)$. Residues commute with $\partial_s$, and the coefficient pairing is horizontal for the two commuting parameter connections, so $P$ is horizontal. This proves the assertion, including nondegeneracy and its sign. □

### VI.2.3 Excluding decomposition on a finite cover

**Lemma VI.2.3.** *Let $G \subseteq \mathrm{GL}(V)$ be an algebraic group acting irreducibly, with $\dim V = r \in \{3,4,5\}$. Suppose $G$ contains a unipotent whose Jordan partition is $[3, 1^{r-3}]$. Then $G^0$ acts irreducibly on $V$.*

*Proof.* The unipotent radical $U$ of $G^0$ is characteristic in $G^0$ and therefore normal in $G$. A unipotent group has a nonzero fixed vector. Its fixed space $V^U$ is $G$-invariant, hence equals $V$. Faithfulness gives $U = 1$, so $G^0$ is reductive and its representation on $V$ is completely reducible.

The isotypic components for $G^0$ are permuted transitively by $G$: otherwise the sum over a proper orbit would contradict irreducibility. If their number is $t$, their common irreducible dimension is $d$, and their common multiplicity is $e$, then $r = tde$. Every unipotent element of $G$ belongs to $G^0$, since the Zariski closure of its cyclic subgroup is the connected one-parameter unipotent group obtained by exponentiating its logarithm. On an isotypic component such an element acts as $A \otimes 1_e$, so its Jordan blocks all have sizes at most $d$. The block of size three forces $d \geq 3$. For $r = 3, 4, 5$, the equality $r = tde$ then forces $t = e = 1$ and $d = r$. Thus $V$ is irreducible under $G^0$. □

This argument deals explicitly with both permutation of components and repeated copies of one component. In particular, it excludes imprimitive finite-cover explanations of the order-five equation.

### VI.2.4 The small orthogonal representations

**Lemma VI.2.4.** *Let $G \subseteq \mathrm{O}_r(\mathbb{C})$ act irreducibly, where $r \in \{3, 4, 5\}$, and contain a unipotent of partition $[3, 1^{r-3}]$. Then $G^0 = \mathrm{SO}_r(\mathbb{C})$.*

*Proof.* By Lemma VI.2.3, $G^0$ is reductive and acts irreducibly. Its center acts by scalars by Schur's lemma; the only scalar isometries are $\pm 1$. Its Lie algebra $\mathfrak{g}$ is therefore semisimple. A maximal torus embeds in one of $\mathrm{SO}_r$, so $\operatorname{rank} \mathfrak{g} \leq 2$.

For $r = 3$, dimension at most $\dim \mathfrak{so}_3 = 3$ leaves only $\mathfrak{sl}_2 = \mathfrak{so}_3$. For $r = 4$, dimension at most six leaves $\mathfrak{sl}_2$ or $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$. The irreducible four-dimensional $\mathfrak{sl}_2$ representation is $\operatorname{Sym}^3 \mathbb{C}^2$; its unique invariant bilinear form is alternating, so it cannot preserve the symmetric form on $V$. The other possibility has dimension six and hence equals $\mathfrak{so}_4$.

For $r = 5$, the rank-at-most-two semisimple possibilities of dimension at most ten are

$$A_1, \quad A_1 \oplus A_1, \quad A_2, \quad B_2 = C_2.$$

The type $G_2$ has dimension fourteen and is too large. A faithful irreducible representation of a product of two simple factors is a tensor product of nontrivial representations, which cannot have prime dimension five. Type $A_2$ has irreducible dimensions

$$\frac{(a+1)(b+1)(a+b+2)}{2}, \qquad a, b \in \mathbb{Z}_{\geq 0},$$

none equal to five. This formula is the $A_2$ case of the Weyl dimension formula; the rank-two root systems and that formula are standard highest-weight theory, see [52]. For type $A_1$, the only five-dimensional irreducible is $\operatorname{Sym}^4 \mathbb{C}^2$. A nonidentity unipotent acts there as a single block of size five: its logarithm is the raising operator on binary quartics, whose nilpotence index is five. Our unipotent has partition $[3, 1, 1]$, a contradiction. Thus $\mathfrak{g}$ is $B_2$ of dimension ten, so it equals $\mathfrak{so}_5$.

In every case a connected closed subgroup of $\mathrm{SO}_r$ with its full Lie algebra is $\mathrm{SO}_r$ itself. □

*Proof of Theorem VI.2.1.* Input (R3) supplies irreducibility and the indicated unipotent for $H_m$. Lemma VI.2.2, with $n = 3$, supplies the orthogonal form. Lemma VI.2.4 gives the claimed connected group. The change $z = s^{-2}$ is a finite algebraic base change and $H_m = s^{-1} F_m(s^{-2})$ differs by a rational gauge. A finite algebraic base change preserves the identity component of a Picard–Vessiot group [43], so the conclusion holds for $F_m$ as well. Finally, after any further finite algebraic base change the Galois group still contains the standard $\mathrm{SO}_r$, which acts irreducibly. Thus the differential module remains irreducible of rank $r$ and its nonzero solution cannot satisfy a smaller-order scalar equation. □

**Corollary VI.2.5** (Exactly one algebraic relation among the first jets)**.** *Suppose $m_1^2, m_2^2, m_3^2$ are pairwise distinct and nonzero. Then*

$$\operatorname{trdeg}_{\mathbb{C}(s)} \mathbb{C}(s)(H_m, H_m', H_m'', H_m''', H_m^{(4)}) = 4.$$

*Their polynomial relations over $\mathbb{C}(s)$ form the principal ideal generated by one nondegenerate quadratic polynomial. For $F_m$ and its first four derivatives the same transcendence degree and single-quadric assertion hold over $\overline{\mathbb{C}(z)}$.*

*Proof.* Choose the cyclic cohomology basis $\omega, \nabla_s\omega, \dots, \nabla_s^4\omega$ in (R), and let $C(s) \in \mathrm{GL}_5(\mathbb{C}(s))$ be the Gram matrix of Lemma VI.2.2. A scalar solution $f$ is a horizontal linear functional in this basis. The dual form gives

$$(f, f', f'', f''', f^{(4)})\, C(s)^{-1} (f, f', f'', f''', f^{(4)})^T = c_f \in \mathbb{C}. \tag{Q}$$

For $f = H_m$ its solution vector is nonzero. The orbit of any nonzero vector under $\mathrm{SO}_5$ has dimension four: it is a nonzero level of a nondegenerate quadratic form, or the punctured isotropic cone. Since the full group lies in $\mathrm{O}_5$, it has the same orbit dimension. Picard–Vessiot Galois correspondence identifies the transcendence degree of the generated solution field with this orbit dimension. The polynomial relation ideal is therefore prime of height one and contains (Q). A nondegenerate quadratic polynomial in five variables, with or without a constant term, is absolutely irreducible: a factorization into affine linear forms would give quadratic rank at most two. Hence (Q) generates the ideal. Finite algebraic extension, the substitution $z = s^{-2}$, and the invertible triangular change between the two derivative vectors preserve these assertions over the algebraic closure. □

**Corollary VI.2.6** (A precise obstruction to elliptic expressions)**.** *In the pairwise-distinct squared-mass case, let $K'$ be any finite algebraic extension of $\mathbb{C}(s)$. Let $E/K'$ be a compositum of finitely many Picard–Vessiot fields of rank-two elliptic Gauss–Manin connections, allowing algebraic pullbacks and finite monodromy factors. Then $H_m$ is not algebraic over $E$. The analogous assertion holds for $F_m$ after a finite algebraic extension of $\mathbb{C}(z)$.*

*Proof.* The connected Galois group of $E$ is a subgroup of a product of elliptic $\mathrm{SL}_2$ groups. Every simple factor of its semisimple Lie quotient is therefore of type $A_1$: a simple factor must project nontrivially to some $\mathfrak{sl}_2$ factor, where it injects and hence equals $\mathfrak{sl}_2$.

Suppose $H_m$ were algebraic over $E$, and let $P$ be the Picard–Vessiot field of its irreducible equation. The compositum $EP$ is Picard–Vessiot over $K'$. All conjugates of $H_m$ under $\mathrm{Gal}(EP/K')$ are algebraic over $E$, because $E$ is stable under this group. Restriction to $\mathrm{Gal}(P/K')$ is surjective, so these conjugates span the whole solution space by irreducibility. A finite solution basis and its derivatives are therefore algebraic over $E$, so $EP/E$ is finite. Restriction from the Galois group of $EP$ to that of $E$ has finite kernel and is surjective, and hence induces an isomorphism of Lie algebras. Restriction to the Galois group of $P$ is surjective. It would follow that $\mathfrak{so}_5$, a simple Lie algebra of type $B_2$, is a quotient of a Lie algebra whose semisimple factors are all $A_1$, which is impossible. The finite base change and gauge relating $F_m$ to $H_m$ give the other assertion. □

This includes algebraic operations on finitely many elliptic period systems and their derivatives. It is not a claim about unrestricted compositions of special functions or about a single numerical value of the period.

## VI.3 Full symplectic Galois groups in every even dimension

Keep the hypercubic series $F_m$ and its Laplace form $H_m$, now for an arbitrary even number $n \geq 2$ of nonzero complex weights. Let the distinct squared weights have multiplicities $n_1, \dots, n_q$, and set

$$R = \prod_{j=1}^{q} (n_j + 1) - [u^{n/2}] \prod_{j=1}^{q} (1 + u + \cdots + u^{n_j}).$$

The all-weight order theorem identifies $R$ as the rank of the irreducible middle connection (R), with no conditions on signed sums.

**Theorem VI.3.1** (All even dimensions and all nonzero complex weights)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois groups of $H_m$ over $\mathbb{C}(s)$ and $F_m$ over $\mathbb{C}(z)$ are $\mathrm{Sp}_R(\mathbb{C})$ in their standard representations. Thus $R$ is even, and for each function $f = H_m$ or $F_m$ the $R$ functions $f, f', \dots, f^{(R-1)}$ are algebraically independent over its coefficient field. These assertions hold even when signed mass sums collide or vanish.*

Orthogonal and symplectic structures for lattice Green differential operators have long been studied; see, for example, [53]. The statement here determines the full group uniformly over every nonzero complex weight tuple in every even dimension, including collided thresholds. No historical-priority claim is inferred from a literature search.

Here $\mathrm{Sp}_R$ denotes the group preserving a nondegenerate alternating form on an $R$-dimensional space. We first isolate the linear-algebra argument that will identify the group.

**Lemma VI.3.2** (One transvection in a connected irreducible group)**.** *Let $(V, \omega)$ be a finite-dimensional complex symplectic space. If a connected algebraic subgroup $G \subseteq \mathrm{Sp}(V)$ acts irreducibly and contains a nonidentity transvection, then $G = \mathrm{Sp}(V)$.*

*Proof.* Write $N_u(v) = \omega(u, v)u$. A symplectic transvection is $1 + aN_u$ for some $u \neq 0$ and $a \neq 0$; this follows directly by writing its rank-one difference as $v \mapsto \ell(v)u$ and using the symplectic identity to conclude $\ell \in \mathbb{C}\omega(u, -)$. Rescaling $u$ absorbs $a$. Its powers are $1 + kN_u$, so their Zariski closure is $\{1 + tN_u : t \in \mathbb{C}\}$. Consequently $N_u \in \mathfrak{g} = \mathrm{Lie}\, G$, as is $N_v$ for every $v \in \mathcal{O} = G \cdot u$. The span of $\mathcal{O}$ is a nonzero invariant space, hence is $V$.

Join two orbit vectors when their symplectic pairing is nonzero. The spans $W_i$ of different connected components of this graph are mutually orthogonal. Their sum is direct: if a finite sum $\sum_i w_i = 0$ with $w_i \in W_i$, then $w_i$ is orthogonal to $W_i$ by this equality and to all other $W_j$ by construction; it is orthogonal to $V$, hence zero. Thus there are finitely many components. The group $G$ permutes their spans. The stabilizer of each span is a closed subgroup of finite index, so connectedness makes every span $G$-invariant. Irreducibility forces a single component. Choose finitely many orbit vectors spanning $V$ and add the finitely many vertices of paths connecting them. This produces a finite connected spanning set $S \subseteq \mathcal{O}$.

For $v, w \in V$ put

$$L_{v,w}(x) = \omega(v, x)w + \omega(w, x)v.$$

The identities

$$[N_v, N_w] = \omega(v, w)L_{v,w}, \qquad [L_{v,w}, N_y] = \omega(v, y)L_{w,y} + \omega(w, y)L_{v,y} \tag{T}$$

follow by direct multiplication. The first gives $L_{v,w} \in \mathfrak{g}$ for edges of the graph, and $L_{v,v} = 2N_v$ is also in $\mathfrak{g}$. Induct on the length of a path from $v$ to $y$. If $\omega(v, y) \neq 0$, the first identity already applies. Otherwise, let $w$ be the penultimate vertex. By induction $L_{v,w} \in \mathfrak{g}$, and the second identity gives $L_{v,y} \in \mathfrak{g}$ because $\omega(w, y) \neq 0$. Hence all $L_{v,w}$ with $v, w \in S$ belong to $\mathfrak{g}$. Since $S$ spans $V$, these span $\mathrm{Sym}^2 V \simeq \mathfrak{sp}(V)$. Thus $\mathfrak{g} = \mathfrak{sp}(V)$, and connectedness proves the lemma. □

### VI.3.1 Formal factors and collided thresholds

We state exactly the local Fourier input. If a holonomic module $\mathcal{K}$ has formal germ at infinity

$$\widehat{\mathcal{K}}_\infty = \bigoplus_{c\in\mathbb{C}} E^{cx}\otimes R_c, \qquad R_c \text{ regular},$$

then stationary phase at the finite transformed point $s = c$ gives

$$\phi_c(\mathcal{F}\mathcal{K}) \simeq \mathcal{F}^{(\infty,0)}(R_c). \tag{LF}$$

Here $\phi_c$ is the formal vanishing-cycle germ, including its regular unipotent part. For a regular germ the local transform on the right is regular of the same rank and preserves its monodromy (up to the harmless inversion convention). In particular a trivial rank-one germ becomes the one-dimensional vanishing cycles of $\delta_0$, with identity monodromy. This statement includes $c = 0$. It follows either by applying the global Fourier transform to the trivial connection or from the inverse regular local-transform conventions. Precise references are Arinkin [77, Corollary 2.3(1), §3.3], García López [40, Proposition 2.1], and Sabbah [41, Remarks 5.2, 5.3(4), 5.7].

For an intermediate extension of positive rank, with nearby monodromy $T$, the canonical map from nearby to vanishing cycles is surjective and the variation map in the opposite direction is injective. Their composites are $T - 1$. Consequently

$$\phi_c \text{ has identity monodromy} \quad\Longrightarrow\quad T \text{ is unipotent and } (T-1)^2 = 0, \qquad \operatorname{rank}(T-1) = \dim \phi_c. \tag{VC}$$

Indeed the vanishing cycles contain every nonidentity generalized eigenspace of $T$ and the image of $T - 1$ on its unipotent part. Thus (VC) does not discard a constant formal factor.

**Lemma VI.3.3.** *For even $n$, the minimal equation of $H_m$ is regular singular everywhere. Each finite local monodromy is unipotent with square-zero difference from the identity. At some finite nonzero point this difference has rank one.*

*Proof.* Let $\mu_j^2$ be the distinct squared weights and put $\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}$. For $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ the Bessel formal solutions at infinity give the direct-sum decomposition

$$\widehat{M}_\infty \simeq \bigoplus_{\lambda\in\Lambda} E^{c_\lambda x}\otimes R_n, \qquad c_\lambda = 2\sum_j \lambda_j\mu_j, \tag{B}$$

where $R_n$ is rank one with a formal solution $x^{-n/2}$. The invertible formal power series in the Bessel expansions are absorbed by rank-one formal gauges. Since $n$ is even, $R_n$ is the trivial formal connection. Distinct indices with equal $c_\lambda$ remain a direct sum of trivial regular factors; collision introduces no extension or logarithm in (B).

The all-weight theorem identifies the equation of $H_m$ with the generic connection of the simple module $\mathcal{F}(j_{!*}M)$. Its positive rank ensures that this is an intermediate extension at every finite point. If $d_c = \#\{\lambda \in \Lambda : c_\lambda = c\}$, (LF) says its formal vanishing cycles at $c$ are regular, of dimension $d_c$, and have identity monodromy. At other points the vanishing cycles are zero. The purely irregular part of a holonomic germ is unchanged by formal vanishing cycles, so regularity of these vanishing cycles excludes an irregular nearby part. By (VC), all these points are regular singular with the asserted unipotence. The infinity statement of the all-weight theorem already gives regularity and unipotence at infinity.

Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every $j$. Define $\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j)$. This is the unique maximizer of the real functional $\operatorname{Re}(\xi c_\lambda)$ on $\Lambda$, and its value is $2\sum_j n_j|\operatorname{Re}(\xi\mu_j)| > 0$. Thus $c_{\lambda^*} \neq 0$ and $d_{c_{\lambda^*}} = 1$. The corresponding local monodromy is a nonidentity transvection. □

*Proof of Theorem VI.3.1 for $H_m$.* The middle pairing in Lemma VI.2.2 is alternating, so the Galois group $G$ lies in $\mathrm{Sp}_R$. The all-weight rank theorem gives irreducibility. By Lemma VI.3.3 the equation is regular singular everywhere, so Schlesinger's density theorem identifies $G$ with the Zariski closure of its monodromy group [43]. The fundamental group of the punctured sphere is generated by the local loops. All their matrices are unipotent, including the one at infinity, so all belong to $G^0$. Thus $G = G^0$. Lemma VI.3.3 supplies one transvection, and Lemma VI.3.2 gives $G = \mathrm{Sp}_R$. □

### VI.3.2 Descent to the ordinary generating function

It remains to exclude a finite similitude multiplier on descending from $s$ to $z = s^{-2}$. This can be checked on the explicit pairing. Let $\omega = [\eta\, dx]$ be the cyclic class evaluating to $H_m$ in (R), where $\eta$ is the even tensor product of the Bessel cyclic vectors. The simultaneous involution

$$J : (s, x) \longmapsto (-s, -x)$$

preserves $N_s$ and induces a semilinear involution of $\mathcal{P}_m$ covering $s \mapsto -s$. It is an isometry for (P): applying $J$ to both entries and then changing $x$ to $-x$ leaves the sum of residues unchanged. On the cyclic class it has $J\omega = -\omega$ because $dx$ changes sign. Therefore $\zeta = s\omega$ is fixed. The parameter derivative

$$\nabla_z = -\frac{s^3}{2}\nabla_s$$

commutes with $J$, so all $\nabla_z^i \zeta$ are fixed. They are a basis by the order theorem and the parity descent for $F_m(s^{-2}) = sH_m(s)$. In this basis the Gram matrix of (P) is fixed by $s \mapsto -s$, and hence belongs to $\mathbb{C}(s^2) = \mathbb{C}(z)$. It is nondegenerate, alternating and $z$-horizontal. Thus the full Galois group of $F_m$ lies in $\mathrm{Sp}_R$. Its identity component is unchanged by the quadratic extension to $\mathbb{C}(s)$, where the group just proved is $\mathrm{Sp}_R$. The full group over $\mathbb{C}(z)$ must therefore be $\mathrm{Sp}_R$ as well.

Finally $\mathrm{Sp}_R$ acts transitively on the nonzero vectors of its standard representation. The orbit of the nonzero solution $H_m$ has dimension $R$, and the same holds for $F_m$. Picard–Vessiot Galois correspondence gives transcendence degree $R$ for the field generated by the first $R$ jets. Since it has exactly $R$ generators, they are algebraically independent. This completes the proof of Theorem VI.3.1.

## VI.4 Removing all threshold restrictions in odd-propagator banana periods

The same transvection argument has a second application to the banana periods in the supplied manuscript [49]. Here the number of Bessel factors is odd, whereas it is even in the hypercubic application. The two integral transforms are different, and so are their parity laws. This section removes the nonzero, distinct-threshold hypothesis from the odd-factor Galois-group theorem in that manuscript. It does not prove the remaining leading-coefficient multiplicity conjecture.

For nonzero complex masses $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \qquad \varpi_m(t) = \sum_{k\geq 0} \frac{c_k(m)}{t^{k+1}}.$$

The series converges near infinity. Group the masses by equality of their squares, with multiplicities $n_1, \dots, n_q$, and write

$$\bigotimes_{j=1}^{q} \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d\geq 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

for restriction to the diagonal $\mathrm{SL}_2$.

**Precise prior input.** The all-mass order theorem of [49] states that the minimal connection of $\varpi_m$ is irreducible, has rank

$$o(m) = \sum_{d\geq 2} (d-1)a_d,$$

and has unipotent monodromy at infinity with blocks $J_{d-1}$, one for each summand with $d \geq 2$. In particular, the block $J_{n-1}$ occurs once. Its middle-cohomology construction gives an invariant nondegenerate alternating form when $n$ is odd. These statements do not require distinct or nonzero signed thresholds. They are used here as previously proved inputs, rather than counted again as new results. The cited manuscript contains their proofs; the new removal of threshold restrictions is proved below.

**Theorem VI.4.1** (Every odd number of propagators, every nonzero mass tuple)**.** *Let $n \geq 3$ be odd and $m_i \in \mathbb{C}^\times$. The differential Galois group over $\mathbb{C}(t)$ of the minimal equation of $\varpi_m$ is*

$$\mathrm{Sp}(o(m), \mathbb{C}).$$

*This holds when nonzero thresholds coincide and when signed mass sums vanish. Consequently*

$$\varpi_m, \varpi_m', \dots, \varpi_m^{(o(m)-1)}$$

*are algebraically independent over $\mathbb{C}(t)$, and remain so over every finite algebraic extension of that differential field.*

### VI.4.1 Why collisions do not obstruct connectedness

Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel connection is

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}, \qquad B_\mu : \quad (\theta_y^2 - \mu^2 y/4)f = 0, \quad \theta_y = y\partial_y.$$

On the formal $u$-cover at infinity, it splits as a direct sum of rank-one factors indexed by

$$\lambda_j \in \{n_j, n_j - 2, \dots, -n_j\}, \qquad c(\lambda) = \sum_j \lambda_j \mu_j.$$

Each factor has the form

$$e^{c(\lambda)u}u^{-n/2}(1+O(u^{-1})).$$

The splitting comes from the two formal Bessel solutions in each factor before taking symmetric and tensor powers. Equality of two exponential factors therefore gives a direct sum with multiplicity, not a new nontrivial formal extension.

For $c \neq 0$, the pair indexed by $\lambda, -\lambda$ descends to

$$A_c(n) = [u \mapsto y = u^2]_+(E^{cu} \otimes U_n), \qquad U_n : \ u^{-n/2}.$$

The standard local Fourier calculation sends this pair, after the first Fourier transform, to

$$E^{c^2/(4s)} \otimes R_c, \qquad R_c \text{ regular of rank one.}$$

The power in its Laplace asymptotic is $s^{(n-3)/2}$: under $y = u^2$, the measure contributes $2u\,du$, evaluation at the critical point $u = c/(2s)$ contributes $s^{n/2-1}$, and the Gaussian fluctuation contributes $s^{-1/2}$. Equivalently this is the formal stationary-phase formula [41, 40]. Thus the monodromy of $R_c$ is $(-1)^{n-3}$, independent of $c$. Only its value 1 for odd $n$ is needed here.

The construction of the banana connection first takes the Fourier transform of $j_{!*}M$, then pulls its restriction to $\mathbb{G}_m$ back by $s = 1/v$, and then takes the Fourier transform of the intermediate extension in $v$. The rational gauge $v^{-1}$ in the scalar integral representation has no effect on monodromy. If $\tau$ is the second Fourier variable, the period normalization is $t = 4\tau$; the factor $E^{c^2v/4}$ is therefore located at $t = c^2$. The factor above produces vanishing cycles at $t = c^2$ with the same monodromy. If several pairs give the same nonzero $c^2$, their vanishing-cycle spaces are the direct sum of these lines. For odd $n$ their monodromy is the identity.

The final Fourier module is simple of positive rank. Hence it is the intermediate extension at each finite point. If $T_a$ denotes its local monodromy at a nonzero threshold $a$, the nearby/vanishing-cycle description gives

$$\phi_a \cong \operatorname{im}(T_a - I), \qquad (T_a - I)^2 = 0.$$

In particular $T_a$ is unipotent, even at a collision. The regular part arising from $c = 0$ is still regular through the two transforms; it can affect the monodromy at zero, but creates no further nonzero singularity. All formal local factors of the final module are regular, so its global equation is regular singular.

The monodromy group is generated by the loops around infinity and the nonzero thresholds: in the fundamental group relation one can omit the loop around zero. Every one of these generators is unipotent. The Zariski closure of the cyclic group generated by a unipotent matrix in characteristic zero is connected. Schlesinger's theorem therefore shows that the differential Galois group $G$ is connected. This argument needs no assertion that the monodromy at zero itself is unipotent.

### VI.4.2 A simple threshold always survives

Choose $\xi \in \mathbb{C}$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every $j$, and maximize $\operatorname{Re}(\xi c(\lambda))$ over the finite index set. There is a unique maximizer, given by

$$\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j).$$

Its value has strictly positive real projection, so $c_* := c(\lambda^*) \neq 0$. The only tuple giving $c_*$ is $\lambda^*$; the only one giving $-c_*$ is $-\lambda^*$. Over $\mathbb{C}$, equality $c(\lambda)^2 = c_*^2$ means $c(\lambda) = c_*$ or $-c_*$. Consequently the squared threshold $c_*^2$ has exactly one contributing pair, even if other thresholds collide or vanish.

Its vanishing-cycle space is one-dimensional with trivial monodromy. Since the final module is an intermediate extension, the corresponding $T_*$ satisfies

$$\operatorname{rank}(T_* - I) = 1, \qquad (T_* - I)^2 = 0.$$

It is a nontrivial symplectic transvection.

*Proof of Theorem VI.4.1.* The prior input gives irreducibility and an invariant alternating form. The preceding arguments give connectedness and a nontrivial transvection. Lemma VI.3.2 therefore gives $G = \mathrm{Sp}(o(m))$.

Let $E$ be a Picard–Vessiot extension and $v$ the nonzero solution vector corresponding to $\varpi_m$. The field generated by its first $o(m)$ derivatives is the fixed field of its stabilizer. Its transcendence degree is consequently the dimension of the $G$-orbit of $v$. A symplectic group is transitive on nonzero vectors, so this dimension is $o(m)$. There are exactly $o(m)$ generators, and hence they are algebraically independent. Algebraic extension of the base field preserves their transcendence degree; equivalently, the connected Galois group is unchanged by finite algebraic base extension. □

For pairwise distinct squared masses, the rank in this theorem is the established formula

$$o(m) = 2^n - \binom{n+1}{\lfloor (n+1)/2 \rfloor}.$$

For all equal squared masses it is $n-1$. The theorem concerns functional algebraic independence, not arithmetic independence of values at a fixed energy. It also does not determine the full Galois group for every even $n$; the reflection case requires a separate argument.

## VI.5 The complete exceptional-pair classification for half-integer beta quantiles

For the strict half-integer beta family, an arithmetic reduction leaves three candidate algebraic probability levels. The remaining issue is to classify the integer shape pairs that actually attain those levels at algebraic quantiles. We prove a closed list: apart from the arcsine family and symmetric medians, the only exceptions are

$$I_{1/4}(3/2, 5/2) = 1/3, \qquad I_{3/4}(5/2, 3/2) = 2/3.$$

The new ingredient is a prime-denominator obstruction for a finite Fourier sum. A central-coefficient product makes that obstruction explicit; Nagura's prime-interval theorem and an elementary probability bound then reduce the unbounded parameter lattice to eleven exact sign comparisons.

### VI.5.1 Statement of the classification

For $a, b > 0$, write

$$B_x(a, b) = \int_0^x t^{a-1}(1-t)^{b-1}\, dt, \qquad I_x(a, b) = \frac{B_x(a, b)}{B_1(a, b)}.$$

The inverse $I^{-1}_{a,b}$ means the strictly increasing inverse in $x \in (0, 1)$; the subscript records the two fixed shapes. For integers $m, n \geq 0$, define

$$\mathcal{E}_{m,n} = \{r \in \overline{\mathbb{Q}} \cap (0, 1) : I^{-1}_{m+1/2,n+1/2}(r) \in \overline{\mathbb{Q}}\}.$$

**Theorem VI.5.1** (Closed classification of all strict half-integer pairs)**.** *For all integers $m, n \geq 0$,*

$$\mathcal{E}_{m,n} = \begin{cases} \mathbb{Q} \cap (0, 1), & m = n = 0, \\ \{1/2\}, & m = n \geq 1, \\ \{1/3\}, & (m, n) = (1, 2), \\ \{2/3\}, & (m, n) = (2, 1), \\ \varnothing, & \textit{otherwise.} \end{cases} \tag{VI.5.1}$$

*The corresponding quantiles are respectively* $\sin^2(\pi r/2)$*,* $1/2$*,* $1/4$*, and* $3/4$*. Every algebraic probability level not listed in* (VI.5.1) *has a transcendental quantile.*

The distinction from the earlier finite arithmetic test is that (VI.5.1) contains no parameter-dependent polynomial evaluation or search. Its proof settles that test simultaneously for every integer pair. We give the arithmetic reduction as well, so that the proof is complete apart from its explicitly cited classical transcendence and prime-interval inputs.

### VI.5.2 Reduction to three algebraic points

Let $w_{m,n}(X) = X^m(1-X)^n$, and put

$$c_{m,n} = \frac{(2m)!(2n)!}{4^{m+n} m!\, n!\, (m+n)!} > 0. \tag{VI.5.2}$$

There is a unique polynomial $R_{m,n} \in \mathbb{Q}[X]$ with

$$w_{m,n}(X) = c_{m,n} + X(1-X)R'_{m,n}(X) + (1/2 - X)R_{m,n}(X). \tag{VI.5.3}$$

Indeed, the operator on a monomial $X^j$ is $(j+1/2)X^j-(j+1)X^{j+1}$, so descending elimination leaves a unique constant. Integration against $[x(1-x)]^{-1/2}$ shows that this constant is (VI.5.2). Consequently

$$I_x(m+1/2,n+1/2)=\frac{2\arcsin\sqrt{x}}{\pi}+\frac{\sqrt{x(1-x)}}{\pi c_{m,n}}R_{m,n}(x). \tag{VI.5.4}$$

In particular,

$$\operatorname{sgn} R_{m,n}(1/4)=\operatorname{sgn}\{I_{1/4}(m+1/2,n+1/2)-1/3\}. \tag{VI.5.5}$$

**Lemma VI.5.2** (Arithmetic and shape reduction)**.** *If $x\in(0,1)$ is algebraic, then*

$$I_x(m+1/2,n+1/2)\in\overline{\mathbb{Q}}\quad\Longleftrightarrow\quad R_{m,n}(x)=0\quad\text{and}\quad\frac{2\arcsin\sqrt{x}}{\pi}\in\mathbb{Q}. \tag{VI.5.6}$$

*If $m,n\geq 1$, $R_{m,n}$ has exactly one root in $(0,1)$, and any algebraic level with an algebraic quantile is one of*

$$(r,x)=(1/3,1/4),\quad(1/2,1/2),\quad(2/3,3/4).$$

*The middle pair occurs if and only if $m=n$. If exactly one of $m,n$ is zero, there are no exceptions.*

*Proof.* Suppose $x$ and $r=I_x(m+1/2,n+1/2)$ are algebraic, and set $\theta=\arcsin\sqrt{x}\in(0,\pi/2)$. Then $\zeta=e^{2i\theta}=1-2x+2i\sqrt{x(1-x)}$ is algebraic. With $\log\zeta=2i\theta$ and $\log(-1)=i\pi$, (VI.5.4) makes

$$\log\zeta-r\log(-1)=-i\frac{\sqrt{x(1-x)}}{c_{m,n}}R_{m,n}(x)$$

algebraic. Baker's theorem on algebraic-coefficient linear forms in logarithms says that such a form is either zero or transcendental [46]. It must vanish. Thus $R_{m,n}(x)=0$ and $\zeta=e^{i\pi r}$. If $r$ were algebraic irrational, the Gelfond–Schneider theorem applied to this value of $(-1)^r$ would contradict the algebraicity of $\zeta$. Hence $r$ is rational. The converse follows from (VI.5.4).

For the root count, put

$$g(\theta)=\sin^{2m}\theta\cos^{2n}\theta,\qquad h(\theta)=\int_0^\theta(g(u)-c_{m,n})\,du.$$

The constant $c_{m,n}$ is the average of $g$ on $[0,\pi/2]$, and (VI.5.4) yields

$$h(\theta)=\tfrac{1}{2}\sqrt{x(1-x)}R_{m,n}(x),\quad x=\sin^2\theta,\qquad h(0)=h(\pi/2)=0. \tag{VI.5.7}$$

For $m,n\geq 1$, $g$ vanishes at both endpoints and has one strict maximum. Its logarithmic derivative in the interior is $2m\cot\theta-2n\tan\theta$, strictly decreasing through zero. Thus $g-c_{m,n}$ has exactly two zeros. The function $h$ first decreases, then increases, then decreases; it is negative near zero and positive just before $\pi/2$. It has exactly one interior zero.

For a rational candidate $r$, let $d$ be the order of $e^{i\pi r}$, so $d\geq 3$. The Galois conjugates of $x=(2-\zeta-\zeta^{-1})/4$ are the distinct numbers $\sin^2(\pi j/d)$, with $(j,d)=1$, identified only under $j\mapsto -j$. All $\varphi(d)/2$ conjugates lie in $(0,1)$ and are roots of the rational polynomial $R_{m,n}$. The one-root count forces $\varphi(d)\leq 2$, hence $d\in\{3,4,6\}$. These give exactly the three displayed pairs.

For the middle pair, symmetry gives $I_{1/2}=1/2$ if $m=n$. If $m>n$, the density at $x\in(0,1/2)$ divided by its density at $1-x$ is $(x/(1-x))^{m-n}<1$, so $I_{1/2}<1/2$. The reverse inequality holds when $m<n$.

If precisely one exponent vanishes, $g$ is strictly monotone, so $g-c_{m,n}$ has exactly one zero. Were $h$ to have an interior zero, its two endpoint zeros would force at least two interior zeros of $h'$ by Rolle's theorem. This is impossible. Equation (VI.5.6) then excludes every exception. □

### VI.5.3 The prime obstruction

**Lemma VI.5.3** (A central-coefficient product)**.** *For every integer $q \geq 0$ and every number $A$, as an identity of polynomial coefficients in $A$,*

$$[z^q](1-z)^A(1+z)^{2q-A} = \frac{(-2)^q}{q!}\prod_{j=1}^{q}(A-(2j-1)). \tag{VI.5.8}$$

*The empty product for $q=0$ equals one. Generalized powers on the left are expanded as formal power series at zero.*

*Proof.* The coefficient is a polynomial in $A$ of degree at most $q$. In

$$(1+z)^{2q}\exp\!\left(A\log\frac{1-z}{1+z}\right), \qquad \log\frac{1-z}{1+z} = -2z + O(z^3),$$

the coefficient of $A^q z^q$ is $(-2)^q/q!$. For each odd integer $A = 1, 3, \ldots, 2q-1$, the expression $(1-z)^A(1+z)^{2q-A}$ is a polynomial of degree $2q$ satisfying $z^{2q}P(1/z) = -P(z)$. Its central coefficient therefore vanishes. These $q$ roots and the leading coefficient prove (VI.5.8). □

**Lemma VI.5.4** (Prime-denominator obstruction)**.** *Let $m, n \geq 1$. If there is a prime $p \geq 5$ such that*

$$\max\{2m, 2n/3\} < p \leq m+n, \tag{VI.5.9}$$

*then $R_{m,n}(1/4) \neq 0$.*

*Proof.* Set $L = m+n$ and write

$$(1-z)^{2m}(1+z)^{2n} = \sum_{j=0}^{2L} a_j z^j.$$

All $a_j$ are integers, and reciprocity gives $a_{2L-j} = a_j$. On setting $z = e^{2i\theta}$, we obtain

$$\sin^{2m}\theta\cos^{2n}\theta = \frac{(-1)^m}{4^L}\left(a_L + 2\sum_{k=1}^{L} a_{L+k}\cos(2k\theta)\right).$$

Its average on $[0, \pi/2]$ is $(-1)^m 4^{-L} a_L = c_{m,n}$. Define the integer-valued function

$$\chi(k) = \frac{2}{\sqrt{3}}\sin\frac{k\pi}{3}; \qquad (\chi(0), \ldots, \chi(5)) = (0, 1, 1, 0, -1, -1),$$

extended periodically modulo six. Integrating the Fourier expansion to $\theta = \pi/6$, and using (VI.5.7), gives the exact identity

$$R_{m,n}(1/4) = (-1)^m 4^{1-L}\sum_{k=1}^{L}\frac{\chi(k)a_{L+k}}{k}. \tag{VI.5.10}$$

The inequality $\max\{2m, 2n/3\} \geq (m+n)/2$ follows by considering $n \leq 3m$ and $n \geq 3m$. Thus (VI.5.9) implies $p > L/2$. Set $q = L - p$, so $0 \leq q < p$. Among the denominators $1, \ldots, L$, exactly $k = p$ is divisible by $p$. Also $\chi(p) \in \{1, -1\}$, since $p \geq 5$. We show that $a_{L+p}$ is not divisible by $p$.

By reciprocity, $a_{L+p} = a_q$. Since $2n = 2p + 2q - 2m$, the identity $(1+z)^p = 1 + z^p$ in $\mathbb{F}_p[[z]]$ shows that

$$a_q \equiv [z^q](1-z)^{2m}(1+z)^{2q-2m} \pmod{p}. \tag{VI.5.11}$$

This remains valid if $2q - 2m < 0$: the power series $1 + z$ is invertible, and $(1 + z)^{2p} = (1 + z^p)^2$ contributes only its constant term through degree $q < p$. By Lemma VI.5.3, the right side is

$$\frac{(-2)^q}{q!} \prod_{j=1}^{q} (2m - (2j - 1)) \pmod{p}.$$

The denominator $q!$ is a unit modulo $p$. Every factor in the product is a nonzero odd integer; its largest possible value is $2m - 1 < p$, and its smallest is

$$2m - 2q + 1 = 2p - 2n + 1 > -p,$$

where the last inequality uses $p > 2n/3$. No factor is divisible by $p$, proving that $a_q$ is a unit. The summand with $k = p$ in (VI.5.10) now has $p$-adic valuation $-1$, while all other summands have nonnegative valuation. The sum is nonzero, as claimed. □

### VI.5.4 Uniform exclusion of the unbounded shape ranges

We use Nagura's theorem in its original real-variable form: for every real $y \geq 25$ there is a prime

$$y < p < \frac{6}{5}y. \tag{VI.5.12}$$

This is the $n = 5$ case of the theorem on page 180 of [54].

**Lemma VI.5.5** (All sufficiently large first exponents)**.** *For every integer $m \geq 13$ and every integer $n \geq 1$, $R_{m,n}(1/4) \neq 0$.*

*Proof.* For fixed $a > 0$ and $x \in (0, 1)$, the identity

$$I_x(a, b + 1) - I_x(a, b) = \frac{x^a(1 - x)^b}{bB_1(a, b)} > 0 \tag{VI.5.13}$$

follows by integration by parts and $B_1(a, b + 1) = bB_1(a, b)/(a + b)$. Thus the beta CDF at $1/4$ strictly increases with $n$ for fixed $m$.

First suppose $n \leq n_0 = \lfloor 7m/5 \rfloor$. It suffices to bound the CDF with $n = n_0$. For $X \sim \mathrm{Beta}(m + 1/2, n_0 + 1/2)$ and $S = m + n_0 + 1$, its mean and variance satisfy

$$\mu = \frac{m + 1/2}{S} > \frac{5}{12}, \qquad \sigma^2 = \frac{(m + 1/2)(n_0 + 1/2)}{S^2(S + 1)} \leq \frac{1}{4(S + 1)} \leq \frac{1}{8m + 8}.$$

Here $S \leq 12m/5 + 1$ proves the strict mean inequality, and $n_0 \geq m$ proves the last variance inequality. The one-sided variance bound

$$\Pr(X - \mu \leq -d) \leq \frac{\sigma^2}{\sigma^2 + d^2} \qquad (d > 0)$$

is obtained by applying Markov's inequality to $(X - \mu - t)^2$ and minimizing $(\sigma^2 + t^2)/(d + t)^2$ over $t \geq 0$; the minimum is at $t = \sigma^2/d$. Taking $d = \mu - 1/4 > 1/6$ gives

$$I_{1/4}(m + 1/2, n_0 + 1/2) < \frac{36}{8m + 44} = \frac{9}{2m + 11} < \frac{1}{3}, \tag{VI.5.14}$$

because $m \geq 13$. Equation (VI.5.5) excludes equality for every $n \leq n_0$.

Next suppose $\lfloor 7m/5 \rfloor < n \leq 3m$. Then $m + n > 12m/5$. Apply (VI.5.12) with $y = 2m \geq 26$ to obtain

$$2m < p < 12m/5 < m + n.$$

Since $n \leq 3m$, this prime satisfies (VI.5.9); Lemma VI.5.4 applies.

Finally, if $n > 3m$, use $y = 2n/3 > 2m \geq 26$. Nagura supplies

$$2n/3 < p < 4n/5 < n < m + n.$$

Again (VI.5.9) holds, and the prime obstruction finishes this last range. □

### VI.5.5 The finite exact certificate and completion

**Theorem VI.5.6** (Unique quarter-point pair)**.** *For all integers $m, n \geq 1$,*

$$R_{m,n}(1/4) = 0 \quad \Longleftrightarrow \quad (m, n) = (1, 2). \tag{VI.5.15}$$

*Equivalently, $I_{1/4}(m + 1/2, n + 1/2) = 1/3$ if and only if $(m, n) = (1, 2)$.*

*Proof.* Lemma VI.5.5 handles $m \geq 13$. For the remaining finitely many first exponents define

$$T_{m,n} = 4^{m+n-1}3^{-n}R_{m,n}(1/4). \tag{VI.5.16}$$

These numbers obey the exact rational recurrences

$$\begin{aligned} T_{0,0} &= 0, \\ T_{0,n} &= \frac{2(2n-1)T_{0,n-1} + 1}{3n} && (n \geq 1), \\ T_{m,n} &= \frac{2(2m-1)T_{m-1,n} - 1}{m+n} && (m \geq 1, n \geq 0). \end{aligned} \tag{VI.5.17}$$

To verify the recurrence, reduction of the incomplete beta integral gives

$$R_{0,n}(X) = \frac{2n-1}{2n}R_{0,n-1}(X) + \frac{(1-X)^{n-1}}{n},$$

and

$$R_{m,n}(X) = \frac{2m-1}{2(m+n)}R_{m-1,n}(X) - \frac{X^{m-1}(1-X)^n}{m+n}.$$

For example, the latter follows by differentiating $x^{m-1/2}(1-x)^{n+1/2}$ and integrating from zero to $x$; the former follows from the same calculation in the other parameter with $m = 0$. Substitution of $x = 1/4$ and (VI.5.16) proves (VI.5.17).

For $m = 1$, direct reduction gives

$$R_{1,2}(X) = -\frac{(2X-3)(4X-1)}{24}, \qquad R_{1,2}(1/4) = 0.$$

Strict monotonicity (VI.5.13) and (VI.5.5) exclude every other $n$ in this row.

For $2 \leq m \leq 12$, Table 1 contains the exact values produced by (VI.5.17) at two consecutive integers. Each first value is negative and each second value positive. By (VI.5.5), the same signs hold for the CDF minus $1/3$. Its strict increase in $n$ excludes every integer $n \geq 1$, including those outside the table. This completes all parameter ranges. □

*Proof of Theorem VI.5.1.* If $m = n = 0$, (VI.5.4) is the arcsine CDF and $R_{0,0} = 0$. Equation (VI.5.6) gives precisely the rational levels, all with quantile $\sin^2(\pi r/2)$. If exactly one exponent is zero, Lemma VI.5.2 excludes exceptions.

For positive exponents, that lemma leaves only the three candidate pairs. Theorem VI.5.6 settles the first candidate. Reflection of the beta density gives

$$I_x(m + 1/2, n + 1/2) = 1 - I_{1-x}(n + 1/2, m + 1/2),$$

so the $3/4$ candidate occurs exactly at $(m, n) = (2, 1)$. The median candidate occurs exactly when $m = n$. These cases are disjoint and exhaust the list. □

**Corollary VI.5.7** (Exact algebraic exceptions for the Jeffreys posterior)**.** *After $m$ successes and $n$ failures, with $m + n \geq 1$, the binomial Jeffreys posterior has an algebraic quantile at an algebraic internal probability level only in the following cases: balanced counts $m = n$ at level $1/2$; counts $(1, 2)$ at level $1/3$; and counts $(2, 1)$ at level $2/3$. All other such posterior quantiles are transcendental.*

*Proof.* The posterior is $\mathrm{Beta}(m + 1/2, n + 1/2)$. Apply Theorem VI.5.1. □

| $m$ | $n_0$ | $T_{m,n_0}$ | $T_{m,n_0+1}$ |
|---|---|---|---|
| 2 | 4 | $-1/45$ | $1/35$ |
| 3 | 6 | $-1/28$ | $1/420$ |
| 4 | 9 | $-2/143$ | $67/6006$ |
| 5 | 12 | $-71/19448$ | $7949/525096$ |
| 6 | 14 | $-2507/167960$ | $179/83980$ |
| 7 | 17 | $-1217/148580$ | $2113/371450$ |
| 8 | 20 | $-3839/1040060$ | $120877/15080870$ |
| 9 | 23 | $-231151/440006560$ | $2112517/220003280$ |
| 10 | 25 | $-11678941/1476337800$ | $103517/57895600$ |
| 11 | 28 | $-4411771/872155860$ | $1079573/306019600$ |
| 12 | 31 | $-1048804991/369448814700$ | $627776153/129013871800$ |

Table 1: Exact rational sign certificate for the eleven remaining rows. Every entry follows from (VI.5.17); no floating-point comparison is used.

**Scope and relation to earlier work.** The new statement is the uniform classification (VI.5.15), which removes the unfinished parameter-dependent test in the preceding polynomial-arcsine analysis. The logarithmic transcendence input, the central-coefficient identity, and Nagura's theorem are not claimed as new. The related paper of Dhillon and Saha [55] concerns arithmetic properties of incomplete beta values. Saha's accompanying conference presentation [56] explicitly discusses unnormalized values with rational nonintegral $a$ and $a + b \in \mathbb{N}$ using Baker's theory. Its stated open question concerning an analogue of Schneider's theorem is distinct from the normalized-CDF pair classification proved here. The paper's publisher abstract and bibliographic details were checked, but a full-text comparison has not been completed. Accordingly no historical-priority claim, nor a claim to resolve a named external conjecture, is made here. The theorem concerns the normalized full-interval beta CDF and every strict half-integer shape pair; it makes no assertion for general rational shapes or arbitrary truncations.

## VI.6 Exact recurrence width for every complex cubic weight

For $\lambda \in \mathbb{C}$, define

$$a_n(\lambda) = \sum_{k=0}^{n} \binom{n}{k}^3 \lambda^k, \qquad E_\lambda(z) = \sum_{n\geq 0} a_{2n}(\lambda) z^n.$$

For $\lambda \neq 0$, its central normalization is

$$B_\lambda(n) = \lambda^{-n} a_{2n}(\lambda) = \sum_{k=-n}^{n} \binom{2n}{n+k}^3 \lambda^k.$$

The *width* of a nonzero eventual recurrence $\sum_{j\in J} q_j(n) f_{n+j} = 0$, with $q_j \in \mathbb{C}(n)$ and finite nonempty $J \subset \mathbb{Z}$, is $\max J - \min J$ after deleting zero coefficients. Geometric multiplication $f_n \mapsto c^n f_n$, $c \neq 0$, does not change the minimum width.

**Theorem VI.6.1** (Complete cubic parameter classification)**.** *For every $\lambda \in \mathbb{C}^*$, the minimum recurrence width of $B_\lambda(n)$, equivalently of $a_{2n}(\lambda)$, over $\mathbb{C}(n)$ is*

$$\begin{cases} 1, & \lambda = -1, \\ 2, & \lambda = 1, \\ 3, & \lambda \notin \{-1, 1\}. \end{cases}$$

*In particular, the collision parameters $\lambda = 8$ and $\lambda = 1/8$ have minimum width three. The conclusion concerns recurrence width, not the differential order of a generating function and not the minimum number of nonzero terms in a recurrence with arbitrary gaps.*

Weighted binomial recurrences were studied by Matala-Aho [26]. The contribution here is the lower bound at every complex specialization, including the collision parameters. We do not identify this precise parameter classification with a named conjecture in the cited literature. McIntosh's alternating problem [29] concerns the different parameter family with fixed weight $-1$ and variable exponent.

### VI.6.1 A hypergeometric period and its monodromy

**Lemma VI.6.2** (Cubic generating function)**.** *With $c = 1 + \lambda$ and $H(u) = {}_2F_1(\frac{1}{3}, \frac{2}{3}; 1; u)$, one has the germ identity*

$$A_\lambda(t) := \sum_{n\geq 0} a_n(\lambda) t^n = \frac{1}{1-ct} H\left(\frac{27\lambda t^2}{(1-ct)^3}\right). \tag{VI.6.1}$$

*Proof.* The terminating Pfaff–Saalschutz identity [57, §16.4.3] gives

$$\sum_{j\geq 0} \frac{(n+j)!}{j!^3 (k-j)!(n-k-j)!} = \binom{n}{k} {}_3F_2\left(\begin{matrix} -k, k-n, n+1 \\ 1, 1 \end{matrix}; 1\right) = \binom{n}{k}^3,$$

where terms with negative factorial arguments are omitted. Taking coefficients of $\lambda^k$ therefore proves

$$a_n(\lambda) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(n+j)!}{j!^3 (n-2j)!} \lambda^j (1+\lambda)^{n-2j}.$$

Summing first over $n - 2j \geq 0$ yields

$$A_\lambda(t) = \sum_{j\geq 0} \frac{(3j)!}{j!^3} \frac{\lambda^j t^{2j}}{(1-ct)^{3j+1}}.$$

Finally $(3j)!/j!^3 = 27^j (\frac{1}{3})_j (\frac{2}{3})_j / (j!)^2$. □

**Lemma VI.6.3** (Irreducibility and genuine finite singularities)**.** *For $\lambda \neq 0$, the rank-two local system obtained by the pullback and rational gauge in (VI.6.1) is irreducible. Its genuine nonzero finite monodromy singularities are exactly the distinct roots of*

$$\Delta_\lambda(t) = (1 - (1+\lambda)t)^3 - 27\lambda t^2. \tag{VI.6.2}$$

*At each such root its local monodromy $T$ is nontrivial unipotent and* $\operatorname{rank}(T - I) = 1$.

*Proof.* The base hypergeometric equation is

$$u(1-u)y'' + (1-2u)y' - \frac{2}{9}y = 0.$$

Its local monodromies at 0 and 1 are nontrivial unipotents with different fixed lines. Indeed, $H(u)$ spans the line of holomorphic solutions at zero, while the logarithmic connection formula [57, §15.8.10] gives

$$H(u) = -\frac{\log(1-u)}{\Gamma(1/3)\Gamma(2/3)} + O(1) \quad (u \longrightarrow 1).$$

The equation is invariant under $u \mapsto 1-u$, which proves the corresponding assertion at zero. Thus the two fixed lines differ; in dimension two this proves irreducibility.

For completeness this irreducibility survives every nonconstant rational pullback. Remove the three base singularities and also the finitely many critical values of the rational map. After removing their preimages, the map is a finite unramified covering. Its fundamental group has finite-index image in the base fundamental group. Consequently its monodromy image has finite index in the original monodromy group. It contains positive powers of each of the two unipotents just described. These powers have the same two different fixed lines, so their common representation is irreducible. The rational map $27\lambda t^2/(1-ct)^3$ is nonconstant for every $\lambda \neq 0$. A nonzero rational scalar gauge does not change its monodromy.

At a root of $\Delta_\lambda$, the rational map takes the value 1. A root of multiplicity $e \geq 1$ therefore gives a conjugate of the $e$th power of the base monodromy at 1, which is still nontrivial unipotent of rank one. There is no cancellation with $1 - ct$ at a root of $\Delta_\lambda$, since $\lambda \neq 0$. If $c \neq 0$, the point $t = 1/c$ maps to infinity with ramification degree three. The base exponents $1/3, 2/3$ have nonintegral difference; after this pullback and the scalar gauge their two local solutions are holomorphic, with leading powers 0 and 1. Their monodromy is therefore the identity. All other finite nonzero points map to ordinary base points, except for the roots already listed. This proves the assertion about genuine monodromy support. □

**Lemma VI.6.4** (The even section generates both period spaces)**.** *For $\lambda \neq 0, -1$, the analytic continuations of*

$$E_\lambda(t^2) = \tfrac{1}{2}\big(A_\lambda(t) + A_\lambda(-t)\big)$$

*span the direct sum of the two rank-two period spaces for $A_\lambda(t)$ and $A_\lambda(-t)$. At a nonzero point $t_0$, the local monodromy on this four-dimensional space consequently has*

$$\operatorname{rank}(T - I) = \mathbf{1}_{\{\Delta_\lambda(t_0)=0\}} + \mathbf{1}_{\{\Delta_\lambda(-t_0)=0\}}.$$

*Here multiplicities of roots of either individual polynomial do not alter their rank-one contributions.*

*Proof.* Work on a common punctured sphere for both equations. Write $S$ for the distinct roots of $\Delta_\lambda$. Its discriminant is

$$\operatorname{disc}_t \Delta_\lambda = -19683\lambda^2(\lambda - 1)^2.$$

If $\lambda \neq 0, \pm 1$, the set $S$ has three nonzero elements, and hence $S \neq -S$. If $\lambda = 1$, then $\Delta_1(t) = (1-8t)(1+t)^2$, so again $S \neq -S$. The two irreducible representations have different genuine monodromy supports and cannot be isomorphic.

Regarded as spaces of local analytic functions at an ordinary base point, their intersection is zero: a nonzero intersection would be an invariant subspace of both irreducible spaces and would identify them. In their direct sum, the monodromy span of the displayed sum projects onto each summand, because both component functions are nonzero and each representation is irreducible. A proper invariant subspace with both projections onto two irreducible summands would be the graph of an isomorphism between them. This is impossible. The span is thus the full direct sum. The rank formula now follows from Lemma VI.6.3 by adding the ranks on the two summands. □

### VI.6.2 Why a collision still consumes two units of width

**Lemma VI.6.5** (A simple leading zero allows at most rank-one monodromy)**.** *Consider a scalar homogeneous differential equation of order $N \geq 1$ with holomorphic coefficients near $s = 0$. If its leading coefficient has a simple zero there, the local monodromy on its solution space satisfies* $\operatorname{rank}(T - I) \leq 1$.

*Proof.* After multiplication by a holomorphic unit, write the equation as

$$sy^{(N)} + \sum_{j=0}^{N-1} b_j(s) y^{(j)} = 0.$$

For a formal solution $y = \sum_{k \geq 0} c_k s^k$, the coefficient of $s^m$ has coefficient

$$\frac{(m+N-1)!}{m!}\bigl(m + b_{N-1}(0)\bigr)$$

on its highest-index unknown $c_{m+N-1}$. All remaining terms involve lower-index coefficients. Initially $c_0, \dots, c_{N-2}$ are free. There is at most one $m \geq 0$ at which the displayed multiplier vanishes. If none exists, these $N-1$ coefficients determine $N-1$ independent formal holomorphic solutions. At the exceptional index, at most one linear compatibility condition is imposed, while $c_{m+N-1}$ becomes a new free coefficient. Hence the dimension is still at least $N-1$. For $N = 1$ the claimed dimension lower bound is zero and is immediate. This is a regular singular equation: after division by $s$, each coefficient has at most a simple pole. The convergent Frobenius theorem therefore makes these formal holomorphic solutions convergent. At least $N-1$ independent local solutions are fixed by monodromy, proving the assertion, including resonant cases. □

**Lemma VI.6.6** (A recurrence gives a finite leading-zero budget)**.** *If a convergent generating-function germ $F(z) = \sum_{n \geq 0} f_n z^n$ has an eventual recurrence of width $L$ over $\mathbb{C}(n)$, it has a scalar homogeneous differential annihilator whose leading coefficient, away from zero, has at most $L$ zeros counted with multiplicity.*

*Proof.* Translate the shifts and clear denominators to write $\sum_{j=0}^{L} p_j(n) f_{n+j} = 0$, with $p_j \in \mathbb{C}[n]$ not all zero. With $\vartheta = z\, d/dz$, coefficient extraction gives

$$\mathcal{D}F = P(z), \qquad \mathcal{D} = \sum_{j=0}^{L} z^{L-j} p_j(\vartheta - j),$$

where $P$ is a polynomial accounting for the finitely many initial indices. If $D = \max_j \deg p_j$, the coefficient of $(d/dz)^D$ in $\mathcal{D}$ is $z^D h(z)$, where

$$h(z) = \sum_{\deg p_j = D} \operatorname{lc}(p_j) z^{L-j}$$

is nonzero and has degree at most $L$. Left-multiplying $\mathcal{D}$ by a sufficiently high power of $d/dz$ kills $P$ and does not change this leading coefficient. This gives the required annihilator. □

### VI.6.3 Proof of the classification

*Proof of Theorem VI.6.1.* We first establish the upper bounds without requiring any recurrence matrix to be invertible. For $\lambda \neq -1$, put

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^3 \lambda^k.$$

Shifting the summation index gives, for every integer $h \geq 0$,

$$L_N\big[k^{3+h} - \lambda(N-k)^3(k+1)^h\big] = 0.$$

The polynomial inside brackets has degree $h+3$ and leading coefficient $1+\lambda \neq 0$. Successive elimination therefore reduces every polynomial moment to the three moments of $1, k, k^2$, with coefficients rational in $N$. Since

$$\binom{N-1}{k}^3 = N^{-3}(N-k)^3 \binom{N}{k}^3,$$

the column $M_N = (L_N[1], L_N[k], L_N[k^2])^{\mathsf{T}}$ satisfies $M_{N-1} = T(N) M_N$ for a rational $3 \times 3$ matrix. It follows that $V_n = M_{2n}$ satisfies $V_{n-1} = T(2n-1)T(2n)V_n$. The first-coordinate functionals expressing $a_{2n}, a_{2n-2}, a_{2n-4}, a_{2n-6}$ in terms of $V_n$ are four rows in a three-dimensional vector space over $\mathbb{C}(n)$. Their linear dependence proves width at most three.

At $\lambda = 1$, the classical cubic Franel recurrence

$$(n+1)^2 a_{n+1}(1) = (7n^2 + 7n + 2)a_n(1) + 8n^2 a_{n-1}(1)$$

(also obtained by substituting (VI.6.1) into the Gauss equation) gives a two-dimensional rational companion system. Taking every second state gives width at most two. At $\lambda = -1$, (VI.6.1) gives

$$A_{-1}(t) = H(-27t^2), \qquad B_{-1}(n) = \frac{(3n)!}{n!^3},$$

which has width one and cannot have width zero.

We next prove all remaining lower bounds. Put $z = t^2$ and compute

$$\begin{aligned} Q_\lambda(z) &= \Delta_\lambda(t)\Delta_\lambda(-t) \\ &= 1 - 3(\lambda^2 + 20\lambda + 1)z \\ &\quad + 3(\lambda^4 - 50\lambda^3 + 141\lambda^2 - 50\lambda + 1)z^2 - (1+\lambda)^6 z^3. \end{aligned}$$

Its discriminant is

$$\operatorname{disc}_z Q_\lambda = -19683\lambda^2(\lambda-8)^2(\lambda-1)^2(\lambda+1)^2(8\lambda-1)^2.$$

For $\lambda \notin \{0, \pm 1, 8, 1/8\}$ it has three distinct nonzero roots. At each such root, the monodromy span of $E_\lambda$ has a nontrivial local monodromy by Lemma VI.6.4. Here the change of variable $z = t^2$ is locally unramified, so its local monodromy is unchanged. Every scalar annihilator of $E_\lambda$ must have a vanishing leading coefficient at each of these three points: at an ordinary point all solutions would be holomorphic and monodromy would be trivial. Lemma VI.6.6 therefore proves width at least three.

For $\lambda = 8$, the factorizations are

$$\Delta_8(t) = (1-27t)(1+27t^2), \qquad Q_8(z) = (1-729z)(1+27z)^2.$$

At $z = -1/27$, both $t$ and $-t$ are roots of $\Delta_8$, and Lemma VI.6.4 gives monodromy rank two on the four-dimensional space generated by the even period. Any scalar annihilator has a local solution space containing this invariant space, so its monodromy rank is at least two. Its leading coefficient can be neither nonzero nor simply zero there, by Lemma VI.6.5; its zero has multiplicity at least two. The point $z = 1/729$ requires a further zero. The budget is again at least three. Reciprocity $a_n(\lambda) = \lambda^n a_n(1/\lambda)$, or equivalently $B_\lambda = B_{1/\lambda}$, proves the same lower bound at $\lambda = 1/8$.

Finally, at $\lambda = 1$ the two distinct points $z = 1/64$ and $z = 1$ coming from $S = \{1/8, -1\}$ both carry nontrivial monodromy of the even period. They require two leading zeros and hence width at least two. Together with the upper bounds this completes the proof. □

*Remark* VI.6.7 (What replaces positivity). The generic distinct-singularity count alone gives no sharp conclusion at $\lambda = 8$. Two independent vanishing cycles become supported at one point of the even generating function. Irreducibility makes both cycles visible in the continuation space of the chosen period; the simple-leading-zero lemma then forces that collision to retain multiplicity two in the differential leading coefficient. Thus the relevant obstruction is local monodromy rank, rather than merely the number of distinct singular points. The argument is specific to the cubic hypergeometric period; it does not assert a classification for all exponents.

## VI.7 Gaussian mixture likelihood across all support and noise scales

Observe $X_i = \Theta_i + \sigma Z_i$, where $\Theta_i$ are independent with law $G$, $Z_i$ are independent standard normal variables, and $\sigma > 0$ is known. The interval $I = [\mu - A, \mu + A]$, $A > 0$, is known. Let $\widehat{G}_I$ be any measurable maximizer of the original Gaussian mixture likelihood over all probability measures on $I$. No bound is imposed on the number of mixing atoms. Set $L = \log(eN)$ and $a = A/\sigma$, and define

$$R(N, A, \sigma) = \sigma \begin{cases} a \min\{1, L^{-1} \log(e + L/a^2)\}, & 0 < a \le \sqrt{L}, \\ \max\{L^{-1/2}, a/\sqrt{N}\}, & a > \sqrt{L}. \end{cases} \tag{VI.7.1}$$

The formulas at the breakpoint agree up to absolute factors.

**Theorem VI.7.1** (The complete Gaussian likelihood curve)**.** *There are absolute constants $c, C > 0$ such that, for every $N \ge 3$, $A, \sigma > 0$, and $\mu \in \mathbb{R}$,*

$$cR(N, A, \sigma) \le \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}(I)} \mathbb{E}_G W_1(\widetilde{G}, G) \le \sup_{G \in \mathcal{P}(I)} \mathbb{E}_G W_1(\widehat{G}_I, G) \le CR(N, A, \sigma).$$

*The infimum permits arbitrary estimators knowing $I$ and $\sigma$. Thus the ordinary constrained NPMLE attains the minimax rate, up to absolute factors, throughout the entire parameter plane.*

For a fixed nonzero $A/\sigma$, the rate is of order $A \log\log N / \log N$. The fixed-compact-support rate was already known for the denoised method of moments; see Wu and Yang [58, Theorem 5 and Proposition 8]. Soloff, Guntuboyina and Sen [64, Theorem 11 and Remark 12] prove a deconvolution guarantee for the unrestricted NPMLE above a support-dependent sample-size threshold, and discuss adaptation as an open direction. The precise claim here is the uniform curve (VI.7.1) for a constrained likelihood fit. It does not resolve every adaptation question for an unrestricted fit. In fact, Proposition VI.7.4 below rules out unrestricted adaptation to this curve for arbitrarily small known intervals.

Translation and scaling reduce the proof to $\mu = 0$, $\sigma = 1$. Write $\varphi, \Phi$ for the standard normal density and CDF, $p_H = H * \varphi$, $q = N^{-1} \sum_i \delta_{X_i}$, and $h(p, r)^2 = \int (\sqrt{p} - \sqrt{r})^2 \, dx$.

### VI.7.1 Likelihood accuracy and a Hermite inverse

We first record a density bound, uniformly for $A \le \sqrt{N}$:

$$\mathbb{E} h(p_{\widehat{G}_I}, p_G) \le \frac{C}{\sqrt{N}} \begin{cases} L, & A \le \sqrt{L}, \\ \sqrt{A}\, L^{3/4}, & \sqrt{L} < A \le \sqrt{N}. \end{cases} \tag{VI.7.2}$$

Here and below $C$ denotes an absolute constant whose value may change. We give the covering argument to make the uniformity clear. Put $D = A \vee 1$. For complex $z$, $\sup_x |\varphi(x - z)| \le C \exp((\operatorname{Im} z)^2/2)$. On the ellipse $z = D\cos(\theta + iu)$, contour shifting bounds the $k$th Chebyshev coefficient, in uniform norm over real $x$, by

$$C \exp\{-c \min(k^2/D^2, k)\}.$$

For $k \le D^2$ take $u$ a sufficiently small multiple of $k/D^2$; for $k > D^2$ take a sufficiently small fixed $u$. Truncating at $d = C\{D\sqrt{\log(C(D+1)/\beta)} + \log(C(D+1)/\beta)\}$ and gridding the $d+1$ Chebyshev moments yields a uniform density cover of radius $\beta$ with $\log M \le Cd \log(C(d+1)/\beta)$. One can take each center to be an actual mixture by choosing one representative from each nonempty grid cell.

Take $U = D + \sqrt{12L}$ and $\beta = N^{-3}/(2U)$. The resulting bound is $\log M \le C(A\sqrt{L} + L)L$. Replace each center $p_j$ by $u_j = p_j + \beta$ on $[-U, U]$ and $u_j(x) = \varphi(|x| - D)$ outside. It dominates every associated mixture and has mass between 1 and $1 + CN^{-3}$. The finite-envelope likelihood lemma, Lemma VI.B.3, gives $\mathbb{E}h^2 \le C(\log M + 1)/N + CN^{-3}$, because the fitted likelihood is at least the true likelihood. This proves (VI.7.2). No empirical-to-density relative entropy is used.

**Lemma VI.7.2** (Uniform small-support inverse)**.** *If $F, H$ are supported on $[-A, A]$, then for every integer $k \ge 1$,*

$$W_1(F, H) \le \frac{CA}{k} + A\{C(1 + \sqrt{k}/A)\}^k h(p_F, p_H). \tag{VI.7.3}$$

*Proof.* For a 1-Lipschitz test $f$ with $f(0) = 0$, approximate $f(Au)/A$ on $[-1, 1]$ by a degree-$k$ polynomial $P(u) = \sum_{j=0}^{k} b_j u^j$ with error $C/k$ and supremum norm at most $C$. Such a polynomial follows from Lemma VI.B.1 by affine rescaling. Expanding $P$ in Chebyshev polynomials, whose coefficients are bounded by twice its supremum norm, gives $\sum_j |b_j| \le C^k$: the sum of absolute monomial coefficients of $T_j$ is at most $3^j$, by its three-term recurrence.

Let $\mathrm{He}_j$ be the probabilists' Hermite polynomial. Its Gaussian expectation equals $\theta^j$, and its translation formula and Gaussian orthogonality give

$$\|\mathrm{He}_j(Z + \theta)\|_2 \le \sum_{r=0}^{j} \binom{j}{r} A^{j-r}\sqrt{r!} \le (A + \sqrt{j})^j \qquad (|\theta| \le A).$$

Writing $p_F - p_H = (\sqrt{p_F} - \sqrt{p_H})(\sqrt{p_F} + \sqrt{p_H})$ and applying Cauchy–Schwarz therefore bounds their $j$th mixing moment difference by $2(A+\sqrt{j})^j h(p_F, p_H)$. Insert these bounds into $A\sum b_j A^{-j}\theta^j$. The constant term cancels, and the approximation error costs $CA/k$. Wasserstein duality proves (VI.7.3). □

### VI.7.2 A boundary drift estimate for the actual constrained fit

The following estimate is the additional ingredient needed to transfer the large-support argument to a constrained NPMLE. Existence of a maximizer follows from weak compactness on $[-A, A]$ and continuity and positivity of the Gaussian likelihood. Write $H = \widehat{G}_I$. The likelihood directional derivative gives

$$S(\theta) = \frac{1}{N}\sum_i \frac{\varphi(X_i - \theta)}{p_H(X_i)} \le 1 \quad (-A \le \theta \le A), \qquad S = 1 \quad H\text{-a.e.}$$

Thus $R_\theta(i) = N^{-1}\varphi(X_i - \theta)/p_H(X_i)$ is a probability law at each contact point, with observation marginal $q$. At an interior contact, differentiation gives $\mathbb{E}_{R_\theta}(X - \theta) = 0$ and $\mathbb{E}_{R_\theta}(X - \theta)^2 \le 1$. At the right endpoint the mean residual is nonnegative, and at the left endpoint it is nonpositive. Define

$$B = H\{A\}\mathbb{E}_{R_A}(X - A) - H\{-A\}\mathbb{E}_{R_{-A}}(X + A), \qquad V = \int \mathbb{E}_{R_\theta}(X - \theta)^2\, dH(\theta),$$

where a missing endpoint atom contributes zero.

**Lemma VI.7.3** (Boundary drift and posterior second moments)**.** *For $A \ge 1$, $B \ge 0$ and*

$$\mathbb{E}B \le C\left\{\mathbb{E}h(p_H, p_G) + \frac{A+1}{\sqrt{N}}\right\}, \qquad \mathbb{E}V \le C. \tag{VI.7.4}$$

*Proof.* Let $w = H\{A\} > 0$ and $\tau(x) = w\varphi(x - A)/p_H(x)$. The function $\tau$ takes values in $[0, 1]$ and is nondecreasing: its derivative is $\tau(x)\{A - \mathbb{E}_H(\Theta \mid X = x)\} \ge 0$. With $f(x) = (x - A)\tau(x)$, the right endpoint contribution is

$$b_+ = wS'(A) = (q - p_H)f = (q - p_G)f + (p_G - p_H)f \ge 0. \tag{VI.7.5}$$

The equality $(p_H)f = 0$ uses the mean of $N(A, 1)$.

We need a weighted monotone empirical-process bound. For every square-integrable real random variable $Y$ and its empirical law $q_Y$, and every constant $a$,

$$\mathbb{E} \sup_{\substack{0\le t\le 1\\ t \text{ nondecreasing}}} |(q_Y - P_Y)[(y-a)t(y)]| \le \frac{C\sqrt{\mathbb{E}(Y-a)^2}}{\sqrt{N}}. \tag{VI.7.6}$$

Indeed a monotone $t$ is a subprobability mixture of threshold indicators $\mathbf{1}_{[s,\infty)}$, with a possible constant term, which is the threshold $s = -\infty$. Symmetrization reduces the supremum to weighted Rademacher sums $N^{-1}\sum_i \varepsilon_i(Y_i - a)\mathbf{1}_{Y_i\ge s}$. Conditional on the data, sort the $Y_i$ in decreasing order. These sums form a subset of the partial sums of independent centered increments $\varepsilon_i(Y_i - a)$. Doob's $L^2$ inequality bounds their expected maximum by $2\{\sum_i (Y_i - a)^2\}^{1/2}$. Jensen's inequality proves (VI.7.6). Applying it with $Y = X$ and $a = A$ bounds the expected absolute empirical term in (VI.7.5) by $C(A + 1)/\sqrt{N}$.

Only an upper bound is needed for its population term. If $f_+ = \max(f, 0)$ and $f_- = \max(-f, 0)$, retaining only positive contributions and using the Hellinger factorization gives

$$(p_G - p_H)f \le 2h(p_G, p_H)\{\|f_-\|_{L^2(p_H)} + \|f_+\|_{L^2(p_G)}\}.$$

Both norms are bounded by absolute constants. For the first, $\tau^2 p_H \le w\varphi(\cdot - A)$ gives $\int f^2 p_H \le w \le 1$. For the second, $f_+ \le (x - A)_+$ and $\Theta \le A$ under the true law, so $\int f_+^2 p_G \le \mathbb{E}Z_+^2 = 1/2$. This proves the required bound for $b_+$; reflection gives the left endpoint bound and hence the first assertion of (VI.7.4).

For the second assertion the interior contribution to $V$ is at most one. The feasible shift from $A$ to $A - 1$ gives

$$\mathbb{E}_{R_A} e^{-(X-A)} = e^{1/2} S(A - 1) \le e^{1/2}.$$

Since $u^2 \le Ce^u$ for $u \ge 0$, the inward residual satisfies $\mathbb{E}_{R_A}(A - X)_+^2 \le C$. The weighted outward residual is at most $w\mathbb{E}_{R_A}(X - A)_+^2 = q[(X - A)_+^2 \tau(X)] \le q[(X - A)_+^2]$, whose expectation is at most $1/2$. The reflected left endpoint gives the same bounds. Summing the three contributions proves $\mathbb{E}V \le C$. □

For a global $C^2$ function $F$ with bounded first two derivatives, Taylor's theorem now gives the deterministic inequality

$$|(H - q)F| \le \|F'\|_\infty B + \tfrac{1}{2}\|F''\|_\infty V. \tag{VI.7.7}$$

This explicitly keeps the boundary term that would be lost by incorrectly treating a constrained posterior as a martingale. The true Gaussian coupling gives $|(G - p_G)F| \le \|F''\|_\infty/2$.

### VI.7.3 A two-scale inverse with boundary terms

For $m \ge 1$, an exact Markov channel sends $N(\theta, 1)$ to $\mathrm{Bin}(m, \phi_m(\theta))$, where $\phi_m(\theta) = \Phi(\theta/\sqrt{m})$. To see this, add to one observation $X$ an independent centered Gaussian vector with covariance $mI_m - \mathbf{1}\mathbf{1}^\mathsf{T}$, and count the nonnegative coordinates of $X\mathbf{1} + \varepsilon$. The resulting coordinates are independent $N(\theta, m)$ variables. The singular covariance of $\varepsilon$ is positive semidefinite and causes no problem. This channel is used only in the proof. For $m = \lceil A^2 \rceil$, $A \ge 1$, the inverse of $\phi_m$ on $[-A, A]$ is $CA$-Lipschitz.

For integers $1 \le S \le K \le m/2$ we claim

$$W_1(H, G) \le C\left\{\frac{A}{K} + \frac{S}{A}(1 + V) + B + W_1(q, p_G) + \frac{A}{S}e^{2K^2/m} h(p_H, p_G)\right\}. \tag{VI.7.8}$$

For a 1-Lipschitz test $f$ on $[-A,A]$, extend $g(u)=f(\phi_m^{-1}(u))/A$ from $\phi_m([-A,A])$ to $[0,1]$ with an absolute Lipschitz constant. Lemma VI.B.1 gives polynomials $P_S,P_K$ approximating $g$ with errors $C/S, C/K$, with first derivatives bounded by $C$ and second derivatives by $CS, CK$, respectively. The function $F_S(\theta)=AP_S(\Phi(\theta/\sqrt{m}))$ is globally $C$-Lipschitz, with

$$F_S''(\theta)=\frac{A}{m}\{P_S''(\Phi(z))\varphi(z)^2-P_S'(\Phi(z))z\varphi(z)\},\qquad z=\theta/\sqrt{m},$$

so $\|F_S''\|_\infty\le CS/A$. Equation (VI.7.7), the true coupling, and empirical Wasserstein duality control this low-degree part. The residual $P_K-P_S$ has norm at most $C/S$ and Bernstein coefficients of degree $m$ at most $Ce^{2K^2/m}/S$, by Lemma VI.B.2. Its expectation difference is thus bounded by that quantity times the predictive $L^1$ distance after the channel, which is at most $2h(p_H,p_G)$ by data processing. The remaining approximation error is $CA/K$. Taking the supremum over $f$ proves (VI.7.8).

The empirical CDF also gives the elementary uniform bound

$$\mathbb{E}W_1(q,p_G)\le C(A+1)/\sqrt{N}. \tag{VI.7.9}$$

Integrate its pointwise standard deviations, using the bound $1/2$ on $[-A,A]$ and the Gaussian tail bound outside that interval. The integral of the square root of a Gaussian tail is finite.

### VI.7.4 Completion of the upper bound

Suppose first $A\le\sqrt{L}$ and put $d=\log(e+L/A^2)$. Choose a sufficiently small absolute $c_0>0$. If $c_0L/d<2$, the deterministic diameter bound $W_1(H,G)\le 2A$ is at most $CA\min(1,d/L)$. Otherwise put $k=\lfloor c_0L/d\rfloor\ge 2$. Then $k\asymp L/d$, and

$$k\log\{C(1+\sqrt{k}/A)\}\le C_1kd\le C_1c_0L\le L/8.$$

Here $k/A^2\le L/A^2$ and $d\ge 1$ justify the first inequality. Equations (VI.7.2) and (VI.7.3) yield

$$\mathbb{E}W_1(H,G)\le CA/k+CAe^{L/8}L/\sqrt{N}\le CA/k\le CAd/L.$$

The absorption is uniform because $k\le L$ and $e^{L/8}L^2/\sqrt{N}$ is bounded for $N\ge 3$.

Next let $\sqrt{L}<A\le\sqrt{N}$ and $m=\lceil A^2\rceil$. For all sufficiently large absolute $N$, choose

$$S=\lfloor A/(16\sqrt{L})\rfloor\vee 1,\qquad K=\left\lfloor\min\{m/2,\sqrt{\alpha mL}\}\right\rfloor,\qquad \alpha=1/64.$$

Then $1\le S\le K\le m/2$, $A/K+S/A\le C/\sqrt{L}$, and $A/S\le C\sqrt{L}$. Apply (VI.7.8), Lemma VI.7.3, and (VI.7.9). The density bound (VI.7.2) controls the remaining terms. In particular the high-degree term is at most $Ce^{2\alpha L}\sqrt{A}L^{5/4}/\sqrt{N}$; its ratio to $L^{-1/2}$ is at most $Ce^{2\alpha L}N^{-1/4}L^{7/4}$, which is uniformly bounded. The term $\mathbb{E}h$ in the boundary bound is smaller still. This proves $\mathbb{E}W_1(H,G)\le C\{L^{-1/2}+A/\sqrt{N}\}$.

If $A>\sqrt{N}$, the empirical-posterior coupling and Lemma VI.7.3 give $\mathbb{E}W_1(H,q)\le\sqrt{\mathbb{E}V}\le C$. Coupling $G$ to $p_G$ costs at most $\mathbb{E}|Z|$, so (VI.7.9) proves the bound $CA/\sqrt{N}$. The same elementary argument handles the bounded set of smaller sample sizes excluded in the preceding choice of $S,K$. This completes the upper bound for all $A>0$ and $N\ge 3$.

### VI.7.5 Matching lower bounds and the cost of unrestricted fitting

For $A^2\le L$ set $d=\log(e+L/A^2)$ and $k=\lceil 100L/d\rceil$. Use the Chebyshev-arcsine pair of Lemma VI.B.4 on $[-A,A]$. Its moments through degree $k-1$ agree, and its Wasserstein separation is at least $A/(\pi k)\ge cA\min(1,d/L)$. The Gaussian identity

$$\int\frac{\varphi(x-\lambda)\varphi(x-\eta)}{\varphi(x)}\,dx=e^{\lambda\eta}$$

therefore gives

$$\int \frac{(p_{F_k^+} - p_{F_k^-})^2}{\varphi} \le 4 \sum_{j \ge k} \frac{A^{2j}}{j!}. \tag{VI.7.10}$$

For completeness put $x = L/A^2 \ge 1$. Since $d \le \log(4x)$ and $\log d \le d/2$,

$$\log \frac{k}{eA^2} \ge \log \frac{100x}{ed} \ge d/2, \qquad k \ge 2eA^2.$$

The series in (VI.7.10) is consequently at most $2(eA^2/k)^k \le 2e^{-50L}$. Cauchy–Schwarz bounds the single-observation total variation by the square root of this expression, and tensorization bounds the $N$-sample total variation by $N$ times that quantity. Le Cam's two-point inequality gives the required small-support lower bound, including the clipped regime $d \gg L$.

For $A > \sqrt{L}$, use instead radius $M = \sqrt{L}/8$ and $k = \lceil 20L \rceil$ in the same construction. Its separation is $c/\sqrt{L}$, while the series in (VI.7.10) is bounded by $4\sum_{j \ge k}(L/64)^j/j! \le Ce^{-40L}$. Finally the two endpoint laws on $\{-A, A\}$ with weights $1/2 \pm c_1/\sqrt{N}$ have Wasserstein separation $cA/\sqrt{N}$. For small absolute $c_1$, the $N$-sample KL divergence of their latent Bernoulli experiments is bounded, and observation can only decrease total variation. Le Cam's inequality gives the second lower bound. Their maximum proves Theorem VI.7.1.

**Proposition VI.7.4** (Failure of unrestricted full-plane adaptation)**.** *Let $\widehat{G}_{\mathbb{R}}$ be any exact unrestricted Gaussian location NPMLE with known noise $\sigma$. Under $G = \delta_\mu$,*

$$\mathbb{E} W_1(\widehat{G}_{\mathbb{R}}, \delta_\mu) \ge \sigma \sqrt{\frac{2}{\pi N}}.$$

*Consequently this estimator cannot attain the known-interval minimax rate uniformly over all positive support radii.*

*Proof.* Every unrestricted maximizer is supported in the sample range: moving a parameter outside that range to its nearest endpoint increases its Gaussian likelihood at every observation. The unrestricted contact condition permits differentiation in both directions, so $\mathbb{E}_{R_\theta} X = \theta$ at every contact. Integrating shows that the fitted mixing mean is exactly $\bar{X}$. Wasserstein duality gives $W_1(\widehat{G}_{\mathbb{R}}, \delta_\mu) \ge |\bar{X} - \mu|$; under the specified truth its expectation is the displayed value. On a known interval $[\mu - A, \mu + A]$, the estimator $\delta_\mu$ has worst-case risk at most $A$. Letting $A\sqrt{N}/\sigma \to 0$ proves the assertion. □

The benchmark in this proposition supplies the interval center. If the center is also unknown, the class itself contains an unknown Gaussian location and has an additional location-estimation cost; the proposition does not rule out adaptation for that different class.

## VI.8 Boundaries of the present extensions

The proofs identify specific places where exceptional parameters can be controlled. Their limits are equally specific. The full hypercubic group in every odd dimension greater than three is not classified here. The banana theorem addresses odd numbers of propagators; the even case and the leading-coefficient multiplicity problem remain separate. Algebraic independence of functions does not imply algebraic independence of their values at a fixed argument.

The beta classification covers all strict half-integer shapes, not all rational shapes or truncated distributions. The cubic recurrence result concerns the exponent three and recurrence width, not all exponents or the number of nonzero terms in a recurrence with gaps. The Gaussian theorem assumes a known support interval and known noise variance and proves rates up to absolute constants, not exact minimax constants.

Two further transfers were explored without a complete theorem. The sharp leading constant for the worst design ratio over all connected graphs remains undetermined: a proposed local variance bound fails on a clique with pendant vertices. A general rational-lemniscate extension also remains open; arbitrary algebraic-curve length arguments are insufficient. Neither attempt is included among the results.

The strongest next verification step is a specialist review of the Fourier intermediate-extension conventions, the all-mass inputs from [032], and the relation to existing classifications. The proofs here are a concrete research draft for that review; a journal-level or historical-significance assessment requires more than internal checks.

## VI.A Previously established hypercubic rank and algebraicity theorem

*Reproduced supporting input from the preceding draft. This appendix is not counted as a new result of the present paper.*

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{VI.A.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{VI.A.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the $n$-dimensional hypercubic lattice when each step in either coordinate direction $i$ has weight $m_i$. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of $\mathrm{SL}_2$, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{VI.A.3}$$

The decomposition in the middle uses the diagonal $\mathrm{SL}_2$; the tensor product itself is irreducible for the product group $\mathrm{SL}_2^q$.

**Theorem VI.A.1** (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle\partial_z\rangle$ annihilating $F_m$ is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{VI.A.4}$$

*The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.*

*More precisely, define*

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

*Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size $d$ for each of the $a_d$ summands with $d \geq 1$ in (VI.A.3).*

**Corollary VI.A.2** (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is 1; with one nonzero weight $m$, it is $(1 - 4m^2 z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (VI.A.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2, 1]$, $[1, 1, 1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem VI.A.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

### VI.A.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection $M$ on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy $T$, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size $d$, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if $M$ has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank}\mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{VI.A.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection $N$ on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero $H^0$ for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

### VI.A.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let $B_\mu$ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{VI.A.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is $\mathrm{SL}_2$; this is the classical non-half-integral Bessel case [42, 43]. At zero, $I_0$ is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ.}$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

**Lemma VI.A.3.** *If the $\mu_j^2$ are pairwise distinct and nonzero, the direct sum of the $B_{\mu_j}$ has differential Galois group $\mathrm{SL}_2^Q$. Consequently*

$$M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$$

*is irreducible of rank $Q$, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities $a_d$ from (VI.A.3); its slopes at infinity are at most one.*

*Proof.* The projections of the Galois group, and of its identity component, onto each $\mathrm{SL}_2$ factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the $I_0$ solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by $g$. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator $e$ on $W$. On $\mathrm{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j-2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (VI.A.3) has the parity of $n$ and contains exactly one weight-zero vector if $n$ is even, or exactly one weight-one vector if $n$ is odd. Thus $b$ is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (VI.A.4). In particular $\mathrm{Sym}^n V$ occurs once, so $r \geq n$.

### VI.A.3 An upper bound that eliminates the endpoint ambiguity

For $\mathrm{Re}\, s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\, dx = \sum_{k\geq 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{VI.A.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist $M$ by $e^{-sx}$; call the resulting connection $N$. On the cyclic vector $\eta$ evaluating to $g$, its $x$-connection is $\nabla_x^M - s$ and its commuting $s$-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice $N_0$ stable under $\theta_x$. Its residue at zero is nilpotent with the same $b$ Jordan blocks as $M$.

The connection $N$ is irreducible of rank $Q \geq 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since $s$ is transcendental

over the mass constants, every formal exponential factor at infinity is nonzero and linear in $x$ after the twist. Thus $\mathrm{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension $b$. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{VI.A.8}$$

Here $\omega = [\eta\, dx]$ belongs to this kernel. To check this without conflating $dx$ and $dx/x$, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator $\theta_x$ acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \geq 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\, dx$ is formally exact at zero. The commuting connection $\nabla_s$ preserves $N_0$, acts on both global and local cohomology, and hence preserves the kernel in (VI.A.8).

It remains to verify that (VI.A.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of $M$, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to $\varepsilon$, circle zero once, and return to infinity on the continued branch. With the factor $e^{-sx}$ this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of $\omega$ along its small circle is $O(\varepsilon(1 + |\log \varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of $K_0$ have at most exponential growth, which is dominated by $e^{-sx}$ in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over $K$ in (VI.A.8), period integration proves

$$\mathrm{ord}_{\mathbb{C}(s)} H_m \leq r. \tag{VI.A.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

### VI.A.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine $x$-line. Lemma VI.A.3 and (VI.A.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank $r$. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes $d$ stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is $M$; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module $\delta_0$. For an entire function $f$ in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\, dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an $A_s$-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{VI.A.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (VI.A.10) is an actual isomorphism.

Since Fourier sends $\delta_0$ to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{FK}$ (at most once) or the trivial affine-line connection.

If the former were absent, $H_m$ would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{FK}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (VI.A.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by $H_m$ is isomorphic to the irreducible generic module of $\mathcal{FK}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

### VI.A.5 Descent and the algebraicity dichotomy

Multiplication by $s$ does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order $r$ over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to $z$ are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem VI.A.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in $W$ supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus $H_m$, and consequently $F_m$, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary VI.A.2.

The argument proves a statement about a function of the variable $z$. It does not assert transcendence of every special value at algebraic $z$, and the rank argument alone does not determine the full differential Galois group. The new group results in Sections VI.2 and VI.3 use the additional arguments given there.

## VI.B Supporting approximation and testing estimates

We collect the approximation and testing estimates used in the proof.

**Lemma VI.B.1** (Polynomial approximation with two controlled derivatives)**.** *For every 1-Lipschitz $g : [0,1] \to \mathbb{R}$ and every integer $k \geq 1$, there is a real polynomial $P_k$ of degree at most $k$ such that*

$$\|g - P_k\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq C, \qquad \|P_k''\|_\infty \leq Ck.$$

*All norms in this display are on* $[0,1]$.

*Proof.* Subtract $g(0)$ and reflect $g$ to a 2-periodic 1-Lipschitz function. Convolve with a positive normalized Jackson trigonometric kernel of degree $2s$. The resulting trigonometric polynomial $Q_s$ approximates $g$ within $C/s$, and $\|Q_s'\|_\infty \leq 1$, because convolution preserves the Lipschitz constant. The trigonometric Bernstein inequality applied to $Q_s'$ gives $\|Q_s''\|_\infty \leq Cs$.

To replace $Q_s$ by an ordinary polynomial, expand its finitely many exponentials $e^{i\pi j(x-1/2)}$, $|j| \leq 2s$, through Taylor degree $100s$. Each Fourier coefficient has magnitude at most the bounded supremum norm of the reflected function. For $|x - 1/2| \leq 1/2$, the exponential-series remainder and its first two derivatives are at most a polynomial in $s$ times $e^{-cs}$, since $|\pi j(x - 1/2)| \leq \pi s$ while the truncation degree is $100s$. Their sum over $O(s)$ modes is still exponentially small. Take the real part and add back $g(0)$. With $s = \lfloor k/100 \rfloor$, this proves the assertions for large $k$; bounded $k$ is handled by a constant polynomial and an enlarged absolute constant. □

**Lemma VI.B.2** (Degree-elevated Bernstein coefficients)**.** *If $P$ is a polynomial of degree $k \leq m/2$ and*

$$P(x) = \sum_{j=0}^{m} b_j \binom{m}{j} x^j (1-x)^{m-j},$$

*then*

$$\max_j |b_j| \leq Ce^{2k^2/m} \|P\|_{[0,1]}.$$

*Proof.* Put $M = \|P\|_{[0,1]}$ and $F(z) = (1+z)^m P(z/(1+z))$; its $j$th coefficient is $\binom{m}{j} b_j$. The maximum principle in the inverse Joukowski variable gives $|P((2 + w + w^{-1})/4)| \leq M|w|^k$ for $|w| \geq 1$. For $0 < j < m$ take $r = j/(m-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}\,|\sin(\phi/2)|$. Direct inversion of the Joukowski map shows that the larger modulus associated with $re^{i\phi}/(1 + re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Thus

$$\frac{|F(re^{i\phi})|}{M(1+r)^m} \leq (1-u^2)^{(m-k)/2}(1+u)^k \leq \exp(2k^2/m)\exp(-mu^2/8).$$

For the final inequality, use $k \leq m/2$, $\log(1+u) \leq u$, $\log(1-u^2) \leq -u^2$, and complete the square in $ku - mu^2/8$. The limiting point $u = 1$ is interpreted by continuity. The average of $\exp(-mu^2/8)$ over $\phi$ is at most $C/\sqrt{1+mv}$, by comparison with a Gaussian integral near zero. Stirling's factorial bounds give

$$\binom{m}{j} \left(\frac{j}{m}\right)^j \left(1 - \frac{j}{m}\right)^{m-j} \geq \frac{c}{\sqrt{1+mv}}.$$

Cauchy's coefficient formula cancels these two square-root factors and proves the claim. The endpoint coefficients are $P(0)$ and $P(1)$. □

**Lemma VI.B.3** (A finite likelihood-envelope bound)**.** *Let $p$ be a probability density and let $u_1, \ldots, u_M$ be nonnegative envelopes with total masses $U_a \in [1, 1+\eta]$. Suppose that, on an event*

*$E$, a fitted density $\widehat{p}$ is dominated by some $u_a$ and satisfies $\prod_i \widehat{p}(X_i) \geq \prod_i p(X_i)$, where the $X_i$ are independent with density $p$. Then, for $x \geq 0$,*

$$\mathbb{P}\left(E \cap \left\{h(p,\widehat{p}) > \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}\right\}\right) \leq e^{-x},$$

*and $\mathbb{E}[\mathbf{1}_E h(p,\widehat{p})^2] \leq C\eta + C(\log M + 1)/N$.*

*Proof.* Put $r_a = u_a/U_a$. If $u_a \geq \widehat{p}$, then $\|r_a - \widehat{p}\|_1 \leq 2\eta$, so $h(r_a, \widehat{p}) \leq \sqrt{2\eta}$, and the likelihood comparison implies $\prod_i r_a(X_i)/p(X_i) \geq e^{-N\eta}$. The expectation of the square root of this product is the $N$th power of the affinity, at most $e^{-Nh(p,r_a)^2/2}$. Markov's inequality and a union bound over the $M$ deterministic envelopes prove the tail inequality. Integrating it proves the second assertion. The event $E$ is only intersected with the bad event; no conditioning or independence assertion on $E$ is used. □

**Lemma VI.B.4** (A quantitative moment-matching pair)**.** *For every interval $[a - M, a + M]$ with $M > 0$ and integer $k \geq 1$, there are probability laws $F_k^+, F_k^-$ on that interval which match moments of every degree less than $k$ and satisfy*

$$W_1(F_k^+, F_k^-) \geq \frac{M}{\pi k}.$$

*Proof.* Push the measures $(1 \pm \frac{1}{2}T_k(x))\,dx/(\pi\sqrt{1-x^2})$ on $[-1,1]$ forward by $x \mapsto a + Mx$. Positivity, normalization and moment matching follow from $|T_k| \leq 1$ and Chebyshev orthogonality. With $x = \cos\theta$, their CDF difference is $\pm\sin(k\theta)/(\pi k)$, with a sign depending only on the orientation. Therefore

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta\, d\theta \geq \frac{M}{\pi k},$$

because $|\sin u| \geq \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1}$. □

# Part VII
# Beyond Generic Parameters: Global Classifications

**Abstract**

We extend three proof mechanisms from preceding manuscripts to parameter regimes in which generic arguments leave exceptional cases unresolved. First, a finite-cover ramification bound, collision multiplicities, and an elementary reflection argument determine the full orthogonal differential Galois group of every odd-dimensional hypercubic period and every even-propagator banana period, for all nonzero complex weights. Combined with the previously established opposite-parity symplectic theorems, this gives both complete parity classifications. Second, a multiplicity-preserving formal-exponential budget proves the minimum recurrence width of every weighted central binomial power sum: it is $\lceil r/2 \rceil$ at weight 1, $\lfloor r/2 \rfloor$ at weight $-1$, and $r$ at every other nonzero complex weight, for all $r \geq 2$. Third, for complementary rational beta exponents other than $1/2$, one explicit coefficient functional of a polynomial weight determines whether every algebraic-level quantile is algebraic or every such quantile is transcendental, on every algebraic truncation interval. Finally, a universal lower bound shows why unrestricted heteroscedastic mixture maximum likelihood cannot have expected transport consistency under arbitrary noise profiles, even when consistent estimation remains possible. Supporting results from the preceding work are identified and partly reproduced. These are mathematical claims with proofs and internal checks, not claims of established historical priority or completed external validation.

**Integration note.** This part completes several parameter classifications and reproduces earlier supporting arguments as identified in its appendices. Part VIII extends its recurrence theorem to arithmetic subsequences and adds different arithmetic applications.

# VII.1 What is completed, and what the proof transfers accomplish

The central issue is uniformity. A theorem for generic parameters does not settle a parameter value where singularities collide, nor does an upper recurrence determine the minimum possible width. Likewise, an algebraic primitive test does not by itself exclude algebraic normalized values when the primitive has a logarithmic part. The following results address these specific obstructions.

## VII.1.1 Two complete group classifications

For a tuple of nonzero complex weights, let $n_j$ denote the multiplicities of its distinct squared weights, and put

$$Q = \prod_j (n_j + 1), \qquad P(u) = \prod_j (1 + u + \cdots + u^{n_j}), \qquad n = \sum_j n_j.$$

For hypercubic periods the rank is

$$R_H = Q - [u^{\lfloor n/2 \rfloor}] P(u).$$

For banana periods with $n \geq 2$, the rank is

$$R_B = \sum_{d \geq 2} (d-1) a_d, \quad \bigotimes_j \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d \geq 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

under the diagonal $\mathrm{SL}_2$. For even $n$, equivalently, $R_B = Q - [u^{n/2}]P - [u^{n/2-1}]P$. The group conclusions, in their standard representations, are

| | $n$ odd | $n$ even |
|---|---|---|
| hypercubic | $\mathrm{O}_{R_H}$ (this paper) | $\mathrm{Sp}_{R_H}$ (prior) |
| banana | $\mathrm{Sp}_{R_B}$ (prior, $n \geq 3$) | $\mathrm{O}_{R_B}$ (this paper). |

There are no additional restrictions on the collisions or vanishing of signed thresholds. Zero weights can be deleted; the rational zero- and one-factor banana cases are recorded separately.

The new ingredient excludes decomposition over finite algebraic covers. Permuting $h$ components of dimension $d$ requires at least $d(2h-2)$ units of negative-eigenspace multiplicity by Riemann–Hurwitz. An exposed reflection cannot contribute to this budget. Concentration bounds and exact small-dimensional arguments rule out the decomposition; a reflection then forces the full orthogonal group.

This also gives precise functional algebraic information. In the symplectic cases the first $R$ jets are algebraically independent. In the orthogonal cases of rank at least three their relation ideal is generated by a single nondegenerate quadratic polynomial, and their transcendence degree is $R - 1$. These are statements about functions, not arithmetic claims about each individual special value.

## VII.1.2 A complete width formula

For

$$B_{r,\lambda}(n) = \lambda^{-n} \sum_{k=0}^{2n} \binom{2n}{k}^r \lambda^k, \quad r \geq 2, \quad \lambda \in \mathbb{C}^*,$$

Theorem VII.4.1 gives

$$\operatorname{minwidth}_{\mathbb{C}(n)} B_{r,\lambda} = \begin{cases} \lceil r/2 \rceil, & \lambda = 1, \\ \lfloor r/2 \rfloor, & \lambda = -1, \\ r, & \lambda \notin \{1, -1\}. \end{cases}$$

The proof counts formal exponential factors with multiplicity after factorial normalization. It therefore retains the information that a count of distinct finite singularities loses at collisions. Katz's classical differential Galois calculation supplies cyclicity; an explicit concomitant handles the adjoint representation at odd $r$ and weight $-1$. Width means largest shift minus smallest shift. It must not be replaced by the number of nonzero terms in an arbitrarily sparse recurrence.

### VII.1.3 An effective arithmetic dichotomy

Fix $r \in \mathbb{Q} \cap (0,1) \setminus \{1/2\}$, algebraic endpoints $0 \le u < v \le 1$, and a real algebraic polynomial $w(x) = \sum_j w_j x^j$ positive on $(u, v)$. Normalize the density $w(x)x^{r-1}(1-x)^{-r}$ on this interval. The single coefficient

$$c_r(w) = \sum_j w_j \frac{(r)_j}{j!}$$

settles every algebraic probability level:

| | |
|---|---|
| $c_r(w) = 0$ | all internal algebraic-level quantiles are algebraic, |
| $c_r(w) \neq 0$ | all internal algebraic-level quantiles are transcendental. |

The exclusion argument combines Baker's independence theorem with coefficient conjugation and strict ordering of complementary beta densities. The argument is uniform in both truncation endpoints. Section VII.5 also gives an explicit polynomial equation in the algebraic case and the equivalent linear-independence statement.

### VII.1.4 A necessary boundary for likelihood methods

For an exact unrestricted Gaussian mixture NPMLE with known noise standard deviations $\sigma_i$, Theorem VII.6.1 proves

$$\mathbb{E}_{\delta_\mu} W_1(\widehat{G}, \delta_\mu) \ge \frac{\max_i \sigma_i}{2\sqrt{2\pi}\, N^2}.$$

With $N - 1$ unit-noise observations and one standard deviation $N^3$, this expected error diverges, although the minimax error on the specified growing compact-support class tends to zero. This is a failure of an unrestricted extension, not a contradiction of the published bounded-covariance theorem. It illustrates why the likelihood-contact principle must be paired with an information condition before a global optimality conclusion is attempted.

### VII.1.5 Dependencies, provenance, and verification status

The new orthogonal group arguments use the prior all-mass rank, irreducibility, pairing, and infinity-monodromy theorems. The hypercubic rank proof, pairing, and opposite-parity group proofs are reproduced in the appendices. The banana rank input is precisely stated and traced to supplied manuscript [032] [49]; its full original development is not reproduced. The arithmetic proof uses classical Baker theorems, and the recurrence proof uses the classical Kloosterman group calculation and existing upper bounds. The proof transfers continue the preceding manuscripts' Bessel and singularity methods, arithmetic separation methods, and likelihood contact methods [65, 66]. Separate internal checks cover the main arguments, exceptional cases, and the invariant-tensor correction. External expert review and full priority assessment remain outstanding.

## VII.2 Full orthogonal groups in every odd hypercubic dimension

Let $n \geq 1$ be odd and $m_1, \dots, m_n \in \mathbb{C}^\times$. Define

$$F_m(z) = \sum_{k \geq 0} (2k)! \left( \sum_{a_1 + \cdots + a_n = k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} \right) z^k, \qquad H_m(s) = s^{-1} F_m(s^{-2}).$$

Group the masses by equality of their squares. Write the distinct squares as $\mu_j^2$ and their multiplicities as $n_j$, $1 \leq j \leq q$. Put

$$Q = \prod_j (n_j + 1), \quad b = [u^{(n-1)/2}] \prod_j (1 + u + \cdots + u^{n_j}), \quad R = Q - b.$$

**Theorem VII.2.1** (Every odd dimension, without threshold restrictions)**.** *The full differential Galois groups of $H_m$ over $\mathbb{C}(s)$ and of $F_m$ over $\mathbb{C}(z)$ are $\mathrm{O}_R(\mathbb{C})$ in their standard representations. This holds for every nonzero complex mass tuple, including all collisions and vanishing of signed thresholds. For $n \geq 3$, the minimal order remains $R$ after every finite algebraic extension of the coefficient field. For either function $f$, the field generated by $f, f', \dots, f^{(R-1)}$ has transcendence degree $R - 1$, and its polynomial relation ideal is generated by one nondegenerate quadratic polynomial.*

The new step is the exclusion of every finite-cover decomposition. A reflection and a nontrivial unipotent in an irreducible orthogonal group would not alone exclude such a decomposition. We use the total collision multiplicity together with Riemann–Hurwitz.

### VII.2.1 Previously proved differential-module input

We use the all-weight rank theorem (Theorem VII.A.1) and the reflected middle-pairing construction (Lemma VII.B.1) proved in the preceding work. In precise terms, put $B_\mu : (\theta_x^2 - 4\mu^2 x^2) y = 0$ and $M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$. The cyclic module of $H_m$ is the simple positive-rank Fourier transform $\mathcal{F}(j_{!*} M)$, equivalently $H^1_{\mathrm{mid}}(M \otimes e^{-sx})$; its rank is $R$. It has a nondegenerate horizontal symmetric form when $n$ is odd. The same form descends to $F_m$: the simultaneous involution $(s, x) \mapsto (-s, -x)$ fixes the cyclic class $s[\eta\, dx]$ and its $z$-derivatives and preserves the residue pairing.

If

$$\bigotimes_j \mathrm{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d \geq 0} (\mathrm{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

under diagonal $\mathrm{SL}_2$, then infinity monodromy of $H_m$ is unipotent with blocks $J_d$ of multiplicities $a_d$, $d \geq 1$. In particular its largest block $J_n$ occurs exactly once. These are established all-weight statements, not generic assumptions.

Here are the finite local data, also fixing the convention at zero. The formal germ of $M$ at infinity is

$$\bigoplus_{\lambda \in \Lambda} E^{c_\lambda x} \otimes R_n, \quad \Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}, \quad c_\lambda = 2 \sum_j \lambda_j \mu_j,$$

where $R_n$ is the rank-one regular germ with solution $x^{-n/2}$. Its monodromy is $-1$. If

$$k_c = \#\{\lambda \in \Lambda : c_\lambda = c\},$$

local Fourier stationary phase gives regular vanishing cycles of dimension $k_c$ with monodromy $-1$ at $s = c$; it gives zero vanishing cycles elsewhere. This follows from the local Fourier formula at every finite point, including zero [77, Corollary 2.3(1), §3.3], and preservation of regular local

monodromy [41, Remarks 5.2, 5.3(4), 5.7]. Colliding exponential factors are a direct sum already before the transform, so their multiplicities add.

The Fourier module is an intermediate extension. Nearby monodromy therefore has no nontrivial unipotent part: such a part would give an eigenvalue-one vanishing cycle. It is an involution $T_c$ with $-1$-eigenspace of dimension $k_c$. The equation is regular singular at every finite point, since the irregular part is unchanged by formal vanishing cycles, and is regular singular at infinity by the rank theorem. In particular

$$T_c^2 = 1, \qquad \operatorname{rank}(T_c - 1) = k_c, \qquad \sum_c k_c = Q. \tag{VII.2.1}$$

Schlesinger's theorem identifies its differential Galois group with the Zariski closure of these local monodromies.

### VII.2.2 Two elementary concentration bounds

**Lemma VII.2.2** (Maximum threshold multiplicity)**.** *For every mass tuple, $k_c \le b$ for every $c$.*

*Proof.* Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every $j$. Change the sign of each $\mu_j$ if necessary so these real parts are positive; this permutes $\Lambda$. Write $\lambda_j = n_j - 2a_j$ with $0 \le a_j \le n_j$. In the product poset $\prod_j \{0, \dots, n_j\}$ with coordinatewise order, a fiber of $\sum_j a_j\mu_j$ is an antichain: strict coordinatewise increase strictly increases its real projection.

A product of finite chains has a symmetric-chain decomposition, which we recall explicitly. A rectangle $\{0, \dots, a\} \times \{0, \dots, d\}$ with $a \le d$ splits into the chains

$$(i, 0), (i, 1), \dots, (i, d-i), (i+1, d-i), \dots, (a, d-i), \quad 0 \le i \le a.$$

Their endpoint ranks add to $a + d$. Taking products of each existing symmetric chain with a new chain and using this rectangle construction proves the assertion by induction. Every chain in the decomposition passes once through rank $(n-1)/2$. Hence the number of chains is $b$, and an antichain meets each at most once. This proves the bound. □

**Lemma VII.2.3** (Uniform chains compared with coin flips)**.** *For every partition $n = \sum_j n_j$,*

$$\frac{b}{Q} \le 2^{-n} \binom{n}{\lfloor n/2 \rfloor}.$$

*Here the lemma allows either parity and defines $b = [u^{\lfloor n/2 \rfloor}] \prod_j (1 + \cdots + u^{n_j})$. More generally, the centered sum of independent uniforms on $\{0, \dots, n_j\}$ has at most the mass of a centered* $\mathrm{Bin}(n, 1/2)$ *in every centered interval. Consequently $Q > 4b$ for every odd $n \ge 9$.*

*Proof.* Let $U_m$ be uniform on $\{0, \dots, m\}$ and let $B_m$ have the binomial distribution with parameters $(m, 1/2)$. Center both at $m/2$. The binomial law places at least as much mass as the uniform law in every centered interval. Indeed its probabilities are symmetric and decrease away from the center, so the average of its central $h$ probabilities is at least the average of all $m + 1$ probabilities, namely $1/(m+1)$.

This comparison is preserved by convolution with a symmetric unimodal law. To see this, for any centered interval $I$, the function $x \mapsto \mathbb{P}(x + Y \in I)$ is symmetric and decreases with $|x|$ when $Y$ has such a law. Summation by parts expresses its expectation as a nonnegative combination of centered-interval probabilities. The statement applies equally to integer and half-integer centered lattices. Symmetric unimodal laws are closed under convolution: each is a nonnegative mixture of uniform laws on centered lattice intervals, and the convolution of two such uniform laws has probabilities proportional to interval-overlap lengths, which are symmetric and unimodal.

Replace the independent variables $U_{n_j}$ one at a time by $B_{n_j}$. The probability in any centered interval cannot decrease. Their final sum has the binomial law $B_n$. For odd $n$, the two central probabilities of the original sum are both $b/Q$; compare the centered interval containing precisely these two lattice points. For even $n$, compare the singleton at the center, whose original probability is $b/Q$. The asserted inequality follows in both parities. At $n = 9$ its right side is $126/512 < 1/4$. For successive odd indices the ratio is $\binom{2h+3}{h+1}2^{-2h-3}/(\binom{2h+1}{h}2^{-2h-1}) = (2h+3)/(2h+4) < 1$, proving the last assertion. □

### VII.2.3 The branch budget of a hypothetical block decomposition

**Lemma VII.2.4** (Isotypic blocks and a ramification budget)**.** *Let a regular-singular connection on the punctured projective line have irreducible monodromy with Zariski closure $G \subseteq \mathrm{GL}(V)$. Suppose infinity monodromy is unipotent with a unique largest Jordan block of size $\ell \geq 2$, and every finite monodromy $T_c$ has eigenvalues in $\{1, -1\}$. Write $k_c$ for its $-1$ generalized eigenspace dimension. Then $G^0$ is reductive and its isotypic components have multiplicity one. If there are $t$ such components of dimension $d$, then*

$$\dim V = td, \qquad d \geq \ell, \qquad d(2t-2) \leq \sum_c k_c.$$

*If $h$ distinct finite local monodromies are reflections, the last right side can be replaced by $\sum_c k_c - h$. If $t > 1$, also $d \leq \max_c k_c$.*

*Proof.* The unipotent radical of $G^0$ is characteristic in $G^0$ and normal in $G$. Its fixed space is nonzero and $G$-invariant, so irreducibility and faithfulness make that radical trivial. Thus $V|_{G^0}$ is completely reducible. Its isotypic components are permuted transitively by $G$; let their common irreducible dimension be $d$ and their common multiplicity be $e$.

Every unipotent element of $G$ belongs to $G^0$, since the Zariski closure of its cyclic subgroup is connected. On each isotypic component its Jordan blocks are repeated $e$ times. The unique largest block at infinity forces $e = 1$ and $d \geq \ell$.

The block permutation action defines a connected degree-$t$ cover of the punctured sphere. Complete it to a map from a compact Riemann surface of genus $g$ to $\mathbb{P}^1$. Infinity is unramified because its unipotent monodromy belongs to $G^0$. For each $T_c$, its unipotent part belongs to $G^0$, so the block action is that of its semisimple part. The latter is an involution. If it swaps $a_c$ disjoint pairs of blocks, each swapped pair contributes $d$ eigenvalues $-1$: its matrix is $\left(\begin{smallmatrix} 0 & A \\ A^{-1} & 0 \end{smallmatrix}\right)$ in suitable bases. Thus $a_c d \leq k_c$. Riemann–Hurwitz, equivalently Euler characteristic for a triangulation with branch values as vertices, gives

$$2t - 2 \leq 2t + 2g - 2 = \sum_c a_c.$$

A reflection has $k_c = 1 < d$, so cannot permute blocks and its contribution can be removed. This proves both budgets. If $t > 1$, some finite monodromy must permute blocks: the finite local loops generate the fundamental group, and the action is transitive. At that point $k_c \geq d$, proving the last claim. □

Assume $n \geq 3$ until the rank-one case in the final proof. Apply the lemma to the equation of $H_m$, with $\ell = n$. Its Galois group $G \subseteq \mathrm{O}_R$ acts irreducibly by the prior rank theorem. If its identity component has $t$ isotypic blocks of dimension $d$, then

$$d \geq n, \qquad R = td, \tag{VII.2.2}$$

and (VII.2.1) gives

$$2t - 2 \leq \frac{Q}{d} = \frac{Qt}{Q-b}. \tag{VII.2.3}$$

We must prove $t = 1$.

At least two finite monodromies are individual reflections. Indeed maximize $\mathrm{Re}(\xi c_\lambda)$ as in Lemma VII.2.2. Its unique maximum is attained by the endpoint sign tuple, has positive projection, and has multiplicity one. Its negative also has multiplicity one. Since $d \geq 3$, neither reflection can permute blocks. Removing their contributions from the branch budget improves (VII.2.3) to

$$d(2t - 2) \leq Q - 2. \tag{VII.2.4}$$

Furthermore, if $t > 1$, some finite local monodromy must permute blocks, since these generate the monodromy group and infinity fixes all blocks. Lemma VII.2.2 then forces

$$d \leq b. \tag{VII.2.5}$$

For odd $n \geq 9$, Lemma VII.2.3 gives $Q > 4b$. Inequality (VII.2.3) rearranges to $t \leq 2(Q - b)/(Q - 2b) < 3$. Thus $t > 1$ would mean $t = 2$ and $d = (Q - b)/2 > b$, contradicting (VII.2.5). For $n = 3$, the possible ranks are $3, 4, 5$, and (VII.2.2) already forces $t = 1$.

For completeness, the remaining finite check for $n = 5, 7$ is shown in the table. Its last column lists every $t \geq 2$ satisfying $R = td$, $d \geq n$, $d \leq b$, and (VII.2.4); it is obtained by checking the integer divisors of the displayed $R$.

| $(n_j)$ | $Q$ | $b$ | $R$ | remaining $t$ |
|---|---|---|---|---|
| $(5)$ | 6 | 1 | 5 | none |
| $(4,1)$ | 10 | 2 | 8 | none |
| $(3,2)$ | 12 | 3 | 9 | none |
| $(3,1,1)$ | 16 | 4 | 12 | none |
| $(2,2,1)$ | 18 | 5 | 13 | none |
| $(2,1,1,1)$ | 24 | 7 | 17 | none |
| $(1^5)$ | 32 | 10 | 22 | none |
| $(7)$ | 8 | 1 | 7 | none |
| $(6,1)$ | 14 | 2 | 12 | none |
| $(5,2)$ | 18 | 3 | 15 | none |
| $(5,1,1)$ | 24 | 4 | 20 | none |
| $(4,3)$ | 20 | 4 | 16 | none |
| $(4,2,1)$ | 30 | 6 | 24 | none |
| $(4,1,1,1)$ | 40 | 8 | 32 | none |
| $(3,3,1)$ | 32 | 7 | 25 | none |
| $(3,2,2)$ | 36 | 8 | 28 | none |
| $(3,2,1,1)$ | 48 | 11 | 37 | none |
| $(3,1,1,1,1)$ | 64 | 15 | 49 | none |
| $(2,2,2,1)$ | 54 | 13 | 41 | none |
| $(2,2,1,1,1)$ | 72 | 18 | 54 | none |
| $(2,1,1,1,1,1)$ | 96 | 25 | 71 | none |
| $(1^7)$ | 128 | 35 | 93 | 3 |

In the last case all seven squared masses are distinct. Choose $\xi$ also to avoid equality between the absolute real projections of any two masses. After changing signs and reordering, write these projections as $0 < a_1 < a_2 < \cdots < a_7$, with oriented masses $\nu_1, \ldots, \nu_7$. Among their subset sums, $0, \nu_1, \nu_2$ are the unique sums with their respective projected values: any other singleton has larger projection than $a_2$, and every subset with two or more entries has projection at least $a_1 + a_2$. Hence the signed exponents

$$2\sum_j \nu_j, \qquad 2\sum_j \nu_j - 4\nu_1, \qquad 2\sum_j \nu_j - 4\nu_2$$

all have multiplicity one. Their negatives do too, and the six are distinct because $\sum_j a_j > 2a_2$. None of these six reflections can permute blocks. The last possibility $t = 3, d = 31$ would therefore require $31(2 \cdot 3 - 2) \le 128 - 6$, or $124 \le 122$, a contradiction. We have proved that $G^0$ acts irreducibly in every odd dimension $n \ge 3$.

### VII.2.4 A reflection forces the full orthogonal Lie algebra

**Lemma VII.2.5.** *Let $G \subseteq \mathrm{O}(V, B)$ be an algebraic subgroup such that $G^0$ acts irreducibly on $V$, with $\dim V \ge 3$. If $G$ contains an orthogonal reflection, then $G^0 = \mathrm{SO}(V, B)$ and $G = \mathrm{O}(V, B)$.*

*Proof.* Write the reflection as $r_u(v) = v - 2B(u, v)u/B(u, u)$, where $B(u, u) \neq 0$. Put $\mathfrak{g} = \operatorname{Lie} G^0$ and $W = \mathfrak{g}u \subseteq u^\perp$. For $v, w \in V$, set $L_{v,w}(x) = B(v, x)w - B(w, x)v$. Since $G^0$ is normal in $G$, $\mathfrak{g}$ is stable under conjugation by $r_u$. For $A \in \mathfrak{g}$, direct calculation gives

$$A - r_u A r_u = \frac{2}{B(u, u)} L_{u,Au}.$$

Thus $L_{u,w} \in \mathfrak{g}$ for all $w \in W$. For $v, w \in W$ the identity $[L_{u,w}, L_{u,v}] = B(u, u)L_{w,v}$ gives $L_{w,v} \in \mathfrak{g}$. For $A \in \mathfrak{g}$ and $w \in W$,

$$[A, L_{u,w}] = L_{Au,w} + L_{u,Aw}.$$

The first term on the right is already in $\mathfrak{g}$, so $L_{u,Aw} \in \mathfrak{g}$. Applying it to $u$ shows $B(u, u)Aw - B(Aw, u)u \in W$. Hence $\mathbb{C}u + W$ is $\mathfrak{g}$-invariant, and therefore $G^0$-invariant since $G^0$ is connected. Irreducibility yields $\mathbb{C}u + W = V$, so $W = u^\perp$. The operators just obtained span all of $\mathfrak{so}(V, B)$. Therefore $G^0 = \mathrm{SO}(V, B)$. The reflection has determinant $-1$, giving the other component. □

*Completion of Theorem VII.2.1.* For $n \ge 3$ we have proved connected irreducibility and exhibited a reflection. The lemma gives the full orthogonal group for $H_m$. The quadratic base change $z = s^{-2}$ and rational gauge retain its identity component. The descended symmetric pairing puts the group of $F_m$ inside $\mathrm{O}_R$. After the quadratic base change its Galois group is a subgroup of the original group, yet it already equals the full $\mathrm{O}_R$ by the result for $H_m$. Hence the original group of $F_m$ also equals $\mathrm{O}_R$. For $n = 1$, the explicit formula $F_m(z) = (1 - 4m_1^2 z)^{-1/2}$ gives $\mathrm{O}_1 = \{\pm 1\}$ for both variables. Its square is rational; explicitly $H_m(s)^2 = (s^2 - 4m_1^2)^{-1}$. These rational functions are not squares in their respective rational function fields. Thus the one-jet field has transcendence degree zero and its relation ideal is generated by the indicated irreducible quadratic over that base field. No absolute irreducibility is asserted in this rank-one case.

For $n \ge 3$, finite algebraic base extension preserves the connected $\mathrm{SO}_R$, which acts irreducibly. Thus the order cannot fall. The orbit of a nonzero vector under $\mathrm{O}_R$ has dimension $R - 1$, whether its quadratic norm is zero or nonzero. The Picard–Vessiot orbit-dimension formula gives the stated transcendence degree. In the cyclic derivative basis, the inverse Gram matrix of the horizontal form gives a quadratic relation $\mathbf{f}^T C^{-1} \mathbf{f} = c$, $c \in \mathbb{C}$. Its quadratic part has rank $R \ge 3$, so this polynomial is absolutely irreducible. The relation ideal is a height-one prime containing this irreducible quadratic, hence is precisely its principal ideal. □

**Corollary VII.2.6** (All dimensions and all weights)**.** *Combining Theorem VII.2.1 with the previously proved all-even-dimensional Theorem VII.C.1 gives, for every $n \ge 1$ and every nonzero complex mass tuple,*

$$G(F_m) = G(H_m) = \begin{cases} \mathrm{O}_R(\mathbb{C}), & n \text{ odd}, \\ \mathrm{Sp}_R(\mathbb{C}), & n \text{ even}, \end{cases} \qquad R = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}).$$

*Zero weights may be deleted before applying this classification; if all weights vanish then $F = 1$ and $H = s^{-1}$ have trivial groups. The new contribution of this section is the odd-dimensional case, including every collided-threshold specialization.*

## VII.3 Full orthogonal groups for every even-propagator banana period

Let $n \geq 2$ be even and let $m_1, \dots, m_n \in \mathbb{C}^\times$. We use the banana period

$$\varpi_m(t) = \sum_{k \geq 0} \frac{c_k(m)}{t^{k+1}}, \qquad c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}.$$

Group the masses by equality of their squares, and denote the multiplicities by $n_1, \dots, n_q$. Put

$$Q = \prod_j (n_j + 1), \qquad B(u) = \prod_j (1 + u + \cdots + u^{n_j}), \qquad b = [u^{n/2}]B(u), \qquad c = [u^{n/2-1}]B(u),$$

and

$$R = Q - b - c, \qquad \delta = \begin{cases} 1 & \text{if every } n_j \text{ is even,} \\ 0 & \text{otherwise,} \end{cases} \qquad \kappa = (-1)^{n/2}, \qquad S = \frac{Q - \kappa\delta}{2}. \quad \text{(VII.3.1)}$$

Here $R$ is the all-mass minimal order established in the supplied manuscript [49]. Indeed, if $\bigotimes_j \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_d (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$ under the diagonal $\mathrm{SL}_2$, then $R = \sum_{d \geq 2}(d-1)a_d = Q - 2b + a_0 = Q - b - c$.

**Theorem VII.3.1** (No exceptional nonzero mass tuples)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois group over $\mathbb{C}(t)$ of the minimal equation of $\varpi_m$ is*

$$\mathrm{O}_R(\mathbb{C}).$$

*There is no distinct-threshold or nonzero-threshold hypothesis. For $n \geq 4$, the minimal equation remains irreducible of order $R$ after every finite algebraic extension of $\mathbb{C}(t)$.*

The new issue is irreducibility under the connected group. Irreducibility under the full group, a reflection, and a nontrivial unipotent do not suffice: an orthogonal group permuting several isomorphic blocks would satisfy those three conditions. We exclude such a decomposition by counting ramification of its finite permutation cover. Collisions contribute to this count, but the total available multiplicity is independent of the collisions.

### VII.3.1 The all-mass local input and its negative-eigenspace budget

We use the previously proved all-mass order and middle-cohomology theorem of [49]: the minimal module is irreducible of rank $R$, preserves a nondegenerate symmetric form, and has unipotent monodromy at infinity with blocks $J_{d-1}$ for the summands with $d \geq 2$. In particular its largest block is $J_{n-1}$ and occurs exactly once. These facts hold without conditions on signed sums. We now compute the finite semisimple local data needed for the extension of the Galois theorem.

Choose representatives $\mu_j$ for the distinct squared masses, and put

$$\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}, \qquad c_\lambda = \sum_j \lambda_j \mu_j, \qquad z_0 = \#\{\lambda \in \Lambda : c_\lambda = 0\}.$$

The coordinatewise negation involution on $\Lambda$ has a fixed point precisely when $\delta = 1$.

**Lemma VII.3.2** (Finite local multiplicities)**.** *The equation is regular singular everywhere. At every nonzero squared threshold $a$, its local monodromy $T_a$ is an involution, whose negative eigenspace has dimension*

$$k_a = \frac{1}{2}\#\{\lambda \in \Lambda : c_\lambda^2 = a\}.$$

*At zero the semisimple part of $T_0$ is an involution whose negative eigenspace has dimension*

$$k_0 = \frac{z_0 - \kappa\delta}{2}.$$

*Consequently*

$$k_0 + \sum_{a\neq 0} k_a = S. \tag{VII.3.2}$$

*At least one nonzero threshold has $k_a = 1$.*

*Proof.* We recall the two-Fourier-transform construction to account for zero as well as nonzero collisions. Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel module $M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$ has, on the formal $u$-cover at infinity, a direct-sum decomposition into rank-one factors

$$E^{c_\lambda u} \otimes u^{-n/2}, \qquad \lambda \in \Lambda.$$

Invertible formal series have been absorbed by rank-one formal gauges. This is a splitting before equal exponential factors are grouped; a collision therefore introduces no formal extension.

For $c_\lambda \neq 0$, the pair $\lambda, -\lambda$ descends to a rank-two ramified factor at $y = \infty$. The first Fourier transform gives $E^{c_\lambda^2/(4s)}$ times a regular rank-one factor with power $s^{(n-3)/2}$ and monodromy $-1$. This is the stationary-phase calculation: the measure $2u\,du$, the value of $u^{1-n/2}$ at $u = c_\lambda/(2s)$, and the Gaussian factor $s^{-1/2}$ produce the displayed power. After inversion $s = 1/v$, the rational gauge $v^{-1}$, and the second Fourier transform, this factor supplies a vanishing-cycle line of monodromy $-1$ at $t = c_\lambda^2$, with the normalization $t = 4\tau$ for the second Fourier variable. Equal squared thresholds give direct sums of these lines. Intermediate extension identifies the nonunit nearby and vanishing-cycle eigenspaces; the remaining nearby eigenspace is trivial. Thus $T_a$ is an involution with the stated negative multiplicity.

For clarity, at exponent zero the original regular formal part has rank $z_0$. Each nonfixed pair $\lambda, -\lambda$ contributes one $+1$ and one $-1$ monodromy eigenvalue upon descent from $u$ to $y$. If the fixed tuple $\lambda = 0$ exists, its factor is $y^{-n/4}$, of monodromy $\kappa$. Hence the negative multiplicity in this original zero-exponent part is $(z_0 - \kappa\delta)/2$.

Local Fourier transform preserves the monodromy of a regular formal germ, up to inversion of the loop convention; this includes the unipotent regular part, in the formal vanishing-cycle convention of [77, Corollary 2.3 and §3.3]. See also [41, 40]. At the first transformed zero, the nonunit $-1$ part of the regular nearby germ is exactly that of its vanishing cycles, so it has this same dimension. Inversion preserves it, and the second local Fourier transform again preserves it. Finally, at the last transformed zero, passing from vanishing to nearby cycles again leaves the $-1$ part unchanged. All other eigenvalues in these regular parts are $+1$; intermediate extension may lengthen their unipotent blocks but cannot create another semisimple eigenvalue. This proves the claim about $T_0$. In particular $T_0^2$ is unipotent. The same local descriptions show regularity at every finite point; regularity at infinity is part of the prior input.

There are $(Q - z_0)/2$ nonzero pairs in total. Adding their negative multiplicities to $k_0$ proves (VII.3.2).

Choose $\xi \in \mathbb{C}^\times$ with $\mathrm{Re}(\xi\mu_j) \neq 0$ for every $j$. The functional $\mathrm{Re}(\xi c_\lambda)$ has the unique maximizer $\lambda_j^* = n_j \operatorname{sign}\mathrm{Re}(\xi\mu_j)$, whose value is strictly positive. Thus $c_* = c_{\lambda^*} \neq 0$. Only $\lambda^*$ and $-\lambda^*$ can have squared exponent $c_*^2$, because the only complex square roots of $c_*^2$ are $c_*$ and $-c_*$. This threshold has $k_{c_*^2} = 1$ and its monodromy is a genuine reflection. □

### VII.3.2 A ramification obstruction to a finite permutation of blocks

**Lemma VII.3.3** (The permutation-cover bound)**.** *Let $n \geq 4$ be even, and let $G$ be the Galois group of the banana equation. If $G^0$ acts reducibly, then there are integers $h \geq 2$ and $d \geq n - 1$*

*with $R = hd$ such that*

$$d(2h-2) \le S-1. \tag{VII.3.3}$$

*In particular $S \le R$ is sufficient for $G^0$ to act irreducibly.*

*Proof.* Apply the general block-budget Lemma VII.2.4 with largest unipotent block size $\ell = n-1$. Lemma VII.3.2 supplies the finite negative-eigenspace budget $S$ and at least one genuine reflection. That general lemma also permits the unipotent part of $T_0$, since only its semisimple eigenvalues enter the budget. It gives $R = hd$, $d \ge n-1$, and $d(2h-2) \le S-1$. If $S \le R = hd$, this would imply $d(h-2) \le -1$, impossible for $h \ge 2$. □

### VII.3.3 The numerical inequality and the two low dimensions

**Lemma VII.3.4.** *For every multiplicity partition of an even $n \ge 8$, $R > Q/2$ and hence $S \le R$.*

*Proof.* Let $X_j$ be independent uniforms on $\{0, \dots, n_j\}$, and center their sum by $n/2$. Its probabilities at zero and at one are $b/Q$ and $c/Q$. The uniform-to-binomial concentration comparison of Lemma VII.2.3 says that its mass in every centered interval is at most the corresponding mass of a centered $\mathrm{Bin}(n, 1/2)$. By symmetry,

$$\begin{aligned}\frac{b+c}{Q} &= \frac{1}{2}\left(\Pr(X_1 + \cdots + X_q = n/2) + \Pr(|X_1 + \cdots + X_q - n/2| \le 1)\right) \\ &\le \frac{\binom{n}{n/2} + \binom{n}{n/2-1}}{2^n} = \frac{\binom{n+1}{n/2}}{2^n}.\end{aligned}$$

At $n = 8$ the last value is $126/256 < 1/2$. For $n = 2k$ its ratio at $k+1$ to its value at $k$ is $(2k+3)/(2k+4) < 1$. Therefore $R/Q = 1 - (b+c)/Q > 1/2$ for every even $n \ge 8$. Since $R$ is an integer and $S \le (Q+1)/2$, this implies $S \le R$. □

For completeness, all smaller multiplicity partitions give the following integer data directly from (VII.3.1).

| $n=4$ partition | $R$ | $S$ | $n=6$ partition | $R$ | $S$ |
|---|---|---|---|---|---|
| $[4]$ | 3 | 2 | $[6]$ | 5 | 4 |
| $[3,1]$ | 4 | 4 | $[5,1]$ | 8 | 6 |
| $[2,2]$ | 4 | 4 | $[4,2]$ | 9 | 8 |
| $[2,1,1]$ | 5 | 6 | $[4,1,1]$ | 12 | 10 |
| $[1,1,1,1]$ | 6 | 8 | $[3,3]$ | 9 | 8 |
| | | | $[3,2,1]$ | 13 | 12 |
| | | | $[3,1,1,1]$ | 17 | 16 |
| | | | $[2,2,2]$ | 14 | 14 |
| | | | $[2,2,1,1]$ | 18 | 18 |
| | | | $[2,1,1,1,1]$ | 23 | 24 |
| | | | $[1,1,1,1,1,1]$ | 29 | 32 |

Every row with $S \le R$ is handled by Lemma VII.3.3. The exceptional ranks $5, 23, 29$ are prime. Since a component dimension must divide the rank and be at least $n-1 \ge 3$, these rows also admit no nontrivial permutation of components. Only $n = 4$ with four distinct squared masses remains.

**Lemma VII.3.5** (Four distinct masses)**.** *For four nonzero masses with pairwise distinct squares, every nonzero squared threshold has multiplicity $k_a \le 2$.*

*Proof.* Fix a nonzero signed exponent $c_*$. Choose a real projection $z \mapsto \mathrm{Re}(\xi z)$ which is nonzero on $c_*$ and on every mass, and gives pairwise distinct absolute values to the masses. Such a choice

excludes only finitely many real lines for $\xi$. Change individual mass signs and reorder so their projected values satisfy $0 < a_1 < a_2 < a_3 < a_4$. All signed tuples giving $c_*$ yield subsets of these four positive numbers with the same sum $v$, where $2v \neq a_1 + a_2 + a_3 + a_4$ because the projected signed exponent is nonzero.

At most one singleton and at most one triple can have sum $v$. Two different equal-sum pairs must be disjoint, because the $a_i$ are distinct; they would be complementary and force $2v = \sum_i a_i$. Thus there is at most one pair. A singleton and a triple of sum $v$ cannot intersect, by positivity, so they too would be complementary and force the forbidden central value. Empty or full subsets cannot share their sum with any other subset. Consequently at most two subsets have sum $v$. The number of signed tuples giving $c_*$ is therefore at most two. Each pair $\lambda, -\lambda$ over the squared threshold contributes exactly one tuple giving $c_*$, which proves the claim. □

In the remaining rank-six case, a nontrivial component decomposition would have two components of dimension three. Lemma VII.3.5 shows that no nonzero-threshold involution can exchange them. The infinity loop also fixes them. These loops generate the punctured-sphere group after omitting the loop at zero; hence the entire permutation action is trivial, contradicting transitivity. This finishes connected irreducibility in every even dimension $n \geq 4$.

*Proof of Theorem VII.3.1.* For $n \geq 4$, the preceding arguments show that $G^0$ is irreducible. The invariant symmetric form places $G$ inside $\mathrm{O}_R$, and Lemma VII.3.2 supplies a reflection. The elementary reflection-normalizer argument in Lemma VII.2.5 now gives $G^0 = \mathrm{SO}_R$ and $G = \mathrm{O}_R$. Finite algebraic base change preserves the identity component; the standard $\mathrm{SO}_R$ representation is irreducible for $R \geq 3$, proving the order assertion.

For $n = 2$, direct summation gives

$$\varpi_m(t) = \frac{1}{\sqrt{(t - (m_1 + m_2)^2)(t - (m_1 - m_2)^2)}}.$$

The two roots are distinct because $m_1 m_2 \neq 0$, so this is a nonrational quadratic algebraic solution. Its rank-one Galois group is $\{\pm 1\} = \mathrm{O}_1$, as required. □

**Corollary VII.3.6** (The unique jet relation)**.** *For even $n \geq 4$ and arbitrary nonzero complex masses, the $R$ functions $\varpi_m, \varpi_m', \ldots, \varpi_m^{(R-1)}$ have transcendence degree $R - 1$ over $\mathbb{C}(t)$. Their polynomial-relation ideal is generated by the one nondegenerate irreducible quadratic relation supplied by the invariant form. The same assertion holds after any finite algebraic extension of the coefficient field.*

*Proof.* The orbit of a nonzero vector in the standard orthogonal representation has dimension $R - 1$, whether its quadratic value is zero or nonzero. The Picard–Vessiot orbit formula therefore gives the stated transcendence degree. The relation ideal is a height-one prime containing the invariant quadratic. For $R \geq 3$ that polynomial is absolutely irreducible, since a product of two affine linear forms has quadratic rank at most two. It thus generates the ideal. After finite base change the connected group is still $\mathrm{SO}_R$ and has the same nonzero-vector orbit dimensions. □

Together with the odd-propagator theorem, this classifies the full banana Galois group for every number of nonzero masses. Zero masses may simply be deleted from the defining series. With no nonzero masses or one nonzero mass, the period is, respectively, $1/t$ or $1/(t - m^2)$ and has trivial Galois group. No assertion about a numerical value at a fixed energy, or about the separate leading-coefficient multiplicity conjecture, is implied.

## VII.4 Minimum recurrence width for every exponent and complex weight

For an integer $r \geq 2$ and $\lambda \in \mathbb{C}^*$, put

$$a_{r,\lambda}(n) = \sum_{k=0}^{n} \binom{n}{k}^r \lambda^k, \qquad B_{r,\lambda}(n) = \lambda^{-n} a_{r,\lambda}(2n).$$

The width of an eventual nonzero recurrence over $\mathbb{C}(n)$ is the difference between its largest and smallest integer shifts with nonzero coefficients. Arbitrary gaps between shifts are permitted. Geometric rescaling of a sequence preserves its minimum width.

**Theorem VII.4.1** (All exponents and all complex weights)**.** *For every integer $r \geq 2$ and every $\lambda \in \mathbb{C}^*$,*

$$\operatorname{minwidth}_{\mathbb{C}(n)} B_{r,\lambda} = \begin{cases} \lceil r/2 \rceil, & \lambda = 1, \\ \lfloor r/2 \rfloor, & \lambda = -1, \\ r, & \lambda \notin \{1, -1\}. \end{cases}$$

*In particular no further exceptional algebraic or transcendental weights occur, and collisions between critical values do not lower the width away from the two symmetry weights.*

The standard upper recurrences at the symmetry weights come from Stoll [28] and McIntosh [29]; weighted recurrences in low degrees were studied by Matala-Aho [26]. The hyper-Bessel product and binomial-period correspondence is also studied by Gaiur, Rubtsov and van Straten [60]. The new step is a uniform lower-bound mechanism using the differential Galois group of a Kloosterman connection. It works even when several formal exponential factors coincide. This is a width theorem, not a theorem about the minimum number of nonzero terms of an arbitrarily sparse recurrence.

### VII.4.1 Hyper-Bessel connections and independence of their scales

Define the entire function

$$\mathcal{F}_r(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^r}.$$

It solves $((x\partial_x)^r - x)y = 0$. Work over $K = \mathbb{C}(t)$, and write $M_a$ for the rank-$r$ differential module whose scalar solutions include

$$f_a(t) = \mathcal{F}_r(at^r), \qquad a \in \mathbb{C}^*.$$

Equivalently its scalar equation is

$$\left\{ \left( \frac{t\partial_t}{r} \right)^r - at^r \right\} y = 0.$$

We use the following precise classical input. The differential Galois group of $M_a$ in its standard representation is

$$G_r = \begin{cases} \mathrm{SL}_r(\mathbb{C}), & r \text{ odd}, \\ \mathrm{Sp}_r(\mathbb{C}), & r \text{ even}. \end{cases} \tag{VII.4.1}$$

This is Katz's Kloosterman-group calculation [59, Corollary 4.4.8]; the connection and its $r$-fold pullback are identified explicitly in [51, Proposition 2.4 and the proof of Proposition 2.7]. Scaling $t$ by a nonzero constant gives the assertion for every $a$. In particular these groups are connected and their standard representations are irreducible.

The formal decomposition at $t = \infty$ consists of $r$ rank-one factors with exponential parts

$$\exp(r\alpha\zeta t), \qquad \zeta^r = 1, \quad \alpha^r = a. \tag{VII.4.2}$$

Their remaining parts are regular singular. Direct substitution gives formal solutions

$$e^{r\alpha\zeta t} t^{-(r-1)/2}(1 + O(t^{-1}));$$

the distinct leading exponential roots recursively determine the formal series. Only the exponential parts, with their multiplicities, will be used below. The duality relation is

$$M_a^* \simeq M_{(-1)^r a}. \tag{VII.4.3}$$

For example, this follows by reversing the cyclic basis for the companion system and alternating its signs, or by formal adjunction with respect to $dx/x$ before the pullback.

**Lemma VII.4.2** (Independence of distinct scale classes)**.** *Choose finitely many nonzero scales $a_1, \dots, a_s$. If $r$ is even, assume they are pairwise distinct. If $r$ is odd, assume $a_i \neq \pm a_j$ for $i \neq j$. The connected differential Galois group of $\bigoplus_i M_{a_i}$ is the full product $G_r^s$.*

*Proof.* The connected joint group projects onto each individual group: its image has finite index in the connected group $G_r$. Its Lie algebra is therefore a subdirect product of simple Lie algebras. By the Lie-algebra form of Goursat's lemma, a proper subdirect product identifies two of these simple factors by an automorphism. Automorphisms of $\mathfrak{sp}_r$ are inner; automorphisms of $\mathfrak{sl}_r$ are inner or the standard duality automorphism. Consequently the two standard representations on the connected joint group would be isomorphic, or in the odd case dual.

Passing to the finite algebraic extension corresponding to the identity component makes this an isomorphism of the associated differential modules. But an isomorphism preserves the formal exponential parts. After any common finite ramification at infinity, (VII.4.2) shows that $M_a$ and $M_b$ can be isomorphic only if $a = b$, and that $M_a$ and $M_b^*$ can be isomorphic only if $a = (-1)^r b$. Indeed equality of the multisets of coefficients of the exponential terms, followed by taking their $r$th powers, gives these equalities. Both possibilities contradict the scale hypotheses. Thus no pair of simple factors is identified and the Lie algebra is the full product. A connected closed subgroup with this Lie algebra is the full product group. □

### VII.4.2 A recurrence budget at irregular infinity

For a nonzero differential operator $P = \sum_i b_i(t)\partial_t^i$ with formal Laurent coefficients at infinity, let $v$ be the largest power of $t$ occurring as a leading power of one of the $b_i$. Define its polynomial principal symbol at infinity by

$$\sigma_\infty(P)(\xi) = \sum_i \left( \lim_{t\to\infty} t^{-v} b_i(t) \right) \xi^i.$$

The limit denotes the constant Laurent coefficient after normalization. The resulting polynomial is nonzero. Since differentiating a Laurent coefficient lowers its leading power of $t$, one has

$$\sigma_\infty(PQ) = \sigma_\infty(P)\sigma_\infty(Q). \tag{VII.4.4}$$

This statement holds up to the harmless nonzero scalar resulting from a different normalization of the leading power.

**Lemma VII.4.3** (Exponential rank is a recurrence-width lower bound)**.** *Suppose*

$$h(t) = \sum_{n\geq 0} \frac{u_n}{((2n)!)^r} t^{2rn}$$

*is a solution of a differential module over $\mathbb{C}(t)$. Suppose the differential module generated by this particular scalar solution has, formally at infinity, a decomposition into rank-one factors with exponential parts $e^{\beta t}$ and regular-singular remaining parts. Let $q$ be the total number of factors with $\beta \neq 0$, counted with multiplicity. Then any eventual recurrence for $u_n$ has width at least $q/(2r)$.*

*Proof.* After translating shifts and clearing denominators, write a proposed recurrence as

$$\sum_{j=0}^{L} p_j(n) u_{n-j} = 0, \qquad p_j \in \mathbb{C}[n].$$

Let $D = \max_j \deg p_j$ and $c_j = \mathrm{lc}(p_j)$ when $\deg p_j = D$. Multiplying the coefficient identity by $((2n)_{\underline{2L}})^r/((2n)!)^r$ shows that $h$ satisfies an inhomogeneous differential equation $Ph = Q(t)$ with polynomial right side and

$$P = \sum_{j=0}^{L} t^{2rj} p_j\left(\frac{t\partial_t}{2r} + j\right) \prod_{\ell=0}^{2(L-j)-1} \left(\frac{t\partial_t}{r} - \ell\right)^r.$$

Empty products are one. The polynomial right side accounts only for finitely many initial coefficients. The largest coefficient power of $t$ in this operator is $D + 2rL$, and its principal symbol is

$$\sigma_\infty(P)(\xi) = \sum_{\deg p_j = D} c_j \left(\frac{\xi}{2r}\right)^D \left(\frac{\xi}{r}\right)^{2r(L-j)}. \tag{VII.4.5}$$

The powers contributed by distinct $j$ differ, so this polynomial is nonzero. Its number of nonzero roots, counted with multiplicity, is at most $2rL$. Left multiplication by a polynomial in $t\partial_t$ killing $Q(t)$ makes the equation homogeneous and only multiplies (VII.4.5) by a constant times a power of $\xi$. Thus the same nonzero-root budget holds for a homogeneous annihilator $P_0$.

Let $R$ be the monic minimal scalar annihilator of $h$. The specified formal decomposition implies a formal first-order factorization

$$R = (\partial_t - \beta_d + O(t^{-1})) \cdots (\partial_t - \beta_1 + O(t^{-1})).$$

One way to see this is to take a rank-one formal submodule, use its solution with logarithmic derivative $\beta_i + O(t^{-1})$ to obtain a right first-order factor, and continue on the quotient. Formal Laurent gauge changes affect the logarithmic derivative by $O(t^{-1})$, so the listed exponential coefficients are preserved. Hence $\sigma_\infty(R) = \prod_i (\xi - \beta_i)$. Minimality gives a right division $P_0 = AR$ over $\mathbb{C}(t)$. Equation (VII.4.4), also valid after formal completion, shows that $\sigma_\infty(R)$ divides $\sigma_\infty(P_0)$. Its $q$ nonzero roots counted with multiplicity therefore satisfy $q \leq 2rL$. □

### VII.4.3 The cyclic period and all parameter strata

The elementary Cauchy-product identity is

$$\mathcal{F}_r(x)\mathcal{F}_r(\lambda x) = \sum_{n\geq 0} \frac{a_{r,\lambda}(n)}{(n!)^r} x^n. \tag{VII.4.6}$$

Consequently the function to which Lemma VII.4.3 applies is

$$h_\lambda(t) = \tfrac{1}{2}\{f_1(t)f_\lambda(t) + f_{-1}(t)f_{-\lambda}(t)\} = \sum_{n\geq 0} \frac{a_{r,\lambda}(2n)}{((2n)!)^r} t^{2rn}. \tag{VII.4.7}$$

Products of solution spaces are interpreted inside a common Picard–Vessiot field. Multiplication is an equivariant linear map from the tensor product of the two representations. If that tensor

product is irreducible, the map is injective because it is nonzero. Also, a vector with a nonzero component in each of two nonisomorphic irreducible summands generates their entire direct sum. These facts identify the module generated by the specific scalar function (VII.4.7), rather than merely an ambient annihilating system.

*Proof of Theorem VII.4.1: lower bounds.* First assume $\lambda \notin \{1, -1\}$. For even $r$, the four scales $1, \lambda, -1, -\lambda$ are distinct; Lemma VII.4.2 gives four independent symplectic factors. The two products in (VII.4.7) belong to nonisomorphic irreducible tensor representations, each of dimension $r^2$. For odd $r$, the two scales $1, \lambda$ give independent special linear factors, while the negative scales give their duals by (VII.4.3). The two products then belong to $V \otimes W$ and $V^* \otimes W^*$. These are nonisomorphic irreducibles since $r \geq 3$ and the standard special-linear representation is not self-dual. In both parities the particular sum generates the full direct sum of dimension $2r^2$.

Every formal exponential part of a product is the sum of one exponential coefficient from each of its two factors. A zero sum for $M_a \otimes M_b$ can occur only when $b = (-1)^r a$. This is excluded in the present case, for both products. Hence all $2r^2$ formal factors are nonzero. Lemma VII.4.3 proves $L \geq r$. Coincidences between nonzero sums are harmless, since the budget retains their multiplicities.

For $\lambda = 1$ and even $r$, the two products are squares for two independent symplectic groups. Their representations are $\operatorname{Sym}^2 V_1$ and $\operatorname{Sym}^2 V_{-1}$, both irreducible and nonisomorphic. In each, the formal exponential coefficients are the unordered sums of two $r$th-root coefficients. Exactly $r/2$ of these sums vanish (the opposite pairs). Each summand thus has $r(r+1)/2 - r/2 = r^2/2$ nonzero formal factors. The total is $r^2$, giving $L \geq r/2$. For odd $r$, the two summands are $\operatorname{Sym}^2 V$ and $\operatorname{Sym}^2 V^*$ for $\mathrm{SL}_r$; they are again nonisomorphic irreducibles. No opposite pair of $r$th roots exists, so their total nonzero formal rank is $r(r+1)$, giving $L \geq (r+1)/2$.

For $\lambda = -1$, the two products in (VII.4.7) coincide. If $r$ is even, $M_1$ and $M_{-1}$ are independent symplectic modules. Their tensor product is irreducible of rank $r^2$ and has no zero exponential part. The resulting lower bound is $L \geq r/2$. If $r$ is odd, the tensor representation is $V \otimes V^* = \mathbf{1} \oplus \mathrm{Ad}$ for $\mathrm{SL}_r$. The multiplication map kills the trivial summand. To check this explicitly, put $\delta = (t/r)\partial_t$. For solutions $y$ of $M_1$ and $z$ of $M_{-1}$, the nondegenerate concomitant

$$\mathcal{B}(y, z) = \sum_{j=0}^{r-1} (-1)^j (\delta^j y)(\delta^{r-1-j} z)$$

is constant, since differentiating telescopes and $r$ is odd. Choose bases $y_i, z_i$ dual for this pairing, and let $Y, Z$ be their jet matrices, with rows indexed by $\delta^j$, $0 \leq j < r$. Then $Y^{\mathsf{T}} J Z = I$, where $J$ is the invertible anti-diagonal matrix with anti-diagonal entries $(-1)^j$. The invariant tensor $\sum_i y_i \otimes z_i$ is consequently sent by multiplication to

$$\sum_i y_i z_i = (Y Z^{\mathsf{T}})_{00} = (J^{-\mathsf{T}})_{00} = 0.$$

The product $f_1 f_{-1}$ is nonzero, so multiplication is nonzero on the irreducible adjoint summand and hence injective there. Its cyclic module is therefore the entire adjoint representation. Its nonzero formal factors are the $r(r-1)$ ordered differences of distinct $r$th-root exponential coefficients. Lemma VII.4.3 gives $L \geq (r-1)/2$. This proves all lower bounds in the theorem. □

*Proof of Theorem VII.4.1: upper bounds.* For $\lambda \notin \{1, -1\}$ put

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k.$$

An index shift gives, for every $h \geq 0$,

$$L_N[k^{r+h} - \lambda (N-k)^r (k+1)^h] = 0.$$

The displayed polynomial has leading coefficient $1-(-1)^r\lambda \neq 0$. Successive elimination expresses every polynomial moment in terms of $L_N[1], \dots, L_N[k^{r-1}]$. Using $\binom{N-1}{k}^r = N^{-r}(N-k)^r\binom{N}{k}^r$ gives a rational $r$-dimensional backward state relation $M_{N-1} = T(N)M_N$. Its two-step relation expresses $r+1$ consecutive even observations as $r+1$ rows on an $r$-dimensional state space. Their rational linear dependence proves width at most $r$; no matrix inverse is needed.

For $\lambda = 1$, the ordinary Franel recurrence of width at most $\lceil r/2 \rceil$ is the known upper bound [28]. Taking every second state of its rational companion system preserves this dimension and gives the same upper bound for the even subsequence. For $\lambda = -1$, McIntosh's construction [29, Theorem 1] directly gives width at most $\lfloor r/2 \rfloor$ for the centered alternating sequence. These upper bounds and the proved lower bounds complete the classification. □

*Remark* VII.4.4 (Why no collision parameters are omitted). Counting only distinct finite critical values loses information when two such values merge. After factorial normalization, their multiplicities become multiplicities of formal exponential factors of a tensor connection. The connected Galois group makes the chosen period cyclic in the relevant representation, and the principal symbol of every annihilator retains each of those factors. This replaces both a generic-parameter specialization argument and any unproved assertion that individual finite singularities remain visible at exceptional weights.

# VII.5 A global algebraic-quantile criterion for complementary beta weights

The arithmetic of a normalized integral can change after truncation, and an algebraic primitive alone does not classify all normalized inverses. For complementary rational beta exponents other than the arcsine exponent, we obtain an exact criterion that is uniform in the truncation interval and in every polynomial weight. One explicit linear functional of the coefficients determines whether all algebraic probability levels have algebraic quantiles or all have transcendental quantiles. There is no intermediate exceptional set.

## VII.5.1 The complete criterion

Fix

$$r \in \mathbb{Q} \cap (0,1), \qquad r \neq \tfrac{1}{2}, \qquad b_r(x) = x^{r-1}(1-x)^{-r}.$$

Let $u, v$ be real algebraic numbers with $0 \le u < v \le 1$. Let $w(X) = \sum_{j=0}^{d} w_j X^j$ have real algebraic coefficients and satisfy $w(x) > 0$ for $u < x < v$. No positivity outside this interval is required. Set

$$F_{r,w;u,v}(x) = \frac{\displaystyle\int_u^x w(t) b_r(t)\,dt}{\displaystyle\int_u^v w(t) b_r(t)\,dt}, \qquad u \le x \le v. \tag{VII.5.1}$$

These integrals converge even when $u = 0$ or $v = 1$. The CDF is continuous and strictly increasing. With $(r)_0 = 1$ and $(r)_j = r(r+1)\cdots(r+j-1)$, define the algebraic number

$$c_r(w) = \sum_{j=0}^{d} w_j \frac{(r)_j}{j!}. \tag{VII.5.2}$$

**Theorem VII.5.1** (Global quantile dichotomy)**.** *Under the preceding hypotheses, the following five statements are equivalent:*

1. $c_r(w) = 0$.
2. $F_{r,w;u,v}(x)$ *is algebraic for some algebraic* $x \in (u,v)$.
3. $F_{r,w;u,v}(x)$ *is algebraic for every algebraic* $x \in (u,v)$.
4. *The quantile* $F^{-1}_{r,w;u,v}(\lambda)$ *is algebraic for some algebraic* $\lambda \in (0,1)$.
5. *The quantile* $F^{-1}_{r,w;u,v}(\lambda)$ *is algebraic for every algebraic* $\lambda \in (0,1)$.

*Thus, if* $c_r(w) \neq 0$*, all internal algebraic-cut cumulative probabilities and all internal algebraic-level quantiles are transcendental. If* $r = p/q$ *in lowest terms and* $c_r(w) = 0$*, the quantile at each algebraic level satisfies an explicit nonconstant polynomial equation of degree* $q \deg w$ *with algebraic coefficients.*

The polynomial reduction used below is the same elementary reduction available for general polynomial beta weights. The substantive arithmetic step is the exclusion when $c_r(w) \neq 0$, for every algebraic truncation and every algebraic level. It follows from a coefficient-conjugation consequence of Baker's theorem and strict ordering of two beta densities.

### VII.5.2 Fixed logarithms and coefficient conjugation

We use two classical consequences of Baker's theorems [61, 46]; the linear-independence statement is also recorded explicitly as Theorem 1.2 of [62]. For any fixed logarithms of nonzero algebraic numbers:

1. a nonzero linear form with algebraic coefficients in those logarithms is transcendental;
2. logarithms linearly independent over $\mathbb{Q}$ remain linearly independent over $\overline{\mathbb{Q}}$.

These are different inputs: the first separates an algebraic correction from a logarithmic expression; the second controls the zero relations.

**Lemma VII.5.2** (Conjugating coefficients in a zero relation)**.** *Let $\ell_1, \dots, \ell_N$ be fixed logarithms of nonzero algebraic numbers, and let $a_1, \dots, a_N \in \overline{\mathbb{Q}}$. Then*

$$\sum_{j=1}^{N} a_j \ell_j = 0 \quad \Longrightarrow \quad \sum_{j=1}^{N} \overline{a_j} \ell_j = 0. \tag{VII.5.3}$$

*The logarithms, their arguments, and their branch choices are held fixed. Only the coefficients are conjugated.*

*Proof.* Choose a $\mathbb{Q}$-basis $\eta_1, \dots, \eta_s$ of the $\mathbb{Q}$-span of the $\ell_j$, taking the basis from the given list. Write $\ell_j = \sum_k q_{jk} \eta_k$ with $q_{jk} \in \mathbb{Q}$. Baker's linear-independence theorem makes the $\eta_k$ independent over $\overline{\mathbb{Q}}$. The initial relation therefore implies $\sum_j a_j q_{jk} = 0$ for every $k$. Conjugating these algebraic equalities, and using that the $q_{jk}$ are rational, proves (VII.5.3). □

For $0 \le y \le 1$ put

$$U_r(y) = \int_0^y b_r(t)\, dt = B_y(r, 1-r).$$

Euler's reflection identity gives

$$U_r(1) = B(r, 1-r) = \frac{\pi}{\sin(\pi r)} = U_{1-r}(1). \tag{VII.5.4}$$

**Lemma VII.5.3** (Transfer between complementary beta integrals)**.** *Let $y_1, \dots, y_N$ be algebraic numbers in $[0,1]$, and let $a_1, \dots, a_N$ be real algebraic numbers. If*

$$\sum_{j=1}^{N} a_j U_r(y_j) = 0, \tag{VII.5.5}$$

*then*

$$\sum_{j=1}^{N} a_j U_{1-r}(y_j) = 0.$$

*Moreover, if the left side of (VII.5.5) is algebraic, it is zero, so the same conclusion holds.*

*Proof.* Write $r = p/q$ in lowest terms, with $1 \le p < q$. For $0 < y < 1$ set

$$s = \left( \frac{y}{1-y} \right)^{1/q} > 0.$$

The substitution $t = z^q/(1+z^q)$ gives

$$U_r(y) = q \int_0^s \frac{z^{p-1}}{1+z^q}\, dz. \tag{VII.5.6}$$

For every $\zeta$ with $\zeta^q = -1$, let $\ell_\zeta(s) = \log(1 - s/\zeta)$ be the continuous branch along real $s \geq 0$ with value zero at $s = 0$. The path has no zero: no such $\zeta$ is positive real. If $\zeta = -1$, the argument is positive real; otherwise it stays strictly in one open half-plane for $s > 0$. Thus these branches are unambiguous. When $y$ is algebraic, $s$ and every $1 - s/\zeta$ are nonzero algebraic numbers.

The residue of $qz^{p-1}/(1 + z^q)$ at $z = \zeta$ is $\zeta^{p-q} = -\zeta^p$. Partial fractions and the common value zero at $s = 0$ give

$$U_r(y) = -\sum_{\zeta^q=-1} \zeta^p \ell_\zeta(s), \qquad U_{1-r}(y) = -\sum_{\zeta^q=-1} \zeta^{q-p} \ell_\zeta(s). \tag{VII.5.7}$$

Conjugating only the coefficients in the first expression produces

$$-\sum_{\zeta^q=-1} \zeta^{-p} \ell_\zeta(s) = \sum_{\zeta^q=-1} \zeta^{q-p} \ell_\zeta(s) = -U_{1-r}(y),$$

because $\zeta^{-p} = -\zeta^{q-p}$.

The endpoint $y = 0$ contributes zero. At $y = 1$, use the finite logarithmic representation

$$U_r(1) = -\frac{i}{\sin(\pi r)} \log(-1), \qquad \log(-1) = i\pi. \tag{VII.5.8}$$

Here $\sin(\pi r)$ is a nonzero real algebraic number. Conjugating its coefficient yields $-U_{1-r}(1)$, by (VII.5.4). This handles both endpoints without taking limits of the individual logarithms in (VII.5.7).

Represent all terms in (VII.5.5) by these finite logarithmic expressions and apply Lemma VII.5.2. The real coefficients $a_j$ are fixed by conjugation, and every $U_r(y_j)$ changes to $-U_{1-r}(y_j)$. This proves the first assertion. For the last assertion the expression is a linear form with algebraic coefficients in fixed logarithms of algebraic numbers. Baker's nonzero-form transcendence theorem forces it to be zero if it is algebraic. □

### VII.5.3 Strict ordering survives every truncation

**Lemma VII.5.4** (No common internal truncated probability)**.** *For any real $0 \leq u < x < v \leq 1$ and $r \neq 1/2$,*

$$\frac{U_r(x) - U_r(u)}{U_r(v) - U_r(u)} \neq \frac{U_{1-r}(x) - U_{1-r}(u)}{U_{1-r}(v) - U_{1-r}(u)}. \tag{VII.5.9}$$

*For $r > 1/2$ the left side is strictly smaller; for $r < 1/2$ it is strictly larger.*

*Proof.* The ratio of the two positive densities is

$$\frac{b_r(t)}{b_{1-r}(t)} = \left(\frac{t}{1-t}\right)^{2r-1},$$

strictly increasing when $r > 1/2$ and strictly decreasing when $r < 1/2$. For the increasing case, set $A_r = \int_u^x b_r$ and $D_r = \int_x^v b_r$, with analogous notation for $1 - r$. Then

$$A_r D_{1-r} - A_{1-r} D_r = \int_u^x \int_x^v \{b_r(t) b_{1-r}(z) - b_{1-r}(t) b_r(z)\}\, dz\, dt < 0.$$

The integrand is strictly negative for interior $t < z$, and all integrals are finite and positive. This proves $A_r/(A_r + D_r) < A_{1-r}/(A_{1-r} + D_{1-r})$. The decreasing case reverses the inequalities. □

### VII.5.4 Polynomial reduction and proof of the criterion

**Lemma VII.5.5** (The single logarithmic coefficient)**.** *There is a unique polynomial* $R \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$ *such that*

$$w(X) = c_r(w) + X(1-X)R'(X) + (r-X)R(X). \tag{VII.5.10}$$

*With*

$$H(x) = x^r(1-x)^{1-r}R(x), \qquad H(0) = H(1) = 0, \tag{VII.5.11}$$

*one has*

$$\int_0^x w(t)b_r(t)\,dt = c_r(w)U_r(x) + H(x). \tag{VII.5.12}$$

*If* $w$ *is nonconstant of degree* $d$*, then* $\deg R = d-1$*. If* $w$ *is constant,* $R = 0$*.*

*Proof.* For $\mathcal{L}_r R = X(1-X)R' + (r-X)R$,

$$\mathcal{L}_r(X^j) = (j+r)X^j - (j+1)X^{j+1}.$$

Descending elimination gives a unique $R$ and constant $c$ with $w = c + \mathcal{L}_r R$, together with the degree assertion. Direct differentiation gives $H' = (\mathcal{L}_r R)b_r$. Since $0 < r < 1$, both endpoint values of $H$ are zero. Integrating over $[0,1]$ gives

$$c = \frac{\int_0^1 w(t)b_r(t)\,dt}{B(r,1-r)} = \sum_j w_j \frac{B(r+j,1-r)}{B(r,1-r)} = \sum_j w_j \frac{(r)_j}{j!} = c_r(w).$$

Integrating to $x$ proves (VII.5.12). The whole-interval integral in this calculation need not be positive when $w$ is positive only on $(u,v)$; accordingly $c_r(w)$ may vanish or have either sign. □

*Proof of Theorem VII.5.1.* Write $c = c_r(w)$ and use $R, H$ from Lemma VII.5.5. Suppose first that $c \neq 0$ and that an algebraic $x \in (u,v)$ has algebraic cumulative probability $\lambda = F_{r,w;u,v}(x)$. The integral equation becomes

$$c\{U_r(x) - U_r(u) - \lambda[U_r(v) - U_r(u)]\} = -H(x) + H(u) + \lambda[H(v) - H(u)]. \tag{VII.5.13}$$

All terms on the right are algebraic: this includes endpoints zero and one by (VII.5.11). By the last assertion of Lemma VII.5.3, the logarithmic expression in braces must vanish. Its coefficients $1, -\lambda, \lambda - 1$ are real algebraic. The same lemma therefore gives

$$U_r(x) - U_r(u) = \lambda[U_r(v) - U_r(u)],$$
$$U_{1-r}(x) - U_{1-r}(u) = \lambda[U_{1-r}(v) - U_{1-r}(u)].$$

This contradicts Lemma VII.5.4. Thus no algebraic $x$ has an algebraic CDF value when $c \neq 0$; equivalently no algebraic level has an algebraic quantile.

If $c = 0$, (VII.5.12) gives

$$F_{r,w;u,v}(x) = \frac{H(x) - H(u)}{H(v) - H(u)}.$$

The denominator is positive because $H' = wb_r > 0$ on $(u,v)$. All these values are algebraic at algebraic $x$. For an algebraic level $\lambda \in (0,1)$, set

$$\eta = H(u) + \lambda[H(v) - H(u)] \in \overline{\mathbb{Q}}.$$

The unique quantile solves $H(x) = \eta$. Writing $r = p/q$ in lowest terms, it is therefore a root of

$$X^p(1-X)^{q-p}R(X)^q - \eta^q = 0. \tag{VII.5.14}$$

Here $w$ cannot be a nonzero constant, since its $c$ would be that constant. Thus $d = \deg w \geq 1$, $R \neq 0$, and the polynomial in (VII.5.14) has degree $q + q\deg R = qd$. In particular it is nonconstant, and its coefficients are algebraic. The quantile is algebraic. Raising to the $q$th power may introduce extra roots, but the original equation $H(x) = \eta$ is strictly monotone on $(u,v)$ and specifies the required root uniquely. All five equivalences now follow. □

### VII.5.5 Consequences and sharpness

**Corollary VII.5.6** (Exact linear-independence formulation)**.** *Under the hypotheses of Theorem VII.5.1, fix any algebraic $x \in (u, v)$ and put*

$$J_1 = \int_u^x w(t) b_r(t)\, dt, \qquad J_2 = \int_u^v w(t) b_r(t)\, dt.$$

*Then*

$$\dim_{\overline{\mathbb{Q}}} \operatorname{span}_{\overline{\mathbb{Q}}}\{1, J_1, J_2\} = \begin{cases} 1, & c_r(w) = 0, \\ 3, & c_r(w) \neq 0. \end{cases}$$

*In particular, for the unweighted complementary beta density, $1$, $B_x(r, 1-r) - B_u(r, 1-r)$, and $B_v(r, 1-r) - B_u(r, 1-r)$ are linearly independent over $\overline{\mathbb{Q}}$ for every algebraic subdivision $0 \le u < x < v \le 1$.*

*Proof.* Let $L_1 = U_r(x) - U_r(u)$ and $L_2 = U_r(v) - U_r(u)$. Any relation $a_0 + a_1 L_1 + a_2 L_2 = 0$ with algebraic coefficients forces $a_0 = 0$ by Baker's nonzero-form theorem. If $a_1 \neq 0$, it gives the algebraic ratio $L_1/L_2 = -a_2/a_1$, contradicting the proof of Theorem VII.5.1 for $w = 1$. If $a_1 = 0$, then $a_2 = 0$ because $L_2 > 0$. Thus $1, L_1, L_2$ are independent. When $c = c_r(w) \neq 0$, the transformation

$$(1, L_1, L_2) \longmapsto (1, cL_1 + H(x) - H(u), cL_2 + H(v) - H(u))$$

is invertible over $\overline{\mathbb{Q}}$, with determinant $c^2$. This gives dimension three. If $c = 0$, both integrals are algebraic, giving dimension one. □

**Corollary VII.5.7** (All positive full-interval polynomial weights)**.** *Let $r \in \mathbb{Q} \cap (0, 1) \setminus \{1/2\}$ and let $w$ be a real-algebraic polynomial positive on $(0, 1)$. Every algebraic-level internal quantile of its normalized beta weighting is transcendental. The same assertion holds after conditioning on any nondegenerate interval with algebraic endpoints in $[0, 1]$.*

*Proof.* The complete integral in Lemma VII.5.5 is positive, so $c_r(w) > 0$. Apply Theorem VII.5.1 to every interval. □

**Corollary VII.5.8** (Every non-half-integral rational integer-sum beta pair)**.** *Let $a, b > 0$ be rational with $a + b \in \mathbb{Z}$ and $a \notin \frac{1}{2}\mathbb{Z}$. Then every algebraic internal probability level of $\mathrm{Beta}(a, b)$ has a transcendental quantile, including after conditioning on any nondegenerate algebraic interval in $[0, 1]$. The conclusion is unchanged by multiplying the density by any real-algebraic polynomial positive on $(0, 1)$.*

*Proof.* Write $a = m + r$, $b = n + 1 - r$, where $m, n \ge 0$ are integers and $r \in \mathbb{Q} \cap (0, 1) \setminus \{1/2\}$. The beta density is $x^m (1 - x)^n b_r(x)$ up to normalization. The polynomial factor is positive throughout $(0, 1)$, so Corollary VII.5.7 applies, also after multiplying by the additional positive polynomial. □

For example, the conclusion holds for every $\mathrm{Beta}(m + 1/3, n + 2/3)$ with integers $m, n \ge 0$, for every algebraic truncation and every algebraic probability level. The prior's two fractions need not have small denominators: the same statement holds for $m + 17/101, n + 84/101$ and, uniformly, for every complementary rational pair other than $1/2, 1/2$. Together with the previously proved closed half-integer classification and the polynomial primitive for integral beta shapes, Corollary VII.5.8 supplies the remaining denominator classes in the full-interval positive rational integer-sum family.

**Example VII.5.9** (Both sides of the criterion occur on the same interval)**.** Take $r = 1/3$ and $[u, v] = [1/3, 1]$. For $w(X) = X - 1/3$, one has $c_r(w) = 0$, $R = -1$, and

$$H(x) = -x^{1/3}(1 - x)^{2/3}.$$

Every algebraic-level quantile is algebraic; more precisely the level-$\lambda$ quantile is the unique $x \in (1/3, 1)$ satisfying

$$x(1 - x)^2 = \frac{4}{27}(1 - \lambda)^3.$$

The left side is strictly decreasing on this interval. For the constant weight $w = 1$ on the very same interval, $c_r(w) = 1$, and every algebraic-level quantile is transcendental.

The restriction $r \neq 1/2$ is essential. The unweighted arcsine CDF is $2 \arcsin \sqrt{x}/\pi$, and every rational level has the algebraic quantile $\sin^2(\pi\lambda/2)$. The coefficient-conjugation transfer then exchanges two identical densities, so the strict-order contradiction correctly disappears. The arcsine exceptional-packet and half-integer pair theorems are separate previously established results, not additional claims of this section.

**Scope and relation to the available literature.** The polynomial elimination and the algebraic-primitive direction are supporting tools already used for other beta parameter families. The new argument here establishes the complementary-exponent obstruction for every algebraic interval and every algebraic probability level, and hence the exact criterion (VII.5.2) for all polynomial weights positive on the selected interval. It does not use an unproved independence statement about arbitrary beta periods.

Dhillon and Saha study arithmetic properties of unnormalized incomplete beta values [55]; the author's conference presentation [56] explicitly includes rational nonintegral $a$ with $a + b \in \mathbb{N}$ and logarithmic methods. The available slides do not state this all-interval normalized inverse criterion. A comparison with the full published article has not been completed, so no historical priority claim is made. The presentation's separate question about a Schneider-type theorem is not claimed to be resolved here.

## VII.6 A single noisy observation can destroy expected transport accuracy

Consider independent observations

$$X_i = \Theta_i + \sigma_i Z_i, \qquad \Theta_i \overset{\text{iid}}{\sim} G, \qquad Z_i \overset{\text{iid}}{\sim} N(0,1), \qquad 1 \le i \le N,$$

with independent latent variables and noises and known positive standard deviations $\sigma_1, \dots, \sigma_N$. Write $\varphi_\sigma(x) = \sigma^{-1}\varphi(x/\sigma)$, where $\varphi$ is the standard normal density, and $p_{H,\sigma} = H * \varphi_\sigma$. The ordinary joint NPMLE is any measurable maximizer

$$\widehat{G} \in \operatorname*{argmax}_{H \in \mathcal{P}(\mathbb{R})} \sum_{i=1}^{N} \log p_{H,\sigma_i}(X_i).$$

In particular, its optimization domain is not restricted to a known support interval.

**Theorem VII.6.1** (A universal noise-outlier lower bound)**.** *For every $N \ge 1$, every positive noise profile, every $\mu \in \mathbb{R}$, and every measurable choice of the exact unrestricted joint NPMLE,*

$$\mathbb{E}_{\delta_\mu} W_1(\widehat{G}, \delta_\mu) \ge \frac{\max_{1 \le i \le N} \sigma_i}{2\sqrt{2\pi}\, N^2}. \tag{VII.6.1}$$

*More precisely, for every data set and every $i$,*

$$W_1(\widehat{G}, \delta_\mu) \ge \frac{|X_i - \mu|}{2N} \left(1 - N \exp\left\{-\frac{(X_i - \mu)^2}{2\sigma_i^2}\right\}\right)_+. \tag{VII.6.2}$$

*Consequently, for a triangular array of noise profiles, expected $W_1$ consistency at any point mass requires $\max_i \sigma_{N,i} = o(N^2)$.*

The condition just obtained is necessary, not sufficient. The theorem concerns expected transport loss. It does not assert failure of convergence in probability, because rare observations may account for the lower bound.

Soloff, Guntuboyina and Sen establish heteroscedastic NPMLE theory, including the likelihood contact condition and deconvolution guarantees [63]. Their stated deconvolution theorem assumes uniform positive lower and upper bounds on the covariance matrices [64, Theorem 11]. The result here rules out an unrestricted extension to arbitrary unbounded noise profiles in expected $W_1$ risk. It does not contradict that bounded-covariance theorem.

### VII.6.1 Proof of the universal lower bound

For completeness, every maximizer is supported on the sample interval $[\min_i X_i, \max_i X_i]$: projecting a parameter outside this interval to its nearest endpoint strictly increases every Gaussian likelihood contribution. Weak compactness of probability measures on this interval and continuity of the likelihood also prove existence.

Let $H$ be a maximizer. Differentiating its likelihood along $(1-\varepsilon)H + \varepsilon\delta_\theta$ at $\varepsilon = 0+$ gives the standard contact inequality

$$S_H(\theta) := \frac{1}{N} \sum_{j=1}^{N} \frac{\varphi_{\sigma_j}(X_j - \theta)}{p_{H,\sigma_j}(X_j)} \le 1 \qquad (\theta \in \mathbb{R}).$$

Evaluating at $\theta = X_i$ and retaining the $i$th nonnegative term yields the fitted-density floor

$$p_{H,\sigma_i}(X_i) \ge \frac{\varphi(0)}{N\sigma_i}. \tag{VII.6.3}$$

We need the following elementary Gaussian estimate: for $z > 0$ and $u \in \mathbb{R}$,

$$e^{-(z-u)^2/2} \le e^{-z^2/2} + \frac{2|u|}{z}. \tag{VII.6.4}$$

If $u < 0$, the exponential decreases. If $u \ge z/2$, the term $2|u|/z$ alone is at least one. In the remaining case $0 \le u < z/2$, the mean-value theorem uses an intermediate $t \in [z-u, z] \subset [z/2, z]$, and

$$zte^{-t^2/2} \le 2t^2 e^{-t^2/2} \le 4/e < 2.$$

This proves (VII.6.4); reflection gives the same estimate for $z < 0$ with $|z|$ in the denominator.

Take $z = (X_i - \mu)/\sigma_i$, $u = (\theta - \mu)/\sigma_i$ and integrate (VII.6.4) against $H$. When $X_i \ne \mu$ it follows that

$$\frac{\sigma_i p_{H,\sigma_i}(X_i)}{\varphi(0)} \le \exp\left\{-\frac{(X_i-\mu)^2}{2\sigma_i^2}\right\} + \frac{2W_1(H,\delta_\mu)}{|X_i - \mu|}.$$

Combining this inequality with (VII.6.3) proves (VII.6.2). If $X_i = \mu$, its right-hand side is zero and the same conclusion is immediate.

Under $G = \delta_\mu$, write $X_i - \mu = \sigma_i Z$ and put $b = \sqrt{2\log N}$, with $b = 0$ when $N = 1$. Direct Gaussian integration gives

$$\mathbb{E}\big[|Z|\mathbf{1}_{\{|Z|>b\}}\big] = \frac{2}{\sqrt{2\pi}N},$$
$$\mathbb{E}\big[|Z|e^{-Z^2/2}\mathbf{1}_{\{|Z|>b\}}\big] = \frac{1}{\sqrt{2\pi}N^2}.$$

Taking expectations in (VII.6.2) therefore yields

$$\mathbb{E}_{\delta_\mu} W_1(H,\delta_\mu) \ge \frac{\sigma_i}{2N}\left\{\frac{2}{\sqrt{2\pi}N} - \frac{1}{\sqrt{2\pi}N}\right\} = \frac{\sigma_i}{2\sqrt{2\pi}N^2}.$$

This holds for each deterministic index $i$, so taking their maximum proves (VII.6.1). □

### VII.6.2 Failure of uniform risk optimality despite many precise observations

The lower bound is not caused by an intrinsically impossible statistical experiment. We give a self-contained benchmark which discards the single imprecise observation. In particular, the comparison requires no assumption about the behavior of another NPMLE.

**Lemma VII.6.2** (A sufficient homoscedastic benchmark)**.** *From $n \ge 2$ independent observations with distribution $G * N(0,1)$, where $G \in \mathcal{P}([-A,A])$ and $A > 0$, there is an estimator $\widetilde{G} \in \mathcal{P}([-A,A])$ such that, for every $T > 0$,*

$$\sup_G \mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C}{T} + CA^{3/2}e^{T^2/2}\sqrt{\frac{T}{n}}, \tag{VII.6.5}$$

*where $C$ is absolute. The estimator may use $A$ and $T$.*

*Proof.* For a finite measure $F$ write $\chi_F(t) = \int e^{itx}F(dx)$, and let $\widehat{\chi}_n(t) = n^{-1}\sum_{j=1}^n e^{itX_j}$. Choose a measurable minimizer, over $H \in \mathcal{P}([-A,A])$, of

$$\int_{-T}^{T} |e^{-t^2/2}\chi_H(t) - \widehat{\chi}_n(t)|^2\, dt.$$

Existence follows from compactness and continuity. Comparing to the true $G$ and applying the triangle inequality in $L^2[-T,T]$ gives

$$\|\chi_{\widetilde{G}} - \chi_G\|_{L^2[-T,T]} \le 2e^{T^2/2}\|\widehat{\chi}_n - \chi_{G*N(0,1)}\|_{L^2[-T,T]}.$$

At each $t$ the empirical characteristic function has variance at most $1/n$. Hence the expected norm on the right is at most $\sqrt{2T/n}$.

Fix a symmetric probability density $K$ whose Fourier transform is supported on $[-1,1]$ and whose first absolute moment is finite. For example, normalize $(\sin(x/4)/(x/4))^4$. Put $K_T(x) = TK(Tx)$. Given a 1-Lipschitz test function $f$ on $[-A,A]$, subtract $f(0)$ and extend it linearly to zero on $[-2A,-A]$ and $[A,2A]$, and then by zero outside $[-2A,2A]$. The resulting $g$ is globally 1-Lipschitz, agrees with $f - f(0)$ on the support, and has $\|g\|_2 \le 2A^{3/2}$. Also $\|g * K_T - g\|_\infty \le C/T$. Fourier inversion, Cauchy–Schwarz and Plancherel yield, uniformly in these tests,

$$\begin{aligned}\left|\int f\, d(\widetilde{G} - G)\right| &\le \frac{C}{T} + \left|\int (g * K_T)\, d(\widetilde{G} - G)\right| \\ &\le \frac{C}{T} + CA^{3/2}\|\chi_{\widetilde{G}} - \chi_G\|_{L^2[-T,T]}.\end{aligned}$$

Here $|\chi_{K_T}| \le 1$ and it vanishes outside $[-T,T]$. The Kantorovich–Rubinstein dual formula and the preceding expectation bound prove (VII.6.5). □

**Corollary VII.6.3** (Divergent MLE risk in a consistently estimable experiment)**.** *Let $N \ge 3$,*

$$A_N = \sqrt{\log(eN)}, \qquad \sigma_1 = \cdots = \sigma_{N-1} = 1, \qquad \sigma_N = N^3, \qquad \mathcal{G}_N = \mathcal{P}([-A_N, A_N]).$$

*Then every measurable unrestricted joint NPMLE satisfies*

$$\sup_{G \in \mathcal{G}_N} \mathbb{E}_G W_1(\widehat{G}, G) \ge \frac{N}{2\sqrt{2\pi}}, \tag{VII.6.6}$$

*whereas the minimax expected loss obeys*

$$\inf_{\widetilde{G}} \sup_{G \in \mathcal{G}_N} \mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C}{\sqrt{\log(eN)}}. \tag{VII.6.7}$$

*Thus ordinary unrestricted joint likelihood is not uniformly optimal over all known heteroscedastic noise profiles, even when all but one observation have unit noise and the true mixing support grows only as $\sqrt{\log N}$.*

*Proof.* The law $\delta_0$ belongs to $\mathcal{G}_N$, and Theorem VII.6.1 gives (VII.6.6). For the upper bound discard observation $N$, take $n = N-1$ and $T = \sqrt{\frac{1}{2}\log n}$ in Lemma VII.6.2. Its second term is at most $C\log(eN)n^{-1/4}$, which is bounded by $C/\sqrt{\log(eN)}$ after increasing the absolute constant. This proves (VII.6.7). □

The benchmark uses a known interval, as is permitted in the minimax comparison. The point-mass lower bound itself needs no support information. No conclusion is drawn here about a likelihood estimator constrained to a supplied interval, or about bounded noise profiles. The mechanism is specific: each datum imposes the density floor (VII.6.3), and an unusually large standardized observation at a very large noise scale forces a nonnegligible transport distance from a point mass. A valid all-profile theorem for ordinary unrestricted likelihood must account for this obstruction.

## VII.7 The exact boundary of the conclusions

The global classifications proved here concern the explicitly defined period families, the stated central binomial power sums, and the specified complementary beta densities. They do not settle a general classification of all periods or of every rational-parameter beta quantile. The excluded arcsine exponent $r = 1/2$ has different arithmetic and was treated separately in the preceding manuscript.

For the banana family, the all-parameter Galois group conclusion is distinct from the leading-coefficient multiplicity question in [032]. Appendix VII.E provides a general local lower bound on leading-coefficient zeros. It does not identify the exact multiplicity at every banana singularity or prove the separate conjecture.

For heteroscedastic likelihood, the necessary condition $\sigma_{\max} = o(N^2)$ is not asserted to be sufficient. The theorem concerns expected $W_1$ loss, and no failure of convergence in probability is deduced from the rare-event lower bound. The minimax comparison specifies the support class and constructs its own comparator, avoiding reliance on an inapplicable homoscedastic extension.

The recurrence theorem counts minimum width, not sparse term count. The group theorems concern differential-algebraic relations between functions, not transcendence of all algebraic-input special values. The arithmetic theorem does address individual algebraic cuts and probability levels, but only under its stated rational-exponent and algebraic-coefficient hypotheses.

**A correction to the preceding prospectus.** The preceding draft described a rational-lemniscate extension too broadly as prospective work. Supplied manuscript [024] already contains the bounded connected rational-lemniscate length result, including its critical-boundary treatment. That statement is not counted as a new result or an unresolved extension here.

## VII.A Previously established hypercubic rank and algebraicity theorem

*Reproduced supporting proof from the preceding manuscript. This appendix is not counted as a new result.*

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{VII.A.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{VII.A.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the $n$-dimensional hypercubic lattice when each step in either coordinate direction $i$ has weight $m_i$. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of $\mathrm{SL}_2$, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{VII.A.3}$$

The decomposition in the middle uses the diagonal $\mathrm{SL}_2$; the tensor product itself is irreducible for the product group $\mathrm{SL}_2^q$.

**Theorem VII.A.1** (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle\partial_z\rangle$ annihilating $F_m$ is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{VII.A.4}$$

*The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.*

*More precisely, define*

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

*Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size $d$ for each of the $a_d$ summands with $d \geq 1$ in* (VII.A.3)*.*

**Corollary VII.A.2** (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is 1; with one nonzero weight $m$, it is $(1 - 4m^2 z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (VII.A.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2, 1]$, $[1, 1, 1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem VII.A.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

### VII.A.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection $M$ on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy $T$, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size $d$, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if $M$ has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank}\mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{VII.A.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection $N$ on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero $H^0$ for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

### VII.A.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let $B_\mu$ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{VII.A.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is $\mathrm{SL}_2$; this is the classical non-half-integral Bessel case [42, 43]. At zero, $I_0$ is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ}.$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

**Lemma VII.A.3.** *If the $\mu_j^2$ are pairwise distinct and nonzero, the direct sum of the $B_{\mu_j}$ has differential Galois group $\mathrm{SL}_2^q$. Consequently*

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$$

*is irreducible of rank $Q$, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities $a_d$ from (VII.A.3); its slopes at infinity are at most one.*

*Proof.* The projections of the Galois group, and of its identity component, onto each $\mathrm{SL}_2$ factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the $I_0$ solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by $g$. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator $e$ on $W$. On $\operatorname{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j-2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (VII.A.3) has the parity of $n$ and contains exactly one weight-zero vector if $n$ is even, or exactly one weight-one vector if $n$ is odd. Thus $b$ is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (VII.A.4). In particular $\operatorname{Sym}^n V$ occurs once, so $r \ge n$.

## VII.A.3 An upper bound that eliminates the endpoint ambiguity

For $\operatorname{Re} s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\,dx = \sum_{k\ge 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{VII.A.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist $M$ by $e^{-sx}$; call the resulting connection $N$. On the cyclic vector $\eta$ evaluating to $g$, its $x$-connection is $\nabla_x^M - s$ and its commuting $s$-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice $N_0$ stable under $\theta_x$. Its residue at zero is nilpotent with the same $b$ Jordan blocks as $M$.

The connection $N$ is irreducible of rank $Q \ge 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since $s$ is transcendental

over the mass constants, every formal exponential factor at infinity is nonzero and linear in $x$ after the twist. Thus $\mathrm{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension $b$. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{VII.A.8}$$

Here $\omega = [\eta\, dx]$ belongs to this kernel. To check this without conflating $dx$ and $dx/x$, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator $\theta_x$ acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \geq 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\, dx$ is formally exact at zero. The commuting connection $\nabla_s$ preserves $N_0$, acts on both global and local cohomology, and hence preserves the kernel in (VII.A.8).

It remains to verify that (VII.A.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of $M$, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to $\varepsilon$, circle zero once, and return to infinity on the continued branch. With the factor $e^{-sx}$ this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of $\omega$ along its small circle is $O(\varepsilon(1 + |\log \varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of $K_0$ have at most exponential growth, which is dominated by $e^{-sx}$ in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over $K$ in (VII.A.8), period integration proves

$$\mathrm{ord}_{\mathbb{C}(s)} H_m \leq r. \tag{VII.A.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

### VII.A.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine $x$-line. Lemma VII.A.3 and (VII.A.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank $r$. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes $d$ stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is $M$; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module $\delta_0$. For an entire function $f$ in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\, dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an $A_s$-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{VII.A.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (VII.A.10) is an actual isomorphism.

Since Fourier sends $\delta_0$ to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{FK}$ (at most once) or the trivial affine-line connection.

If the former were absent, $H_m$ would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{FK}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (VII.A.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by $H_m$ is isomorphic to the irreducible generic module of $\mathcal{FK}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

### VII.A.5 Descent and the algebraicity dichotomy

Multiplication by $s$ does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order $r$ over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to $z$ are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem VII.A.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in $W$ supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus $H_m$, and consequently $F_m$, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary VII.A.2.

The argument proves a statement about a function of the variable $z$. It does not assert transcendence of every special value at algebraic $z$, and the rank argument alone does not determine the full differential Galois group. The group classifications use the additional arguments in Sections VII.2 and VII.C.

## VII.B Previously established reflected Poincaré pairing

*Reproduced supporting proof from the preceding manuscript.*

The next construction works in every dimension and for every squared-mass multiplicity pattern. We give it directly rather than infer the sign of a form from a Fourier-transform convention.

**Lemma VII.B.1** (A form for every choice of weights)**.** *For $n$ nonzero complex weights, the middle connection $\mathcal{P}_m$ has a nondegenerate horizontal bilinear form of parity $(-1)^{n+1}$. Here parity $+1$ means symmetric. Thus its Galois group is contained in an orthogonal group for odd $n$ and a symplectic group for even $n$.*

*Proof.* In the Bessel basis $b_\mu, \theta_x b_\mu$, the form $B_\mu(b_\mu, \theta_x b_\mu) = 1$ is alternating, nondegenerate and horizontal. Indeed, $\theta_x^2 b_\mu = 4\mu^2 x^2 b_\mu$ makes its connection matrix infinitesimally symplectic. Symmetric powers and tensor products give $M$ a horizontal nondegenerate form $B$ of parity $\epsilon = (-1)^n$.

Let $\iota : x \mapsto -x$. Its canonical action fixes each Bessel basis vector $b_\mu, \theta_x b_\mu$; the defining matrix is even in $x$. This identifies $\iota^* M$ with $M$ and preserves $B$. Consequently

$$\iota^* N_s = M \otimes e^{sx} \simeq N_s^\vee,$$

compatibly both with the $x$-connection and with the $s$-connections $\partial_s + x$ on the left and the dual of $\partial_s - x$ on the right. Curve Poincaré duality therefore supplies a perfect pairing of $\mathcal{P}_m$ with itself. The use of the middle image makes this pairing nondegenerate: the annihilator of the image of $H_c^1 \to H^1$ is the kernel of the dual map, so the perfect compact-support pairing descends to the two images. This is the same duality-and-involution construction used for exponential connections in [51, §2, especially (2.12) and Corollary 2.15].

For completeness, its sign can be checked by residues. Represent $a, b \in \mathcal{P}_m$ by $N_s$-valued one-forms $\alpha, \beta$, with formal primitives $f_a, f_b$ at 0 and $\infty$. The sum of local residues gives

$$P(a,b) = \sum_{v=0,\infty} \operatorname{Res}_v B(f_a, \iota^* \beta). \tag{P}$$

Local exactness at both points is precisely the middle condition; at infinity the twisted connection has no local cohomology. Formula (P) is independent of the choices by the compact-support duality just described. Pullback by $\iota$ preserves the sum of residues, and $\iota$ is an isometry for $B$. Hence

$$P(b,a) = \sum_v \operatorname{Res}_v B(\iota^* f_b, \alpha) = \epsilon \sum_v \operatorname{Res}_v B(\alpha, \iota^* f_b).$$

The one-form

$$dB(f_a, \iota^* f_b) = B(\alpha, \iota^* f_b) + B(f_a, \iota^* \beta)$$

has zero residues. It follows that $P(b,a) = -\epsilon P(a,b)$. Residues commute with $\partial_s$, and the coefficient pairing is horizontal for the two commuting parameter connections, so $P$ is horizontal. This proves the assertion, including nondegeneracy and its sign. □

## VII.C Previously established even-dimensional symplectic theorem

*Reproduced supporting proof; the new opposite-parity result is Theorem VII.2.1.*

Keep the hypercubic series $F_m$ and its Laplace form $H_m$, now for an arbitrary even number $n \geq 2$ of nonzero complex weights. Let the distinct squared weights have multiplicities $n_1, \dots, n_q$, and set

$$R = \prod_{j=1}^{q} (n_j + 1) - [u^{n/2}] \prod_{j=1}^{q} (1 + u + \cdots + u^{n_j}).$$

The all-weight order theorem identifies $R$ as the rank of the irreducible middle connection (VII.A.8), with no conditions on signed sums.

**Theorem VII.C.1** (All even dimensions and all nonzero complex weights)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois groups of $H_m$ over $\mathbb{C}(s)$ and $F_m$ over $\mathbb{C}(z)$ are $\mathrm{Sp}_R(\mathbb{C})$ in their standard representations. Thus $R$ is even, and for each function $f = H_m$ or $F_m$ the $R$ functions $f, f', \dots, f^{(R-1)}$ are algebraically independent over its coefficient field. These assertions hold even when signed mass sums collide or vanish.*

Orthogonal and symplectic structures for lattice Green differential operators have long been studied; see, for example, [53]. The statement here determines the full group uniformly over every nonzero complex weight tuple in every even dimension, including collided thresholds. No historical-priority claim is inferred from a literature search.

Here $\mathrm{Sp}_R$ denotes the group preserving a nondegenerate alternating form on an $R$-dimensional space. We first isolate the linear-algebra argument that will identify the group.

**Lemma VII.C.2** (One transvection in a connected irreducible group)**.** *Let $(V, \omega)$ be a finite-dimensional complex symplectic space. If a connected algebraic subgroup $G \subseteq \mathrm{Sp}(V)$ acts irreducibly and contains a nonidentity transvection, then $G = \mathrm{Sp}(V)$.*

*Proof.* Write $N_u(v) = \omega(u, v)u$. A symplectic transvection is $1 + aN_u$ for some $u \neq 0$ and $a \neq 0$; this follows directly by writing its rank-one difference as $v \mapsto \ell(v)u$ and using the symplectic identity to conclude $\ell \in \mathbb{C}\omega(u, -)$. Rescaling $u$ absorbs $a$. Its powers are $1 + kN_u$, so their Zariski closure is $\{1 + tN_u : t \in \mathbb{C}\}$. Consequently $N_u \in \mathfrak{g} = \mathrm{Lie}\, G$, as is $N_v$ for every $v \in \mathcal{O} = G \cdot u$. The span of $\mathcal{O}$ is a nonzero invariant space, hence is $V$.

Join two orbit vectors when their symplectic pairing is nonzero. The spans $W_i$ of different connected components of this graph are mutually orthogonal. Their sum is direct: if a finite sum $\sum_i w_i = 0$ with $w_i \in W_i$, then $w_i$ is orthogonal to $W_i$ by this equality and to all other $W_j$ by construction; it is orthogonal to $V$, hence zero. Thus there are finitely many components. The group $G$ permutes their spans. The stabilizer of each span is a closed subgroup of finite index, so connectedness makes every span $G$-invariant. Irreducibility forces a single component. Choose finitely many orbit vectors spanning $V$ and add the finitely many vertices of paths connecting them. This produces a finite connected spanning set $S \subseteq \mathcal{O}$.

For $v, w \in V$ put

$$L_{v,w}(x) = \omega(v, x)w + \omega(w, x)v.$$

The identities

$$[N_v, N_w] = \omega(v, w)L_{v,w}, \qquad [L_{v,w}, N_y] = \omega(v, y)L_{w,y} + \omega(w, y)L_{v,y} \tag{T}$$

follow by direct multiplication. The first gives $L_{v,w} \in \mathfrak{g}$ for edges of the graph, and $L_{v,v} = 2N_v$ is also in $\mathfrak{g}$. Induct on the length of a path from $v$ to $y$. If $\omega(v, y) \neq 0$, the first identity already applies. Otherwise, let $w$ be the penultimate vertex. By induction $L_{v,w} \in \mathfrak{g}$, and the second identity gives $L_{v,y} \in \mathfrak{g}$ because $\omega(w, y) \neq 0$. Hence all $L_{v,w}$ with $v, w \in S$ belong to $\mathfrak{g}$. Since $S$ spans $V$, these span $\mathrm{Sym}^2 V \simeq \mathfrak{sp}(V)$. Thus $\mathfrak{g} = \mathfrak{sp}(V)$, and connectedness proves the lemma. □

### VII.C.1 Formal factors and collided thresholds

We state exactly the local Fourier input. If a holonomic module $\mathcal{K}$ has formal germ at infinity

$$\widehat{\mathcal{K}}_\infty = \bigoplus_{c\in\mathbb{C}} E^{cx}\otimes R_c, \qquad R_c \text{ regular},$$

then stationary phase at the finite transformed point $s = c$ gives

$$\phi_c(\mathcal{F}\mathcal{K}) \simeq \mathcal{F}^{(\infty,0)}(R_c). \tag{LF}$$

Here $\phi_c$ is the formal vanishing-cycle germ, including its regular unipotent part. For a regular germ the local transform on the right is regular of the same rank and preserves its monodromy (up to the harmless inversion convention). In particular a trivial rank-one germ becomes the one-dimensional vanishing cycles of $\delta_0$, with identity monodromy. This statement includes $c = 0$. It follows either by applying the global Fourier transform to the trivial connection or from the inverse regular local-transform conventions. Precise references are Arinkin [77, Corollary 2.3(1), §3.3], García López [40, Proposition 2.1], and Sabbah [41, Remarks 5.2, 5.3(4), 5.7].

For an intermediate extension of positive rank, with nearby monodromy $T$, the canonical map from nearby to vanishing cycles is surjective and the variation map in the opposite direction is injective. Their composites are $T - 1$. Consequently

$$\phi_c \text{ has identity monodromy} \quad\Longrightarrow\quad T \text{ is unipotent and } (T-1)^2 = 0, \qquad \operatorname{rank}(T-1) = \dim \phi_c. \tag{VC}$$

Indeed the vanishing cycles contain every nonidentity generalized eigenspace of $T$ and the image of $T - 1$ on its unipotent part. Thus (VC) does not discard a constant formal factor.

**Lemma VII.C.3.** *For even $n$, the minimal equation of $H_m$ is regular singular everywhere. Each finite local monodromy is unipotent with square-zero difference from the identity. At some finite nonzero point this difference has rank one.*

*Proof.* Let $\mu_j^2$ be the distinct squared weights and put $\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}$. For $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ the Bessel formal solutions at infinity give the direct-sum decomposition

$$\widehat{M}_\infty \simeq \bigoplus_{\lambda\in\Lambda} E^{c_\lambda x}\otimes R_n, \qquad c_\lambda = 2\sum_j \lambda_j\mu_j, \tag{B}$$

where $R_n$ is rank one with a formal solution $x^{-n/2}$. The invertible formal power series in the Bessel expansions are absorbed by rank-one formal gauges. Since $n$ is even, $R_n$ is the trivial formal connection. Distinct indices with equal $c_\lambda$ remain a direct sum of trivial regular factors; collision introduces no extension or logarithm in (B).

The all-weight theorem identifies the equation of $H_m$ with the generic connection of the simple module $\mathcal{F}(j_{!*}M)$. Its positive rank ensures that this is an intermediate extension at every finite point. If $d_c = \#\{\lambda \in \Lambda : c_\lambda = c\}$, (LF) says its formal vanishing cycles at $c$ are regular, of dimension $d_c$, and have identity monodromy. At other points the vanishing cycles are zero. The purely irregular part of a holonomic germ is unchanged by formal vanishing cycles, so regularity of these vanishing cycles excludes an irregular nearby part. By (VC), all these points are regular singular with the asserted unipotence. The infinity statement of the all-weight theorem already gives regularity and unipotence at infinity.

Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every $j$. Define $\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j)$. This is the unique maximizer of the real functional $\operatorname{Re}(\xi c_\lambda)$ on $\Lambda$, and its value is $2\sum_j n_j|\operatorname{Re}(\xi\mu_j)| > 0$. Thus $c_{\lambda^*} \neq 0$ and $d_{c_{\lambda^*}} = 1$. The corresponding local monodromy is a nonidentity transvection. □

*Proof of Theorem VII.C.1 for $H_m$.* The middle pairing in Lemma VII.B.1 is alternating, so the Galois group $G$ lies in $\mathrm{Sp}_R$. The all-weight rank theorem gives irreducibility. By Lemma VII.C.3 the equation is regular singular everywhere, so Schlesinger's density theorem identifies $G$ with the Zariski closure of its monodromy group [43]. The fundamental group of the punctured sphere is generated by the local loops. All their matrices are unipotent, including the one at infinity, so all belong to $G^0$. Thus $G = G^0$. Lemma VII.C.3 supplies one transvection, and Lemma VII.C.2 gives $G = \mathrm{Sp}_R$. □

### VII.C.2 Descent to the ordinary generating function

It remains to exclude a finite similitude multiplier on descending from $s$ to $z = s^{-2}$. This can be checked on the explicit pairing. Let $\omega = [\eta\, dx]$ be the cyclic class evaluating to $H_m$ in (VII.A.8), where $\eta$ is the even tensor product of the Bessel cyclic vectors. The simultaneous involution

$$J : (s, x) \longmapsto (-s, -x)$$

preserves $N_s$ and induces a semilinear involution of $\mathcal{P}_m$ covering $s \mapsto -s$. It is an isometry for (P): applying $J$ to both entries and then changing $x$ to $-x$ leaves the sum of residues unchanged. On the cyclic class it has $J\omega = -\omega$ because $dx$ changes sign. Therefore $\zeta = s\omega$ is fixed. The parameter derivative

$$\nabla_z = -\frac{s^3}{2}\nabla_s$$

commutes with $J$, so all $\nabla_z^i \zeta$ are fixed. They are a basis by the order theorem and the parity descent for $F_m(s^{-2}) = sH_m(s)$. In this basis the Gram matrix of (P) is fixed by $s \mapsto -s$, and hence belongs to $\mathbb{C}(s^2) = \mathbb{C}(z)$. It is nondegenerate, alternating and $z$-horizontal. Thus the full Galois group of $F_m$ lies in $\mathrm{Sp}_R$. Its identity component is unchanged by the quadratic extension to $\mathbb{C}(s)$, where the group just proved is $\mathrm{Sp}_R$. The full group over $\mathbb{C}(z)$ must therefore be $\mathrm{Sp}_R$ as well.

Finally $\mathrm{Sp}_R$ acts transitively on the nonzero vectors of its standard representation. The orbit of the nonzero solution $H_m$ has dimension $R$, and the same holds for $F_m$. Picard–Vessiot Galois correspondence gives transcendence degree $R$ for the field generated by the first $R$ jets. Since it has exactly $R$ generators, they are algebraically independent. This completes the proof of Theorem VII.C.1.

## VII.D Previously established odd-propagator symplectic theorem

*Reproduced supporting proof; the new opposite-parity result is Theorem VII.3.1.*

The same transvection argument has a second application to the banana periods in the supplied manuscript [49]. Here the number of Bessel factors is odd, whereas it is even in the hypercubic application. The two integral transforms are different, and so are their parity laws. This section removes the nonzero, distinct-threshold hypothesis from the odd-factor Galois-group theorem in that manuscript. It does not prove the remaining leading-coefficient multiplicity conjecture.

For nonzero complex masses $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \qquad \varpi_m(t) = \sum_{k\geq 0} \frac{c_k(m)}{t^{k+1}}.$$

The series converges near infinity. Group the masses by equality of their squares, with multiplicities $n_1, \dots, n_q$, and write

$$\bigotimes_{j=1}^{q} \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d\geq 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

for restriction to the diagonal $\mathrm{SL}_2$.

**Precise prior input.** The all-mass order theorem of [49] states that the minimal connection of $\varpi_m$ is irreducible, has rank

$$o(m) = \sum_{d\geq 2} (d-1)a_d,$$

and has unipotent monodromy at infinity with blocks $J_{d-1}$, one for each summand with $d \geq 2$. In particular, the block $J_{n-1}$ occurs once. Its middle-cohomology construction gives an invariant nondegenerate alternating form when $n$ is odd. These statements do not require distinct or nonzero signed thresholds. They are used here as previously proved inputs, rather than counted again as new results. The cited manuscript contains their proofs; the new removal of threshold restrictions is proved below.

**Theorem VII.D.1** (Every odd number of propagators, every nonzero mass tuple)**.** *Let $n \geq 3$ be odd and $m_i \in \mathbb{C}^\times$. The differential Galois group over $\mathbb{C}(t)$ of the minimal equation of $\varpi_m$ is*

$$\mathrm{Sp}(o(m), \mathbb{C}).$$

*This holds when nonzero thresholds coincide and when signed mass sums vanish. Consequently*

$$\varpi_m, \varpi_m', \dots, \varpi_m^{(o(m)-1)}$$

*are algebraically independent over $\mathbb{C}(t)$, and remain so over every finite algebraic extension of that differential field.*

### VII.D.1 Why collisions do not obstruct connectedness

Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel connection is

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}, \qquad B_\mu : \quad (\theta_y^2 - \mu^2 y/4)f = 0, \quad \theta_y = y\partial_y.$$

On the formal $u$-cover at infinity, it splits as a direct sum of rank-one factors indexed by

$$\lambda_j \in \{n_j, n_j - 2, \dots, -n_j\}, \qquad c(\lambda) = \sum_j \lambda_j \mu_j.$$

Each factor has the form

$$e^{c(\lambda)u} u^{-n/2}(1 + O(u^{-1})).$$

The splitting comes from the two formal Bessel solutions in each factor before taking symmetric and tensor powers. Equality of two exponential factors therefore gives a direct sum with multiplicity, not a new nontrivial formal extension.

For $c \neq 0$, the pair indexed by $\lambda, -\lambda$ descends to

$$A_c(n) = [u \mapsto y = u^2]_+(E^{cu} \otimes U_n), \qquad U_n : \ u^{-n/2}.$$

The standard local Fourier calculation sends this pair, after the first Fourier transform, to

$$E^{c^2/(4s)} \otimes R_c, \qquad R_c \text{ regular of rank one.}$$

The power in its Laplace asymptotic is $s^{(n-3)/2}$: under $y = u^2$, the measure contributes $2u\,du$, evaluation at the critical point $u = c/(2s)$ contributes $s^{n/2-1}$, and the Gaussian fluctuation contributes $s^{-1/2}$. Equivalently this is the formal stationary-phase formula [41, 40]. Thus the monodromy of $R_c$ is $(-1)^{n-3}$, independent of $c$. Only its value 1 for odd $n$ is needed here.

The construction of the banana connection first takes the Fourier transform of $j_{!*}M$, then pulls its restriction to $\mathbb{G}_m$ back by $s = 1/v$, and then takes the Fourier transform of the intermediate extension in $v$. The rational gauge $v^{-1}$ in the scalar integral representation has no effect on monodromy. If $\tau$ is the second Fourier variable, the period normalization is $t = 4\tau$; the factor $E^{c^2 v/4}$ is therefore located at $t = c^2$. The factor above produces vanishing cycles at $t = c^2$ with the same monodromy. If several pairs give the same nonzero $c^2$, their vanishing-cycle spaces are the direct sum of these lines. For odd $n$ their monodromy is the identity.

The final Fourier module is simple of positive rank. Hence it is the intermediate extension at each finite point. If $T_a$ denotes its local monodromy at a nonzero threshold $a$, the nearby/vanishing-cycle description gives

$$\phi_a \cong \operatorname{im}(T_a - I), \qquad (T_a - I)^2 = 0.$$

In particular $T_a$ is unipotent, even at a collision. The regular part arising from $c = 0$ is still regular through the two transforms; it can affect the monodromy at zero, but creates no further nonzero singularity. All formal local factors of the final module are regular, so its global equation is regular singular.

The monodromy group is generated by the loops around infinity and the nonzero thresholds: in the fundamental group relation one can omit the loop around zero. Every one of these generators is unipotent. The Zariski closure of the cyclic group generated by a unipotent matrix in characteristic zero is connected. Schlesinger's theorem therefore shows that the differential Galois group $G$ is connected. This argument needs no assertion that the monodromy at zero itself is unipotent.

### VII.D.2 A simple threshold always survives

Choose $\xi \in \mathbb{C}$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every $j$, and maximize $\operatorname{Re}(\xi c(\lambda))$ over the finite index set. There is a unique maximizer, given by

$$\lambda_j^* = n_j \operatorname{sign} \operatorname{Re}(\xi\mu_j).$$

Its value has strictly positive real projection, so $c_* := c(\lambda^*) \neq 0$. The only tuple giving $c_*$ is $\lambda^*$; the only one giving $-c_*$ is $-\lambda^*$. Over $\mathbb{C}$, equality $c(\lambda)^2 = c_*^2$ means $c(\lambda) = c_*$ or $-c_*$.

Consequently the squared threshold $c_*^2$ has exactly one contributing pair, even if other thresholds collide or vanish.

Its vanishing-cycle space is one-dimensional with trivial monodromy. Since the final module is an intermediate extension, the corresponding $T_*$ satisfies

$$\operatorname{rank}(T_* - I) = 1, \qquad (T_* - I)^2 = 0.$$

It is a nontrivial symplectic transvection.

*Proof of Theorem VII.D.1.* The prior input gives irreducibility and an invariant alternating form. The preceding arguments give connectedness and a nontrivial transvection. Lemma VII.C.2 therefore gives $G = \mathrm{Sp}(o(m))$.

Let $E$ be a Picard–Vessiot extension and $v$ the nonzero solution vector corresponding to $\varpi_m$. The field generated by its first $o(m)$ derivatives is the fixed field of its stabilizer. Its transcendence degree is consequently the dimension of the $G$-orbit of $v$. A symplectic group is transitive on nonzero vectors, so this dimension is $o(m)$. There are exactly $o(m)$ generators, and hence they are algebraically independent. Algebraic extension of the base field preserves their transcendence degree; equivalently, the connected Galois group is unchanged by finite algebraic base extension. □

For pairwise distinct squared masses, the rank in this theorem is the established formula

$$o(m) = 2^n - \binom{n+1}{\lfloor (n+1)/2 \rfloor}.$$

For all equal squared masses it is $n-1$. The theorem concerns functional algebraic independence, not arithmetic independence of values at a fixed energy. The separate reflection argument for even $n$ is given in Section VII.3.

# VII.E Leading-coefficient multiplicity bounds monodromy variation

The following local statement applies to arbitrary singularities, including irregular ones. It does not infer convergence of formal solutions at an irregular point.

**Lemma VII.E.1** (A simple leading zero)**.** *Let*

$$ty^{(N)} + \sum_{j=0}^{N-1} b_j(t)y^{(j)} = 0, \qquad b_j \in \mathbb{C}\{t\}, \quad N \geq 1.$$

*Its local monodromy $T$ on the $N$-dimensional solution space satisfies* $\operatorname{rank}(T - I) \leq 1$.

*Proof.* For $N = 1$ the assertion is immediate. Suppose $N \geq 2$ and substitute $y = \sum_{n\geq 0} c_n t^n$. At coefficient $t^k$, the coefficient multiplying the new unknown $c_{k+N-1}$ is

$$\frac{(k+N-1)!}{k!}\big(k + b_{N-1}(0)\big).$$

All remaining terms involve earlier coefficients. The multiplier has at most one zero for $k \geq 0$; there the recurrence imposes at most one constraint and introduces one new free coefficient. Thus the formal Taylor solution space has dimension at least $N - 1$. After division by $t$, each pole order is at most $1 \leq N - j$, so the equation is regular singular. Frobenius convergence gives $N - 1$ single-valued solutions and hence $\dim \ker(T - I) \geq N - 1$. □

**Theorem VII.E.2** (Multiplicity bound, with no regularity hypothesis)**.** *Let*

$$L = \sum_{j=0}^{N} a_j(t)\partial_t^j, \qquad a_j \in \mathbb{C}\{t\}, \quad a_N \not\equiv 0,$$

*and let $m = \operatorname{ord}_0(a_N)$. If $T$ is the local monodromy of $Ly = 0$ around zero, then*

$$\operatorname{rank}(T - I) \leq \min\{m, N\}.$$

*Proof.* The case $m = 0$ is ordinary. For $m \geq 1$, choose a small closed disk $D$ centered at zero with no other zero of $a_N$ and no zero of $a_N'$ away from zero; this is possible since $a_N = t^m u(t)$ with $u(0) \neq 0$. Replace only the leading coefficient:

$$L_\varepsilon = (a_N(t) - \varepsilon)\partial_t^N + \sum_{j=0}^{N-1} a_j(t)\partial_t^j.$$

For small nonzero $\varepsilon$, Rouché's theorem gives exactly $m$ zeros of $a_N - \varepsilon$ in $D$, all simple by the derivative condition, and none on $\partial D$.

At each of these $m$ points, division by a holomorphic unit puts the operator into Lemma VII.E.1. The corresponding local monodromy has variation rank at most one. Monodromy $T_\varepsilon$ around $\partial D$ is a product of conjugates of these local monodromies. The identity $AB - I = A(B - I) + (A - I)$ gives $\operatorname{rank}(T_\varepsilon - I) \leq m$.

Normalize the companion fundamental matrix to the identity at a fixed point of the circle. Its coefficient matrix converges uniformly there to that of $L$, so parameter dependence of solutions gives $T_\varepsilon \to T$. Vanishing of the $(m + 1)$-minors is closed; therefore $\operatorname{rank}(T - I) \leq m$. The bound by $N$ is automatic. □

**Corollary VII.E.3** (Several singular points)**.** *If a scalar operator with polynomial coefficients annihilates a local system whose monodromies $T_c$ at distinct finite points $c$ are visible on its solution space, then its leading coefficient has degree at least*

$$\sum_c \operatorname{rank}(T_c - I).$$

*Here "visible" may mean restriction to a monodromy-invariant subspace; restriction cannot increase the rank of $T_c - I$.*

*Proof.* Apply Theorem VII.E.2 at every $c$ and sum the required zero multiplicities of the leading coefficient. □

## Part VIII

# Entropy–Transport, Critical Values, and Laplace Monodromy

**Abstract**

This expanded revision supplies proofs for several remaining ranges and extends the structural conclusions of the preceding manuscript. For Chebyshev coefficients in the Bernstein basis, we prove that $\log 2$ is the optimal universal exponential constant when an absolute multiplicative prefactor is allowed, and determine the exact exponential profile in proportional and moderate degrees. Incorporating the previously established full estimation curves closes the Poisson support transition and the worst-null binomial testing transition. We classify the minimum recurrence width of every arithmetic subsequence of every nontrivially weighted binomial-power sum, and identify the entire space of algebraic-number-valued Apéry limits. A local Fourier-transform argument replaces an insufficiently specified global-period input and gives an iterated monodromy annihilator bound. Applications determine exact Bessel determinant modules and algebraic relations among initial derivatives; for bounded-height tableau exponential generating functions, the arithmetic conclusions hold at every nonzero algebraic argument. Finally, for positive rational beta parameters with integral sum, arbitrary algebraic cuts, and positive algebraic polynomial tilts, a two-interval Angelesco polynomial has algebraic coefficients exactly in the polynomial weight case or on the diagonal of its multi-index. The earlier results used in the revision are distinguished from the new extensions, and the unresolved statements are listed explicitly.

**Integration note.** This is the latest expanded revision, incorporating the complete Poisson and binomial estimation curves and the repaired local Fourier argument. Its new classifications and its explicit remaining questions are preserved without altering their hypotheses.

## VIII.1 What this revision establishes

The starting point is the thirty-page draft entitled *Entropy–Transport, Critical Values and Laplace Monodromy*. This revision retains its principal conclusions with their precise hypotheses, replaces the global-period dependency in its Laplace argument, and incorporates relevant results proved in the intervening development. Every unpublished supporting argument needed here is reproduced; published structural inputs are identified where they enter. No earlier local file or separate bibliography is needed to compile or read this manuscript.

| Topic | Starting point | Conclusion in this revision |
|---|---|---|
| Chebyshev coefficients | Constant-one bound $e^{m^2/t}$ and a bulk energy bound | Sharp fixed-prefactor exponent $\log 2$; exact proportional and moderate exponential profiles. |
| Poisson mixtures | A remaining window between polynomial support scales and $B \geq N$ | Full known-support minimax curve, incorporating the earlier all-support proof; crossover $B \asymp N/\log(eN)$. |
| Binomial testing | Near-linear replication ranges | Full worst-null separation curve for all $N, t \geq 1$, by the complete estimation theorem. |
| Franel recurrences | Ordinary and alternating minimum widths | All complex weights $\lambda \neq 0$ and all progressions $qn + b$, including every parity exception. |
| Apéry limits | Rational-valued eventual uniqueness | A complete $K$-linear classification for every $\mathbb{Q} \subseteq K \subseteq \overline{\mathbb{Q}}$ and every progression. |
| Laplace monodromy | A rapid-decay period input needing clarification | A local Fourier proof and the iterated bound $(T_r - I)^r p(T_r) = 0$. |
| Bessel determinants | Three ordinary-generating-function algebraicity thresholds | Exact modules and jet relations; tableau specialization at every nonzero algebraic point. |
| Angelesco polynomials | Arithmetic obstruction for incomplete beta integrals | An exact algebraicity criterion for every two-component multi-index, with polynomial tilts. |

Here "global" refers to the explicitly quantified parameter family in each theorem. It does not remove hypotheses concerning known support, algebraic parameters, positivity, or regular singularities. In particular, the arithmetic uniqueness statements concern eventual solutions with algebraic-number values; they do not assert uniqueness in the complex solution space. The new Angelesco theorem requires integral parameter sum, and the sharp coefficient theorem allows a fixed prefactor.

Three mechanisms connect the results. Integrating a coefficient contour retains the Gaussian localization lost by a supremum bound. Likelihood scores produce an exact posterior coupling, allowing different polynomial scales to use different statistical norms. In the differential arguments, formal exponential multiplicities and local Jordan blocks retain information that distinct critical values alone can lose. Arithmetic specialization then uses classical transcendence theorems with their exceptional-point hypotheses checked explicitly.

## VIII.2 The constant-one Chebyshev–Bernstein theorem

This section retains the attached draft's constant-one theorem and energy refinement. To make its finite verification fully reproducible, we use the earlier conversation manuscript's rational certificate, printed in full below, in place of an unprinted interval-arithmetic check. The next section gives a new sharp exponent theorem with a fixed universal prefactor.

Write $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and define $C(t,m,j)$ by

$$T_m(2x-1) = \sum_{j=0}^{t} C(t,m,j)B_j^t(x), \qquad 0 \le m \le t.$$

**Theorem VIII.2.1** (The coefficient conjecture)**.** *For all integers $t \ge 1$, $0 \le m \le t$, and $0 \le j \le t$,*

$$|C(t,m,j)| \le \exp(m^2/t). \tag{VIII.2.1}$$

*The inequality is strict for $m > 0$. The proof below is a finite-certificate proof: its only computational part uses exact rational arithmetic for $t < 512$. The complete verifier is included in the text.*

This is the precise constant-one assertion posed in [70, Supplement, Remark 4.1, equation (23)], rather than a bound with an unspecified multiplicative constant. The distinction was retained in the subsequent discussion [71, Appendix C.2] and in our preceding manuscript. The universal exponent with an unspecified fixed prefactor is determined in Theorem VIII.3.1; the prefactor-one optimum remains separate.

**Lemma VIII.2.2** (Three complementary coefficient bounds)**.** *For $1 \le m \le t$,*

$$|C(t,m,j)| \le 2^m, \tag{VIII.2.2}$$

$$|C(t,m,j)| \le (t+1)\exp\{tH(m/t)\} \le (t+1)\exp\{(\log 2)m^2/t\}, \tag{VIII.2.3}$$

*where $H(a) = \frac{1}{2}[(1+a)\log(1+a) + (1-a)\log(1-a)]$ and $0\log 0 = 0$. For $m < t$ there is also the bound*

$$|C(t,m,j)| \le D(t,m) := \prod_{l=1}^{m-1} \frac{t+l}{t-l}. \tag{VIII.2.4}$$

*In particular, $D(t,m) \le e^{m^2/t}$ whenever*

$$(m+6)(m-1)^2 \le 6t^2. \tag{VIII.2.5}$$

*Proof.* At degree $m$ the coefficients are $(-1)^{m-i}\binom{2m}{2i}/\binom{m}{i}$. The successive absolute-value ratio is $(2m-2i-1)/(2i+1)$, so their maximum occurs at a middle index. Vandermonde's identity bounds the numerator at that index by $2^m \max_i \binom{m}{i}$, proving (VIII.2.2) at degree $m$. Degree elevation replaces coefficients by convex combinations and proves the bound at every $t \ge m$.

For (VIII.2.3), put

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t} \binom{t}{j} C(t,m,j)z^j.$$

Endpoint coefficients have absolute value one. For $0 < j < t$, set $r = j/(t-j)$ and, on $z = re^{i\theta}$, set $u = 2\sqrt{r}\,|\sin(\theta/2)|/(1+r)$. The larger inverse Joukowski modulus of $(z-1)/(z+1)$ is $\sqrt{(1+u)/(1-u)}$. Consequently

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le (1-u^2)^{t/2}\left(\frac{1+u}{1-u}\right)^{m/2} \le \exp\{tH(m/t)\}.$$

The last maximum is attained at $u = m/t$; limiting values handle $u = 1$. Cauchy's formula and $\binom{t}{j} r^j/(1+r)^t \ge 1/(t+1)$ give the first inequality in (VIII.2.3). This probability is a mode of a Binomial$(t, j/t)$ law, which proves the stated lower bound. Finally

$$H(a) = \sum_{k \ge 1} \frac{a^{2k}}{2k(2k-1)} \le a^2 \sum_{k \ge 1} \frac{1}{2k(2k-1)} = a^2 \log 2.$$

To apply the older bound (VIII.2.4), combine the hypergeometric formula with the identity

$$x^r = \sum_j (j)_{\underline{r}} B_j^t(x)/(t)_{\underline{r}}.$$

This gives

$$C(t,m,j) = (-1)^m \, {}_3F_2\left(\begin{matrix} -m, m, -j \\ 1/2, -t \end{matrix}; 1\right) = (-1)^m Q_m(j; -1/2, -1/2, t).$$

Dette's Hahn bound [73, Corollary 3.6(b)] is exactly (VIII.2.4). Its stipulated range $m < t$ is respected here. Thus this inequality itself is prior art. For the final assertion, using the positive series for arctanh gives

$$\begin{aligned} \log D(t,m) &= 2 \sum_{l=1}^{m-1} \operatorname{arctanh}(l/t) \\ &\le \frac{m(m-1)}{t} + \frac{m^2(m-1)^2}{6t^3\{1 - (m-1)^2/t^2\}}. \end{aligned}$$

The second term is at most $m/t$ precisely under (VIII.2.5). □

*Proof of Theorem VIII.2.1.* The case $m = 0$ is immediate. Suppose first $t \ge 512$. If $m^2/t \ge \frac{10}{3}\log(t+1)$, the contour bound proves the assertion because $\log 2 < 7/10$. Otherwise,

$$m^2 < \frac{10}{3} t \log(t+1) \le \frac{25}{9} t^{4/3}, \qquad m < \frac{5}{3} t^{2/3}.$$

Here $\log(t+1) \le (5/6)t^{1/3}$ for $t \ge 512$: at 512 it follows from $513 < 729$, $\log 3 < 11/10$, and $33/5 < 20/3$; the difference $(5/6)t^{1/3} - \log(t+1)$ has positive derivative thereafter. In particular $m < t$, and

$$(m+6)(m-1)^2 \le m^3 + 6m^2 \le \left(\frac{125}{27} + \frac{25}{96}\right) t^2 < 6t^2.$$

The last estimate uses $t^{2/3} \ge 64$. The Hahn bound therefore applies. The elementary logarithmic inequalities just used can, for example, be verified by positive Taylor lower bounds for $e^{7/10}$ and $e^{11/10}$.

For $1 \le t < 512$, the following verifier covers every pair $(t, m)$. Its first branch uses (VIII.2.5); its second uses $2^m \le e^{7m/10} \le e^{m^2/t}$; its third uses the contour bound with $\log 2 < 7/10$. Every other pair is checked with the exact identity

$$C(t,m,j) = \frac{1}{\binom{t}{j}} \sum_{i=\max(0, j-t+m)}^{\min(m,j)} (-1)^{m-i} \binom{2m}{2i} \binom{t-m}{j-i}. \tag{VIII.2.6}$$

The function `exp_lower` returns a rational number no larger than $e^x$ for $x \ge 0$, so each successful assertion is a rigorous inequality in the required direction. There are no floating-point comparisons. The complete output is

$$(34784, 39518, 56329, 185), \qquad 12734 \text{ direct coefficient comparisons}, \qquad \max t_{\mathrm{direct}} = 145.$$

The four counts sum to $511 \cdot 512/2$, as required. □

```
from fractions import Fraction
from math import comb

def exp_lower(x, degree=32):
    term = total = Fraction(1)
    for k in range(1, degree + 1):
        term *= x / k
        total += term
    return total

counts = [0, 0, 0, 0]
comparisons = largest_direct = 0
for t in range(1, 512):
    for m in range(1, t + 1):
        if m < t and (m+6)*(m-1)**2 <= 6*t*t:
            counts[0] += 1
            continue
        if 10*m >= 7*t:
            counts[1] += 1
            continue
        if exp_lower(Fraction(3*m*m, 10*t)) >= t+1:
            counts[2] += 1
            continue
        lower = exp_lower(Fraction(m*m, t))
        for j in range(t + 1):
            numerator = sum(
                (-1)**(m-i)*comb(2*m, 2*i)*comb(t-m, j-i)
                for i in range(max(0,j-t+m), min(m,j)+1)
            )
            assert Fraction(abs(numerator), comb(t,j)) <= lower
            comparisons += 1
        counts[3] += 1
        largest_direct = max(largest_direct, t)
print(counts, comparisons, largest_direct)
assert sum(counts) == 511*512//2
```

### VIII.2.1 Krawtchouk identities and the energy refinement

The following identities also justify the weighted norm used in the next section. Define

$$K_k(x;N)=\sum_{h}(-1)^h\binom{x}{h}\binom{N-x}{k-h},\qquad \sum_{k=0}^{N}K_k(x;N)u^k=(1-u)^x(1+u)^{N-x}.$$

**Lemma VIII.2.3.** *For $0\le m,j\le t$,*

$$\binom{t}{j}C(t,m,j)=(-1)^{m+j}K_{2j}(t+m;2t).$$

*Moreover*

$$\sum_{x=0}^{N}\binom{N}{x}K_k(x;N)K_\ell(x;N)=2^N\binom{N}{k}\mathbf{1}_{k=\ell}.$$

*For fixed $t,j$, write $y=2j/t-1$ and $C_m=C(t,m,j)$. Then*

$$C_0=1,\quad C_1=y,\qquad (t-m)C_{m+1}=2tyC_m-(t+m)C_{m-1}\quad(1\le m\le t-1).$$

*Proof.* In the coefficient generating polynomial $F(z)$ above, put $z=w^2$. The even part of $(1+iw)^{t+m}(1-iw)^{t-m}$ is

$$\frac{1}{2}\{(1+iw)^{t+m}(1-iw)^{t-m}+(1-iw)^{t+m}(1+iw)^{t-m}\}.$$

Multiplication by $(-1)^m$ gives $F(z)$. Comparing its even powers with the Krawtchouk generating function, and using $K_k(N-x;N) = (-1)^k K_k(x;N)$, gives the first identity. Multiplying two generating functions and summing with weight $\binom{N}{x}$ gives $(2+2uv)^N$; comparison of coefficients gives orthogonality. Finally, the same generating function verifies

$$(N-x)K_k(x+1;N) + xK_k(x-1;N) = (N-2k)K_k(x;N).$$

Take $N = 2t$, $k = 2j$, $x = t+m$ and insert the signs in the first identity to obtain the recurrence. □

The one-term consequence of this orthogonality identity is the exact squared bound used in the attached draft,

$$C(t,m,j)^2 \le \frac{4^t\binom{2t}{2j}}{\binom{2t}{t+m}\binom{t}{j}^2}.$$

It is not needed by the rational-certificate proof above.

**Proposition VIII.2.4** (The energy bound retained from the draft)**.** *For fixed $t \ge 2, j$, set $y = 2j/t - 1$, $C_m = C(t,m,j)$, and*

$$\alpha_0 = 1, \qquad \alpha_{m+1} = \alpha_m \frac{(t-m)^2}{(t+m)(t-m+1)}.$$

*For $1 \le m \le t$ define*

$$Q_m = \alpha_m C_m^2 + \alpha_{m-1}C_{m-1}^2 - \frac{2t\alpha_m}{t-m+1} y C_m C_{m-1}, \qquad \kappa_m = \frac{t^2}{(t+m-1)(t-m+2)}.$$

*Then, for $1 \le m < t$,*

$$Q_{m+1} - Q_m = -\frac{2(m-1)\alpha_{m-1}}{(t+m-1)(t-m+2)} C_{m-1}^2 \le 0, \qquad Q_1 = 1 - \frac{t}{t+1}y^2.$$

*If $\kappa_m y^2 < 1$, this yields*

$$C(t,m,j)^2 \le D(t,m)\frac{t+1}{t-m+1}\frac{1-ty^2/(t+1)}{1-\kappa_m y^2},$$

*where $D(t,m) = \prod_{i=1}^{m-1}(t+i)/(t-i)$ also at $m = t$. In particular, for each fixed $0 < \delta < 1$, uniformly for $|y| \le 1-\delta$ and $m \le \delta t/2$,*

$$|C(t,m,j)| \le K_\delta \exp\{m(m-1)/(2t) + O(m^4/t^3)\}.$$

*Proof.* Substitute the three-term recurrence into $Q_{m+1}$ and cancel its $y^2 C_m^2$ terms. The result is

$$Q_{m+1} = \alpha_m C_m^2 - \frac{2t\alpha_m}{t-m+1} y C_m C_{m-1} + \alpha_m \frac{t+m}{t-m+1} C_{m-1}^2.$$

Subtracting $Q_m$ and using the ratio of consecutive $\alpha$'s gives the displayed difference. Completing the square gives

$$Q_m = \alpha_{m-1}\left(C_{m-1} - \frac{t\alpha_m y}{(t-m+1)\alpha_{m-1}} C_m\right)^2 + \alpha_m C_m^2(1-\kappa_m y^2).$$

Telescoping the defining product gives $\alpha_m^{-1} = D(t,m)(t+1)/(t-m+1)$. These identities and $Q_m \le Q_1$ prove the bound. In the stated bulk range, $(t+m-1)(t-m+2) \ge t^2 - m^2$, so $\kappa_m y^2 \le (1-\delta)^2/(1-\delta^2/4) < 1$ uniformly. The remaining rational factors are bounded, and the positive series for artanh gives $\log D(t,m) = m(m-1)/t + O(m^4/t^3)$ for $m \le t/2$. □

## VIII.3 The optimal exponent with a universal prefactor

Let $M_{t,m} = \max_{0\le j\le t} |C(t,m,j)|$ and put

$$H(a) = \tfrac{1}{2}\{(1+a)\log(1+a) + (1-a)\log(1-a)\}, \qquad 0 \le a \le 1,$$

where $0\log 0 = 0$. The following strengthens the coefficient results of the attached draft. Its distinction between a fixed multiplicative constant and multiplicative constant one is essential.

**Theorem VIII.3.1** (A sharp exponential constant)**.** *There is an absolute constant $K$ such that, for all $t \ge 1$, $0 \le m \le t$ and $0 \le j \le t$,*

$$|C(t,m,j)| \le K(1+m^2/t)^{1/4}\exp\{tH(m/t)\}, \tag{VIII.3.1}$$

$$|C(t,m,j)| \le K\,2^{m^2/t}. \tag{VIII.3.2}$$

*The exponent $\log 2$ in the second inequality is optimal among inequalities $|C(t,m,j)| \le K_c\exp(cm^2/t)$ with a finite constant $K_c$ independent of $t,m,j$. This does not prove the same inequality with prefactor exactly one.*

The improvement comes from integrating the coefficient contour, rather than replacing its integrand by its supremum. The resulting prefactor depends on $m^2/t$, instead of on $t$ alone.

**Lemma VIII.3.2** (A Gaussian integral against the arcsine law)**.** *For every $A, B \ge 0$,*

$$(1+B)\int_0^{\pi/2} \exp\{-\tfrac{1}{2}(B\sin v - A)^2\}\,dv \le K\sqrt{1+A}.$$

*Proof.* The case $B \le 1$ is immediate. Suppose $B > 1$. The substitution $y = B\sin v$ gives the measure $dy/\sqrt{B^2-y^2}$ on $[0,B]$. If $B \le 2(A+1)$, use $\sqrt{B^2-y^2} \ge \sqrt{B(B-y)}$ to bound the integral by

$$B^{-1/2}\int_0^B e^{-(y-A)^2/2}(B-y)^{-1/2}\,dy \le B^{-1/2}(2+\sqrt{2\pi}).$$

For the last inequality put $u = B-y$, split at $u=1$, and use $u^{-1/2} \le 1$ on the remaining interval. Multiplication by $1+B \le 2B$ proves the claim in this case.

If $B > 2(A+1)$, the part $0 \le y \le 3B/4$ is at most $K/B$, because the denominator is at least $\sqrt{7}B/4$ and the Gaussian integral is bounded. On $3B/4 \le y \le B$ we have $y - A > B/4$; this part is at most $(\pi/2)e^{-B^2/32}$. Multiplying by $1+B$ gives a uniform constant, which is at most $K\sqrt{1+A}$. □

*Proof of* (VIII.3.1). Endpoint coefficients have absolute value one. For $0 < j < t$, put $p = j/t$, $r = p/(1-p)$, $v = p(1-p)$, $a = m/t$, and

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t}\binom{t}{j}C(t,m,j)z^j.$$

For $z = re^{i\theta}$ set $u = 2\sqrt{v}\,|\sin(\theta/2)|$. The inverse Joukowski formula and $|T_m(w)| \le |\zeta|^m$ when $w = (\zeta+\zeta^{-1})/2$, $|\zeta| \ge 1$, give

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le \exp\{tf_a(u)\}, \qquad f_a(u) = \tfrac{1}{2}\log(1-u^2) + a\,\operatorname{artanh} u.$$

Explicitly, one may take $\zeta = (\sqrt{z}+i)/(\sqrt{z}-i)$ or its reciprocal; choosing the larger modulus gives $|\zeta| = \sqrt{(1+u)/(1-u)}$. Limits cover the removable endpoint configurations. Since

$f_a'(u) = (a-u)/(1-u^2)$ and $f_a(a) = H(a)$, integration of this derivative on the interval between $u$ and $a$ shows

$$f_a(u) \le H(a) - \tfrac{1}{2}(u-a)^2. \tag{*}$$

This uses $1/(1-w^2) \ge 1$ and also holds at $a = 1$ by a limit.

The binomial mass $\pi_j = \binom{t}{j} p^j (1-p)^{t-j}$ is a mode of $\mathrm{Bin}(t,p)$. Chebyshev's probability inequality on the interval of radius $2\sqrt{tv}$ around its mean $j$ implies

$$\pi_j \ge \frac{3/4}{4\sqrt{tv}+1} \ge \frac{c}{1+\sqrt{tv}}.$$

Cauchy's coefficient formula, (*) and Lemma VIII.3.2, with $A = m/\sqrt{t}$ and $B = 2\sqrt{tv}$, therefore yield

$$|C(t,m,j)| \le K e^{tH(a)} \sqrt{1+m/\sqrt{t}} \le K(1+m^2/t)^{1/4} e^{tH(a)}.$$

The final inequality follows from $(1+u)^2 \le 2(1+u^2)$. □

*Proof of* (VIII.3.2) *and optimality.* Write $x = m^2/t$ and $\Delta(a) = (\log 2)a^2 - H(a)$. The positive power series

$$H(a) = \sum_{h \ge 1} \frac{a^{2h}}{2h(2h-1)}$$

shows that $H(a)/a^2$ increases strictly on $(0,1)$ and tends to $\log 2$ at one. Hence for $0 \le a \le 99/100$ there is an absolute $c_0 > 0$ such that $t\Delta(a) \ge c_0 x$. The factor $(1+x)^{1/4} e^{-c_0 x}$ is bounded, proving the assertion in this range.

Suppose $99/100 \le a < 1$ and write $\delta = 1-a$, $k = t-m$. Then $k$ is an integer with $1 \le k \le t/100$. Since $H'(a) = \operatorname{artanh} a$,

$$\begin{aligned} \Delta(1-\delta) &\ge -2(\log 2)\delta + \tfrac{1}{2}\int_0^{\delta} \log(1/u)\,du \\ &= \delta\{\tfrac{1}{2}\log(1/\delta) + \tfrac{1}{2} - 2\log 2\} \ge \tfrac{1}{4}\delta \log(1/\delta). \end{aligned}$$

The last inequality follows from $\delta \le 1/100$; for example $\log 100 > 4$ and $\log 2 < 7/10$ suffice. The function $k \mapsto k\log(t/k)$ increases for $1 \le k \le t/100$, so

$$t\Delta(a) \ge \tfrac{1}{4} k \log(t/k) \ge \tfrac{1}{4}\log t.$$

Since $x \le t$, this absorbs the prefactor in (VIII.3.1). At $m = t$, the elementary degree bound $|C(t,t,j)| \le 2^t$ proves the result directly.

Finally, for even $t$ the exact central coefficient is

$$|C(t,t,t/2)| = \frac{\binom{2t}{t}}{\binom{t}{t/2}} = 2^{t-1/2}(1+o(1)).$$

Thus no fixed prefactor permits an exponent $c < \log 2$. □

### VIII.3.1 Exact weighted energy and matching exponential profiles

The next identity is the classical Hahn orthogonality in this normalization. We prove it directly from the Krawtchouk identities; its sharp-profile consequences are the additional conclusions needed here. Put

$$w_{t,j} = 4^{-t}\binom{2j}{j}\binom{2t-2j}{t-j}, \qquad A_{t,m} = \frac{(t+m)!(t-m)!}{(t!)^2}.$$

**Proposition VIII.3.3** (Exact weighted inverse norm)**.** *The weights $w_{t,j}$ sum to one. If $P(x)=\sum_{m=0}^{t} a_m T_m(2x-1)=\sum_{j=0}^{t} b_j B_j^t(x)$, with real coefficients, then*

$$\sum_{j=0}^{t} w_{t,j} b_j^2 = a_0^2 + \frac{1}{2}\sum_{m=1}^{t} A_{t,m} a_m^2.$$

*In particular, for $m \geq 1$,*

$$M_{t,m} \geq \sqrt{A_{t,m}/2}. \tag{VIII.3.3}$$

*Proof.* The generating series of $4^{-j}\binom{2j}{j}$ is $(1-z)^{-1/2}$, so convolution proves the assertion about the weights. Write $K_k(x;N)$ for the Krawtchouk polynomials defined in the preceding section. Their combinatorial formula gives the duality

$$\frac{K_k(x;N)}{\binom{N}{k}} = \frac{K_x(k;N)}{\binom{N}{x}}.$$

Indeed, multiplying either side by $\binom{N}{x}\binom{N}{k}$ counts ordered subsets of sizes $x, k$, with the sign of their intersection. Together with the orthogonality identity from that section this gives

$$\sum_{k=0}^{N} \frac{K_k(x;N)K_k(y;N)}{\binom{N}{k}} = \frac{2^N}{\binom{N}{x}}\mathbf{1}_{x=y}.$$

Using $K_k(N-y;N)=(-1)^k K_k(y;N)$, its even-$k$ part is half the sum of the right sides for $y$ and $N-y$.

Take $N=2t$, $x=t+m$, $y=t+\ell$, and use $\binom{t}{j}C(t,m,j)=(-1)^{m+j}K_{2j}(t+m;2t)$ and

$$\frac{w_{t,j}}{\binom{t}{j}^2} = \frac{(2t)!}{4^t (t!)^2}\frac{1}{\binom{2t}{2j}}.$$

For $0 \leq m, \ell \leq t$, the resulting weighted inner product is zero unless $m=\ell$. Its value is one for $m=\ell=0$ and $A_{t,m}/2$ otherwise. Expanding $b_j=\sum_m a_m C(t,m,j)$ proves the identity. Averaging $C(t,m,j)^2 \leq M_{t,m}^2$ proves the lower bound. □

**Corollary VIII.3.4** (Sharp proportional-degree and moderate-degree scales)**.** *If $t\to\infty$ and $m/t \to a \in [0,1]$, then*

$$\frac{1}{t}\log M_{t,m} \longrightarrow H(a).$$

*If instead $m=o(t)$ and $m^2/t \to \infty$, then*

$$\log M_{t,m} = (\tfrac{1}{2}+o(1))m^2/t.$$

*The latter maximum is over every coefficient, without a restriction keeping $j/t$ away from zero and one.*

*Proof.* The elementary factorial estimate $\log(n!)=n\log n - n + O(\log(n+1))$, valid also at $n=0$ with $0\log 0=0$, gives $\log A_{t,m}=2tH(m/t)+O(\log(t+1))$ uniformly. Combine (VIII.3.3) with (VIII.3.1); the case $m=0$ is immediate. For $m=o(t)$, the product

$$A_{t,m}=\prod_{i=1}^{m}\frac{t+i}{t-i+1}$$

and $\log(1+u)=u+O(u^2)$ give $\log A_{t,m}=m^2/t+O(m^3/t^2)$. Also $tH(m/t)=m^2/(2t)+O(m^4/t^3)$. The logarithm of the prefactor in (VIII.3.1) is $O(1+\log(1+m^2/t))=o(m^2/t)$, proving the second assertion. □

*Remark* VIII.3.5 (What remains open in this argument). The conjecturally stronger statement $M_{t,m} \le 2^{m^2/t}$ with prefactor one is not proved here. Conversely, a global exponent $1/2$ with either a fixed or polynomial prefactor is impossible: the degree-$t$ central coefficient grows as $2^{t-1/2}$. The exponent $1/2$ is the sharp moderate-degree limit, whereas $\log 2$ is the sharp universal exponent when a fixed prefactor is allowed.

## VIII.4 The entropy–transport transfer retained from the source

Let $\mathcal{X}$ be a countable metric space, let $K_\theta$ be component probability kernels, and write $p_H = \int K_\theta \, dH(\theta)$. For an empirical law $q$, assume the exact mixing NPMLE $H$ exists and has positive probability on the observed support. Zero empirical cells are omitted in score sums.

**Theorem VIII.4.1** (Component concentration transfers through likelihood contact)**.** *Suppose that $\psi : [0,\infty) \to [0,\infty)$ is nondecreasing and concave and that*

$$W_1(R, K_\theta) \le \psi(D(R\|K_\theta))$$

*for every component and every probability law with finite entropy. Then*

$$W_1(q, p_H) \le \psi(D(q\|p_H)) \le \psi(D(q\|p_G))$$

*for every admissible comparator mixing law $G$ for which the final entropy is finite.*

*Proof.* The directional derivative toward a point mass gives $S(\theta) = \sum_x q(x)K_\theta(x)/p_H(x) \le 1$. Its $H$-integral equals one, so $S = 1$ almost surely under $H$. Thus $R_\theta(x) = q(x)K_\theta(x)/p_H(x)$ is a probability kernel $H$-almost surely, and $\int R_\theta \, dH = q$. The entropy chain rule gives

$$\int D(R_\theta\|K_\theta) \, dH(\theta) = D(q\|p_H).$$

Mixing couplings, applying the component bound, and then Jensen's inequality prove the first conclusion. Likelihood optimality gives $D(q\|p_H) \le D(q\|p_G)$, proving the second. All applications below have finite right sides; truncation of transport costs gives the same conclusion before passing to an unbounded cost. □

This is the source manuscript's transfer principle. Its application to binomial kernels has $\psi(u) = \sqrt{u/(2t)}$ for normalized count cost. For Poisson mixtures, a parameter-free occupancy channel will import that same inequality while preserving the original likelihood fit.

The argument does not normalize every stratum of an arbitrary heterogeneous joint likelihood: only the aggregate score is normalized. Consequently an arbitrary-profile theorem does not follow verbatim. The preceding work's separate two-depth result uses an affine posterior repair; the present full-plane results concern common known trial counts or common known exposures.

## VIII.5 Closure of the Poisson critical window

Consider the experiment

$$p_G(j) = \int_0^B e^{-\lambda} \frac{\lambda^j}{j!}\, dG(\lambda), \qquad X_1, \dots, X_N \overset{\text{iid}}{\sim} p_G,$$

where the support radius $B > 0$ is known. Let $\widehat{G}$ be any measurable maximizer of the original Poisson mixture likelihood over all probability laws on $[0, B]$. There is no restriction on the number of mixing atoms. Set $L = \log(eN)$ and, for $N \geq 1$, define

$$r(N, B) = \begin{cases} B \min\{1, L^{-1} \log(e + L/B)\}, & 0 < B \leq L, \\ \max\{\sqrt{B/L}, B/\sqrt{N}\}, & B > L. \end{cases} \tag{VIII.5.1}$$

The two expressions at the breakpoint agree up to absolute constants.

**Theorem VIII.5.1** (Global Poisson population recovery)**.** *There exist absolute constants $0 < c < C < \infty$ such that, for every $N \geq 1$ and every $B > 0$,*

$$c\, r(N, B) \leq \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\, r(N, B).$$

*In particular, the ordinary NPMLE is minimax up to absolute factors throughout the entire support/sample-size plane. For observations $X_i \mid \Theta_i \sim \mathrm{Poi}(t\Theta_i)$ with $G \in \mathcal{P}([0, M])$, the corresponding rate is $r(N, tM)/t$.*

The exact baseline in the supplied draft is its Theorem B:

$$\sup_G \mathbb{E}_G W_1(\widehat{G}, G) \leq C \left\{ 1 + \sqrt{\frac{B}{\log(1 + N/B)}} \right\}.$$

Together with the lower bounds cited there and its Theorem C, this proves optimality for $c_0 \log N \leq B \leq N^{1-\varepsilon}$ and for $B \geq N$, but leaves the transition window recorded in its Section 14. Theorem VIII.5.1 closes that window for the same ordinary likelihood estimator. Its growing-support crossover is $B \asymp N/\log(eN)$: below that scale the deconvolution term $\sqrt{B/L}$ dominates, and above it the latent-sampling term $B/\sqrt{N}$ dominates. There is no residual logarithmic loss.

Miao et al. [78, Theorem 6.1(a)] established the small-support rate for the NPMLE, and their Theorem 6.1(b) gives $\sqrt{B/\log N}$ under $B \leq C_0 N^{1/3-\varepsilon_0}$. The present revision incorporates the all-support argument established in the preceding development, rather than counting the same theorem as a second discovery. It also includes the diameter clipping needed for arbitrarily small $B$; an unclipped lower-rate formula cannot be extrapolated beyond the maximum possible loss $B$. All ingredients used below are proved here, including the two-scale binomial inverse inequality in Proposition VIII.5.6. The estimator is never replaced by a transformed-data MLE.

For $N \in \{1, 2\}$ the rate in (VIII.5.1) is comparable to $B$. The diameter bound and the two-point latent-sampling lower bound given below therefore suffice. In the proof of the nontrivial regimes we may assume $N \geq 3$.

### VIII.5.1 A parameter-free occupancy channel

For an integer $m \geq 1$, define a Markov kernel $A_m$ as follows. Given a count $x$, independently assign $x$ balls uniformly to $m$ boxes, and output the number of occupied boxes. Poisson splitting proves

$$A_m \,\mathrm{Poi}(\lambda) = \mathrm{Bin}(m, \phi_m(\lambda)), \qquad \phi_m(\lambda) = 1 - e^{-\lambda/m}. \tag{VIII.5.2}$$

Indeed the box counts are independent Poisson random variables of mean $\lambda/m$, so their nonempty indicators are independent Bernoulli variables. In particular, if $F = \phi_m \# G$, then $A_m p_G$ is exactly the binomial mixture with mixing law $F$. On $[0, \phi_m(B)]$ the inverse map has derivative at most $me^{B/m}$, and consequently

$$W_1(G, H) \le me^{B/m} W_1(\phi_m \# G, \phi_m \# H). \tag{VIII.5.3}$$

This channel is a proof device. We do not transform the data before fitting, or assert that a transformed MLE remains an MLE.

We use $W_{1,m}$ for transportation on $\{0, \dots, m\}$ with cost $|i-j|/m$, and $h(p,q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

**Lemma VIII.5.2** (Empirical entropy on the infinite alphabet)**.** *Let $q$ be the empirical law of $N$ observations from $p_G$. Uniformly over $G \in \mathcal{P}([0, B])$,*

$$\mathbb{E} D(q \| p_G) \le C(B + L)/N. \tag{VIII.5.4}$$

*Proof.* For every coordinate with probability $p_j > 0$, convexity and $\log u \le u - 1$ give

$$0 \le \mathbb{E}\{q_j \log(q_j/p_j) - q_j + p_j\} \le \min\{(1 - p_j)/N,\ p_j \log(1/p_j)\}.$$

The second bound also uses $q_j \log q_j \le 0$. The expression inside braces is nonnegative, so its infinite sum can be averaged by Tonelli's theorem. Coordinates with $p_j = 0$ contribute zero almost surely. For $j \ge B$, the Poisson mass is increasing in its mean throughout $[0, B]$, and hence

$$p_G(j) \le e^{-B} B^j / j! \le (eB/j)^j \le e^{-j} \quad (j \ge e^2 B).$$

Take $J = \lceil e^2 B + 3L \rceil$. The first $J+1$ coordinates contribute at most $(J+1)/N$. Since $u \log(1/u)$ is increasing on $(0, e^{-1}]$, the remaining contribution is bounded by $\sum_{j>J} j e^{-j} \le C(J+1)e^{-J} \le C/N$. This proves (VIII.5.4). Notice that the expected empirical chi-square divergence on the full infinite alphabet would not give this conclusion. □

**Lemma VIII.5.3** (Root-$N$ prediction after occupancy)**.** *For $m = \lceil \max(B, L) \rceil$ and every Poisson NPMLE $\widehat{G}$,*

$$\mathbb{E} W_{1,m}(A_m p_{\widehat{G}}, A_m p_G) \le C/\sqrt{N}. \tag{VIII.5.5}$$

*Proof.* Existence follows from compactness of the mixing parameter space and upper semicontinuity of the likelihood; the admissible fit $\delta_B$ is strictly positive at every observed count. Write $H = \widehat{G}$. The likelihood directional derivative gives

$$S(\lambda) = \sum_{j : q_j > 0} \frac{q_j K_\lambda(j)}{p_H(j)} \le 1 \quad (0 \le \lambda \le B).$$

Since $\int S \, dH = 1$, equality holds $H$-almost everywhere. Thus $R_\lambda(j) = q_j K_\lambda(j)/p_H(j)$ is a probability kernel $H$-almost everywhere and $\int R_\lambda \, dH = q$. Its entropy chain rule and the likelihood comparison yield

$$\int D(R_\lambda \| K_\lambda) \, dH(\lambda) = D(q \| p_H) \le D(q \| p_G). \tag{VIII.5.6}$$

The binomial transport-entropy inequality is

$$W_{1,m}(R, \mathrm{Bin}(m, p))^2 \le D(R \| \mathrm{Bin}(m, p))/(2m).$$

For completeness, a normalized-count 1-Lipschitz function changes by at most $1/m$ on changing one of the $m$ Bernoulli bits. The conditional Hoeffding bound gives centered log-mgf at most

$u^2/(8m)$; entropy duality and Wasserstein duality give the displayed inequality, including degenerate endpoints whenever the entropy is finite. Apply it to $A_m R_\lambda$, use entropy data processing, and integrate (VIII.5.6). Convexity of transportation under mixtures and Cauchy–Schwarz give

$$W_{1,m}(A_m q, A_m p_H) \le \sqrt{D(q\|p_G)/(2m)}.$$

For each count threshold, the CDF of $A_m q$ is an average of independent $[0,1]$-valued functions of the $X_i$. Its expected absolute error is at most $1/(2\sqrt{N})$. The CDF formula for $W_{1,m}$ therefore gives $\mathbb{E}W_{1,m}(A_m q, A_m p_G) \le 1/(2\sqrt{N})$. Now use Lemma VIII.5.2 and $(B+L)/m \le 2$. □

### VIII.5.2 A growing-support Hellinger bound

**Lemma VIII.5.4** (Poisson mixture entropy and likelihood control)**.** *For every fixed absolute $b_0 > 0$, uniformly over $b_0 L \le B \le N$,*

$$\mathbb{E}h(p_{\widehat{G}}, p_G) \le C_{b_0} B^{1/4} L^{3/4}/\sqrt{N}. \tag{VIII.5.7}$$

*Proof.* For complex $z$ the vector $K_z = (e^{-z} z^j/j!)_{j\ge 0}$ belongs to $\ell_1$ and satisfies $\|K_z\|_1 = e^{|z|-\operatorname{Re} z}$. On the ellipse $z = B(1+\cos(\theta+iu))/2$, direct calculation gives

$$|z| - \operatorname{Re} z = \tfrac{B}{2}(\cosh u - 1)(1-\cos\theta) \le B(\cosh u - 1).$$

The $k$th vector Chebyshev coefficient $v_k$ of $x \mapsto K_{Bx}$ therefore satisfies, by contour shifting,

$$\|v_k\|_1 \le 2\exp\{B(\cosh u - 1) - ku\} \le 2\exp\{-c\min(k^2/B, k)\}.$$

For the second inequality take $u = k/(2B)$ if $k \le 2B$, using $\cosh u - 1 \le u^2$ for $0 \le u \le 1$, and take $u = 1$ otherwise. It follows by summing the Gaussian and exponential tails that truncating the expansion at

$$d = \left\lceil C\left\{\sqrt{B\log(C(B+1)/\beta)} + \log(C(B+1)/\beta)\right\}\right\rceil$$

has uniform $\ell_1$ error at most $\beta/8$. Every Chebyshev moment of a probability law on $[0,B]$ belongs to $[-1,1]$, and $\|v_k\|_1 \le 2$ also follows directly from its real integral formula. Grid the first $d$ moments with mesh $\beta/(16(d+1))$. Choose one actual mixture in each occupied grid box. This gives an $\ell_1$ cover of radius $\beta$, with centers in the mixture class and cardinality $M$ satisfying

$$\log M \le Cd\log(C(d+1)/\beta). \tag{VIII.5.8}$$

Take $J = \lceil e^2 B + 6L\rceil$ and $\beta = N^{-3}/(J+1)$. In the indicated range, $d \le C_{b_0}\sqrt{BL}$ and $\log M \le C_{b_0}\sqrt{B}L^{3/2}$. For every cover center $p_a$, define the pointwise envelope

$$u_a(j) = \begin{cases} p_a(j) + \beta, & j \le J, \\ e^{-j}, & j > J. \end{cases}$$

Every mixture is bounded coordinatewise by its associated envelope, whose total mass $U_a$ lies in $[1, 1+\eta]$ for $\eta = CN^{-3}$, whenever that envelope is used for at least one mixture. Unused envelopes can be discarded. Normalize $r_a = u_a/U_a$. An associated mixture $p$ satisfies $h(p, r_a) \le \sqrt{2\eta}$.

Here is the elementary likelihood argument, to specify the treatment of the infinite alphabet. If $\widehat{p}$ is a likelihood maximizer and $u_a \ge \widehat{p}$, then

$$\prod_{i=1}^{N} \frac{r_a(X_i)}{p_G(X_i)} \ge e^{-N\eta}.$$

Under $p_G$, the expectation of the square root of this product is the $N$th power of the affinity, at most $\exp\{-Nh(p_G, r_a)^2/2\}$. Markov's inequality followed by a union bound shows, with probability at least $1 - e^{-x}$,

$$h(p_G, \widehat{p}) \le \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}.$$

Integration gives $\mathbb{E}h(p_G, \widehat{p})^2 \le C\eta + C(\log M + 1)/N$, which proves (VIII.5.7). □

### VIII.5.3 The two-scale argument closes every growing-support regime

The binomial inverse inequality proved below states that if $F, H$ are mixing laws on $[0, 1]$, $b_F, b_H$ their $m$-trial binomial mixtures, and $1 \le S \le K \le m/2$, then

$$W_1(F, H) \le C/K + Ce^{4S^2/m} W_{1,m}(b_F, b_H) + Ce^{2K^2/m} h(b_F, b_H)/S. \tag{VIII.5.9}$$

Its small high-frequency amplitude $1/S$ is indispensable. The proof approximates a Lipschitz test function at two polynomial degrees, preserves a bounded derivative at the lower degree, and uses the degree-elevated Bernstein coefficient bounds. This is Proposition VIII.5.6.

Suppose $b_0 L \le B \le N$, with $b_0 > 0$ an absolute constant to be fixed below. Set $m = \lceil \max(B, L) \rceil$, so $m \asymp_{b_0} B$. For sufficiently large absolute $N$, take

$$S = \lfloor \sqrt{m}/16 \rfloor, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These are admissible, $S \asymp \sqrt{m}$, $e^{4S^2/m} \le C$, and $m/K \le C\sqrt{m/L}$ because $m \ge L$. Apply (VIII.5.9) to the pushforwards under $\phi_m$, then use (VIII.5.3), Lemmas VIII.5.3 and VIII.5.4, and Hellinger data processing. This gives

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \left\{ \sqrt{B/L} + \frac{B}{\sqrt{N}} + \frac{e^{2aL} B^{3/4} L^{3/4}}{\sqrt{N}} \right\}.$$

The ratio of the last term to $\sqrt{B/L}$ is at most $Ce^{2aL} N^{-1/4} L^{5/4}$, which is uniformly bounded since $B \le N$ and $2a < 1/4$. We have proved

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \{ \sqrt{B/L} + B/\sqrt{N} \}. \tag{VIII.5.10}$$

Bounded $N$ is absorbed in the absolute constant by the diameter bound; this is uniform in $B$ because $B \ge b_0 L$ in this step.

If $B > N$, no Hellinger estimate or approximation is necessary. For any binomial mixing law, coupling its parameter $p$ with its normalized count gives expected cost at most $1/(2\sqrt{m})$. Thus

$$W_1(F, H) \le m^{-1/2} + W_{1,m}(b_F, b_H).$$

Together with (VIII.5.3) and Lemma VIII.5.3, this gives $\mathbb{E}W_1(\widehat{G}, G) \le C(\sqrt{B} + B/\sqrt{N}) \le CB/\sqrt{N}$. Every $B \ge L$ is now covered.

### VIII.5.4 The small-support regime, including arbitrarily small radii

We include a direct proof to ensure uniformity as $B \downarrow 0$. Let $f$ be 1-Lipschitz on $[0, B]$ and normalize $f(0) = 0$. Jackson approximation gives a degree-$k$ polynomial $P(\lambda) = \sum_{l=0}^{k} a_l \lambda^l$ such that

$$\|f - P\|_{[0,B]} \le CB/k, \qquad \|P\|_{[0,B]} \le CB.$$

Its monomial coefficients satisfy

$$|a_l| \le BC_1^k B^{-l}. \tag{VIII.5.11}$$

Indeed expand $P(Bx)$ in shifted Chebyshev polynomials: each Chebyshev coefficient has magnitude at most $2\|P\|$, and the sum of absolute monomial coefficients of $T_j(2x-1)$ is at most $7^j$ by its three-term recurrence. Summing the finite geometric series proves (VIII.5.11) after enlarging $C_1$.

Take $k \ge 2eB$ and put $Q_k(\lambda) = \sum_{r=0}^{k} \lambda^r/r!$. The exponential remainder implies

$$\|e^{-\lambda}P(\lambda)Q_k(\lambda) - f(\lambda)\|_{[0,B]} \le CB/k.$$

For example, $0 \le 1 - e^{-\lambda}Q_k(\lambda) \le C(eB/(k+1))^{k+1} \le C2^{-k}$, which suffices. Write this weighted polynomial as $\sum_{j=0}^{2k} b_j K_\lambda(j)$. Since

$$b_j = \sum_{\substack{0\le l\le k\\ 0\le j-l\le k}} a_l \frac{j!}{(j-l)!},$$

equation (VIII.5.11) gives

$$\max_{0\le j\le 2k} |b_j| \le B(C_2 k/B)^k. \tag{VIII.5.12}$$

The factors $k+1$ and $C_1^k$ have been absorbed into $C_2^k$; $2k/B \ge 1$ by hypothesis. Wasserstein duality consequently gives the deterministic inequality

$$W_1(F,H) \le CB/k + B(C_2k/B)^k \|p_F - p_H\|_1. \tag{VIII.5.13}$$

For the MLE, likelihood comparison, $h^2 \le D$, the Hellinger triangle inequality and $\|p-q\|_1 \le 2h(p,q)$ imply

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_1 \le 4\sqrt{\mathbb{E}D(q\|p_G)} \le C\sqrt{L/N} \qquad (B \le L).$$

Put $d_0 = \log(e + L/B)$ and $k = \lfloor \gamma L/d_0 \rfloor$, where $\gamma > 0$ is a sufficiently small absolute constant. If $\gamma L/d_0 < 2$, the diameter bound $W_1 \le B$ is already at most a constant times $B\min(1, d_0/L)$. Otherwise $k \ge \gamma L/(2d_0)$. Choose a fixed $b_0 > 0$ sufficiently small that $B \le b_0 L$ implies $\gamma(L/B)/(2\log(e+L/B)) \ge 2e$; then $k \ge 2eB$. Furthermore

$$k\log(C_2k/B) \le C_3\gamma L \le L/8$$

after decreasing $\gamma$ once, since $\log(C_2\gamma(L/B)/d_0) \le C_3 d_0$. The stochastic term in (VIII.5.13) is therefore at most $CBe^{L/8}\sqrt{L/N} \le CB/k$: here $k \le \gamma L$ and $e^{-3L/8}L^{3/2}$ is bounded. This proves the required rate for $B \le b_0 L$. On $b_0 L \le B \le L$, equation (VIII.5.10) is bounded by an absolute constant, whereas $B\min(1, d_0/L)$ is bounded below by a positive absolute constant. This proves the upper bound in Theorem VIII.5.1 for every $B$.

### VIII.5.5 Matching lower bounds without support-range restrictions

We first record a convenient moment-matching pair. On an interval $[a-M, a+M]$, push forward the probability measures

$$(1 \pm \tfrac{1}{2}T_k(x))\frac{dx}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

under $x \mapsto a + Mx$. Denote them by $F_k^+, F_k^-$. They match all moments of degree less than $k$. Their CDF difference is, up to orientation, $\sin(k\theta)/(\pi k)$ for $x = \cos\theta$. Consequently

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \ge \frac{M}{\pi k}. \tag{VIII.5.14}$$

The last inequality follows from $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1}$.

For $B \le L$, use this pair on $[0, B]$. Expanding $e^{-\lambda}$ and using the matched moments gives the elementary bound

$$\|p_{F_k^+} - p_{F_k^-}\|_1 \le 2 \sum_{r \ge k} \frac{(2B)^r}{r!}. \tag{VIII.5.15}$$

Indeed the term with observed count $j$ and exponential-expansion index $l$ vanishes if $j + l < k$; summing the absolute values of all remaining terms gives (VIII.5.15). Set $d_0 = \log(e + L/B)$ and $k = \lceil 100L/d_0 \rceil$. Writing $t = L/B \ge 1$, we have $t = e^{d_0} - e \ge e^{d_0}/(e+1)$ and $e^{d_0/2}/d_0 \ge e/2$. These inequalities show

$$\log \frac{k}{2eB} \ge \frac{d_0}{2}, \qquad k \log \frac{k}{2eB} \ge 50L.$$

They also ensure $k \ge 4B$. Thus the tail in (VIII.5.15) is at most $4(2eB/k)^k \le 4e^{-50L}$. The two $N$-sample laws have TV at most $2Ne^{-50L}$, whereas (VIII.5.14) is at least $cB \min\{1, d_0/L\}$. Le Cam's two-point inequality proves the small-support lower bound, including arbitrarily tiny $B$.

For $B \ge L$, set $a = B/2$, $M = \sqrt{BL}/8$, and $k = \lceil 20L \rceil$. The moment-matching interval lies inside $[0, B]$. The exact Poisson identity

$$\sum_{x=0}^{\infty} \frac{K_\lambda(x) K_\mu(x)}{K_a(x)} = \exp\{(\lambda - a)(\mu - a)/a\}$$

shows, upon integrating against the signed difference $F_k^+ - F_k^-$, that

$$\sum_x \frac{(p_{F_k^+}(x) - p_{F_k^-}(x))^2}{K_a(x)} \le 4 \sum_{r \ge k} \frac{(M^2/a)^r}{r!} = 4 \sum_{r \ge k} \frac{(L/32)^r}{r!}.$$

All exchanges are justified by boundedness of $\lambda, \mu$ on the integration interval and the absolutely convergent exponential series. The last tail is at most $Ce^{-40L}$, for instance by $r! \ge (r/e)^r$ and a geometric-tail bound. Cauchy–Schwarz bounds the predictive TV by half the square root of this quantity. Le Cam's inequality and (VIII.5.14) therefore give the lower rate $c\sqrt{B/L}$.

Finally consider mixing laws on $\{0, B\}$ with respective weights $1/2 \pm c_0/\sqrt{N}$ at $B$, for a sufficiently small absolute $c_0 > 0$. Their latent $N$-sample laws have TV bounded away from one, by the Bernoulli KL formula and Pinsker's inequality. The Poisson sample is a Markov image of that latent sample, so its TV is no larger. The mixing-law distance is $2c_0 B/\sqrt{N}$. Le Cam's inequality yields the remaining lower rate $cB/\sqrt{N}$. Taking the maximum completes the proof of Theorem VIII.5.1.

The proof uses a common known exposure and a known compact support. It makes no assertion for heterogeneous exposures, unknown support radii, or an MLE constrained to a prescribed number of components. The gain over a direct one-scale inverse estimate is structural: the occupancy channel imports a binomial concentration inequality, likelihood contact supplies posterior entropy, and the two-scale inverse places predictive transportation and high-frequency Hellinger error in different terms.

### VIII.5.6 Self-contained binomial inversion tools

These approximation tools are reproduced to make the closure proof independent of the earlier supplement. In this subsection write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and $p_F(j) = \int B_j^t \, dF$. The integer $t$ is a binomial trial count, $W_{1,t}$ uses cost $|i - j|/t$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

**Lemma VIII.5.5** (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute $C$ with the following properties. For every 1-Lipschitz $f : [0, 1] \to \mathbb{R}$ and integer $k \ge 1$, there is a polynomial $P_k$ of degree at most $k$ such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

*If $P=\sum_{j=0}^{t} b_j B_j^t$ has degree $k\le t/2$, then*

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{VIII.5.16}$$

*For the derivative-stable $P_k$, its coefficient vector also satisfies*

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{VIII.5.17}$$

*Proof.* Subtract $f(0)$, reflect $f$ on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial $Q_s$ has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most $e^{-100s}$. Indeed $|\pi l(x-1/2)|\le \pi s$ and the factorial inequality $M!\ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s=\lfloor k/100\rfloor$ for $k\ge 100$; use a constant polynomial and enlarge $C$ for smaller $k$.

For (VIII.5.16), put $M=\|P\|_\infty$ and $F(z)=(1+z)^tP(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)|\le M|w|^k$ for $|w|\ge 1$. For $0<j<t$, set $r=j/(t-j)$, $v=r/(1+r)^2$, and $u=2\sqrt{v}|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k\le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j}\ge c/\sqrt{1+tv}$, proves (VIII.5.16). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k'=t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (VIII.5.16) at degree $k-1\le (t-1)/2$ and use $2(k-1)^2/(t-1)\le 4k^2/t$. □

**Proposition VIII.5.6** (Two-scale deterministic inversion)**.** *For integers $1\le s\le k\le t/2$ and any two mixing laws $G,H$,*

$$W_1(G,H)\le \frac{C}{k}+Ce^{4s^2/t}W_{1,t}(p_G,p_H)+\frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{VIII.5.18}$$

*Proof.* For any 1-Lipschitz test $f$, take the polynomials $P_k,P_s$ from Lemma VIII.5.5. The approximation error in integrating $P_k$ instead of $f$ against $G-H$ is at most $C/k$. The Bernstein coefficients of $P_s$ have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k-P_s\|_\infty \le C/k+C/s\le C/s.$$

Thus the Bernstein coefficients of $P_k-P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test $P_s$ by count transportation and $P_k-P_s$ by $\|p_G-p_H\|_1\le 2h(p_G,p_H)$. All constants are uniform over $f$, so taking the Wasserstein dual supremum proves the result. □

## VIII.6 Closing the entire binomial random-effects testing transition

The attached manuscript left open the testing transition $t \asymp N/\log N$. Its sharp deterministic inverse modulus also showed that global count-Wasserstein error alone cannot close that transition. The previously proved complete estimation theorem uses two different statistical norms on two different parts of a test polynomial. The low-degree part is controlled by likelihood transportation. The remaining polynomial has small amplitude, and is controlled by a predictive Hellinger bound. This yields the full replication curve, including its transition and its parametric plateau.

For a probability measure $G$ on $[0,1]$, put

$$K_x(j) = B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

We observe $N$ independent counts from $p_G$. Their empirical law is $q$. The unrestricted mixing-measure MLE is any maximizer $\widehat{G}$ of $\ell(H) = \sum_j q_j \log p_H(j)$. Write $\widehat{p} = p_{\widehat{G}}$, $L_N = \log(eN)$, and

$$\mathcal{R}_{N,t} = \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G), \qquad r_{N,t} = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{N}}\right\}.$$

The infimum allows randomized estimators. Wasserstein distance on the count space uses the normalized cost $|j-l|/t$ and is denoted $W_{1,t}$. Our Hellinger convention is $h(p,v)^2 = \sum_j (\sqrt{p_j} - \sqrt{v_j})^2$, also for nonnormalized nonnegative vectors $v$.

**Theorem VIII.6.1** (All-replication optimality of the unrestricted MLE)**.** *There are absolute constants $0 < c < C < \infty$ such that, for every pair of integers $N, t \ge 1$,*

$$c r_{N,t} \le \mathcal{R}_{N,t} \le \sup_G \mathbb{E}_G W_1(\widehat{G}, G) \le C r_{N,t}. \tag{VIII.6.1}$$

*The upper bound holds for every measurable choice of an unrestricted MLE. In particular, at $t \asymp N/\log(eN)$ its risk is $O(N^{-1/2})$, and the same order holds for every larger $t$.*

The statistical question extends the replication-range conjectures of Vinayak, Kong, Valiant and Kakade [70] and the discussion in Section 6.2 of Musco, Musco, Rosenblatt and Singh [71]. The preceding near-linear result already answered the fixed-polynomial-range question in that discussion. The estimation theorem is reproduced from the preceding compilation; its consequence for the full testing curve is derived below. It is not counted as a new estimation result here.

### VIII.6.1 An elementary finite-envelope bound for predictive likelihood

The following entropy argument is stated with its details because it must apply even when the true distribution has zero cells. Neither an interior mixing density nor a regular parametric likelihood is assumed.

**Lemma VIII.6.2** (Small covers of the binomial moment body)**.** *For $0 < \beta < 1$ the set $\{p_G : G \in \mathcal{P}([0,1])\}$ has a finite $\ell_1$ cover of radius $\beta$, with centers in that same set, whose cardinality $M$ satisfies*

$$\log M \le m \log \frac{C(m+1)}{\beta}, \qquad m = \min\left\{t, \left\lceil \sqrt{2t \log(16t/\beta)} \right\rceil\right\}. \tag{VIII.6.2}$$

*Proof.* View $K_x = (B_0^t(x), \dots, B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x = \sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k \ge 1$. Indeed, the even periodic function $F(z) = K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z = \theta + iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right| + \left|\frac{1-\cos z}{2}\right| = \cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1 = (\cosh u)^t \le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u = k/t$. This proves the claim. In particular, truncating after degree $m < t$ has vector $\ell_1$ error at most $2te^{-(m+1)^2/(2t)} \le \beta/8$, uniformly in $x$. For $m = t$ the error is zero.

Each mixing law contributes its first $m$ shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If $G, H$ lie in one cell, their truncated transforms differ in $\ell_1$ norm by at most $2m\beta/[8(m+1)] \le \beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most $\beta$. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

**Lemma VIII.6.3** (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let $p$ be the true law on a finite alphabet with $d$ cells, and suppose a model admits an $\ell_1$ cover of radius $\eta/d$, with $M$ probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

*For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most $e^{-x}$,*

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{VIII.6.3}$$

*Consequently*

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{VIII.6.4}$$

*Proof.* For each center $p_a$, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one $u_a$, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of $p$, where all sampled observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}
\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\
&= e^{N\varepsilon/2}\left(1 + \frac{\eta}{2} - \frac{h(p,u_a)^2}{2}\right)^N \\
&\le \exp\{N(\varepsilon + \eta - r^2)/2\}
\end{aligned}$$

when $h(p, u_a) > r$. Taking the union over the $M$ envelopes proves (VIII.6.3). Integrating that tail, and using $(a+b)^2 \le 2a^2 + 2b^2$, proves (VIII.6.4). □

**Corollary VIII.6.4** (A sufficient predictive Hellinger rate)**.** *For $1 \le t \le N$, every exact MLE satisfies*

$$\mathbb{E}_G h(p_G, \widehat{p}) \le \frac{C t^{1/4} L_N^{3/4}}{\sqrt{N}}. \tag{VIII.6.5}$$

*More generally, if $\ell(\widetilde{G}) \ge \ell^* - \varepsilon$, the right side can be increased by $C\sqrt{\varepsilon}$.*

*Proof.* Set $\eta = (eN)^{-2}$ and $\beta = \eta/(t+1)$ in the preceding lemmas. For $t \le N$, (VIII.6.2) gives $m \le C\sqrt{tL_N}$ and $\log M \le C\sqrt{t}\, L_N^{3/2}$. The MLE likelihood is at least that of the true law. The approximate likelihood condition loses at most $N\varepsilon$. Apply (VIII.6.4) and Jensen's inequality. □

The logarithmic factors in (VIII.6.5) need not be sharp. The argument below only needs the strict improvement from effective dimension $t$ to a quantity of order $\sqrt{t}$ up to logarithms.

**Lemma VIII.6.5** (Likelihood transport, recalled)**.** *For every binomial-mixture MLE,*

$$W_{1,t}(q, \widehat{p})^2 \le \frac{D(q\|\widehat{p})}{2t} \le \frac{D(q\|p_G)}{2t}. \tag{VIII.6.6}$$

*Consequently*

$$\mathbb{E}_G W_{1,t}(\widehat{p}, p_G) \le C/\sqrt{N}. \tag{VIII.6.7}$$

*Proof.* In score sums, terms with $q_j = 0$ are set to zero; every observed cell has positive fitted probability. The likelihood score condition gives $S(x) = \sum_j q_j K_x(j)/\widehat{p}(j) \le 1$, and its integral against $\widehat{G}$ is one. Thus $S = 1$ almost surely under $\widehat{G}$. The posterior vectors $R_x(j) = q_j K_x(j)/\widehat{p}(j)$ are therefore probability laws, with mixture $q$, and

$$\int D(R_x\|K_x)\, d\widehat{G}(x) = D(q\|\widehat{p}).$$

Every normalized-count binomial kernel satisfies $W_{1,t}(R, K_x)^2 \le D(R\|K_x)/(2t)$: a 1-Lipschitz test of the normalized sum of $t$ Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8t)$ by bounded differences; entropy duality and optimization over $\lambda$ give the inequality. Mix the couplings, use Cauchy–Schwarz, and then use likelihood optimality to prove (VIII.6.6). Also $\mathbb{E}D(q\|p_G) \le t/N$, by $D(q\|p_G) \le \sum_{j:p_G(j)>0}(q_j - p_G(j))^2/p_G(j)$. Finally $\mathbb{E}W_{1,t}(q, p_G) \le 1/(2\sqrt{N})$ by integrating the standard deviation of the empirical distribution function. The triangle inequality proves (VIII.6.7). □

*Upper bound in Theorem VIII.6.1.* It suffices first to treat $t \le N$. Bounded $N$ or bounded $t$ are handled by the loss bound one. Fix a sufficiently small absolute $a > 0$, for example $a = 1/64$, and take

$$k = \left\lfloor \min\{t/2, \sqrt{atL_N}\} \right\rfloor, \qquad s = \lfloor \sqrt{t}/4 \rfloor.$$

For sufficiently large $N$ and $t$, these satisfy $1 \le s \le k \le t/2$, $s \asymp \sqrt{t}$, and

$$1/k \le C\{1/t + (tL_N)^{-1/2}\}, \qquad e^{4s^2/t} \le C.$$

Use Proposition VIII.5.6, Lemma VIII.6.5, and Corollary VIII.6.4. This gives

$$\mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{k} + \frac{C}{\sqrt{N}} + \frac{C e^{2k^2/t} L_N^{3/4}}{\sqrt{N}\, t^{1/4}}. \tag{VIII.6.8}$$

The final term, multiplied by $\sqrt{tL_N}$, is at most

$$C(eN)^{2a} N^{-1/2} t^{1/4} L_N^{5/4} \le C(eN)^{2a} N^{-1/4} L_N^{5/4} \le C.$$

The last bound is uniform because $2a < 1/4$. This proves the desired upper bound when $t \le N$.

For $t \ge N$, couple every mixing parameter $x$ to its normalized binomial count. The expected absolute difference is at most $\sqrt{x(1-x)/t} \le 1/(2\sqrt{t})$. Hence, after identifying count laws with measures on $[0,1]$,

$$W_1(\widehat{G}, G) \le 1/\sqrt{t} + W_{1,t}(\widehat{p}, p_G).$$

Take expectations and apply (VIII.6.7). This is $O(N^{-1/2})$ for $t \ge N$, and completes the proof of the upper bound. □

### VIII.6.2 Matching lower bounds

*Lower bound in Theorem VIII.6.1.* The $N^{-1/2}$ term follows from mixing laws supported on $\{0,1\}$ with weights $1/2 \pm c/\sqrt{N}$. Each count then reveals exactly one endpoint label, regardless of $t$. Their $N$-sample total variation stays bounded away from one for sufficiently small absolute $c$.

For the other terms let $\pi(dx) = dx/[\pi\sqrt{x(1-x)}]$ and $h_k(x) = T_k(2x-1)$. Define $G_k^{\pm} = (1 \pm h_k/2)\pi$. Integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp 1/k.$$

If $k = t+1$, the count laws are identical, proving the $1/t$ lower bound. For $k \le t$, draw $X \sim \pi$, $J \mid X \sim \mathrm{Binomial}(t, X)$ and put $Qh(j) = \mathbb{E}[h(X) \mid J = j]$. The posterior is $\mathrm{Beta}(j+1/2, t-j+1/2)$. The self-adjoint operator $Q^*Q$ preserves polynomial degrees and has leading degree-$k$ coefficient

$$\lambda_{t,k} = \prod_{i=0}^{k-1} \frac{t-i}{t+1+i}.$$

Orthogonality of the Chebyshev polynomials therefore gives $Q^*Qh_k = \lambda_{t,k}h_k$ and $\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k}/2$. This is the classical beta–binomial diagonalization [72, Section 4.1]. The count laws are $p_\pi(1 \pm Qh_k/2)$, so

$$\chi^2(p_{G_k^+} \| p_{G_k^-}) \le 2\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k} \le e^{-k^2/(2t)}.$$

Take $k = \lceil\sqrt{8t\log(8N)}\rceil$ if this is at most $t$. The chi-squared divergence of the $N$-fold products is $(1+\chi^2)^N - 1 \le e^{N(8N)^{-4}} - 1$, so their total variation is bounded by an absolute constant smaller than one. The two-point testing bound yields risk at least $c/k \ge c'/\sqrt{tL_N}$. If this choice has $k > t$, then $t \le CL_N$ apart from bounded $t$, and $1/\sqrt{tL_N} \le C'/t$; the exact moment-matching lower bound already suffices. The maximum of the three lower bounds is comparable to their sum clipped at one, proving (VIII.6.1). □

### VIII.6.3 The full worst-null testing curve

Fix $\alpha, \beta \in (0,1)$ with $\alpha + \beta < 1$. A null-dependent test $\psi_{G_0}$ has level $\alpha$ if $\mathbb{E}_{G_0}\psi_{G_0} \le \alpha$. Define $\varepsilon^*_{N,t}(\alpha,\beta)$ as the infimum of the separations $\varepsilon$ for which such tests exist for every $G_0$, with $\mathbb{E}_G(1-\psi_{G_0}) \le \beta$ whenever $W_1(G, G_0) \ge \varepsilon$. Thus the rate is uniform in the null law; it need not describe the difficulty at each particular fixed null.

**Theorem VIII.6.6** (All trial counts, including the critical window)**.** *For all integers $N, t \ge 1$,*

$$\varepsilon^*_{N,t}(\alpha,\beta) \asymp_{\alpha,\beta} \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}}\right\}.$$

*An upper bound is attained by the plug-in test based on any unrestricted NPMLE. In particular the parametric plateau begins at $t \asymp N/\log(eN)$, and the window left in the attached manuscript has no remaining logarithmic gap.*

*Proof.* Let $r = r_{N,t}$ and let $C$ be the upper-bound constant in Theorem VIII.6.1. Reject when $W_1(\widehat{G}, G_0) > Cr/\alpha$. Markov's inequality gives level $\alpha$. If $W_1(G, G_0) > Cr(\alpha^{-1} + \beta^{-1})$, the triangle inequality bounds acceptance probability by $\Pr_G\{W_1(\widehat{G}, G) > Cr/\beta\} \le \beta$. Using a slightly larger constant handles non-strict separation.

For the lower bound use the same three pairs constructed in the lower proof of Theorem VIII.6.1, taking one member as the null and the other as the alternative. The degree-$t+1$ pair gives identical sample laws and separation at least $c/t$. For the middle term choose $k = \lceil \sqrt{C_{\alpha,\beta} t \log(eN)} \rceil$ when $k \le t$; the displayed chi-square estimate makes the product-law total variation smaller than $1 - \alpha - \beta$ by taking $C_{\alpha,\beta}$ large. If $k > t$, the $1/t$ pair already gives the required order. Finally choose the endpoint-weight perturbation of size $c_{\alpha,\beta}/\sqrt{N}$ with $c_{\alpha,\beta}$ small, giving the $N^{-1/2}$ obstruction. For any two sample laws $P_0, P_1$, the sum of type I and type II errors is at least $1 - \|P_0 - P_1\|_{\mathrm{TV}}$. These pairs therefore prove the maximum of the three lower bounds. Their maximum is comparable to their sum clipped at one, as required. □

Kania, Wasserman and Balakrishnan [69, Section 3, Theorem 3] give the corresponding worst-null lower curve and discuss the gap in the intermediate replication range. The attached manuscript's Corollary D used only the near-linear estimation bound. Theorem VIII.6.6 closes its remaining range by inserting the complete estimation theorem; the testing transfer itself is the same elementary Markov argument.

# VIII.7 Global Franel recurrences and arithmetic Apéry limits

The positive-moment argument in the attached draft explains why definite critical values force recurrence complexity. We retain that criterion, but remove it from the logical dependencies of the Franel classification. The replacement uses multiplicities of formal exponential factors and therefore covers complex weights and colliding critical values. We then extend the classification to every arithmetic subsequence and identify the entire space of Apéry limits over the algebraic numbers.

## VIII.7.1 The critical-value criterion and its scope

The width of an eventual nonzero recurrence $\sum_{j\in J} p_j(n)u_{n+j} = 0$, with $p_j \in \mathbb{C}(n)$, is $\max J - \min J$ after zero coefficients are removed. Here $J$ is a finite set of integers; gaps are allowed. The zero sequence has minimum width zero. A sequence with positive minimum width is understood modulo finitely many initial terms. Width is not the number of its nonzero terms and is not the order of an ordinary differential equation.

**Proposition VIII.7.1** (Visibility of same-side definite critical values)**.** *Let $\Phi$ be real analytic on a compact real analytic manifold of positive dimension, equipped with a positive analytic density $d\mu$. Set $u_n = \int \Phi^n \, d\mu$. Suppose that a nonzero critical value $v$ has finitely many critical points above it, all nondegenerate local maxima, or all nondegenerate local minima. Regular points in the same fiber are allowed. Then some analytic continuation of $\sum_{n\geq 0} u_n z^n$ is singular at $z = 1/v$. Consequently, $s$ distinct such nonzero critical values force minimum recurrence width at least $s$.*

*Proof.* The analytic Morse lemma and radial integration give, near $v$, a pushforward density with singular part $\mathbf{1}_{\{\epsilon u>0\}}(\epsilon u)^{d/2-1}h(\epsilon u)$, where $u = x - v$, $d$ is the manifold dimension, $\epsilon$ is the common side and $h$ is analytic with $h(0) > 0$. The regular contribution is smooth. Positivity and the common side prevent cancellation of the first nonsmooth term. This remains true when $d/2-1$ is an integer: the one-sided polynomial then has a derivative jump. The Cauchy transform of this measure cannot be holomorphic through $v$, because its boundary imaginary part is a nonzero constant multiple of this nonsmooth density. Inverting the Cauchy variable gives the asserted singularity of the moment generating function.

For the recurrence assertion, translate shifts to $0, \dots, L$ and clear denominators. With $\theta = z\partial_z$, the coefficient recurrence gives an inhomogeneous differential equation whose operator has the form $\sum_{j=0}^{L} z^{L-j} p_j(\theta - j)$. Its highest derivative coefficient is $z^D H(z)$ with $\deg H \leq L$. An operator polynomial in $\theta$ removes the polynomial inhomogeneous term without creating new nonzero finite zeros in that leading coefficient. Every branch is regular away from those zeros and zero. Thus at most $L$ distinct nonzero finite singularities are possible. □

This proposition requires the stated real positivity and Hessian conditions. It does not assert that every complex critical value is visible. None of those conditions is needed in the next theorem.

## VIII.7.2 Every weight and every arithmetic subsequence

For $r \geq 2$ and $\lambda \in \mathbb{C}^*$ write

$$a_{r,\lambda}(N) = \sum_{k=0}^{N} \binom{N}{k}^r \lambda^k, \qquad u_{r,\lambda;q,b}(n) = a_{r,\lambda}(qn + b), \quad q \geq 1, \quad 0 \leq b < q.$$

**Theorem VIII.7.2** (Complete subsequence classification)**.** *For every integer $r \geq 2$, every $\lambda \in \mathbb{C}^*$, and all $q, b$ as above, the minimum eventual recurrence width of $u_{r,\lambda;q,b}$ over $\mathbb{C}(n)$ is*

| *parameter stratum* | *minimum width* |
|---|---|
| $\lambda \notin \{1, -1\}$ | $r$ |
| $\lambda = 1$ | $\lceil r/2 \rceil$ |
| $\lambda = -1,\ q$ *odd* | $2\lfloor r/2 \rfloor$ |
| $\lambda = -1,\ q$ *even*, $b$ *even* | $\lfloor r/2 \rfloor$ |
| $\lambda = -1,\ q$ *even*, $b$ *odd* | 0. |

*In the last case the sequence is identically zero. The same minimum holds over every field $K(n)$ with $\mathbb{Q}(\lambda) \subseteq K \subseteq \mathbb{C}$.*

The $q = 2, b = 0$ case contains the all-weight centered classification. For arbitrary $q, b$, the conclusion also rules out a loss of width under sampling, except for the stated alternating parity symmetry. Stoll's and McIntosh's constructions provide the symmetry-weight upper bounds [28, 29]. Weighted low-degree recurrences and the higher-Bessel product interpretation have established precedents [26, 60]. The theorem above is proved here by a uniform lower bound, including multiplicity at exceptional weights; it is not presented as a previously named conjecture about subsequences.

### VIII.7.3 The differential Galois input

Define the entire function

$$\mathcal{F}_r(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^r}.$$

It solves $((x\partial_x)^r - x)y = 0$. Work over $K = \mathbb{C}(t)$, and write $M_a$ for the rank-$r$ differential module whose scalar solutions include

$$f_a(t) = \mathcal{F}_r(at^r), \qquad a \in \mathbb{C}^*.$$

Equivalently its scalar equation is

$$\left\{ \left( \frac{t\partial_t}{r} \right)^r - at^r \right\} y = 0.$$

We use the following precise classical input. The differential Galois group of $M_a$ in its standard representation is

$$G_r = \begin{cases} \mathrm{SL}_r(\mathbb{C}), & r \text{ odd}, \\ \mathrm{Sp}_r(\mathbb{C}), & r \text{ even}. \end{cases} \tag{VIII.7.1}$$

This is Katz's Kloosterman-group calculation [59, Corollary 4.4.8]; the connection and its $r$-fold pullback are identified explicitly in [51, Proposition 2.4 and the proof of Proposition 2.7]. Scaling $t$ by a nonzero constant gives the assertion for every $a$. In particular these groups are connected and their standard representations are irreducible.

The formal decomposition at $t = \infty$ consists of $r$ rank-one factors with exponential parts

$$\exp(r\alpha\zeta t), \qquad \zeta^r = 1, \quad \alpha^r = a. \tag{VIII.7.2}$$

Their remaining parts are regular singular. Direct substitution gives formal solutions

$$e^{r\alpha\zeta t} t^{-(r-1)/2} (1 + O(t^{-1}));$$

the distinct leading exponential roots recursively determine the formal series. Only the exponential parts, with their multiplicities, will be used below. The duality relation is

$$M_a^* \simeq M_{(-1)^r a}. \tag{VIII.7.3}$$

For example, this follows by reversing the cyclic basis for the companion system and alternating its signs, or by formal adjunction with respect to $dx/x$ before the pullback.

**Lemma VIII.7.3** (Independence of distinct scale classes)**.** *Choose finitely many nonzero scales $a_1, \dots, a_s$. If $r$ is even, assume they are pairwise distinct. If $r$ is odd, assume $a_i \neq \pm a_j$ for $i \neq j$. The connected differential Galois group of $\bigoplus_i M_{a_i}$ is the full product $G_r^s$.*

*Proof.* The connected joint group projects onto each individual group: its image has finite index in the connected group $G_r$. Its Lie algebra is therefore a subdirect product of simple Lie algebras. By the Lie-algebra form of Goursat's lemma, a proper subdirect product identifies two of these simple factors by an automorphism. Automorphisms of $\mathfrak{sp}_r$ are inner; automorphisms of $\mathfrak{sl}_r$ are inner or the standard duality automorphism. Consequently the two standard representations on the connected joint group would be isomorphic, or in the odd case dual.

Passing to the finite algebraic extension corresponding to the identity component makes this an isomorphism of the associated differential modules. But an isomorphism preserves the formal exponential parts. After any common finite ramification at infinity, (VIII.7.2) shows that $M_a$ and $M_b$ can be isomorphic only if $a = b$, and that $M_a$ and $M_b^*$ can be isomorphic only if $a = (-1)^r b$. Indeed equality of the multisets of coefficients of the exponential terms, followed by taking their $r$th powers, gives these equalities. Both possibilities contradict the scale hypotheses. Thus no pair of simple factors is identified and the Lie algebra is the full product. A connected closed subgroup with this Lie algebra is the full product group. □

### VIII.7.4 A factorial-normalization budget with an arbitrary sampling interval

For a nonzero differential operator $P = \sum_i b_i(t)\partial_t^i$ with formal Laurent coefficients at infinity, let $v$ be the largest leading power of $t$ among its coefficients and put

$$\sigma_\infty(P)(\xi) = \sum_i \left( \lim_{t\to\infty} t^{-v} b_i(t) \right) \xi^i.$$

The limit means the constant coefficient after Laurent normalization. Differentiating a Laurent coefficient lowers its degree, so

$$\sigma_\infty(PQ) = \sigma_\infty(P)\sigma_\infty(Q) \tag{VIII.7.4}$$

(up to a harmless nonzero constant if another normalization is used).

**Lemma VIII.7.4** (Nonzero exponential multiplicities bound width)**.** *Let*

$$H(t) = \sum_{n\geq 0} \frac{u_n}{((qn+b)!)^r} t^{r(qn+b)}.$$

*Suppose the differential module generated by this scalar function has at infinity a decomposition into rank-one factors with exponential parts $e^{\beta t}$ and regular-singular remaining parts. If $\nu$ of these factors have $\beta \neq 0$, counted with multiplicity, then every eventual recurrence of $u_n$ has width at least $\nu/(qr)$.*

*Proof.* Write a proposed width-$L$ recurrence as $\sum_{j=0}^{L} p_j(n)u_{n-j} = 0$, with $p_j \in \mathbb{C}[n]$. Let $D = \max \deg p_j$ and let $c_j$ be the leading coefficient when $\deg p_j = D$. Multiplication by $((qn+b)_{\underline{qL}})^r/((qn+b)!)^r$ gives $PH = Q(t)$, with $Q$ polynomial and

$$P = \sum_{j=0}^{L} t^{qrj} p_j \left(\frac{t\partial_t}{qr} + j - \frac{b}{q}\right)^{q(L-j)-1} \prod_{\ell=0} \left(\frac{t\partial_t}{r} - \ell\right)^r.$$

Indeed the coefficient contributed by the $j$th term at exponent $r(qn+b)$ is $p_j(n)u_{n-j}/((qn+b-qL)!)^r$ for sufficiently large $n$. The finitely many other coefficients account for $Q$. The largest coefficient degree in $t$ is $D + qrL$, and

$$\sigma_\infty(P)(\xi) = \sum_{\deg p_j = D} c_j \left(\frac{\xi}{qr}\right)^D \left(\frac{\xi}{r}\right)^{qr(L-j)}. \tag{VIII.7.5}$$

Distinct $j$ give distinct powers, so this is nonzero and has at most $qrL$ nonzero roots with multiplicity. Left multiplication by a polynomial in $t\partial_t$ killing $Q$ makes the equation homogeneous and adds only zero roots to the principal symbol.

Let $R$ be the monic minimal scalar annihilator of $H$. Its formal rank-one decomposition gives a first-order factorization with factors $\partial_t - \beta_i + O(t^{-1})$: take a formal rank-one solution, right-divide by its logarithmic-derivative factor, and repeat on the quotient. Laurent gauge changes add only $O(t^{-1})$. Thus $\sigma_\infty(R) = \prod_i(\xi - \beta_i)$. The homogeneous annihilator is a left multiple of $R$. Equation (VIII.7.4) then forces its principal symbol to contain all $\nu$ nonzero factors of $R$, with multiplicity. Consequently $\nu \le qrL$. □

### VIII.7.5 Cyclicity after root-of-unity filtering

The Cauchy-product identity gives

$$\mathcal{F}_r(x)\mathcal{F}_r(\lambda x) = \sum_{N\ge 0} \frac{a_{r,\lambda}(N)}{(N!)^r} x^N.$$

Writing $\mu_q = \{\epsilon : \epsilon^q = 1\}$, its root-of-unity filter is

$$H_{q,b}(t) = \frac{1}{q} \sum_{\epsilon\in\mu_q} \epsilon^{-b} f_\epsilon(t) f_{\lambda\epsilon}(t) = \sum_{n\ge 0} \frac{a_{r,\lambda}(qn+b)}{((qn+b)!)^r} t^{r(qn+b)}. \tag{VIII.7.6}$$

All products lie in one Picard–Vessiot field. Multiplication of scalar solutions is an equivariant linear map from their tensor product. On an irreducible representation a nonzero such map is injective. Distinct nonisomorphic irreducible images have direct sum, and a vector with a nonzero component in each generates the whole direct sum. These facts establish cyclicity for the specific filtered function, not just for an ambient system that annihilates it.

*Lower bounds in Theorem VIII.7.2.* First suppose $\lambda \ne \pm 1$. Each product in (VIII.7.6) belongs to an irreducible rank-$r^2$ tensor representation. For even $r$ the two scales are distinct independent symplectic factors. For odd $r$ they are distinct modulo sign, and hence independent special-linear factors. Use Lemma VIII.7.3 for the finite set of all scale classes occurring in the filter.

The $q$ product representations are pairwise nonisomorphic even when their scales overlap. For even $r$, their two-factor supports agree only if the unordered pairs $\{\epsilon, \lambda\epsilon\}$ agree. For odd $r$, a representation also records whether it is standard or dual at each signed scale class: these are inequivalent for odd $r \ge 3$. Hence isomorphism again requires equality of the unordered pairs of actual scales. For two distinct indices $\epsilon, \eta$, such equality would require $\eta = \lambda\epsilon$ and $\lambda\eta = \epsilon$, forcing $\lambda^2 = 1$, which is excluded. Since every coefficient $\epsilon^{-b}/q$ is nonzero, the filter is cyclic

in all $q$ irreducible summands. A formal exponential coefficient in $M_a \otimes M_c$ can vanish only if $c = (-1)^r a$. This too is excluded. Therefore $\nu = qr^2$ and the budget gives $L \geq r$.

For $\lambda = 1$, the summands are $\operatorname{Sym}^2 M_\epsilon$. Their representations are irreducible and pairwise nonisomorphic. For even $r$ this follows from distinct symplectic scale factors. For odd $r$, opposite scales if present give $\operatorname{Sym}^2 V$ and $\operatorname{Sym}^2 V^*$, which are nonisomorphic for $r \geq 3$. In each summand the formal exponents are the unordered sums of two $r$th-root exponents. There are $r/2$ zero sums when $r$ is even, and none when $r$ is odd. The nonzero rank per summand is thus $r^2/2$ for even $r$, and $r(r+1)/2$ for odd $r$. The $q$ nonzero filter components give respectively $L \geq r/2$ and $L \geq (r+1)/2$.

Now let $\lambda = -1$. The products indexed by $\epsilon$ and $-\epsilon$ coincide. If $q$ is even, their filter coefficients sum to zero for odd $b$, and to $2\epsilon^{-b}/q$ for even $b$. This proves the asserted vanishing in the first case. In the second case there are $q/2$ surviving products. If $q$ is odd, none of these pairs repeats and there are $q$ surviving products.

For even $r$, each surviving product is an irreducible tensor of two independent symplectic modules. Distinct surviving products have disjoint scale pairs, so they are nonisomorphic. Each has $r^2$ nonzero formal factors, since zero would require the scales to be equal rather than opposite.

For odd $r$, each product lies in $V \otimes V^* = \mathbf{1} \oplus \mathrm{Ad}$ for the special-linear factor indexed by the class $\{\epsilon, -\epsilon\}$. The multiplication map kills the trivial summand. To check this without assuming that invariant scalar functions are constant, put $\delta = (t/r)\partial_t$. For solutions $y, z$ of $\delta^r y = \epsilon t^r y$, $\delta^r z = -\epsilon t^r z$, the nondegenerate concomitant

$$\mathcal{B}(y,z) = \sum_{j=0}^{r-1} (-1)^j (\delta^j y)(\delta^{r-1-j} z)$$

is constant. Indeed its derivative telescopes to zero. Choose dual solution bases $y_i, z_i$ for this pairing and jet matrices $Y, Z$ whose rows are $\delta^j$, $0 \leq j < r$. Then $Y^{\mathsf{T}} J Z = I$, with $J$ invertible and anti-diagonal, whence

$$\sum_i y_i z_i = (Y Z^{\mathsf{T}})_{00} = (J^{-\mathsf{T}})_{00} = 0 \qquad (r \geq 2).$$

The selected product is nonzero, so the multiplication map is injective on the irreducible adjoint summand. Different surviving products use different independent scale classes. Each therefore contributes the $r(r-1)$ nonzero ordered differences of distinct $r$th-root exponents. There are $q$ such products for odd $q$, or $q/2$ for even $q$ and even $b$. Dividing their nonzero ranks by $qr$ gives respectively $2\lfloor r/2 \rfloor$ and $\lfloor r/2 \rfloor$, in both parities of $r$. □

*Upper bounds and coefficient fields.* For $\lambda \neq \pm 1$ set $\mathcal{L}_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k$. An index shift gives

$$\mathcal{L}_N[k^{r+h} - \lambda (N-k)^r (k+1)^h] = 0 \qquad (h \geq 0).$$

The polynomial displayed here has leading coefficient $1 - (-1)^r \lambda \neq 0$. Successive elimination reduces every moment to $\mathcal{L}_N[1], \ldots, \mathcal{L}_N[k^{r-1}]$. The identity $\binom{N-1}{k}^r = N^{-r}(N-k)^r \binom{N}{k}^r$ therefore gives a backward $r$-dimensional state relation over $\mathbb{Q}(\lambda)(N)$. Apply it $q$ times between observations: $r+1$ consecutive observations on the progression $qn + b$ are rows on the same $r$-dimensional state, so they satisfy a recurrence of width at most $r$. This argument needs no matrix inverse.

For $\lambda = 1$, Stoll's ordinary Franel recurrence has width at most $m = \lceil r/2 \rceil$ over $\mathbb{Q}(n)$ [28]. A rational companion system has dimension $m$; its $q$-step system gives the same upper bound on every progression. For $\lambda = -1$, symmetry gives $a_{r,-1}(N) = 0$ for odd $N$. McIntosh's recurrence [29, Theorem 1] gives width at most $h = \lfloor r/2 \rfloor$ for $A_r(n) = (-1)^n a_{r,-1}(2n)$. Geometric rescaling preserves width, so even progressions have the claimed upper bound $h$. Interlacing this recurrence

with the zero odd sequence gives a width-$2h$ recurrence for $a_{r,-1}(N)$: a recurrence on its even entries with shifts $j$ becomes one with shifts $2j$, valid also on the zero odd entries. Taking a $q$-step companion system gives the upper bound $2h$ for odd $q$. All upper constructions have coefficients in $\mathbb{Q}(\lambda)(n)$. Since the lower bounds hold over $\mathbb{C}(n)$, extending the coefficient field anywhere between these fields cannot change the minimum. □

**Corollary VIII.7.5** (The Franel and McIntosh statements of the draft)**.** *For every $r \geq 2$,*

$$\operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=0}^{n} \binom{n}{k}^r = \lceil r/2 \rceil, \qquad \operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=-n}^{n} (-1)^k \binom{2n}{n+k}^r = \lfloor r/2 \rfloor.$$

*More generally, the centered weighted sequence $\lambda^{-n} a_{r,\lambda}(2n)$ has width $\lceil r/2 \rceil$ at $\lambda = 1$, $\lfloor r/2 \rfloor$ at $\lambda = -1$, and $r$ elsewhere.*

*Proof.* Use Theorem VIII.7.2 with $(q, b) = (1, 0)$ or $(2, 0)$. The centered alternating sum equals $(-1)^n a_{r,-1}(2n)$, and geometric rescaling preserves width. □

The alternating conclusion is Theorem E of the attached draft, now independent of a critical-value visibility argument. The unweighted conclusion removes the restriction to telescoping recurrences in [74, Theorem 1.1]. Critical-value collisions do not create exceptions in Theorem VIII.7.2: the budget counts formal exponential factors with multiplicity, not distinct critical values.

### VIII.7.6 The whole arithmetic space of Apéry limits

Put $A_r(N) = a_{r,1}(N)$ and $m = \lceil r/2 \rceil$. In this subsection the notation $A_r$ denotes the ordinary, not the alternating, Franel sequence. Define rational sequences $A_{r,j}(N)$ by

$$\sum_{j \geq 0} A_{r,j}(N) z^{2j} = \sum_{k=0}^{N} \binom{N}{k}^r \left\{ \prod_{\ell=1}^{k} \left(1 - \frac{z}{\ell}\right) \prod_{\ell=1}^{N-k} \left(1 + \frac{z}{\ell}\right) \right\}^{-r}. \tag{VIII.7.7}$$

The right side is even by replacing $k$ with $N - k$, and $A_{r,0} = A_r$. Let

$$\phi_{r,j} = [z^{2j}] \left(\frac{z}{\sin z}\right)^r \in \mathbb{Q}_{>0}.$$

The input from Straub–Zudilin [74, Theorem 1.3] is that $A_{r,j}$, $0 \leq j < m$, satisfy every telescoping recurrence for $\binom{N}{k}^r$ on a sufficiently late tail, and have limits

$$\lim_{N \to \infty} \frac{A_{r,j}(N)}{A_r(N)} = \phi_{r,j} \pi^{2j}. \tag{VIII.7.8}$$

Van Hoeij's invariant-submodule calculation [76, Remark 6] gives a telescoper of order $m$ for this summand. These are published inputs; the transfer to the unrestricted minimal recurrence is supplied by the preceding theorem.

For $q \geq 1$ and $0 \leq b < q$, let $L_{r;q,b}$ denote the monic minimal eventual recurrence of $A_r(qn+b)$, with shifts normalized to $0, \ldots, m$. If $K$ is a field between $\mathbb{Q}$ and $\overline{\mathbb{Q}}$, write $\mathcal{S}_{r;q,b}(K)$ for the $K$-valued solutions on a sufficiently late regular tail, identifying sequences that agree eventually.

**Theorem VIII.7.6** (Complete arithmetic Apéry-limit classification)**.** *For every $r \geq 2$, $q \geq 1$, $0 \leq b < q$ and $\mathbb{Q} \subseteq K \subseteq \overline{\mathbb{Q}}$, the following statements hold.*

1. *$L_{r;q,b}$ has order $m$ and coefficients in $\mathbb{Q}(n)$. For $(q, b) = (1, 0)$ it is also the monic minimal telescoper for the summand $\binom{n}{k}^r$.*

2. *The sequences $A_{r,j}(qn+b)$, $0 \le j < m$, are a basis of $\mathcal{S}_{r;q,b}(K)$. Every complex tail solution also has a unique expansion in this same basis, with complex constants.*

3. *Every $X \in \mathcal{S}_{r;q,b}(K)$ has an Apéry limit, and*

$$X \longmapsto \lim_{n\to\infty} \frac{X(n)}{A_r(qn+b)} \quad \text{is a } K\text{-linear isomorphism onto} \quad \bigoplus_{j=0}^{m-1} K\pi^{2j}. \tag{VIII.7.9}$$

*In particular two algebraic-number-valued tail solutions with the same limit are equal eventually.*

*Proof.* For $q = 1, b = 0$, the minimal recurrence has order $m$ by Corollary VIII.7.5. Stoll provides an annihilator of this order over $\mathbb{Q}(n)$, so its monic normalization is the unique minimal operator even over $\mathbb{C}(n)$. The minimal telescoper supplied by van Hoeij has the same order and annihilates the sequence. Right division by the monic minimal recurrence leaves a lower-order annihilator as remainder, necessarily zero. Equality of orders thus makes the two monic operators equal. Equation (VIII.7.8) therefore applies to solutions of this minimal recurrence.

Pass to a tail beyond all poles and integer zeros of its leading and trailing coefficients, and beyond the finitely many exceptional indices for the $A_{r,j}$. A companion system has dimension $m$. Its $q$-step system yields one common rational-coefficient recurrence of order at most $m$ for every sampled solution. By Theorem VIII.7.2, the sampled sequence $A_r(qn+b)$ has minimum order exactly $m$. Hence this common operator has order $m$ and, after normalization, is $L_{r;q,b}$.

The $m$ rational sequences $A_{r,j}(qn+b)$ are linearly independent over $\mathbb{C}$. Indeed a complex dependence among rational columns of an infinite matrix implies a rational dependence: the row space is finite-dimensional and can be spanned by finitely many rational rows, whose nullspace is defined over $\mathbb{Q}$. A rational dependence on any tail would, after division by $A_r(qn+b)$ and passage to the limit, contradict the transcendence of $\pi$ in (VIII.7.8). The solution space of a regular order-$m$ tail recurrence has dimension $m$, so these columns are a basis over $\mathbb{C}$.

Evaluation at $m$ consecutive sufficiently late indices is an isomorphism from this solution space to $\mathbb{C}^m$. The resulting basis matrix has rational entries and nonzero determinant, hence rational inverse. A solution taking values in $K$ consequently has all its basis coefficients in $K$, proving the basis assertion over $K$. Its limit is $\sum_{j=0}^{m-1} c_j \phi_{r,j} \pi^{2j}$. The numbers $1, \pi^2, \dots, \pi^{2m-2}$ are linearly independent over $\overline{\mathbb{Q}}$, and every $\phi_{r,j}$ is nonzero rational. This proves injectivity and the stated image. □

An algebraic-number-valued solution need not be assumed in advance to lie in one number field: its $m$ initial tail values generate a finite extension of $\mathbb{Q}$, and the rational recurrence then keeps its entire tail in that field. Thus the theorem covers all such solutions, including those initially described only as $\overline{\mathbb{Q}}$-valued.

**Corollary VIII.7.7** (The normalized limits in Theorem F, over algebraic numbers)**.** *For $1 \le j < m$ put*

$$D_{r,j}(N) = \frac{A_{r,j}(N)}{2^{\binom{r+2j-1}{2j}}}.$$

*Then $D_{r,j}(0) = 0$, $D_{r,j}(1) = 1$, and it is a solution of the minimal Franel recurrence on a tail, with*

$$\lim_{N\to\infty} \frac{D_{r,j}(N)}{A_r(N)} = \frac{\phi_{r,j}\pi^{2j}}{2^{\binom{r+2j-1}{2j}}}.$$

*It is the unique algebraic-number-valued tail solution with that limit. The same uniqueness holds for its sampled sequence $D_{r,j}(qn+b)$ relative to $L_{r;q,b}$ and $A_r(qn+b)$. In particular, for $r \ge 3$*

*and $r \geq 5$, respectively,*

$$\lim \frac{D_{r,1}(N)}{A_r(N)} = \frac{\zeta(2)}{r+1}, \qquad \lim \frac{D_{r,2}(N)}{A_r(N)} = \frac{3(5r+2)\zeta(4)}{(r+1)(r+2)(r+3)}.$$

*Proof.* At $N = 0$ the deformation is 1, while at $N = 1$ it is $(1-z)^{-r}+(1+z)^{-r}$. The stated initial values follow by coefficient extraction. The limits and uniqueness follow from Theorem VIII.7.6. Finally $\phi_{r,1} = r/6$ and $\phi_{r,2} = r(5r+2)/360$ give the two special formulas, already computed in [74, Examples 1.5 and 1.7]. □

### VIII.7.7 Exact scope of the conjectural statements

Chamberland–Straub [75, Conjectures 9 and 11] state the normalized $\zeta(2)$ and $\zeta(4)$ limits, respectively for $r \geq 3$ and $r \geq 5$, and include uniqueness. Their wording does not specify a coefficient field for the secondary solution. The numerical values of the limits and the deformed rational sequences are already in Straub–Zudilin. What the argument here supplies is their membership in the unrestricted minimal recurrence, complete uniqueness over $\overline{\mathbb{Q}}$, and the same conclusions for every arithmetic subsequence. It does not establish unrestricted complex uniqueness.

In fact, on the complex tail solution space the limit map has a kernel of dimension $m-1$. For $r \geq 7$ this obstruction persists even with the two displayed initial normalizations in the eventual sense. To make it explicit, set $c_j = A_{r,j}(1)$ and $\ell_j = \phi_{r,j}\pi^{2j}$, $j = 1, 2, 3$, and put

$$Z = (c_2\ell_3 - c_3\ell_2)A_{r,1} + (c_3\ell_1 - c_1\ell_3)A_{r,2} + (c_1\ell_2 - c_2\ell_1)A_{r,3}.$$

Then $Z(0) = Z(1) = 0$, $Z/A_r \to 0$, and $Z$ is not the zero tail solution: its coefficient of $A_{r,3}$ is nonzero by the transcendence of $\pi$, and the basis is independent. Thus $D_{r,1} + cZ$ has the same two initial values and limit for every $c \in \mathbb{C}$, and satisfies the recurrence eventually. An assertion requiring the recurrence to hold at every small nonnegative index needs a separate regularity argument; no such assertion is inferred here from eventual annihilation. The coefficient-degree prediction in [74, Remark 4.1] is likewise not settled by a width bound.

## VIII.8 A local Fourier proof and sharpening of the Laplace–Jordan principle

Theorem J in the attached draft can be proved without identifying global Fourier solutions with rapid-decay periods or rotating their integration chains. The replacement uses formal stationary phase and the elementary nearby-cycle/vanishing-cycle description at one point. It also identifies which eigenvalue can acquire an additional Jordan step.

**Lemma VIII.8.1** (Nearby and vanishing monodromy)**.** *Let $V, W$ be finite-dimensional complex vector spaces, with maps $u : V \to W$ and $v : W \to V$, and suppose*

$$T = I_V + vu, \qquad S = I_W + uv$$

*are invertible. For every polynomial $p \in \mathbb{C}[z]$,*

$$(T - I_V)p(T) = v\,p(S)u. \tag{VIII.8.1}$$

*In particular, $p(S) = 0$ implies $(T - I_V)p(T) = 0$. For each $\lambda \neq 1$, the restrictions of $T$ and $S$ to their $\lambda$-generalized eigenspaces are isomorphic. Only the 1-eigenspace can acquire a further Jordan step in passing from $S$ to $T$.*

*Proof.* The relation $uT = Su$ implies $up(T) = p(S)u$. Multiplication by $v$ proves (VIII.8.1). On a $\lambda$-generalized eigenspace with $\lambda \neq 1$, $T - I_V$ and $S - I_W$ are invertible. Thus $u$ and $v$ give mutually inverse isomorphisms after composing one of them with the appropriate inverse of $T - I_V$ or $S - I_W$. The identity then proves the assertion at $\lambda = 1$ as well. □

We use local monodromy at infinity in the positive local coordinate $1/s$. With this convention, regular local Fourier transform preserves its monodromy conjugacy class. A change of Fourier sign is a homothety of the target and does not alter that class.

**Proposition VIII.8.2** (The Fourier monodromy bound)**.** *Let $\mathcal{M}$ be a holonomic module on the complex affine $s$-line, whose generic connection is regular singular at infinity. Let $M$ be the monodromy of that regular germ. Write $T$ for the monodromy at $x = 0$ of the generic connection underlying its Fourier transform. Then the transformed connection has no singularity at a finite $x \neq 0$, is regular singular or ordinary at 0, and*

$$p(M) = 0 \quad \Longrightarrow \quad (T - I)p(T) = 0.$$

*The same annihilating polynomial applies to the monodromy of every differential subquotient of the transformed generic connection. No middle-extension hypothesis on $\mathcal{M}$ is required.*

*Proof.* Formal stationary phase gives

$$\Phi_x(\mathcal{F}\mathcal{M}) \simeq \mathcal{F}^{(\infty,x)}\big((\Psi_\infty \mathcal{M})^{\leq 1,(x)}\big)$$

for every finite $x$; this is [77, Corollary 2.3(1)]. Because $\Psi_\infty\mathcal{M}$ is regular, its exponential-type summand is zero unless $x = 0$. At $x = 0$ its local Fourier transform is again regular and has monodromy $M$: [41, Remark 5.2 and Remark 5.3(4)]. These statements concern the extended formal local Fourier functor, which includes trivial regular summands. For example, a trivial rank-one infinity germ contributes the trivial rank-one vanishing cycles of a delta module; it is not discarded.

For a holonomic module on a formal disc, its irregular part is the same in nearby and vanishing cycles. Thus the regularity of the vanishing cycles proves regularity of the nearby germ, and their vanishing at $x \neq 0$ proves smoothness there. On the regular part, nearby and vanishing cycles

have the description $(V, W, u, v)$ with monodromies $I + vu$ and $I + uv$, respectively; see [77, §3.3, formula (3.3)]. The vanishing monodromy $S$ is conjugate to $M$. Lemma VIII.8.1 therefore proves the asserted identity and bound. Restricting an annihilating polynomial to an invariant subspace or its quotient preserves the identity. The argument also works for the dual local system of solutions, since local Fourier transform is compatible with duality and both source and target monodromies are dualized. □

**Theorem VIII.8.3** (Eigenvalue-resolved iterated Borel bound)**.** *Let $A_0(t) = \sum_{n\ge 0} a_n t^n$ be a nonzero D-finite germ with positive radius of convergence. Suppose its minimal equation is regular singular or ordinary at 0, and denote its monodromy by $M$. For $r \ge 0$, put*

$$A_r(x) = \sum_{n\ge 0} \frac{a_n x^n}{(n!)^r}.$$

*Each minimal equation is regular singular or ordinary at 0. If $p(M) = 0$, then its monodromy $T_r$ satisfies*

$$(T_r - I)^r p(T_r) = 0. \tag{VIII.8.2}$$

*In particular, nonunit monodromy eigenvalues acquire no larger Jordan blocks under any number of Borel transforms. The largest Jordan-block size at eigenvalue 1 increases by at most $r$. If $A_0$ is algebraic, every nonunit block of $T_r$ is semisimple, and every block at eigenvalue 1 has size at most $r + 1$.*

*Proof.* It suffices to prove the assertion for one transform and iterate. Put $f(s) = s^{-1}A_0(s^{-1})$ and $\mathcal{M} = \mathbb{C}[s]\langle\partial_s\rangle \cdot f$. Its generic connection is the rational gauge and variable transform of the minimal equation of $A_0$. It is regular at infinity, and its local monodromy in the coordinate $1/s$ is $M$.

Here is the exact relation with the series transform. On $\mathbb{C}[s] \oplus s^{-1}\mathbb{C}[[s^{-1}]]$, define

$$\mathcal{B}(s^j) = 0 \quad (j \ge 0), \qquad \mathcal{B}(s^{-n-1}) = x^n/n! \quad (n \ge 0).$$

Direct checking on monomials gives $\mathcal{B}(sf) = \partial_x\mathcal{B}(f)$ and $\mathcal{B}(\partial_s f) = -x\mathcal{B}(f)$. Consequently, with Fourier convention $s \mapsto \partial_x$, $\partial_s \mapsto -x$, there is a surjection of holonomic modules

$$\mathcal{F}(\mathcal{M}) \longrightarrow \mathbb{C}[x]\langle\partial_x\rangle \cdot A_1.$$

The solution local system of the minimal equation of $A_1$ is therefore a subquotient of the Fourier solution local system. Proposition VIII.8.2 gives $(T_1 - I)p(T_1) = 0$ and regularity. Applying the same argument successively with the polynomial $(z - 1)^j p(z)$ proves (VIII.8.2).

For algebraic $A_0$, the local monodromy of its minimal equation has finite order: its solutions are spanned by finitely many algebraic conjugates. Its minimal polynomial is a product of distinct roots-of-unity factors. Multiplication by $(z - 1)^r$ therefore gives exactly the stated eigenvalue-specific bounds. □

Theorem J of the attached draft follows by taking $p(z) = (z^N - 1)^b$ and $r = 1$, since $(z^N - 1)^{b+1}$ is divisible by $(z - 1)(z^N - 1)^b$. Its algebraicity consequences use $r = 1$ and $r = 2$. More generally, a block of size $L$ at eigenvalue 1 in $A_r$ forces a block of size at least $L - r$ at eigenvalue 1 in $A_0$, whenever the regularity hypothesis on $A_0$ holds. In particular, a block of size $r + 2$ excludes algebraicity of $A_0$. This replaces the rapid-decay-period input for the attached algebraicity classification; it does not supply any assertion about numerical values of the ordinary generating functions.

## VIII.9 Bessel determinants and the three algebraicity classifications

This section supplies the combinatorial definitions, Bessel reductions, and scalar ordinary-point result used in the differential and arithmetic theorems of Section VIII.10. The Laplace input is Theorem VIII.8.3. In particular, no unpublished tableau operator theorem is needed for the arithmetic specialization.

### VIII.9.1 Definitions and classical determinant identities

Let $y_k(n)$ count standard Young tableaux of size $n$ with at most $k$ rows, and let $u_k(n)$ count permutations of $\{1,\dots,n\}$ whose longest increasing subsequence has length at most $k$. Let $f_k(n)$ count perfect matchings of $\{1,\dots,2n\}$ with no $k$ mutually crossing arcs. Thus a forbidden crossing consists of arcs $(a_i,b_i)$ with $a_1<\cdots<a_k<b_1<\cdots<b_k$. All three empty objects have count 1. Define the ordinary generating functions

$$\widetilde{Y}_k(t)=\sum_{n\ge0}y_k(n)t^n,\qquad \widetilde{U}_k(t)=\sum_{n\ge0}u_k(n)t^n,\qquad \widetilde{F}_k(t)=\sum_{n\ge0}f_k(n)t^n,$$

and the normalized functions

$$Y_k(x)=\sum_{n\ge0}y_k(n)\frac{x^n}{n!},\quad U_k(s)=\sum_{n\ge0}u_k(n)\frac{s^n}{(n!)^2},\quad F_k(x)=\sum_{n\ge0}f_k(n)\frac{x^{2n}}{(2n)!}.$$

Put $b_j(x)=I_j(2x)$, with $b_{-j}=b_j$. The classical Gordon–Gessel and Weyl-chamber determinant identities are

$$Y_{2r}(x)=\det[b_{i-j}(x)+b_{i+j-1}(x)]_{i,j=1}^{r}, \tag{VIII.9.1}$$
$$Y_{2r+1}(x)=e^x\det[b_{i-j}(x)-b_{i+j}(x)]_{i,j=1}^{r}, \tag{VIII.9.2}$$
$$U_k(s)=\det[I_{i-j}(2\sqrt{s})]_{i,j=1}^{k}, \tag{VIII.9.3}$$
$$F_k(x)=\det[b_{i-j}(x)-b_{i+j}(x)]_{i,j=1}^{k-1}. \tag{VIII.9.4}$$

Empty determinants equal 1. The tableau formulas are stated in [80, §1.2, equations (6)–(7)], with their attribution to Gordon and Gessel; the permutation formula is Gessel's [79]. Formula (VIII.9.4) is the Grabiner–Magyar formula [81], also displayed with exactly this normalization in [82, equation (1.4)]. These enumerative identities are the imported combinatorial inputs.

### VIII.9.2 Reduction to two Bessel coordinates

Write

$$y(x)=I_0(2x)=\sum_{m\ge0}\frac{x^{2m}}{(m!)^2},\qquad y_s(s)=I_0(2\sqrt{s})=\sum_{m\ge0}\frac{s^m}{(m!)^2}.$$

Their scalar equations and logarithmic companion solutions are

$$y''=4y-x^{-1}y',\qquad \kappa(x):=K_0(2x)=-(\log x)y(x)+h(x), \tag{VIII.9.5}$$
$$y_s''=s^{-1}(y_s-y_s'),\qquad \kappa_s(s):=K_0(2\sqrt{s})=-\tfrac12(\log s)y_s(s)+h_s(s), \tag{VIII.9.6}$$

where $h,h_s$ are holomorphic at zero. The integer-order Bessel recurrences and these local expansions may be found in [57, §§10.29, 10.31].

**Lemma VIII.9.1** (Bessel group and independence)**.** *The two Bessel equations have differential Galois group* $\mathrm{SL}_2$ *over their rational function fields. The pairs* $(y, y')$ *and* $(y_s, y_s')$ *are algebraically independent over* $\mathbb{C}(x)$ *and* $\mathbb{C}(s)$*, respectively.*

*Proof.* Kolchin's Bessel-group theorem gives $\mathrm{SL}_2$ for the ordinary Bessel equation of order zero [42]; the precise non-half-integer criterion is also stated in [83, Example 1.3.32]. A nonzero linear change of variable gives the modified equation (VIII.9.5). Equation (VIII.9.6) pulls back to it under $s = x^2$. Its Galois group contains that of this finite algebraic pullback and is contained in $\mathrm{SL}_2$ because its Wronskian is a constant times $1/s$. It is therefore $\mathrm{SL}_2$ as well.

For either equation, let $v$ be the analytic Bessel solution and $w$ an independent solution in a Picard–Vessiot extension. If $R(v, v') = 0$ with rational-function coefficients, the group $\mathrm{SL}_2$, transitive on nonzero solution vectors, gives $R(c_1 v + c_2 w, c_1 v' + c_2 w') = 0$ for all constants $c_1, c_2$. The fundamental matrix is invertible. A polynomial over the Picard–Vessiot field which vanishes for all $(c_1, c_2) \in \mathbb{C}^2$ is zero. Hence $R = 0$. □

**Lemma VIII.9.2** (Determinants are homogeneous forms)**.** *Each determinant in* (VIII.9.1)–(VIII.9.4) *is a nonzero homogeneous polynomial in the corresponding Bessel pair, with rational Laurent-polynomial coefficients. The degrees are* $\lfloor k/2 \rfloor$ *for* $e^{-\varepsilon x} Y_k(x)$*, where* $k = 2\lfloor k/2 \rfloor + \varepsilon$*;* $k$ *for* $U_k(s)$*; and* $k - 1$ *for* $F_k(x)$*.*

*Proof.* The recurrences

$$b_0 = y, \qquad b_1 = y'/2, \qquad b_{j+1} = b_{j-1} - (j/x) b_j$$

express every $b_j$ as a rational Laurent-polynomial linear form in $y, y'$. Thus a determinant of size $d$ is homogeneous of degree $d$.

For the $s$-variable put $J_j(s) = s^{-j/2} I_j(2\sqrt{s})$ for $j \geq 0$. Then $J_0 = y_s$, $J_1 = y_s'$ and $s J_{j+1} = J_{j-1} - j J_j$. Multiplying row $i$ of (VIII.9.3) by $s^{-i/2}$ and column $j$ by $s^{j/2}$ does not change its determinant, and changes its entry to $s^{\max(0, j-i)} J_{|i-j|}(s)$. All coefficients are therefore in $\mathbb{Q}[s, s^{-1}]$. Each resulting form is nonzero because the normalized counting series has constant coefficient 1. □

**Lemma VIII.9.3** (The logarithmic degree is exact)**.** *Let* $v$ *be either analytic Bessel solution above, in variable* $z$*, and let* $\Phi \in \mathbb{C}(z)[X_0, X_1]$ *be nonzero homogeneous of degree* $d \geq 1$*. The function* $\Phi(v, v')$ *has minimal differential order* $d + 1$*, and its minimal equation at zero has one unipotent Jordan block of size* $d + 1$*. Multiplication by* $e^{\varepsilon z}$ *does not change these two assertions.*

*Proof.* Extend differentiation to the polynomial ring in $X_0, X_1$ using the relevant Bessel equation. If a rational differential operator $L$ annihilates $\Phi(v, v')$, then the polynomial $L\Phi$ vanishes at $(v, v')$. Lemma VIII.9.1 implies $L\Phi = 0$ as a polynomial. Thus $L$ also annihilates $\Phi(w, w')$ for the logarithmic companion $w$.

By (VIII.9.5) or (VIII.9.6), this latter solution has the convergent expansion

$$\Phi(w, w') = \sum_{j=0}^{d} (\log z)^j g_j(z), \qquad g_d = c^d \Phi(v, v') \neq 0, \quad c = -1 \text{ or } -\tfrac{1}{2},$$

where the $g_j$ are meromorphic at zero. Therefore $(T - I)^d \Phi(w, w') = d!(2\pi i)^d g_d \neq 0$. On the other hand, all derivatives of $\Phi(v, v')$ belong to the $(d + 1)$-dimensional rational span of $v^d, v^{d-1} v', \ldots, (v')^d$. Its minimal order is at most $d + 1$, so the block just found occupies the entire solution space and the order is exactly $d + 1$. The exponential factor gives an invertible rational differential gauge $\partial_z \mapsto \partial_z - \varepsilon$ and is single-valued at zero. □

### VIII.9.3 Ordinary generating functions

**Theorem VIII.9.4** (The three complete algebraicity classifications)**.** *For positive integer parameters:*

1. *$\widetilde{Y}_k$ is algebraic if and only if $k \le 3$;*
2. *the ordinary generating function for permutations avoiding an increasing subsequence of length $\ell$ is algebraic if and only if $\ell \le 3$;*
3. *$\widetilde{F}_k$ is algebraic if and only if $k \le 2$.*

*Proof.* The small tableau cases are

$$\widetilde{Y}_1(t) = \frac{1}{1-t}, \qquad \widetilde{Y}_2(t) = \frac{\sqrt{(1+2t)/(1-2t)}-1}{2t}, \qquad \widetilde{Y}_3(t) = \frac{1-t-\sqrt{1-2t-3t^2}}{2t^2}.$$

The apparent denominators at zero are removable in the chosen formal branches. The latter two formulas follow respectively from $y_2(n) = \binom{n}{\lfloor n/2 \rfloor}$ and the Motzkin enumeration [80, equation (3)]. For $k \ge 4$, Lemmas VIII.9.2–VIII.9.3 give a block of size $\lfloor k/2 \rfloor + 1 \ge 3$ in $Y_k$. If $\widetilde{Y}_k$ were algebraic, the onefold Borel bound of Theorem VIII.8.3 would make every block at eigenvalue 1 have size at most two, a contradiction.

For permutations, avoidance of length $\ell$ is counted by $u_{\ell-1}(n)$ when $\ell \ge 2$. The cases $\ell = 1, 2, 3$ give respectively 1, $(1-t)^{-1}$, and the Catalan series $(1-\sqrt{1-4t})/(2t)$. For $\ell \ge 4$, put $k = \ell - 1 \ge 3$. The series $U_k$ has a block of size $k+1 \ge 4$, whereas algebraicity of $\widetilde{U}_k$ and the twofold Borel bound permit size at most three.

For matchings, $k = 1$ allows only the empty matching and $k = 2$ gives the Catalan series. If $k \ge 3$ and $\widetilde{F}_k$ were algebraic, then $\widetilde{F}_k(t^2)$ would be algebraic. Its exponential generating function, obtained by padding the sequence with zeros at odd indices, is $F_k(x)$. The onefold Borel bound gives block size at most two, contrary to its size $k \ge 3$ from Lemma VIII.9.3.

In each contradiction the assumed algebraic power series is convergent near zero and its minimal equation is regular singular there, as its solutions are spanned by algebraic conjugates with convergent Puiseux expansions. Thus all hypotheses of Theorem VIII.8.3 are satisfied. □

### VIII.9.4 A published Jacobi system and a scalar tableau equation

The arithmetic specialization below requires every nonzero point to be ordinary for the minimal tableau scalar equation. We now prove that property directly, using an explicit scalar elimination of a published Jacobi system.

For $N \ge 1$, $\alpha > -1$, $b > 0$, $\beta > 0$, and $0 \le p \le N$, let

$$H_p(z) = \binom{N}{p}^{-1} \int_{[0,1]^N} e^{-z\sum u_i} \prod_i u_i^{\alpha}(1-u_i)^{b-1} \prod_{i<j} |u_i - u_j|^{\beta} \cdot e_p(1-u_1, \dots, 1-u_N)\, du,$$

where $e_p$ is the elementary symmetric polynomial and $e_0 = 1$. The Forrester–Kumar system [84, Corollary 2.5, equations (2.19)–(2.20)] states that

$$(N-p)zH_{p+1} = ((N-p)z + B_p)H_p + zH_p' - D_pH_{p-1}, \tag{VIII.9.7}$$

where

$$B_p = p\left(\alpha + b + 1 + \frac{\beta}{2}(2N-p-1)\right), \qquad D_p = p\left(b + \frac{\beta}{2}(N-p)\right).$$

Terms outside the range $0 \le p \le N$ are omitted. This is the only published differential-system input to the construction; the elimination and its specialization are given explicitly next.

**Lemma VIII.9.5** (Scalar elimination)**.** *There is a scalar differential operator* $\mathcal{A} = \sum_{m=0}^{N+1} a_m(z)\partial_z^m$ *annihilating* $H_0$ *such that*

$$a_m \in \mathbb{R}[z], \qquad a_{N+1} = z^N, \qquad \deg a_m \le N, \qquad z^{m-1} \mid a_m \quad (m \ge 1).$$

*The coefficients are rational when the parameters are rational.*

*Proof.* Write $D = \partial_z$ and construct rational operators

$$P_0 = 1, \qquad P_1 = 1 + D/N,$$

and, for $1 \le p < N$,

$$(N-p)P_{p+1} = \left(D + N - p + \frac{B_p}{z}\right) P_p - \frac{D_p}{z} P_{p-1}. \tag{VIII.9.8}$$

Equation (VIII.9.7) gives $H_p = P_p H_0$. Write $P_p = \sum_{m=0}^{p} q_{p,m} D^m$ for $p \ge 1$. Induction using $D \circ (qD^m) = qD^{m+1} + q'D^m$ gives

$$q_{p,p} = \frac{(N-p)!}{N!}, \qquad q_{p,m} \in \operatorname{span}\{z^{-(p-1)}, \dots, 1\}, \qquad \operatorname{ord}_0 q_{p,m} \ge m - p.$$

For the induction, differentiation or division by $z$ lowers the least Laurent exponent by at most one, and no positive exponent is introduced. The highest derivative comes only from $D \circ P_p$. The coefficient of $D^m$ in that product is $q'_{p,m} + q_{p,m-1}$, whose order is at least $m - p - 1$; the other terms in (VIII.9.8) have the same bound. The case $p = 1$ involving $P_0$ is immediate.

The $p = N$ equation in (VIII.9.7) now shows that

$$\mathcal{A} = N! z^{N-1}\bigl(zD \circ P_N + B_N P_N - D_N P_{N-1}\bigr) \tag{VIII.9.9}$$

annihilates $H_0$. The bracket has Laurent exponents between $-(N-1)$ and 1, so multiplication by $z^{N-1}$ makes its coefficients polynomials of degree at most $N$. Its top coefficient is $z/N!$, giving $a_{N+1} = z^N$. The coefficient of $D^m$ in the bracket has order at least $m - N$, proving the stated divisibility after multiplication. The formulas also prove rationality. □

**Proposition VIII.9.6** (No nonzero tableau scalar singularities)**.** *Let* $k = 2r + \varepsilon \ge 2$*, with* $\varepsilon \in \{0, 1\}$*. There is an order-*$r+1$ *operator over* $\mathbb{Q}[x]$ *annihilating* $Y_k$ *whose leading coefficient is* $x^r$*. The minimal monic operator has order* $r + 1$ *and is ordinary at every* $x \neq 0$*.*

*Proof.* First derive the required Jacobi integral from the determinant. The Bessel integral

$$b_j(x) = \frac{1}{\pi} \int_0^{\pi} e^{2x\cos\theta} \cos(j\theta)\, d\theta$$

and the identities

$$\cos((i-j)\theta) + \cos((i+j-1)\theta) = 2\cos((i - \tfrac{1}{2})\theta)\cos((j - \tfrac{1}{2})\theta),$$

$$\cos((i-j)\theta) - \cos((i+j)\theta) = 2\sin(i\theta)\sin(j\theta)$$

convert the two tableau matrices into Gram matrices. The quotients $\cos((i - \frac{1}{2})\theta)/\cos(\theta/2)$ and $\sin(i\theta)/\sin\theta$ are monic polynomials of degree $i - 1$ in $u = 2\cos\theta$. Putting $a = \varepsilon - \frac{1}{2}$ and $w(u) = (2-u)^a(2+u)^{1/2}$, both Gram matrices have entries

$$\frac{1}{2\pi} \int_{-2}^{2} e^{xu} Q_i(u) Q_j(u) w(u)\, du.$$

Expanding the determinant and applying Fubini, or equivalently using Andreief's identity, gives

$$Y_k(x) = \frac{e^{\varepsilon x}}{r!(2\pi)^r} \int_{[-2,2]^r} e^{x\sum u_i} \Delta(u)^2 \prod_i w(u_i)\, du. \tag{VIII.9.10}$$

Here the determinants of the monic polynomial bases are the Vandermonde $\Delta$.

Substitute $u_i = 4v_i - 2$ in (VIII.9.10) and take

$$N = r, \qquad \alpha = \tfrac{1}{2}, \qquad b = \varepsilon + \tfrac{1}{2}, \qquad \beta = 2$$

in the preceding Jacobi system. The precise identity is

$$Y_k(x) = \frac{4^{r(r+\varepsilon)}}{r!(2\pi)^r} e^{(\varepsilon-2r)x} H_0(-4x).$$

Indeed the squared Vandermonde, weights, and Jacobian contribute $4^{r(r-1)}$, $4^{\varepsilon r}$, and $4^r$. In the operator of Lemma VIII.9.5, substitute

$$z = -4x, \qquad \partial_z = -\tfrac{1}{4}(\partial_x - (\varepsilon - 2r)).$$

After multiplication by $-4$, its leading coefficient is $x^r$. All coefficients are rational polynomials and its order is $r+1$. Lemma VIII.9.3 already proves that this is the minimal order. Dividing this operator by $x^r$ therefore gives the minimal monic operator. Its coefficients are holomorphic at every nonzero point, proving the assertion. □

Only the ordinary-point conclusion of Proposition VIII.9.6 is needed in Theorem VIII.10.4. Possible cancellation of a common power of $x$ in a primitive polynomial normalization is irrelevant to that conclusion. This explicit construction supplies the required statement without invoking a separate internal minimal-operator manuscript.

# VIII.10 Exact Bessel modules, algebraic base changes, and tableau values

The algebraicity classifications of Theorem VIII.9.4 admit stronger differential and arithmetic conclusions. The general homogeneous-form theorem below determines the normalized equations, their complete local partition, and all algebraic relations between their initial jets. The ordinary-point result needed for arithmetic specialization was proved explicitly in Proposition VIII.9.6.

## VIII.10.1 A homogeneous-form theorem

Let $K = \mathbb{C}(z)$ and let $y$ denote either

$$y(z) = I_0(2z), \qquad y'' = 4y - z^{-1}y', \quad \text{or} \quad y(z) = I_0(2\sqrt{z}), \qquad y'' = z^{-1}(y - y').$$

Both rank-two connections have differential Galois group $\mathrm{SL}_2$. Their local monodromy at zero is a nontrivial unipotent block of size two. These Bessel inputs, including their primary references, are established in Lemma VIII.9.1. For $d \geq 1$, let $\Phi \in K[X_0, X_1]$ be a nonzero homogeneous form of degree $d$, and put

$$F(z) = e^{\varepsilon z}\Phi(y(z), y'(z)), \qquad \varepsilon \in \{0, 1\}.$$

**Theorem VIII.10.1** (Exact module and stability)**.** *The minimal differential operator of $F$ over $K$ is irreducible of order $d+1$. Its differential Galois group, in its solution representation, is*

$$\begin{cases} \mathrm{Sym}^d(\mathrm{SL}_2), & \varepsilon = 0, \\ \{c\,\mathrm{Sym}^d(g) : c \in \mathbb{C}^*,\ g \in \mathrm{SL}_2\}, & \varepsilon = 1. \end{cases}$$

*In the first case this is isomorphic to $\mathrm{SL}_2$ for odd $d$ and to $\mathrm{PSL}_2$ for even $d$. At zero the complete local monodromy partition consists of one unipotent block $J_{d+1}(1)$.*

*All these orders and irreducibility assertions remain valid over every finite algebraic extension of $K$. After a ramified algebraic change of variable, the local block at a point above zero still has size $d+1$.*

*Proof.* Let $V$ be the two-dimensional Bessel solution space in a Picard–Vessiot extension. Polarizing $u \mapsto \Phi(u, u')$ gives a $\mathrm{SL}_2$-equivariant map from $\mathrm{Sym}^d V$ to the differential field. It is nonzero because $y, y'$ are algebraically independent over $K$, by Lemma VIII.9.1. The representation $\mathrm{Sym}^d V$ is irreducible, so this map is injective. The Galois orbit of any nonzero image spans the entire image. The standard Galois-orbit characterization of a minimal scalar operator therefore gives the claimed order, irreducibility, and solution representation.

For $\varepsilon = 1$, the compositum of the Bessel Picard–Vessiot extension and that of $u' = u$ has group $\mathrm{SL}_2 \times \mathbb{G}_m$. Indeed, a proper subdirect product would give a nontrivial common quotient of $\mathrm{SL}_2$ and $\mathbb{G}_m$, which is impossible. The preceding argument applies to the irreducible representation $\mathrm{Sym}^d V$ tensored with the standard character of $\mathbb{G}_m$.

The symmetric power of a nontrivial $J_2(1)$ is a single $J_{d+1}(1)$: in the usual degree-$d$ polynomial representation, its logarithm acts by a nonzero multiple of $X_0\partial_{X_1}$, with one chain of length $d+1$. The exponential twist is single-valued and nonvanishing at zero and does not alter this partition.

Both displayed Galois groups are connected. Finite algebraic extension of the base leaves the identity component unchanged, and hence leaves the full connected group and its irreducible representation unchanged. A ramification of index $e$ replaces the local monodromy by its $e$th power; a unipotent Jordan block in characteristic zero retains its size under this operation. □

**Corollary VIII.10.2** (The three determinant families)**.** *With the notation of Section VIII.9, the following orders are exact:*

| *function* | $d$ | *minimal order* | *monodromy at zero* |
|---|---|---|---|
| $Y_k(x),\ k \geq 2$ | $\lfloor k/2 \rfloor$ | $\lfloor k/2 \rfloor + 1$ | $J_{\lfloor k/2 \rfloor + 1}(1)$ |
| $U_k(s),\ k \geq 1$ | $k$ | $k+1$ | $J_{k+1}(1)$ |
| $F_k(x),\ k \geq 2$ | $k-1$ | $k$ | $J_k(1)$. |

*The permutation and matching groups are respectively* $\mathrm{Sym}^k(\mathrm{SL}_2)$ *and* $\mathrm{Sym}^{k-1}(\mathrm{SL}_2)$*. For tableaux, write* $k = 2d+\varepsilon$*; the group is the corresponding group in Theorem VIII.10.1. Every listed minimal order persists after finite algebraic extension of the coefficient field.*

*Proof.* The determinant reductions in Lemma VIII.9.2 give nonzero homogeneous forms of exactly the displayed degrees. Their nonvanishing also follows from the constant term 1 of each generating function. Apply Theorem VIII.10.1. □

### VIII.10.2 The complete algebraic relations among initial jets

Write $\mathbf{J}_F = (F, F', \dots, F^{(d)})^{\mathsf{T}}$ and

$$\mathbf{m}_d = (y^d, y^{d-1}y', \dots, (y')^d)^{\mathsf{T}}.$$

Differentiating and reducing $y''$ by the Bessel equation produces a rational matrix $P(z)$ with

$$\mathbf{J}_F = e^{\varepsilon z} P(z) \mathbf{m}_d. \tag{VIII.10.1}$$

**Theorem VIII.10.3** (Jet cone)**.** *The matrix* $P$ *belongs to* $\mathrm{GL}_{d+1}(K)$*. After the invertible rational change of jet coordinates* $\mathbf{Z} = P^{-1}\mathbf{J}_F$*, the ideal of all polynomial relations over* $K$ *is generated by*

$$Z_i Z_{j+1} - Z_{i+1} Z_j, \qquad 0 \leq i < j \leq d-1. \tag{VIII.10.2}$$

*For* $d = 1$ *the ideal is zero. Consequently*

$$\mathrm{trdeg}_K\, K(F, F', \dots, F^{(d)}) = 2.$$

*Every two distinct entries of this jet vector are algebraically independent over* $K$*, and remain so over every finite algebraic extension of* $K$*. When* $\varepsilon = 0$*, one has the further exact field identity*

$$K(F, F', \dots, F^{(d)}) = K(y^d, y^{d-1}y', \dots, (y')^d) = K(y, y')^{\mu_d},$$

*where* $\mu_d$ *acts by simultaneous scalar multiplication on* $y, y'$*. Thus its index in* $K(y, y')$ *is exactly* $d$*.*

*Proof.* If the rows of $P$ were dependent, (VIII.10.1) would give a nonzero rational differential equation for $F$ of order at most $d$, contradicting Theorem VIII.10.1. Therefore $P$ is invertible.

The kernel of the substitution $Z_j \mapsto X_0^{d-j} X_1^j$ is the homogeneous ideal (VIII.10.2) of the affine cone over the degree-$d$ rational normal curve. For completeness, these minors identify all products of $Z_j$ having the same number of factors and the same sum of indices; their images are the same monomial in $X_0, X_1$. Grouping monomials by those two integers proves that they generate the entire kernel. Algebraic independence of $y, y'$ shows that this is exactly the relation ideal for $\varepsilon = 0$. For $\varepsilon = 1$, the three functions $y, y', e^z$ are algebraically independent: the group $\mathrm{SL}_2 \times \mathbb{G}_m$ acts transitively on the nonzero Bessel solution vector and independently on the exponential coordinate. Multiplying every degree-$d$ monomial by the same independent scalar preserves the same homogeneous kernel. This proves the ideal assertion and its dimension.

Two different rows of $P$ are linearly independent, so their degree-$d$ forms $A, B$ are not proportional. Two nonproportional homogeneous forms of the same positive degree in two variables are algebraically independent: their ratio is a nonconstant function of $X_1/X_0$, while simultaneous scaling changes their common magnitude. Equivalently, their Jacobian is nonzero. The same conclusion holds after multiplying both by the independent scalar $e^z$. This proves pairwise independence; an algebraic extension of the coefficient field cannot decrease transcendence degree. Finally the displayed monomial field is the fixed field of the faithful scalar action of $\mu_d$ on the rational field $K(y, y')$, proving the index statement. □

### VIII.10.3 Arithmetic specialization without exceptional nonzero points

For tableaux Proposition VIII.9.6 supplies an additional essential input: an order-$d+1$ annihilator of $Y_k$ has leading coefficient $x^d$, where $d = \lfloor k/2 \rfloor$, and $d+1$ is the exact minimal order. Thus every $\alpha \neq 0$ is an ordinary point of the minimal scalar equation. This property is stronger than having a connection which is merely gauge-equivalent to one regular at $\alpha$; it is what rules out exceptional zeros of the scalar determinant form.

**Theorem VIII.10.4** (All nonzero algebraic tableau arguments)**.** *Let $k \geq 2$, $d = \lfloor k/2 \rfloor$, and $\alpha \in \overline{\mathbb{Q}}^*$. Then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(Y_k(\alpha), Y_k'(\alpha), \dots, Y_k^{(d)}(\alpha)) = 2.$$

*Every two distinct numbers in this displayed tuple are algebraically independent over $\overline{\mathbb{Q}}$. All their algebraic relations are obtained from (VIII.10.2) by the invertible algebraic change of coordinates $\mathbf{Z} = P(\alpha)^{-1}\mathbf{J}_{Y_k}(\alpha)$. In particular $Y_k(\alpha)$ is transcendental and nonzero. For $k = 1$, $Y_1(\alpha) = e^\alpha$ is transcendental.*

*Proof.* Put $y = I_0(2x)$. The functions $y, y', e^x$ are $E$-functions over $\mathbb{Q}$; their first-order system is

$$\frac{d}{dx}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 4 & -1/x & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix}.$$

They are algebraically independent over $\mathbb{C}(x)$ by the group argument in Theorem VIII.10.3. The Siegel–Shidlovskii theorem in [68, Theorem 1.1], with common denominator $x$, therefore says that $y(\alpha), y'(\alpha), e^\alpha$ are algebraically independent over $\overline{\mathbb{Q}}$ for every nonzero algebraic $\alpha$.

The determinant reductions use coefficients in $\mathbb{Q}[x, x^{-1}]$, so $P$ in (VIII.10.1) has no pole at $\alpha$. Moreover $P(\alpha)$ is invertible. Indeed the symmetric-power Bessel system is ordinary there, and so is the minimal scalar equation of $Y_k$ by the preceding input. Evaluation of its first $d+1$ jets is an isomorphism from its solution space to $\mathbb{C}^{d+1}$. The injective polarized solution map of Theorem VIII.10.1 is an isomorphism between solution spaces of that same dimension. Their evaluation matrices are related by $P(\alpha)$, forcing its determinant to be nonzero.

We may therefore specialize the monomial-coordinate argument of Theorem VIII.10.3. The independence of the three basic values gives exactly the Veronese-cone relation ideal and transcendence degree two. Different rows of the invertible $P(\alpha)$ are nonproportional homogeneous forms, giving pairwise algebraic independence by the same Jacobian argument. The case $k = 1$ is the classical transcendence of $e^\alpha$. □

*Remark* VIII.10.5 (Scope of the arithmetic statement). No corresponding absence of exceptional nonzero scalar points is assumed here for the permutation and matching determinants. Their functional order, Galois, and jet-cone conclusions hold for all parameters as stated. The arithmetic theorem above uses Proposition VIII.9.6 explicitly; functional Galois information alone would not justify its all-points specialization.

## VIII.11 An arithmetic classification for split beta orthogonality

The critical-value argument uses positivity to prevent cancellation of local contributions. A different use of positivity gives an arithmetic classification for multiple orthogonal polynomials: once a logarithmic moment identity forces orthogonality on both sides of a cut, the required sign changes can exceed the degree. The classification below is uniform in both orthogonality indices, in the algebraic cut, and in every positive algebraic polynomial modification of the weight.

### VIII.11.1 The complete parameter and index classification

Let $a, b > 0$ be rational with $a + b \in \mathbb{Z}$, let $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, and let $w \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$ satisfy $w(x) > 0$ for $0 < x < 1$. Define the positive integrable weight

$$\rho(x) = w(x)x^{a-1}(1 - x)^{b-1}.$$

For $m, n \geq 0$, the monic type-II Angelesco polynomial $P_{m,n}$ has degree $m + n$ and satisfies

$$\int_0^\xi P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < m), \tag{VIII.11.1}$$

$$\int_\xi^1 P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < n). \tag{VIII.11.2}$$

Empty lists of conditions are allowed, and $P_{0,0} = 1$. Existence and uniqueness will be proved below.

**Theorem VIII.11.1** (Exact algebraic-coefficient classification)**.** *For the preceding family, the following are equivalent:*

1. *Every coefficient of $P_{m,n}$ is algebraic.*
2. *Every zero of $P_{m,n}$ is algebraic.*
3. *Either $a, b \in \mathbb{Z}$, or $m = n$.*

*In the algebraic cases the coefficients belong to*

$$K = \mathbb{Q}(\xi, w_0, \ldots, w_d), \qquad w(X) = \sum_{j=0}^{d} w_j X^j.$$

*Consequently, when $a, b$ are nonintegers and $m \neq n$, $P_{m,n}$ has at least one transcendental coefficient and at least one transcendental zero. All its zeros are nevertheless real and simple, with precisely $m$ in $(0, \xi)$ and $n$ in $(\xi, 1)$.*

The rationality of $a, b$ and the integer-sum condition are essential to the stated arithmetic reduction. The theorem includes the arcsine pair $a = b = 1/2$; no exclusion of that pair is needed. We assert that at least one zero is transcendental off the diagonal, not that every zero is.

### VIII.11.2 Positivity and normality

**Lemma VIII.11.2** (Sign changes and existence)**.** *For a positive integrable weight on $(0, 1)$, each system (VIII.11.1)–(VIII.11.2) has a unique monic solution of degree $m + n$. That solution has exactly the zero distribution stated in Theorem VIII.11.1.*

*Proof.* A nonzero real polynomial $Q$ orthogonal to $1, x, \dots, x^{k-1}$ on an interval with a positive weight has at least $k$ sign changes in its interior. Indeed, if the sign-change points were $t_1, \dots, t_s$ with $s < k$, the polynomial $q(x) = \prod_{i=1}^{s}(x - t_i)$ would make $Qq$ have one constant nonzero sign except at finitely many points. Its weighted integral could not vanish, contradicting orthogonality to $q$.

A nonzero polynomial satisfying both lists therefore has at least $m + n$ distinct real zeros in the two disjoint interiors. In particular, there is no such polynomial of degree at most $m + n - 1$. The linear map from that $(m+n)$-dimensional coefficient space to its $m+n$ moments is injective, hence bijective. It follows that the lower coefficients of a monic degree-$m + n$ solution exist uniquely. The same sign-change count exhausts its degree, proving simplicity and the precise locations of all zeros. The case $m + n = 0$ is immediate. □

### VIII.11.3 A logarithmic moment forces orthogonality on both sides

The transcendence input is the classical consequence of Baker's theorem: a nonzero linear form, with algebraic coefficients, in fixed logarithms of nonzero algebraic numbers is transcendental [46]. We reproduce the needed integral reduction, so the argument does not require any unproved period-independence assertion.

For $r \in \mathbb{Q} \cap (0, 1)$ write

$$b_r(x) = x^{r-1}(1 - x)^{-r}, \qquad U_r(y) = \int_0^y b_r(x)\, dx.$$

**Lemma VIII.11.3** (Two transcendental base integrals)**.** *If $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, then both $U_r(\xi)$ and $\int_\xi^1 b_r(x)\, dx$ are transcendental. This includes $r = 1/2$.*

*Proof.* Write $r = p/q$ with $1 \le p < q$ and put $s = (\xi/(1 - \xi))^{1/q} > 0$. The substitution $x = z^q/(1 + z^q)$ and partial fractions give

$$U_r(\xi) = q \int_0^s \frac{z^{p-1}}{1 + z^q}\, dz = -\sum_{\zeta^q = -1} \zeta^p \log(1 - s/\zeta).$$

For each root $\zeta$, take the continuous logarithm along positive real $s$, starting at zero when $s = 0$. Its argument never vanishes because no root of $\zeta^q = -1$ is positive real. The displayed identity follows by differentiation and the common value at $s = 0$. All logarithm arguments at the specified $s$ are nonzero algebraic numbers, and all coefficients are algebraic. The integral is strictly positive, so Baker's theorem proves its transcendence. Finally,

$$\int_\xi^1 b_r(x)\, dx = U_{1-r}(1 - \xi),$$

and the same argument applies. □

**Lemma VIII.11.4** (Arithmetic synchronization of moments)**.** *Let $r \in \mathbb{Q} \cap (0, 1)$ and $S \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$. Then*

$$\int_0^\xi S(x) b_r(x)\, dx = 0 \quad \Longleftrightarrow \quad \int_\xi^1 S(x) b_r(x)\, dx = 0.$$

*More precisely, define the unique constant $c(S)$ and polynomial $R_S$ by*

$$S(X) = c(S) + X(1 - X)R_S'(X) + (r - X)R_S(X). \tag{VIII.11.3}$$

*Either vanishing condition is equivalent to the two algebraic linear conditions*

$$c(S) = 0, \qquad R_S(\xi) = 0. \tag{VIII.11.4}$$

*If $S$ has coefficients in a field $K \supset \mathbb{Q}$ containing $\xi$, then $c(S)$ and all coefficients of $R_S$ belong to $K$.*

*Proof.* The operator in (VIII.11.3) sends $X^j$ to $(j+r)X^j - (j+1)X^{j+1}$. Descending elimination therefore gives a unique $R_S$ and constant $c(S)$, over the coefficient field. Put $H_S(x) = x^r(1-x)^{1-r}R_S(x)$. Direct differentiation and $H_S(0) = H_S(1) = 0$ give

$$\int_0^\xi Sb_r = c(S)U_r(\xi) + H_S(\xi),$$
$$\int_\xi^1 Sb_r = c(S)\{U_r(1) - U_r(\xi)\} - H_S(\xi).$$

Here $H_S(\xi)$ is algebraic. Either integral can vanish only if $c(S) = 0$, by Lemma VIII.11.3. It then forces $H_S(\xi) = 0$, equivalently $R_S(\xi) = 0$ since $0 < \xi < 1$. Conversely, these two conditions make both integrals zero. All operations in descending elimination take place over $K$. □

### VIII.11.4 Proof of the classification

*Proof of Theorem VIII.11.1.* Suppose first that $a, b$ are integers. Since they are positive, $\rho$ is a polynomial with coefficients in $K$. Each moment on either interval belongs to $K$, by termwise integration and $\xi \in K$. The invertible linear system in Lemma VIII.11.2 consequently has its monic solution in $K[X]$.

Otherwise neither shape is an integer, since their sum is an integer. Write

$$a = A + r, \qquad b = B + 1 - r, \qquad A, B \in \mathbb{Z}_{\geq 0}, \quad r \in \mathbb{Q} \cap (0,1).$$

Then $\rho = gb_r$, where $g(X) = w(X)X^A(1-X)^B \in K[X]$ is positive on $(0,1)$.

Assume that $P_{m,n}$ has algebraic coefficients and set $h = \max(m,n)$. For every $0 \leq j < h$, at least one of (VIII.11.1) and (VIII.11.2) holds. Apply Lemma VIII.11.4 to the algebraic polynomial $S = X^j P_{m,n} g$. Both integrals then vanish. Thus $P_{m,n}$ satisfies $h$ orthogonality conditions on each interval. Positivity forces at least $2h$ distinct zeros, so

$$m + n = \deg P_{m,n} \geq 2\max(m,n).$$

This is possible only when $m = n$.

Conversely let $m = n = k$. The $2k$ equations

$$c(X^j P g) = 0, \qquad R_{X^j P g}(\xi) = 0, \qquad 0 \leq j < k, \tag{VIII.11.5}$$

are homogeneous linear equations over $K$ in the $2k+1$ coefficients of $P$ of degree at most $2k$. They have a nonzero solution in $K[X]$. Lemma VIII.11.4 shows that this solution satisfies both orthogonality lists. By the sign-change argument it has degree exactly $2k$. Divide by its nonzero leading coefficient. Uniqueness identifies the resulting monic polynomial with $P_{k,k}$, proving $P_{k,k} \in K[X]$.

Finally, a monic polynomial has all roots algebraic if and only if all its coefficients are algebraic: one direction uses elementary symmetric functions, and the other uses the algebraic closedness of $\overline{\mathbb{Q}}$. The zero distribution follows from Lemma VIII.11.2. □

### VIII.11.5 Quadrature, an explicit example, and the literature boundary

**Corollary VIII.11.5** (Algebraic cuts destroy all-algebraic Gaussian nodes)**.** *If $a, b > 0$ are rational nonintegers with integer sum, then for every algebraic $\xi \in (0,1)$ and every $N \geq 1$, the $N$-node Gaussian quadrature rule for $\rho$ restricted to either $[0,\xi]$ or $[\xi,1]$ has at least one transcendental node. In contrast, all zeros of every balanced $P_{N,N}$ are algebraic. These assertions remain true under every positive algebraic polynomial factor $w$.*

*Proof.* The Gaussian nodes are the zeros of the degree-$N$ monic orthogonal polynomial, namely $P_{N,0}$ or $P_{0,N}$. Apply the theorem. The balanced assertion is the case $(m, n) = (N, N)$. □

For $w = 1$ and $(a, b) = (r, 1 - r)$, the classical diagonal Rodrigues formula specializes to

$$P_{k,k}(x) = \frac{(-1)^k}{(2k)_k\, b_r(x)} \frac{d^k}{dx^k} \left[ x^{k+r-1}(1-x)^{k-r}(x-\xi)^k \right], \qquad k \geq 1. \tag{VIII.11.6}$$

Leibniz' rule makes the right side a polynomial of degree at most $2k$. Integration by parts $k$ times on each interval proves the orthogonality: derivatives of order at most $k - 1$ of the expression in brackets vanish at all relevant endpoints, and $\deg(x^j) < k$. Its leading coefficient before normalization is $(-1)^k(2k)_k$, as can be read from the highest-degree terms. This establishes (VIII.11.6) and its monic normalization. For example,

$$P_{1,1}(x) = x^2 - \frac{1 + r + \xi}{2} x + \frac{r\xi}{2}.$$

At $r = 1/3$ and $\xi = 1/2$ its two zeros are

$$\frac{11 - \sqrt{73}}{24}, \qquad \frac{11 + \sqrt{73}}{24},$$

one on each side of the cut. Every off-diagonal polynomial for the same measures has a transcendental coefficient and a transcendental zero, including the degree-one polynomial for either single interval.

The diagonal formula is classical. After an affine change of variable, this is the Jacobi–Angelesco system with zero exponent at the common endpoint; see Van Assche and Coussement [67, §3.5, Eq. (3.45)]. Their equations (3.48)–(3.50) also express near-diagonal polynomials through incomplete moment ratios. We do not count the diagonal Rodrigues formula, Angelesco normality, polynomial beta reduction, or Baker's theorem as new. The addition proved here is the exact arithmetic exclusion of every unbalanced index throughout the specified parameter family, including positive polynomial modifications. We make no historical-priority claim. Nor does this result assert algebraic independence of different zeros or settle an unrestricted problem for arbitrary rational beta shapes with noninteger sum.

## VIII.12 Exact scope and remaining questions

The gaps in the source draft are not all of the same kind. The Poisson and binomial transition gaps are removed by incorporating the complete estimation results and proving the testing consequence. The local Fourier argument supplies a precise replacement for the rapid-decay period assertion. The Franel theorem goes beyond the original central sums to all arithmetic subsequences. The Angelesco and tableau-value theorems are additional arithmetic applications rather than assertions about unnamed classical conjectures.

The following distinctions remain part of the mathematical conclusions.

1. The inequality $|C(t, m, j)| \leq 2^{m^2/t}$ with prefactor exactly one is not proved. The exponent $\log 2$ is optimal when an absolute prefactor is allowed. An exponent $1/2$ is valid in the moderate-degree asymptotic regime but impossible globally with a polynomial prefactor.

2. The mixture theorems use a known support interval and an ordinary unrestricted maximum-likelihood maximizer. They do not establish an unknown-support adaptation theorem. The testing rate is worst-null, not a claim that every fixed null is equally difficult. Arbitrary heterogeneous observation kernels still require a separate normalization or comparison argument; the homogeneous posterior identity cannot simply be applied within each stratum.

3. Recurrence width and differential order are different invariants. The Franel results classify width over rational-function coefficient fields. They do not determine the least possible coefficient degrees of every minimum-width recurrence.

4. The Apéry results concern regular eventual tails. Initial singular indices of an arbitrary chosen polynomial recurrence are not silently filled in. Complex linear combinations can share a limit; injectivity is asserted over subfields of the algebraic numbers.

5. The iterated Fourier statement assumes the specified regularity at the source point. It is not a bound for arbitrary irregular source connections. For Bessel determinant values, absence of exceptional nonzero points is proved for tableaux; it is not asserted for every rational homogeneous form or every scalar determinant realization.

6. For Angelesco polynomials, a noninteger parameter sum, an arbitrary nonpolynomial tilt, or more than two intervals lies outside the theorem. In the nonalgebraic cases the proof gives at least one transcendental coefficient and at least one transcendental zero, not transcendence of every zero.

The finite verification in Section VIII.2 is a proof certificate, not numerical evidence: every comparison uses rational arithmetic and a one-sided Taylor bound. It checks all 130816 pairs with $1 \leq m \leq t < 512$ through four exhaustive branches, including 12734 direct coefficient comparisons. Symbolic checks of representative Angelesco formulas supplement, but do not replace, the all-parameter proofs. The structural and statistical arguments underwent separate internal reviews. These checks improve the manuscript's auditability; they do not constitute external peer review or establish historical priority.

# Part IX

# Exact Cone Gluing, Collision Tests, and Algebraicity Obstructions

**Abstract**

We develop the endpoint analysis of earlier Bessel–Jacobi work into an exact local contour identity. The discrepancy between the two physical germs across a cone vertex is an outward-cone period with a fixed phase. The proof works at resonance without dividing by exponent differences, and without first constructing a differential equation. A polyhedral analyticity argument applies this identity to positive weighted traces with heterogeneous endpoint powers, nonnegative pairwise repulsion exponents, and a positive analytic tilt. Coincident vertex heights are handled by adding their full germs. Further applications give the exact logarithmic and differentiability classification at every ordinary Jacobi trace knot; a global endpoint expansion for sums of independent beta variables; and a finite polynomial criterion for all subset-sum collisions in integer-parameter Jacobi transforms. In the last setting the minimal differential order equals the number of genuine density knots. A three-variable example shows that interaction restores a frequency whose leading corner amplitudes cancel. Classical beta convolutions, contour continuation, and spline theory are explicitly acknowledged; these results are not advertised as the resolution of a newly identified historical conjecture.

**Integration note.** This part contains the complete later cone-gluing manuscript and the subsequent proved algebraicity corollaries. The all-parameter ordinary Jacobi identity agrees with Part IV; the broader graphical class and polynomial collision criterion extend its scope.

## IX.1 What is extended, and what is not claimed

The supplied Bessel–tableaux–banana manuscript [032] analyzes Bessel and tableau integrals by their corner expansions and positive Laguerre–Selberg amplitudes. Part I of the subsequent collected volume develops complete endpoint Fourier expansions for Jacobi transforms. Parts II and IV use a published Forrester–Kumar differential system to identify physical connection coefficients, first off resonance and then at every real integrable parameter. The present draft changes the logical order: an exact contour calculation proves the local connection first; Fourier asymptotics then describe its singularity and detect cancellations.

The following distinctions delimit the conclusions.

1. The graphical interaction theorem gives exact local discrepancies at *every* real knot, with no generic-position assumption. For an arbitrary analytic tilt it does not assert global complex continuation, holonomicity, or a full connection matrix for arbitrary bases.

2. The independent-beta formula is global along the real support and uses explicitly specified upper boundary values. Special beta convolution and spline formulas are already classical. The examples illustrate the general normalization and its cancellations, rather than establish priority for those distributions.

3. The finite collision theorem gives an exact differential order for every positive weight vector when the density weight is polynomial. Its proof is a finite endpoint calculation; its relationship with classical box splines is retained rather than suppressed.

4. The ordinary Jacobi regularity theorem is a consequence of the explicit nonzero connection coefficient. It adds a complete local singularity classification to the earlier connection formula.

The known Forrester–Kumar system and integer-parameter connection problem are described in [84]. The earlier local identity can be recovered from the general cone theorem, but none of the present all-parameter contour arguments requires that system.

For Jacobi weights we use exponents $a_i, b_i > -1$ in $x_i^{a_i}(1-x_i)^{b_i}$. In the independent-beta section, the letters $a_i, b_i > 0$ instead denote *shape parameters*, so the corresponding endpoint exponents are $a_i - 1, b_i - 1$. This change is stated again there. An upper continuation passes a real singular point through the upper half-plane with no additional winding. Thus $(-s)^\lambda$ continued from negative to positive $s$ becomes $e^{-\pi i\lambda}s^\lambda$.

## IX.2 An exact local gluing identity for analytic cone amplitudes

The argument below isolates the local step needed in a weighted-trace connection formula. It uses elementary contour deformation, rather than a differential equation or a nonresonance assumption. It does not, by itself, prove global continuation of a weighted-trace density. The beta integral and its analytic continuation are classical; see, for example, the defining integral and hypergeometric representation in `https://dlmf.nist.gov/8.17.E1` and `https://dlmf.nist.gov/8.17.E7`. No priority claim is made for the local identity.

### IX.2.1 The radial statement

Fix real numbers $\alpha, \eta > 0$ and put $D = \alpha + \eta$. Let $a(r, q)$ be holomorphic on the bidisc $|r|, |q| < R$. Choose $0 < \delta < R/4$ and define, for real $|s| < \delta$,

$$I(s) = \int_{\substack{r,q\geq 0\\ r+q\leq\delta}} \delta_0(s - r + q) r^{\alpha-1} q^{\eta-1} a(r, q)\, dr\, dq. \tag{IX.2.1}$$

Here $\delta_0$ is the Dirac delta. Equivalently, with $Q(s) = (\delta - s)/2$,

$$I_+(s) = \int_0^{Q(s)} (q+s)^{\alpha-1} q^{\eta-1} a(q+s, q)\, dq, \qquad 0 < s < \delta, \tag{IX.2.2}$$

$$I_-(s) = \int_{-s}^{Q(s)} (q+s)^{\alpha-1} q^{\eta-1} a(q+s, q)\, dq, \qquad -\delta < s < 0. \tag{IX.2.3}$$

These are real-analytic on their respective open intervals. Let $I_-^{\uparrow}$ denote continuation of the negative-side germ to positive $s$ through the upper half-plane, with no additional winding around zero. Thus the continuation of $(-s)^\lambda$ is $e^{-\pi i\lambda} s^\lambda$.

**Theorem IX.2.1** (Exact cone gluing)**.** *The continuation just described exists for sufficiently small $|s| > 0$. For positive $s$ sufficiently close to zero,*

$$\boxed{I_+(s) - I_-^{\uparrow}(s) = e^{-\pi i\eta} F(s), \qquad F(s) = s^{D-1} \int_0^1 (1-v)^{\alpha-1} v^{\eta-1} a\big(s(1-v), -sv\big)\, dv.} \tag{IX.2.4}$$

*There is no restriction on whether $D$ is an integer. The analytic factor $s^{1-D}F(s)$ is holomorphic at zero. The same identity holds for complex $\alpha, \eta$ with positive real parts, using the continued positive-axis branches.*

*Proof.* For $s \neq 0$, the two branch points of the integrand in the $q$-plane are $-s$ and 0. Start with negative real $s$, with the integration path in (IX.2.3) and positive real values of its powers. As $s$ moves through the upper half-plane, its endpoint $-s$ moves through the lower half-plane. Transport the integration contour continuously without crossing either moving branch point. The upper endpoint $Q(s)$ remains near $\delta/2$ and causes no obstruction. All contours can be kept in a fixed small bidisc in the variables $(q+s, q)$. This defines holomorphic continuation locally along the path: after straightening a small segment at a moving endpoint, its integrand is a holomorphic bounded factor times a fixed integrable power. Compact subarcs away from the endpoints present only ordinary holomorphic parameter integrals.

For positive real $s$, the transported contour can be deformed to the segment from $-s$ to 0 approached below, followed by the segment from 0 to $Q(s)$. In making this deformation, small arcs about $-s$ and 0 give vanishing contributions, respectively $O(\varepsilon^{\operatorname{Re}\alpha})$ and $O(\varepsilon^{\operatorname{Re}\eta})$, for each fixed $s \neq 0$. On the first segment $q + s$ is positive, while the continued argument of $q$ is $-\pi$. Consequently

$$I_-^{\uparrow}(s) = I_+(s) + e^{-\pi i(\eta-1)} \int_{-s}^{0} (q+s)^{\alpha-1} (-q)^{\eta-1} a(q+s, q)\, dq.$$

The substitution $q = -sv$ turns the final integral into $F(s)$. Since $e^{-\pi i(\eta-1)} = -e^{-\pi i\eta}$, this proves (IX.2.4). Uniform convergence of the integral defining its analytic factor proves holomorphy at zero. The argument only requires positive real parts of the two exponents, so also proves the final assertion. In particular, no division by $\sin(\pi D)$ and no limit through nonresonant parameters is required. □

*Remark* IX.2.2 (Empty clusters)*.* If the negative cluster is absent, replace its radial measure by a unit point mass at zero. The physical left germ is zero and the jump is simply the positive-cone density $F(s)$; the phase is one. If the positive cluster is absent, the physical right germ is zero and the same calculation gives $-I_-^{\uparrow} = e^{-\pi i\eta} F$. These cases correspond formally to $\eta = 0$ or $\alpha = 0$, but are not obtained by inserting zero into the beta integral. The case in which both clusters are absent is a point mass, not a density, and is excluded.

### IX.2.2 Homogeneous cone measures and the simplex normalization

Let $K_+, K_-$ be cones with positive linear radial functions $r(u)$ and $q(v)$, whose sections $r = 1$ and $q = 1$ are compact. Assume their measures have the polar decompositions

$$d\mu_+(u) = r^{\alpha-1} dr\, d\nu_+(\omega), \quad u = r\omega, \qquad d\mu_-(v) = q^{\eta-1} dq\, d\nu_-(\zeta), \quad v = q\zeta,$$

where $\nu_\pm$ are finite measures. Let $A(u, v)$ be analytic under complex radial scaling near the origin, uniformly on these compact angular sections. More precisely, assume $A(r\omega, q\zeta)$ is holomorphic for $|r|, |q| < R$ and is uniformly bounded on each smaller bidisc. Then

$$a(r, q) = \int A(r\omega, q\zeta)\, d\nu_+(\omega) d\nu_-(\zeta)$$

is holomorphic by dominated integration. Applying Theorem IX.2.1 to this function proves the identity for the pushforward of $A(u, v)d\mu_+(u)d\mu_-(v)$ by $r(u) - q(v)$, truncated by $r(u) + q(v) \le \delta$.

Its canonical outward-cone germ is

$$F(s) = s^{D-1} \int_{r(u)+q(v)=1} A(su, -sv)\, d\sigma_\mu(u, v). \tag{IX.2.5}$$

The measure in this formula is explicitly defined by

$$\int B\, d\sigma_\mu = \int_0^1 \int B((1 - v)\omega, v\zeta)(1 - v)^{\alpha-1} v^{\eta-1}\, d\nu_+(\omega) d\nu_-(\zeta)\, dv.$$

Thus there is no Euclidean surface-measure ambiguity. If $A(0, 0) = 1$, the positive normalization constant is

$$Z = \nu_+(K_+ \cap \{r = 1\})\nu_-(K_- \cap \{q = 1\}) \frac{\Gamma(\alpha)\Gamma(\eta)}{\Gamma(D)}, \qquad F(s) = Zs^{D-1}U(s), \quad U(0) = 1. \tag{IX.2.6}$$

Consequently the coefficient of the normalized corner germ $s^{D-1}U(s)$ is $e^{-\pi i\eta}Z$.

### IX.2.3 Every Taylor coefficient and the complete Fourier symbol

Write the convergent expansion $a(r, q) = \sum_{k,l\ge0} a_{kl} r^k q^l$. Termwise integration gives the convergent series

$$F(s) = s^{D-1} \sum_{j\ge0} f_j s^j, \qquad f_j = \sum_{k+l=j} (-1)^l a_{kl} \frac{\Gamma(\alpha + k)\Gamma(\eta + l)}{\Gamma(D + j)}. \tag{IX.2.7}$$

This is the all-coefficient content of the jump formula. In particular, it is the replacement $v \mapsto -v$ in the analytic amplitude that produces the sign $(-1)^l$; using $A(su, sv)$ would be incorrect.

For a smooth cutoff $\chi$ equal to one near zero and supported in a sufficiently small interval, the local Fourier transform has expansion

$$\int e^{iRs}\chi(s)I(s)\, ds \sim \sum_{j\ge0} \Gamma(D + j)e^{\pi i(\alpha-\eta+j)/2} f_j R^{-D-j}, \qquad R \to +\infty. \tag{IX.2.8}$$

Indeed, the monomial $r^k q^l$ contributes

$$a_{kl}\Gamma(\alpha + k)\Gamma(\eta + l)e^{\pi i(\alpha+k-\eta-l)/2} R^{-D-k-l}.$$

Summing over $k + l = j$ gives (IX.2.8). Taylor remainders give a complete expansion by the one-dimensional endpoint argument in each radial variable. The truncation boundary $r + q = \delta$ causes no other local contribution: its phase $r - q$ has nonzero tangential derivative, and the cutoff removes its endpoints $s = \pm\delta$. Repeated integration by parts there gives arbitrary decay.

### IX.2.4 Collisions, resonance, and the exact scope of gluing

**Corollary IX.2.3** (Several corners at one knot)**.** *Suppose a density near a knot $\tau$ is the sum of finitely many local cone densities $I_h(t-\tau)$ of the preceding form and a single function $B(t)$ holomorphic through $\tau$. Different corners may have different $\alpha_h, \eta_h$, and their values $D_h = \alpha_h + \eta_h$ may coincide or differ by integers. Then*

$$\rho_+(\tau+s) - \rho_-^{\uparrow}(\tau+s) = \sum_h e^{-\pi i \eta_h} F_h(s). \tag{IX.2.9}$$

*No distinctness or nonresonance condition is required.*

*Proof.* Continue each of the finitely many identities (IX.2.4); the continuation of $B$ has no discrepancy. □

The condition on $B$ is essential for this exact conclusion. When the $D_h$ are nonintegral and a prior argument gives decompositions into holomorphic backgrounds and convergent noninteger-power series, a complete Fourier expansion containing only the noninteger scales forces the two backgrounds to agree: the first unmatched Taylor coefficient would create an integer Fourier power. If several $D_h$ agree modulo integers, Fourier matching identifies only their combined coefficients. Their separately normalized contributions are still supplied by the individual cone integrals in (IX.2.9). When a $D_h$ is integral, integer Fourier powers cannot separate a background mismatch from that corner's contribution. The direct contour argument remains valid regardless.

One cannot infer exact analytic gluing from full Fourier asymptotics alone. For example, let $\chi$ be a compactly supported smooth cutoff equal to one near zero and set

$$h(s) = \begin{cases} \chi(s) e^{-1/s^2}, & s > 0, \\ 0, & s \le 0. \end{cases}$$

This is smooth, and its Fourier transform decays faster than every inverse power. Adding it changes no coefficient in any complete local Fourier power expansion. Yet its right germ is nonzero, whereas its left germ and every continuation of that zero germ are zero. Thus a smooth localization proves a singular expansion only modulo a smooth remainder, unless analyticity or an appropriate finite-dimensional differential system is established separately.

Changing the radial truncation from $\delta$ to another nearby positive value changes $I$ by a function holomorphic for small $s$: the difference of (IX.2.2) is an integral over a segment bounded away from both moving branch points, with holomorphic endpoints. Hence the corner jump is independent of this particular analytic truncation.

### IX.2.5 Application data for a weighted Jacobi corner

For $c_i > 0$, $a_i, b_i > -1$, and $\beta \ge 0$, consider the weight

$$|\Delta(x)|^{\beta} \prod_{i=1}^{N} x_i^{a_i} (1-x_i)^{b_i}, \qquad t = \sum_{i=1}^{N} c_i x_i.$$

At the corner indexed by $S \subseteq \{1, \dots, N\}$, put $x_i = u_i$ for $i \notin S$ and $x_j = 1 - v_j$ for $j \in S$. The knot and radial functions are

$$\tau_S = \sum_{j \in S} c_j, \quad r = \sum_{i \notin S} c_i u_i, \quad q = \sum_{j \in S} c_j v_j,$$

and the two homogeneity parameters are

$$\alpha_S = \sum_{i\notin S}(a_i+1) + \beta\binom{N-|S|}{2},$$
$$\eta_S = \sum_{j\in S}(b_j+1) + \beta\binom{|S|}{2}.$$

The homogeneous measures are

$$|\Delta(u)|^\beta \prod_{i\notin S} u_i^{a_i}\,du, \qquad |\Delta(v)|^\beta \prod_{j\in S} v_j^{b_j}\,dv.$$

Their angular measures are finite: the positive $c_i$ give compact simplexes, endpoint powers are integrable, and $\beta \geq 0$ causes no collision blow-up. The analytic amplitude is

$$A_S(u,v) = \prod_{i\notin S}(1-u_i)^{b_i} \prod_{j\in S}(1-v_j)^{a_j} \prod_{\substack{i\notin S\\ j\in S}}(1-u_i-v_j)^\beta.$$

Every power uses its branch equal to one at the corner. Thus the local outward germ is obtained by replacing this amplitude by

$$A_S(su,-sv) = \prod_{i\notin S}(1-su_i)^{b_i} \prod_{j\in S}(1+sv_j)^{a_j} \prod_{\substack{i\notin S\\ j\in S}}(1+s(v_j-u_i))^\beta.$$

The exact local phase is $e^{-\pi i\eta_S}$. Its normalization is the weighted-simplex integral of the two homogeneous measures; there need not be a Selberg product for unequal $c_i, a_i, b_i$. If several subset sums $\tau_S$ coincide, their contributions add as in Corollary IX.2.3.

The next argument proves the required analyticity of the complement. It also allows nonuniform interaction exponents and an analytic tilt. Global continuation along arbitrary complex paths is a separate issue and is not asserted.

### IX.2.6 Analyticity of regular polytope slices

**Lemma IX.2.4** (Analytic slices)**.** *Let $P \subset \mathbb{R}^N$ be a full-dimensional compact polytope and let $L \neq 0$ be a linear functional. Suppose $t_0$ is not the value of $L$ at any vertex of $P$. Let $\ell_1, \dots, \ell_M$ be affine functions nonnegative on $P$, let $e_j$ be real, and let $H$ be holomorphic near $P$. If*

$$\int_{P\cap\{L=t_0\}} \prod_j \ell_j(x)^{e_j}\,\frac{dS(x)}{|\nabla L|} < \infty,$$

*then the density*

$$J(t) = \int_{P\cap\{L=t\}} H(x)\prod_j \ell_j(x)^{e_j},\frac{dS(x)}{|\nabla L|}$$

*has a holomorphic germ at $t_0$. Factors identically zero on $P$ are excluded; zero exponents may be omitted.*

*Proof.* The assertion is immediate when the slice is empty. Otherwise, in a sufficiently small real interval about $t_0$, the edges of $P$ crossed by $L = t$ are fixed. Each slice vertex is the intersection of one such edge with $L = t$, so is an affine function of $t$. The face incidences of the slice are fixed on this interval. Fix an ordering of its vertices and use the resulting pulling triangulation for all these slices. Its nondegenerate simplices vary affinely in their vertices and remain nondegenerate

after the interval is shortened. Their intersections have measure zero and hence do not affect the integral.

Choose fixed linear coordinates on $\ker L$. On each simplex the map from a fixed standard simplex has a Jacobian determinant holomorphic in $t$, nonzero at $t_0$, and of fixed real sign near $t_0$. The density factor from the coarea formula is a fixed constant. Write its vertices as $v_0(t), \dots, v_{N-1}(t)$. For barycentric coordinates $\lambda_k \ge 0$, $\sum_k \lambda_k = 1$,

$$\ell_j\Big(\sum_k \lambda_k v_k(t)\Big) = \sum_k \lambda_k a_{jk}(t), \qquad a_{jk}(t) = \ell_j(v_k(t)).$$

Every $a_{jk}$ is affine in $t$. It is either identically zero or strictly positive at $t_0$: the vertex $v_k(t_0)$ lies in the relative interior of an original edge of $P$, and a nonnegative affine function can vanish there only if it vanishes on that whole edge. Thus, outside the fixed zero faces, the ratio of the preceding linear form to its value at $t_0$ is a convex combination of the ratios $a_{jk}(t)/a_{jk}(t_0)$ over the nonzero coefficients. For complex $t$ in a small disc, all these ratios, uniformly in the barycentric coordinates, lie in a fixed disc about one disjoint from zero. Their powers $e_j$ therefore have a common holomorphic branch and a uniform bound. The integrand is dominated by a fixed constant times its integrable absolute weight at $t_0$. Holomorphic dominated integration on each simplex proves the claim. For $N = 1$, the slice is a point on an interval; the same argument is simply substitution with the constant coarea factor. □

### IX.2.7 All-parameter local connections with graphical interactions

Fix $N \ge 1$, numbers $c_i > 0$, $a_i, b_i > -1$, and $\beta_{ij} \ge 0$ for $1 \le i < j \le N$. Let $H$ be holomorphic on a complex neighborhood of $[0,1]^N$, and real and strictly positive on that cube. Put

$$\begin{aligned} w(x) &= H(x) \prod_i x_i^{a_i}(1-x_i)^{b_i} \prod_{i<j} |x_i - x_j|^{\beta_{ij}}, \\ Z &= \int_{[0,1]^N} w(x)\,dx > 0, \qquad L(x) = \sum_i c_i x_i, \\ \rho(t) &= Z^{-1} \int_{[0,1]^N} \delta_0(t - L(x)) w(x)\,dx. \end{aligned} \tag{IX.2.10}$$

For a subset $S$, let $v_S$ be the cube vertex whose coordinates in $S$ are one, and write $T = S^c$ and $\tau_S = L(v_S)$. Define

$$\begin{aligned} \alpha_S &= \sum_{i \in T} (a_i + 1) + \sum_{\substack{i<j \\ i,j \in T}} \beta_{ij}, \\ \eta_S &= \sum_{i \in S} (b_i + 1) + \sum_{\substack{i<j \\ i,j \in S}} \beta_{ij}, \qquad D_S = \alpha_S + \eta_S. \end{aligned}$$

An empty sum is zero. The two weighted Laplace integrals are

$$\begin{aligned} W_T^- &= \int_{[0,\infty)^T} e^{-\sum_{i \in T} c_i u_i} \prod_{i \in T} u_i^{a_i} \prod_{\substack{i<j \\ i,j \in T}} |u_i - u_j|^{\beta_{ij}}\,du, \\ W_S^+ &= \int_{[0,\infty)^S} e^{-\sum_{j \in S} c_j v_j} \prod_{j \in S} v_j^{b_j} \prod_{\substack{i<j \\ i,j \in S}} |v_i - v_j|^{\beta_{ij}}\,dv. \end{aligned}$$

Empty integrals are one. These integrals are finite and positive: endpoint powers are integrable, collision factors are locally bounded, and the exponential controls polynomial growth at infinity. Set

$$Z_S = \frac{W_T^- W_S^+}{\Gamma(D_S)}.$$

On the simplex $\sum_{i\in T} c_i u_i + \sum_{j\in S} c_j v_j = 1$, let $d\sigma_S$ be the delta measure multiplied by

$$\prod_{i\in T} u_i^{a_i} \prod_{j\in S} v_j^{b_j} \prod_{\substack{i<j\\ i,j\in T}} |u_i-u_j|^{\beta_{ij}} \prod_{\substack{i<j\\ i,j\in S}} |v_i-v_j|^{\beta_{ij}}.$$

Its total mass is $Z_S$ by the Laplace radial decomposition. This identity also covers an empty cluster without introducing $\Gamma(0)$. For $N=1$ the delta measure is a point mass of size $1/c_1$ multiplied by the displayed homogeneous weight. Define $x_S(s;u,v)$ by $x_i = su_i$ for $i\in T$ and $x_j = 1+sv_j$ for $j\in S$, and put

$$U_S(s) = \frac{1}{Z_S}\int \frac{H(x_S(s;u,v))}{H(v_S)} \prod_{i\in T}(1-su_i)^{b_i} \prod_{j\in S}(1+sv_j)^{a_j} \cdot \prod_{\substack{i\in T\\ j\in S}} (1+s(v_j-u_i))^{\beta_{\min(i,j),\max(i,j)}}\, d\sigma_S(u,v), \tag{IX.2.11}$$

$$F_S(\tau_S+s) = s^{D_S-1}U_S(s), \qquad s>0. \tag{IX.2.12}$$

All analytic factors use their branches equal to one at $s=0$. The simplex is compact, its weight is integrable, and $H$ extends past the corner. Therefore $U_S$ is holomorphic near zero and $U_S(0)=1$.

**Theorem IX.2.5** (Weighted graphical Jacobi local connections)**.** *The density* (IX.2.10) *is real-analytic away from the finite set of distinct subset sums* $\{\tau_S\}$*. At every such knot* $\tau$*, its left germ continues to its right through a sufficiently small upper semicircle. On the right of the knot,*

$$\boxed{\rho_+(\tau+s) - \rho_-^{\uparrow}(\tau+s) = \sum_{S:\,\tau_S=\tau} e^{-\pi i\eta_S} \frac{H(v_S)W_T^-W_S^+}{Z\Gamma(D_S)} F_S(\tau+s), \qquad 0<s\ll 1.} \tag{IX.2.13}$$

*The exterior physical density is zero at the two extreme endpoints. The formula includes every resonance and every collision among the subset sums. It determines physical local discrepancies in the specified normalization; it does not assert a full connection matrix between arbitrary bases of solutions of a differential system.*

*Proof.* Divide the cube into its $N!$ closed order chambers. Each is a simplex whose vertices are a nested chain of cube vertices. On each chamber every absolute difference is a nonnegative affine function after choosing its sign. The only vertex heights are subset sums. At a height avoiding them, Lemma IX.2.4 therefore proves analyticity once the slice weight is known to be integrable.

Here is a direct integrability check, also used below. A compact slice avoiding all cube vertices can be covered by finitely many regions on each of which one coordinate $x_i$ belongs to $[\varepsilon, 1-\varepsilon]$ for a fixed $\varepsilon>0$. Eliminate this coordinate in the equation $L(x)=t$, with delta Jacobian $1/c_i$. Its endpoint factors are bounded. Every interaction factor is bounded on the cube because its exponent is nonnegative. The remaining weight is bounded by a constant times $\prod_{j\neq i} x_j^{a_j}(1-x_j)^{b_j}$, whose integral on the remaining cube is finite. This proves slice integrability.

Fix a knot $\tau$. For every cube vertex $v_S$ with $\tau_S=\tau$, remove the open corner neighborhood

$$P_S(\delta) = \{x\in[0,1]^N : \sum_i c_i|x_i-(v_S)_i| < \delta\}.$$

Choose $\delta>0$ small enough that their closures are disjoint and each corner neighborhood meets only order chambers containing that corner. The latter is possible because there are finitely many closed chambers, and a vertex has positive distance from every chamber not containing it. We may also require $\delta < \min_i c_i$. It may also be chosen so that no new vertex of the complementary chamber polytopes has height $\tau$. To see this explicitly, consider an edge of an order-chamber

simplex incident to $v_S$. Its other endpoint is nested with $v_S$, so all changing coordinates move either from zero towards one, or from one towards zero. Consequently, on that edge,

$$|L(x) - \tau_S| = \sum_i c_i |x_i - (v_S)_i|.$$

Its intersection with the new truncation face has height $\tau_S + \delta$ or $\tau_S - \delta$. A cut sufficiently close to a simplex vertex creates only these edge-intersection vertices. Disjointness of the removed corner closures prevents intersections between their truncation faces inside a chamber. The other vertices are original cube vertices with heights different from $\tau$. Thus all complementary chamber polytopes have no vertex at height $\tau$.

Their slice weights are integrable by the preceding check, since the truncated complement contains no cube vertex of height $\tau$. Lemma IX.2.4 shows that their sum is a single holomorphic background through $\tau$. Boundaries between chambers and between truncated pieces have zero slice measure for sufficiently small nonzero $s = t - \tau$ and do not change the decomposition.

In $P_S(\delta)$ use $x_i = u_i$ for $i \in T$ and $x_j = 1 - v_j$ for $j \in S$. Then $t - \tau_S = r - q$ and its truncation is exactly $r + q < \delta$. The two homogeneous measures have parameters $\alpha_S, \eta_S$. The analytic amplitude is $H(v_S)$ times the one whose outward value appears in (IX.2.11). Theorem IX.2.1, the normalization (IX.2.6), and the separate empty-cluster calculation therefore give its discrepancy as

$$e^{-\pi i \eta_S} H(v_S) Z_S F_S.$$

Sum over the corners at $\tau$, cancel the common holomorphic background, and divide by the global normalization $Z$. This gives (IX.2.13). The same argument supplies the asserted local upper continuation, including at the extreme endpoints. In dimension one it reduces directly to the one-sided beta-weight calculation, with delta Jacobian $1/c_1$; no chamber singularities or interactions are present. □

The theorem is local at every knot. For an arbitrary analytic tilt $H$, its possible singularities away from the original cube depend on the chosen complex extension of $H$. Consequently this proof does not assert continuation of every $F_S$ along every path in the complement of the subset sums. Such a global assertion requires a separate argument and potentially a restriction on $H$.

## IX.3 Weighted sums of beta variables: exact corner germs and collided knots

Let $m \geq 1$ and let $X_i$ be independent Beta$(a_i, b_i)$ variables, $a_i, b_i > 0$, and let $T = \sum_{i=1}^m c_i X_i$, where $c_i > 0$. Write $B_i = B(a_i, b_i)$. All assertions below concern real shape parameters; no integrality or nonresonance assumption is imposed. The exact density of two beta variables and their linear combinations is classical (Pham-Gia–Turkkan, 1998). Consequently the density formula below is not presented as a first-discovery claim. Its explicit connection normalization and collision criterion are the points to compare with existing general-sum formulas.

For a subset $S \subseteq \{1, \dots, m\}$ put

$$v_S = \sum_{i \in S} c_i, \quad d_{S,i} = \begin{cases} b_i & i \in S, \\ a_i & i \notin S, \end{cases} \quad e_{S,i} = a_i + b_i - d_{S,i}, \quad \epsilon_{S,i} = \begin{cases} -1 & i \in S, \\ 1 & i \notin S, \end{cases}$$

$$D_S = \sum_i d_{S,i}, \qquad \eta_S = \sum_{i \in S} b_i, \qquad k_S = \frac{\prod_i c_i^{-d_{S,i}} \Gamma(d_{S,i})/B_i}{\Gamma(D_S)}.$$

On the standard simplex $\Delta_{m-1}=\{u_i\ge 0,\ \sum_i u_i=1\}$, use the measure obtained by eliminating the last coordinate. Define

$$U_S(s)=\frac{\Gamma(D_S)}{\prod_i\Gamma(d_{S,i})}\int_{\Delta_{m-1}}\prod_i u_i^{d_{S,i}-1}\,(1-\epsilon_{S,i}su_i/c_i)^{e_{S,i}-1}\,du. \tag{IX.3.1}$$

This is analytic for $|s|<\min_i c_i$, is real on a real neighborhood of zero, and satisfies $U_S(0)=1$. Set $F_S(s)=s^{D_S-1}U_S(s)$, with the positive determination for small $s>0$. The superscript $+$ below means continuation above the positive real singular points.

**Theorem IX.3.1** (Global corner representation)**.** *Away from the finite set of subset sums $\{v_S\}$, the density of $T$ is*

$$f_T(t)=\sum_{S:\ v_S<t} e^{-\pi i\eta_S}k_S F_S^{+}(t-v_S). \tag{IX.3.2}$$

*The formula holds for every $a_i,b_i,c_i>0$, including coincident subset sums and integral $D_S$. It also gives zero outside the support $[0,\sum_i c_i]$. No limiting substitution in a formula with sine denominators is used.*

*The normalized germ has the explicit Lauricella representation*

$$U_S(s)=F_B^{(m)}((d_{S,i})_i,(1-e_{S,i})_i;D_S;(\epsilon_{S,i}s/c_i)_i)\,,$$

*where the convention for $F_B$ is*

$$F_B^{(m)}(\mathbf{p},\mathbf{q};D;\mathbf{z})=\sum_{\mathbf{n}\ge 0}\frac{\prod_i(p_i)_{n_i}(q_i)_{n_i}}{(D)_{|\mathbf{n}|}}\prod_i\frac{z_i^{n_i}}{n_i!}.$$

*Proof.* The density of $c_iX_i$ on $(0,c_i)$ is $c_i^{-a_i}x^{a_i-1}(1-x/c_i)^{b_i-1}/B_i$. On the positive half-line define its lower-endpoint extension

$$g_{i,0}(x)=\frac{c_i^{-a_i}}{B_i}x^{a_i-1}(1-(x+i0)/c_i)^{b_i-1}$$

and the upper-endpoint function

$$g_{i,1}(s)=\frac{c_i^{-b_i}}{B_i}s^{b_i-1}(1+s/c_i)^{a_i-1}\quad(s>0).$$

These are locally integrable because the shapes are positive. For $x>c_i$, the upper continuation of the first expression equals $e^{-\pi i(b_i-1)}g_{i,1}(x-c_i)$, and hence

$$f_{c_iX_i}(x)=\mathbf{1}_{x>0}g_{i,0}(x)+e^{-\pi ib_i}\mathbf{1}_{x>c_i}g_{i,1}(x-c_i).$$

This is an exact identity of locally integrable functions. Convolve these $m$ identities and expand the finite product. For the term indexed by $S$, put $x_i=(t-v_S)u_i$. The power and Jacobian give $(t-v_S)^{D_S-1}$ and the remaining simplex integral is exactly $k_SU_S(t-v_S)$.

For $\operatorname{Im}s>0$ the simplex factors have no zeros in its interior, so the same integral is a holomorphic continuation of its germ. Its boundary values give the specified upper continuations; local integrability holds away from subset sums. This proves (IX.3.2). Its support and reality also follow immediately from the original convolution identity. Finally expand each factor in (IX.3.1) in its convergent binomial series and use the Dirichlet integral to obtain the stated $F_B$ series. □

**Theorem IX.3.2** (Exact connection at every collided knot)**.** *Fix a subset sum $v$. The physical density to its right minus the upper continuation of the physical density to its left is, for small $s = t - v > 0$, exactly*

$$J_v(s) = \sum_{S : v_S = v} e^{-\pi i \eta_S} k_S s^{D_S - 1} U_S(s). \tag{IX.3.3}$$

*In particular contributions must be summed before deciding whether the knot is a genuine singularity.*

*Proof.* In (IX.3.2), the subsets with $v_S < v$ represent the density on the preceding interval. Continue each of these terms along the same upper path. Immediately to the right of $v$, precisely the subsets with $v_S = v$ have been added. Their difference is (IX.3.3). This argument does not divide by differences of subset sums or by trigonometric expressions in the shapes. □

### IX.3.1 A coefficient criterion for cancellation and regularity

**Lemma IX.3.3** (Moderate growth from an elementary radial integral)**.** *The local complex branches of a beta-sum density have moderate growth at every subset sum. Its local singular part is a finite sum of radial corner integrals*

$$I_+(s) = \int_0^{(\delta - s)/2} (q + s)^{\alpha - 1} q^{\eta - 1} A(q + s, q)\, dq, \qquad \alpha, \eta > 0,$$

*and their upper continuations, with $A$ holomorphic near the origin. An empty positive or negative cluster instead gives a one-sided power times a holomorphic function.*

*Proof.* Near a corner, set $x_i = u_i$ at a lower endpoint and $x_i = c_i - v_i$ at an upper endpoint. Group their positive distances into $r = \sum u_i$ and $q = \sum v_i$. Dirichlet integration within the two clusters gives the radial powers with $\alpha = \sum_{i \notin S} a_i$ and $\eta = \sum_{i \in S} b_i$; the remaining factors are holomorphic under radial scaling, uniformly on the compact angular simplexes. Truncating by $r + q < \delta$ gives the displayed integral after imposing $s = r - q$.

For completeness the remaining part has a holomorphic density germ. Subdivide each coordinate interval into two short endpoint intervals and a middle interval. Choose their interior division points so that no artificial sum of division endpoints equals the knot under study; there are only finitely many forbidden equalities. In each resulting box, the moving hyperplane has constant incidence with all faces near that knot unless it meets one of the original corner vertices already isolated above. Triangulate its fiber with vertices affine in $s$. Pullback to fixed simplexes leaves fixed integrable endpoint powers times holomorphic factors bounded uniformly on a complex neighborhood. Dominated integration proves holomorphy. Replacing a small corner box by the radial truncation changes only such a transverse boundary contribution.

It remains to bound the radial integral. Replace its upper endpoint by a fixed small $\delta_0 > 0$; the intervening integral is holomorphic and bounded near zero. Put $\rho = |s|$ and split the real integration segment at $2\rho$. On $q \geq 2\rho$, $|q + s| \geq q/2$, so the absolute integral is bounded by

$$C \int_{2\rho}^{\delta_0} q^{\alpha + \eta - 2}\, dq \leq C' \{1 + \rho^{\alpha + \eta - 1} + |\log \rho|\}.$$

On $0 \leq q \leq 2\rho$, scale $q = \rho u$, $s = \rho e^{i\theta}$. The remaining integral

$$\int_0^2 u^{\eta - 1} |u + e^{i\theta}|^{\alpha - 1}\, du$$

is bounded uniformly for $0 \leq \theta \leq \pi$: its possible singularities at $u = 0$ and $u = 1$ are separate and integrable, because $\alpha, \eta > 0$. Thus this part is $O(\rho^{\alpha + \eta - 1})$. Contour continuation of the

negative-side branch differs by an integral from $-s$ to zero; scaling that segment gives the same power bound. Empty-cluster terms satisfy the assertion directly. These estimates establish moderate growth without a differential equation or a nonresonance assumption. □

The following elementary local reconstruction is useful. Suppose a real density, analytic on both punctured sides of a point, has an upper connection jump

$$J(s) = \sum_{\nu} j_\nu s^\nu, \qquad \nu > -1,$$

where the exponents lie in a finite union of unit-spaced progressions and the grouped series converge. Assume its complex branches have moderate growth at the point. Modulo a function analytic through zero, its real local form is the convergent sum of

$$(\operatorname{Re} j_\nu + \cot(\pi\nu)\operatorname{Im} j_\nu)\, s_+^\nu + \frac{\operatorname{Im} j_\nu}{\sin(\pi\nu)}(-s)_+^\nu, \qquad \nu \notin \mathbb{Z}, \tag{IX.3.4}$$

and

$$\operatorname{Re} j_q\, s_+^q + \frac{\operatorname{Im} j_q}{\pi} s^q \log|s|, \qquad q \in \mathbb{Z}_{\geq 0}. \tag{IX.3.5}$$

Here $s_+^\nu$ is zero for $s < 0$ and $s^\nu$ for $s > 0$. For (IX.3.4), upper continuation takes $(-s)^\nu$ to $e^{-\pi i\nu}s^\nu$; for (IX.3.5), it takes $\log(-s)$ to $\log s - \pi i$. These observations verify the asserted jump term by term. The sine denominators are bounded away from zero within each noninteger progression, so convergence is preserved. Subtract this explicit local form. The remaining upper branch has real boundary values on both sides and zero discrepancy. Schwarz reflection extends it to a holomorphic function on a punctured disk. Moderate growth makes the isolated singularity a pole or a removable singularity. Local integrability on the real line excludes every pole, proving the claim.

Lemma IX.3.3 verifies this hypothesis for the beta convolutions, including the endpoint and one-variable cases.

Expand each $U_S(s) = \sum_{n\geq 0} u_{S,n} s^n$ using Theorem IX.3.1, and group all equal powers in (IX.3.3):

$$j_{v,\nu} = \sum_{\substack{v_S = v,\ n\geq 0 \\ D_S - 1 + n = \nu}} e^{-\pi i \eta_S} k_S u_{S,n}. \tag{IX.3.6}$$

**Theorem IX.3.4** (Exact cancellation and regularity)**.** *A real knot is analytic exactly when all coefficients $j_{v,\nu}$ vanish. Otherwise let $\nu_0$ be the first exponent with nonzero coefficient. If $\nu_0 > 0$, the density admits a local $C^k$ representative at that knot for exactly the nonnegative integers $k < \nu_0$. It is not $C^{\lceil \nu_0 \rceil}$. If $\nu_0 \leq 0$, it is not continuous. A nonzero imaginary coefficient at an integer exponent means a logarithmic singularity, whereas a real coefficient there means a one-sided polynomial discrepancy. Moreover its exact local Sobolev threshold is*

$$f_T \in H^s_{\mathrm{loc}} \text{ at } v \quad \Longleftrightarrow \quad s < \nu_0 + \tfrac{1}{2}.$$

*If all jump coefficients vanish, it belongs locally to every $H^s$. This is a coefficient criterion, not a claim of a finite uniform zero-testing algorithm.*

*Proof.* The convergent reconstruction (IX.3.4)–(IX.3.5) and Lemma IX.3.3 give the analytic and $C^k$ assertions. Only nonnegative integers $k$ occur in the $C^k$ assertion; a negative first exponent gives an unbounded one-sided power, while exponent zero gives a jump or a logarithm.

Take a smooth cutoff $\chi$ equal to one near $v$. The complete localized Fourier expansion, in the convention $e^{iR(t-v)}$, is

$$\int e^{iR(t-v)}\chi(t) f_T(t)\, dt \sim \sum_{\nu} \Gamma(\nu+1) e^{i\pi(\nu+1)/2} j_{v,\nu} R^{-\nu-1}, \qquad R \to +\infty.$$

One can verify this termwise in the displayed local reconstruction. Alternatively a monomial $r^k q^l$ in the radial amplitude contributes $\Gamma(\alpha+k)\Gamma(\eta+l)e^{i\pi(\alpha+k-\eta-l)/2}R^{-\alpha-\eta-k-l}$. Its beta-integral jump coefficient has the factor $e^{-\pi i\eta}(-1)^l\Gamma(\alpha+k)\Gamma(\eta+l)/\Gamma(\alpha+\eta+k+l)$, which gives exactly the universal coefficient above. Taylor remainders and integration by parts away from the corner justify the complete expansion. Reality determines the negative-frequency expansion by conjugation. The first nonzero coefficient therefore gives decay of exact magnitude $R^{-\nu_0-1}$ in both directions. The defining weighted Fourier integral for $H^s$ converges exactly when $2s-2\nu_0-2<-1$, proving the threshold, including failure at equality. □

### IX.3.2 Exact examples

**A completely cancelled subset-sum knot.** For three independent uniforms with $c=(1,1,2)$,

$$f_T(t)=\frac{1}{4}\{t_+^2-2(t-1)_+^2+2(t-3)_+^2-(t-4)_+^2\}.$$

The two contributions at $v=2$ are $S=\{3\}$ and $S=\{1,2\}$. They have opposite signs and identical constant normalized germs, so $J_2=0$: the subset sum 2 is not a singularity at all. This uniform special case is the classical spline phenomenon.

**Partial cancellation raises differentiability.** Let $X_1\sim\mathrm{Beta}(1,1)$ and $X_2\sim\mathrm{Beta}(2,2)$ with $c_1=c_2=1$. Direct convolution gives

$$f_T(t)=\begin{cases}3t^2-2t^3, & 0<t<1,\\ 1-3(t-1)^2+2(t-1)^3, & 1<t<2,\\ 0, & t\notin[0,2].\end{cases}$$

Both corners at 1 have $D_S=3$, but their order-two coefficients cancel. The exact discrepancy is $J_1(s)=4s^3$. Consequently the density is $C^2$ but not $C^3$ there; using only the smallest uncombined corner exponent would miss this improvement.

**A resonant logarithmic example.** For $X_1\sim\mathrm{Beta}(1/2,1/2)$ and $X_2\sim\mathrm{Beta}(3/2,3/2)$, again with unit weights, both corners at 1 have $D_S=2$ and $k_S=4/\pi$. Their phases are $-i$ and $i$. The two normalized germs satisfy $U_{\{2\}}(s)=U_{\{1\}}(-s)$ and $U'_{\{1\}}(0)=-1/2$. Hence

$$J_1(s)=\frac{4i}{\pi}s^2+O(s^4).$$

The local density therefore has the term $4s^2\log|s|/\pi^2$ and is $C^1$ but not $C^2$ at the center. The value at the center is $f_T(1)=8/\pi^2$, directly from convolution.

**Scope and provenance.** The exact formulas and examples above are proved here, but no claim of historical first discovery is made. In particular, the two-variable density is covered by T. Pham-Gia and N. Turkkan, *Distribution of the linear combination of two general beta variables and applications*, Communications in Statistics–Theory and Methods 27 (1998), 1851–1869, doi:10.1080/03610929808832194. A recent weighted-sum expansion by Lee–Park, AIMS Mathematics 11 (2026), 28348–28382, doi:10.3934/math.20261129, Theorem 3.5, already gives a finite truncated-polynomial density formula for arbitrary positive weights when all variables have shape $\mathrm{Beta}(1,L-1)$ for a common integer $L\geq 2$; that formula permits subset-sum collisions. Their Lauricella $F_A$ expression concerns a subsequent gamma mixture, not arbitrary positive real beta shapes. The all-shape connection normalization and exact collision criterion are therefore the more specific comparison targets; the short literature check does not establish their historical priority.

## IX.4 All collided graphical knots: exact singularity criteria

The reconstruction argument in the preceding section depends only on an analytic cone amplitude, integrable homogeneous angular measures, and a holomorphic transverse contribution. It therefore applies to the full graphical family of Theorem IX.2.5, including its analytic tilt. This gives an exact criterion even when the finite polynomial test of the last section is unavailable.

**Theorem IX.4.1** (Grouped germs determine local regularity)**.** *Use the notation of Theorem IX.2.5. At a fixed subset-sum knot $\tau$, expand its complete discrepancy as*

$$J_\tau(s) = \sum_{S:\tau_S=\tau} C_S s^{D_S-1} U_S(s) = \sum_\nu j_{\tau,\nu} s^\nu, \qquad C_S = e^{-\pi i \eta_S} \frac{H(v_S) W_{S^c}^- W_S^+}{Z\Gamma(D_S)}.$$

*Terms with the same exponent are combined. This is a convergent series whose exponents belong to finitely many translates of the nonnegative integers, and all exponents exceed $-1$. The density is analytic across $\tau$ if and only if every grouped coefficient $j_{\tau,\nu}$ is zero. Otherwise let $\nu_\tau$ be the least exponent with a nonzero coefficient. Then, for integers $k \geq 0$,*

$$\rho \text{ admits a } C^k \text{ representative near } \tau \quad \Longleftrightarrow \quad k < \nu_\tau.$$

*For a smooth cutoff $\chi$ equal to one near $\tau$ and supported sufficiently close to that knot, and any real $\sigma$,*

$$\chi\rho \in H^\sigma(\mathbb{R}) \quad \Longleftrightarrow \quad \sigma < \nu_\tau + \tfrac{1}{2}.$$

*When $\nu_\tau$ is a nonnegative integer, its nonzero imaginary coefficient is precisely a logarithmic term with coefficient $\operatorname{Im} j_{\tau,\nu_\tau}/\pi$; its real part is the one-sided power discrepancy. At a knot belonging to exactly one corner, $\nu_\tau = D_S - 1$ and that knot is never removable.*

*Proof.* The proof of Lemma IX.3.3 is unchanged for the two homogeneous cone measures in Theorem IX.2.5: their finite angular integrals give a holomorphic bounded radial amplitude, and the radial exponents are $\alpha_S, \eta_S > 0$ for nonempty clusters. Lemma IX.2.4 supplies the common holomorphic remainder. Empty clusters give one-sided powers directly. Thus the moderate-growth hypothesis and the local reconstruction formulas (IX.3.4)–(IX.3.5) apply. They prove the analyticity and classical differentiability assertions.

Alternatively, the exact Fourier relation makes the sharp Sobolev threshold explicit. By (IX.2.8), a coefficient $j_{\tau,\nu}$ of the connection germ contributes

$$e^{i\tau R}\Gamma(\nu+1)e^{\pi i(\nu+1)/2} j_{\tau,\nu} R^{-\nu-1}$$

to the localized positive-frequency Fourier expansion. Indeed $j_{\tau,\nu} = e^{-\pi i \eta_S} f_{S,n}$ and $\nu+1 = D_S+n$ on each corner, so the factor is independent of the corner after equal exponents are combined. The first surviving term is nonzero; reality gives the conjugate behavior at negative frequency. Integrating its squared modulus with weight $(1+R^2)^\sigma$ proves the sharp threshold, including failure at equality. For a single corner $C_S \neq 0$ and $U_S(0) = 1$, so its leading exponent survives. □

The global real-line thresholds follow by a finite partition of unity: take the minimum of $\nu_\tau$ over the nonremovable knots. There are at least the two support endpoints. The criterion is exact, but for an arbitrary analytic tilt it is a test of a whole convergent germ, not a finite algorithm. The polynomial theorem below supplies a finite test in its specified integer-parameter regime.

## IX.5 Resonance and sharp regularity of Jacobi traces

This section refines the all-parameter physical connection theorem. Its additional conclusions are the exact logarithmic term at resonance, its absence criterion, and sharp regularity. These are consequences of the connection theorem and its complete local Fourier expansion. No independent historical-priority claim is made for the regularity of power singularities.

### IX.5.1 Notation and inputs

Fix $N \ge 1$, $a, b > -1$, and $\beta \ge 0$. Let

$$\rho(t) = S_N^{-1} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx,$$
$$S_N = \int_{[0,1]^N} |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx.$$

Extend the density by zero outside $[0, N]$. For a knot $p \in \{0, \dots, N\}$, put $m = N - p$ and

$$\alpha = m(a+1) + \tfrac{\beta}{2} m(m-1), \quad \eta = p(b+1) + \tfrac{\beta}{2} p(p-1), \quad D = \alpha + \eta > 0.$$

Set

$$W_r(q) = \prod_{j=0}^{r-1} \frac{\Gamma(q+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(1+\beta/2)}, \qquad W_0(q) = 1.$$

$$K = \binom{N}{p} \frac{W_m(a) W_p(b)}{S_N} > 0.$$

These symbols always refer to the fixed knot $p$.

For completeness, the canonical analytic factor is the following simplex integral. On $\sum_i u_i + \sum_j v_j = 1$, with $m$ lower and $p$ upper coordinates, define

$$w(u,v) = |\Delta(u)|^\beta |\Delta(v)|^\beta \prod_i u_i^a \prod_j v_j^b, \qquad Z = W_m(a) W_p(b) / \Gamma(D).$$

With Dirac-delta simplex measure $d\sigma$, let

$$U(s) = Z^{-1} \int w(u,v) \prod_i (1 - s u_i)^b \prod_j (1 + s v_j)^a \prod_{i,j} (1 + s(v_j - u_i))^\beta \, d\sigma.$$

The branches equal one at zero. The function $U$ is holomorphic for $|s| < 1$, real on the real axis, and $U(0) = 1$. Empty products are one. Write $F(p+s) = s^{D-1} U(s)$, initially for $s > 0$.

Theorem IX.2.5, specialized to equal weights, common endpoint parameters, a complete interaction graph, and analytic factor one, gives the physical upper continuation jump

$$\rho_{\text{right}} - \rho_{\text{left}}^{\uparrow} = CF, \qquad C = K e^{-\pi i \eta} / \Gamma(D). \tag{IX.5.1}$$

The $\binom{N}{p}$ corners at the same knot have identical normalized functions, so their coefficients add to this nonzero value. Equation (IX.2.8) gives, for a smooth cutoff $\chi$ supported close to $p$ and equal to one there, the complete expansion

$$\int_{\mathbb{R}} e^{iRt} \chi(t) \rho(t) \, dt \sim e^{ipR} R^{-D} \sum_{\ell \ge 0} A_\ell R^{-\ell}, \quad A_0 = K e^{\pi i (\alpha - \eta)/2} \ne 0, \quad R \longrightarrow +\infty. \tag{IX.5.2}$$

No $\log R$ terms occur. Both formulas include endpoints and $\beta = 0$.

The moderate-growth proof of Lemma IX.3.3 applies unchanged to homogeneous cone measures with integrable angular weights and an analytic amplitude: radial scaling replaces the Dirichlet angular integrals by the two cluster integrals, leaving exactly the same one-dimensional estimates. Here those weights include the within-cluster Vandermonde powers, which are integrable for the stated parameters. The analytic part away from the corners is supplied by Theorem IX.2.5. Thus the local reconstruction following Lemma IX.3.3 applies to the Jacobi density without any differential-system input.

### IX.5.2 Exact resonance

**Theorem IX.5.1** (Complete local singularity classification)**.** *There is a real analytic germ $A$ at zero with the following properties. If $D \notin \mathbb{Z}$, then, for sufficiently small real $s \neq 0$,*

$$\rho(p+s) = A(s) + \begin{cases} c_+ s^{D-1} U(s), & s > 0, \\ c_- (-s)^{D-1} U(s), & s < 0, \end{cases} \qquad c_+ = \frac{K \sin(\pi\alpha)}{\Gamma(D)\sin(\pi D)}, \qquad c_- = \frac{K \sin(\pi\eta)}{\Gamma(D)\sin(\pi D)}. \tag{IX.5.3}$$

*At least one of $c_+, c_-$ is nonzero.*

*If $D = d \in \mathbb{Z}_{\geq 1}$, put $k = d-1$. Then*

$$\boxed{\rho(p+s) = A(s) + L\, s^k U(s) \log|s| + J\, \mathbf{1}_{\{s>0\}} s^k U(s)} \tag{IX.5.4}$$

*where*

$$L = -\frac{K \sin(\pi\eta)}{\pi\Gamma(d)} = \frac{(-1)^d K \sin(\pi\alpha)}{\pi\Gamma(d)}, \qquad J = \frac{K\cos(\pi\eta)}{\Gamma(d)}. \tag{IX.5.5}$$

*The entire logarithmic coefficient is $Ls^k U(s)$.*

*At resonance there is no logarithmic term if and only if $\alpha, \eta$ are both integers. In that case $J = (-1)^\eta K/\Gamma(d) \neq 0$. No knot is a point of real analyticity.*

*Proof.* Consider first noninteger $D$. The proposed piecewise fractional-power term in (IX.5.3) has upper continuation discrepancy

$$(c_+ - c_- e^{-\pi i(D-1)}) s^{D-1} U(s) = C s^{D-1} U(s),$$

as follows by taking real and imaginary parts and using $D = \alpha + \eta$. Its coefficients are real and at least one is nonzero: their simultaneous vanishing would force both $\alpha$ and $\eta$ to be integers.

At resonance $D = d$, the function $s^k U(s)$ is holomorphic and real on the real axis. Upper continuation takes $\log(-s)$ to $\log s - i\pi$. Therefore the proposed real function

$$Q(s) = L s^k U(s) \log|s| + J \mathbf{1}_{\{s>0\}} s^k U(s)$$

has upper discrepancy $(J + i\pi L) s^k U(s)$. Taking $J = \operatorname{Re} C$ and $L = \operatorname{Im} C/\pi$ gives exactly (IX.5.5) and matches (IX.5.1).

In either case subtract the proposed singular term from the physical density. The remaining upper branch has zero discrepancy and real boundary values on both punctured sides. Schwarz reflection gives a holomorphic function on a punctured disk. The radial moderate-growth bound makes its singularity a pole or a removable singularity. The physical density and the subtracted terms are locally integrable because $D > 0$, so no pole is possible. This proves that the remainder is a common real analytic germ $A$, exactly as in the local reconstruction following Lemma IX.3.3.

At integer $D$, $\sin(\pi\eta) = 0$ is equivalent to integrality of both $\eta$ and $\alpha$. If $L = 0$, then $J \neq 0$. A nonzero fractional power, logarithm, or analytic jump rules out analyticity across the knot. □

Here an analytic jump means a difference between two analytic germs after removing the common logarithmic term. It need not be a polynomial: integer $\alpha, \eta$ do not force $U$ to be polynomial.

**Corollary IX.5.2** (Renormalized derivative jump)**.** *At resonance, subtract $Ls^k U(s)\log|s|$. The resulting function has matching derivatives below order $k$; its derivative of order $k$ has right-minus-left jump*

$$k!J = K\cos(\pi\eta).$$

*The leading logarithm in the unrenormalized $k$-th derivative is*

$$k!L\log|s| = \frac{(-1)^d K\sin(\pi\alpha)}{\pi}\log|s|.$$

*The renormalized function is analytic across the knot precisely when $\cos(\pi\eta) = 0$. For $k = 0$, the derivative jump means the function jump.*

### IX.5.3 Sharp regularity

**Theorem IX.5.3** (Local and global thresholds)**.** *For every integer $j \geq 0$, the density has a $C^j$ representative near $p$ if and only if*

$$j < D_p - 1.$$

*For noninteger $D_p > 1$, put $j = \lfloor D_p - 1\rfloor$ and $\theta = D_p - 1 - j$. It is $C^{j,\theta}$, but not $C^{j,\theta'}$ for any $\theta < \theta' \leq 1$. If $D_p = k+1$, $k \geq 1$, it is $C^{k-1,1}$ when $L = 0$. When $L \neq 0$, it is $C^{k-1,\theta}$ for every $0 < \theta < 1$ but not $C^{k-1,1}$.*

*For every real $\sigma$, with a cutoff as above,*

$$\chi\rho \in H^\sigma(\mathbb{R}) \quad\Longleftrightarrow\quad \sigma < D_p - \tfrac{1}{2}. \tag{IX.5.6}$$

*Here $H^\sigma$ is the $L^2$ Bessel-potential space. For the entire zero-extended density,*

$$\rho \in H^\sigma(\mathbb{R}) \quad\Longleftrightarrow\quad \sigma < \min_{0\leq p\leq N} D_p - \tfrac{1}{2}. \tag{IX.5.7}$$

*Its global $C^j$ threshold is $j < \min_p D_p - 1$. Equality at every stated threshold is excluded.*

*Proof.* Differentiate the local formulas, using $U(0) = 1$. A noninteger power has the stated exact Hölder regularity. The $(k-1)$-st derivative of $s^k\log|s|$ has a nonzero $s\log|s|$ term, which is Hölder of every order below one but not Lipschitz. If $L = 0$, the nonzero $J$ yields a Lipschitz $(k-1)$-st derivative and a jump in derivative $k$. For $D_p < 1$ the density is unbounded on at least one side. For $D_p = 1$ there is either a logarithmic divergence or a nonzero jump.

By (IX.5.2), the localized Fourier transform has modulus asymptotic to $KR^{-D_p}$ at positive infinity. Reality gives the same modulus at negative infinity. Therefore the Sobolev condition is exactly convergence of $\int_1^\infty R^{2\sigma-2D_p}\,dR$. At equality this integral diverges logarithmically. A smooth partition of unity about the finitely many knots and their complement proves the global sufficient condition. Necessity follows by localizing at a minimizing knot, because multiplication by a smooth cutoff is bounded on $H^\sigma$ for every real $\sigma$. Away from the knots the density is analytic. The classical global threshold follows in the same way. □

The minimizing exponent is explicit:

$$D_p = \beta(p - N/2)^2 + (b-a)(p - N/2) + \frac{N(a+b+2)}{2} + \frac{\beta N(N-2)}{4}.$$

For $\beta > 0$, choose the integer $p \in [0, N]$ nearest to $N/2 + (a-b)/(2\beta)$, clipping to the endpoints. For $\beta = 0$, $\min_p D_p = N\min(a+1, b+1)$. Ties do not alter the threshold.

### IX.5.4 Symmetric midpoint

**Corollary IX.5.4** (Midpoint parity dichotomy)**.** *Suppose $a = b$, $N$ is even, and $p = N/2$. Then $U$ is even, $\alpha = \eta = D/2$, and the density is even about $p$. If $D \notin \mathbb{Z}$, its nonanalytic part is*

$$\frac{K}{2\Gamma(D)\cos(\pi D/2)}|s|^{D-1}U(s).$$

*If $D = d$ is odd, its nonanalytic part is $Ls^{d-1}U(s)\log|s|$, with $L \neq 0$ and $J = 0$. If $D = d$ is even, its nonanalytic part modulo an even analytic germ is*

$$\frac{(-1)^{d/2}K}{2\Gamma(d)}|s|^{d-1}U(s).$$

*Thus symmetry never removes an integer midpoint singularity. For odd $N$, the midpoint is not a knot and the density is analytic there.*

*Proof.* Interchanging the two simplex clusters takes $s$ to $-s$; hence $U$ is even. Reflection $x_i \mapsto 1 - x_i$ proves symmetry of the density. Substitute $\alpha = \eta = D/2$ in the formulas above. For even $d$, the integer $k = d - 1$ is odd and $\mathbf{1}_{\{s>0\}}s^k = (s^k + |s|^k)/2$. Absorb the analytic odd term into the background; symmetry makes the remaining background even. □

## IX.6 Exact collision tests for polynomial Jacobi weights

This section gives a finite exact test at every subset-sum collision for integer endpoint powers and even Dyson index. It also identifies the test with removal of a knot in the trace density. Polynomial densities on slices of cubes belong to the classical spline framework; see [87, 88]. Finite density formulas for positive weighted independent beta variables are also available; for example, [89, Theorem 3.5] treats independent $\mathrm{Beta}(1, \ell - 1)$ variables for integer $\ell \geq 2$ without requiring distinct subset sums. The assertion here is the explicit specialization to Jacobi weights, together with the exact minimal rational differential order and a collision example in which a vanished leading corner term does not remove its frequency. No claim that the underlying polynomial integration or spline representation is new is made.

Let $N \geq 1$, $c_i > 0$, $a_i, b_i, m \in \mathbb{Z}_{\geq 0}$, and put

$$P(u) = \prod_{i<j}(u_j - u_i)^{2m}\prod_{i=1}^{N}u_i^{a_i}(1-u_i)^{b_i}, \qquad D = \sum_i(a_i + b_i) + mN(N-1), \qquad L = N + D.$$

Define the unnormalized transform

$$Z(t) = \int_{[0,1]^N} e^{t\sum_i c_i u_i}P(u)\,du.$$

It is entire and $Z(0) > 0$. Constant normalization by $Z(0)$ has no effect on the differential orders below. For $S \subseteq [N]$, let $v_S$ be its zero-one indicator vector and $\lambda_S = \sum_{i\in S} c_i$. Write $\Lambda = \{\lambda_S : S \subseteq [N]\}$, with repetitions removed. Multi-indices have nonnegative integer components.

**Theorem IX.6.1** (Complete finite collision criterion)**.** *For each $\lambda \in \Lambda$ define the polynomial*

$$Q_\lambda(t) = \sum_{\substack{S\subseteq[N]\\ \lambda_S=\lambda}}(-1)^{N-|S|}\sum_{|\alpha|\leq D}\frac{(-1)^{|\alpha|}(\partial^\alpha P)(v_S)}{\prod_i c_i^{\alpha_i+1}}\,t^{D-|\alpha|}. \tag{IX.6.1}$$

*Then the following statements hold.*

1. *The exact identity*
$$t^L Z(t) = \sum_{\lambda \in \Lambda} e^{\lambda t} Q_\lambda(t) \tag{IX.6.2}$$
*holds for every complex* $t$.

2. *The minimal order over* $\mathbb{C}(t)$ *of a nonzero homogeneous linear differential operator annihilating* $Z$ *is*
$$M = \#\{\lambda \in \Lambda : Q_\lambda \not\equiv 0\}. \tag{IX.6.3}$$
*In particular, a frequency is removed if and only if, for every* $0 \le k \le D$,
$$\sum_{\substack{S \subseteq [N] \\ \lambda_S = \lambda}} (-1)^{N-|S|+k} \sum_{|\alpha|=k} \frac{(\partial^\alpha P)(v_S)}{\prod_i c_i^{\alpha_i+1}} = 0. \tag{IX.6.4}$$

3. *These tests give a finite semialgebraic partition of the positive weight space* $(0,\infty)^N$ *on which the exact order is constant. For algebraic positive weights, they are exact algebraic decision tests.*

*Proof.* For a polynomial $p$ of degree at most $D$, repeated integration by parts gives, for $t \neq 0$,
$$\int_0^1 e^{ctu} p(u)\, du = \sum_{k=0}^{D} \frac{(-1)^k}{(ct)^{k+1}} \bigl(e^{ct} p^{(k)}(1) - p^{(k)}(0)\bigr).$$
Apply this identity successively in every coordinate. Derivatives of total order greater than $D$ vanish. Choosing the upper endpoint on $S$ and the lower endpoint on its complement gives exactly the sign and denominator in (IX.6.1). Grouping equal values of $\lambda_S$ proves (IX.6.2) away from zero. Both sides are entire, so the identity extends to zero.

For each surviving $\lambda$, set $y_\lambda = e^{\lambda t} Q_\lambda(t)/t^L$. Its logarithmic derivative is rational. Thus the $\mathbb{C}(t)$-span of the $M$ functions $y_\lambda$ is closed under differentiation and contains $Z$; a linear dependence among $Z, Z', \dots, Z^{(M)}$ gives an annihilator of order at most $M$.

Distinct real exponentials are linearly independent over $\mathbb{C}(t)$. Indeed, after clearing denominators, a purported dependence is a polynomial-exponential sum. Along the positive real axis, divide by its largest exponential with a nonzero polynomial coefficient. A nonzero polynomial cannot equal an exponentially decaying sum, a contradiction. Now suppose a rational operator $A$ annihilates $Z$. Every $Ay_\lambda$ is $e^{\lambda t}$ times a rational function. The same independence gives $Ay_\lambda = 0$ for every surviving $\lambda$. The functions $y_\lambda$ are also linearly independent over the constants. On a disk avoiding the poles of the coefficients and the zeros of the leading coefficient, an operator of order $r$ has a solution space of dimension $r$. Consequently $r \ge M$, proving (IX.6.3).

Coefficient comparison gives (IX.6.4). There are finitely many possible equality patterns among the linear forms $\lambda_S$. Fix such a pattern. Each coefficient in (IX.6.1) is a rational function of $c$ with rational coefficients and denominators consisting only of powers of the positive $c_i$. Multiplication, for example, by $\prod_i c_i^{D+1}$ clears them. Equality or inequality to zero of these finitely many polynomials, together with the subset-sum equality pattern, gives the claimed finite semialgebraic partition. □

**Lemma IX.6.2** (An individual corner never disappears)**.** *Let* $Q_S$ *denote the single-*$S$ *summand in* (IX.6.1)*, before colliding frequencies are grouped. Then* $Q_S \neq 0$*. Consequently pairwise distinct subset sums give order* $2^N$*.*

*Proof.* Use local coordinates $u_i = x_i$ on $S^c$ and $u_i = 1 - x_i$ on $S$. The lowest homogeneous term of $P$ is
$$H(x) = \Delta(x_{S^c})^{2m} \prod_{i \notin S} x_i^{a_i} \Delta(x_S)^{2m} \prod_{i \in S} x_i^{b_i}.$$

It has separate degrees $h_- = \sum_{i\notin S} a_i + m|S^c|(|S^c|-1)$ and $h_+ = \sum_{i\in S} b_i + m|S|(|S|-1)$ in the two clusters. The coefficient of the highest power $t^{D-h_--h_+}$ in $Q_S$ is therefore

$$(-1)^{N-|S|+h_-}\int_{(0,\infty)^N} e^{-\sum_i c_i x_i} H(x)\,dx.$$

Indeed, each monomial in the local coordinates contributes its factorial moment, and the endpoint changes of sign leave precisely the displayed common sign. The integral is finite and strictly positive. The highest coefficient is consequently nonzero. □

*Remark* IX.6.3 (Constructing a minimal operator). The theorem is constructive even when the weights are not rational. For the surviving frequencies form the Wronskian of the $y_\lambda$. It is nonzero: after factoring out each column's $y_\lambda$, its leading determinant at infinity is the Vandermonde in the distinct $\lambda$. The monic Wronskian operator with these $M$ solutions has rational coefficients, because exponential factors cancel from its coefficient ratios. It is a minimal operator for $Z$. Individual $y_\lambda$ can have poles at zero; the complete sum $Z$ has a removable singularity there. No conclusion about absence of apparent singularities of the scalar operator is needed.

### IX.6.1 The same test removes precisely the density knots

Let $\rho$ be the compactly supported density defined by

$$\int_{\mathbb{R}} \varphi(x)\rho(x)\,dx = \int_{[0,1]^N} \varphi\Big(\sum_i c_i u_i\Big) P(u)\,du.$$

Such an integrable density exists by integrating first in $u_1$, since $c_1 > 0$. It is not normalized; its integral is $Z(0)$.

**Theorem IX.6.4** (Differential order equals the number of genuine knots)**.** *The density $\rho$ is a polynomial of degree at most $L-1$ on each component of $\mathbb{R}\setminus\Lambda$. With $\mathcal{D} = d/dx$, its distributional derivative satisfies*

$$\mathcal{D}^L\rho = (-1)^L \sum_{\lambda\in\Lambda} Q_\lambda(-\mathcal{D})\delta_\lambda. \tag{IX.6.5}$$

*A candidate knot $\lambda$ is removable, meaning that one polynomial agrees almost everywhere with $\rho$ on a neighborhood of $\lambda$, if and only if $Q_\lambda = 0$. Hence the minimal order in* (IX.6.3) *equals the number of genuine knots, including the two support endpoints.*

*Proof.* For a compactly supported distribution, the transform used here sends $\mathcal{D}$ to multiplication by $-t$ and sends $\delta_\lambda$ to $e^{\lambda t}$. Thus the transforms of both sides of (IX.6.5) agree by (IX.6.2). Injectivity of the Fourier transform on compactly supported distributions proves the identity. On an interval disjoint from $\Lambda$, it gives $\mathcal{D}^L\rho = 0$, so $\rho$ is a polynomial distribution of degree less than $L$.

In a neighborhood containing no other candidate knot, the right side is zero exactly when $Q_\lambda = 0$, since the derivatives of a Dirac mass are linearly independent. Vanishing is equivalent to one polynomial on that neighborhood. Finally, the support is exactly $[0, \sum_i c_i]$: $P$ is positive on the interior away from the diagonals and coordinate endpoints, and every sufficiently small interval inside the image has positive mass. The density is zero outside its support. A polynomial agreeing across either support endpoint would therefore be identically zero near that endpoint, a contradiction. Both endpoints are genuine knots. □

### IX.6.2 A leading cancellation that does not remove a frequency

**Proposition IX.6.5** (Uniform and interacting three-variable laws)**.** *Let $N = 3$, $a_i = b_i = 0$, and $c = (p, q, p+q)$ with $p, q > 0$. For $\beta = 0$ the exact differential order is six if $p \neq q$ and four if $p = q$. For $\beta = 2$ it is seven if $p \neq q$ and five if $p = q$.*

*Proof.* For $\beta = 0$, independence gives

$$Z(t) = \frac{(e^{pt}-1)(e^{qt}-1)(e^{(p+q)t}-1)}{pq(p+q)t^3}.$$

The terms at frequency $p+q$ cancel. If $p \neq q$, the remaining frequencies are $0, p, q, 2p+q, p+2q, 2p+2q$, all distinct with nonzero coefficients. If $p = q$, the numerator reduces to $e^{4pt} - 2e^{3pt} + 2e^{pt} - 1$, giving four frequencies.

For $\beta = 2$, $D = 6$ and $L = 9$. The two corners at the central frequency are $(0, 0, 1)$ and $(1, 1, 0)$. Substitution in (IX.6.1) gives

$$Q_{p+q}(t) = \frac{24(p^4+q^4)}{p^4q^4(p+q)^4} t((p+q)^2t^2 + 6), \tag{IX.6.6}$$

which is nonzero. To verify this finite calculation, at the first corner substitute $(u_1, u_2, u_3) = (x, y, 1-z)$ into $P$:

$$P = (y-x)^2(1-z-x)^2(1-z-y)^2.$$

Expand this polynomial and use (IX.6.1); the coefficients of $t^4, t^2, t^0$ in the sum of the two central corners are zero, while those of $t^3$ and $t$ are respectively

$$\frac{24(p^4+q^4)}{p^4q^4(p+q)^2}, \qquad \frac{144(p^4+q^4)}{p^4q^4(p+q)^4}.$$

All other frequencies have one corner if $p \neq q$, and its polynomial is nonzero by Lemma IX.6.2. When $p = q$, the only further collisions pair corners related by interchanging the first two coordinates. Their complete corner polynomials are equal, and hence add rather than cancel. The frequency counts are consequently seven and five. □

For example, the independent uniform laws have

$$Z_{(1,1,2),0}(t) = \frac{e^{4t} - 2e^{3t} + 2e^t - 1}{2t^3}, \qquad Z_{(1,2,3),0}(t) = \frac{e^{6t} - e^{5t} - e^{4t} + e^{2t} + e^t - 1}{6t^3}.$$

Their orders are four and six. At Dyson index two, the central polynomials are respectively $6t(2t^2+3)$ and $17t(3t^2+2)/18$, and the orders increase to five and seven.

The structural reason that a leading cancellation can be misleading is particularly transparent here. Whenever $P(1-u) = P(u)$ and $C = \sum_i c_i$, the substitution $u \mapsto 1-u$ yields

$$Z(t) = e^{Ct}Z(-t), \qquad Q_{C-\lambda}(t) = (-1)^L Q_\lambda(-t).$$

For symmetric integer endpoint powers, $D$ is even. With odd $N$, the central polynomial must therefore be odd. For independent uniform variables it is constant and hence vanishes; for $\Delta^2$ it can be a nonzero odd polynomial, as (IX.6.6) shows.

*Remark* IX.6.6 (Scope). The finite criterion covers arbitrary positive weights, all their subset-sum degeneracies, and nonhomogeneous nonnegative integer endpoint powers. It is an exact algorithmic classification, not a short combinatorial list of every possible weight relation. It does not resolve colliding frequencies for arbitrary noninteger endpoint powers or for non-even Dyson index. The elementary polynomial regime is also not evidence for irreducibility or a full differential Galois group.

## IX.7 Algebraicity obstructions for physical Jacobi densities

This section records the two consequences discussed after the standalone cone-gluing manuscript. They follow from its explicit nonzero connections and logarithmic classification. They concern algebraicity of functions over $\mathbb{C}(t)$, not the arithmetic nature of their individual numerical values. No separate historical-priority claim is attached to these corollaries.

For fixed $N, a, b, \beta$ in the range of Section IX.5, call the physical density *piecewise algebraic* if, on each interval $(j, j+1)$, $0 \le j < N$, its analytic germ is algebraic over $\mathbb{C}(t)$. The polynomial relation is allowed to depend on the interval. Constants are in $\mathbb{C}$, so no arithmetic condition on the Selberg normalization is implicit in this definition.

We use the classical Newton–Puiseux theorem: an algebraic germ at a finite point has a convergent Laurent–Puiseux expansion with rational exponents and finitely many local branches; see [90], Theorem VII.7. In particular, its leading order is rational, and a nonzero logarithmic coefficient is impossible.

**Theorem IX.7.1** (Rational parameters are necessary)**.** *Let $N \ge 2$, $a, b > -1$, and $\beta \ge 0$. If the ordinary Jacobi trace density is piecewise algebraic, then*

$$a, b, \beta \in \mathbb{Q}.$$

*Equivalently, if one of these three parameters is irrational, at least one physical interval germ is transcendental over $\mathbb{C}(t)$.*

*Proof.* At the knot $p \in \{0, \dots, N\}$ the exact connection is

$$\rho_{p,+} - \rho_{p,-}^{\uparrow} = C_p s^{D_p - 1} U_p(s), \qquad C_p \neq 0, \quad U_p(0) = 1.$$

At the two endpoints the exterior germ is the zero function, which is algebraic. Analytic continuation preserves an algebraic relation, and the difference of two algebraic germs is algebraic. Thus the displayed nonzero germ is algebraic at every $p$. Its Puiseux order is $D_p - 1$, so $D_p \in \mathbb{Q}$ for all $p$.

The exponents are the quadratic polynomial

$$D_p = N(a+1) + \frac{\beta N(N-1)}{2} + p\big(b - a - \beta N\big) + \beta p^2.$$

Since $N \ge 2$, its second difference gives

$$\beta = \frac{D_2 - 2D_1 + D_0}{2} \in \mathbb{Q}.$$

The endpoint identities then imply

$$a = \frac{D_0 - \beta N(N-1)/2}{N} - 1 \in \mathbb{Q}, \qquad b = \frac{D_N - \beta N(N-1)/2}{N} - 1 \in \mathbb{Q}.$$

No algebraicity assertion about the constants $C_p$ over $\mathbb{Q}$ is needed. □

**Theorem IX.7.2** (A nonzero resonant logarithm excludes algebraicity)**.** *At an interior knot $p$, suppose that $D_p = d$ is a positive integer and $\eta_p \notin \mathbb{Z}$. Both adjacent physical interval germs are transcendental over $\mathbb{C}(t)$ and have infinitely many local branches under repeated continuation around $p$.*

*Proof.* By Theorem IX.5.1, with $s = t - p$,

$$\rho(p+s) = A(s) + LF(s)\log|s| + J\mathbf{1}_{\{s>0\}}F(s), \quad F(s) = s^{d-1}U_p(s), \quad L = -\frac{K_p \sin(\pi\eta_p)}{\pi\Gamma(d)} \neq 0.$$

Here $A$ and $F$ are single-valued holomorphic germs and $F$ is not identically zero. Complexifying either side replaces the real logarithm by $\log s$ or $\log(-s)$ on a chosen branch. After $n$ positive loops around zero, the germ changes by $2\pi i n L F(s)$. These are distinct germs for distinct integers $n$. Algebraic germs have finitely many local branches, so neither adjacent germ is algebraic. □

**Corollary IX.7.3** (An all-repulsion nonalgebraic family)**.** *For $N = 2$, $a = b = -1/2$, and every real $\beta \geq 0$, both physical density germs on $(0, 1)$ and $(1, 2)$ are transcendental over $\mathbb{C}(t)$. More precisely, at $t = 1$,*

$$\rho(1 + s) = A(s) - \frac{2}{S_2} U_1(s) \log |s|, \qquad U_1(0) = 1, \quad S_2 > 0.$$

*Proof.* At $p = 1$ the two clusters each contain one variable, so $\alpha_1 = \eta_1 = 1/2$ and $D_1 = 1$, independently of $\beta$. Moreover $W_1(-1/2) = \Gamma(1/2) = \sqrt{\pi}$, whence $K_1 = 2\pi/S_2$. Formula (IX.5.5) gives $L = -2/S_2$ and $J = 0$. Apply Theorem IX.7.2. □

The converse of Theorem IX.7.1 is false: the last corollary with $\beta = 0$ already gives rational parameters with nonalgebraic density germs. Conversely, when $a, b$ are nonnegative integers and $\beta$ is an even nonnegative integer, the polynomial slice construction in Section IX.6 makes each interval density a polynomial, hence algebraic. The necessary rationality test, the logarithmic obstruction, and that sufficient polynomial regime do not constitute a complete classification. Rational nonpolynomial cases with no local logarithmic obstruction require additional global analysis.

## IX.8 A concrete comparison and the remaining global question

The three-variable family in Proposition IX.6.5 gives a small exact diagnostic of the method. For weights $(1, 2, 3)$ the candidate heights are $0, 1, 2, 3, 4, 5, 6$. Under the independent uniform law, the central height 3 is removable and the transform has minimal order six. Under the normalized Jacobi weight $360 \prod_{i<j}(x_i - x_j)^2$, the central height is genuine and the minimal order is seven. The leading corner contributions at that height still cancel. The complete calculation detects the surviving term.

For the latter law, its central polynomial is

$$Q_3(z) = \frac{17}{18} z(3z^2 + 2), \qquad L = 9, \qquad Z(0) = \frac{1}{360}.$$

Equation (IX.6.5), or direct inverse Laplace transformation, gives the normalized local polynomial discrepancy

$$\rho(3 + s)_{\text{right}} - \rho(3 + s)^{\uparrow}_{\text{left}} = \frac{17}{2} s^5 + \frac{17}{126} s^7.$$

Consequently the density is $C^4$ but not $C^5$ at 3; its fifth derivative has right-minus-left jump 1020. Symmetry about 3 allows the entire nonanalytic part to be written

$$\frac{17}{4}|s|^5 + \frac{17}{252}|s|^7$$

modulo an even analytic germ. This contrasts with complete removal under the independent law. The coefficient signs use the convention $Z(z) = \int e^{zt} \rho(t)\, dt$, so a term $z^{-2r}$ corresponds to $t_+^{2r-1}/(2r - 1)!$.

A main question left by this part is a genuinely global one: already for $H = 1$ and arbitrary noninteger repulsion exponents, find an explicit minimal continuation system and an effective finite criterion for complete cancellation at every collided height. Existence of holonomic systems

for hypergeometric integrals is a classical input, not a new open claim here. The local formula determines the whole germ, but does not by itself decide whether an arbitrary analytic germ vanishes from finitely many coefficients. In particular, a scalar leading amplitude cannot supply that criterion, as the preceding example proves. A separate task is to determine which of these normalizations and classifications already occur in the extensive literature on Selberg integrals, hypergeometric connections, and splines. Neither task is replaced by the exact calculations here.

## References


[1] T. D. Noe, *On the divisibility of generalized central trinomial coefficients*, Journal of Integer Sequences **9** (2006), Article 06.2.7. https://cs.uwaterloo.ca/journals/JIS/VOL9/Noe/noe35.html.

[2] H. Liang, Y. Wang and Y. Wang, *Analytic aspects of generalized central trinomial coefficients*, arXiv:2303.06922v2 (2023). https://arxiv.org/abs/2303.06922.

[3] A. Huber and G. Wüstholz, *Transcendence and Linear Relations of 1-Periods*, Cambridge Tracts in Mathematics 227, Cambridge University Press, 2022. Preprint https://arxiv.org/abs/1805.10104, Theorem 6.2.

[4] J. C. del Valle and A. V. Turbiner, *Power-like potentials: from the Bohr–Sommerfeld energies to exact ones*, International Journal of Modern Physics A **36** (2021), 2150221. https://arxiv.org/abs/2108.00327.

[5] T. Leung, X. Li, and Z. Wang, *Optimal Starting-Stopping and Switching of a CIR Process with Fixed Costs*, author manuscript, arXiv:1411.6080, version dated 30 August 2018. https://arxiv.org/pdf/1411.6080, Theorem 3.1 and equation (3.7).

[6] E. Gassiat, A. Cleynen, and S. Robin, *Finite state space non parametric Hidden Markov Models are in general identifiable*, author preprint, arXiv:1306.4657 (2013). https://arxiv.org/abs/1306.4657.

[7] E. Gassiat, S. Le Corff, and L. Lehéricy, *Identifiability and consistent estimation of nonparametric translation hidden Markov models with general state space*, author preprint, arXiv:1902.01070 (2019). https://arxiv.org/abs/1902.01070.

[8] KEIO, *Minimal Recurrence Orders and Integral Solution Lattices for Franel Sequences*, supplied unpublished manuscript [006] (2026).

[9] KEIO, *One Missing Edge: The Exact $K_n - e$ Cone, Strict Support, and Sharp Defect Laws*, supplied unpublished manuscript [015] (2026).

[10] KEIO, *Gamma-Type Functions: Arithmetic and Differential Independence, Probability Laws, and Interpolation*, supplied unpublished manuscript [025] (2026).

[11] KEIO, *Observation and Recovery in Arithmetic and Probability*, supplied unpublished manuscript [028] (2026).

[12] KEIO, *Arithmetic and Observation: Classifications, Recovery, and Transcendence Barriers*, supplied unpublished manuscript [030] (2026).

[13] A. Suliman and O. Weinstein, *Infinite Lewis Weights in Spectral Graph Theory*, arXiv:2302.05966v1 (2023). https://arxiv.org/abs/2302.05966. Theorem 1.3, Conjectures 1.5 and 4.6, and Section 4.1. Additional source link: https://arxiv.org/abs/2302.05966v1.

[14] K. Abraham, E. Gassiat, and Z. Naulet, *Frontiers to the learning of nonparametric hidden Markov models*, Journal of Machine Learning Research **26**, no. 155 (2025), 1–75. https://jmlr.org/papers/volume26/24-2230/24-2230.pdf.

[15] KEIO, *Exact Complexity and Arithmetic Transfers: Spectral Design, Hamiltonian Dynamics, Optimal Stopping, and Hidden-State Recovery*, unpublished research draft, 19 October 2026, 32 pp. Reproduced as Part I of the present omnibus; retained here as a historical provenance reference.

[16] C. Musco, C. Musco, L. Rosenblatt and A. V. Singh, Sharper bounds for Chebyshev moment matching, with applications, *Proceedings of COLT*, PMLR **291** (2025), 4309–4358. Author version arXiv:2408.12385v3, 18 May 2026, Section 6. https://arxiv.org/html/2408.12385v3.

[17] M. Chen, C. Gao and Z. Ren, A general decision theory for Huber's $\varepsilon$-contamination model, *Electronic Journal of Statistics* **10** (2016), 3752–3774. https://arxiv.org/abs/1511.04144.

[18] B. G. Lindsay, The geometry of mixture likelihoods: a general theory, *Annals of Statistics* **11** (1983), 86–94. https://doi.org/10.1214/aos/1176346059.

[19] S. G. Bobkov and F. Götze, Exponential integrability and transportation cost related to logarithmic Sobolev inequalities, *Journal of Functional Analysis* **163** (1999), 1–28. https://doi.org/10.1006/jfan.1998.3326.

[20] KEIO, supplied integrated research compilation, Parts VI–VIII (2026). Part VI proves the complete clean binomial-mixture likelihood transition; Part VII proves the TV inverse modulus and Huber minimax curve; Part VIII proves polynomial-time robust recovery under data-dependent report replacement and explicitly leaves robustness of the unmodified NPMLE open. AI-assisted research manuscript, not externally peer reviewed.

[21] J. Lee, V. Baćak and E. H. Kennedy, Learning smooth populations of parameters with trial heterogeneity, arXiv:2507.23140v2, 9 October 2026. https://arxiv.org/abs/2507.23140v2.

[22] J. Lee and E. H. Kennedy, Minimax rates for learning smooth populations of parameters, arXiv:2610.04135v2, 9 October 2026. https://arxiv.org/html/2610.04135v2.

[23] P. Rigollet and J. Weed, Entropic optimal transport is maximum-likelihood deconvolution, *Comptes Rendus Mathématique* **356** (2018), 1228–1235. https://doi.org/10.1016/j.crma.2018.10.010.

[24] J. Niles-Weed, Reweighted information inequalities, arXiv:2603.13135v1 (2026). https://arxiv.org/html/2603.13135v1.

[25] KEIO, *Sharp Bounds and Connection Formulas Beyond Existing Restrictions: Four Proof Transfers in Spectral Design, Binomial Sums, Random Matrices, and Hidden-State Statistics*, unpublished AI-assisted research draft (2026), 23 pp. Not externally peer reviewed. Reproduced as Part II of the present omnibus; retained here as a historical provenance reference.

[26] T. Matala-Aho, *On recurrences for some hypergeometric type polynomials*, Astérisque **198–199–200** (1991), 237–244. https://www.numdam.org/item/AST_1991__198-199-200__237_0/.

[27] Y. Simsek, *Generating functions for finite sums involving higher powers of binomial coefficients: Analysis of hypergeometric functions including new families of polynomials and numbers*, Journal of Mathematical Analysis and Applications **477** (2019), no. 2, 1328–1352. doi:10.1016/j.jmaa.2019.05.015.

[28] M. Stoll, *Bounds for the length of recurrence relations for convolutions of P-recursive sequences*, European Journal of Combinatorics **18** (1997), no. 6, 707–712. doi:10.1006/eujc.1996.0123.

[29] R. J. McIntosh, *Recurrences for alternating sums of powers of binomial coefficients*, Journal of Combinatorial Theory, Series A **63** (1993), no. 2, 223–233. doi:10.1016/0097-3165(93)90058-G. The cited inputs include Theorems 1 and 2.

[30] KEIO, *Observation and Recovery: Geometry, Mixtures, and Spectral Inversion*, supplied unpublished manuscript [029], 16 October 2026. Part VI, "Boundary Laws for Observation and Recovery," section "The complete binomial-mixture likelihood transition." AI-assisted research manuscript, not externally peer reviewed.

[31] KEIO, *Entropy, Likelihood Geometry, and Sharp Proof Transfers*, unpublished AI-assisted research draft, 19 October 2026, 34 pp. Not externally peer reviewed. Reproduced as Part III of the present omnibus; retained here as a historical provenance reference.

[32] Y. Baraud and L. Birgé, *$\rho$-estimators revisited: general theory and applications*, arXiv:1605.05051, version 5 (2017). https://arxiv.org/abs/1605.05051.

[33] H. Fujisawa and S. Eguchi, *Robust estimation in the normal mixture model*, Journal of Statistical Planning and Inference **136** (2006), 3989–4011. https://doi.org/10.1016/j.jspi.2005.03.008.

[34] A. Lecestre, *Robust Estimation in Finite Mixture Models*, arXiv:2107.00587 (2021). https://arxiv.org/abs/2107.00587.

[35] C. Banderier and M. Drmota, *Formulae and asymptotics for coefficients of algebraic functions*, Combinatorics, Probability and Computing **24** (2015), no. 1, 1–53. doi:10.1017/S0963548314000728. Author version: https://lipn.fr/~banderier/Papers/Alg.pdf.

[36] B. Adamczewski, J. P. Bell, and É. Delaygue, *Algebraic independence of G-functions and congruences "à la Lucas"*, Annales scientifiques de l'École normale supérieure **52** (2019), 515–559. doi:10.24033/asens.2392. https://arxiv.org/abs/1603.04187.

[37] A. J. Guttmann, *Lattice Green functions in all dimensions*, J. Phys. A: Math. Theor. **43** (2010), 305205. https://arxiv.org/abs/1004.1435.

[38] R. T. Delves and G. S. Joyce, *On the Green function for the anisotropic simple cubic lattice*, Ann. Phys. **291** (2001), 71 ff. https://www.sciencedirect.com/science/article/pii/S0003491601961482.

[39] N. Elkies, T. W. Grimm and M. D. Schwartz, *The anisotropic Watson integral*, announced as in preparation on the authors' project website, consulted 19 October 2026. https://bootloops.ai/summaries/watson.html. The public summary, rather than an available full proof, is cited only to identify the prior generic three-dimensional claim.

[40] R. García López, *Microlocalization and stationary phase*, Asian J. Math. **8** (2004), 747–768. https://arxiv.org/abs/math/0307366.

[41] C. Sabbah, *An explicit stationary phase formula for the local formal Fourier–Laplace transform*, in Singularities I, Contemp. Math. **474** (2008), 309–330, especially Remark 5.7. https://arxiv.org/abs/0706.3570. Other cited inputs are Remarks 5.2 and 5.3(4).

[42] E. R. Kolchin, *Algebraic groups and algebraic dependence*, Amer. J. Math. **90** (1968), 1151–1164.

[43] M. van der Put and M. F. Singer, *Galois Theory of Linear Differential Equations*, Grundlehren Math. Wiss. **328**, Springer, 2003.

[44] N. M. Katz, *Exponential Sums and Differential Equations*, Ann. of Math. Stud. **124**, Princeton University Press, 1990.

[45] B. Malgrange, *Équations différentielles à coefficients polynomiaux*, Progr. Math. **96**, Birkhäuser, 1991.

[46] A. Baker, *Linear forms in the logarithms of algebraic numbers (III)*, Mathematika **14** (1967), 220–228. https://doi.org/10.1112/S0025579300003843.

[47] Y. Polyanskiy and M. Sellke, *Nonparametric MLE for Gaussian location mixtures: Certified computation and generic behavior*, arXiv:2503.20193v1 (2025), Proposition 1.1.

[48] A. Ghosh, S. Boyd, and A. Saberi, *Minimizing effective resistance of a graph*, SIAM Review **50** (2008), no. 1, 37–66. doi:10.1137/050645452. Additional source link: https://web.stanford.edu/~boyd/papers/pdf/eff_res.pdf.

[49] KEIO, *Minimal Differential and Recurrence Orders: Bessel Periods, Bounded-Height Tableaux, and Banana Integrals*, supplied unpublished manuscript [032], 18 October 2026. Not externally peer reviewed.

[50] KEIO, *Transcendence of Algebraic Arc Lengths and Inverse Periods: Division Dichotomies, Beta Quantiles, and Arithmetic Transfers*, supplied unpublished manuscript [024], 12 October 2026. Not externally peer reviewed.

[51] J. Fresán, C. Sabbah and J.-D. Yu, *Hodge theory of Kloosterman connections*, Duke Math. J. **171** (2022), 1649–1747. https://arxiv.org/abs/1810.06454. Additional source link: https://perso.pages.math.cnrs.fr/users/claude.sabbah/articles/kloosterman-revised10.pdf.

[52] P. Etingof, *Lie Groups and Lie Algebras I*, MIT 18.745, 2020, Lectures 25–26. Highest-weight theory and the Weyl dimension formula. MIT OpenCourseWare, Lecture 26.

[53] S. Boukraa, S. Hassani, J.-M. Maillard and J.-A. Weil, *Differential algebra on lattice Green functions and Calabi–Yau operators (unabridged version)*, 2014. https://arxiv.org/abs/1311.2470.

[54] J. Nagura, *On the interval containing at least one prime number*, Proceedings of the Japan Academy **28** (1952), 177–181. https://doi.org/10.3792/pja/1195570997. The theorem on page 180 includes $x < p < 6x/5$ for every real $x \geq 25$.

[55] S. Dhillon and E. Saha, *On arithmetic nature of special values of the incomplete beta function*, Acta Arithmetica **218** (2025), 273–284. https://doi.org/10.4064/aa240610-6-10.

[56] E. Saha, *Arithmetic nature of the special values of the incomplete beta function*, conference presentation, Recurrence, Transcendence, and Diophantine Approximation, 14 July 2025, slides 1–4. Included in the collected speed-talk slides at https://emresertoz.com/wp-content/uploads/2025/07/speed_talks.pdf.

[57] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, §§8.17, 10.29, 10.31, 10.32, 15.8, 15.10, 16.4. https://dlmf.nist.gov/. Incomplete beta functions: https://dlmf.nist.gov/8.17. Bessel functions: https://dlmf.nist.gov/10.

[58] Y. Wu and P. Yang, *Optimal estimation of Gaussian mixtures via denoised method of moments*, Annals of Statistics **48** (2020), 1981–2007. arXiv:1807.07237, Theorem 5, Proposition 8 and Lemma 24.

[59] N. M. Katz, On the calculation of some differential Galois groups, *Invent. Math.* **87** (1987), 13–61, Corollary 4.4.8. https://web.math.princeton.edu/~nmk/old/calcdiffgal.pdf.

[60] I. Gaiur, V. Rubtsov and D. van Straten, Product formulas for the higher Bessel functions, *Selecta Math. (N.S.)* **32** (2026), article 12. https://doi.org/10.1007/s00029-025-01114-w. Preprint: https://arxiv.org/abs/2405.03015.

[61] A. Baker, *Linear forms in the logarithms of algebraic numbers*, Mathematika **13** (1966), 204–216. https://doi.org/10.1112/S0025579300003971.

[62] S. Dasgupta, *Ranks of matrices of logarithms of algebraic numbers I: the theorems of Baker and Waldschmidt–Masser*, Essential Number Theory **2** (2023), 93–138. https://doi.org/10.2140/ent.2023.2.93. Author version: https://arxiv.org/abs/2303.02037.

[63] J. A. Soloff, A. Guntuboyina and B. Sen, *Multivariate, heteroscedastic empirical Bayes via nonparametric maximum likelihood*, Journal of the Royal Statistical Society Series B **87** (2025), 1–32. https://doi.org/10.1093/jrsssb/qkae040.

[64] J. A. Soloff, A. Guntuboyina and B. Sen, *Multivariate, heteroscedastic empirical Bayes via nonparametric maximum likelihood*, arXiv:2109.03466v2, version 2 dated 24 August 2026. The cited theorem numbering follows this updated preprint: Theorem 11, Remark 12, and Appendices D.1 and E. https://arxiv.org/html/2109.03466v2. Related journal version: Journal of the Royal Statistical Society, Series B **87** (2025), 1–32, https://doi.org/10.1093/jrsssb/qkae040; its deconvolution result is numbered Theorem 10.

[65] KEIO, *Global Algebraicity and Exact Complexity*, preceding unpublished research manuscript, 19 October 2026. Reproduced as Part V of the present omnibus; retained here as a historical provenance reference.

[66] KEIO, *Global Galois Groups and Complete Parameter Classifications*, preceding unpublished research manuscript, 19 October 2026. Reproduced as Part VI of the present omnibus; retained here as a historical provenance reference.

[67] W. Van Assche and E. Coussement, Some classical multiple orthogonal polynomials, *Journal of Computational and Applied Mathematics* **127** (2001), 317–347. doi:10.1016/S0377-0427(00)00503-3. Author manuscript: https://wis.kuleuven.be/analyse/members/walter/clasmult.pdf.

[68] F. Beukers, *A refined version of the Siegel–Shidlovskii theorem*, Ann. of Math. (2) **163** (2006), no. 1, 369–379. See Theorem 1.1 on p. 370. https://annals.math.princeton.edu/wp-content/uploads/annals-v163-n1-p08.pdf The cited inputs also include Corollary 1.4. Additional source link: https://doi.org/10.4007/annals.2006.163.369.

[69] L. Kania, L. Wasserman and S. Balakrishnan, *Testing Random Effects for Binomial Data*, arXiv:2504.13977v3, 2025, Section 3 and Theorem 3. https://arxiv.org/pdf/2504.13977.

[70] R. Korlakai Vinayak, W. Kong, G. Valiant and S. Kakade, *Maximum likelihood estimation for learning populations of parameters*, Proceedings of ICML 2019, PMLR **97**, 6448–6457. https://proceedings.mlr.press/v97/vinayak19a.html.

[71] C. Musco, C. Musco, L. Rosenblatt and A. V. Singh, *Sharper Bounds for Chebyshev Moment Matching, with Applications*, Proceedings of COLT 2025, PMLR **291**, 4309–4358. https://proceedings.mlr.press/v291/musco25a.html.

[72] P. Diaconis, K. Khare and L. Saloff-Coste, *Gibbs sampling, exponential families and orthogonal polynomials*, Statistical Science **23** (2008), 151–178.

[73] H. Dette, *New bounds for Hahn and Krawtchouk polynomials*, SIAM Journal on Mathematical Analysis **26** (1995), 1647–1659. https://arxiv.org/abs/math/9406223; Corollary 3.6(b).

[74] A. Straub and W. Zudilin, Sums of powers of binomials, their Apéry limits, and Franel's suspicions, *Int. Math. Res. Not.* **2023** (2023), no. 11, 9861–9879. https://arminstraub.com/downloads/pub/franel-aperylimits.pdf. The earlier citation used the author manuscript dated 19 March 2022 at the same URL.

[75] M. Chamberland and A. Straub, Apéry limits: experiments and proofs, *Amer. Math. Monthly* **128** (2021), no. 9, 811–824. https://arminstraub.com/downloads/pub/aperylimits.pdf.

[76] M. van Hoeij, Submodule approach to creative telescoping, preprint, 2024, Remark 6. https://arxiv.org/abs/2401.08455.

[77] D. Arinkin, *Fourier transform and middle convolution for irregular D-modules*, 2008, especially Corollary 2.3 and §3.3. https://arxiv.org/abs/0808.0699.

[78] Z. Miao, W. Kong, R. Korlakai Vinayak, W. Sun and F. Han, *Fisher–Pitman permutation tests based on nonparametric Poisson mixtures with application to single cell genomics*, author version dated 8 June 2021, arXiv:2106.03022v1, Theorems 6.1–6.2 and Remark 6.1. https://ramyakv.github.io/testing-poisson-npmle.pdf. The support-growth comparisons refer specifically to this author version. Additional source link: https://arxiv.org/abs/2106.03022v1.

[79] I. M. Gessel, *Symmetric functions and P-recursiveness*, J. Combin. Theory Ser. A **53** (1990), 257–285.

[80] M. Mishna, *On standard Young tableaux of bounded height*, in *Recent Trends in Algebraic Combinatorics*, Assoc. Women Math. Ser. 16, Springer, 2019, 281–303; https://arxiv.org/abs/1805.08130.

[81] D. J. Grabiner and P. Magyar, *Random walks in Weyl chambers and the decomposition of tensor powers*, J. Algebraic Combin. **2** (1993), 239–260.

[82] E. Y. Jin, C. M. Reidys, and R. R. Wang, *Asymptotic analysis of k-noncrossing matchings*, arXiv:0803.0848 (2008), equation (1.4). https://arxiv.org/abs/0803.0848.

[83] M. F. Singer, *Introduction to the Galois theory of linear differential equations*, in *Algebraic Theory of Differential Equations*, London Math. Soc. Lecture Note Ser. 357, Cambridge University Press, 2009, Example 1.3.32; https://arxiv.org/abs/0712.4124.

[84] P. J. Forrester and S. Kumar, *Differential recurrences for the distribution of the trace of the β-Jacobi ensemble*, Physica D **434** (2022), 133220. https://doi.org/10.1016/j.physd.2022.133220. Preprint arXiv:2011.00787v2, Corollary 2.5, equations (2.19)–(2.20). https://arxiv.org/abs/2011.00787v2. Additional source link: https://arxiv.org/html/2011.00787v2.

[85] T. Pham-Gia and N. Turkkan, *Distribution of the linear combination of two general beta variables and applications*, Communications in Statistics–Theory and Methods **27** (1998), 1851–1869. https://doi.org/10.1080/03610929808832194.

[86] C. De Concini and C. Procesi, *The algebra of the box spline*. https://arxiv.org/abs/math/0602019.

[87] J.-L. Marichal and M. J. Mossinghoff, *Slices, slabs, and sections of the unit hypercube*, Online Journal of Analytic Combinatorics, Issue 3 (2008), article 1. https://arxiv.org/abs/math/0607715.

[88] Z. Xu, *Multivariate splines and polytopes*, Journal of Approximation Theory **163** (2011), 377–387. https://doi.org/10.1016/j.jat.2010.10.005. Author version: https://arxiv.org/abs/0806.1127.

[89] K.-H. Lee and N.-J. Park, *An exact gamma–beta scale-mixture law for the squared norm of a complex-Gaussian-scaled sum of hyperspherical random vectors*, AIMS Mathematics **11** (2026), no. 9, 28348–28382, Theorem 3.5, equation (3.26). https://doi.org/10.3934/math.20261129.

[90] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Theorem VII.7 (Newton–Puiseux expansions at a singularity). https://ac.cs.princeton.edu/home/AC.pdf.